

fong-spivak-invitation: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_F00001	17	0.998	a	a	Connections are symmetric, so if a is connected to b , then b is connected to a . Connec-
fong-spivak-invitation_F00002	17	0.998	b	b	Connections are symmetric, so if a is connected to b , then b is connected to a . Connec-
fong-spivak-invitation_F00003	53	1.000	$\mathcal{V}$	$\mathcal{V}$	preorders $\mathcal{V}$.
fong-spivak-invitation_F00004	15	1.000	$f: X \rightarrow Y$	$f: X \rightarrow Y$	maps. A map $f: X \rightarrow Y$ is a kind of observation of object X via a specified relationship
fong-spivak-invitation_F00005	15	1.000	X	X	maps. A map $f: X \rightarrow Y$ is a kind of observation of object X via a specified relationship
fong-spivak-invitation_F00006	15	1.000	Y	Y	it has with another object, Y . For example, think of X as the subject of an experiment
fong-spivak-invitation_F00007	15	1.000	f	f	Asking which aspects of X one wants to preserve under the observation f becomes
fong-spivak-invitation_F00008	15	1.000	$\mathbb{R}$	$\mathbb{R}$	tions f from $\mathbb{R}$ to $\mathbb{R}$ (where $\mathbb{R}$ is the set of real numbers), and we can think of them
fong-spivak-invitation_F00009	15	0.980	x	x	as observations: rather than view x “directly,” we only observe $f(x)$. Out of all the
fong-spivak-invitation_F00010	15	0.980	$f(x)$	$f(x)$	as observations: rather than view x “directly,” we only observe $f(x)$. Out of all the
fong-spivak-invitation_F00011	16	1.000	$f: \mathbb{R} \rightarrow \mathbb{R}$	$f: \mathbb{R} \rightarrow \mathbb{R}$	functions $f: \mathbb{R} \rightarrow \mathbb{R}$, only some of them preserve the order of numbers, only some of
fong-spivak-invitation_F00012	16	0.412	$x \leq y$	$x \leq y$	(a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$, for all $x, y \in \mathbb{R}$; ¹
fong-spivak-invitation_F00013	16	0.412	$f(x) \leq f(y)$	$f(x) \leq f(y)$	(a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$, for all $x, y \in \mathbb{R}$; ¹
fong-spivak-invitation_F00014	16	0.412	$x, y \in \mathbb{R}$	$x, y \in \mathbb{R}$	(a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$, for all $x, y \in \mathbb{R}$; ¹
fong-spivak-invitation_F00015	16	1.000	$ x - y = f(x) - f(y) $	$ x - y = f(x) - f(y) $	(b) <i>metric-preserving</i> if $ x - y = f(x) - f(y) $;
fong-spivak-invitation_F00016	16	1.000	$f(x + y) = f(x) + f(y)$	$f(x + y) = f(x) + f(y)$	(c) <i>addition-preserving</i> if $f(x + y) = f(x) + f(y)$.
fong-spivak-invitation_F00017	16	—	$\square$	$\square$	—
fong-spivak-invitation_F00018	16	0.942	(x, y)	(x, y)	¹ We are often taught to view functions $f: \mathbb{R} \rightarrow \mathbb{R}$ as plots in the (x, y) -coordinate system, where x is the
fong-spivak-invitation_F00019	16	1.000	y	y	domain (independent) variable and y is the codomain (dependent) variable. In this book, we do not adhere
fong-spivak-invitation_F00020	16	1.000	$f(x)=x^2$	$f(x) = x^2$	example consider the function $f(x) = x^2$. Then f being order-preserving would say that, for any
fong-spivak-invitation_F00021	16	1.000	$x, y \in \mathbb{R}$	$x, y \in \mathbb{R}$	$x, y \in \mathbb{R}$, if $x \leq y$ then $x^2 \leq y^2$; is that true?
fong-spivak-invitation_F00022	16	1.000	$x^2 \leq y^2$	$x^2 \leq y^2$	$x, y \in \mathbb{R}$, if $x \leq y$ then $x^2 \leq y^2$; is that true?
fong-spivak-invitation_F00023	17	1.000	c	c	tions are also transitive, meaning that if a is connected to b , and b is connected to c , then

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fong-spivak-invitation_F00024	17	■ 1.000	a, b	a, b	a is connected to c ; that is, all a, b , and c are connected. Friendship is not transitive – my
fong-spivak-invitation_F00025	17	■ 1.000	Φ	Φ	is observing this system. Her observation of interest, which we call Φ , extracts a single
fong-spivak-invitation_F00026	17	■ 1.000	A	A	Joining two systems A and B is performed simply by combining their connections. That
fong-spivak-invitation_F00027	17	■ 1.000	B	B	Joining two systems A and B is performed simply by combining their connections. That
fong-spivak-invitation_F00028	17	■ 1.000	$A \vee B$	$A \vee B$	is, we shall say the <i>join</i> of systems A and B , denoted $A \vee B$, has a connection between
fong-spivak-invitation_F00029	17	■ 1.000	$z_{\{1\}}, \ldots, z_{\{n\}}$	$z_1, \dots, z_n$	points x and y if there are some points $z_1, \dots, z_n$ such that each of the following is
fong-spivak-invitation_F00030	17	■ 0.999	$B: x$	$B: x$	true in at least one of A or B : x is connected to z_1 , z_i is connected to z_{i+1} , and z_n is
fong-spivak-invitation_F00031	17	■ 0.999	$z_{\{1\}}, z_{\{i\}}$	z_1, z_i	true in at least one of A or B : x is connected to z_1 , z_i is connected to z_{i+1} , and z_n is
fong-spivak-invitation_F00032	17	■ 0.999	$z_{\{i+1\}}$	z_{i+1}	true in at least one of A or B : x is connected to z_1 , z_i is connected to z_{i+1} , and z_n is
fong-spivak-invitation_F00033	17	■ 0.999	$z_{\{n\}}$	z_n	true in at least one of A or B : x is connected to z_1 , z_i is connected to z_{i+1} , and z_n is
fong-spivak-invitation_F00034	18	■ 0.493	$\Phi(\left(\{ \}^{\odot}\right)=\Phi(\left(\{ \}^{\odot} 6 \right)=$	$\Phi(\odot) = \Phi(\odot 6) =$	We see above in Eq. (1.1) that $\Phi(\mathbb{V}) = \Phi(\mathbb{V}) = \text{false}$: in both cases $\bullet$ is not
fong-spivak-invitation_F00035	18	■ 0.999	$*$	$*$	connected to $*$. On the other hand, when we join these two systems as in Eq. (1.2), we
fong-spivak-invitation_F00036	18	■ 0.444	$\Phi(\left(\oplus^{\odot} \vee\right)=\Phi(\left(\odot \oplus\right)=$	$\Phi(\oplus \vee) = \Phi(\odot \oplus) =$	see that $\Phi(\mathbb{V} \vee \mathbb{V}) = \Phi(\mathbb{V}) = \text{true}$: in the joined system, $\bullet$ is connected to $*$. The
fong-spivak-invitation_F00037	18	■ 0.444	$\bullet$	$\bullet$	see that $\Phi(\mathbb{V} \vee \mathbb{V}) = \Phi(\mathbb{V}) = \text{true}$: in the joined system, $\bullet$ is connected to $*$. The
fong-spivak-invitation_F00038	19	■ 1.000	$A \leq B$	$A \leq B$	tems A and B , we say that $A \leq B$ if, whenever x is connected to y in A , then x is
fong-spivak-invitation_F00039	19	■ 1.000	$\leq$	$\leq$	This notion of $\leq$ leads to the following diagram:
fong-spivak-invitation_F00040	19	■ 1.000	$A \leq (A \vee B)$	$A \leq (A \vee B)$	system that is bigger than both A and B . That is, $A \leq (A \vee B)$ and $B \leq (A \vee B)$, and
fong-spivak-invitation_F00041	19	■ 1.000	$B \leq (A \vee B)$	$B \leq (A \vee B)$	system that is bigger than both A and B . That is, $A \leq (A \vee B)$ and $B \leq (A \vee B)$, and
fong-spivak-invitation_F00042	19	■ 0.996	C	C	for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-invitation_F00043	19	■ 0.996	$A \leq C$	$A \leq C$	for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-invitation_F00044	19	■ 0.996	$B \leq C$	$B \leq C$	for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-invitation_F00045	19	■ 0.996	$(A \vee B) \leq C$	$(A \vee B) \leq C$	for any C , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$. Let’s walk through this with an
fong-spivak-invitation_F00046	19	■ 0.999	$\{\bullet, *\}$	$\{\bullet, *\}$	1. Write down all the partitions of a two-element set $\{\bullet, *\}$, order them as above, and
fong-spivak-invitation_F00047	20	■ 0.883	$\mathbb{B}=\{$	$\mathbb{B} = \{$	The set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans also has an order, $\text{false} \leq \text{true}$:
fong-spivak-invitation_F00048	20	■ 0.883	$\}$	$\}$	The set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans also has an order, $\text{false} \leq \text{true}$:
fong-spivak-invitation_F00049	20	■ 0.997	$\not \leq$	$\not \leq$	Thus $\text{false} \leq \text{false}$, $\text{false} \leq \text{true}$, and $\text{true} \leq \text{true}$, but $\text{true} \not \leq \text{false}$.
fong-spivak-invitation_F00050	20	■ 0.999	$B.^2$	$B.^2$	In other words, $A \leq B$ if A implies B . ²
fong-spivak-invitation_F00051	20	■ 1.000	A, B	A, B	For any A, B in $\mathbb{B}$, we can again write $A \vee B$ to mean the least element that is greater

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fong-spivak-invitation_F00052	20	■ 1.000	$\mathbb{B}$	$\mathbb{B}$	For any A, B in $\mathbb{B}$, we can again write $A \vee B$ to mean the least element that is greater
fong-spivak-invitation_F00053	20	■ 1.000	$\Phi(A)=\Phi(B)=$	$\Phi(A) = \Phi(B) =$	with $\Phi(A) = \Phi(B) = \text{false}$, so $\Phi(A) \vee \Phi(B) = \text{false}$, but where $\Phi(A \vee B) =$
fong-spivak-invitation_F00054	20	■ 1.000	$\Phi(A) \vee \Phi(B)=$	$\Phi(A) \vee \Phi(B) =$	with $\Phi(A) = \Phi(B) = \text{false}$, so $\Phi(A) \vee \Phi(B) = \text{false}$, but where $\Phi(A \vee B) =$
fong-spivak-invitation_F00055	20	■ 1.000	$\Phi(A \vee B)=$	$\Phi(A \vee B) =$	with $\Phi(A) = \Phi(B) = \text{false}$, so $\Phi(A) \vee \Phi(B) = \text{false}$, but where $\Phi(A \vee B) =$
fong-spivak-invitation_F00056	20	■ 1.000	P	P	² In mathematical logic, <code>false</code> implies <code>true</code> but <code>true</code> does not imply <code>false</code> . That is “ P implies Q ”
fong-spivak-invitation_F00057	20	■ 1.000	Q	Q	² In mathematical logic, <code>false</code> implies <code>true</code> but <code>true</code> does not imply <code>false</code> . That is “ P implies Q ”
fong-spivak-invitation_F00058	21	■ 1.000	a, b, c	a, b, c	tree, or the names of your favorite fruits, or simply some symbols a, b, c . For example,
fong-spivak-invitation_F00059	21	■ 1.000	$A=\{h, 1\}$	$A = \{h, 1\}$	we write $A = \{h, 1\}$ to denote the set, called A , that contains exactly two elements, one
fong-spivak-invitation_F00060	21	■ 0.935	h	h	called h and one called 1. The set $\{h, h, 1, h, 1\}$ is exactly the same as A because they
fong-spivak-invitation_F00061	21	■ 0.935	$\{h, h, 1, h, 1\}$	$\{h, h, 1, h, 1\}$	called h and one called 1. The set $\{h, h, 1, h, 1\}$ is exactly the same as A because they
fong-spivak-invitation_F00062	21	■ 1.000	$x \in X$	$x \in X$	the notation doesn’t change the set. ³ For an arbitrary set X , we write $x \in X$ if x is an
fong-spivak-invitation_F00063	21	■ 1.000	$h \in A$	$h \in A$	element of X ; so we have $h \in A$ and $1 \in A$, but $0 \notin A$.
fong-spivak-invitation_F00064	21	■ 1.000	$1 \in A$	$1 \in A$	element of X ; so we have $h \in A$ and $1 \in A$, but $0 \notin A$.
fong-spivak-invitation_F00065	21	■ 1.000	$0 \notin A$	$0 \notin A$	element of X ; so we have $h \in A$ and $1 \in A$, but $0 \notin A$.
fong-spivak-invitation_F00066	21	■ 1.000	$\varnothing$	$\varnothing$	• $\varnothing$ denotes the empty set; it has no elements.
fong-spivak-invitation_F00067	21	■ 1.000	$\mathbb{N}$	$\mathbb{N}$	• $\mathbb{N}$ denotes the set of <i>natural numbers</i> ; it has elements $0, 1, 2, 3, \dots, 90^{717}, \dots$
fong-spivak-invitation_F00068	21	■ 1.000	$0, 1, 2, 3, \ldots, 90^{717}, \ldots$	$0, 1, 2, 3, \dots, 90^{717}, \dots$	• $\mathbb{N}$ denotes the set of <i>natural numbers</i> ; it has elements $0, 1, 2, 3, \dots, 90^{717}, \dots$.
fong-spivak-invitation_F00069	21	■ 0.996	$\underline{n}$	$\underline{n}$	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the <i>nth ordinal</i> ; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-invitation_F00070	21	■ 0.996	$n \in \mathbb{N}$	$n \in \mathbb{N}$	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the <i>nth ordinal</i> ; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-invitation_F00071	21	■ 0.996	n	n	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the <i>nth ordinal</i> ; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-invitation_F00072	21	■ 0.996	$1, 2, \ldots, n$	$1, 2, \dots, n$	• $\underline{n}$, for any $n \in \mathbb{N}$, denotes the <i>nth ordinal</i> ; it has n elements $1, 2, \dots, n$. For example,
fong-spivak-invitation_F00073	21	■ 0.999	$\underline{0}=\varnothing, \underline{1}=\{1\}$	$\underline{0} = \varnothing, \underline{1} = \{1\}$	$\underline{0} = \varnothing, \underline{1} = \{1\}$, and $\underline{5} = \{1, 2, 3, 4, 5\}$.
fong-spivak-invitation_F00074	21	■ 0.999	$\underline{5}=\{1, 2, 3, 4, 5\}$	$\underline{5} = \{1, 2, 3, 4, 5\}$	$\underline{0} = \varnothing, \underline{1} = \{1\}$, and $\underline{5} = \{1, 2, 3, 4, 5\}$.
fong-spivak-invitation_F00075	21	■ 0.398	$\mathbb{Z}$	$\mathbb{Z}$	• $\mathbb{Z}$, the set of <i>integers</i> ; it has elements $\dots, -2, -1, 0, 1, 2, \dots, 90^{717}, \dots$
fong-spivak-invitation_F00076	21	■ 0.398	$90^{717}, \ldots$	$90^{717}, \dots$	• $\mathbb{Z}$, the set of <i>integers</i> ; it has elements $\dots, -2, -1, 0, 1, 2, \dots, 90^{717}, \dots$
fong-spivak-invitation_F00077	21	■ 1.000	$\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457$	$\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457$	• $\mathbb{R}$, the set of <i>real numbers</i> ; it has elements like $\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457$,
fong-spivak-invitation_F00078	21	■ 0.977	90^{717}	90^{717}	90^{717} , etc.
fong-spivak-invitation_F00079	21	■ 1.000	$X \subseteq Y$	$X \subseteq Y$	Given sets X and Y , we say that X is a <i>subset</i> of Y , and write $X \subseteq Y$, if every element

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fong-spivak-invitation_F00080	21	■ 1.000	$\{h\} \subseteq A$	$\{h\} \subseteq A$	in X is also in Y . For example $\{h\} \subseteq A$. Note that the empty set $\emptyset := \{\}$ is a subset
fong-spivak-invitation_F00081	21	■ 1.000	$\varnothing := \{\}$	$\emptyset := \{\}$	in X is also in Y . For example $\{h\} \subseteq A$. Note that the empty set $\emptyset := \{\}$ is a subset
fong-spivak-invitation_F00082	21	■ 1.000	$\{y \in Y \mid P(y)\}$	$\{y \in Y \mid P(y)\}$	element of Y , we write $\{y \in Y \mid P(y)\}$ to mean the subset of those y 's that satisfy P .
fong-spivak-invitation_F00083	21	■ 1.000	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$	1. Is it true that $\mathbb{N} = \{n \in \mathbb{Z} \mid n > 0\}$?
fong-spivak-invitation_F00084	21	■ 0.741	$h, 1$	$h, 1$	³ If you want a notion where “ $h, 1$ ” is different from “ $h, h, 1, h, 1,$ ” you can use something called <i>bags</i> ,
fong-spivak-invitation_F00085	21	■ 0.741	$h, h, 1, h, 1$	$h, h, 1, h, 1$	³ If you want a notion where “ $h, 1$ ” is different from “ $h, h, 1, h,$ ” you can use something called <i>bags</i> ,
fong-spivak-invitation_F00086	21	■ 0.995	$Z :=$	$Z :=$	⁴ When we write $Z := f \circ \circ$, it means “assign the meaning $f \circ \circ$ to variable Z ,” whereas $Z = f \circ \circ$ means
fong-spivak-invitation_F00087	21	■ 0.995	Z	Z	⁴ When we write $Z := f \circ \circ$, it means “assign the meaning $f \circ \circ$ to variable Z ,” whereas $Z = f \circ \circ$ means
fong-spivak-invitation_F00088	21	■ 0.995	$Z =$	$Z =$	⁴ When we write $Z := f \circ \circ$, it means “assign the meaning $f \circ \circ$ to variable Z ,” whereas $Z = f \circ \circ$ means
fong-spivak-invitation_F00089	21	■ 1.000	$3+2:=5$	$3+2:=5$	$Z = f \circ \circ$ but not vice versa; indeed it <i>would not</i> be proper to write $3+2:=5$ or $\{\} := \emptyset$.
fong-spivak-invitation_F00090	21	■ 1.000	$\} := \varnothing$	$\} := \emptyset$	$Z = f \circ \circ$ but not vice versa; indeed it <i>would not</i> be proper to write $3+2:=5$ or $\{\} := \emptyset$.
fong-spivak-invitation_F00091	22	■ 1.000	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$	$\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$	2. Is it true that $\mathbb{N} = \{n \in \mathbb{Z} \mid n > 1\}$?
fong-spivak-invitation_F00092	22	■ 1.000	$\varnothing = \{n \in \mathbb{Z} \mid 1 < n < 2\}$	$\emptyset = \{n \in \mathbb{Z} \mid 1 < n < 2\}$	3. Is it true that $\emptyset = \{n \in \mathbb{Z} \mid 1 < n < 2\}$?
fong-spivak-invitation_F00093	22	■ 1.000	$X_{\{1\}}$	X_1	If both X_1 and X_2 are subsets of Y , their <i>union</i> , denoted $X_1 \cup X_2$, is also a subset of Y ,
fong-spivak-invitation_F00094	22	■ 1.000	$X_{\{2\}}$	X_2	If both X_1 and X_2 are subsets of Y , their <i>union</i> , denoted $X_1 \cup X_2$, is also a subset of Y ,
fong-spivak-invitation_F00095	22	■ 1.000	$X_{\{1\}} \cup X_{\{2\}}$	$X_1 \cup X_2$	If both X_1 and X_2 are subsets of Y , their <i>union</i> , denoted $X_1 \cup X_2$, is also a subset of Y ,
fong-spivak-invitation_F00096	22	■ 1.000	$Y = \{1, 2, 3, 4\}$	$Y = \{1, 2, 3, 4\}$	example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-invitation_F00097	22	■ 1.000	$X_{\{1\}} = \{1, 2\}$	$X_1 = \{1, 2\}$	example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-invitation_F00098	22	■ 1.000	$X_{\{2\}} = \{2, 4\}$	$X_2 = \{2, 4\}$	example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-invitation_F00099	22	■ 1.000	$X_{\{1\}} \cup X_{\{2\}} = \{1, 2, 4\}$	$X_1 \cup X_2 = \{1, 2, 4\}$	example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$, then $X_1 \cup X_2 = \{1, 2, 4\}$.
fong-spivak-invitation_F00100	22	■ 1.000	$\varnothing \cup X = X$	$\emptyset \cup X = X$	Note that $\emptyset \cup X = X$ for any $X \subseteq Y$.
fong-spivak-invitation_F00101	22	■ 1.000	$X_{\{1\}} \cap X_{\{2\}}$	$X_1 \cap X_2$	Similarly, if both X_1 and X_2 are subsets of Y , then their <i>intersection</i> , denoted $X_1 \cap X_2$,
fong-spivak-invitation_F00102	22	■ 1.000	$\{1, 2, 3\} \cap \{2, 5\} = \{2\}$	$\{1, 2, 3\} \cap \{2, 5\} = \{2\}$	X_1 and in X_2 , and no others. So $\{1, 2, 3\} \cap \{2, 5\} = \{2\}$.
fong-spivak-invitation_F00103	22	■ 1.000	$X_{\{\emptyset\}} = \varnothing, X_{\{1\}} = \{1\}, X_{\{2\}} = \{1, 2\}$	$X_0 = \emptyset, X_1 = \{1\}, X_2 = \{1, 2\}$	the sets $X_0 = \emptyset, X_1 = \{1\}, X_2 = \{1, 2\}$, etc. as subsets of $\mathbb{N}$, and we want to know
fong-spivak-invitation_F00104	22	■ 1.000	$\bigcup_{n \in \mathbb{N}} X_n$	$\bigcup_{n \in \mathbb{N}} X_n$	what the union of all of them is. This union is written $\bigcup_{n \in \mathbb{N}} X_n$, and it is the subset of $\mathbb{N}$
fong-spivak-invitation_F00105	22	■ 0.994	$X_{\{n\}}$	X_n	that contains every element of every X_n , but no others. Namely, $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$
fong-spivak-invitation_F00106	22	■ 0.994	$\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$	$\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$	that contains every element of every X_n , but no others. Namely, $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$
fong-spivak-invitation_F00107	22	■ 1.000	$n \geq 1\}$	$n \geq 1\}$	$n \geq 1\}$. Similarly one can write $\bigcap_{n \in \mathbb{N}} X_n$ for the intersection of all of them, which will

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fong-spivak-invitation_F00108	22	■ 1.000	$\bigcap_{n \in \mathbb{N}} X_n$	$\bigcap_{n \in \mathbb{N}} X_n$	$n \geq 1$). Similarly one can write $\bigcap_{n \in \mathbb{N}} X_n$ for the intersection of all of them, which will
fong-spivak-invitation_F00109	22	■ 0.998	$X \times Y$	$X \times Y$	Given two sets X and Y , the <i>product</i> $X \times Y$ of X and Y is the set of pairs (x, y) ,
fong-spivak-invitation_F00110	22	■ 1.000	$y \in Y$	$y \in Y$	where $x \in X$ and $y \in Y$.
fong-spivak-invitation_F00111	22	■ 1.000	$X \sqcup Y$	$X \sqcup Y$	in common. Given two sets X and Y , their <i>disjoint union</i> $X \sqcup Y$ is the set of pairs of the
fong-spivak-invitation_F00112	22	■ 1.000	$(x, 1)$	$(x, 1)$	form $(x, 1)$ or $(y, 2)$, where $x \in X$ and $y \in Y$.
fong-spivak-invitation_F00113	22	■ 1.000	$(y, 2)$	$(y, 2)$	form $(x, 1)$ or $(y, 2)$, where $x \in X$ and $y \in Y$.
fong-spivak-invitation_F00114	22	■ 1.000	$A := \{h, 1\}$	$A := \{h, 1\}$	Exercise 1.7. Let $A := \{h, 1\}$ and $B := \{1, 2, 3\}$.
fong-spivak-invitation_F00115	22	■ 1.000	$B := \{1, 2, 3\}$	$B := \{1, 2, 3\}$	Exercise 1.7. Let $A := \{h, 1\}$ and $B := \{1, 2, 3\}$.
fong-spivak-invitation_F00116	22	■ 1.000	$A \times B$	$A \times B$	3. There are six elements in $A \times B$; write them out.
fong-spivak-invitation_F00117	22	■ 1.000	$A \sqcup B$	$A \sqcup B$	4. There are five elements of $A \sqcup B$; write them out.
fong-spivak-invitation_F00118	22	■ 0.711	$\{h, 1, 2, 3\}$	$\{h, 1, 2, 3\}$	5. If we consider A and B as subsets of the set $\{h, 1, 2, 3\}$, there are four elements of
fong-spivak-invitation_F00119	22	■ 1.000	$A \cup B$	$A \cup B$	$A \cup B$; write them out.
fong-spivak-invitation_F00120	22	■ 0.998	$R \subseteq X \times Y$	$R \subseteq X \times Y$	$R \subseteq X \times Y$. A <i>binary relation on X</i> is a relation between X and X , i.e. a subset
fong-spivak-invitation_F00121	22	■ 1.000	$R \subseteq X \times X$	$R \subseteq X \times X$	$R \subseteq X \times X$.
fong-spivak-invitation_F00122	22	■ 1.000	$R \subseteq A \times A$	$R \subseteq A \times A$	It is convenient to use something called <i>infix notation</i> for binary relations $R \subseteq A \times A$.
fong-spivak-invitation_F00123	22	■ 0.992	$a \star b$	$a \star b$	This means one picks a symbol, say $\star$, and writes $a \star b$ to mean $(a, b) \in R$.
fong-spivak-invitation_F00124	22	■ 0.992	$(a, b) \in R$	$(a, b) \in R$	This means one picks a symbol, say $\star$, and writes $a \star b$ to mean $(a, b) \in R$.
fong-spivak-invitation_F00125	22	■ 1.000	$(5, 6) \in R$	$(5, 6) \in R$	$(5, 6) \in R$, we write $5 \leq 6$.
fong-spivak-invitation_F00126	22	■ 1.000	$5 \leq 6$	$5 \leq 6$	$(5, 6) \in R$, we write $5 \leq 6$.
fong-spivak-invitation_F00127	23	■ 0.997	$=, \approx, <, >$	$=, \approx, <, >$	Other examples of infix notation for relations are $=, \approx, <, >$. In number theory, we
fong-spivak-invitation_F00128	23	■ 1.000	$p \in P$	$p \in P$	Definition 1.10. If A is a set, a <i>partition</i> of A consists of a set P and, for each $p \in P$,
fong-spivak-invitation_F00129	23	■ 1.000	$A_p \subseteq A$	$A_p \subseteq A$	a nonempty subset $A_p \subseteq A$, such that
fong-spivak-invitation_F00130	23	■ 1.000	$\{A_p\}_{p \in P}$	$\{A_p\}_{p \in P}$	We may denote the partition by $\{A_p\}_{p \in P}$. We refer to P as the set of <i>part labels</i> and if
fong-spivak-invitation_F00131	23	■ 1.000	A_p	A_p	$p \in P$ is a part label, we refer to A_p as the <i>pth part</i> . The condition (1.5) says that each
fong-spivak-invitation_F00132	23	■ 1.000	p	p	$p \in P$ is a part label, we refer to A_p as the <i>pth part</i> . The condition (1.5) says that each
fong-spivak-invitation_F00133	23	■ 1.000	$a \in A$	$a \in A$	element $a \in A$ is in exactly one part.
fong-spivak-invitation_F00134	23	■ 1.000	$\{A_{p'}^{'}\}_{p' \in P'}$	$\{A_{p'}^{'}\}_{p' \in P'}$	We consider two different partitions $\{A_p\}_{p \in P}$ and $\{A_{p'}^{'}\}_{p' \in P'}$ of A to be the same if

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fong-spivak-invitation_F00135	23	■ 1.000	$p^{\{\backslash prime\}} \in P^{\{\backslash prime\}}$	$p' \in P'$	for each $p \in P$ there exists a $p' \in P'$ with $A_p = A'_{p'}$. In other words, if two ways
fong-spivak-invitation_F00136	23	■ 1.000	$A_{\{p\}}=A_{\{p^{\{\backslash prime\}}\}^{\{\backslash prime\}}}$	$A_p = A'_{p'}$	for each $p \in P$ there exists a $p' \in P'$ with $A_p = A'_{p'}$. In other words, if two ways
fong-spivak-invitation_F00137	23	■ 1.000	$a, b \in \{11, 12, 13, 21, 22, 23\}$	$a, b \in \{11, 12, 13, 21, 22, 23\}$	For any two elements <u>$a, b \in \{11, 12, 13, 21, 22, 23\}$</u> , let's allow ourselves to write a
fong-spivak-invitation_F00138	23	■ 1.000	$a \sim b$	$a \sim b$	twiddle (tilde) symbol $a \sim b$ between them if a and b are both in the same part. Write
fong-spivak-invitation_F00139	23	■ 0.923	(a, b)	(a, b)	down every pair of elements (a, b) that are in the same part. There should be $10.^5$
fong-spivak-invitation_F00140	23	■ 0.923	$10.^{\{5\}}$	$10.^5$	down every pair of elements (a, b) that are in the same part. There should be $10.^5$
fong-spivak-invitation_F00141	24	■ 0.998	$\sim$	$\sim$	give it infix notation $\sim$, satisfying the following three properties:
fong-spivak-invitation_F00142	24	■ 1.000	$a \sim a$	$a \sim a$	(a) $a \sim a$, for all $a \in A$,
fong-spivak-invitation_F00143	24	■ 1.000	$\{ \}^{\{6\}} b \sim a$	${}^6b \sim a$	(b) $a \sim b$ iff ⁶ $b \sim a$, for all $a, b \in A$,
fong-spivak-invitation_F00144	24	■ 1.000	$a, b \in A$	$a, b \in A$	(b) $a \sim b$ iff ⁶ $b \sim a$, for all $a, b \in A$,
fong-spivak-invitation_F00145	24	■ 1.000	$b \sim c$	$b \sim c$	(c) if $a \sim b$ and $b \sim c$ then $a \sim c$, for all $a, b, c \in A$.
fong-spivak-invitation_F00146	24	■ 1.000	$a \sim c$	$a \sim c$	(c) if $a \sim b$ and $b \sim c$ then $a \sim c$, for all $a, b, c \in A$.
fong-spivak-invitation_F00147	24	■ 1.000	$a, b, c \in A$	$a, b, c \in A$	(c) if $a \sim b$ and $b \sim c$ then $a \sim c$, for all $a, b, c \in A$.
fong-spivak-invitation_F00148	24	■ 1.000	$a \in A_{\{p\}}$	$a \in A_p$	some $p \in P$ such that $a \in A_p$ and $b \in A_p$. It is obvious that a is in the same part as
fong-spivak-invitation_F00149	24	■ 1.000	$b \in A_{\{p\}}$	$b \in A_p$	some $p \in P$ such that $a \in A_p$ and $b \in A_p$. It is obvious that a is in the same part as
fong-spivak-invitation_F00150	24	■ 0.945	$X \subseteq A$	$X \subseteq A$	saying what the parts are. Say that a subset $X \subseteq A$ is $(\sim)$ -closed if, for every $x \in X$
fong-spivak-invitation_F00151	24	■ 0.980	$x^{\{\backslash prime\}} \sim x$	$x' \sim x$	and $x' \sim x$, we have $x' \in X$. Say that a subset $X \subseteq A$ is $(\sim)$ -connected if it is
fong-spivak-invitation_F00152	24	■ 0.980	$x^{\{\backslash prime\}} \in X$	$x' \in X$	and $x' \sim x$, we have $x' \in X$. Say that a subset $X \subseteq A$ is $(\sim)$ -connected if it is
fong-spivak-invitation_F00153	24	■ 0.840	$x \sim y$	$x \sim y$	nonempty and $x \sim y$ for every $x, y \in X$. Then the parts corresponding to $\sim$ are exactly
fong-spivak-invitation_F00154	24	■ 0.840	$x, y \in X$	$x, y \in X$	nonempty and $x \sim y$ for every $x, y \in X$. Then the parts corresponding to $\sim$ are exactly
fong-spivak-invitation_F00155	24	■ 0.998	$(\sim)$	$(\sim)$	$(\sim)$ -closed and $(\sim)$ -connected subsets $\{A_p\}_{p \in P}$.
fong-spivak-invitation_F00156	24	■ 1.000	$p \neq q$	$p \neq q$	2. Show that if $p \neq q$, i.e. if A_p and A_q are not exactly the same set, then $A_p \cap A_q = \emptyset$.
fong-spivak-invitation_F00157	24	■ 1.000	$A_{\{q\}}$	A_q	2. Show that if $p \neq q$, i.e. if A_p and A_q are not exactly the same set, then $A_p \cap A_q = \emptyset$.
fong-spivak-invitation_F00158	24	■ 1.000	$A_{\{p\}} \cap A_{\{q\}}=\varnothing$	$A_p \cap A_q = \emptyset$	2. Show that if $p \neq q$, i.e. if A_p and A_q are not exactly the same set, then $A_p \cap A_q = \emptyset$.
fong-spivak-invitation_F00159	24	■ 1.000	$A=\bigcup_{p \in P} A_{\{p\}}$	$A = \bigcup_{p \in P} A_p$	3. Show that $A = \bigcup_{p \in P} A_p$.
fong-spivak-invitation_F00160	24	■ 1.000	$A / \sim$	$A / \sim$	<i>quotient</i> $A / \sim$ of A under $\sim$ is the set of parts of the corresponding partition.

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fong-spivak-invitation_F00161	25	■ 1.000	S	S	Definition 1.17. Let S and T be sets. A function from S to T is a subset $F \subseteq S \times T$
fong-spivak-invitation_F00162	25	■ 1.000	T	T	Definition 1.17. Let S and T be sets. A function from S to T is a subset $F \subseteq S \times T$
fong-spivak-invitation_F00163	25	■ 1.000	$F \subseteq S \times T$	$F \subseteq S \times T$	Definition 1.17. Let S and T be sets. A function from S to T is a subset $F \subseteq S \times T$
fong-spivak-invitation_F00164	25	■ 1.000	$s \in S$	$s \in S$	such that for all $s \in S$, there exists a unique $t \in T$ with $(s, t) \in F$.
fong-spivak-invitation_F00165	25	■ 1.000	$t \in T$	$t \in T$	such that for all $s \in S$, there exists a unique $t \in T$ with $(s, t) \in F$.
fong-spivak-invitation_F00166	25	■ 1.000	$(s, t) \in F$	$(s, t) \in F$	such that for all $s \in S$, there exists a unique $t \in T$ with $\boxed{(s, t) \in F}$.
fong-spivak-invitation_F00167	25	■ 1.000	F	F	The function F is often denoted $F: S \rightarrow T$. From now on, we write $F(s) = t$, or
fong-spivak-invitation_F00168	25	■ 1.000	$F: S \rightarrow T$	$F: S \rightarrow T$	The function F is often denoted $F: S \rightarrow T$. From now on, we write $F(s) = t$, or
fong-spivak-invitation_F00169	25	■ 1.000	$F(s)=t$	$F(s) = t$	The function F is often denoted $F: S \rightarrow T$. From now on, we write $F(s) = t$, or
fong-spivak-invitation_F00170	25	■ 1.000	$s \mapsto t$	$s \mapsto t$	sometimes $s \mapsto t$, to mean $(s, t) \in F$. For any $t \in T$, the <i>preimage of t along F</i> is the
fong-spivak-invitation_F00171	25	■ 1.000	t	t	sometimes $s \mapsto t$, to mean $(s, t) \in F$. For any $t \in T$, the <i>preimage of t along F</i> is the
fong-spivak-invitation_F00172	25	■ 1.000	$f^{-1}(t) := \{s \in S \mid F(s)=t\}$	$f^{-1}(t) := \{s \in S \mid F(s) = t\}$	subset $\boxed{f^{-1}(t) := \{s \in S \mid F(s) = t\}}$.
fong-spivak-invitation_F00173	25	■ 1.000	$s_1, s_2 \in S$	$s_1, s_2 \in S$	$F(s) = t$. A function is called <i>injective</i> , or an <i>injection</i> , if for all $t \in T$ and $\boxed{s_1, s_2 \in S}$
fong-spivak-invitation_F00174	25	■ 1.000	$F(s_1)=t$	$F(s_1) = t$	with $F(s_1) = t$ and $F(s_2) = t$, we have $s_1 = s_2$. A function is called <i>bijective</i> if it is
fong-spivak-invitation_F00175	25	■ 1.000	$F(s_2)=t$	$F(s_2) = t$	with $F(s_1) = t$ and $F(s_2) = t$, we have $s_1 = s_2$. A function is called <i>bijective</i> if it is
fong-spivak-invitation_F00176	25	■ 1.000	$s_1=s_2$	$s_1 = s_2$	with $F(s_1) = t$ and $F(s_2) = t$, we have $s_1 = s_2$. A function is called <i>bijective</i> if it is
fong-spivak-invitation_F00177	25	■ 0.918	$\rightarrow, \rightarrowtail, \twoheadrightarrow, \stackrel{\cong}{\rightarrow}$	$\rightarrow, \rightarrowtail, \twoheadrightarrow, \stackrel{\cong}{\rightarrow}$	We use various decorations on arrows, $\rightarrow, \rightarrowtail, \twoheadrightarrow, \stackrel{\cong}{\rightarrow}$ to denote these special sorts of
fong-spivak-invitation_F00178	25	■ 0.680	id_X	id_X	a set X , denoted id_X . It is the bijective function $\mathrm{id}_X(x) = x$.
fong-spivak-invitation_F00179	25	■ 0.680	$\mathrm{id}_X(x)=x$	$\mathrm{id}_X(x) = x$	a set X , denoted id_X . It is the bijective function $\mathrm{id}_X(x) = x$.
fong-spivak-invitation_F00180	25	■ 1.000	$f: A \rightarrow B$	$f: A \rightarrow B$	1. Find two sets A and B and a function $\boxed{f: A \rightarrow B}$ that is injective but not surjective.
fong-spivak-invitation_F00181	26	■ 1.000	$f: A \rightarrow \varnothing$	$f: A \rightarrow \varnothing$	Exercise 1.20. Suppose that A is a set and $f: A \rightarrow \varnothing$ is a function to the empty set.
fong-spivak-invitation_F00182	26	■ 1.000	$f: A \rightarrow P$	$f: A \rightarrow P$	functions out of A . Given a surjective function $f: A \twoheadrightarrow P$, where P is any other set,
fong-spivak-invitation_F00183	26	■ 1.000	$f^{-1}(p) \subseteq A$	$f^{-1}(p) \subseteq A$	the preimages $f^{-1}(p) \subseteq A$, one for each element $p \in P$, form a partition of A . Here is
fong-spivak-invitation_F00184	26	■ 1.000	$S:=\{11,12,13,21,22,23\}$	$S := \{11, 12, 13, 21, 22, 23\}$	Consider the partition of $\boxed{S := \{11, 12, 13, 21, 22, 23\}}$ shown below:
fong-spivak-invitation_F00185	26	■ 1.000	$P=\{a, b, c, d\}$	$P = \{a, b, c, d\}$	It has been partitioned into four parts, so let $P = \{a, b, c, d\}$ and let $f: S \twoheadrightarrow P$ be
fong-spivak-invitation_F00186	26	■ 1.000	$f: S \rightarrow P$	$f: S \rightarrow P$	It has been partitioned into four parts, so let $P = \{a, b, c, d\}$ and let $f: S \twoheadrightarrow P$ be
fong-spivak-invitation_F00187	26	■ 1.000	$F: X \rightarrow Y$	$F: X \rightarrow Y$	Definition 1.23. If $F: X \rightarrow Y$ is a function and $G: Y \rightarrow Z$ is a function, their <i>com-</i>

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fong-spivak-invitation_F00188	26	■ 1.000	$G: Y \rightarrow Z$	$G: Y \rightarrow Z$	Definition 1.23. If $F: X \rightarrow Y$ is a function and $G: Y \rightarrow Z$ is a function, their <i>com-</i>
fong-spivak-invitation_F00189	26	■ 0.999	$X \rightarrow Z$	$X \rightarrow Z$	<i>posite</i> is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$. It is often denoted
fong-spivak-invitation_F00190	26	■ 0.999	$G(F(x))$	$G(F(x))$	<i>posite</i> is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$. It is often denoted
fong-spivak-invitation_F00191	26	■ 0.996	$G \circ F$	$G \circ F$	$G \circ F$, but we prefer to denote it $F \circ G$. It takes any element $x \in X$, evaluates F to get
fong-spivak-invitation_F00192	26	■ 0.996	$F; G$	$F; G$	$G \circ F$, but we prefer to denote it $F \circ G$. It takes any element $x \in X$, evaluates F to get
fong-spivak-invitation_F00193	26	■ 1.000	$F(x) \in Y$	$F(x) \in Y$	an element $F(x) \in Y$ and then evaluates G to get an element $G(F(x))$.
fong-spivak-invitation_F00194	26	■ 1.000	G	G	an element $F(x) \in Y$ and then evaluates G to get an element $G(F(x))$.
fong-spivak-invitation_F00195	26	■ 0.997	$\{1\} \rightarrow X$	$\{1\} \rightarrow X$	$\{1\} \rightarrow X$, namely the function sending 1 to x . For example, the three functions $\{1\} \rightarrow$
fong-spivak-invitation_F00196	26	■ 0.997	$\{1\} \rightarrow$	$\{1\} \rightarrow$	$\{1\} \rightarrow X$, namely the function sending 1 to x . For example, the three functions $\{1\} \rightarrow$
fong-spivak-invitation_F00197	27	■ 1.000	$x: \{1\} \rightarrow X$	$x: \{1\} \rightarrow X$	function $x: \{1\} \rightarrow X$. Then evaluating F at x is given by the composite $F(x) = x \circ F$.
fong-spivak-invitation_F00198	27	■ 1.000	$F(x)=x; F$	$F(x) = x; F$	function $x: \{1\} \rightarrow X$. Then evaluating F at x is given by the composite $F(x) = x \circ F$.
fong-spivak-invitation_F00199	27	■ 1.000	$x \leq x$	$x \leq x$	(a) $x \leq x$; and
fong-spivak-invitation_F00200	27	■ 1.000	$y \leq z$	$y \leq z$	(b) if $x \leq y$ and $y \leq z$, then $x \leq z$.
fong-spivak-invitation_F00201	27	■ 1.000	$x \leq z$	$x \leq z$	(b) if $x \leq y$ and $y \leq z$, then $x \leq z$.
fong-spivak-invitation_F00202	27	■ 0.837	$y \leq x$	$y \leq x$	$y \leq x$, we write $x \cong y$ and say x and y are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting
fong-spivak-invitation_F00203	27	■ 0.837	$x \cong y$	$x \cong y$	$y \leq x$, we write $x \cong y$ and say x and y are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting
fong-spivak-invitation_F00204	27	■ 0.837	$X, \leq$	$X, \leq$	$y \leq x$, we write $x \cong y$ and say x and y are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting
fong-spivak-invitation_F00205	27	■ 0.981	$(X, =)$	$(X, =)$	$(X, =)$. This means that the only order relationships on X are of the form $x \leq x$; if
fong-spivak-invitation_F00206	27	■ 1.000	$x \neq y$	$x \neq y$	$x \neq y$ then neither $x \leq y$ nor $y \leq x$ hold.
fong-spivak-invitation_F00207	27	■ 0.931	$X \times X \subseteq X \times X$	$X \times X \subseteq X \times X$	preorder $(X, \leq)$ by equipping it with the total binary relation $X \times X \subseteq X \times X$. This is a
fong-spivak-invitation_F00208	28	■ 0.999	$x=y$	$x = y$	(c) $x \cong y$ implies $x = y$.
fong-spivak-invitation_F00209	28	■ 1.000	x, y	x, y	any two elements x, y for which $x \cong y$, i.e. for which $x \leq y$ and $y \leq x$.
fong-spivak-invitation_F00210	28	■ 1.000	$G=(V, A, s, t)$	$G = (V, A, s, t)$	Definition 1.31. A graph $G = (V, A, s, t)$ consists of a set V whose elements are called
fong-spivak-invitation_F00211	28	■ 1.000	V	V	Definition 1.31. A graph $G = (V, A, s, t)$ consists of a set V whose elements are called
fong-spivak-invitation_F00212	28	■ 1.000	$s, t: A \rightarrow V$	$s, t: A \rightarrow V$	<i>vertices</i> , a set A whose elements are called <i>arrows</i> , and two functions $s, t: A \rightarrow V$
fong-spivak-invitation_F00213	28	■ 1.000	$s(a)=v$	$s(a) = v$	known as the <i>source</i> and <i>target</i> functions respectively. Given $a \in A$ with $s(a) = v$ and
fong-spivak-invitation_F00214	28	■ 1.000	$t(a)=w$	$t(a) = w$	$t(a) = w$, we say that a is an arrow from v to w .

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fong-spivak-invitation_F00215	28	■ 1.000	v	v	$t(a) = w$, we say that a is an arrow from v to w .
fong-spivak-invitation_F00216	28	■ 1.000	w	w	$t(a) = w$, we say that a is an arrow from v to w .
fong-spivak-invitation_F00217	28	■ 1.000	$V=\{1,2,3,4\}$	$V = \{1, 2, 3, 4\}$	It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$. The source and target functions,
fong-spivak-invitation_F00218	28	■ 1.000	$A=\{a, b, c, d, e\}$	$A = \{a, b, c, d, e\}$	It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$. The source and target functions,
fong-spivak-invitation_F00221	120	■ 0.999	e	e	The functor Π_F puts into Airline Seat the set of those pairs (e, f) where e is an Econ-
fong-spivak-invitation_F00222	29	■ 0.928	$1 \rightarrow 2$	$1 \rightarrow 2$	of length 0. There are infinitely many paths $1 \rightarrow 2$ because one can loop and loop and
fong-spivak-invitation_F00223	29	■ 1.000	d	d	loop through d as many times as one pleases.
fong-spivak-invitation_F00224	29	■ 0.531	$P, \leq$	$P, \leq$	graph $G = (V, A, s, t)$ that gives a <i>presentation</i> of a preorder $(P, \leq)$. The elements of
fong-spivak-invitation_F00225	29	■ 1.000	$v \leq w$	$v \leq w$	P are the vertices V in G , and the order $\leq$ is given by $v \leq w$ iff there is a path $v \rightarrow w$.
fong-spivak-invitation_F00226	29	■ 1.000	$v \rightarrow w$	$v \rightarrow w$	P are the vertices V in G , and the order $\leq$ is given by $v \leq w$ iff there is a path $v \rightarrow w$.
fong-spivak-invitation_F00227	29	■ 1.000	$v \rightarrow v$	$v \rightarrow v$	For any vertex v , there is always a path $v \rightarrow v$, and this translates into the reflexivity
fong-spivak-invitation_F00228	29	■ 1.000	$u \rightarrow v$	$u \rightarrow v$	law from Definition 1.25. The fact that paths $u \rightarrow v$ and $v \rightarrow w$ can be concatenated to
fong-spivak-invitation_F00229	29	■ 1.000	$u \rightarrow w$	$u \rightarrow w$	a path $u \rightarrow w$ translates into the transitivity law.
fong-spivak-invitation_F00230	29	■ 0.998	$(P, \leq)$	$(P, \leq)$	Exercise 1.35. What preorder relation $(P, \leq)$ is depicted by the graph G in Exam-
fong-spivak-invitation_F00231	29	■ 0.898	$\left(p_{\{1\}}, p_{\{2\}}\right)$	(p_1, p_2)	ple 1.32? That is, write down the elements of P and write down every pair (p_1, p_2) for
fong-spivak-invitation_F00232	29	■ 1.000	$p_{\{1\}} \leq p_{\{2\}}$	$p_1 \leq p_2$	which $p_1 \leq p_2$.
fong-spivak-invitation_F00233	29	■ 1.000	$\{\bullet, \circ, *\}$	$\{\bullet, \circ, *\}$	Exercise 1.37. Let X be the set of partitions of $\{\bullet, \circ, *\}$; it has five elements and an
fong-spivak-invitation_F00234	29	■ 1.000	$\mathbb{N}:=\{0,1,2,3,\ldots\}$	$\mathbb{N} := \{0, 1, 2, 3, \dots\}$	Example 1.40 (Natural numbers). The natural numbers $\mathbb{N} := \{0, 1, 2, 3, \dots\}$ are a pre-
fong-spivak-invitation_F00235	29	■ 1.000	$0 \leq 1$	$0 \leq 1$	order with the order given by the usual size ordering, e.g. $0 \leq 1$ and $5 \leq 100$. This is
fong-spivak-invitation_F00236	29	■ 1.000	$5 \leq 100$	$5 \leq 100$	order with the order given by the usual size ordering, e.g. $0 \leq 1$ and $5 \leq 100$. This is
fong-spivak-invitation_F00237	29	■ 1.000	$m \leq n$	$m \leq n$	a total order: either $m \leq n$ or $n \leq m$ for all m, n . One can see that its Hasse diagram
fong-spivak-invitation_F00238	29	■ 1.000	$n \leq m$	$n \leq m$	a total order: either $m \leq n$ or $n \leq m$ for all m, n . One can see that its Hasse diagram
fong-spivak-invitation_F00239	29	■ 1.000	m, n	m, n	a total order: either $m \leq n$ or $n \leq m$ for all m, n . One can see that its Hasse diagram
fong-spivak-invitation_F00240	30	■ 0.989	$a \leq b$	$a \leq b$	and b – namely all those in the middle row – which satisfy neither $a \leq b$ nor $b \leq a$.
fong-spivak-invitation_F00241	30	■ 0.989	$b \leq a$	$b \leq a$	and b – namely all those in the middle row – which satisfy neither $a \leq b$ nor $b \leq a$.
fong-spivak-invitation_F00242	30	■ 1.000	$n=m$	$n = m$	$n = m$. Another ordering is the reverse ordering, such as $5 \leq 3$ and $3 \leq 2$, like how
fong-spivak-invitation_F00243	30	■ 1.000	$5 \leq 3$	$5 \leq 3$	$n = m$. Another ordering is the reverse ordering, such as $5 \leq 3$ and $3 \leq 2$, like how
fong-spivak-invitation_F00244	30	■ 1.000	$3 \leq 2$	$3 \leq 2$	$n = m$. Another ordering is the reverse ordering, such as $5 \leq 3$ and $3 \leq 2$, like how

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fong-spivak-invitation_F00245	30	■ 1.000	m	m	m without remainder. In this ordering $2 \leq 4$, for example, but $2 \not\leq 3$, since there is a
fong-spivak-invitation_F00246	30	■ 1.000	$2 \leq 4$	$2 \leq 4$	m without remainder. In this ordering $2 \leq 4$, for example, but $2 \not\leq 3$, since there is a
fong-spivak-invitation_F00247	30	■ 1.000	$2 \not\leq 3$	$2 \not\leq 3$	m without remainder. In this ordering $2 \leq 4$, for example, but $2 \not\leq 3$, since there is a
fong-spivak-invitation_F00248	30	■ 0.994	$1, 2, \ldots, 10$	$1, 2, \dots, 10$	Exercise 1.41. Write down the numbers 1, 2, ..., 10 and draw an arrow $a \rightarrow b$ if a
fong-spivak-invitation_F00249	30	■ 0.994	$a \rightarrowtail b$	$a \rightarrow b$	Exercise 1.41. Write down the numbers $1, 2, \dots, 10$ and draw an arrow $a \rightarrow b$ if a
fong-spivak-invitation_F00250	30	■ 0.999	$-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100 / 3$	$-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100/3$	ordering,” e.g. $-500 < -499 < 0 < \sqrt{2} < 100/3$.
fong-spivak-invitation_F00251	30	■ 1.000	$\mathrm{P}(X)$	$P(X)$	set of X ; we denote it $P(X)$. The power set can naturally be given an order by inclusion
fong-spivak-invitation_F00252	31	■ 1.000	$X = \{0, 1, 2\}$	$X = \{0, 1, 2\}$	For example, taking $X = \{0, 1, 2\}$, we depict $P(X)$ as
fong-spivak-invitation_F00253	31	■ 0.933	$\mathrm{P}(\{1, 2, \ldots, n\})$	$P\{1, 2, \dots, n\}$, ⁷	See the cube? The Hasse diagram for the power set of a finite set, say $P\{1, 2, \dots, n\}$, ⁷
fong-spivak-invitation_F00254	31	■ 0.998	$\mathrm{P}(\varnothing), \mathrm{P}\{1\}$	$P(\varnothing), P\{1\}$	Exercise 1.46. Draw the Hasse diagrams for $P(\varnothing)$, $P\{1\}$, and $P\{1, 2\}$.
fong-spivak-invitation_F00255	31	■ 0.998	$\mathrm{P}\{1, 2\}$	$P\{1, 2\}$	Exercise 1.46. Draw the Hasse diagrams for $P(\varnothing)$, $P\{1\}$, and $P\{1, 2\}$.
fong-spivak-invitation_F00256	31	■ 0.981	$\operatorname{Prt}(A)$	$\operatorname{Prt}(A)$	think about how we might order the set $\operatorname{Prt}(A)$ of <i>all partitions</i> of A , for some set A . In
fong-spivak-invitation_F00257	31	■ 0.999	$q \in Q$	$q \in Q$	partition P is <i>finer</i> than a partition Q if, for every part $p \in P$, there is a part $q \in Q$
fong-spivak-invitation_F00258	31	■ 1.000	$A_p \subseteq A_q$	$A_p \subseteq A_q$	such that $A_p \subseteq A_q$. We could also say that Q is <i>coarser</i> than P .
fong-spivak-invitation_F00259	31	■ 1.000	$g: A \rightarrowtail Q$	$g: A \rightarrow Q$	tions out of A . Then $f: A \twoheadrightarrow P$ is finer than $g: A \twoheadrightarrow Q$ if there is a function $h: P \rightarrow Q$
fong-spivak-invitation_F00260	31	■ 1.000	$h: P \rightarrowtail Q$	$h: P \rightarrow Q$	tions out of A . Then $f: A \twoheadrightarrow P$ is finer than $g: A \twoheadrightarrow Q$ if there is a function $h: P \rightarrow Q$
fong-spivak-invitation_F00261	31	■ 1.000	$f; h = g$	$f; h = g$	such that $f \circ h = g$.
fong-spivak-invitation_F00262	31	■ 0.519	U	U	Example 1.49 (Upper sets). Given a preorder $(P, \leq)$, an <i>upper set</i> in P is a subset U of
fong-spivak-invitation_F00263	31	■ 1.000	$p \in U$	$p \in U$	P satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element
fong-spivak-invitation_F00264	31	■ 1.000	$p \leq q$	$p \leq q$	P satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element
fong-spivak-invitation_F00265	31	■ 1.000	$q \in U$	$q \in U$	P satisfying the condition that if $p \in U$ and $p \leq q$, then $q \in U$. “If p is an element
fong-spivak-invitation_F00266	31	■ 0.913	$U(P)$	$U(P)$	then so is anything bigger.” Write $U(P)$ for the set of upper sets in P . We can give the
fong-spivak-invitation_F00267	31	■ 0.906	$U \leq V$	$U \leq V$	set U an order by letting $U \leq V$ if U is contained in V .
fong-spivak-invitation_F00268	31	■ 0.973	$(\mathbb{B}, \leq)$	$(\mathbb{B}, \leq)$	For example, if $(\mathbb{B}, \leq)$ is the booleans (Example 1.29), then its preorder of upper sets
fong-spivak-invitation_F00269	31	■ 0.997	$U(\mathbb{B})$	$U(\mathbb{B})$	$U(\mathbb{B})$ is
fong-spivak-invitation_F00270	31	■ 1.000	$\mathrm{P} X$	PX	⁷ Note that we omit the parentheses here, writing PX instead of $P(X)$; throughout this book we will omit
fong-spivak-invitation_F00271	32	■ 0.845	$\subseteq \mathbb{B}$	$\subseteq \mathbb{B}$	The subset $\{\text{false}\} \subseteq \mathbb{B}$ is not an upper set, because $\text{false} \leq \text{true}$ and $\text{true} \notin$
fong-spivak-invitation_F00272	32	■ 0.845	$\notin$	$\notin$	The subset $\{\text{false}\} \subseteq \mathbb{B}$ is not an upper set, because $\text{false} \leq \text{true}$ and $\text{true} \notin$

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fong-spivak-invitation_F00273	32	■ 0.996	$(Q, \leq)$	$(Q, \leq)$	Example 1.51 (Product preorder). Given preorders $(P, \leq)$ and $(Q, \leq)$, we may define a
fong-spivak-invitation_F00274	32	■ 1.000	$P \times Q$	$P \times Q$	preorder structure on the product set $P \times Q$ by setting $(p, q) \leq (p', q')$ if and only if
fong-spivak-invitation_F00275	32	■ 1.000	$(p, q) \leq (p', q')$	$(p, q) \leq (p', q')$	preorder structure on the product set $P \times Q$ by setting $\boxed{p, q} < \boxed{p', q'}$ if and only if
fong-spivak-invitation_F00276	32	■ 1.000	$p \leq p'$	$p \leq p'$	$p \leq p'$ and $q \leq q'$. We call this the <i>product preorder</i> . This is a basic example of a more
fong-spivak-invitation_F00277	32	■ 1.000	$q \leq q'$	$q \leq q'$	$p \leq p'$ and $q \leq q'$. We call this the <i>product preorder</i> . This is a basic example of a more
fong-spivak-invitation_F00278	32	■ 0.763	$\left(P, \leq^{\mathrm{op}}\right)$	$(P, \leq^{\mathrm{op}})$	preorder $(P, \leq^{\mathrm{op}})$ to have the same set of elements, but with $p \leq^{\mathrm{op}} q$ if and only if
fong-spivak-invitation_F00279	32	■ 0.763	$p \leq^{\mathrm{op}} q$	$p \leq^{\mathrm{op}} q$	preorder $(P, \leq^{\mathrm{op}})$ to have the same set of elements, but with $p \leq^{\mathrm{op}} q$ if and only if
fong-spivak-invitation_F00280	32	■ 1.000	$q \leq p$	$q \leq p$	$q \leq p$.
fong-spivak-invitation_F00281	32	■ 0.999	$A, \leq_A$	$A, \leq_A$	Definition 1.54. A monotone map between preorders $(A, \leq_A)$ and $(B, \leq_B)$ is a function
fong-spivak-invitation_F00282	32	■ 0.999	$B, \leq_B$	$B, \leq_B$	Definition 1.54. A monotone map between preorders $(A, \leq_A)$ and $(B, \leq_B)$ is a function
fong-spivak-invitation_F00283	32	■ 0.993	$x, y \in A$	$x, y \in A$	$f: A \rightarrow B$ such that, for all elements $x, y \in A$, if $x \leq_A y$ then $f(x) \leq_B f(y)$.
fong-spivak-invitation_F00284	32	■ 0.993	$x \leq_A y$	$x \leq_A y$	$f: A \rightarrow B$ such that, for all elements $x, y \in A$, if $x \leq_A y$ then $f(x) \leq_B f(y)$.
fong-spivak-invitation_F00285	32	■ 0.993	$f(x) \leq_B f(y)$	$f(x) \leq_B f(y)$	$f: A \rightarrow B$ such that, for all elements $x, y \in A$, if $x \leq_A y$ then $\boxed{f(x) \leq_B f(y)}$.
fong-spivak-invitation_F00286	32	■ 1.000	$A \rightarrow B$	$A \rightarrow B$	A monotone map $A \rightarrow B$ between two preorders associates to each element of pre-
fong-spivak-invitation_F00287	33	■ 1.000	$x \in A$	$x \in A$	each element $x \in A$ to its image $f(x) \in B$. Note that the order must be preserved in
fong-spivak-invitation_F00288	33	■ 1.000	$f(x) \in B$	$f(x) \in B$	each element $x \in A$ to its image $f(x) \in B$. Note that the order must be preserved in
fong-spivak-invitation_F00289	33	■ 1.000	$f(y)$	$f(y)$	hand preorder A , then the image $f(x)$ will be above the image $f(y)$ in the right-hand
fong-spivak-invitation_F00290	33	■ 1.000	$\mathbb{B} \rightarrow \mathbb{N}$	$\mathbb{B} \rightarrow \mathbb{N}$	the preorder of natural numbers from Example 1.40. The map $\mathbb{B} \rightarrow \mathbb{N}$ sending <i>false</i>
fong-spivak-invitation_F00291	34	■ 1.000	$x \rightarrow y$	$x \rightarrow y$	because it preserves order: whenever there is a path $x \rightarrow y$ upstairs, there is a path
fong-spivak-invitation_F00292	34	■ 1.000	$F(x) \rightarrow F(y)$	$F(x) \rightarrow F(y)$	$\boxed{F(x) \rightarrow F(y)}$ downstairs.
fong-spivak-invitation_F00293	34	■ 0.998	$ \cdot : \mathcal{P}(X) \rightarrow \mathbb{N}$	$ \cdot : \mathcal{P}(X) \rightarrow \mathbb{N}$	tion from Example 1.45. The map $ \cdot : \mathcal{P}(X) \rightarrow \mathbb{N}$ sending each subset S to its number
fong-spivak-invitation_F00294	34	■ 0.996	$ S $	$ S $	of elements $ S $, also called its <i>cardinality</i> , is a monotone map.
fong-spivak-invitation_F00295	34	■ 1.000	$0 \leq 1 \leq 2 \leq 3$	$0 \leq 1 \leq 2 \leq 3$	2. Draw the Hasse diagram for the preorder $0 \leq 1 \leq 2 \leq 3$.
fong-spivak-invitation_F00296	34	■ 0.948	$ \cdot $	$ \cdot $	3. Draw the cardinality map $ \cdot $ from Example 1.57 as dashed lines between them.
fong-spivak-invitation_F00297	34	■ 0.993	$\mathcal{U}(P) \rightarrow \mathcal{P}(P)$	$\mathcal{U}(P) \rightarrow \mathcal{P}(P)$	$(P, \leq)$, the map $\mathcal{U}(P) \rightarrow \mathcal{P}(P)$ sending each upper set of $(P, \leq)$ to itself – considered
fong-spivak-invitation_F00298	34	■ 0.998	$\mathcal{P}(\mathbb{B})$	$\mathcal{P}(\mathbb{B})$	Example 1.49, and you drew the Hasse diagram for $\mathcal{P}(\mathbb{B})$ in Exercise 1.46. Now draw
fong-spivak-invitation_F00299	34	■ 1.000	$\uparrow p := \{p' \in P \mid p \leq p'\}$	$\uparrow p := \{p' \in P \mid p \leq p'\}$	1. Show that the set $\boxed{\uparrow p := \{p' \in P \mid p \leq p'\}}$ is an upper set, for any $p \in P$.

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fong-spivak-invitation_F00300	34	■ 0.801	$\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$	$\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$	2. Show that this construction defines a monotone map $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$.
fong-spivak-invitation_F00301	34	■ 0.999	$\uparrow(p') \subseteq \uparrow(p)$	$\uparrow(p') \subseteq \uparrow(p)$	3. Show that $p \leq p'$ in P if and only if $\uparrow(p') \subseteq \uparrow(p)$.
fong-spivak-invitation_F00302	34	■ 0.952	$b \geq a \leq c$	$b \geq a \leq c$	4. Draw a picture of the map $\uparrow$ in the case where P is the preorder ($b \geq a \leq c$) from
fong-spivak-invitation_F00303	34	■ 1.000	$f: (P, \leq_P) \rightarrow (Q, \leq_Q)$	$f: (P, \leq_P) \rightarrow (Q, \leq_Q)$	Exercise 1.62. As you know, a monotone map $f: (P, \leq_P) \rightarrow (Q, \leq_Q)$ consists of a
fong-spivak-invitation_F00304	34	■ 0.999	$f: P \rightarrow Q$	$f: P \rightarrow Q$	function $f: P \rightarrow Q$ that satisfies a “monotonicity” property. Show that when $(P, \leq_P)$
fong-spivak-invitation_F00305	34	■ 0.999	$(P, \leq_P)$	$(P, \leq_P)$	function $f: P \rightarrow Q$ that satisfies a “monotonicity” property. Show that when $(P, \leq_P)$
fong-spivak-invitation_F00306	34	■ 1.000	$P \rightarrow Q$	$P \rightarrow Q$	is a discrete preorder, then <i>every</i> function $P \rightarrow Q$ satisfies the monotonicity property,
fong-spivak-invitation_F00307	34	■ 1.000	$\leq_Q$	$\leq_Q$	regardless of the order $\leq_Q$.
fong-spivak-invitation_F00308	35	■ 1.000	$\mathrm{Prt}(X)$	$\mathrm{Prt}(X)$	Example 1.63. Recall from Example 1.47 that given a set X we define $\mathrm{Prt}(X)$ to be the
fong-spivak-invitation_F00309	35	■ 1.000	$s: X \rightarrow P$	$s: X \rightarrow P$	$s: X \twoheadrightarrow P$ for some set P .
fong-spivak-invitation_F00310	35	■ 1.000	$f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$	$f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$	Any surjective function $f: X \twoheadrightarrow Y$ induces a monotone map $f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$,
fong-spivak-invitation_F00311	35	■ 1.000	$s: Y \rightarrow P$	$s: Y \rightarrow P$	going “backwards.” It is defined by sending a partition $s: Y \twoheadrightarrow P$ to the composite
fong-spivak-invitation_F00312	35	■ 0.727	s	s	$f \circ s: X \twoheadrightarrow P$. ⁸
fong-spivak-invitation_F00313	35	■ 0.727	$X \rightarrow P$	$X \rightarrow P$	$f \circ s: X \twoheadrightarrow P$. ⁸
fong-spivak-invitation_F00314	35	■ 1.000	$f^*(P)$	$f^*(P)$	partitions P and Q of Y , and then find $f^*(P)$ and $f^*(Q)$.
fong-spivak-invitation_F00315	35	■ 1.000	$f^*(Q)$	$f^*(Q)$	partitions P and Q of Y , and then find $f^*(P)$ and $f^*(Q)$.
fong-spivak-invitation_F00316	35	■ 0.865	$P, \leq_P$	$P, \leq_P$	Proposition 1.65. For any preorder $(P, \leq_P)$, the identity function is monotone.
fong-spivak-invitation_F00317	35	■ 0.794	$(Q, \leq_Q)$	$(Q, \leq_Q)$	If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,
fong-spivak-invitation_F00318	35	■ 0.794	$(R, \leq_R)$	$(R, \leq_R)$	If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,
fong-spivak-invitation_F00319	35	■ 0.794	$g: Q \rightarrow R$	$g: Q \rightarrow R$	If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,
fong-spivak-invitation_F00320	35	■ 0.991	$(f; g): P \rightarrow R$	$(f; g): P \rightarrow R$	then $(f \circ g): P \rightarrow R$ is also monotone.
fong-spivak-invitation_F00321	35	■ 0.514	$\mathrm{id}_P: P \rightarrow P$	$\mathrm{id}_P: P \rightarrow P$	identity function $\mathrm{id}_P: P \rightarrow P$ is a monotone map $(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$ if and only if
fong-spivak-invitation_F00322	35	■ 0.514	$(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$	$(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$	identity function $\mathrm{id}_P: P \rightarrow P$ is a monotone map $(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$ if and only if
fong-spivak-invitation_F00323	35	■ 1.000	$p, q \in P$	$p, q \in P$	for all $p, q \in P$ we have $q \leq p$ whenever $p \leq q$. For historical reasons connected to
fong-spivak-invitation_F00324	35	■ 0.795	$p_1 = p_2$	$p_1 = p_2$	be made into a discrete preorder ($p_1 \leq p_2$ iff $p_1 = p_2$), and the round-trips return us
fong-spivak-invitation_F00325	36	■ 1.000	$g: Q \rightarrow P$	$g: Q \rightarrow P$	$f: P \rightarrow Q$ is called an <i>isomorphism</i> if there exists a monotone function $g: Q \rightarrow P$
fong-spivak-invitation_F00326	36	■ 0.957	$f; g = \mathrm{id}_P$	$f; g = \mathrm{id}_P$	such that $f \circ g = \mathrm{id}_P$ and $g \circ f = \mathrm{id}_Q$. This means that, for any $p \in P$ and $q \in Q$, we

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fong-spivak-invitation_F00327	36	■ 0.957	$g ; f=\operatorname{id}_{\{Q\}}$	$g; f = \operatorname{id}_Q$	such that $f \circ g = \operatorname{id}_P$ and $g \circ f = \operatorname{id}_Q$. This means that, for any $p \in P$ and $q \in Q$, we
fong-spivak-invitation_F00328	36	■ 1.000	g	g	We refer to g as the <i>inverse</i> of f , and vice versa: f is the inverse of g .
fong-spivak-invitation_F00329	36	■ 1.000	P, Q	P, Q	Example 1.71. Here are the Hasse diagrams for three preorders P, Q , and R , all of
fong-spivak-invitation_F00330	36	■ 1.000	R	R	Example 1.71. Here are the Hasse diagrams for three preorders P, Q , and R , all of
fong-spivak-invitation_F00331	36	■ 1.000	$f(a)=v, f(b)=w, f(c)=x, f(d)=y$	$f(a) = v, f(b) = w, f(c) = x, f(d) = y$	The map $f: P \rightarrow Q$ given by $f(a) = v, f(b) = w, f(c) = x, f(d) = y$, and
fong-spivak-invitation_F00332	36	■ 1.000	$f(e)=z$	$f(e) = z$	$f(e) = z$ has an inverse.
fong-spivak-invitation_F00333	36	■ 0.743	$x \rightarrowtail z$	$x \rightarrow z$	is an arrow $x \rightarrow z$ in the Hasse diagram for R and not one in Q , but in fact this arrow
fong-spivak-invitation_F00334	36	■ 1.000	$v \rightarrowtail y$	$v \rightarrow y$	drawn an arrow $v \rightarrow y$ in either Q or R and it would not have changed the preorder.
fong-spivak-invitation_F00335	36	■ 0.657	$\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrowtail \mathbb{B}$	$\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$	“Is $\bullet$ connected to $*$?” is a monotone map $\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$; see also Eq. (1.3).
fong-spivak-invitation_F00336	36	■ 1.000	$P \rightarrowtail \mathbb{B}$	$P \rightarrow \mathbb{B}$	Proposition 1.73. Let P be a preorder. Monotone maps $P \rightarrow \mathbb{B}$ are in one-to-one
fong-spivak-invitation_F00337	37	■ 1.000	$f: P \rightarrowtail \mathbb{B}$	$f: P \rightarrow \mathbb{B}$	<i>Proof.</i> Let $f: P \rightarrow \mathbb{B}$ be a monotone map. We will show that the subset
fong-spivak-invitation_F00338	37	■ 1.000	f^{-1}	f^{-1}	$f^{-1}(\operatorname{true}) \subseteq P$ is an upper set. Suppose $p \in f^{-1}(\operatorname{true})$ and $p \leq q$; then
fong-spivak-invitation_F00339	37	■ 1.000	$\subseteq P$	$\subseteq P$	$f^{-1}(\operatorname{true}) \subseteq P$ is an upper set. Suppose $p \in f^{-1}(\operatorname{true})$ and $p \leq q$; then
fong-spivak-invitation_F00340	37	■ 1.000	$p \in f^{-1}$	$p \in f^{-1}$	$f^{-1}(\operatorname{true}) \subseteq P$ is an upper set. Suppose $p \in f^{-1}(\operatorname{true})$ and $p \leq q$; then
fong-spivak-invitation_F00341	37	■ 0.617	$=f(p) \leq f(q)$	$= f(p) \leq f(q)$	$\operatorname{true} = f(p) \leq f(q)$. But in $\mathbb{B}$, if $\operatorname{true} \leq f(q)$ then $\operatorname{true} = f(q)$. This implies
fong-spivak-invitation_F00342	37	■ 0.617	$\leq f(q)$	$\leq f(q)$	$\operatorname{true} = f(p) \leq f(q)$. But in $\mathbb{B}$, if $\operatorname{true} \leq f(q)$ then $\operatorname{true} = f(q)$. This implies
fong-spivak-invitation_F00343	37	■ 0.617	$=f(q)$	$= f(q)$	$\operatorname{true} = f(p) \leq f(q)$. But in $\mathbb{B}$, if $\operatorname{true} \leq f(q)$ then $\operatorname{true} = f(q)$. This implies
fong-spivak-invitation_F00344	37	■ 1.000	$q \in f^{-1}$	$q \in f^{-1}$	$q \in f^{-1}(\operatorname{true})$ and thus shows that $f^{-1}(\operatorname{true})$ is an upper set.
fong-spivak-invitation_F00345	37	■ 0.522	$f_{\{U\}}: P \rightarrowtail \mathbb{B}$	$f_U: P \rightarrow \mathbb{B}$	Conversely, if U is an upper set in P , define $f_U: P \rightarrow \mathbb{B}$ such that $f_U(p) = \operatorname{true}$
fong-spivak-invitation_F00346	37	■ 0.522	$f_{\{U\}}(p)=\operatorname{true}$	$f_U(p) = \operatorname{true}$	Conversely, if U is an upper set in P , define $f_U: P \rightarrow \mathbb{B}$ such that $f_U(p) = \operatorname{true}$
fong-spivak-invitation_F00347	37	■ 1.000	$f_{\{U\}}(p)=$	$f_U(p) =$	when $p \in U$, and $f_U(p) = \operatorname{false}$ when $p \notin U$. This is a monotone map, because
fong-spivak-invitation_F00348	37	■ 1.000	$p \notin U$	$p \notin U$	when $p \in U$, and $f_U(p) = \operatorname{false}$ when $p \notin U$. This is a monotone map, because
fong-spivak-invitation_F00349	37	■ 1.000	$f_{\{U\}}(p)=\operatorname{true}=f(q)$	$f_U(p) = \operatorname{true} = f(q)$	if $p \leq q$, then either $p \in U$, so $q \in U$ and $f_U(p) = \operatorname{true} = f(q)$, or $p \notin U$, so
fong-spivak-invitation_F00350	37	■ 1.000	$\leq f_{\{U\}}(q)$	$\leq f_U(q)$	$f_U(p) = \operatorname{false} \leq f_U(q)$.
fong-spivak-invitation_F00351	37	■ 0.890	$f^*: \operatorname{U}(Q) \rightarrowtail \operatorname{U}(P)$	$f^*: \operatorname{U}(Q) \rightarrow \operatorname{U}(P)$	map. Then we can define a monotone map $f^*: \operatorname{U}(Q) \rightarrow \operatorname{U}(P)$ sending an upper set
fong-spivak-invitation_F00352	37	■ 1.000	$U \subseteqtail Q$	$U \subseteq Q$	$U \subseteq Q$ to the upper set $f^{-1}(U) \subseteq P$. We call this the <i>pullback along f</i> .
fong-spivak-invitation_F00353	37	■ 1.000	$f^{-1}(U) \subseteqtail P$	$f^{-1}(U) \subseteq P$	$U \subseteq Q$ to the upper set $f^{-1}(U) \subseteq P$. We call this the <i>pullback along f</i> .

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fong-spivak-invitation_F00354	37	■ 1.000	$f^{\{*\}}$	f^*	understood in terms of composition. Indeed, show that f^* is defined by taking $u: Q \rightarrow$
fong-spivak-invitation_F00355	37	■ 1.000	$u: Q \rightarrow$	$u: Q \rightarrow$	understood in terms of composition. Indeed, show that f^* is defined by taking $u: Q \rightarrow$
fong-spivak-invitation_F00356	37	■ 1.000	$(f; u): P \rightarrow \mathbb{B}$	$(f; u): P \rightarrow \mathbb{B}$	$\mathbb{B}$ to $(f; u): P \rightarrow \mathbb{B}$.
fong-spivak-invitation_F00357	37	■ 0.798	$(\mathbb{R}, \leq)$	$(\mathbb{R}, \leq)$	Consider the preorder $(\mathbb{R}, \leq)$ of real numbers ordered in the usual way. The subset
fong-spivak-invitation_F00358	37	■ 1.000	$\mathbb{N} \subseteq \mathbb{R}$	$\mathbb{N} \subseteq \mathbb{R}$	$\mathbb{N} \subseteq \mathbb{R}$ has many lower bounds, namely -1.5 is a lower bound: every element of $\mathbb{N}$ is
fong-spivak-invitation_F00359	37	■ 1.000	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$	1. Why is 0 a lower bound for $\{\frac{1}{n+1} \mid n \in \mathbb{N}\} \subseteq \mathbb{R}$?
fong-spivak-invitation_F00360	38	■ 0.752	$A \subseteq P$	$A \subseteq P$	Definition 1.76. Let $(P, \leq)$ be a preorder, and let $A \subseteq P$ be a subset. We say that an
fong-spivak-invitation_F00361	38	■ 1.000	$p \leq a$	$p \leq a$	(a) for all $a \in A$, we have $p \leq a$,
fong-spivak-invitation_F00362	38	■ 1.000	q	q	(b) for all q such that $q \leq a$ for all $a \in A$, we have that $q \leq p$.
fong-spivak-invitation_F00363	38	■ 1.000	$q \leq a$	$q \leq a$	(b) for all q such that $q \leq a$ for all $a \in A$, we have that $q \leq p$.
fong-spivak-invitation_F00364	38	■ 0.998	$p = \bigwedge A, p = \bigwedge_{a \in A} a$	$p = \bigwedge A, p = \bigwedge_{a \in A} a$	We write $p = \bigwedge A$, $p = \bigwedge_{a \in A} a$, or, if the dummy variable a is clear from context,
fong-spivak-invitation_F00365	38	■ 0.999	$p = \bigwedge_A a$	$p = \bigwedge_A a$	just $p = \bigwedge_A a$. If A just consists of two elements, say $A = \{a, b\}$, we can denote $\bigwedge A$
fong-spivak-invitation_F00366	38	■ 0.999	$A = \{a, b\}$	$A = \{a, b\}$	just $p = \bigwedge_A a$. If A just consists of two elements, say $A = \{a, b\}$, we can denote $\bigwedge A$
fong-spivak-invitation_F00367	38	■ 0.999	$\bigwedge A$	$\bigwedge A$	just $p = \bigwedge_A a$. If A just consists of two elements, say $A = \{a, b\}$, we can denote $\bigwedge A$
fong-spivak-invitation_F00368	38	■ 1.000	$a \wedge b$	$a \wedge b$	simply by $a \wedge b$.
fong-spivak-invitation_F00369	38	■ 1.000	$a \leq p$	$a \leq p$	(a) for all $a \in A$ we have $a \leq p$,
fong-spivak-invitation_F00370	38	■ 1.000	$a \leq q$	$a \leq q$	(b) for all q such that $a \leq q$ for all $a \in A$, we have that $p \leq q$.
fong-spivak-invitation_F00371	38	■ 0.777	$p = \bigvee A$	$p = \bigvee A$	We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$, or when $A = \{a, b\}$ we may simply write
fong-spivak-invitation_F00372	38	■ 0.777	$p = \bigvee_{a \in A} a$	$p = \bigvee_{a \in A} a$	We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$, or when $A = \{a, b\}$ we may simply write
fong-spivak-invitation_F00373	38	■ 1.000	$p = a \vee b$	$p = a \vee b$	$p = a \vee b$.
fong-spivak-invitation_F00374	38	■ 1.000	$p = \bigwedge A$	$p = \bigwedge A$	$A \subseteq P$, say $p = \bigwedge A$ and $q = \bigwedge A$ with $p \neq q$, which does not make sense if
fong-spivak-invitation_F00375	38	■ 1.000	$q = \bigwedge A$	$q = \bigwedge A$	$A \subseteq P$, say $p = \bigwedge A$ and $q = \bigwedge A$ with $p \neq q$, which does not make sense if
fong-spivak-invitation_F00376	38	■ 1.000	p, q	p, q	meets p, q are automatically isomorphic: the very definition of meet forces both $p \leq$
fong-spivak-invitation_F00377	38	■ 1.000	$p \leq q$	$p \leq q$	meets p, q are automatically isomorphic: the very definition of meet forces both $p \leq$
fong-spivak-invitation_F00378	38	■ 1.000	$p \cong q$	$p \cong q$	q and $q \leq p$, and thus we have $p \cong q$. So, for any $x \in P$, we have $p \leq x$ iff
fong-spivak-invitation_F00379	38	■ 1.000	$x \in P$	$x \in P$	q and $q \leq p$, and thus we have $p \cong q$. So, for any $x \in P$, we have $p \leq x$ iff
fong-spivak-invitation_F00380	38	■ 1.000	$p \leq x$	$p \leq x$	q and $q \leq p$, and thus we have $p \cong q$. So, for any $x \in P$, we have $p \leq x$ iff

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fong-spivak-invitation_F00381	38	■ 1.000	$q \leq x$	$q \leq x$	$q \leq x$ and $x \leq p$ iff $x \leq q$. Thus as long as we are only interested in elements of P
fong-spivak-invitation_F00382	38	■ 1.000	$x \leq p$	$x \leq p$	$q \leq x$ and $x \leq p$ iff $x \leq q$. Thus as long as we are only interested in elements of P
fong-spivak-invitation_F00383	38	■ 1.000	$x \leq q$	$x \leq q$	$q \leq x$ and $x \leq p$ iff $x \leq q$. Thus as long as we are only interested in elements of P
fong-spivak-invitation_F00384	38	■ 1.000	$P=\{p, q, r\}$	$P = \{p, q, r\}$	subset A need not have a meet or a join. Consider the three-element set $P = \{p, q, r\}$
fong-spivak-invitation_F00385	38	■ 1.000	$A=\{p, q\}$	$A = \{p, q\}$	with the discrete ordering. The set $A = \{p, q\}$ does not have a join in P because if x
fong-spivak-invitation_F00386	39	■ 0.630	$a, b\}$	$a, b\}$	Let A be the subset $\{a, b\}$ in the preorder specified by this Hasse diagram. Then both c
fong-spivak-invitation_F00387	39	■ 0.992	$c \leq d$	$c \leq d$	less than d . Note that, as in Remark 1.77, $c \leq d$ and $d \leq c$, so $c \cong d$. Such will always
fong-spivak-invitation_F00388	39	■ 0.992	$d \leq c$	$d \leq c$	less than d . Note that, as in Remark 1.77, $c \leq d$ and $d \leq c$, so $c \cong d$. Such will always
fong-spivak-invitation_F00389	39	■ 0.992	$c \cong d$	$c \cong d$	less than d . Note that, as in Remark 1.77, $c \leq d$ and $d \leq c$, so $c \cong d$. Such will always
fong-spivak-invitation_F00390	39	■ 1.000	$A=\{p\}$	$A = \{p\}$	$A = \{p\}$ with one element.
fong-spivak-invitation_F00391	39	■ 1.000	$\wedge A \cong p$	$\wedge A \cong p$	1. Show that $\wedge A \cong p$.
fong-spivak-invitation_F00392	39	■ 1.000	$\wedge A = p$	$\wedge A = p$	2. Show that if P is in fact a partial order, then $\wedge A = p$.
fong-spivak-invitation_F00393	39	■ 0.994	$\wedge$	$\wedge$	3. Are the analogous facts true when $\wedge$ is replaced by $\vee$?
fong-spivak-invitation_F00394	39	■ 0.994	$\vee$	$\vee$	3. Are the analogous facts true when $\wedge$ is replaced by $\vee$?
fong-spivak-invitation_F00395	39	■ 1.000	$p \vee p = p \wedge p = p$	$p \vee p = p \wedge p = p$	Example 1.81. In any partial order P , we have $p \vee p = p \wedge p = p$. The reason is
fong-spivak-invitation_F00396	39	■ 0.996	$p \vee p$	$p \vee p$	that our notation says $p \vee p$ means $\vee\{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so
fong-spivak-invitation_F00397	39	■ 0.996	$\bigvee\{p, p\}$	$\bigvee\{p, p\}$	that our notation says $p \vee p$ means $\vee\{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so
fong-spivak-invitation_F00398	39	■ 0.996	$\{p, p\}=\{p\}$	$\{p, p\} = \{p\}$	that our notation says $p \vee p$ means $\vee\{p, p\}$. But $\{p, p\} = \{p\}$ (see Section 1.2.1), so
fong-spivak-invitation_F00399	39	■ 1.000	$p \vee p = p$	$p \vee p = p$	$p \vee p = p$ by Exercise 1.80.
fong-spivak-invitation_F00400	39	■ 1.000	$A, B \subseteq X$	$A, B \subseteq X$	Example 1.82. In a power set $P(X)$, the meet of a collection of subsets, say $A, B \subseteq X$,
fong-spivak-invitation_F00401	39	■ 1.000	$A \wedge B = A \cap B$	$A \wedge B = A \cap B$	is their intersection $A \wedge B = A \cap B$, while the join is their union, $A \vee B =$
fong-spivak-invitation_F00402	39	■ 1.000	$A \vee B =$	$A \vee B =$	is their intersection $A \wedge B = A \cap B$, while the join is their union, $A \vee B =$
fong-spivak-invitation_F00403	40	■ 1.000	$n \mid m$	$n \mid m$	Exercise 1.85. Recall the division ordering on $\mathbb{N}$ from Example 1.40: we write $n \mid m$ if
fong-spivak-invitation_F00404	40	■ 0.924	$A \subseteq B \subseteq P$	$A \subseteq B \subseteq P$	Proposition 1.86. Suppose $(P, \leq)$ is a preorder and $A \subseteq B \subseteq P$ are subsets that have
fong-spivak-invitation_F00405	40	■ 1.000	$\wedge B \leq \wedge A$	$\wedge B \leq \wedge A$	meets. Then $\wedge B \leq \wedge A$.
fong-spivak-invitation_F00406	40	■ 1.000	$\bigvee A \leq \bigvee B$	$\bigvee A \leq \bigvee B$	Similarly, if A and B have joins, then $\bigvee A \leq \bigvee B$.
fong-spivak-invitation_F00407	40	■ 1.000	$m = \wedge A$	$m = \wedge A$	<i>Proof.</i> Let $m = \wedge A$ and $n = \wedge B$. Then for any $a \in A$ we also have $a \in B$, so

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fong-spivak-invitation_F00408	40	■ 1.000	$n = \wedge B$	$n = \wedge B$	<i>Proof.</i> Let $m = \bigwedge A$ and $n = \bigwedge B$. Then for any $a \in A$ we also have $a \in B$, so
fong-spivak-invitation_F00409	40	■ 1.000	$a \in B$	$a \in B$	<i>Proof.</i> Let $m = \bigwedge A$ and $n = \bigwedge B$. Then for any $a \in A$ we also have $a \in B$, so
fong-spivak-invitation_F00410	40	■ 1.000	$n \leq a$	$n \leq a$	$n \leq a$ because n is a lower bound for B . Thus n is also a lower bound for A and
fong-spivak-invitation_F00411	40	■ 1.000	$\Phi: P \rightarrow Q$	$\Phi: P \rightarrow Q$	$\Phi: P \rightarrow Q$ is a phenomenon (we might say “feature”) of P as observed by Q . Adam
fong-spivak-invitation_F00412	40	■ 1.000	$f(a \wedge$	$f(a \wedge$	Definition 1.87. We say that a monotone map $f: P \rightarrow Q$ <i>preserves meets</i> if $f(a \wedge$
fong-spivak-invitation_F00413	40	■ 1.000	$b) \cong f(a) \wedge f(b)$	$b) \cong f(a) \wedge f(b)$	$b) \cong f(a) \wedge f(b)$ for all $a, b \in P$. We similarly say f <i>preserves joins</i> if $f(a \vee b) \cong$
fong-spivak-invitation_F00414	40	■ 1.000	$a, b \in P$	$a, b \in P$	$b) \cong f(a) \wedge f(b)$ for all $a, b \in P$. We similarly say f <i>preserves joins</i> if $f(a \vee b) \cong$
fong-spivak-invitation_F00415	40	■ 1.000	$f(a \vee b) \cong$	$f(a \vee b) \cong$	$b) \cong f(a) \wedge f(b)$ for all $a, b \in P$. We similarly say f <i>preserves joins</i> if $f(a \vee b) \cong$
fong-spivak-invitation_F00416	40	■ 1.000	$f(a) \vee f(b)$	$f(a) \vee f(b)$	$f(a) \vee f(b)$ for all $a, b \in P$.
fong-spivak-invitation_F00417	40	■ 1.000	$f(a \vee b)$	$f(a \vee b)$	side $f(a \vee b)$ is the observation of the combined system $a \vee b$. The inequality implies
fong-spivak-invitation_F00418	40	■ 1.000	$a \vee b$	$a \vee b$	side $f(a \vee b)$ is the observation of the combined system $a \vee b$. The inequality implies
fong-spivak-invitation_F00419	41	■ 1.000	$f(a \vee$	$f(a \vee$	Exercise 1.89. In Definition 1.88, we defined generativity of f as the inequality $f(a \vee$
fong-spivak-invitation_F00420	41	■ 1.000	$b) \neq f(a) \vee f(b)$	$b) \neq f(a) \vee f(b)$	$b) \neq f(a) \vee f(b)$, but in the subsequent text we seemed to imply there would be not
fong-spivak-invitation_F00421	41	■ 1.000	$f(a), f(b) \in Q$	$f(a), f(b) \in Q$	$f(a), f(b) \in Q$ have a join, then indeed $f(a) \vee f(b) \leq f(a \vee b)$.
fong-spivak-invitation_F00422	41	■ 1.000	$f(a) \vee f(b) \leq f(a \vee b)$	$f(a) \vee f(b) \leq f(a \vee b)$	$f(a), f(b) \in Q$ have a join, then indeed $f(a) \vee f(b) < f(a \vee b)$.
fong-spivak-invitation_F00423	41	■ 0.927	$(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$	$(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$	Example 1.91. Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$, which
fong-spivak-invitation_F00424	41	■ 0.927	$x \in \mathbb{Z}$	$x \in \mathbb{Z}$	Example 1.91. Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$, which
fong-spivak-invitation_F00425	41	■ 0.927	$3x$	$3x$	Example 1.91. Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$, which
fong-spivak-invitation_F00426	41	■ 1.000	$3x \in \mathbb{Z} \subseteq \mathbb{R}$	$3x \in \mathbb{Z} \subseteq \mathbb{R}$	we can consider as a real number $3x \in \mathbb{Z} \subseteq \mathbb{R}$. Let’s find a left adjoint for the map
fong-spivak-invitation_F00427	42	■ 0.509	$\lceil z \rceil$	$\lceil z \rceil$	Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$, and write $\lfloor z \rfloor$ for the largest
fong-spivak-invitation_F00428	42	■ 0.509	$z \in \mathbb{R}$	$z \in \mathbb{R}$	Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$, and write $\lfloor z \rfloor$ for the largest
fong-spivak-invitation_F00429	42	■ 0.509	$\lfloor z \rfloor$	$\lfloor z \rfloor$	Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$, and write $\lfloor z \rfloor$ for the largest
fong-spivak-invitation_F00430	42	■ 0.937	$z \in \mathbb{R}$	$z \in \mathbb{R}$	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ⁹ As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let’s
fong-spivak-invitation_F00431	42	■ 0.937	$\lceil 3.14 \rceil = 4$	$\lceil 3.14 \rceil = 4$	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ⁹ As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let’s
fong-spivak-invitation_F00432	42	■ 0.937	$\lfloor 3.14 \rfloor = 3$	$\lfloor 3.14 \rfloor = 3$. ⁹	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ⁹ As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let’s
fong-spivak-invitation_F00433	42	■ 0.937	$\mathbb{R} \rightarrow \mathbb{Z}$	$\mathbb{R} \rightarrow \mathbb{Z}$	integer below $z \in \mathbb{R}$, e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$. ⁹ As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$, let’s
fong-spivak-invitation_F00434	42	■ 0.994	$\lceil -/3 \rceil$	$\lceil -/3 \rceil$	see if $\lceil -/3 \rceil$ works.

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fong-spivak-invitation_F00435	42	■ 0.651	$(3 \times -)$	$(3 \times -)$	Success! Thus we have a Galois connection between $\lceil -/3 \rceil$ and $(3 \times -)$.
fong-spivak-invitation_F00436	42	■ 1.000	$P=Q=\underline{3}$	$P = Q = \underline{3}$	Exercise 1.93. Consider the preorder $P = Q = \underline{3}$.
fong-spivak-invitation_F00437	42	■ 1.000	f, g	f, g	1. Let f, g be the monotone maps shown below:
fong-spivak-invitation_F00438	42	■ 1.000	$1 \leq p, q \leq 3$	$1 \leq p, q \leq 3$	Is it the case that f is left adjoint to g ? Check that for each $1 \leq p, q \leq 3$, one has
fong-spivak-invitation_F00439	42	■ 1.000	$f(p) \leq q$	$f(p) \leq q$	$f(p) \leq q$ iff $p \leq g(q)$.
fong-spivak-invitation_F00440	42	■ 1.000	$p \leq g(q)$	$p \leq g(q)$	$f(p) \leq q$ iff $p \leq g(q)$.
fong-spivak-invitation_F00441	42	■ 1.000	$L: \mathbb{Z} \rightarrow \mathbb{R}$	$L: \mathbb{Z} \rightarrow \mathbb{R}$	1. Does $\lceil -/3 \rceil$ have a left adjoint $L: \mathbb{Z} \rightarrow \mathbb{R}$?
fong-spivak-invitation_F00442	42	■ 1.000	$\lfloor 3 \rfloor = \lceil 3 \rceil$	$\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$	mouthful. Anyway, $\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$.
fong-spivak-invitation_F00443	43	■ 1.000	$\text{Prt}(S)$	$\text{Prt}(S)$	Recall from Example 1.47 that we can understand the set $\text{Prt}(S)$ of partitions on a set S
fong-spivak-invitation_F00444	43	■ 1.000	$g: S \rightarrow T$	$g: S \rightarrow T$	Suppose we are given any function $g: S \rightarrow T$. We will show that this function
fong-spivak-invitation_F00445	43	■ 0.882	$g!: \text{Prt}(S) \rightleftarrows \text{Prt}(T): g^*$	$g!: \text{Prt}(S) \rightleftarrows \text{Prt}(T): g^*$	g induces a Galois connection $g!: \text{Prt}(S) \rightleftarrows \text{Prt}(T): g^*$ between the preorder of
fong-spivak-invitation_F00446	43	■ 0.993	$g!: \text{Prt}(S) \rightarrow \text{Prt}(T)$	$g!: \text{Prt}(S) \rightarrow \text{Prt}(T)$	Start with $g: S \rightarrow T$; we first want to understand $g!: \text{Prt}(S) \rightarrow \text{Prt}(T)$. So start with
fong-spivak-invitation_F00447	43	■ 1.000	$\sim_S$	$\sim_S$	a partition $\sim_S$ of S . To begin the process of obtaining a partition $\sim_T$ on T , say that
fong-spivak-invitation_F00448	43	■ 1.000	$\sim_T$	$\sim_T$	a partition $\sim_S$ of S . To begin the process of obtaining a partition $\sim_T$ on T , say that
fong-spivak-invitation_F00449	43	■ 0.983	$t_1, t_2 \in T$	$t_1, t_2 \in T$	two elements $t_1, t_2 \in T$ are in the same part, $t_1 \sim_T t_2$, if there exist $s_1, s_2 \in S$ such
fong-spivak-invitation_F00450	43	■ 0.983	$t_1 \sim_T t_2$	$t_1 \sim_T t_2$	two elements $t_1, t_2 \in T$ are in the same part, $t_1 \sim_T t_2$, if there exist $s_1, s_2 \in S$ such
fong-spivak-invitation_F00451	43	■ 0.997	$s_1 \sim_S s_2$	$s_1 \sim_S s_2$	that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$. However, the result of doing so will
fong-spivak-invitation_F00452	43	■ 0.997	$g(s_1) = t_1$	$g(s_1) = t_1$	that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$. However, the result of doing so will
fong-spivak-invitation_F00453	43	■ 0.997	$g(s_2) = t_2$	$g(s_2) = t_2$	that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$. However, the result of doing so will
fong-spivak-invitation_F00454	43	■ 0.769	$t_2 \sim_T t_3$	$t_2 \sim_T t_3$	not necessarily be transitive – you may get $t_1 \sim_T t_2$ and $t_2 \sim_T t_3$ without $t_1 \stackrel{?}{\sim}_T t_3$ –
fong-spivak-invitation_F00455	43	■ 0.769	$t_1 \stackrel{?}{\sim}_T t_3$	$t_1 \stackrel{?}{\sim}_T t_3$	not necessarily be transitive – you may get $t_1 \sim_T t_2$ and $t_2 \sim_T t_3$ without $t_1 \stackrel{?}{\sim}_T t_3$ –
fong-spivak-invitation_F00456	43	■ 1.000	$g!(\sim_S) := \sim_T$	$g!(\sim_S) := \sim_T$	pieces (take the transitive closure). The result is $g!(\sim_S) := \sim_T$.
fong-spivak-invitation_F00457	43	■ 0.998	g^*	g^*	Again starting with g , we want to get the right adjoint $g^*: \text{Prt}(T) \rightarrow \text{Prt}(S)$. So start
fong-spivak-invitation_F00458	43	■ 0.998	$\text{Prt}(T) \rightarrow \text{Prt}(S)$	$\text{Prt}(T) \rightarrow \text{Prt}(S)$	Again starting with g , we want to get the right adjoint $g^*: \text{Prt}(T) \rightarrow \text{Prt}(S)$. So start
fong-spivak-invitation_F00459	43	■ 1.000	$g(s_1) \sim_T$	$g(s_1) \sim_T$	with a partition $\sim_T$ of T . Get a partition $\sim_S$ on S by saying that $s_1 \sim_S s_2$ iff $g(s_1) \sim_T$
fong-spivak-invitation_F00460	43	■ 1.000	$g(s_2)$	$g(s_2)$	$g(s_2)$. The result is $g^*(\sim_T) := \sim_S$.
fong-spivak-invitation_F00461	43	■ 1.000	$g^*(\sim_T) := \sim_S$	$g^*(\sim_T) := \sim_S$	$g(s_2)$. The result is $g^*(\sim_T) := \sim_S$.

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fong-spivak-invitation_F00462	43	■ 1.000	$S=\{1,2,3,4\}, T=\{12,3,4\}$	$S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$	Example 1.96. Let $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$, and define $g: S \rightarrow T$ by $g(1) :=$
fong-spivak-invitation_F00463	43	■ 1.000	$g(1) :=$	$g(1) :=$	Example 1.96. Let $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$, and define $g: S \rightarrow T$ by $g(1) :=$
fong-spivak-invitation_F00464	43	■ 1.000	$g(2) := 12, g(3) := 3$	$g(2) := 12, g(3) := 3$	$g(2) := 12, g(3) := 3$, and $g(4) := 4$. The partition shown left below is translated by g !
fong-spivak-invitation_F00465	43	■ 1.000	$g(4) := 4$	$g(4) := 4$	$g(2) := 12, g(3) := 3$, and $g(4) := 4$. The partition shown left below is translated by g !
fong-spivak-invitation_F00466	44	■ 0.872	$c: S \rightarrowtail P$	$c: S \rightarrow P$	four different ones and, for each one, call it $c: S \rightarrowtail P$, find $g_!(c)$, where S, T , and
fong-spivak-invitation_F00467	44	■ 0.872	$g_!{}(c)$	$g_!(c)$	four different ones and, for each one, call it $c: S \rightarrowtail P$, find $g_!(c)$, where S, T , and
fong-spivak-invitation_F00468	44	■ 0.872	S, T	S, T	four different ones and, for each one, call it $c: S \rightarrowtail P$, find $g_!(c)$, where S, T , and
fong-spivak-invitation_F00469	44	■ 1.000	$T =$	$T =$	Exercise 1.99. There are five partitions possible on a set with three elements, say $T =$
fong-spivak-invitation_F00470	44	■ 0.746	$\{12, 3, 4\}$	$\{12, 3, 4\}$	$\{12, 3, 4\}$. Using the same S and $g: S \rightarrow T$ as in Example 1.96, determine the partition
fong-spivak-invitation_F00471	44	■ 1.000	$g^*{}(c)$	$g^*(c)$	$g^*(c)$ on S for each of the five partitions $c: T \rightarrowtail P$.
fong-spivak-invitation_F00472	44	■ 1.000	$c: T \rightarrowtail P$	$c: T \rightarrow P$	$g^*(c)$ on S for each of the five partitions $c: T \rightarrowtail P$.
fong-spivak-invitation_F00473	44	■ 1.000	$g^*{}: \operatorname{Prt}(T) \rightarrowtail \operatorname{Prt}(S)$	$g^*: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$	really is left adjoint to $g^*: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$ would take too much time for this sketch.
fong-spivak-invitation_F00474	44	■ 1.000	$d: T \rightarrowtail P'$	$d: T \rightarrow P'$	2. Choose any coarser partition $d: T \rightarrowtail P'$, i.e. where $g_!(c) \leq d$.
fong-spivak-invitation_F00475	44	■ 1.000	$g_!{}(c) \leq d$	$g_!(c) \leq d$	2. Choose any coarser partition $d: T \rightarrowtail P'$, i.e. where $g_!(c) \leq d$.
fong-spivak-invitation_F00476	44	■ 0.999	$e: T \rightarrowtail Q$	$e: T \rightarrow Q$	3. Choose any non-coarser partition $e: T \rightarrowtail Q$, i.e. where $g_!(c) \not\leq e$. (If you can't do
fong-spivak-invitation_F00477	44	■ 0.999	$g_!{}(c) \not\leq e$	$g_!(c) \not\leq e$	3. Choose any non-coarser partition $e: T \rightarrowtail Q$, i.e. where $g_!(c) \not\leq e$. (If you can't do
fong-spivak-invitation_F00478	44	■ 1.000	$g^*{}(d)$	$g^*(d)$	4. Find $g^*(d)$ and $g^*(e)$.
fong-spivak-invitation_F00479	44	■ 1.000	$g^*{}(e)$	$g^*(e)$	4. Find $g^*(d)$ and $g^*(e)$.
fong-spivak-invitation_F00480	44	■ 1.000	$c \leq g^*{}(d)$	$c \leq g^*(d)$	we should have $c \leq g^*(d)$ and $c \not\leq g^*(e)$. Show that this is true.
fong-spivak-invitation_F00481	44	■ 1.000	$c \not\leq g^*{}(e)$	$c \not\leq g^*(e)$	we should have $c \leq g^*(d)$ and $c \not\leq g^*(e)$. Show that this is true.
fong-spivak-invitation_F00482	45	■ 1.000	$q := f(p)$	$q := f(p)$	<i>Proof.</i> Suppose f is left adjoint to g . Take any $p \in P$, and let $q := f(p)$. By reflexiv-
fong-spivak-invitation_F00483	45	■ 1.000	$p \leq g(f(p))$	$p \leq g(f(p))$	but this means $p \leq g(f(p))$. The proof that $f(g(q)) \leq q$ is similar.
fong-spivak-invitation_F00484	45	■ 1.000	$f(g(q)) \leq q$	$f(g(q)) \leq q$	but this means $p \leq g(f(p))$. The proof that $f(g(q)) \leq q$ is similar.
fong-spivak-invitation_F00485	45	■ 1.000	$g(f(p)) \leq$	$g(f(p)) \leq$	$f(p) \leq q$ iff $p \leq g(q)$. Suppose $f(p) \leq q$; then since g is monotonic, $g(f(p)) \leq$
fong-spivak-invitation_F00486	45	■ 1.000	$g(q)$	$g(q)$	$g(q)$, but $p \leq g(f(p))$ so $p \leq g(q)$. The proof that $p \leq g(q)$ implies $f(p) \leq q$ is
fong-spivak-invitation_F00487	45	■ 1.000	$g^{\prime}{}(q)$	g'	That means, for any other right adjoint g' , we have $g(q) \cong g'(q)$ for all $q \in Q$.
fong-spivak-invitation_F00488	45	■ 1.000	$g(q) \cong g^{\prime}{}(q)$	$g(q) \cong g'(q)$	That means, for any other right adjoint g' , we have $g(q) \cong g'(q)$ for all $q \in Q$.

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fong-spivak-invitation_F00489	45	■ 1.000	$A \subseteq Q$	$A \subseteq Q$	$g: Q \rightarrow P$. Suppose that $A \subseteq Q$ is any subset, and let $g(A) := \{g(a) \mid a \in A\}$ be
fong-spivak-invitation_F00490	45	■ 1.000	$g(A) := \{g(a) \mid a \in A\}$	$g(A) := \{g(a) \mid a \in A\}$	$g: Q \rightarrow P$. Suppose that $A \subseteq Q$ is any subset, and let $g(A) := \{g(a) \mid a \in A\}$ be
fong-spivak-invitation_F00491	45	■ 1.000	$\bigwedge A \in Q$	$\bigwedge A \in Q$	its image. Then if A has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in P , and we
fong-spivak-invitation_F00492	45	■ 1.000	$g(A)$	$g(A)$	its image. Then if A has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in P , and we
fong-spivak-invitation_F00493	45	■ 1.000	$\bigwedge g(A)$	$\bigwedge g(A)$	its image. Then if A has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in P , and we
fong-spivak-invitation_F00494	45	■ 1.000	$\bigvee A \in P$	$\bigvee A \in P$	is any subset that has a join $\bigvee A \in P$, then $f(A)$ has a join $\bigvee f(A)$ in Q , and we
fong-spivak-invitation_F00495	45	■ 1.000	$f(A)$	$f(A)$	is any subset that has a join $\bigvee A \in P$, then $f(A)$ has a join $\bigvee f(A)$ in Q , and we
fong-spivak-invitation_F00496	45	■ 1.000	$\bigvee f(A)$	$\bigvee f(A)$	is any subset that has a join $\bigvee A \in P$, then $f(A)$ has a join $\bigvee f(A)$ in Q , and we
fong-spivak-invitation_F00497	45	■ 1.000	$m := \bigwedge A$	$m := \bigwedge A$	to f . Let $A \subseteq Q$ be any subset and let $m := \bigwedge A$ be its meet. Then since g is monotone
fong-spivak-invitation_F00498	45	■ 1.000	$g(m) \leq g(a)$	$g(m) \leq g(a)$	$g(m) \leq g(a)$ for all $a \in A$, so $g(m)$ is a lower bound for the set $g(A)$. We will be done
fong-spivak-invitation_F00499	45	■ 1.000	$g(m)$	$g(m)$	$g(m) \leq g(a)$ for all $a \in A$, so $g(m)$ is a lower bound for the set $g(A)$. We will be done
fong-spivak-invitation_F00500	45	■ 1.000	$b \leq g(a)$	$b \leq g(a)$	have $b \leq g(a)$ and we want to show that $b \leq g(m)$. Then by definition of g being a
fong-spivak-invitation_F00501	45	■ 1.000	$b \leq g(m)$	$b \leq g(m)$	have $b \leq g(a)$ and we want to show that $b \leq g(m)$. Then by definition of g being a
fong-spivak-invitation_F00502	45	■ 0.999	$f(b) \leq a$	$f(b) \leq a$	right adjoint (Definition 1.90), we also have $f(b) \leq a$. This means that $f(b)$ is a lower
fong-spivak-invitation_F00503	45	■ 0.999	$f(b)$	$f(b)$	right adjoint (Definition 1.90), we also have $f(b) \leq a$. This means that $f(b)$ is a lower
fong-spivak-invitation_F00504	45	■ 1.000	$f(b) \leq m$	$f(b) \leq m$	bound for A in Q . Since the meet m is the greatest lower bound, we have $f(b) \leq m$.
fong-spivak-invitation_F00505	46	■ 1.000	$f(3.9) := 4$	$f(3.9) := 4$	far as possible but with $f(3.9) := 4$. Both f and g are monotonic, and one can check
fong-spivak-invitation_F00506	46	■ 1.000	$1 \vee 2 = 4$	$1 \vee 2 = 4$	$1 \vee 2 = 4$ holds in Q , whereas $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$ in P .
fong-spivak-invitation_F00507	46	■ 1.000	$g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$	$g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$	$1 \vee 2 = 4$ holds in Q , whereas $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$ in P .
fong-spivak-invitation_F00508	47	■ 1.000	$\left\{q^{\prime} \in Q \mid p^{\prime} \leq g\left(q^{\prime}\right)\right\} \subseteq \left\{q \in Q \mid p \leq g(q)\right\}$	$\left\{q^{\prime} \in Q \mid p^{\prime} \leq g\left(q^{\prime}\right)\right\} \subseteq \left\{q \in Q \mid p \leq g(q)\right\}$	need to show it is monotone. So suppose that $p \leq p'$. Then $\{q' \in Q \mid p' \leq g(q')\} \subseteq$
fong-spivak-invitation_F00509	47	■ 1.000	$\left\{q \in Q \mid p \leq g(q)\right\}$	$\left\{q \in Q \mid p \leq g(q)\right\}$	$\{q \in Q \mid p \leq g(q)\}$. By Proposition 1.86, this implies $f(p) \leq f(p')$. Thus f is
fong-spivak-invitation_F00510	47	■ 1.000	$f(p) \leq f\left(p^{\prime}\right)$	$f(p) \leq f\left(p^{\prime}\right)$	$\{q \in Q \mid p \leq g(q)\}$. By Proposition 1.86, this implies $f(p) \leq f(p')$. Thus f is
fong-spivak-invitation_F00511	47	■ 1.000	$p_{\{0\}} \leq g\left(f\left(p_{\{0\}}\right)\right)$	$p_0 \leq g\left(f\left(p_0\right)\right)$	By Proposition 1.104, it suffices to show that $p_0 \leq g\left(f\left(p_0\right)\right)$ and that $f\left(g\left(q_0\right)\right) \leq q_0$
fong-spivak-invitation_F00512	47	■ 1.000	$f\left(g\left(q_{\{0\}}\right)\right) \leq q_{\{0\}}$	$f\left(g\left(q_0\right)\right) \leq q_0$	By Proposition 1.104, it suffices to show that $p_0 \leq g\left(f\left(p_0\right)\right)$ and that $f\left(g\left(q_0\right)\right) \leq q_0$
fong-spivak-invitation_F00513	47	■ 1.000	$p_{\{0\}} \in P$	$p_0 \in P$	for all $p_0 \in P$ and $q_0 \in Q$. For the first, we have
fong-spivak-invitation_F00514	47	■ 1.000	$q_{\{0\}} \in Q$	$q_0 \in Q$	for all $p_0 \in P$ and $q_0 \in Q$. For the first, we have
fong-spivak-invitation_F00515	47	■ 1.000	$p_{\{0\}}$	p_0	where the first inequality follows from the fact that if p_0 is below every element of a set,

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fong-spivak-invitation_F00516	47	■ 1.000	$\left\{q_{\{0\}}\right\} \subseteq \left\{q \in Q \mid g\left(q_{\{0\}}\right) \leq \left(q_{\{0\}}\right)\right\}.$	$\left\{q_0\right\} \subseteq\left\{q \in Q \mid g\left(q_0\right) \leq\right.$	where the first inequality follows from Proposition 1.86 since $\boxed{\{q_0\} \subseteq \{q \in Q \mid g(q_0) \preceq$
fong-spivak-invitation_F00517	47	■ 0.990	$g(q)\backslash\}$	$g(q)\}$	$g(q)\}$, and the fact that $\bigwedge\left\{q_0\right\}=q_0$.
fong-spivak-invitation_F00518	47	■ 0.990	$\wedge\left\{q_{\{0\}}\right\}=q_{\{0\}}$	$\wedge\left\{q_0\right\}=q_0$	$g(q)\}$, and the fact that $\bigwedge\left\{q_0\right\}=q_0$.
fong-spivak-invitation_F00519	47	■ 0.999	$f^{\ast}: \mathrm{P}(B) \rightarrow \mathrm{P}(A)$	$f^{\ast}: P(B) \rightarrow P(A)$	Then we have the monotone map $f^{\ast}: P(B) \rightarrow P(A)$ that category theorists call “pull-
fong-spivak-invitation_F00520	47	■ 1.000	$B^{\prime} \subseteq B$	$B' \subseteq B$	back along f .” This map takes a subset $B' \subseteq B$ to its preimage $f^{-1}\left(B'\right) \subseteq A$; that is,
fong-spivak-invitation_F00521	47	■ 1.000	$f^{-1}\left(B^{\prime}\right) \subseteq A$	$f^{-1}\left(B'\right) \subseteq A$	back along f .” This map takes a subset $B' \subseteq B$ to its preimage $f^{-1}\left(B'\right) \subseteq A$; that is,
fong-spivak-invitation_F00522	47	■ 1.000	$B^{\prime}$	B'	it takes a collection B' of buckets, and tells you all the apples that they contain in total.
fong-spivak-invitation_F00523	47	■ 1.000	$f_!(A)$	$f_!(A)$	The left adjoint $f_!(A)$ is given by the direct image: it maps a subset $A' \subseteq A$ to
fong-spivak-invitation_F00524	47	■ 1.000	$A^{\prime} \subseteq A$	$A' \subseteq A$	The left adjoint $f_!(A)$ is given by the direct image: it maps a subset $A' \subseteq A$ to
fong-spivak-invitation_F00525	47	■ 1.000	$A^{\prime}$	A'	This map takes a set A' of apples, and tells you all the buckets that contain at least one
fong-spivak-invitation_F00526	47	■ 1.000	$f_{\ast}$	$f_{\ast}$	The right adjoint $f_{\ast}$ maps a subset $A' \subseteq A$ to
fong-spivak-invitation_F00527	47	■ 0.923	$f^{\ast}, f_!,$	$f^{\ast}, f_!$	whose only apples are from A' . But we did not invent these mappings $f^{\ast}, f_!$, and
fong-spivak-invitation_F00528	48	■ 1.000	$B_{\{1\}}, B_{\{2\}} \subseteq Y$	$B_1, B_2 \subseteq Y$	1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^{\ast}\left(B_1\right)$ and $f^{\ast}\left(B_2\right)$.
fong-spivak-invitation_F00529	48	■ 1.000	$f^{\ast}\left(B_{\{1\}}\right)$	$f^{\ast}\left(B_1\right)$	1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^{\ast}\left(B_1\right)$ and $f^{\ast}\left(B_2\right)$.
fong-spivak-invitation_F00530	48	■ 1.000	$f^{\ast}\left(B_{\{2\}}\right)$	$f^{\ast}\left(B_2\right)$	1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^{\ast}\left(B_1\right)$ and $f^{\ast}\left(B_2\right)$.
fong-spivak-invitation_F00531	48	■ 1.000	$A_{\{1\}}, A_{\{2\}} \subseteq X$	$A_1, A_2 \subseteq X$	2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!\left(A_1\right)$ and $f_!\left(A_2\right)$.
fong-spivak-invitation_F00532	48	■ 1.000	$f_!\left(A_{\{1\}}\right)$	$f_!\left(A_1\right)$	2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!\left(A_1\right)$ and $f_!\left(A_2\right)$.
fong-spivak-invitation_F00533	48	■ 1.000	$f_!\left(A_{\{2\}}\right)$	$f_!\left(A_2\right)$	2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!\left(A_1\right)$ and $f_!\left(A_2\right)$.
fong-spivak-invitation_F00534	48	■ 1.000	$f_{\ast}\left(A_{\{1\}}\right)$	$f_{\ast}\left(A_1\right)$	3. With the same $A_1, A_2 \subseteq X$, find $f_{\ast}\left(A_1\right)$ and $f_{\ast}\left(A_2\right)$.
fong-spivak-invitation_F00535	48	■ 1.000	$f_{\ast}\left(A_{\{2\}}\right)$	$f_{\ast}\left(A_2\right)$	3. With the same $A_1, A_2 \subseteq X$, find $f_{\ast}\left(A_1\right)$ and $f_{\ast}\left(A_2\right)$.
fong-spivak-invitation_F00536	48	■ 1.000	$f ; g: P \rightarrow P$	$f ; g: P \rightarrow P$	f and g to arrive at a monotone map $f \circ g: P \rightarrow P$ from preorder P to itself. This
fong-spivak-invitation_F00537	48	■ 0.999	$p \leq(f \circ g)(p)$	$p \leq(f \circ g)(p)$	monotone map has the property that $p \leq(f \circ g)(p)$, and that $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$
fong-spivak-invitation_F00538	48	■ 0.999	$(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$	$(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$	monotone map has the property that $p \leq(f \circ g)(p)$, and that $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$
fong-spivak-invitation_F00539	48	■ 1.000	$(f ; g ; f ; g)(p) \cong(f ; g)(p)$	$(f ; g ; f ; g)(p) \cong(f ; g)(p)$	2. $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$. To prove this, show inequalities in both directions, $\leq$
fong-spivak-invitation_F00540	48	■ 1.000	$\geq$	$\geq$	and $\geq$.
fong-spivak-invitation_F00541	48	■ 1.000	$j: P \rightarrow P$	$j: P \rightarrow P$	Definition 1.112. A closure operator $j: P \rightarrow P$ on a preorder P is a monotone map
fong-spivak-invitation_F00542	48	■ 0.999	$p \leq j(p)$	$p \leq j(p)$	(a) $p \leq j(p)$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_F00543	48	■ 1.000	$j(j(p)) \backslash \text{cong } j(p)$	$j(j(p)) \cong j(p)$	(b) $j(j(p)) \cong j(p)$.
fong-spivak-invitation_F00544	48	■ 1.000	$7+2+3$	$7 + 2 + 3$	expressions. For example, a computer can rewrite the expression $7 + 2 + 3$ as the expres-
fong-spivak-invitation_F00545	48	■ 0.936	$j: \text{exp} \backslash \text{rightarrow exp}$	$j : \text{exp} \rightarrow \text{exp}$	and reducing it to another expression. So it is a map $j: \text{exp} \rightarrow \text{exp}$. It furthermore
fong-spivak-invitation_F00546	48	■ 1.000	$j(x)$	$j(x)$	means that if an expression x can be rewritten into expression y , then the reduction $j(x)$
fong-spivak-invitation_F00547	48	■ 1.000	$j(y)$	$j(y)$	can be rewritten into $j(y)$. Moreover, the requirement $x \leq j(x)$ implies that j can only
fong-spivak-invitation_F00548	48	■ 1.000	$x \backslash \text{leq } j(x)$	$x \leq j(x)$	can be rewritten into $j(y)$. Moreover, the requirement $x \leq j(x)$ implies that j can only
fong-spivak-invitation_F00549	48	■ 1.000	j	j	can be rewritten into $j(y)$. Moreover, the requirement $x \leq j(x)$ implies that j can only
fong-spivak-invitation_F00550	48	■ 1.000	$j(j(x))=j(x)$	$j(j(x)) = j(x)$	$j(j(x)) = j(x)$ implies that if you try to reduce an expression that has already been
fong-spivak-invitation_F00551	48	■ 0.906	$(g \backslash \text{circ } f)(q) \backslash \text{leq } q$	$(g \circ f)(q) \leq q$	¹⁰ The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$
fong-spivak-invitation_F00552	48	■ 0.906	$(g \backslash \text{circ } f \backslash \text{circ } g \backslash \text{circ } f)(q) \backslash \text{cong}(g \backslash \text{circ } f)(q)$	$(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$	¹⁰ The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$
fong-spivak-invitation_F00553	49	■ 0.673	$_{{j}}$	j	fix_j to have elements the fixed points of j ; that is,
fong-spivak-invitation_F00554	49	■ 0.506	$\mathrm{fix}_{{j}}$	fix_j	This is a subset of P and inherits an order as a result; hence fix_j is a sub-preorder of P .
fong-spivak-invitation_F00555	49	■ 1.000	$j(p)$	$j(p)$	Note that $j(p)$ is a fixed point for all $p \in P$, since $j(j(p)) \cong j(p)$.
fong-spivak-invitation_F00556	49	■ 1.000	$j: P \backslash \text{rightarrow } \mathrm{fix}_{{j}}$	$j : P \rightarrow \text{fix}_j$	We define an adjunction with left adjoint $j: P \rightarrow \text{fix}_j$ sending p to $j(p)$, and right
fong-spivak-invitation_F00557	49	■ 0.888	$g:$	$g:$	adjoint $g: \text{fix}_j \rightarrow P$ simply the inclusion of the sub-preorder. To see it’s really an
fong-spivak-invitation_F00558	49	■ 0.888	$_{{j}} \backslash \text{rightarrow } P$	$j \rightarrow P$	adjoint $g: \text{fix}_j \rightarrow P$ simply the inclusion of the sub-preorder. To see it’s really an
fong-spivak-invitation_F00559	49	■ 0.979	$q \in \operatorname{fix}_{{j}}$	$q \in \text{fix}_j$	adjunction, we need to see that for any $p \in P$ and $q \in \text{fix}_j$, we have $j(p) \leq q$
fong-spivak-invitation_F00560	49	■ 0.979	$j(p) \backslash \text{leq } q$	$j(p) \leq q$	adjunction, we need to see that for any $p \in P$ and $q \in \text{fix}_j$, we have $j(p) \leq q$
fong-spivak-invitation_F00561	49	■ 1.000	$j(p) \backslash \text{leq } j(q) \backslash \text{cong } q$	$j(p) \leq j(q) \cong q$	$j(p) \leq j(q) \cong q$.
fong-spivak-invitation_F00562	49	■ 1.000	$p \backslash \text{Rrightarrow } q$	$p \Rightarrow q$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this
fong-spivak-invitation_F00563	49	■ 1.000	$p \backslash \text{Rrightarrow } p$	$p \Rightarrow p$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this
fong-spivak-invitation_F00564	49	■ 1.000	$q \backslash \text{Rrightarrow } r$	$q \Rightarrow r$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this
fong-spivak-invitation_F00565	49	■ 1.000	$p \backslash \text{Rrightarrow } r$	$p \Rightarrow r$	$p \Rightarrow q$. Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$, we also have $p \Rightarrow r$, this
fong-spivak-invitation_F00566	49	■ 1.000	$p \backslash \text{Rrightarrow } j(p)$	$p \Rightarrow j(p)$	propositions to propositions, for which $p \Rightarrow j(p)$ and $j(j(p)) = j(p)$. An example
fong-spivak-invitation_F00567	49	■ 1.000	$j(j(p))=j(p)$	$j(j(p)) = j(p)$	propositions to propositions, for which $p \Rightarrow j(p)$ and $j(j(p)) = j(p)$. An example
fong-spivak-invitation_F00568	49	■ 1.000	$B \backslash \text{Rrightarrow } p$	$B \Rightarrow p$	“assuming Bob is in San Diego, p ” means $B \Rightarrow p$. Let’s see why $B \Rightarrow -$ is a closure
fong-spivak-invitation_F00569	49	■ 1.000	$B \backslash \text{Rrightarrow -}$	$B \Rightarrow -$	“assuming Bob is in San Diego, p ” means $B \Rightarrow p$. Let’s see why $B \Rightarrow -$ is a closure

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fong-spivak-invitation_F00570	50	■ 0.951	$\mathbf{Rel}(S)$	$\mathbf{Rel}(S)$	Given any set S , there is a set $\mathbf{Rel}(S)$ of binary relations on S . An element $R \in \mathbf{Rel}(S)$
fong-spivak-invitation_F00571	50	■ 0.951	$R \in \mathbf{Rel}(S)$	$R \in \mathbf{Rel}(S)$	Given any set S , there is a set $\mathbf{Rel}(S)$ of binary relations on S . An element $R \in \mathbf{Rel}(S)$
fong-spivak-invitation_F00572	50	■ 0.958	$R \subseteq S \times S$	$R \subseteq S \times S$	is formally a subset $R \subseteq S \times S$. The set $\mathbf{Rel}(S)$ can be given an order via the subset
fong-spivak-invitation_F00573	50	■ 0.958	$\mathbf{Rel}(S)$	$\mathbf{Rel}(S)$	is formally a subset $R \subseteq S \times S$. The set $\mathbf{Rel}(S)$ can be given an order via the subset
fong-spivak-invitation_F00574	50	■ 1.000	$R \subseteq R'$	$R \subseteq R'$	relation, $R \subseteq R'$, i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$.
fong-spivak-invitation_F00575	50	■ 1.000	$R(s_1, s_2)$	$R(s_1, s_2)$	relation, $R \subseteq R'$, i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$.
fong-spivak-invitation_F00576	50	■ 1.000	$R'(s_1, s_2)$	$R'(s_1, s_2)$	relation, $R \subseteq R'$, i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$.
fong-spivak-invitation_F00577	50	■ 0.999	$\mathbf{Rel}(\{1\})$	$\mathbf{Rel}(\{1\})$	For example, the Hasse diagram for $\mathbf{Rel}(\{1\})$ is:
fong-spivak-invitation_F00578	50	■ 0.948	$\mathbf{Rel}(\{1, 2\})$	$\mathbf{Rel}(\{1, 2\})$	Exercise 1.116. Draw the Hasse diagram for the preorder $\mathbf{Rel}(\{1, 2\})$ of all binary
fong-spivak-invitation_F00579	50	■ 1.000	$\mathbf{Pos}(S)$	$\mathbf{Pos}(S)$	For any set S , there is also a set $\mathbf{Pos}(S)$, consisting of all the preorder relations on S .
fong-spivak-invitation_F00580	50	■ 0.936	$\sqsubseteq$	$\sqsubseteq$	In fact there is a preorder structure $\sqsubseteq$ on $\mathbf{Pos}(S)$, again given by inclusion: $\leq$ is below $\leq'$
fong-spivak-invitation_F00581	50	■ 0.936	$\leq'$	$\leq'$	In fact there is a preorder structure $\sqsubseteq$ on $\mathbf{Pos}(S)$, again given by inclusion: $\leq$ is below $\leq'$
fong-spivak-invitation_F00582	50	■ 0.989	$\leq \sqsubseteq \leq'$	$\leq \sqsubseteq \leq'$	(we'll write $\leq \sqsubseteq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-invitation_F00583	50	■ 0.989	$a \leq' b$	$a \leq' b$	(we'll write $\leq \sqsubseteq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-invitation_F00584	50	■ 0.989	$a, b \in S$	$a, b \in S$	(we'll write $\leq \sqsubseteq \leq'$) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$. A preorder of preorder
fong-spivak-invitation_F00585	50	■ 1.000	$\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$	$\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$	$\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$. This is the right adjoint of a Galois connection. Its left adjoint is
fong-spivak-invitation_F00586	50	■ 0.995	$\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$	$\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$	a monotone map $\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$ given by taking any relation R , writing it in
fong-spivak-invitation_F00587	50	■ 1.000	$s \leq s$	$s \leq s$	infix notation using $\leq$, and taking the reflexive and transitive closure, i.e. adding $s \leq s$
fong-spivak-invitation_F00588	50	■ 1.000	$s \leq u$	$s \leq u$	for every s and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$.
fong-spivak-invitation_F00589	50	■ 1.000	$s \leq t$	$s \leq t$	for every s and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$.
fong-spivak-invitation_F00590	50	■ 1.000	$t \leq u$	$t \leq u$	for every s and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$.
fong-spivak-invitation_F00591	50	■ 1.000	$S = \{1, 2, 3\}$	$S = \{1, 2, 3\}$	Exercise 1.117. Let $S = \{1, 2, 3\}$. Let's try to understand the adjunction discussed
fong-spivak-invitation_F00592	50	■ 1.000	$U(\leq)$	$U(\leq)$	1. Come up with any preorder relation $\leq$ on S , and define $U(\leq)$ to be the subset
fong-spivak-invitation_F00593	50	■ 1.000	$U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$	$U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$	$U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$, i.e. $U(\leq)$ is the image of $\leq$ under the
fong-spivak-invitation_F00594	50	■ 1.000	$Q \subseteq S \times S$	$Q \subseteq S \times S$	2. Come up with any two binary relations $Q \subseteq S \times S$ and $Q' \subseteq S \times S$ such that
fong-spivak-invitation_F00595	50	■ 1.000	$Q' \subseteq S \times S$	$Q' \subseteq S \times S$	2. Come up with any two binary relations $Q \subseteq S \times S$ and $Q' \subseteq S \times S$ such that
fong-spivak-invitation_F00596	50	■ 1.000	$Q \subseteq U(\leq)$	$Q \subseteq U(\leq)$	$Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$. Note that your choice of Q, Q' do not have to come

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fong-spivak-invitation_F00597	50	■ 1.000	$Q^{\prime} \not\subseteq U(\leq)$	$Q' \not\subseteq U(\leq)$	$Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$. Note that your choice of Q , Q' do not have to come
fong-spivak-invitation_F00598	50	■ 1.000	$Q, Q^{\prime}$	Q, Q'	$Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$. Note that your choice of Q , Q' do not have to come
fong-spivak-invitation_F00599	50	■ 0.825	$\mathrm{Cl}(Q) \sqsubseteq \leq$	$\mathrm{Cl}(Q) \sqsubseteq \leq$	that $\mathrm{Cl}(Q) \sqsubseteq \leq$, where $\sqsubseteq$ is the order on $\mathbf{Pos}(S)$, defined immediately above this
fong-spivak-invitation_F00600	50	■ 0.825	$\operatorname{Pos}(S)$	$\operatorname{Pos}(S)$	that $\mathrm{Cl}(Q) \sqsubseteq \leq$, where $\sqsubseteq$ is the order on $\mathbf{Pos}(S)$, defined immediately above this
fong-spivak-invitation_F00601	50	■ 0.508	$\mathrm{Cl}(Q^{\prime}) \not\subseteq \leq$	$\mathrm{Cl}(Q') \not\subseteq \leq$	4. Concretely show that $\mathrm{Cl}(Q') \not\subseteq \leq$.
fong-spivak-invitation_F00602	51	■ 1.000	$a \otimes b$	$a \otimes b$	a monoidal structure on preorders. This allows us to construct an element $a \otimes b$ of a
fong-spivak-invitation_F00603	53	■ 1.000	$\boldsymbol{a}$	$\boldsymbol{a}$	2.1 Getting from a to b
fong-spivak-invitation_F00604	53	■ 1.000	$\boldsymbol{b}$	$\boldsymbol{b}$	2.1 Getting from a to b
fong-spivak-invitation_F00605	53	■ 1.000	$d \in [0, \infty]$	$d \in [0, \infty]$	the question of getting from a to b has an answer $d \in [0, \infty]$, a cost.
fong-spivak-invitation_F00606	54	■ 0.994	$(X, \leq)$	$(X, \leq)$	In Section 1.2.2 we introduced preorders. The notation for a preorder, namely $(X, \leq)$,
fong-spivak-invitation_F00607	54	■ 1.000	$I \in X$	$I \in X$	(i) an element $I \in X$, called the <i>monoidal unit</i> ,
fong-spivak-invitation_F00608	54	■ 1.000	$\otimes: X \times X \rightarrow X$	$\otimes: X \times X \rightarrow X$	(ii) a function $\otimes: X \times X \rightarrow X$, called the <i>monoidal product</i> .
fong-spivak-invitation_F00609	54	■ 0.974	$\otimes(x_1, x_2) =$	$\otimes(x_1, x_2) =$	These constituents must satisfy the following properties, where we write $\otimes(x_1, x_2) =$
fong-spivak-invitation_F00610	54	■ 1.000	$x_1 \otimes x_2$	$x_1 \otimes x_2$	$x_1 \otimes x_2$:
fong-spivak-invitation_F00611	54	■ 1.000	$x_1, x_2, y_1, y_2 \in X$	$x_1, x_2, y_1, y_2 \in X$	(a) for all $x_1, x_2, y_1, y_2 \in X$ if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-invitation_F00612	54	■ 1.000	$x_1 \leq y_1$	$x_1 \leq y_1$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-invitation_F00613	54	■ 1.000	$x_2 \leq y_2$	$x_2 \leq y_2$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-invitation_F00614	54	■ 1.000	$x_1 \otimes x_2 \leq y_1 \otimes y_2$	$x_1 \otimes x_2 \leq y_1 \otimes y_2$	(a) for all $x_1, x_2, y_1, y_2 \in X$, if $x_1 \leq y_1$ and $x_2 \leq y_2$, then $x_1 \otimes x_2 \leq y_1 \otimes y_2$,
fong-spivak-invitation_F00615	54	■ 1.000	$I \otimes x = x$	$I \otimes x = x$	(b) for all $x \in X$, the equations $I \otimes x = x$ and $x \otimes I = x$ hold,
fong-spivak-invitation_F00616	54	■ 1.000	$x \otimes I = x$	$x \otimes I = x$	(b) for all $x \in X$, the equations $I \otimes x = x$ and $x \otimes I = x$ hold,
fong-spivak-invitation_F00617	54	■ 1.000	$x, y, z \in X$	$x, y, z \in X$	(c) for all $x, y, z \in X$, the equation $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ holds,
fong-spivak-invitation_F00618	54	■ 1.000	$(x \otimes y) \otimes z = x \otimes (y \otimes z)$	$(x \otimes y) \otimes z = x \otimes (y \otimes z)$	(c) for all $x, y, z \in X$, the equation $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ holds,
fong-spivak-invitation_F00619	54	■ 1.000	$x \otimes y = y \otimes x$	$x \otimes y = y \otimes x$	(d) for all $x, y \in X$, the equation $x \otimes y = y \otimes x$ holds.
fong-spivak-invitation_F00620	54	■ 0.828	$X, \leq, I, \otimes$	$X, \leq, I, \otimes$	tively. A preorder equipped with a symmetric monoidal structure, $(X, \leq, I, \otimes)$, is called
fong-spivak-invitation_F00621	55	■ 1.000	I	I	Anyone can propose a set X , an order $\leq$ on X , an element I in X , and a binary
fong-spivak-invitation_F00622	55	■ 0.487	$\cong$	$\cong$	Remark 2.2. It is often useful to replace $=$ with $\cong$ throughout Definition 2.1. The
fong-spivak-invitation_F00623	55	■ 0.962	$\{*\}$	$\{*\}$	units we have seen include I (as in the definition), 0 , 1 , true , false , $\{*\}$, and more.

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fong-spivak-invitation_F00624	55	0.572	$+, *, \wedge, \vee$	$+, *, \wedge, \vee$	Monoidal products we have seen include $\otimes$ (as in the definition), $+$, $*$, $\wedge$, $\vee$, and $\times$. The
fong-spivak-invitation_F00625	55	0.572	$\times$	$\times$	Monoidal products we have seen include $\otimes$ (as in the definition), $+$, $*$, $\wedge$, $\vee$, and $\times$. The
fong-spivak-invitation_F00626	55	1.000	$-5 \leq \sqrt{2}$	$-5 \leq \sqrt{2}$	numbers; e.g. $-5 < \sqrt{2}$. We propose 0 as a monoidal unit and $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as a
fong-spivak-invitation_F00627	55	1.000	$+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	$+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	numbers; e.g. $-5 \leq \sqrt{2}$. We propose 0 as a monoidal unit and $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as a
fong-spivak-invitation_F00628	55	0.887	$\mathbb{R}, \leq, 0, +$	$\mathbb{R}, \leq, 0, +$	monoidal product. Does $(\mathbb{R}, \leq, 0, +)$ satisfy the conditions of Definition 2.1?
fong-spivak-invitation_F00629	55	1.000	$x_1 + x_2 \leq y_1 + y_2$	$x_1 + x_2 \leq y_1 + y_2$	If $x_1 \leq y_1$ and $x_2 \leq y_2$, it is true that $x_1 + x_2 \leq y_1 + y_2$. It is also true that $0 + x = x$
fong-spivak-invitation_F00630	55	1.000	$0 + x = x$	$0 + x = x$	If $x_1 \leq y_1$ and $x_2 \leq y_2$, it is true that $x_1 + x_2 \leq y_1 + y_2$. It is also true that $0 + x = x$
fong-spivak-invitation_F00631	55	0.988	$x + 0 = x$	$x + 0 = x$	and $x + 0 = x$, that $(x + y) + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-invitation_F00632	55	0.988	$(x + y) + z = x + (y + z)$	$(x + y) + z = x + (y + z)$	and $x + 0 = x$, that $x + y + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-invitation_F00633	55	0.988	$x + y = y + x$	$x + y = y + x$	and $x + 0 = x$, that $(x + y) + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-invitation_F00634	55	0.988	$(\mathbb{R}, \leq, 0, +)$	$(\mathbb{R}, \leq, 0, +)$	and $x + 0 = x$, that $(x + y) + z = x + (y + z)$, and that $x + y = y + x$. Thus $(\mathbb{R}, \leq, 0, +)$
fong-spivak-invitation_F00635	55	0.842	$\mathbb{R}, \leq$	$\mathbb{R}, \leq$	Exercise 2.4. Consider again the preorder $(\mathbb{R}, \leq)$ from Example 2.3. Someone pro-
fong-spivak-invitation_F00636	55	0.975	M	M	Example 2.5. A <i>monoid</i> consists of a set M , a function $*$: $M \times M \rightarrow M$ called the
fong-spivak-invitation_F00637	55	0.975	$*$: $M \times M \rightarrow M$	$*$: $M \times M \rightarrow M$	Example 2.5. A <i>monoid</i> consists of a set M , a function $*$: $M \times M \rightarrow M$ called the
fong-spivak-invitation_F00638	55	1.000	$e \in M$	$e \in M$	<i>monoid multiplication</i> , and an element $e \in M$ called the <i>monoid unit</i> , such that, when
fong-spivak-invitation_F00639	55	1.000	$*(m, n)$	$*(m, n)$	you write $*(m, n)$ as $m * n$, i.e. using infix notation, the equations
fong-spivak-invitation_F00640	55	1.000	$m * n$	$m * n$	you write $*(m, n)$ as $m * n$, i.e. using infix notation, the equations
fong-spivak-invitation_F00641	55	1.000	$m, n, p \in M$	$m, n, p \in M$	hold for all $m, n, p \in M$. It is called <i>commutative</i> if also $m * n = n * m$.
fong-spivak-invitation_F00642	55	1.000	$m * n = n * m$	$m * n = n * m$	hold for all $m, n, p \in M$. It is called <i>commutative</i> if also $m * n = n * m$.
fong-spivak-invitation_F00643	55	0.481	$_S$	S	Every set S determines a discrete preorder Disc _{S} (where $m \leq n$ iff $m = n$; see
fong-spivak-invitation_F00644	55	0.481	$m = n$	$m = n$	Every set S determines a discrete preorder Disc _{S} (where $m \leq n$ iff $m = n$; see
fong-spivak-invitation_F00645	55	0.769	$(M, e, *)$	$(M, e, *)$	Example 1.27), and it is easy to check that if $(M, e, *)$ is a commutative monoid then
fong-spivak-invitation_F00646	55	0.570	$(\text{Disc}_M, =, e, *)$	$(\text{Disc}_M, =, e, *)$	$(\text{Disc}_M, =, e, *)$ is a symmetric monoidal preorder.
fong-spivak-invitation_F00647	55	0.570	$(\text{Disc}_M, =, e, *)$	$(\text{Disc}_M, =, e, *)$	$(\text{Disc}_M, =, e, *)$ is a symmetric monoidal preorder.
fong-spivak-invitation_F00648	55	0.932	$(M, *, e)$	$(M, *, e)$	Exercise 2.6. We said it was easy to check that if $(M, *, e)$ is a commutative monoid
fong-spivak-invitation_F00649	55	0.366	$(\text{Disc}_M, =, *, e)$	$(\text{Disc}_M, =, *, e)$	then $(\text{Disc}_M, =, *, e)$ is a symmetric monoidal preorder. Were we telling the truth? $\diamond$
fong-spivak-invitation_F00650	55	1.000	H	H	Let H be the set of all poker hands, where a hand means a choice of five cards

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fong-spivak-invitation_F00651	56	■ 1.000	$h \leq h^{\prime}$	$h \leq h'$	from the standard 52-card deck. As an order, put $h \leq h'$ if h' beats or equals h in
fong-spivak-invitation_F00652	56	■ 1.000	$h^{\prime}$	h'	from the standard 52-card deck. As an order, put $h \leq h'$ if h' beats or equals h in
fong-spivak-invitation_F00653	56	■ 1.000	$\otimes: H \times H \rightarrow H$	$\otimes: H \times H \rightarrow H$	One could propose a monoidal product $\otimes: H \times H \rightarrow H$ by assigning $h_1 \otimes h_2$ to be
fong-spivak-invitation_F00654	56	■ 1.000	$h_{\{1\}} \otimes h_{\{2\}}$	$h_1 \otimes h_2$	One could propose a monoidal product $\otimes: H \times H \rightarrow H$ by assigning $h_1 \otimes h_2$ to be
fong-spivak-invitation_F00655	56	■ 1.000	$h_{\{1\}}$	h_1	“the best hand one can form out of the ten cards in h_1 and h_2 .” If some cards are in both
fong-spivak-invitation_F00656	56	■ 1.000	$h_{\{2\}}$	h_2	“the best hand one can form out of the ten cards in h_1 and h_2 .” If some cards are in both
fong-spivak-invitation_F00657	56	■ 0.179	$\{2 \odot, 5 \curvearrowright, 6 \odot, 6 \infty, 7 \rightarrow\} = \{2 \odot, 3 \odot, 4 \curvearrowright, 5 \curvearrowright, 6 \bigodot\}$	$\{2 \odot, 5 \curvearrowright, 6 \odot, 6 \infty, 7 \rightarrow\} = \{2 \odot, 3 \odot, 4 \curvearrowright, 5 \curvearrowright, 6 \bigodot\}$	$\{2 \heartsuit, 5 \heartsuit, 6 \heartsuit, 6 \clubsuit, 7 \clubsuit\} = \{2 \heartsuit, 3 \heartsuit, 4 \heartsuit, 5 \heartsuit, 6 \heartsuit\}$ because the latter is the best
fong-spivak-invitation_F00658	56	■ 1.000	$h_{\{1\}} \leq i_{\{1\}}$	$h_1 \leq i_1$	could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$, and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$.
fong-spivak-invitation_F00659	56	■ 1.000	$h_{\{2\}} \leq i_{\{2\}}$	$h_2 \leq i_2$	could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$, and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$.
fong-spivak-invitation_F00660	56	■ 1.000	$h_{\{1\}} \otimes h_{\{2\}} \leq i_{\{1\}} \otimes i_{\{2\}}$	$h_1 \otimes h_2 \leq i_1 \otimes i_2$	could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$, and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$.
fong-spivak-invitation_F00661	56	■ 0.201	$h_{\{1\}} \otimes h_{\{2\}} = \{10 \wedge, \mathrm{~J} \in, \mathrm{Q}, \mathrm{K}, \mathrm{A}\}$	$h_1 \otimes h_2 = \{10 \wedge, \mathrm{~J} \in, \mathrm{Q}, \mathrm{K}, \mathrm{A}\}$	Here, $h_1 \leq i_1$ and $h_2 \leq i_2$, but $h_1 \otimes h_2 = \{10 \clubsuit, \mathrm{~J} \clubsuit, \mathrm{~O} \clubsuit, \mathrm{~K} \clubsuit, \mathrm{~A} \clubsuit\}$ is the best
fong-spivak-invitation_F00662	56	■ 0.409	$i_{\{1\}} \otimes i_{\{2\}} = \{5 \bullet, 5 \oslash, 6 \circ, 6 \diamond, \mathrm{Q} \diamond\}$	$i_1 \otimes i_2 = \{5 \bullet, 5 \oslash, 6 \circ, 6 \diamond, \mathrm{Q} \diamond\}$	possible hand and beats $i_1 \otimes i_2 = \{5 \clubsuit, 5 \heartsuit, 6 \heartsuit, 6 \diamond, \mathrm{Q} \diamond\}$.
fong-spivak-invitation_F00663	57	■ 0.689	$(x \in X)$	$(x \in X)$	$(x \in X)$, relationship ($\leq$), and combination (using transitivity and $\otimes$), and so do wiring
fong-spivak-invitation_F00664	57	■ 0.689	$(\leq)$	$(\leq)$	$(x \in X)$, relationship ($\leq$), and combination (using transitivity and $\otimes$), and so do wiring
fong-spivak-invitation_F00665	57	■ 0.921	$(X, \leq I, \otimes)$	$(X, \leq I, \otimes)$	$(X, \leq, I, \otimes)$ as in Definition 2.1. Wiring diagrams have wires on the left and the right.
fong-spivak-invitation_F00666	57	■ 1.000	$x \otimes y$	$x \otimes y$	draw one wire labeled $x \otimes y$.
fong-spivak-invitation_F00667	58	■ 0.999	$x_{\{1\}} \otimes x_{\{2\}} \leq y_{\{1\}} \otimes y_{\{2\}} \otimes y_{\{3\}}$	$x_1 \otimes x_2 \leq y_1 \otimes y_2 \otimes y_3$	The validity of the second box corresponds to the inequality $x_1 \otimes x_2 \leq y_1 \otimes y_2 \otimes y_3$.
fong-spivak-invitation_F00668	58	■ 0.933	$4 \leq 7$	$4 \leq 7$	And here we express a couple of facts about $(\mathbb{R}, \leq, 0, +)$ in this language: $4 \leq 7$ and
fong-spivak-invitation_F00669	58	■ 1.000	$2+5 \leq -1+5+3$	$2+5 \leq -1+5+3$	$2+5 \leq -1+5+3$.
fong-spivak-invitation_F00670	58	■ 1.000	$x_{\{1\}} \otimes x_{\{2\}} \leq$	$x_1 \otimes x_2 \leq$	structure (Definition 2.1). Property (a) says that if $x_1 \leq y_1$ and $x_2 \leq y_2$ then $x_1 \otimes x_2 \leq$
fong-spivak-invitation_F00671	58	■ 1.000	$y_{\{1\}} \otimes y_{\{2\}}$	$y_1 \otimes y_2$	$y_1 \otimes y_2$. This corresponds to the idea that stacking any two valid boxes in parallel is still
fong-spivak-invitation_F00672	59	■ 0.977	$\mathcal{X} = (X, \leq, I, \otimes)$	$\mathcal{X} = (X, \leq, I, \otimes)$	Given a monoidal preorder $\mathcal{X} = (X, \leq, I, \otimes)$, a wiring diagram is a graphical proof of
fong-spivak-invitation_F00673	59	■ 0.973	$\mathcal{X}$	$\mathcal{X}$	something about $\mathcal{X}$. Each box in the diagram has a left side and a right side, say x and
fong-spivak-invitation_F00674	60	■ 1.000	$v+w+u$	$v+w+u$	technically there is no such thing as $v+w+u$, for example. Instead, there is $(v+w)+u$
fong-spivak-invitation_F00675	60	■ 1.000	$(v+w)+u$	$(v+w)+u$	technically there is no such thing as $v+w+u$, for example. Instead, there is $(v+w)+u$
fong-spivak-invitation_F00676	60	■ 1.000	$v+(w+u)$	$v+(w+u)$	and $v+(w+u)$, and so on.
fong-spivak-invitation_F00677	61	■ 1.000	$2 \mathrm{~H}_{\{2\}} + 2 \mathrm{~Na}$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na}$	The collection on the left, $2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na}$, is called the <i>reactant</i> , and the collection on the

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fong-spivak-invitation_F00678	61	■ 1.000	$2 \mathrm{NaOH} + \mathrm{H}_2$	$2\mathrm{NaOH} + \mathrm{H}_2$	right, $2\mathrm{NaOH} + \mathrm{H}_2$, is called the <i>product</i> .
fong-spivak-invitation_F00679	61	■ 0.688	$\mathrm{NaCl} \in \mathrm{Mat}$	$\mathrm{NaCl} \in \mathrm{Mat}$	tions of atoms and molecules, sometimes called <i>materials</i> . So we have $\mathrm{NaCl} \in \mathrm{Mat}$ and
fong-spivak-invitation_F00680	61	■ 0.484	$4 \mathrm{H}_2\mathrm{O} + 6 \mathrm{Ne} \in \mathrm{Mat}$	$4\mathrm{H}_2\mathrm{O} + 6\mathrm{Ne} \in$	$4\mathrm{H}_2\mathrm{O} + 6\mathrm{Ne} \in$ Mat.
fong-spivak-invitation_F00681	62	■ 0.725	$\otimes$	$\otimes$	in the chemistry setting, taking the place of $\otimes$. While it does not come up in practice,
fong-spivak-invitation_F00682	62	■ 1.000	k	k	Here k is the catalyst because it is found in both the reactant and the product of the
fong-spivak-invitation_F00683	62	■ 1.000	$x + y \rightarrow z$	$x + y \rightarrow z$	reaction. It is said to catalyze the reaction $x + y \rightarrow z$. The idea is that the reaction
fong-spivak-invitation_F00684	62	■ 1.000	$x + y + k \rightarrow z + k$	$x + y + k \rightarrow z + k$	corresponds to the composite reaction $x + y + k \rightarrow z + k$.
fong-spivak-invitation_F00685	63	■ 1.000	$x \leq I$	$x \leq I$	(e) $x \leq I$ for all $x \in X$.
fong-spivak-invitation_F00686	63	■ 1.000	$x \rightarrow 0$	$x \rightarrow 0$	chemistry section, we would write instead $x \rightarrow 0$: any x yields nothing. But this is
fong-spivak-invitation_F00687	63	■ 1.000	10^{23}	10^{23}	manipulate – each item produced. In chemistry, when you have 10^{23} of substance A
fong-spivak-invitation_F00688	64	■ 1.000	$x \leq x + x$	$x \leq x + x$	(f) $x \leq x + x$ for all $x \in X$
fong-spivak-invitation_F00689	65	■ 0.953	$=0$	$= 0$	<code>false = 0</code> and <code>true = 1</code> , then $\wedge$ corresponds to the usual multiplication operation
fong-spivak-invitation_F00690	65	■ 0.953	$\operatorname{true}=1$	<code>true = 1</code>	<code>false = 0</code> and <code>true = 1</code> , then $\wedge$ corresponds to the usual multiplication operation
fong-spivak-invitation_F00691	65	■ 0.905	$= (\mathbb{B}, \leq$	$= (\mathbb{B}, \leq$	preorder which we denote Bool $:= (\mathbb{B}, \leq, \text{true}, \wedge)$.
fong-spivak-invitation_F00692	65	■ 0.905	$\wedge)$	$\wedge)$	preorder which we denote Bool $:= (\mathbb{B}, \leq, \text{true}, \wedge)$.
fong-spivak-invitation_F00693	65	■ 0.975	$\mathcal{V} = (V, \leq, I, \otimes)$	$\mathcal{V} = (V, \leq, I, \otimes)$	monoidal preorder $\mathcal{V} = (V, \leq, I, \otimes)$ means “letting $\mathcal{V}$ structure the question of get-
fong-spivak-invitation_F00694	66	■ 0.921	$\mathbb{B}, \leq$	$\mathbb{B}, \leq$	Exercise 2.13. Let $(\mathbb{B}, \leq)$ be as above, but now consider the monoidal product to be $\vee$
fong-spivak-invitation_F00695	66	■ 0.850	$(\mathbb{N}, \leq)$	$(\mathbb{N}, \leq)$	preorder $(\mathbb{N}, \leq)$ of natural numbers, where $\leq$ is the usual ordering ($0 \leq 1$ and $5 \leq 16$).
fong-spivak-invitation_F00696	66	■ 0.850	$5 \leq 16$	$5 \leq 16$	preorder $(\mathbb{N}, \leq)$ of natural numbers, where $\leq$ is the usual ordering ($0 \leq 1$ and $5 \leq 16$).
fong-spivak-invitation_F00697	66	■ 0.999	$6 + 4 = 10$	$6 + 4 = 10$	where the monoidal unit is 0 and the monoidal product is $+$, i.e. $6 + 4 = 10$. It is easy
fong-spivak-invitation_F00698	66	■ 1.000	$6 * 4 = 24$	$6 * 4 = 24$	product is $*$, i.e. $6 * 4 = 24$. What should the monoidal unit be?
fong-spivak-invitation_F00699	66	■ 1.000	$m \mid n$	$m \mid n$	“divisibility” order on $\mathbb{N}$: we write $m \mid n$ to mean that m divides into n without remainder.
fong-spivak-invitation_F00700	66	■ 0.999	$1 \mid m$	$1 \mid m$	So $1 \mid m$ for all m and $4 \mid 12$.
fong-spivak-invitation_F00701	66	■ 0.999	$4 \mid 12$	$4 \mid 12$	So $1 \mid m$ for all m and $4 \mid 12$.
fong-spivak-invitation_F00702	66	■ 0.970	$(\mathbb{N}, \mid)$	$(\mathbb{N}, \mid)$	There is a monoidal structure on $(\mathbb{N}, \mid)$, where the monoidal unit is 1 and the
fong-spivak-invitation_F00703	66	■ 1.000	$x_1 \mid y_1$	$x_1 \mid y_1$	monoidal product is $*$, i.e. $6 * 4 = 24$. Then if $x_1 \mid y_1$ and $x_2 \mid y_2$, then $(x_1 * x_2) \mid (y_1 * y_2)$.
fong-spivak-invitation_F00704	66	■ 1.000	$x_2 \mid y_2$	$x_2 \mid y_2$	monoidal product is $*$, i.e. $6 * 4 = 24$. Then if $x_1 \mid y_1$ and $x_2 \mid y_2$, then $(x_1 * x_2) \mid (y_1 * y_2)$.
fong-spivak-invitation_F00705	66	■ 1.000	$\left(x_1 * x_2\right) \mid \left(y_1 * y_2\right)$	$\left(x_1 * x_2\right) \mid \left(y_1 * y_2\right)$	monoidal product is $*$, i.e. $6 * 4 = 24$. Then if $x_1 \mid y_1$ and $x_2 \mid y_2$, then $(x_1 * x_2) \mid (y_1 * y_2)$.

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fong-spivak-invitation_F00706	66	■ 1.000	$p_{\{1\}}, p_{\{2\}} \in \mathbb{N}$	$p_1, p_2 \in \mathbb{N}$	Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$, then
fong-spivak-invitation_F00707	66	■ 1.000	$x_{\{1\}} * p_{\{1\}} = y_{\{1\}}$	$x_1 * p_1 = y_1$	Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$, then
fong-spivak-invitation_F00708	66	■ 1.000	$x_{\{2\}} * p_{\{2\}} = y_{\{2\}}$	$x_2 * p_2 = y_2$	Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$, then
fong-spivak-invitation_F00709	66	■ 1.000	$\left(p_{\{1\}} * p_{\{2\}}\right) * \left(x_{\{1\}} * x_{\{2\}}\right) = y_{\{1\}} * y_{\{2\}}$	$(p_1 * p_2) * (x_1 * x_2) = y_1 * y_2$	$(p_1 * p_2) * (x_1 * x_2) = y_1 * y_2$.
fong-spivak-invitation_F00710	67	■ 1.000	$\mathrm{P}(S)$	$P(S)$	Exercise 2.19. Let S be a set and let $P(S)$ be its power set, the set of all subsets of
fong-spivak-invitation_F00711	67	■ 1.000	$\varnothing \subseteq S$	$\emptyset \subseteq S$	S , including the empty subset, $\emptyset \subseteq S$, and the “everything” subset, $S \subseteq S$. We can
fong-spivak-invitation_F00712	67	■ 1.000	$S \subseteq S$	$S \subseteq S$	S , including the empty subset, $\emptyset \subseteq S$, and the “everything” subset, $S \subseteq S$. We can
fong-spivak-invitation_F00713	67	■ 1.000	$A \subseteq B$	$A \subseteq B$	give $P(S)$ an order: $A \leq B$ is given by the subset relation $A \subseteq B$, as discussed in
fong-spivak-invitation_F00714	67	■ 1.000	$A \cap B$	$A \cap B$	S and monoidal product given by intersection $A \cap B$.
fong-spivak-invitation_F00715	67	■ 0.958	$\{\}^{\mathbb{N}}$	$\mathbb{N}$	Exercise 2.20. Let $\mathrm{Prop}^{\mathbb{N}}$ denote the set of all mathematical statements one can make
fong-spivak-invitation_F00716	67	■ 0.422	$\operatorname{Prop}^{\mathbb{N}}$	$\mathrm{Prop}^{\mathbb{N}}$	if and only if the other is true. For example “ n is prime” is an element of $\mathrm{Prop}^{\mathbb{N}}$, and
fong-spivak-invitation_F00717	67	■ 1.000	$n=2$	$n = 2$	so are “ $n = 2$ ” and “ $n \geq 11$.” The statements “ $n + 2 = 5$ ” and “ n is the least odd
fong-spivak-invitation_F00718	67	■ 1.000	$n \geq 11$	$n \geq 11$	so are “ $n = 2$ ” and “ $n \geq 11$.” The statements “ $n + 2 = 5$ ” and “ n is the least odd
fong-spivak-invitation_F00719	67	■ 1.000	$n+2=5$	$n + 2 = 5$	so are “ $n = 2$ ” and “ $n \geq 11$.” The statements “ $n + 2 = 5$ ” and “ n is the least odd
fong-spivak-invitation_F00720	67	■ 0.998	$P, Q \in \operatorname{Prop}^{\mathbb{N}}$	$P, Q \in \mathrm{Prop}^{\mathbb{N}}$	prime” are considered the same. Given $P, Q \in \mathrm{Prop}^{\mathbb{N}}$, we say $P \leq Q$ if for all $n \in \mathbb{N}$,
fong-spivak-invitation_F00721	67	■ 0.998	$P \leq Q$	$P \leq Q$	prime” are considered the same. Given $P, Q \in \mathrm{Prop}^{\mathbb{N}}$, we say $P \leq Q$ if for all $n \in \mathbb{N}$,
fong-spivak-invitation_F00722	67	■ 1.000	$P(n)$	$P(n)$	whenever $P(n)$ is true, so is $Q(n)$.
fong-spivak-invitation_F00723	67	■ 1.000	$Q(n)$	$Q(n)$	whenever $P(n)$ is true, so is $Q(n)$.
fong-spivak-invitation_F00724	67	■ 1.000	$[0, \infty]$	$[0, \infty]$	Example 2.21 (Lawvere’s monoidal preorder, Cost). Let $[0, \infty]$ denote the set of nonneg-
fong-spivak-invitation_F00725	67	■ 0.820	$0, 1, 15.33 \overline{3}$	$0, 1, 15.33\overline{3}$	ative real numbers – such as $0, 1, 15.33\overline{3}$, and 2π – together with ∞ . Consider the
fong-spivak-invitation_F00726	67	■ 0.820	2π	2π	ative real numbers – such as $0, 1, 15.33\overline{3}$, and 2π – together with ∞ . Consider the
fong-spivak-invitation_F00727	67	■ 0.820	∞	∞	ative real numbers – such as $0, 1, 15.33\overline{3}$, and 2π – together with ∞ . Consider the
fong-spivak-invitation_F00728	67	■ 0.972	$[0, \infty], \geq$	$[0, \infty], \geq$	preorder $([0, \infty], \geq)$, with the usual notion of $\geq$, where of course $\infty \geq x$ for all
fong-spivak-invitation_F00729	67	■ 0.972	$\infty \geq x$	$\infty \geq x$	preorder $([0, \infty], \geq)$, with the usual notion of $\geq$, where of course $\infty \geq x$ for all
fong-spivak-invitation_F00730	67	■ 1.000	$x \in [0, \infty]$	$x \in [0, \infty]$	$x \in [0, \infty]$.
fong-spivak-invitation_F00731	67	■ 0.993	$x + \infty = \infty$	$x + \infty = \infty$	monoidal product is $+$. In particular, $x + \infty = \infty$ for any $x \in [0, \infty]$. Let’s call this
fong-spivak-invitation_F00732	67	■ 1.000	$(X, \leq)^{\mathrm{op}} := (X, \geq)$	$(X, \leq)^{\mathrm{op}} := (X, \geq)$	One can take the opposite of any preorder, just flip the order: $(X, \leq)^{\mathrm{op}} := (X, \geq)$; see

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fong-spivak-invitation_F00733	68	■ 0.514	$\mathcal{X}=(X, \leq)$	$\mathcal{X} = (X, \leq)$	Proposition 2.22. Suppose $\mathcal{X} = (X, \leq)$ is a preorder and $\mathcal{X}^{\text{op}} = (X, \geq)$ is its
fong-spivak-invitation_F00734	68	■ 0.514	$X^{\{\mathrm{op}\}}=(X, \geq)$	$X^{\text{op}} = (X, \geq)$	Proposition 2.22. Suppose $\mathcal{X} = (X, \leq)$ is a preorder and $\mathcal{X}^{\text{op}} = (X, \geq)$ is its
fong-spivak-invitation_F00735	68	■ 0.840	$(X, \leq, I, \otimes)$	$(X, \leq, I, \otimes)$	opposite. If $(X, \leq, I, \otimes)$ is a symmetric monoidal preorder then so is its opposite,
fong-spivak-invitation_F00736	68	■ 1.000	$(X, \geq, I, \otimes)$	$(X, \geq, I, \otimes)$	$(X, \geq, I, \otimes)$.
fong-spivak-invitation_F00737	68	■ 0.481	$x_{\{1\}} \geq y_{\{1\}}$	$x_1 \geq y_1$	<i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\text{op}}$; we need
fong-spivak-invitation_F00738	68	■ 0.481	$x_{\{2\}} \geq y_{\{2\}}$	$x_2 \geq y_2$	<i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\text{op}}$; we need
fong-spivak-invitation_F00739	68	■ 0.481	$\mathcal{X}^{\{\text{op}\}}$	$\mathcal{X}^{\text{op}}$	<i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\text{op}}$; we need
fong-spivak-invitation_F00740	68	■ 1.000	$x_{\{1\}} \otimes x_{\{2\}} \geq y_{\{1\}} \otimes y_{\{2\}}$	$x_1 \otimes x_2 \geq y_1 \otimes y_2$	to show that $x_1 \otimes x_2 \geq y_1 \otimes y_2$. But by definition of opposite order, we have $y_1 \leq x_1$
fong-spivak-invitation_F00741	68	■ 1.000	$y_{\{1\}} \leq x_{\{1\}}$	$y_1 \leq x_1$	to show that $x_1 \otimes x_2 \geq y_1 \otimes y_2$. But by definition of opposite order, we have $y_1 \leq x_1$
fong-spivak-invitation_F00742	68	■ 0.925	$y_{\{2\}} \leq x_{\{2\}}$	$y_2 \leq x_2$	and $y_2 \leq x_2$ in $\mathcal{X}$, and thus $y_1 \otimes y_2 \leq x_1 \otimes x_2$ in $\mathcal{X}$. Thus indeed $x_1 \otimes x_2 \geq y_1 \otimes y_2$ in
fong-spivak-invitation_F00743	68	■ 0.925	$y_{\{1\}} \otimes y_{\{2\}} \leq x_{\{1\}} \otimes x_{\{2\}}$	$y_1 \otimes y_2 \leq x_1 \otimes x_2$	and $y_2 \leq x_2$ in $\mathcal{X}$, and thus $y_1 \otimes y_2 \leq x_1 \otimes x_2$ in $\mathcal{X}$. Thus indeed $x_1 \otimes x_2 \geq y_1 \otimes y_2$ in
fong-spivak-invitation_F00744	68	■ 0.989	$\{\}^{\{\mathrm{op}\}}$	$^{\text{op}}$	Cost ^{op} is too.
fong-spivak-invitation_F00745	68	■ 1.000	$x \cong x^{\{\prime\}}$	$x \cong x'$	relation $\cong$ on X , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$.
fong-spivak-invitation_F00746	68	■ 1.000	$x \leq x^{\{\prime\}}$	$x \leq x'$	relation $\cong$ on X , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$.
fong-spivak-invitation_F00747	68	■ 1.000	$x^{\{\prime\}} \leq x$	$x' \leq x$	relation $\cong$ on X , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$.
fong-spivak-invitation_F00748	68	■ 0.904	$\mathcal{P}=\left(P, \leq_{\{P\}}, I_{\{P\}}, \otimes_{\{P\}}\right)$	$\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$	Definition 2.25. Let $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$ and $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$ be monoidal
fong-spivak-invitation_F00749	68	■ 0.904	$\mathcal{Q}=\left(Q, \leq_{\{Q\}}, I_{\{Q\}}, \otimes_{\{Q\}}\right)$	$\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$	Definition 2.25. Let $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$ and $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$ be monoidal
fong-spivak-invitation_F00750	68	■ 0.945	$\mathcal{P}$	$\mathcal{P}$	preorders. A <i>monoidal monotone</i> from $\mathcal{P}$ to $\mathcal{Q}$ is a monotone map $f: (P, \leq_P) \rightarrow$
fong-spivak-invitation_F00751	68	■ 0.945	$f:\left(P, \leq_{\{P\}}\right) \rightarrow$	$f: (P, \leq_P) \rightarrow$	preorders. A <i>monoidal monotone</i> from $\mathcal{P}$ to $\mathcal{Q}$ is a monotone map $f: (P, \leq_P) \rightarrow$
fong-spivak-invitation_F00752	68	■ 1.000	$I_{\{Q\}} \leq_{\{Q\}} f\left(I_{\{P\}}\right)$	$I_Q \leq_Q f(I_P)$	(a) $I_Q \leq_Q f(I_P)$,
fong-spivak-invitation_F00753	68	■ 1.000	$f\left(p_{\{1\}}\right) \otimes_{\{Q\}} f\left(p_{\{2\}}\right) \leq_{\{Q\}} f\left(p_{\{1\}} \otimes_{\{P\}} p_{\{2\}}\right)$	$f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$	(b) $f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$;
fong-spivak-invitation_F00754	68	■ 1.000	$p_{\{1\}}, p_{\{2\}} \in P$	$p_1, p_2 \in P$	for all $p_1, p_2 \in P$.
fong-spivak-invitation_F00755	68	■ 0.970	$I_{\{Q\}} \cong f\left(I_{\{P\}}\right)$	$I_Q \cong f(I_P)$	(a') $I_Q \cong f(I_P)$,
fong-spivak-invitation_F00756	68	■ 1.000	$f\left(p_{\{1\}}\right) \otimes_{\{Q\}} f\left(p_{\{2\}}\right) \cong f\left(p_{\{1\}} \otimes_{\{P\}} p_{\{2\}}\right)$	$f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$	(b') $f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$;
fong-spivak-invitation_F00757	68	■ 0.507	$\left(\mathrm{a}^{\prime}\right) I_{\{Q\}}=f\left(I_{\{P\}}\right)$	(a'') $I_Q = f(I_P)$	(a'') $I_Q = f(I_P)$,
fong-spivak-invitation_F00758	68	■ 0.998	$f\left(p_{\{1\}}\right) \otimes_{\{Q\}} f\left(p_{\{2\}}\right)=f\left(p_{\{1\}} \otimes_{\{P\}} p_{\{2\}}\right)$	$f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$	(b'') $f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$.
fong-spivak-invitation_F00759	69	■ 1.000	$i:\left(\mathbb{Z}, \leq, 0,+\right) \rightarrow\left(\mathbb{R}, \leq, 0,+\right)$	$i: (\mathbb{Z}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$	Example 2.26. There is a monoidal monotone $i: (\mathbb{Z}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$, where

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fong-spivak-invitation_F00760	69	■ 1.000	$i(n)=n$	$i(n) = n$	$i(n) = n$ for all $n \in \mathbb{Z}$. It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$. It is even
fong-spivak-invitation_F00761	69	■ 1.000	$n \in \mathbb{Z}$	$n \in \mathbb{Z}$	$i(n) = n$ for all $n \in \mathbb{Z}$. It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$. It is even
fong-spivak-invitation_F00762	69	■ 1.000	$i(m) \leq i(n)$	$i(m) \leq i(n)$	$i(n) = n$ for all $n \in \mathbb{Z}$. It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$. It is even
fong-spivak-invitation_F00763	69	■ 0.997	$i(0)=0$	$i(0) = 0$	strict monoidal because $i(0) = 0$ and $i(m+n) = i(m) + i(n)$.
fong-spivak-invitation_F00764	69	■ 0.997	$i(m+n)=i(m)+i(n)$	$i(m+n) = i(m) + i(n)$	strict monoidal because $i(0) = 0$ and $i(m+n) = i(m) + i(n)$.
fong-spivak-invitation_F00765	69	■ 1.000	$f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{Z}, \leq, 0, +)$	$f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{Z}, \leq, 0, +)$	There is also a monoidal monotone $f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{Z}, \leq, 0, +)$ going the other
fong-spivak-invitation_F00766	69	■ 1.000	$f(x) := \lfloor x \rfloor$	$f(x) := \lfloor x \rfloor$	way. Here $f(x) := \lfloor x \rfloor$ is the floor function, e.g. $f(3.14) = 3$. It is monotonic because
fong-spivak-invitation_F00767	69	■ 1.000	$f(3.14)=3$	$f(3.14) = 3$	way. Here $f(x) := \lfloor x \rfloor$ is the floor function, e.g. $f(3.14) = 3$. It is monotonic because
fong-spivak-invitation_F00768	69	■ 1.000	$f(0)=0$	$f(0) = 0$	$x \leq y$ implies $f(x) \leq f(y)$. Also $f(0) = 0$ and $f(x) + f(y) \leq f(x+y)$, so it is
fong-spivak-invitation_F00769	69	■ 1.000	$f(x)+f(y) \leq f(x+y)$	$f(x) + f(y) \leq f(x+y)$	$x \leq y$ implies $f(x) \leq f(y)$. Also $f(0) = 0$ and $f(x) + f(y) \leq f(x+y)$, so it is
fong-spivak-invitation_F00770	69	■ 1.000	$f(0.5)+f(0.5) \neq$	$f(0.5) + f(0.5) \neq$	a monoidal monotone. But it is not strict or even strong because $f(0.5) + f(0.5) \neq$
fong-spivak-invitation_F00771	69	■ 1.000	$f(0.5+0.5)$	$f(0.5 + 0.5)$	$f(0.5 + 0.5)$.
fong-spivak-invitation_F00772	69	■ 0.970	$=([0, \infty], \geq, 0, +)$	$= ([0, \infty], \geq, 0, +)$	Recall Bool = $(\mathbb{B}, \leq, \text{true}, \wedge)$ from Example 2.12 and Cost = $([0, \infty], \geq, 0, +)$
fong-spivak-invitation_F00773	69	■ 0.835	$:=\infty$	$:= \infty$	$g(\text{false}) := \infty$ and $g(\text{true}) := 0$.
fong-spivak-invitation_F00774	69	■ 0.835	$:=0$	$:= 0$	$g(\text{false}) := \infty$ and $g(\text{true}) := 0$.
fong-spivak-invitation_F00775	69	■ 0.483	$g: (\mathbb{B}, \leq$	$g: (\mathbb{B}, \leq$	1. Check that the map $g: (\mathbb{B}, \leq, \text{true}, \wedge) \rightarrow ([0, \infty], \geq, 0, +)$ presented above
fong-spivak-invitation_F00776	69	■ 0.483	$\wedge) \rightarrow ([0, \infty], \geq, 0, +)$	$\wedge) \rightarrow ([0, \infty], \geq, 0, +)$	1. Check that the map $g: (\mathbb{B}, \leq, \text{true}, \wedge) \rightarrow ([0, \infty], \geq, 0, +)$ presented above
fong-spivak-invitation_F00777	69	■ 1.000	$d, u: [0, \infty] \rightarrow \mathbb{B}$	$d, u: [0, \infty] \rightarrow \mathbb{B}$	functions $d, u: [0, \infty] \rightarrow \mathbb{B}$ defined as follows:
fong-spivak-invitation_F00778	69	■ 1.000	u	u	4. Is u monotonic?
fong-spivak-invitation_F00779	69	■ 0.860	$(\mathbb{N}, \leq, 1, *)$	$(\mathbb{N}, \leq, 1, *)$	1. Is $(\mathbb{N}, \leq, 1, *)$ a monoidal preorder, where $*$ is the usual multiplication of natural
fong-spivak-invitation_F00780	69	■ 1.000	$(\mathbb{N}, \leq, 0, +) \rightarrow$	$(\mathbb{N}, \leq, 0, +) \rightarrow$	2. If not, why not? If so, does there exist a monoidal monotone $(\mathbb{N}, \leq, 0, +) \rightarrow$
fong-spivak-invitation_F00781	69	■ 0.988	$\mathbb{Z}, \leq, 1, *$	$\mathbb{Z}, \leq, 1, *$	3. Is $(\mathbb{Z}, \leq, 1, *)$ a monoidal preorder?
fong-spivak-invitation_F00782	70	■ 0.763	$\text{Ob}(\mathcal{X})$	$\text{Ob}(\mathcal{X})$	(i) one specifies a set $\text{Ob}(\mathcal{X})$, elements of which are called <i>objects</i> ;
fong-spivak-invitation_F00783	70	■ 1.000	$\mathcal{X}(x, y) \in V$	$\mathcal{X}(x, y) \in V$	(ii) for every two objects x, y , one specifies an element $\mathcal{X}(x, y) \in V$, called the <i>hom-</i>
fong-spivak-invitation_F00784	70	■ 0.622	$x \in \text{Ob}(\mathcal{X})$	$x \in \text{Ob}(\mathcal{X})$	(a) for every object $x \in \text{Ob}(\mathcal{X})$ we have $I \leq \mathcal{X}(x, x)$,
fong-spivak-invitation_F00785	70	■ 0.622	$I \leq \mathcal{X}(x, x)$	$I \leq \mathcal{X}(x, x)$	(a) for every object $x \in \text{Ob}(\mathcal{X})$ we have $I \leq \mathcal{X}(x, x)$,
fong-spivak-invitation_F00786	70	■ 0.288	$x, y, z \in \text{Ob}(\mathcal{X})$	$x, y, z \in \text{Ob}(\mathcal{X})$	(b) for every three objects $x, y, z \in \text{Ob}(\mathcal{X})$, we have $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$.

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fong-spivak-invitation_F00787	70	■ 0.288	$\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	$\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathfrak{X}(x, z)$	(b) for every three objects $x, y, z \in \text{Ob}(\mathcal{X})$, we have $\mathfrak{X}(x, y) \otimes \mathfrak{X}(y, z) \leq \mathfrak{X}(x, z)$.
fong-spivak-invitation_F00788	70	■ 0.848	$\mathfrak{X}$	$\mathfrak{X}$	How does this correspond to a Bool -category $\mathfrak{X}$? Well, the objects of $\mathfrak{X}$ are simply
fong-spivak-invitation_F00789	70	■ 0.751	$\text{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$	$\text{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$	the elements of the preorder, i.e. $\text{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$. Next, for every pair of objects
fong-spivak-invitation_F00790	70	■ 0.940	$\mathcal{X}(x, y)$	$\mathcal{X}(x, y)$	name for $\mathfrak{X}(x, y)$ might be <i>mapping object</i> , but we use “hom” mainly because it is an important jargon
fong-spivak-invitation_F00791	71	■ 0.998	$t \not\leq s$	$t \not\leq s$	false otherwise. So, for example, since $s \leq t$ and $t \not\leq s$, we have $\mathfrak{X}(s, t) = \text{true}$
fong-spivak-invitation_F00792	71	■ 0.998	$\mathcal{X}(s, t) =$	$\mathcal{X}(s, t) =$	false otherwise. So, for example, since $s \leq t$ and $t \not\leq s$, we have $\mathfrak{X}(s, t) = \text{true}$
fong-spivak-invitation_F00793	71	■ 0.998	$\mathcal{X}(t, s) =$	$\mathcal{X}(t, s) =$	and $\mathfrak{X}(t, s) = \text{false}$. Recalling from Example 2.12 that the monoidal unit I of Bool
fong-spivak-invitation_F00794	132	■ 1.000	r	r	infeasible – for that pair. In other words, “yes I can provide p given r ” or “no, I cannot
fong-spivak-invitation_F00795	71	■ 0.538	$\text{Ob}(\mathfrak{X})$	$\text{Ob}(\mathfrak{X})$	(i) a set $\text{Ob}(\mathfrak{X})$,
fong-spivak-invitation_F00796	71	■ 0.656	$x, y \in \text{Ob}(\mathfrak{X})$	$x, y \in \text{Ob}(\mathfrak{X})$	(ii) for every $x, y \in \text{Ob}(\mathfrak{X})$ an element $\mathfrak{X}(x, y) \in \mathbb{B}$, i.e. the element $\mathfrak{X}(x, y)$ is either
fong-spivak-invitation_F00797	71	■ 0.656	$\mathcal{X}(x, y) \in \mathbb{B}$	$\mathcal{X}(x, y) \in \mathbb{B}$	(ii) for every $x, y \in \text{Ob}(\mathfrak{X})$ an element $\mathfrak{X}(x, y) \in \mathbb{B}$, i.e. the element $\mathfrak{X}(x, y)$ is either
fong-spivak-invitation_F00798	71	■ 0.510	$X := \text{Ob}(\mathfrak{X})$	$X := \text{Ob}(\mathfrak{X})$	of $\mathfrak{X}$. So let’s call the preorder $(X, \leq)$, and let $X := \text{Ob}(\mathfrak{X})$. For the $\leq$ relation, let’s
fong-spivak-invitation_F00799	71	■ 0.932	$\mathcal{X}(x, y) =$	$\mathcal{X}(x, y) =$	declare $x \leq y$ iff $\mathfrak{X}(x, y) = \text{true}$. We have the makings of a preorder, but for it to
fong-spivak-invitation_F00800	72	■ 0.767	$\leq \mathcal{X}(x, x)$	$\leq \mathcal{X}(x, x)$	(a) for every element $x \in X$ we have $\text{true} \leq \mathfrak{X}(x, x)$,
fong-spivak-invitation_F00801	72	■ 0.835	$\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	$\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$	(b) for every three elements $x, y, z \in X$ we have $\mathfrak{X}(x, y) \wedge \mathfrak{X}(y, z) \leq \mathfrak{X}(x, z)$.
fong-spivak-invitation_F00802	72	■ 0.943	$b \in$	$b \in$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathfrak{X}(x, x) =$
fong-spivak-invitation_F00803	72	■ 0.943	$\leq b$	$\leq b$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathfrak{X}(x, x) =$
fong-spivak-invitation_F00804	72	■ 0.943	$b =$	$b =$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathfrak{X}(x, x) =$
fong-spivak-invitation_F00805	72	■ 0.943	$\mathcal{X}(x, x) =$	$\mathcal{X}(x, x) =$	For $b \in \mathbf{Bool}$, if $\text{true} \leq b$ then $b = \text{true}$, so the first statement says $\mathfrak{X}(x, x) =$
fong-spivak-invitation_F00806	72	■ 0.555	$b \in \mathbb{B}$	$b \in \mathbb{B}$	$\text{false} \leq b$ for all $b \in \mathbb{B}$, the only way statement (b) has any force is if $\mathfrak{X}(x, y) = \text{true}$
fong-spivak-invitation_F00807	72	■ 0.555	$\mathcal{X}(x, y) = \text{tr}$	$\mathcal{X}(x, y) = \text{tr}$	$\text{false} \leq b$ for all $b \in \mathbb{B}$, the only way statement (b) has any force is if $\mathfrak{X}(x, y) = \text{true}$
fong-spivak-invitation_F00808	72	■ 0.994	$\mathcal{X}(y, z) =$	$\mathcal{X}(y, z) =$	and $\mathfrak{X}(y, z) = \text{true}$, in which case it forces $\mathfrak{X}(x, z) = \text{true}$. This condition exactly
fong-spivak-invitation_F00809	72	■ 0.994	$\mathcal{X}(x, z) =$	$\mathcal{X}(x, z) =$	and $\mathfrak{X}(y, z) = \text{true}$, in which case it forces $\mathfrak{X}(x, z) = \text{true}$. This condition exactly
fong-spivak-invitation_F00810	72	■ 0.777	X, d	X, d	Definition 2.34. A metric space (X, d) consists of:
fong-spivak-invitation_F00811	72	■ 1.000	$d: X \times X \rightarrow \mathbb{R}_{\geq 0}$	$d: X \times X \rightarrow \mathbb{R}_{\geq 0}$	(ii) a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$, where $d(x, y)$ is called the <i>distance between</i> x
fong-spivak-invitation_F00812	72	■ 1.000	$d(x, y)$	$d(x, y)$	(ii) a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$, where $d(x, y)$ is called the <i>distance between</i> x
fong-spivak-invitation_F00813	72	■ 1.000	$d(x, x) = 0$	$d(x, x) = 0$	(a) for every $x \in X$, we have $d(x, x) = 0$,

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fong-spivak-invitation_F00814	72	■ 1.000	$d(x, y)=0$	$d(x, y) = 0$	(b) for every $x, y \in X$, if $d(x, y) = 0$ then $x = y$,
fong-spivak-invitation_F00815	72	■ 1.000	$d(x, y)=d(y, x)$	$d(x, y) = d(y, x)$	(c) for every $x, y \in X$, we have $d(x, y) = d(y, x)$,
fong-spivak-invitation_F00816	72	■ 1.000	$d(x, y)+d(y, z) \geq d(x, z)$	$d(x, y) + d(y, z) \geq d(x, z)$	(d) for every $x, y, z \in X$, we have $d(x, y) + d(y, z) \geq d(x, z)$.
fong-spivak-invitation_F00817	72	■ 1.000	$d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$	$d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$	If we ask instead in (ii) for a function $d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$ we call
fong-spivak-invitation_F00818	72	■ 0.672	(X, d)	(X, d)	(X, d) an <i>extended metric space</i> .
fong-spivak-invitation_F00819	73	■ 1.000	z	z	The triangle inequality says that when plotting a route from x to z , the distance is
fong-spivak-invitation_F00820	73	■ 1.000	$x \rightarrow y \rightarrow z$	$x \rightarrow y \rightarrow z$	always at most what you get by choosing an intermediate point y and going $x \rightarrow y \rightarrow z$.
fong-spivak-invitation_F00821	73	■ 1.000	$3+5 \geq 7.2$	$3 + 5 \geq 7.2$	It can be invoked in three different ways in the above picture: $3 + 5 \geq 7.2$, but also
fong-spivak-invitation_F00822	73	■ 0.997	$5+7.2 \geq 3$	$5 + 7.2 \geq 3$	$5 + 7.2 \geq 3$, and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2$, $7.2 + 5 \geq 3$, and $7.2 + 3 \geq 5$.
fong-spivak-invitation_F00823	73	■ 0.997	$3+7.2 \geq 5$	$3 + 7.2 \geq 5$	$5 + 7.2 \geq 3$, and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2$, $7.2 + 5 \geq 3$, and $7.2 + 3 \geq 5$.
fong-spivak-invitation_F00824	73	■ 0.997	$5+3 \geq 7.2, 7.2+5 \geq 3$	$5 + 3 \geq 7.2, 7.2 + 5 \geq 3$	$5 + 7.2 \geq 3$, and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2, 7.2 + 5 \geq 3$, and $7.2 + 3 \geq 5$.
fong-spivak-invitation_F00825	73	■ 0.997	$7.2+3 \geq 5$	$7.2 + 3 \geq 5$	$5 + 7.2 \geq 3$, and $3 + 7.2 \geq 5$. Oh yeah, and $5 + 3 \geq 7.2, 7.2 + 5 \geq 3$, and $7.2 + 3 \geq 5$.
fong-spivak-invitation_F00826	73	■ 0.528	$d(x, y)=? d(y, x)$	$d(x, y) =? d(y, x)$	are any hills, the symmetry axiom, $d(x, y) =? d(y, x)$, fails: it's easier to get from x
fong-spivak-invitation_F00827	73	■ 0.358	$d($	$d($	It also violates another condition, because $d(\text{Boston}, \text{US}) = 0$. No matter where you
fong-spivak-invitation_F00828	73	■ 0.374	$) \neq 0$	$) \neq 0$	$d(\text{US}, \text{Boston}) \neq 0$.
fong-spivak-invitation_F00829	73	■ 0.982	$U \subseteq X$	$U \subseteq X$	³ The Hausdorff distance gives a metric on the set of all subsets $U \subseteq X$ of a given metric space (X, d) . One
fong-spivak-invitation_F00830	73	■ 1.000	$d(U, V) := \max(d_L(U, V), d_L(V, U))$	$d(U, V) := \max(d_L(U, V), d_L(V, U))$	formula and symmetrize it: $d(U, V) := \max(d_L(U, V), d_L(V, U))$. We happen to see the unsymmetrized
fong-spivak-invitation_F00831	74	■ 0.958	$\{r \in \mathbb{R} \mid r < 0\}$	$\{r \in \mathbb{R} \mid r < 0\}$	two regions is often infinite, e.g. the distance between $\{r \in \mathbb{R} \mid r < 0\}$ and $\{0\}$ as subsets
fong-spivak-invitation_F00832	74	■ 0.958	$\{0\}$	$\{0\}$	two regions is often infinite, e.g. the distance between $\{r \in \mathbb{R} \mid r < 0\}$ and $\{0\}$ as subsets
fong-spivak-invitation_F00833	74	■ 0.725	$(\mathbb{R}, d)$	$(\mathbb{R}, d)$	of $(\mathbb{R}, d)$ is infinite.
fong-spivak-invitation_F00834	74	■ 0.578	$x, y \in \text{Ob}(\mathcal{X})$	$x, y \in \text{Ob}(\mathcal{X})$	(ii) for every $x, y \in \text{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in [0, \infty]$.
fong-spivak-invitation_F00835	74	■ 0.578	$\mathcal{X}(x, y) \in [0, \infty]$	$\mathcal{X}(x, y) \in [0, \infty]$	(ii) for every $x, y \in \text{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in [0, \infty]$.
fong-spivak-invitation_F00836	74	■ 1.000	$0 \geq d(x, x)$	$0 \geq d(x, x)$	(a) $0 \geq d(x, x)$ for all $x \in X$,
fong-spivak-invitation_F00837	74	■ 1.000	$d(x, x) \in [0, \infty]$	$d(x, x) \in [0, \infty]$	Since $d(x, x) \in [0, \infty]$, if $0 \geq d(x, x)$ then $d(x, x) = 0$. So the first condition is
fong-spivak-invitation_F00838	74	■ 1.000	$d(x, y) := y - x $	$d(x, y) := y - x $	hence a Lawvere metric space structure. Namely $d(x, y) := y - x $, the absolute value
fong-spivak-invitation_F00839	74	■ 1.000	$d(3, 7)=4$	$d(3, 7) = 4$	of the difference. So $d(3, 7) = 4$.
fong-spivak-invitation_F00840	74	■ 0.953	$\mathbb{R}_{\geq 0}, \geq, 0, +$	$\mathbb{R}_{\geq 0}, \geq, 0, +$	Exercise 2.38. Consider the symmetric monoidal preorder $(\mathbb{R}_{>0}, \geq, 0, +)$, which is

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fong-spivak-invitation_F00841	74	■ 0.994	$\left(\mathbb{R}_{\geq 0}, \geq, 0, +\right)$	$(\mathbb{R}_{\geq 0}, \geq, 0, +)$	difference between a Lawvere metric space and a $(\mathbb{R}_{\geq 0}, \geq, 0, +)$ -category in the sense
fong-spivak-invitation_F00842	74	■ 0.998	$w \geq 0$	$w \geq 0$	graph – a graph whose edges are labeled with numbers $w \geq 0$ – into a Lawvere metric
fong-spivak-invitation_F00843	75	■ 1.000	$d_{\{X\}}$	d_X	Given a weighted graph, one forms a metric d_X on its set X of vertices by setting $d(p, q)$
fong-spivak-invitation_F00844	75	■ 1.000	$d(p, q)$	$d(p, q)$	Given a weighted graph, one forms a metric d_X on its set X of vertices by setting $d(p, q)$
fong-spivak-invitation_F00846	106	■ 1.000	D	D	C , and D' to D . That is, once we have made this decision about how F acts on objects,
fong-spivak-invitation_F00847	75	■ 1.000	$M_{\{G\}}$	M_G	G and getting a square matrix M_G , whose rows and columns are indexed by the vertices
fong-spivak-invitation_F00848	76	■ 1.000	$d_{\{Y\}}$	d_Y	that the more difficult “distance matrices” d_Y , such as (2.19), can be obtained from
fong-spivak-invitation_F00849	76	■ 1.000	$M_{\{Y\}}$	M_Y	the easy graph matrices M_Y , such as (2.20), by repeating a certain sort of “matrix
fong-spivak-invitation_F00850	76	■ 1.000	$M_{\{X\}}$	M_X	Exercise 2.40. Copy and complete the matrix M_X associated to the graph X in
fong-spivak-invitation_F00852	76	■ 0.731	$\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$	$\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$	Exercise 2.42. Let M be a set and let $\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$ be the monoidal preorder
fong-spivak-invitation_F00853	76	■ 1.000	$\mathcal{M}$	$\mathcal{M}$	of modes of transportation (e.g. car, boat, foot). Then an $\mathcal{M}$ -category $\mathcal{X}$ tells you
fong-spivak-invitation_F00854	76	■ 1.000	$a, b \in$	$a, b \in$	all the modes that will get you from a all the way to b , for any two points $a, b \in$
fong-spivak-invitation_F00855	77	■ 0.981	$M =$	$M =$	$M = \{\text{car, boat, foot}\}$.
fong-spivak-invitation_F00856	77	■ 0.925	$\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$	$\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$	Exercise 2.43. Consider the monoidal preorder $\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$.
fong-spivak-invitation_F00857	77	■ 1.000	$\mathbb{N} \cup \{\infty\}$	$\mathbb{N} \cup \{\infty\}$	1. Draw a small graph labeled by elements of $\mathbb{N} \cup \{\infty\}$.
fong-spivak-invitation_F00858	77	■ 1.000	$\mathcal{W}$	$\mathcal{W}$	3. Prove that this matrix is the matrix of hom-objects for a $\mathcal{W}$ -category. This will give
fong-spivak-invitation_F00859	77	■ 0.998	$f: \mathcal{V} \rightarrow \mathcal{W}$	$f: \mathcal{V} \rightarrow \mathcal{W}$	base. This allows us to use a monoidal monotone $f: \mathcal{V} \rightarrow \mathcal{W}$ to construct $\mathcal{W}$ -categories
fong-spivak-invitation_F00860	77	■ 1.000	$\mathcal{V} \rightarrow \mathcal{W}$	$\mathcal{V} \rightarrow \mathcal{W}$	Any monoidal monotone $\mathcal{V} \rightarrow \mathcal{W}$ between symmetric monoidal preorders lets us
fong-spivak-invitation_F00861	77	■ 1.000	$\mathcal{C}$	$\mathcal{C}$	Construction 2.44. Let $f: \mathcal{V} \rightarrow \mathcal{W}$ be a monoidal monotone. Given a $\mathcal{V}$ -category $\mathcal{C}$,
fong-spivak-invitation_F00862	77	■ 1.000	$\mathcal{C}_f$	$\mathcal{C}_f$	one forms the associated $\mathcal{W}$ -category, say $\mathcal{C}_f$, as follows.
fong-spivak-invitation_F00863	77	■ 0.999	$\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$	$\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$	(i) We take the same objects: $\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$.
fong-spivak-invitation_F00864	77	■ 0.979	$c, d \in \mathrm{Ob}(\mathcal{C})$	$c, d \in \mathrm{Ob}(\mathcal{C})$	(ii) For any $c, d \in \mathrm{Ob}(\mathcal{C})$, put $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$.
fong-spivak-invitation_F00865	77	■ 0.979	$\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$	$\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$	(ii) For any $c, d \in \mathrm{Ob}(\mathcal{C})$, put $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$.
fong-spivak-invitation_F00866	78	■ 1.000	$c \in \mathcal{C}$	$c \in \mathcal{C}$	(a) for every $c \in \mathcal{C}$, we have
fong-spivak-invitation_F00867	78	■ 0.945	$c, d, e \in \mathrm{Ob}(\mathcal{C})$	$c, d, e \in \mathrm{Ob}(\mathcal{C})$	(b) for every $c, d, e \in \mathrm{Ob}(\mathcal{C})$, we have
fong-spivak-invitation_F00868	78	■ 0.998	$f: [0, \infty] \rightarrow \{\text{true, false}\}$	$f: [0, \infty] \rightarrow \{\text{true, false}\}$	Example 2.45. As an example, consider the function $f: [0, \infty] \rightarrow \{\text{true, false}\}$
fong-spivak-invitation_F00869	78	■ 0.831	$x_f \neq x_g$	$x_f \neq x_g$	$\mathcal{X}_f \neq \mathcal{X}_g$.

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fong-spivak-invitation_F00870	79	0.703	$\mathcal{Y}$	$\mathcal{Y}$	$\mathcal{V}$ -categories. A $\mathcal{V}$ -functor from $\mathcal{X}$ to $\mathcal{Y}$, denoted $F: \mathcal{X} \rightarrow \mathcal{Y}$, consists of one constituent
fong-spivak-invitation_F00871	79	0.703	$F: \mathcal{X} \rightarrow \mathcal{Y}$	$F: \mathcal{X} \rightarrow \mathcal{Y}$	$\mathcal{V}$ -categories. A $\mathcal{V}$ -functor from $\mathcal{X}$ to $\mathcal{Y}$, denoted $F: \mathcal{X} \rightarrow \mathcal{Y}$, consists of one constituent
fong-spivak-invitation_F00872	79	0.752	$F: \mathcal{O}b(\mathcal{X}) \rightarrow \mathcal{O}b(\mathcal{Y})$	$F: \mathcal{O}b(\mathcal{X}) \rightarrow \mathcal{O}b(\mathcal{Y})$	(i) a function $F: \mathcal{O}b(\mathcal{X}) \rightarrow \mathcal{O}b(\mathcal{Y})$
fong-spivak-invitation_F00873	79	0.907	$x_1, x_2 \in \mathcal{O}b(\mathcal{X})$	$x_1, x_2 \in \mathcal{O}b(\mathcal{X})$	(a) for all $x_1, x_2 \in \mathcal{O}b(\mathcal{X})$, one has $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$.
fong-spivak-invitation_F00874	79	0.907	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$	$\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$	(a) for all $x_1, x_2 \in \mathcal{O}b(\mathcal{X})$, one has $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$.
fong-spivak-invitation_F00875	79	0.931	$\mathcal{X}(x_1, x_2) =$	$\mathcal{X}(x_1, x_2) =$	preorders are Bool -categories, where $\mathcal{X}(x_1, x_2) = \text{true}$ is denoted $x_1 \leq x_2$. One
fong-spivak-invitation_F00876	79	0.931	$x_1 \leq x_2$	$x_1 \leq x_2$	preorders are Bool -categories, where $\mathcal{X}(x_1, x_2) = \text{true}$ is denoted $x_1 \leq x_2$. One
fong-spivak-invitation_F00877	79	0.968	$(X, \leq_X) \rightarrow (Y, \leq_Y)$	$(X, \leq_X) \rightarrow (Y, \leq_Y)$	functors, and that's true. A monotone map $(X, \leq_X) \rightarrow (Y, \leq_Y)$ is a function $F: X \rightarrow Y$
fong-spivak-invitation_F00878	79	0.998	$x_1, x_2 \in X$	$x_1, x_2 \in X$	such that, for every $x_1, x_2 \in X$, if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$. In other words, we
fong-spivak-invitation_F00879	79	0.998	$x_1 \leq_X x_2$	$x_1 \leq_X x_2$	such that, for every $x_1, x_2 \in X$, if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$. In other words, we
fong-spivak-invitation_F00880	79	0.998	$F(x_1) \leq_Y F(x_2)$	$F(x_1) \leq_Y F(x_2)$	such that, for every $x_1, x_2 \in X$, if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$. In other words, we
fong-spivak-invitation_F00881	79	0.803	$\mathcal{V} = \mathbf{Bool}$	$\mathcal{V} = \mathbf{Bool}$	where the above $\leq$ takes place in the enriching category $\mathcal{V} = \mathbf{Bool}$; this is exactly the
fong-spivak-invitation_F00882	79	0.999	(X, d_X)	(X, d_X)	points does not increase. That is, given Lawvere metric spaces (X, d_X) and (Y, d_Y) , a
fong-spivak-invitation_F00883	79	0.999	(Y, d_Y)	(Y, d_Y)	points does not increase. That is, given Lawvere metric spaces (X, d_X) and (Y, d_Y) , a
fong-spivak-invitation_F00884	79	0.999	$d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$	$d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$	have $d_Y(x_1, x_2) > d_Y(F(x_1), F(x_2))$.
fong-spivak-invitation_F00885	79	0.888	$\mathcal{X}^{\text{op}}$	$\mathcal{X}^{\text{op}}$	category $\mathcal{X}$ is denoted $\mathcal{X}^{\text{op}}$ and is defined by
fong-spivak-invitation_F00886	79	0.625	$\mathcal{O}b(\mathcal{X}^{\text{op}}) := \mathcal{O}b(\mathcal{X})$	$\mathcal{O}b(\mathcal{X}^{\text{op}}) := \mathcal{O}b(\mathcal{X})$	(i) $\mathcal{O}b(\mathcal{X}^{\text{op}}) := \mathcal{O}b(\mathcal{X})$, and
fong-spivak-invitation_F00887	79	0.688	$x, y \in \mathcal{X}$	$x, y \in \mathcal{X}$	(ii) for all $x, y \in \mathcal{X}$, we have $\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$.
fong-spivak-invitation_F00888	79	0.688	$\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$	$\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$	(ii) for all $x, y \in \mathcal{X}$, we have $\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$.
fong-spivak-invitation_F00889	79	1.000	$\dagger: \mathcal{X} \rightarrow$	$\dagger: \mathcal{X} \rightarrow$	A $\mathcal{V}$ -category $\mathcal{X}$ is a <i>dagger</i> $\mathcal{V}$ -category if the identity function is a $\mathcal{V}$ -functor $\dagger: \mathcal{X} \rightarrow$
fong-spivak-invitation_F00890	79	0.154	$\mathcal{X}^{\text{Op}}$	$\mathcal{X}^{\text{Op}}$	$\mathcal{X}^{\text{op}}$. And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$, then
fong-spivak-invitation_F00891	79	0.154	$I \leq \mathcal{X}(x, y)$	$I \leq \mathcal{X}(x, y)$	$\mathcal{X}^{\text{op}}$. And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$, then
fong-spivak-invitation_F00892	79	0.154	$I \leq \mathcal{X}(y, x)$	$I \leq \mathcal{X}(y, x)$	$\mathcal{X}^{\text{op}}$. And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$, then
fong-spivak-invitation_F00893	80	0.797	$\mathcal{X} \times \mathcal{Y}$	$\mathcal{X} \times \mathcal{Y}$	<i>product</i> , to be the $\mathcal{V}$ -category $\mathcal{X} \times \mathcal{Y}$ with
fong-spivak-invitation_F00894	80	0.435	$\mathcal{O}b(\mathcal{X} \times \mathcal{Y}) := \mathcal{O}b(\mathcal{X}) \times \mathcal{O}b(\mathcal{Y})$	$\mathcal{O}b(\mathcal{X} \times \mathcal{Y}) := \mathcal{O}b(\mathcal{X}) \times \mathcal{O}b(\mathcal{Y})$	(i) $\mathcal{O}b(\mathcal{X} \times \mathcal{Y}) := \mathcal{O}b(\mathcal{X}) \times \mathcal{O}b(\mathcal{Y})$
fong-spivak-invitation_F00895	80	0.978	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$	(ii) $(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$.

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fong-spivak-invitation_F00896	80	■ 0.977	$\left(x^{\wedge\{\text{prime}\}}, y^{\wedge\{\text{prime}\}}\right)$	(x', y')	for two objects (x, y) and (x', y') in $\text{Ob}(\mathcal{X} \times \mathcal{Y})$.
fong-spivak-invitation_F00897	80	■ 0.977	$\operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$	$\text{Ob}(\mathcal{X} \times \mathcal{Y})$	for two objects (x, y) and (x', y') in $\text{Ob}(\mathcal{X} \times \mathcal{Y})$.
fong-spivak-invitation_F00898	80	■ 0.722	$(x, y) \in \operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$	$(x, y) \in \text{Ob}(\mathcal{X} \times \mathcal{Y})$	1. Check that for every object $(x, y) \in \text{Ob}(\mathcal{X} \times \mathcal{Y})$ we have $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$.
fong-spivak-invitation_F00899	80	■ 0.722	$I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$	$I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$	1. Check that for every object $(x, y) \in \text{Ob}(\mathcal{X} \times \mathcal{Y})$ we have $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$.
fong-spivak-invitation_F00900	80	■ 1.000	$\left(x_1, y_1\right), \left(x_2, y_2\right)$	$(x_1, y_1), (x_2, y_2)$	2. Check that for every three objects $(x_1, y_1), (x_2, y_2)$, and (x_3, y_3) , we have
fong-spivak-invitation_F00901	80	■ 1.000	$\left(x_3, y_3\right)$	(x_3, y_3)	2. Check that for every three objects $(x_1, y_1), (x_2, y_2)$, and (x_3, y_3) , we have
fong-spivak-invitation_F00902	80	■ 1.000	$\left(P, \leq_P\right) \times \left(Q, \leq_Q\right)$	$(P, \leq_P) \times (Q, \leq_Q)$	When taking the product of two preorders $(P, \leq_P) \times (Q, \leq_Q)$, as first described in
fong-spivak-invitation_F00903	80	■ 0.879	$\left(p_1, q_1\right) \leq \left(p_2, q_2\right)$	$(p_1, q_1) \leq (p_2, q_2)$	Example 1.51, we say that $(p_1, q_1) \leq (p_2, q_2)$ iff both $p_1 \leq p_2$ AND $q_1 \leq q_2$; the AND
fong-spivak-invitation_F00904	80	■ 0.879	$q_1 \leq q_2$	$q_1 \leq q_2$	Example 1.51, we say that $(p_1, q_1) \leq (p_2, q_2)$ iff both $p_1 \leq p_2$ AND $q_1 \leq q_2$; the AND
fong-spivak-invitation_F00905	81	■ 0.996	$d_{X \times Y}\left((x, y), \left(x^{\wedge\{\text{prime}\}}, y^{\wedge\{\text{prime}\}}\right)\right)$	$d_{X \times Y}((x, y), (x', y'))$	objects in $\mathcal{X} \times \mathcal{Y}$. And the distance $d_{X \times Y}((x, y), (x', y'))$ between any two points is
fong-spivak-invitation_F00906	81	■ 1.000	$d_X\left(x, x^{\wedge\{\text{prime}\}}\right)+d_Y\left(y, y^{\wedge\{\text{prime}\}}\right)$	$d_X(x, x') + d_Y(y, y')$	given by the sum $d_X(x, x') + d_Y(y, y')$.
fong-spivak-invitation_F00907	81	■ 0.999	$\mathbb{R} \times \mathbb{R}$	$\mathbb{R} \times \mathbb{R}$	Example 2.37). Form the Cost -product $\mathbb{R} \times \mathbb{R}$. What is the distance from $(5, 6)$ to
fong-spivak-invitation_F00908	81	■ 0.999	$(5, 6)$	$(5, 6)$	Example 2.37). Form the Cost -product $\mathbb{R} \times \mathbb{R}$. What is the distance from $(5, 6)$ to
fong-spivak-invitation_F00909	81	■ 0.991	$(-1, 4)$	$(-1, 4)$	$(-1, 4)$? Hint: apply Definition 2.53; the answer is not $\sqrt{40}$.
fong-spivak-invitation_F00910	81	■ 0.991	$\sqrt{40}$	$\sqrt{40}$	$(-1, 4)$? Hint: apply Definition 2.53; the answer is not $\sqrt{40}$.
fong-spivak-invitation_F00911	125	■ 1.000	$x \times y$	$x \times y$	corresponding category. Show that the product $x \times y$ in $\mathcal{P}$ agrees with their meet $x \wedge y$
fong-spivak-invitation_F00921	82	■ 0.788	$\mathcal{V}=(V, \leq I, \otimes)$	$\mathcal{V} = (V, \leq I, \otimes)$	Definition 2.57. A symmetric monoidal preorder $\mathcal{V} = (V, \leq, I, \otimes)$ is called <i>symmetric</i>
fong-spivak-invitation_F00922	82	■ 1.000	$v, w \in V$	$v, w \in V$	<i>monoidal closed</i> (or just <i>closed</i>) if, for every two elements $v, w \in V$, there is an element
fong-spivak-invitation_F00923	82	■ 0.984	$v \multimap w$	$v \multimap w$	$v \multimap w$ in $\mathcal{V}$, called the <i>hom-element</i> , with the property
fong-spivak-invitation_F00924	82	■ 1.000	$a, v, w \in V$	$a, v, w \in V$	for all $a, v, w \in V$.
fong-spivak-invitation_F00925	82	■ 1.000	$v \in V$	$v \in V$	preorder is monoidal closed iff, given any $v \in V$, the map $(- \otimes v): V \rightarrow V$ given by
fong-spivak-invitation_F00926	82	■ 1.000	$(- \otimes v): V \rightarrow V$	$(- \otimes v): V \rightarrow V$	preorder is monoidal closed iff, given any $v \in V$, the map $(- \otimes v): V \rightarrow V$ given by
fong-spivak-invitation_F00927	82	■ 0.756	$(v \multimap -): V \rightarrow V$	$(v \multimap -): V \rightarrow V$	multiplying with v has a right adjoint. We write this right adjoint $(v \multimap -): V \rightarrow V$.
fong-spivak-invitation_F00928	82	■ 1.000	$(- \otimes v)$	$(- \otimes v)$	1. Using Definition 2.1, show that $(- \otimes v)$ is monotone.
fong-spivak-invitation_F00929	82	■ 1.000	$((v \multimap w) \otimes v) \leq w$	$((v \multimap w) \otimes v) \leq w$	2. Supposing that $\mathcal{V}$ is closed, show that for all $v, w \in V$ we have $((v \multimap w) \otimes v) \leq w$.
fong-spivak-invitation_F00930	82	■ 0.771	$(v \multimap -)$	$(v \multimap -)$	3. Using part 2, show that $(v \multimap -)$ is monotone.
fong-spivak-invitation_F00931	83	■ 1.000	$x, y \in [0, \infty]$	$x, y \in [0, \infty]$	Indeed, for any $x, y \in [0, \infty]$, define $x \multimap y := \max(0, y - x)$. Then, for any $a, x, y \in$

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fong-spivak-invitation_F00932	83	■ 1.000	$x \multimap y := \max(0, y - x)$	$x \multimap y := \max(0, y - x)$	Indeed, for any $x, y \in [0, \infty]$, define $x \multimap y := \max(0, y - x)$. Then, for any $a, x, y \in$
fong-spivak-invitation_F00933	83	■ 1.000	$a, x, y \in$	$a, x, y \in$	Indeed, for any $x, y \in [0, \infty]$, define $x \multimap y := \max(0, y - x)$. Then, for any $a, x, y \in$
fong-spivak-invitation_F00934	83	■ 0.543	$(\mathbb{B}, \leq$	$(\mathbb{B}, \leq$	Example 2.62. A non-example is $(\mathbb{B}, \leq, \text{false}, \vee)$. Indeed, suppose we had a $\multimap$ oper-
fong-spivak-invitation_F00935	83	■ 0.543	$\vee)$	$\vee)$	Example 2.62. A non-example is $(\mathbb{B}, \leq, \text{false}, \vee)$. Indeed, suppose we had a $\multimap$ oper-
fong-spivak-invitation_F00936	83	■ 0.812	$\leq p \multimap q$	$\leq p \multimap q$	ator as in Definition 2.57. Note that $\text{false} \leq p \multimap q$, for any p, q no matter what $\multimap$
fong-spivak-invitation_F00937	83	■ 0.812	$\longrightarrow$	$\longrightarrow$	ator as in Definition 2.57. Note that $\text{false} \leq p \multimap q$, for any p, q no matter what $\multimap$
fong-spivak-invitation_F00938	83	■ 1.000	$a =$	$a =$	is, because false is less than everything. But using $a = \text{false}$, $p = \text{true}$, and
fong-spivak-invitation_F00939	83	■ 1.000	$p =$	$p =$	is, because false is less than everything. But using $a = \text{false}$, $p = \text{true}$, and
fong-spivak-invitation_F00940	83	■ 0.927	$q =$	$q =$	$q = \text{false}$, we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p \multimap q)$.
fong-spivak-invitation_F00941	83	■ 0.927	$(a \vee p) \not\leq q$	$(a \vee p) \not\leq q$	$q = \text{false}$, we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p \multimap q)$.
fong-spivak-invitation_F00942	83	■ 0.927	$a \leq (p \multimap q)$	$a \leq (p \multimap q)$	$q = \text{false}$, we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p \multimap q)$.
fong-spivak-invitation_F00943	83	■ 0.969	c, d	c, d	that for every two material collections c, d one can form a material collection $c \multimap d$
fong-spivak-invitation_F00944	83	■ 0.969	$c \longrightarrow d$	$c \rightarrow d$	that for every two material collections c, d one can form a material collection $c \multimap d$
fong-spivak-invitation_F00945	83	■ 1.000	$2 \text{H}_2\text{O} + 2 \text{Na} \rightarrow 2 \text{NaOH} + \text{H}_2$	$2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$	Or, more down to earth, since we have the reaction $2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$, we
fong-spivak-invitation_F00946	83	■ 0.975	$2 \text{Na} \multimap (2 \text{NaOH} + \text{H}_2)$	$2\text{Na} \multimap (2\text{NaOH} + \text{H}_2)$	But it is not so far-fetched: this hypothetical new substance $(2\text{Na} \multimap (2\text{NaOH} + \text{H}_2))$
fong-spivak-invitation_F00947	83	■ 0.977	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$	Proposition 2.64. Suppose $\mathcal{V} = (V, <, I, \otimes, \multimap)$ is a symmetric monoidal preorder
fong-spivak-invitation_F00948	83	■ 1.000	$\multimap v : (V, \leq) \rightarrow (V, \leq)$	$\multimap v : (V, \leq) \rightarrow (V, \leq)$	(a) For every $v \in V$, the monotone map $\multimap v : (V, \leq) \rightarrow (V, \leq)$ is left adjoint to
fong-spivak-invitation_F00949	83	■ 0.779	$v \multimap - : (V, \leq) \rightarrow (V, \leq)$	$v \multimap - : (V, \leq) \rightarrow (V, \leq)$	$v \multimap - : (V, <) \rightarrow (V, <)$.
fong-spivak-invitation_F00950	84	■ 1.000	$A \subseteq V$	$A \subseteq V$	(b) For any element $v \in V$ and set of elements $A \subseteq V$, if the join $\bigvee_{a \in A} a$ exists then
fong-spivak-invitation_F00951	84	■ 1.000	$\bigvee_{a \in A} a$	$\bigvee_{a \in A} a$	(b) For any element $v \in V$ and set of elements $A \subseteq V$, if the join $\bigvee_{a \in A} a$ exists then
fong-spivak-invitation_F00952	84	■ 1.000	$\bigvee_{a \in A} v \otimes a$	$\bigvee_{a \in A} v \otimes a$	so does $\bigvee_{a \in A} v \otimes a$ and we have
fong-spivak-invitation_F00953	84	■ 1.000	$v \otimes (v \multimap w) \leq w$	$v \otimes (v \multimap w) \leq w$	(c) For any $v, w \in V$, we have $v \otimes (v \multimap w) \leq w$.
fong-spivak-invitation_F00954	84	■ 1.000	$v \cong (I \multimap v)$	$v \cong (I \multimap v)$	(d) For any $v \in V$, we have $v \cong (I \multimap v)$.
fong-spivak-invitation_F00955	84	■ 1.000	$u, v, w \in V$	$u, v, w \in V$	(e) For any $u, v, w \in V$, we have $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$.
fong-spivak-invitation_F00956	84	■ 1.000	$(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$	$(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$	(e) For any $u, v, w \in V$, we have $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$.
fong-spivak-invitation_F00957	84	■ 1.000	$(v \multimap w) \leq (v \multimap w)$	$(v \multimap w) \leq (v \multimap w)$	Proposition 1.101. More concretely, from reflexivity $(v \multimap w) \leq (v \multimap w)$, we
fong-spivak-invitation_F00958	84	■ 1.000	$(v \multimap w) \otimes v \leq w$	$(v \multimap w) \otimes v \leq w$	obtain $(v \multimap w) \otimes v \leq w$ Eq. (2.23), and we are done by symmetry, which says

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fong-spivak-invitation_F00959	84	■ 1.000	$v \otimes (v \multimap w) = (v \multimap w) \otimes v$	$v \otimes (v \multimap w) = (v \multimap w) \otimes v$	$v \otimes (v \multimap w) = (v \multimap w) \otimes v$.
fong-spivak-invitation_F00960	84	■ 1.000	$v \otimes I = v \leq v$	$v \otimes I = v \leq v$	(d) Since $v \otimes I = v \leq v$, Eq. (2.23) says $v \leq (I \multimap v)$. For the other direction, we have
fong-spivak-invitation_F00961	84	■ 1.000	$v \leq (I \multimap v)$	$v \leq (I \multimap v)$	(d) Since $v \otimes I = v \leq v$, Eq. (2.23) says $v \leq (I \multimap v)$. For the other direction, we have
fong-spivak-invitation_F00962	84	■ 1.000	$(I \multimap v) = I \otimes (I \multimap v) \leq v$	$(I \multimap v) = I \otimes (I \multimap v) \leq v$	$(I \multimap v) = I \otimes (I \multimap v) < v$ by (c).
fong-spivak-invitation_F00963	84	■ 1.000	$u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$	$u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$	(e) To obtain this inequality, we just need $u \otimes (u \multimap v) \otimes (v \multimap w) < w$. But this
fong-spivak-invitation_F00964	84	■ 1.000	$v, w \in$	$v, w \in$	Remark 2.65. We can consider $\mathcal{V}$ to be enriched in itself. That is, for every $v, w \in$
fong-spivak-invitation_F00965	84	■ 0.999	$\operatorname{Ob}(\mathcal{V})$	$\operatorname{Ob}(\mathcal{V})$	$\operatorname{Ob}(\mathcal{V})$, we can define $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$. For this to really be an enrichment,
fong-spivak-invitation_F00966	84	■ 0.999	$\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$	$\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$	$\operatorname{Ob}(\mathcal{V})$, we can define $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$. For this to really be an enrichment,
fong-spivak-invitation_F00967	84	■ 1.000	$I \leq$	$I \leq$	we just need to check the two conditions of Definition 2.30. The first condition $I \leq$
fong-spivak-invitation_F00968	84	■ 0.688	$\mathcal{X}(x, x) = (x \multimap x)$	$\mathcal{X}(x, x) = (x \multimap x)$	$\mathcal{X}(x, x) = (x \multimap x)$ is satisfied because $I \otimes x \leq x$. The second condition is satisfied by
fong-spivak-invitation_F00969	84	■ 0.688	$I \otimes x \leq x$	$I \otimes x \leq x$	$\mathcal{X}(x, x) = (x \multimap x)$ is satisfied because $I \otimes x < x$. The second condition is satisfied by
fong-spivak-invitation_F00970	84	■ 0.901	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$	$\mathcal{V} = (V, \leq, I, \otimes, \multimap)$	order $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$ that has all joins: $\bigvee A$ exists for every $A \subseteq V$. In particular,
fong-spivak-invitation_F00971	84	■ 0.901	$\bigvee A$	$\bigvee A$	order $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$ that has all joins: $\bigvee A$ exists for every $A \subseteq V$. In particular,
fong-spivak-invitation_F00972	84	■ 0.928	$0 := \bigvee \varnothing$	$0 := \bigvee \varnothing$	we often denote the empty join by $0 := \bigvee \varnothing$.
fong-spivak-invitation_F00973	85	■ 1.000	$A \subseteq [0, \infty]$	$A \subseteq [0, \infty]$	Cost is a quantale, we take an arbitrary set of elements $A \subseteq [0, \infty]$ and ask if it has a
fong-spivak-invitation_F00974	85	■ 0.999	$a \geq \bigvee A$	$a \geq \bigvee A$	a. $a \geq \bigvee A$ for all $a \in A$, and
fong-spivak-invitation_F00975	85	■ 1.000	$b \in [0, \infty]$	$b \in [0, \infty]$	b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$, then $\bigvee A \geq b$.
fong-spivak-invitation_F00976	85	■ 1.000	$a \geq b$	$a \geq b$	b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$, then $\bigvee A \geq b$.
fong-spivak-invitation_F00977	85	■ 1.000	$\bigvee A \geq b$	$\bigvee A \geq b$	b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$, then $\bigvee A \geq b$.
fong-spivak-invitation_F00978	85	■ 0.544	$A \cdot \{ \}^{\{5\}}$	$A \cdot \{ \}^{\{5\}}$	bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$. We have joins for infinite
fong-spivak-invitation_F00979	85	■ 0.544	$A = \{2, 3\}$	$A = \{2, 3\}$	bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$. We have joins for infinite
fong-spivak-invitation_F00980	85	■ 0.544	$\bigvee A = 2$	$\bigvee A = 2$	bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$. We have joins for infinite
fong-spivak-invitation_F00981	85	■ 0.996	$B = \{2.5, 2.05, 2.005, \dots\}$	$B = \{2.5, 2.05, 2.005, \dots\}$	sets too: if $B = \{2.5, 2.05, 2.005, \dots\}$, its infimum is 2. Finally, in order to say that
fong-spivak-invitation_F00982	85	■ 0.997	$([0, \infty], \geq)$	$([0, \infty], \geq)$	$([0, \infty], \geq)$ has all joins, we need a join to exist for the empty set $A = \varnothing$ too. The first
fong-spivak-invitation_F00983	85	■ 0.997	$A = \varnothing$	$A = \varnothing$	$([0, \infty], \geq)$ has all joins, we need a join to exist for the empty set $A = \varnothing$ too. The first
fong-spivak-invitation_F00984	85	■ 0.999	$\bigvee \varnothing \geq b$	$\bigvee \varnothing \geq b$	for any $b \in [0, \infty]$ we have $\bigvee \varnothing \geq b$; this means $\bigvee \varnothing = \infty$.
fong-spivak-invitation_F00985	85	■ 0.999	$\bigvee \varnothing = \infty$	$\bigvee \varnothing = \infty$	for any $b \in [0, \infty]$ we have $\bigvee \varnothing \geq b$; this means $\bigvee \varnothing = \infty$.

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fong-spivak-invitation_F00986	85	■ 0.934	$\bigvee \varnothing$	$\bigvee \emptyset$	1. What is $\bigvee \emptyset$, which we generally denote 0, in the case
fong-spivak-invitation_F00987	85	■ 0.921	$\mathcal{V}=\mathbf{Bool}=(\mathbb{B}, \leq$	$\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq$	a. $\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$?
fong-spivak-invitation_F00988	85	■ 0.735	$\mathcal{V}=\mathbf{Cost}=(\mathbb{R}_{\geq 0}, \leq,$	$\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$	b. $\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$?
fong-spivak-invitation_F00989	85	■ 1.000	$x \vee y$	$x \vee y$	2. What is the join $x \vee y$ in the case
fong-spivak-invitation_F00990	85	■ 0.802	$\mathcal{V} =$	$\mathcal{V} =$	a. $\mathcal{V} = \mathbf{Bool}$, and $x, y \in \mathbb{B}$ are booleans?
fong-spivak-invitation_F00991	85	■ 0.802	$x, y \in \mathbb{B}$	$x, y \in \mathbb{B}$	a. $\mathcal{V} = \mathbf{Bool}$, and $x, y \in \mathbb{B}$ are booleans?
fong-spivak-invitation_F00992	85	■ 0.422	$\mathbf{Bool}=(\mathbb{B}, \leq$	$\mathbf{Bool} = (\mathbb{B}, \leq$	Exercise 2.69. Show that $\mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$ is a quantale.
fong-spivak-invitation_F00993	85	■ 0.737	$P(S), \subseteq, S, \cap$	$P(S), \subseteq, S, \cap$	Exercise 2.70. Let S be a set and recall the power set monoidal preorder $(P(S), \subseteq, S, \cap)$
fong-spivak-invitation_F00994	85	■ 0.720	$\{3.1, 3.01, 3.001, \dots\}$	$\{3.1, 3.01, 3.001, \dots\}$	number that is $\leq$ everything in A . For example, the infimum of $\{3.1, 3.01, 3.001, \dots\}$ is 3. But note that
fong-spivak-invitation_F00995	86	■ 1.000	$\mathcal{P}=(P, \leq)$	$\mathcal{P} = (P, \leq)$	Proposition 2.72. Let $\mathcal{P} = (P, \leq)$ be a preorder. It has all joins iff it has all meets.
fong-spivak-invitation_F00996	86	■ 0.931	$\mathcal{P}^{\text{op}}$	$\mathcal{P}^{\text{op}}$	<i>Proof.</i> The joins (resp. meets) in $\mathcal{P}$ are the meets (resp. joins) in $\mathcal{P}^{\text{op}}$, so the two claims
fong-spivak-invitation_F00997	86	■ 1.000	$A \subseteq \mathcal{P}$	$A \subseteq \mathcal{P}$	Suppose $\mathcal{P}$ has all joins and suppose that $A \subseteq \mathcal{P}$ is a subset for which we want the
fong-spivak-invitation_F00998	86	■ 1.000	$M_A := \{p \in P \mid p \leq a$	$M_A := \{p \in P \mid p \leq a$	meet. Consider the set $M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$ of elements below
fong-spivak-invitation_F00999	86	■ 1.000	$a \in A\}$	$a \in A\}$	meet. Consider the set $M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$ of elements below
fong-spivak-invitation_F01000	86	■ 1.000	$m_A := \bigvee_{p \in M_A} p$	$m_A := \bigvee_{p \in M_A} p$	everything in A . Let $m_A := \bigvee_{p \in M_A} p$ be their join. We claim that m_A is a meet for A .
fong-spivak-invitation_F01001	86	■ 1.000	m_A	m_A	everything in A . Let $m_A := \bigvee_{p \in M_A} p$ be their join. We claim that m_A is a meet for A .
fong-spivak-invitation_F01002	86	■ 1.000	$m_A \leq a$	$m_A \leq a$	We first need to know that for any $a \in A$ we have $m_A \leq a$, but this is by definition
fong-spivak-invitation_F01003	86	■ 1.000	$p \in M_A$	$p \in M_A$	of join: since all $p \in M_A$ satisfy $p \leq a$, so does their join $m_A \leq a$. We second need
fong-spivak-invitation_F01004	86	■ 1.000	$m' \in P$	$m' \in P$	to know that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every
fong-spivak-invitation_F01005	86	■ 1.000	$m' \leq a$	$m' \leq a$	to know that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every
fong-spivak-invitation_F01006	86	■ 1.000	$m' \leq m$	$m' \leq m$	to know that for any $m' \in P$ with $m' \leq a$ for all $a \in A$, we have $m' \leq m$. But every
fong-spivak-invitation_F01007	86	■ 1.000	m'	m'	such m' is actually an element of M_A and m is their join, so $m' \leq m$. This completes the
fong-spivak-invitation_F01008	86	■ 1.000	M_A	M_A	such m' is actually an element of M_A and m is their join, so $m' \leq m$. This completes the
fong-spivak-invitation_F01009	86	■ 1.000	$V \subseteq X$	$V \subseteq X$	and $V \subseteq X$, we can generalize the usual Hausdorff distance, on the left below, to the
fong-spivak-invitation_F01010	86	■ 0.877	$u \in U$	$u \in U$	answers the question “can I get into V from every $u \in U$,” i.e. $\forall u \in U. \exists v \in V. u \leq v$. ⁶ Or
fong-spivak-invitation_F01011	86	■ 0.877	$\forall u \in U. \exists v \in V. u \leq v$	$\forall u \in U. \exists v \in V. u \leq v$	answers the question “can I get into V from every $u \in U$,” i.e. $\forall u \in U. \exists v \in V. u \leq v$. ⁶ Or
fong-spivak-invitation_F01012	86	■ 1.000	$\mathcal{V}=\mathbf{P}(M)$	$\mathcal{V} = \mathbf{P}(M)$	for another example, use $\mathcal{V} = \mathbf{P}(M)$ with its interpretation as modes of transportation,

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fong-spivak-invitation_F01013	86	■ 1.000	$d(U, V) \in \mathrm{P}(M)$	$d(U, V) \in \mathrm{P}(M)$	as in Exercise 2.42. Then the Hausdorff distance $d(U, V) \in \mathrm{P}(M)$ tells us those modes
fong-spivak-invitation_F01014	86	■ 1.000	$-\otimes v: V \rightarrow V$	$-\otimes v: V \rightarrow V$	Eq. (2.24) holds. We need to show that if Eq. (2.24) holds then $-\otimes v: V \rightarrow V$ has a
fong-spivak-invitation_F01015	86	■ 0.999	$v \multimap -$	$v \multimap -$	right adjoint $v \multimap -$. This is just the adjoint functor theorem, Theorem 1.108. It says
fong-spivak-invitation_F01016	86	■ 1.000	$\forall$	$\forall$	⁶ Here $\forall$ is the universal quantifier, “for all,” and $\exists$ is the existential quantifier, “there exists.”
fong-spivak-invitation_F01017	86	■ 1.000	$\exists$	$\exists$	⁶ Here $\forall$ is the universal quantifier, “for all,” and $\exists$ is the existential quantifier, “there exists.”
fong-spivak-invitation_F01018	87	■ 0.959	$\sum$	$\sum$	We will get a formula where joins stand in for the sum operation $\sum$, and $\otimes$ stands in for
fong-spivak-invitation_F01019	87	■ 1.000	$\mathcal{V}=(V, \leq, \otimes, I)$	$\mathcal{V}=(V, \leq, \otimes, I)$	Definition 2.75. Let $\mathcal{V}=(V, <, \otimes, I)$ be a quantale. Given sets X and Y , a <i>matrix with</i>
fong-spivak-invitation_F01020	87	■ 1.000	$M: X \times Y \rightarrow V$	$M: X \times Y \rightarrow V$	<i>entries in $\mathcal{V}$</i> , or simply a <i>$\mathcal{V}$-matrix</i> , is a function $M: X \times Y \rightarrow V$. For any $x \in X$ and
fong-spivak-invitation_F01021	87	■ 0.991	$M(x, y)$	$M(x, y)$	$y \in Y$, we call $M(x, y)$ the <i>(x, y)-entry</i> .
fong-spivak-invitation_F01022	87	■ 1.000	$N: Y \times Z \rightarrow V$	$N: Y \times Z \rightarrow V$	Here is how you multiply $\mathcal{V}$ -matrices $M: X \times Y \rightarrow V$ and $N: Y \times Z \rightarrow V$. Their
fong-spivak-invitation_F01023	87	■ 1.000	$(M * N): X \times Z \rightarrow V$	$(M * N): X \times Z \rightarrow V$	product is defined to be the matrix $(M * N): X \times Z \rightarrow V$, whose entries are given by
fong-spivak-invitation_F01024	87	■ 0.999	$M * N$	$M * N$	Example 2.76. Let $\mathcal{V} = \mathbf{Bool}$. Here is an example of matrix multiplication $M * N$.
fong-spivak-invitation_F01025	87	■ 1.000	$X=\{1,2,3\}, Y=\{1,2\}$	$X=\{1,2,3\}, Y=\{1,2\}$	Here $X=\{1,2,3\}$, $Y=\{1,2\}$, and $Z=\{1,2,3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$
fong-spivak-invitation_F01026	87	■ 1.000	$Z=\{1,2,3\}$	$Z=\{1,2,3\}$	Here $X=\{1,2,3\}$, $Y=\{1,2\}$, and $Z=\{1,2,3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$
fong-spivak-invitation_F01027	87	■ 1.000	$M: X \times Y \rightarrow \mathbb{B}$	$M: X \times Y \rightarrow \mathbb{B}$	Here $X=\{1,2,3\}$, $Y=\{1,2\}$, and $Z=\{1,2,3\}$, matrices $M: X \times Y \rightarrow \mathbb{B}$
fong-spivak-invitation_F01028	87	■ 1.000	$N: Y \times Z \rightarrow \mathbb{B}$	$N: Y \times Z \rightarrow \mathbb{B}$	and $N: Y \times Z \rightarrow \mathbb{B}$ are shown to the left below, and their product is shown to the
fong-spivak-invitation_F01029	87	■ 1.000	$I_X: X \times X \rightarrow V$	$I_X: X \times X \rightarrow V$	The identity V -matrix on a set X is $I_X: X \times X \rightarrow V$ given by
fong-spivak-invitation_F01030	87	■ 0.984	(2×2)	(2×2)	Exercise 2.77. Write down the (2×2) -identity matrix for each of the quantales
fong-spivak-invitation_F01031	88	■ 1.000	$I_X * M = M$	$I_X * M = M$	$I_X * M = M$.
fong-spivak-invitation_F01032	88	■ 1.000	$M: W \times X \rightarrow V, N: X \times Y \rightarrow V$	$M: W \times X \rightarrow V, N: X \times Y \rightarrow V$	2. Prove the <i>associative law</i> : for any matrices $M: W \times X \rightarrow V, N: X \times Y \rightarrow V$, and
fong-spivak-invitation_F01033	88	■ 1.000	$P: Y \times Z \rightarrow V$	$P: Y \times Z \rightarrow V$	$P: Y \times Z \rightarrow V$, one has $(M * N) * P = M * (N * P)$.
fong-spivak-invitation_F01034	88	■ 1.000	$(M * N) * P = M * (N * P)$	$(M * N) * P = M * (N * P)$	$P: Y \times Z \rightarrow V$, one has $(M * N) * P = M * (N * P)$.
fong-spivak-invitation_F01035	88	■ 1.000	M_Y^2	M_Y^2	When we multiply M_Y by itself using the formula Eq. (2.26), the result M_Y^2 tells us
fong-spivak-invitation_F01036	88	■ 1.000	M_Y^3	M_Y^3	the length of the shortest path traversing two edges or fewer. Similarly M_Y^3 tells us about
fong-spivak-invitation_F01039	88	■ 1.000	$M_Y^2 = M_Y^3$	$M_Y^2 = M_Y^3$	One sees that the powers stabilize: $M_Y^2 = M_Y^3$; as soon as that happens one has the
fong-spivak-invitation_F01040	88	■ 0.999	M_Y^n	M_Y^n	matrix of distances, d_Y . Indeed M_Y^n records the lengths of the shortest path traverse n
fong-spivak-invitation_F01041	88	■ 1.000	M_X^2, M_X^3	M_X^2, M_X^3	culate M_X^2 , M_X^3 , and M_X^4 . Check that M_X^4 is what you got for the distance matrix in

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fong-spivak-invitation_F01042	88	■ 1.000	M_X^4	M_X^4	culate M_X^2 , M_X^3 , and M_X^4 . Check that M_X^4 is what you got for the distance matrix in
fong-spivak-invitation_F01043	91	■ 1.000	$\$ 2.5$	\$2.5	ogy) budgets, and that the market for data integration software was \$2.5 billion in 2007
fong-spivak-invitation_F01044	93	■ 1.000	$\mathcal{C} \rightarrow$	$\mathcal{C} \rightarrow$	given by a “set-valued” functor $\mathcal{C} \rightarrow \mathbf{Set}$, and that databases can be mapped to each
fong-spivak-invitation_F01045	93	■ 0.999	$\mathcal{C} \rightarrow \mathcal{D}$	$\mathcal{C} \rightarrow \mathcal{D}$	other via functors $\mathcal{C} \rightarrow \mathcal{D}$. In other words, there is a relatively large overlap between
fong-spivak-invitation_F01046	95	■ 1.000	$\{ \}^2 \mathrm{Ob}(\mathcal{C})$	${}^2\mathrm{Ob}(\mathcal{C})$	(i) one specifies a collection ² $\mathrm{Ob}(\mathcal{C})$, elements of which are called <i>objects</i> ;
fong-spivak-invitation_F01047	95	■ 0.959	$\mathcal{C}(c, d), \{ \}^3$	$\mathcal{C}(c, d), {}^3$	(ii) for every two objects c, d , one specifies a set $\mathcal{C}(c, d)$, ³ elements of which are called
fong-spivak-invitation_F01048	95	■ 0.901	$c \in \mathrm{Ob}(\mathcal{C})$	$c \in \mathrm{Ob}(\mathcal{C})$	(iii) for every object $c \in \mathrm{Ob}(\mathcal{C})$, one specifies a morphism $\mathrm{id}_c \in \mathcal{C}(c, c)$, called the
fong-spivak-invitation_F01049	95	■ 0.901	$\mathrm{id}_c \in \mathcal{C}(c, c)$	$\mathrm{id}_c \in \mathcal{C}(c, c)$	(iii) for every object $c \in \mathrm{Ob}(\mathcal{C})$, one specifies a morphism $\mathrm{id}_c \in \mathcal{C}(c, c)$, called the
fong-spivak-invitation_F01050	95	■ 0.791	$f \in \mathcal{C}(c, d)$	$f \in \mathcal{C}(c, d)$	(iv) for every three objects $c, d, e \in \mathrm{Ob}(\mathcal{C})$ and morphisms $f \in \mathcal{C}(c, d)$ and $g \in \mathcal{C}(d, e)$,
fong-spivak-invitation_F01051	95	■ 0.791	$g \in \mathcal{C}(d, e)$	$g \in \mathcal{C}(d, e)$	(iv) for every three objects $c, d, e \in \mathrm{Ob}(\mathcal{C})$ and morphisms $f \in \mathcal{C}(c, d)$ and $g \in \mathcal{C}(d, e)$,
fong-spivak-invitation_F01052	95	■ 0.983	$f ; g \in \mathcal{C}(c, e)$	$f ; g \in \mathcal{C}(c, e)$	one specifies a morphism $f \circ g \in \mathcal{C}(c, e)$, called <i>the composite of f and g</i> .
fong-spivak-invitation_F01053	95	■ 0.843	$c \in \operatorname{Ob}(\mathcal{C})$	$c \in \mathrm{Ob}(\mathcal{C})$	We will sometimes write an object $c \in \mathcal{C}$, instead of $c \in \mathrm{Ob}(\mathcal{C})$. It will also be conve-
fong-spivak-invitation_F01054	95	■ 0.874	$f: c \rightarrow d$	$f: c \rightarrow d$	nient to denote elements $f \in \mathcal{C}(c, d)$ as $f: c \rightarrow d$. Here, c is called the <i>domain</i> of f ,
fong-spivak-invitation_F01055	95	■ 0.664	$\mathrm{id}_c ; f = f$	$\mathrm{id}_c; f = f$	nothing: $\mathrm{id}_c \circ f = f$ and $f \circ \mathrm{id}_d = f$;
fong-spivak-invitation_F01056	95	■ 0.664	$f ; \mathrm{id}_d = f$	$f; \mathrm{id}_d = f$	nothing: $\mathrm{id}_c \circ f = f$ and $f \circ \mathrm{id}_d = f$;
fong-spivak-invitation_F01057	95	■ 1.000	$f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$	$f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$	(b) <i>associativity</i> : for any three morphisms $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$, and $h: c_2 \rightarrow c_3$,
fong-spivak-invitation_F01058	95	■ 1.000	$h: c_2 \rightarrow c_3$	$h: c_2 \rightarrow c_3$	(b) <i>associativity</i> : for any three morphisms $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$, and $h: c_2 \rightarrow c_3$,
fong-spivak-invitation_F01059	95	■ 0.998	$(f ; g) ; h = f ; (g ; h)$	$(f;g);h = f;(g;h)$	the following are equal: $(f \circ g) \circ h = f \circ (g \circ h)$. We write this composite simply as
fong-spivak-invitation_F01060	95	■ 0.957	$\mathcal{C}(c, d)$	$\mathcal{C}(c, d)$	³ This set $\mathcal{C}(c, d)$ is often denoted $\mathrm{Hom}_{\mathcal{C}}(c, d)$, and called the “hom-set from c to d .” The word “hom” stands
fong-spivak-invitation_F01061	95	■ 0.957	$\operatorname{Hom}_{\mathcal{C}}(c, d)$	$\mathrm{Hom}_{\mathcal{C}}(c, d)$	³ This set $\mathcal{C}(c, d)$ is often denoted $\mathrm{Hom}_{\mathcal{C}}(c, d)$, and called the “hom-set from c to d .” The word “hom” stands
fong-spivak-invitation_F01062	96	■ 1.000	$p: v \rightarrow w$	$p: v \rightarrow w$	$p: v \rightarrow w$. To every vertex v , there is a trivial path, containing no arrows, starting
fong-spivak-invitation_F01063	96	■ 0.999	id_v	id_v	and ending at v ; we often denote it by id_v or simply by v . We may also concatenate
fong-spivak-invitation_F01064	96	■ 1.000	$q: w \rightarrow x$	$q: w \rightarrow x$	paths: given $p: v \rightarrow w$ and $q: w \rightarrow x$, their concatenation is denoted $p \circ q$, and it
fong-spivak-invitation_F01065	96	■ 1.000	$p ; q$	$p; q$	paths: given $p: v \rightarrow w$ and $q: w \rightarrow x$, their concatenation is denoted $p \circ q$, and it
fong-spivak-invitation_F01066	96	■ 1.000	$v \rightarrow x$	$v \rightarrow x$	goes $\Rightarrow$.
fong-spivak-invitation_F01067	96	■ 0.658	$(V, \leq)$	$(V, \leq)$	In Chapter 1, we used graphs to depict preorders $(V, \leq)$: the vertices form the ele-
fong-spivak-invitation_F01068	96	■ 0.977	$\operatorname{Free}(G)$	$\mathrm{Free}(G)$	Definition 3.3. For any graph $G = (V, A, s, t)$, we can define a category $\mathbf{Free}(G)$,

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fong-spivak-invitation_F01069	96	0.651	$v_{\{1\}}$	v_1	It has two objects v_1 and v_2 , and three morphisms: $\mathrm{id}_{v_1} : v_1 \rightarrow v_1$, $f_1 : v_1 \rightarrow v_2$, and
fong-spivak-invitation_F01070	96	0.651	$v_{\{2\}}$	v_2	It has two objects v_1 and v_2 , and three morphisms: $\mathrm{id}_{v_1} : v_1 \rightarrow v_1$, $f_1 : v_1 \rightarrow v_2$, and
fong-spivak-invitation_F01071	96	0.651	$\mathrm{id}_{v_{\{1\}}} : v_{\{1\}} \rightarrow v_{\{1\}}$, $f_{\{1\}} : v_{\{1\}} \rightarrow v_{\{2\}}$	$\mathrm{id}_{v_1} : v_1 \rightarrow v_1, f_1 : v_1 \rightarrow v_2$	It has two objects v_1 and v_2 , and three morphisms: $\mathrm{id}_{v_1} : v_1 \rightarrow v_1$, $f_1 : v_1 \rightarrow v_2$, and
fong-spivak-invitation_F01072	96	0.992	$\mathrm{id}_{v_{\{2\}}} : v_{\{2\}} \rightarrow v_{\{2\}}$	$\mathrm{id}_{v_2} : v_2 \rightarrow v_2$	$\mathrm{id}_{v_2} : v_2 \rightarrow v_2$. Here id_{v_1} is the path of length 0 starting and ending at v_1 , f_1 is the path
fong-spivak-invitation_F01073	96	0.992	$\mathrm{id}_{v_{\{1\}}}$	id_{v_1}	$\mathrm{id}_{v_2} : v_2 \rightarrow v_2$. Here id_{v_1} is the path of length 0 starting and ending at v_1 , f_1 is the path
fong-spivak-invitation_F01074	96	0.992	$v_{\{1\}}$, $f_{\{1\}}$	v_1, f_1	$\mathrm{id}_{v_2} : v_2 \rightarrow v_2$. Here id_{v_1} is the path of length 0 starting and ending at v_1 , f_1 is the path
fong-spivak-invitation_F01075	96	0.999	$f_{\{1\}}$	f_1	of length 1 consisting of just the arrow f_1 , and id_{v_2} is the length 0 path at v_2 . As our
fong-spivak-invitation_F01076	96	0.999	$\mathrm{id}_{v_{\{2\}}}$	id_{v_2}	of length 1 consisting of just the arrow f_1 , and id_{v_2} is the length 0 path at v_2 . As our
fong-spivak-invitation_F01077	96	0.424	$f_{\{1\}}=f_{\{1\}}$	$f_1 = f_1$	v_2 . As composition is given by concatenation, we have, for example, $\mathrm{id}_{v_1} \circ f_1 = f_1$,
fong-spivak-invitation_F01078	96	0.596	$\mathrm{id}_{v_{\{2\}}} \circ \mathrm{id}_{v_{\{2\}}} = \mathrm{id}_{v_{\{2\}}}$	$\mathrm{id}_{v_2} \circ \mathrm{id}_{v_2} = \mathrm{id}_{v_2}$	$\mathrm{id}_{v_2} \circ \mathrm{id}_{v_2} = \mathrm{id}_{v_2}$, and so on.
fong-spivak-invitation_F01079	96	0.821	$\mathrm{Free}(G)$	$\mathbf{Free}(G)$	free category $\mathbf{Free}(G)$, at least when the one we mean is clear enough from context.
fong-spivak-invitation_F01080	97	0.850	$\mathbf{0}$	$\mathbf{0}$	2. What is the category $\mathbf{0}$?
fong-spivak-invitation_F01081	97	0.994	$\mathbf{n}$	$\mathbf{n}$	3. What is the formula for the number of morphisms in $\mathbf{n}$ for arbitrary $n \in \mathbb{N}$?
fong-spivak-invitation_F01082	97	0.337	$\mathrm{id}_{\{z, s, (s \circ s) \circ s\}}$	$= \{z, s, (s \circ s) \circ s\}$	a unique path of length n for every natural number $n \in \mathbb{N}$. That is, $\mathrm{Path} = \{z, s, (s \circ s) \circ s\}$
fong-spivak-invitation_F01083	97	0.999	id_z	id_z	id_z . There is a one-to-one correspondence between Path and the natural numbers, $\mathbb{N} =$
fong-spivak-invitation_F01084	97	0.999	$\mathbb{N}$	$\mathbb{N}$	id_z . There is a one-to-one correspondence between Path and the natural numbers, $\mathbb{N} =$
fong-spivak-invitation_F01085	97	0.567	$\{0, 1, 2, 3, \dots\}$	$\{0, 1, 2, 3, \dots\}$	$\{0, 1, 2, 3, \dots\}$.
fong-spivak-invitation_F01086	97	1.000	$m * 1 = m = 1 * m$	$m * 1 = m = 1 * m$	$m * 1 = m = 1 * m$ and $(m * n) * p = m * (n * p)$.
fong-spivak-invitation_F01087	97	1.000	$(m * n) * p = m * (n * p)$	$(m * n) * p = m * (n * p)$	$m * 1 = m = 1 * m$ and $(m * n) * p = m * (n * p)$.
fong-spivak-invitation_F01088	97	0.996	$M := \mathcal{C}(\bullet, \bullet)$	$M := \mathcal{C}(\bullet, \bullet)$	nection. Given a category $\mathcal{C}$ with one object, say $\bullet$, let $M := \mathcal{C}(\bullet, \bullet)$, let $e = \mathrm{id}_\bullet$,
fong-spivak-invitation_F01089	97	0.996	$e = \mathrm{id}_\bullet$	$e = \mathrm{id}_\bullet$	nection. Given a category $\mathcal{C}$ with one object, say $\bullet$, let $M := \mathcal{C}(\bullet, \bullet)$, let $e = \mathrm{id}_\bullet$,
fong-spivak-invitation_F01090	97	0.896	$* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$	$* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$	and let $* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$ be the composition operation $* = \circ$. The
fong-spivak-invitation_F01091	97	0.896	$* = \circ$	$* = \circ$	and let $* : \mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$ be the composition operation $* = \circ$. The
fong-spivak-invitation_F01092	97	1.000	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$	numbers $n \in \mathbb{N}$. Paths can be concatenated. Given numbers $m, n \in \mathbb{N}$, what number
fong-spivak-invitation_F01093	98	0.170	i	i	because $f \circ h$ and $g \circ i$ are made equal. This category is called the “commutative square.”
fong-spivak-invitation_F01094	98	0.251	$g \circ i$	$g \circ i$	One might say “the missing one is $g \circ i$,” but that is not quite right: $g \circ i$ is there too,
fong-spivak-invitation_F01095	98	0.251	$g \circ i$	$g \circ i$	One might say “the missing one is $g \circ i$,” but that is not quite right: $g \circ i$ is there too,

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fong-spivak-invitation_F01096	98	0.696	$f ; h$	$f;h$	because it is equal to $f \circ h$. As usual, A denotes id_A , etc.
fong-spivak-invitation_F01097	98	0.696	$\operatorname{id}_{\{A\}}$	id_A	because it is equal to $f \circ h$. As usual, A denotes id_A , etc.
fong-spivak-invitation_F01098	99	0.255	$s ; s=z$	$s; s = z$	In $\mathcal{C}$ we have the equation $s \circ s = z$. But this implies $s \circ s \circ s = z \circ s = s$! And similarly
fong-spivak-invitation_F01099	99	0.255	$s ; s ; s=z$	$s; s; s = z$	In $\mathcal{C}$ we have the equation $s \circ s = z$. But this implies $s \circ s \circ s = z \circ s = s$! And similarly
fong-spivak-invitation_F01100	99	0.153	$s=z$	$s = z$	we have $s \circ s \circ s \circ s = z \circ z = z$. The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$, with
fong-spivak-invitation_F01101	99	0.153	$z=z$	$z = z$	we have $s \circ s \circ s \circ s = z \circ z = z$. The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$, with
fong-spivak-invitation_F01102	99	0.153	$\{z, s\}$	$\{z, s\}$	we have $s \circ s \circ s \circ s = z \circ z = z$. The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$, with
fong-spivak-invitation_F01103	99	0.217	$=z$	$= z$	composition described by $s \circ s = z \circ z = z$, and $z \circ s = s \circ z = s$. In group theory, one
fong-spivak-invitation_F01104	99	0.217	$s=s$	$s = s$	composition described by $s \circ s = z \circ z = z$, and $z \circ s = s \circ z = s$. In group theory, one
fong-spivak-invitation_F01105	99	0.217	$z=s$	$z = s$	composition described by $s \circ s = z \circ z = z$, and $z \circ s = s \circ z = s$. In group theory, one
fong-spivak-invitation_F01106	99	1.000	$\mathbb{Z} / 2 \mathbb{Z}$	$\mathbb{Z}/2\mathbb{Z}$	would speak of a group called $\mathbb{Z}/2\mathbb{Z}$.
fong-spivak-invitation_F01107	99	1.000	$\mathcal{D}$	$\mathcal{D}$	Exercise 3.12. Write down all the morphisms in the category $\mathcal{D}$ from Example 3.11. $\diamond$
fong-spivak-invitation_F01108	99	0.991	$\operatorname{Ob}(\mathcal{P})=P$	$\text{Ob}(\mathcal{P}) = P$	are precisely the elements of P ; that is, $\text{Ob}(\mathcal{P}) = P$. As for morphisms, $\mathcal{P}$ has exactly
fong-spivak-invitation_F01109	99	1.000	$p \rightarrow q$	$p \rightarrow q$	one morphism $p \rightarrow q$ if $p \leq q$ and no morphisms $p \rightarrow q$ if $p \not\leq q$. The fact that
fong-spivak-invitation_F01110	99	1.000	$p \not \leq q$	$p \not\leq q$	one morphism $p \rightarrow q$ if $p \leq q$ and no morphisms $p \rightarrow q$ if $p \not\leq q$. The fact that
fong-spivak-invitation_F01111	100	0.921	$(C, \leq)$	$(C, \leq)$	Given any category $\mathcal{C}$, one can obtain a preorder $(C, \leq)$ from it by destroying the dis-
fong-spivak-invitation_F01112	100	0.931	$C:=\operatorname{Ob}(\mathcal{C})$	$C := \text{Ob}(\mathcal{C})$	tinction between any two parallel morphisms. That is, let $C := \text{Ob}(\mathcal{C})$, and put $c_1 \leq c_2$
fong-spivak-invitation_F01113	100	0.931	$c_{\{1\}} \leq c_{\{2\}}$	$c_1 \leq c_2$	tinction between any two parallel morphisms. That is, let $C := \text{Ob}(\mathcal{C})$, and put $c_1 \leq c_2$
fong-spivak-invitation_F01114	100	0.999	$\mathcal{C}(c_{\{1\}}, c_{\{2\}}) \neq \varnothing$	$\mathcal{C}(c_1, c_2) \neq \emptyset$	iff $\mathcal{C}(c_1, c_2) \neq \emptyset$. If there is one, or two, or fifty, or infinitely many morphisms $c_1 \rightarrow c_2$
fong-spivak-invitation_F01115	100	0.999	$c_{\{1\}} \rightarrow c_{\{2\}}$	$c_1 \rightarrow c_2$	iff $\mathcal{C}(c_1, c_2) \neq \emptyset$. If there is one, or two, or fifty, or infinitely many morphisms $c_1 \rightarrow c_2$
fong-spivak-invitation_F01116	101	0.878	$\boldsymbol{\operatorname{Set}}(S, T)=\{f: S \rightarrow T \mid f$	$\mathbf{Set}(S, T) = \{f : S \rightarrow T \mid f$	(ii) If S and T are sets, then $\mathbf{Set}(S, T) = \{f : S \rightarrow T \mid f \text{ is a function}\}$.
fong-spivak-invitation_F01117	101	1.000	$\operatorname{id}_{\{S\}}: S \rightarrow S$	$\text{id}_S : S \rightarrow S$	(iii) For each set S , the identity morphism is the function $\text{id}_S : S \rightarrow S$ given by
fong-spivak-invitation_F01118	101	0.979	$\operatorname{id}_{\{S\}}(s):=s$	$\text{id}_S(s) := s$	$\text{id}_S(s) := s$ for each $s \in S$.
fong-spivak-invitation_F01119	101	1.000	$f: S \rightarrow T$	$f : S \rightarrow T$	(iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$, their composite is the function $f \circ g : S \rightarrow U$
fong-spivak-invitation_F01120	101	1.000	$g: T \rightarrow U$	$g : T \rightarrow U$	(iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$, their composite is the function $f \circ g : S \rightarrow U$
fong-spivak-invitation_F01121	101	1.000	$f ; g: S \rightarrow U$	$f; g : S \rightarrow U$	(iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$, their composite is the function $f \circ g : S \rightarrow U$
fong-spivak-invitation_F01122	101	1.000	$(f ; g)(s):=g(f(s))$	$(f; g)(s) := g(f(s))$	given by $(f \circ g)(s) := g(f(s))$.

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fong-spivak-invitation_F01123	101	■ 1.000	$\underline{2}=\{1,2\}$	$\underline{2} = \{1, 2\}$	Exercise 3.18. Let $\underline{2} = \{1, 2\}$ and $\underline{3} = \{1, 2, 3\}$. These are objects in the category
fong-spivak-invitation_F01124	101	■ 1.000	$\underline{3}=\{1,2,3\}$	$\underline{3} = \{1, 2, 3\}$	Exercise 3.18. Let $\underline{2} = \{1, 2\}$ and $\underline{3} = \{1, 2, 3\}$. These are objects in the category
fong-spivak-invitation_F01125	102	■ 0.881	$\mathcal{C}^{\mathrm{op}}$	$\mathcal{C}^{\mathrm{op}}$	Example 3.20. Let $\mathcal{C}$ be a category. Its <i>opposite</i> , denoted $\mathcal{C}^{\mathrm{op}}$, is the category with the
fong-spivak-invitation_F01126	102	■ 0.488	$\operatorname{Ob}(\mathcal{C}^{\mathrm{op}}) := \operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{C}^{\mathrm{op}}) := \operatorname{Ob}(\mathcal{C})$	same objects, $\operatorname{Ob}(\mathcal{C}^{\mathrm{op}}) := \operatorname{Ob}(\mathcal{C})$, and for any two objects $c, d \in \operatorname{Ob}(\mathcal{C})$, one has
fong-spivak-invitation_F01127	102	■ 0.968	$\mathcal{C}^{\mathrm{op}}(c, d) := \mathcal{C}(d, c)$	$\mathcal{C}^{\mathrm{op}}(c, d) := \mathcal{C}(d, c)$	$\mathcal{C}^{\mathrm{op}}(c, d) := \mathcal{C}(d, c)$. Identities and composition are as in $\mathcal{C}$.
fong-spivak-invitation_F01128	102	■ 0.107	$\{\boxed{}, \sim\} \square$	$\{\boxed{}, \square\}$	in the category Set , one can exchange the set $\{\blacksquare, \square\}$ for the set $\{0, 1\}$ and everything
fong-spivak-invitation_F01129	102	■ 1.000	$a \cong b$	$a \cong b$	with elements a, b , such that $a \leq b$ and $b \leq a$, i.e. $a \cong b$, then a and b are essentially
fong-spivak-invitation_F01130	102	■ 1.000	$c \leq a$	$c \leq a$	$c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes this notion
fong-spivak-invitation_F01131	102	■ 1.000	$c \leq b$	$c \leq b$	$c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes this notion
fong-spivak-invitation_F01132	102	■ 1.000	$c \geq a$	$c \geq a$	$c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes this notion
fong-spivak-invitation_F01133	102	■ 1.000	$c \geq b$	$c \geq b$	$c \leq a$ iff $c \leq b$, and $c \geq a$ iff $c \geq b$. The notion of isomorphism formalizes this notion
fong-spivak-invitation_F01134	102	■ 0.456	$g: B \rightarrow A$	$g: B \rightarrow A$	morphism $g: B \rightarrow A$ satisfying $f \circ g = \operatorname{id}_A$ and $g \circ f = \operatorname{id}_B$. In this case we call f
fong-spivak-invitation_F01135	102	■ 0.456	$f \circ g = \operatorname{id}_A$	$f \circ g = \operatorname{id}_A$	morphism $g: B \rightarrow A$ satisfying $f \circ g = \operatorname{id}_A$ and $g \circ f = \operatorname{id}_B$. In this case we call f
fong-spivak-invitation_F01136	102	■ 0.456	$g \circ f = \operatorname{id}_B$	$g \circ f = \operatorname{id}_B$	morphism $g: B \rightarrow A$ satisfying $f \circ g = \operatorname{id}_A$ and $g \circ f = \operatorname{id}_B$. In this case we call f
fong-spivak-invitation_F01137	102	■ 1.000	$g = f^{-1}$	$g = f^{-1}$	and g <i>inverses</i> , and we often write $g = f^{-1}$, or equivalently $f = g^{-1}$. We also say that
fong-spivak-invitation_F01138	102	■ 1.000	$f = g^{-1}$	$f = g^{-1}$	and g <i>inverses</i> , and we often write $g = f^{-1}$, or equivalently $f = g^{-1}$. We also say that
fong-spivak-invitation_F01139	102	■ 1.000	$A := \{a, b, c\}$	$A := \{a, b, c\}$	Example 3.22. The set $A := \{a, b, c\}$ and the set $\underline{3} = \{1, 2, 3\}$ are isomorphic, i.e. there
fong-spivak-invitation_F01140	102	■ 1.000	$f: A \rightarrow \underline{3}$	$f: A \rightarrow \underline{3}$	exists an isomorphism $f: A \rightarrow \underline{3}$ given by $f(a) = 2, f(b) = 1, f(c) = 3$. The
fong-spivak-invitation_F01141	102	■ 1.000	$f(a)=2, f(b)=1, f(c)=3$	$f(a) = 2, f(b) = 1, f(c) = 3$	exists an isomorphism $f: A \rightarrow \underline{3}$ given by $f(a) = 2, f(b) = 1, f(c) = 3$. The
fong-spivak-invitation_F01142	103	■ 0.979	$A \in$	$A \in$	understood in terms of isomorphisms in FinSet . Namely, for any finite set $A \in \mathbf{FinSet}$,
fong-spivak-invitation_F01143	103	■ 1.000	$A \cong \underline{n}$	$A \cong \underline{n}$	its cardinality is the number $n \in \mathbb{N}$ such that there exists an isomorphism $A \cong \underline{n}$. Georg
fong-spivak-invitation_F01144	103	■ 1.000	$f^{-1}: \underline{3} \rightarrow A$	$f^{-1}: \underline{3} \rightarrow A$	1. What is the inverse $f^{-1}: \underline{3} \rightarrow A$ of the function f given in Example 3.22?
fong-spivak-invitation_F01145	103	■ 1.000	$A \rightarrow \underline{3}$	$A \rightarrow \underline{3}$	2. How many distinct isomorphisms are there $A \rightarrow \underline{3}$?
fong-spivak-invitation_F01146	103	■ 0.998	id_c	id_c	identity id_c is an isomorphism.
fong-spivak-invitation_F01147	103	■ 1.000	$f: \underline{2} \rightarrow \underline{3}$	$f: \underline{2} \rightarrow \underline{3}$	Consider the functions $f: \underline{2} \rightarrow \underline{3}$ and $g: \underline{3} \rightarrow \underline{2}$ drawn below:
fong-spivak-invitation_F01148	103	■ 1.000	$g: \underline{3} \rightarrow \underline{2}$	$g: \underline{3} \rightarrow \underline{2}$	Consider the functions $f: \underline{2} \rightarrow \underline{3}$ and $g: \underline{3} \rightarrow \underline{2}$ drawn below:
fong-spivak-invitation_F01149	103	■ 0.180	$f \circ g = \operatorname{id}_{\underline{2}}$	$f \circ g = \operatorname{id}_{\underline{2}}$	Then the reader should be able to check instantly that $f \circ g = \operatorname{id}_{\underline{2}}$ but $g \circ f \neq \operatorname{id}_{\underline{3}}$. Thus

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fong-spivak-invitation_F01150	103	■ 0.180	$g: f \not\equiv \mathrm{id}_{\underline{3}}$	$g: f \neq \mathrm{id}_3$	Then the reader should be able to check instantly that $f \circ g = \mathrm{id}_2$ but $g \circ f \neq \mathrm{id}_3$. Thus
fong-spivak-invitation_F01151	105	■ 1.000	$F: \mathcal{C} \rightarrow \mathcal{D}$	$F: \mathcal{C} \rightarrow \mathcal{D}$	$F: \mathcal{C} \rightarrow \mathcal{D}$,
fong-spivak-invitation_F01152	105	■ 0.802	$F(c) \in \mathrm{Ob}(\mathcal{D})$	$F(c) \in \mathrm{Ob}(\mathcal{D})$	(i) for every object $c \in \mathrm{Ob}(\mathcal{C})$, one specifies an object $F(c) \in \mathrm{Ob}(\mathcal{D})$;
fong-spivak-invitation_F01153	105	■ 1.000	$f: c_1 \rightarrow c_2$	$f: c_1 \rightarrow c_2$	(ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$, one specifies a morphism $F(f): F(c_1) \rightarrow$
fong-spivak-invitation_F01154	105	■ 1.000	$F(f): F(c_1) \rightarrow$	$F(f): F(c_1) \rightarrow$	(ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$, one specifies a morphism $F(f): F(c_1) \rightarrow$
fong-spivak-invitation_F01155	105	■ 1.000	$F(c_2)$	$F(c_2)$	$F(c_2)$ in $\mathcal{D}$.
fong-spivak-invitation_F01156	105	■ 0.306	$F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$	$F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$	(a) for every object $c \in \mathrm{Ob}(\mathcal{C})$, we have $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$;
fong-spivak-invitation_F01157	105	■ 0.620	$c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$	$c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$	(b) for every three objects $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$ and two morphisms $f \in \mathcal{C}(c_1, c_2), g \in$
fong-spivak-invitation_F01158	105	■ 0.620	$f \in \mathcal{C}(c_1, c_2), g \in$	$f \in \mathcal{C}(c_1, c_2), g \in$	(b) for every three objects $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$ and two morphisms $f \in \mathcal{C}(c_1, c_2), g \in$
fong-spivak-invitation_F01159	105	■ 1.000	$\mathcal{C}(c_2, c_3)$	$\mathcal{C}(c_2, c_3)$	$\mathcal{C}(c_2, c_3)$, the equation $F(f \circ g) = F(f) \circ F(g)$ holds in $\mathcal{D}$.
fong-spivak-invitation_F01160	105	■ 1.000	$F(f \circ g) = F(f) \circ F(g)$	$F(f \circ g) = F(f) \circ F(g)$	$\mathcal{C}(c_2, c_3)$, the equation $F(f \circ g) = F(f) \circ F(g)$ holds in $\mathcal{D}$.
fong-spivak-invitation_F01161	105	■ 0.979	$F: \mathbf{2} \rightarrow \mathbf{3}$	$F: \mathbf{2} \rightarrow \mathbf{3}$	Example 3.29. For example, here we draw three functors $F: \mathbf{2} \rightarrow \mathbf{3}$:
fong-spivak-invitation_F01162	106	■ 1.000	n_0	n_0	path to the trivial path, i.e. the identity on n_0 . In the middle diagram $F(m_0) = n_0$,
fong-spivak-invitation_F01163	106	■ 1.000	$F(m_0) = n_0$	$F(m_0) = n_0$	path to the trivial path, i.e. the identity on n_0 . In the middle diagram $F(m_0) = n_0$,
fong-spivak-invitation_F01164	106	■ 1.000	$F(f_1) = g_1$	$F(f_1) = g_1$	$F(f_1) = g_1$, and $F(m_1) = n_1$. In the right-hand diagram, $F(m_0) = n_0, F(m_1) = n_2$,
fong-spivak-invitation_F01165	106	■ 1.000	$F(m_1) = n_1$	$F(m_1) = n_1$	$F(f_1) = g_1$, and $F(m_1) = n_1$. In the right-hand diagram, $F(m_0) = n_0, F(m_1) = n_2$,
fong-spivak-invitation_F01166	106	■ 1.000	$F(m_0) = n_0, F(m_1) = n_2$	$F(m_0) = n_0, F(m_1) = n_2$	$F(f_1) = g_1$, and $F(m_1) = n_1$. In the right-hand diagram, $F(m_0) = n_0, F(m_1) = n_2$,
fong-spivak-invitation_F01167	106	■ 0.917	g_2	g_2	and $F(f_1) = g_1 \circ g_2$.
fong-spivak-invitation_F01168	106	■ 0.974	$2 \rightarrow 3$	$2 \rightarrow 3$	Exercise 3.30. Above we wrote down three functors $2 \rightarrow 3$. Find and write down all
fong-spivak-invitation_F01169	106	■ 0.915	$\mathbf{2} \rightarrow \mathbf{3}$	$\mathbf{2} \rightarrow \mathbf{3}$	the remaining functors $2 \rightarrow 3$.
fong-spivak-invitation_F01170	106	■ 1.000	$\mathcal{F}$	$\mathcal{F}$	There are lots of functors from the free square category (let's call it $\mathcal{F}$) to the
fong-spivak-invitation_F01171	106	■ 1.000	$F: \mathcal{F} \rightarrow \mathcal{C}$	$F: \mathcal{F} \rightarrow \mathcal{C}$	However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends A' to A , B' to B , C' to
fong-spivak-invitation_F01172	106	■ 1.000	A, B'	A, B'	However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends A' to A , B' to B , C' to
fong-spivak-invitation_F01173	106	■ 1.000	B, C'	B, C'	However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends A' to A , B' to B , C' to
fong-spivak-invitation_F01174	106	■ 1.000	D'	D'	C , and D' to D . That is, once we have made this decision about how F acts on objects,
fong-spivak-invitation_F01175	106	■ 0.851	$\mathcal{C} = \bullet \rightarrow \bullet$	$\mathcal{C} = \bullet \rightarrow \bullet$	Exercise 3.33. Consider the free categories $\mathcal{C} = \boxed{\bullet \rightarrow \bullet}$ and $\mathcal{D} = \boxed{\bullet \rightrightarrows \bullet}$.
fong-spivak-invitation_F01176	106	■ 0.851	$\mathcal{D} = \bullet \rightrightarrows \bullet$	$\mathcal{D} = \bullet \rightrightarrows \bullet$	Exercise 3.33. Consider the free categories $\mathcal{C} = \boxed{\bullet \rightarrow \bullet}$ and $\mathcal{D} = \boxed{\bullet \rightrightarrows \bullet}$.

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fong-spivak-invitation_F01177	106	■ 1.000	$F, G: \mathcal{C} \rightarrow \mathcal{D}$	$F, G: \mathcal{C} \rightarrow \mathcal{D}$	Give two functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$ that act the same on objects but differently on
fong-spivak-invitation_F01178	106	■ 1.000	$A^{\prime}, B$	A', B	gory $\mathcal{C}$ to the free square category $\mathcal{F}$, but <i>none</i> that sends A to A' , B to B' , C to C' ,
fong-spivak-invitation_F01179	106	■ 1.000	$B^{\prime}, C$	B', C	gory $\mathcal{C}$ to the free square category $\mathcal{F}$, but <i>none</i> that sends A to A' , B to B' , C to C' ,
fong-spivak-invitation_F01180	106	■ 1.000	$C^{\prime}$	C'	gory $\mathcal{C}$ to the free square category $\mathcal{F}$, but <i>none</i> that sends A to A' , B to B' , C to C' ,
fong-spivak-invitation_F01181	106	■ 0.961	$f; h = g; i$	$f; h = g; i$	and D to D' . The reason is that if F were such a functor, then since $f \circ h = g \circ i$ in $\mathcal{C}$, we
fong-spivak-invitation_F01182	107	■ 1.000	$F(f; h) = F(g; i)$	$F(f; h) = F(g; i)$	would have $F(f \circ h) = F(g \circ i)$, but then the rules of functors would let us reason as
fong-spivak-invitation_F01183	107	■ 0.600	$f^{\prime}; h^{\prime} = g^{\prime}; i^{\prime}$	$f'; h' = g'; i'$	The resulting equation, $f' \circ h' = g' \circ i'$ does not hold in $\mathcal{F}$ because it is a free category
fong-spivak-invitation_F01184	107	■ 0.998	$\mathcal{N}$	$\mathcal{N}$	For example, consider the preorder $(\mathbb{N}, \leq)$ considered as a category $\mathcal{N}$ with objects
fong-spivak-invitation_F01185	107	■ 0.998	$\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$	$\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$	$\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$. A functor $F: \mathcal{N} \rightarrow \mathcal{N}$
fong-spivak-invitation_F01186	107	■ 0.998	$m \rightarrow n$	$m \rightarrow n$	$\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$. A functor $F: \mathcal{N} \rightarrow \mathcal{N}$
fong-spivak-invitation_F01187	107	■ 0.998	$F: \mathcal{N} \rightarrow \mathcal{N}$	$F: \mathcal{N} \rightarrow \mathcal{N}$	$\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$. A functor $F: \mathcal{N} \rightarrow \mathcal{N}$
fong-spivak-invitation_F01188	107	■ 1.000	$F(n) \in \mathbb{N}$	$F(n) \in \mathbb{N}$	sends each object $n \in \mathbb{N}$ to an object $F(n) \in \mathbb{N}$. It must send morphisms in $\mathcal{N}$ to
fong-spivak-invitation_F01189	107	■ 1.000	$F(m) \rightarrow F(n)$	$F(m) \rightarrow F(n)$	also be a morphism $F(m) \rightarrow F(n)$. In other words, if $m \leq n$, then we must also have
fong-spivak-invitation_F01190	107	■ 1.000	$F(m) \leq F(n)$	$F(m) \leq F(n)$	$F(m) \leq F(n)$. But as long as $m \leq n$ implies $F(m) \leq F(n)$, we have a functor.
fong-spivak-invitation_F01191	107	■ 1.000	$\mathbb{N} \rightarrow \mathbb{N}$	$\mathbb{N} \rightarrow \mathbb{N}$	Thus a functor $F: \mathcal{N} \rightarrow \mathcal{N}$ and a monotone map $\mathbb{N} \rightarrow \mathbb{N}$ are the same thing.
fong-spivak-invitation_F01192	107	■ 0.908	$\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$	$\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$	1. Given any category $\mathcal{C}$, show that there exists a functor $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$, known as the
fong-spivak-invitation_F01193	107	■ 1.000	$\operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{C})$	Note that a functor $\mathcal{C} \rightarrow \mathcal{D}$ consists of a function from $\operatorname{Ob}(\mathcal{C})$ to $\operatorname{Ob}(\mathcal{D})$ and for each
fong-spivak-invitation_F01194	107	■ 1.000	$\operatorname{Ob}(\mathcal{D})$	$\operatorname{Ob}(\mathcal{D})$	Note that a functor $\mathcal{C} \rightarrow \mathcal{D}$ consists of a function from $\operatorname{Ob}(\mathcal{C})$ to $\operatorname{Ob}(\mathcal{D})$ and for each
fong-spivak-invitation_F01195	107	■ 0.965	$c_1, c_2 \in \mathcal{C}$	$c_1, c_2 \in \mathcal{C}$	pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$.
fong-spivak-invitation_F01196	107	■ 0.965	$\mathcal{C}(c_1, c_2)$	$\mathcal{C}(c_1, c_2)$	pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$.
fong-spivak-invitation_F01197	107	■ 0.965	$\mathcal{D}(F(c_1), F(c_2))$	$\mathcal{D}(F(c_1), F(c_2))$	pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$.
fong-spivak-invitation_F01198	107	■ 0.907	$G: \mathcal{D} \rightarrow \mathcal{E}$	$G: \mathcal{D} \rightarrow \mathcal{E}$	2. Show that given $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$, we can define a new functor $(F \circ$
fong-spivak-invitation_F01199	107	■ 1.000	$G): \mathcal{C} \rightarrow \mathcal{E}$	$G): \mathcal{C} \rightarrow \mathcal{E}$	$G): \mathcal{C} \rightarrow \mathcal{E}$ just by composing functions.
fong-spivak-invitation_F01200	107	■ 1.000	$F: \mathcal{C} \rightarrow$	$F: \mathcal{C} \rightarrow$	Let $\mathcal{C}$ be a category, and recall the category Set from Definition 3.17. A functor $F: \mathcal{C} \rightarrow$
fong-spivak-invitation_F01201	107	■ 0.817	$I: \mathcal{C} \rightarrow$	$I: \mathcal{C} \rightarrow$	the database – is given by a set-valued functor $I: \mathcal{C} \rightarrow \mathbf{Set}$. The only additional detail
fong-spivak-invitation_F01202	108	■ 1.000	$c = \stackrel{\text{string}}{\circ}$	$c = \stackrel{\text{string}}{\circ}$	is that for any white node, such as $c = \stackrel{\text{string}}{\circ}$, we want to force I to map to the set of
fong-spivak-invitation_F01203	108	■ 0.694	$\mathcal{C}^5$	$\mathcal{C}^5$	functor $I: \mathcal{C} \rightarrow \mathbf{Set}$. ⁵

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fong-spivak-invitation_F01204	108	■ 0.991	$\mathrm{id}_{\mathbf{1}}$	$\mathrm{id}_{\mathbf{1}}$	phism $\mathrm{id}_{\mathbf{1}}$, and no other morphisms. For any functor $F: \mathbf{1} \rightarrow \mathbf{Set}$ one can extract a set
fong-spivak-invitation_F01205	108	■ 0.991	$F: \mathbf{1} \rightarrow$	$F: \mathbf{1} \rightarrow$	phism $\mathrm{id}_{\mathbf{1}}$, and no other morphisms. For any functor $F: \mathbf{1} \rightarrow \mathbf{Set}$ one can extract a set
fong-spivak-invitation_F01206	108	■ 0.941	$F(1)$	$F(1)$	$F(1)$. Show that, for any set S , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$. $\diamond$
fong-spivak-invitation_F01207	108	■ 0.941	$F_{\mathbf{S}}: \mathbf{1} \rightarrow$	$F_S: \mathbf{1} \rightarrow$	$F(1)$. Show that, for any set S , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$. $\diamond$
fong-spivak-invitation_F01208	108	■ 0.941	$F_{\mathbf{S}}(1)=S$	$F_S(1) = S$	$F(1)$. Show that, for any set S , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$. $\diamond$
fong-spivak-invitation_F01209	108	■ 0.998	$Z:=F(z)$	$Z := F(z)$	What is a set-valued functor $F: \mathcal{C} \rightarrow \mathbf{Set}$? It will consist of a set $Z := F(z)$ and a
fong-spivak-invitation_F01210	108	■ 0.993	$S:=F(s): Z \rightarrow Z$	$S := F(s): Z \rightarrow Z$	function $S := F(s): Z \rightarrow Z$, subject to the requirement that $S \circ S = S$. Here are some
fong-spivak-invitation_F01211	108	■ 0.993	$S ; S=S$	$S; S = S$	function $S := F(s): Z \rightarrow Z$, subject to the requirement that $S \circ S = S$. Here are some
fong-spivak-invitation_F01212	108	■ 0.934	$Z=\mathbb{N}$	$Z = \mathbb{N}$	• $Z = \mathbb{N}$ is the set of natural numbers and S sends each number to 0. In particular, 0
fong-spivak-invitation_F01213	108	■ 0.255	$13+(2 * 4)$	$13 + (2 * 4)$	• Z is the set of all well-formed arithmetic expressions, such as $13 + (2 * 4)$ or -5 , that
fong-spivak-invitation_F01214	108	■ 1.000	$Z=\mathbb{N}_{\geq 2}$	$Z = \mathbb{N}_{\geq 2}$	• $Z = \mathbb{N}_{\geq 2}$, and S sends n to its smallest prime factor. The smallest prime factor of a
fong-spivak-invitation_F01215	108	■ 1.000	$r \in I(c)$	$r \in I(c)$	row $r \in I(c)$.
fong-spivak-invitation_F01217	110	■ 0.997	$\alpha: F \rightarrow G$	$\alpha: F \Rightarrow G$	specify a natural transformation $\alpha: F \Rightarrow G$.
fong-spivak-invitation_F01218	110	■ 0.999	$\alpha_c: F(c) \rightarrow G(c)$	$\alpha_c: F(c) \rightarrow G(c)$	(i) for each object $c \in \mathcal{C}$, one specifies a morphism $\alpha_c: F(c) \rightarrow G(c)$ in $\mathcal{D}$, called the
fong-spivak-invitation_F01219	110	■ 1.000	α	α	c -component of α .
fong-spivak-invitation_F01220	110	■ 1.000	$\alpha: F \rightarrow G$	$\alpha: F \rightarrow G$	A natural transformation $\alpha: F \rightarrow G$ is called a natural isomorphism if each
fong-spivak-invitation_F01221	110	■ 0.986	α_c	α_c	component α_c is an isomorphism in $\mathcal{D}$.
fong-spivak-invitation_F01222	110	■ 0.854	$D: \mathcal{J} \rightarrow \mathcal{C}$	$D: \mathcal{J} \rightarrow \mathcal{C}$	Definition 3.42. A diagram D in $\mathcal{C}$ is a functor $D: \mathcal{J} \rightarrow \mathcal{C}$ from any category $\mathcal{J}$, called
fong-spivak-invitation_F01223	110	■ 0.854	$\mathcal{J}$	$\mathcal{J}$	Definition 3.42. A diagram D in $\mathcal{C}$ is a functor $D: \mathcal{J} \rightarrow \mathcal{C}$ from any category $\mathcal{J}$, called
fong-spivak-invitation_F01224	110	■ 1.000	$D(f)=D(f')$	$D(f) = D(f')$	the indexing category of the diagram D . We say that D commutes if $D(f) = D(f')$
fong-spivak-invitation_F01225	110	■ 0.961	$f, f': a \rightarrow b$	$f, f': a \rightarrow b$	holds for every parallel pair of morphisms $f, f': a \rightarrow b$ in $\mathcal{J}$.
fong-spivak-invitation_F01226	110	■ 0.961	$\mathcal{J}^7$	$\mathcal{J}^7$	holds for every parallel pair of morphisms $f, f': a \rightarrow b$ in $\mathcal{J}$.
fong-spivak-invitation_F01227	110	■ 1.000	$F(c) \Rightarrow G(d)$	$F(c) \Rightarrow G(d)$	In terms of Eq. (3.9), the only case of two parallel morphisms is that of $F(c) \Rightarrow G(d)$,
fong-spivak-invitation_F01228	110	■ 0.920	$F(f) \circ \alpha_d = \alpha_c \circ G(f)$	$F(f) \circ \alpha_d = \alpha_c \circ G(f)$	so to say that the diagram commutes is to say that $F(f) \circ \alpha_d = \alpha_c \circ G(f)$. This is exactly
fong-spivak-invitation_F01229	111	■ 1.000	$\alpha_1: v \rightarrow x$	$\alpha_1: v \rightarrow x$	ural transformation $\alpha: F \Rightarrow G$ is given by choosing components $\alpha_1: v \rightarrow x$ and
fong-spivak-invitation_F01230	111	■ 1.000	$\alpha_2: w \rightarrow y$	$\alpha_2: w \rightarrow y$	$\alpha_2: w \rightarrow y$. We have highlighted the only choice for each in green: namely, $\alpha_1 = c$ and
fong-spivak-invitation_F01231	111	■ 1.000	$\alpha_1=c$	$\alpha_1 = c$	$\alpha_2: w \rightarrow y$. We have highlighted the only choice for each in green: namely, $\alpha_1 = c$ and

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fong-spivak-invitation_F01232	111	■ 0.998	$\alpha_2 = g$	$\alpha_2 = g$	$\alpha_2 = g.$
fong-spivak-invitation_F01233	111	■ 0.476	$\mathcal{C} \xrightarrow[G]{F} \mathcal{D}$	$\mathcal{C} \xrightarrow[G]{F} \mathcal{D}$	notation $\mathcal{C} \begin{array}{c} \xrightarrow{F} \\ \alpha \Downarrow \\ \xrightarrow{G} \end{array} \mathcal{D}?$
fong-spivak-invitation_F01234	111	■ 1.000	$A, B: \mathbf{1} \rightarrow$	$A, B: \mathbf{1} \rightarrow$	fied with a set. So suppose A and B are sets considered as functors $A, B: \mathbf{1} \rightarrow$
fong-spivak-invitation_F01235	111	■ 0.852	$\mathcal{D}^{\mathcal{C}}$	$\mathcal{D}^{\mathcal{C}}$	Definition 3.45. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We denote by $\mathcal{D}^{\mathcal{C}}$ the category whose
fong-spivak-invitation_F01236	111	■ 0.997	$\mathcal{D}^{\mathcal{C}}(F, G)$	$\mathcal{D}^{\mathcal{C}}(F, G)$	objects are functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and whose morphisms $\mathcal{D}^{\mathcal{C}}(F, G)$ are the natural trans-
fong-spivak-invitation_F01237	111	■ 0.998	$F \in \mathcal{D}^{\mathcal{P}}$	$F \in \mathcal{D}^{\mathcal{P}}$	2. Propose an identity natural transformation on any object $F \in \mathcal{D}^{\mathcal{C}}$, and check that it
fong-spivak-invitation_F01238	111	■ 0.748	$\mathbf{1}$	$\mathbf{1}$	Example 3.47. In our new language, Example 3.44 says that $\mathbf{Set}^{\mathbf{1}}$ is isomorphic to $\mathbf{Set}$.
fong-spivak-invitation_F01239	111	■ 0.701	$F_0, F_1, F_2, \dots$	$F_0, F_1, F_2, \dots$	sequence $(F_0, F_1, F_2, \dots)$ of natural numbers, i.e. $F_0 \leq F_1 \leq F_2 \leq \dots$. If G is another
fong-spivak-invitation_F01240	111	■ 0.701	$F_0 \leq F_1 \leq F_2 \leq \dots$	$F_0 \leq F_1 \leq F_2 \leq \dots$	sequence $(F_0, F_1, F_2, \dots)$ of natural numbers, i.e. $F_0 < F_1 < F_2 < \dots$. If G is another
fong-spivak-invitation_F01241	111	■ 1.000	$\alpha_n: F_n \rightarrow G_n$	$\alpha_n: F_n \rightarrow G_n$	nent $\alpha_n: F_n \rightarrow G_n$ has no data, it just tells us a fact: that $F_n \leq G_n$. And the naturality
fong-spivak-invitation_F01242	111	■ 1.000	$F_n \leq G_n$	$F_n \leq G_n$	nent $\alpha_n: F_n \rightarrow G_n$ has no data, it just tells us a fact: that $F_n \leq G_n$. And the naturality
fong-spivak-invitation_F01243	112	■ 1.000	$F \Rightarrow G$	$F \Rightarrow G$	$F \Rightarrow G$ are the same. In other words, the category $\mathcal{N}^{\mathcal{N}}$ is itself a preorder, namely the
fong-spivak-invitation_F01244	112	■ 1.000	$\mathcal{N}^{\mathcal{N}}$	$\mathcal{N}^{\mathcal{N}}$	$F \Rightarrow G$ are the same. In other words, the category $\mathcal{N}^{\mathcal{N}}$ is itself a preorder, namely the
fong-spivak-invitation_F01245	112	■ 1.000	$F, G: \mathcal{C} \rightarrow \mathcal{P}$	$F, G: \mathcal{C} \rightarrow \mathcal{P}$	1. For any two functors $F, G: \mathcal{C} \rightarrow \mathcal{P}$, there is at most one natural transformation
fong-spivak-invitation_F01246	112	■ 1.000	$F \rightarrow G$	$F \rightarrow G$	$F \Rightarrow G$.
fong-spivak-invitation_F01247	112	■ 1.000	$F, G: \mathcal{P} \rightarrow \mathcal{C}$	$F, G: \mathcal{P} \rightarrow \mathcal{C}$	2. For any two functors $F, G: \mathcal{P} \rightarrow \mathcal{C}$, there is at most one natural transformation
fong-spivak-invitation_F01248	112	■ 0.689	$F:$	$F:$	these two categories equivalent because there exist functors $F: \mathbf{PrO} \rightarrow \mathbf{Bool-Cat}$ and
fong-spivak-invitation_F01249	112	■ 0.795	$G \cong \mathrm{id}_{\text{Pro}}$	$G \cong \mathrm{id}_{\text{Pro}}$	$G: \mathbf{Bool-Cat} \rightarrow \mathbf{PrO}$ such that there exist natural isomorphisms $F \circ G \cong \mathrm{id}_{\mathbf{PrO}}$ and
fong-spivak-invitation_F01250	112	■ 0.563	$G: F \cong \mathrm{id}_{\text{Bool-Cat}}$	$G: F \cong \mathrm{id}_{\text{Bool-Cat}}$	$G \circ F \cong \mathrm{id}_{\mathbf{Bool-Cat}}$ in the sense of Definition 3.41.
fong-spivak-invitation_F01251	112	■ 1.000	$I, J: \mathcal{C} \rightarrow$	$I, J: \mathcal{C} \rightarrow$	Definition 3.51. Suppose that $\mathcal{C}$ is a database schema and $I, J: \mathcal{C} \rightarrow \mathbf{Set}$ are
fong-spivak-invitation_F01252	112	■ 0.949	$\alpha: I \rightarrow J$	$\alpha: I \rightarrow J$	tion $\alpha: I \rightarrow J$. Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in
fong-spivak-invitation_F01253	112	■ 0.949	$:=$	$:=$	tion $\alpha: I \rightarrow J$. Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in
fong-spivak-invitation_F01254	112	■ 0.949	$\mathcal{C}$	$\mathcal{C}$	tion $\alpha: I \rightarrow J$. Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in
fong-spivak-invitation_F01255	113	■ 0.993	$I: \mathbf{Gr} \rightarrow$	$I: \mathbf{Gr} \rightarrow$	Here is an example of a $\mathbf{Gr}$ -instance, i.e. a set-valued functor $I: \mathbf{Gr} \rightarrow \mathbf{Set}$, in table
fong-spivak-invitation_F01256	113	■ 0.508	$=\{a, b, c, d, e\}$	$=\{a, b, c, d, e\}$	Here $I(\text{Arrow}) = \{a, b, c, d, e\}$, and $I(\text{Vertex}) = \{1, 2, 3, 4\}$. One can draw the
fong-spivak-invitation_F01257	113	■ 0.508	$=\{1, 2, 3, 4\}$	$=\{1, 2, 3, 4\}$	Here $I(\text{Arrow}) = \{a, b, c, d, e\}$, and $I(\text{Vertex}) = \{1, 2, 3, 4\}$. One can draw the

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fong-spivak-invitation_F01258	113	■ 0.987	$G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$	$G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$	are called <i>graph homomorphisms</i> . Let’s unwind this. Suppose that $G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$
fong-spivak-invitation_F01259	113	■ 0.990	$G, H \in \mathbf{Gr}$	$G, H \in \mathbf{Gr}$	are functors (i.e. Gr -instances); that is, they are objects $G, H \in \mathbf{Gr-Inst}$. A mor-
fong-spivak-invitation_F01260	113	■ 1.000	$G \rightarrow H$	$G \rightarrow H$	phism $G \rightarrow H$ is a natural transformation $\alpha: G \rightarrow H$ between them; what does that
fong-spivak-invitation_F01261	113	■ 1.000	$\alpha: G \rightarrow H$	$\alpha: G \rightarrow H$	phism $G \rightarrow H$ is a natural transformation $\alpha: G \rightarrow H$ between them; what does that
fong-spivak-invitation_F01262	113	■ 0.981	$\mathbf{Gr}$	Gr	By Definition 3.41, since Gr has two objects, α consists of two components,
fong-spivak-invitation_F01263	114	■ 1.000	α_{Vertex}	α_{Vertex}	tions α_{Vertex} and α_{Arrow} to respect source and targets in G and H . The left diagram
fong-spivak-invitation_F01264	114	■ 1.000	α_{Arrow}	α_{Arrow}	tions α_{Vertex} and α_{Arrow} to respect source and targets in G and H . The left diagram
fong-spivak-invitation_F01265	114	■ 1.000	G, H	G, H	Exercise 3.54. We claim that – with G, H as in Example 3.53 – there is exactly one
fong-spivak-invitation_F01266	114	■ 0.991	$\alpha_{\text{Arrow}}(a) = d$	$\alpha_{\text{Arrow}}(a) = d$	graph homomorphism $\alpha: G \rightarrow H$ such that $\alpha_{\text{Arrow}}(a) = d$.
fong-spivak-invitation_F01267	115	■ 0.994	$F: \mathbf{Gr} \rightarrow$	$F: \mathbf{Gr} \rightarrow$	The reader can create any functor she likes, but we will use a specific functor $F: \mathbf{Gr} \rightarrow$
fong-spivak-invitation_F01268	116	■ 0.156	$I:$	$I:$	$I: \mathbf{Gr} \rightarrow \mathbf{Set}$. Both F and I respect identities and composition, so $F \circ I$ does too. Thus
fong-spivak-invitation_F01269	116	■ 0.156	$\rightarrow$	$\rightarrow$	$I: \mathbf{Gr} \rightarrow \mathbf{Set}$. Both F and I respect identities and composition, so $F \circ I$ does too. Thus
fong-spivak-invitation_F01270	116	■ 0.156	$F \circ I$	$F \circ I$	$I: \mathbf{Gr} \rightarrow \mathbf{Set}$. Both F and I respect identities and composition, so $F \circ I$ does too. Thus
fong-spivak-invitation_F01271	117	■ 0.649	$G: \mathbf{Gr} \rightarrow \mathbf{DDS}$	$G: \mathbf{Gr} \rightarrow \mathbf{DDS}$	Exercise 3.55. Consider the functor $G: \mathbf{Gr} \rightarrow \mathbf{DDS}$ given by sending “source” to
fong-spivak-invitation_F01272	117	■ 0.858	$I: \mathcal{D} \rightarrow$	$I: \mathcal{D} \rightarrow$	set-valued functor $I: \mathcal{D} \rightarrow \mathbf{Set}$, we refer to the composite functor $F \circ I: \mathcal{C} \rightarrow \mathbf{Set}$ as
fong-spivak-invitation_F01273	117	■ 0.858	$F: I: \mathcal{C} \rightarrow$	$F: I: \mathcal{C} \rightarrow$	set-valued functor $I: \mathcal{D} \rightarrow \mathbf{Set}$, we refer to the composite functor $F \circ I: \mathcal{C} \rightarrow \mathbf{Set}$ as
fong-spivak-invitation_F01274	117	■ 0.957	$\alpha: I \Rightarrow J$	$\alpha: I \Rightarrow J$	the <i>pullback of I along F</i> . Given a natural transformation $\alpha: I \Rightarrow J$, there is a natural
fong-spivak-invitation_F01275	117	■ 0.587	$\alpha_F: F \circ I \Rightarrow F \circ J$	$\alpha_F: F \circ I \Rightarrow F \circ J$	transformation $\alpha_F: F \circ I \Rightarrow F \circ J$, whose component $(F \circ I)(c) \rightarrow (F \circ J)(c)$ for any
fong-spivak-invitation_F01276	117	■ 0.587	$(F \circ I)(c) \rightarrow (F \circ J)(c)$	$(F \circ I)(c) \rightarrow (F \circ J)(c)$	transformation $\alpha_F: F \circ I \Rightarrow F \circ J$, whose component $(F \circ I)(c) \rightarrow (F \circ J)(c)$ for any
fong-spivak-invitation_F01277	117	■ 0.677	$(\alpha_F)_c := \alpha_{Fc}$	$(\alpha_F)_c := \alpha_{Fc}$	$c \in \text{Ob}(\mathcal{C})$ is given by $(\alpha_F)_c := \alpha_{Fc}$.
fong-spivak-invitation_F01278	117	■ 0.970	$\Delta_F: \mathcal{D}$	$\Delta_F: \mathcal{D}$	This uses the data of F to define a functor $\Delta_F: \mathcal{D-Inst} \rightarrow \mathcal{C-Inst}$.
fong-spivak-invitation_F01279	117	■ 1.000	Δ	Δ	in general. The relevance to databases is that the data migration functor Δ from Defi-
fong-spivak-invitation_F01280	117	■ 1.000	Σ	Σ	nition 3.56 always has two adjoints of its own: a left adjoint which we denote Σ and a
fong-spivak-invitation_F01281	117	■ 1.000	Π	Π	right adjoint which we denote Π .
fong-spivak-invitation_F01282	117	■ 1.000	$P(a, b)$	$P(a, b)$	Recall from Section 3.2.3 that in a preorder P , a hom-set $P(a, b)$ has one element
fong-spivak-invitation_F01283	117	■ 1.000	$Q(f(p), q) \cong P(p, g(q))$	$Q(f(p), q) \cong P(p, g(q))$	isomorphism of sets $Q(f(p), q) \cong P(p, g(q))$: either both are one-element sets
fong-spivak-invitation_F01284	118	■ 1.000	$L: \mathcal{C} \rightarrow \mathcal{D}$	$L: \mathcal{C} \rightarrow \mathcal{D}$	Definition 3.57. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be functors.

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fong-spivak-invitation_F01285	118	■ 1.000	$R: \mathcal{D} \rightarrow \mathcal{C}$	$R: \mathcal{D} \rightarrow \mathcal{C}$	Definition 3.57. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be functors.
fong-spivak-invitation_F01286	118	■ 1.000	L	L	We say that L is <i>left adjoint to</i> R (and that R is <i>right adjoint to</i> L) if, for any $c \in \mathcal{C}$ and
fong-spivak-invitation_F01287	118	■ 0.990	$d \in \mathcal{D}$	$d \in \mathcal{D}$	$d \in \mathcal{D}$, there is an isomorphism of hom-sets
fong-spivak-invitation_F01288	118	■ 0.798	$d.^8$	$d.^8$	that is natural in c and d . ⁸
fong-spivak-invitation_F01289	118	■ 1.000	$f: c \rightarrow R(d)$	$f: c \rightarrow R(d)$	Given a morphism $f: c \rightarrow R(d)$ in $\mathcal{C}$, its image $g := \alpha_{c,d}(f)$ is called its <i>mate</i> .
fong-spivak-invitation_F01290	118	■ 1.000	$g := \alpha_{c,d}(f)$	$g := \alpha_{c,d}(f)$	Given a morphism $f: c \rightarrow R(d)$ in $\mathcal{C}$, its image $g := \alpha_{c,d}(f)$ is called its <i>mate</i> .
fong-spivak-invitation_F01291	118	■ 1.000	$g: L(c) \rightarrow d$	$g: L(c) \rightarrow d$	Similarly, the mate of $g: L(c) \rightarrow d$ is f .
fong-spivak-invitation_F01292	118	■ 1.000	$L \dashv R$	$L \dashv R$	To denote an adjunction we write $L \dashv R$, or in diagrams
fong-spivak-invitation_F01293	118	■ 0.528	$B \in \operatorname{Ob}$	$B \in \operatorname{Ob}$	Example 3.59. Let $B \in \operatorname{Ob}(\mathbf{Set})$ be any set. There is an adjunction called “currying B .”
fong-spivak-invitation_F01294	118	■ 1.000	A, C	A, C	Abstractly we write it as on the left, but what this means is that, for any sets A, C , there
fong-spivak-invitation_F01295	118	■ 0.997	$B \rightarrow C$	$B \rightarrow C$	C are sets. Then the set of functions $B \rightarrow C$ is also a set; let’s denote it C^B . It’s written
fong-spivak-invitation_F01296	118	■ 0.997	C^B	C^B	C are sets. Then the set of functions $B \rightarrow C$ is also a set; let’s denote it C^B . It’s written
fong-spivak-invitation_F01297	118	■ 1.000	10^3	10^3	this way because if C has 10 elements and B has 3 elements then C^B has 10^3 elements,
fong-spivak-invitation_F01298	118	■ 1.000	$ \left C^B\right = C ^{ B }$	$ C^B = C ^{ B }$	and more generally for any two finite sets $ C^B = C ^{ B }$.
fong-spivak-invitation_F01299	118	■ 1.000	$(A \times B) \rightarrow C$	$(A \times B) \rightarrow C$	dence between functions $(A \times B) \rightarrow C$ and functions $A \rightarrow C^B$. Intuitively, if I have a
fong-spivak-invitation_F01300	118	■ 1.000	$A \rightarrow C^B$	$A \rightarrow C^B$	dence between functions $(A \times B) \rightarrow C$ and functions $A \rightarrow C^B$. Intuitively, if I have a
fong-spivak-invitation_F01301	118	■ 0.522	$\mathcal{C}^{\operatorname{op}} \times \mathcal{D} \rightarrow$	$\mathcal{C}^{\operatorname{op}} \times \mathcal{D} \rightarrow$	⁸ This naturality is between functors $\mathcal{C}^{\operatorname{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$. It says that for any morphisms $f: c' \rightarrow c$ in $\mathcal{C}$ and
fong-spivak-invitation_F01302	118	■ 0.522	$f: c^{\prime} \rightarrow c$	$f: c' \rightarrow c$	⁸ This naturality is between functors $\mathcal{C}^{\operatorname{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$. It says that for any morphisms $f: c' \rightarrow c$ in $\mathcal{C}$ and
fong-spivak-invitation_F01303	118	■ 1.000	$g: d \rightarrow d^{\prime}$	$g: d \rightarrow d'$	$g: d \rightarrow d'$ in $\mathcal{D}$, the following diagram commutes:
fong-spivak-invitation_F01304	119	■ 1.000	$b \multimap c$	$b \multimap c$	objects and what we called hom-elements $b \multimap c$ in Definition 2.57.
fong-spivak-invitation_F01305	119	■ 1.000	$- \times B$	$- \times B$	Exercise 3.60. In Example 3.59, we discussed an adjunction between functors $- \times B$
fong-spivak-invitation_F01306	119	■ 1.000	$(-)^B$	$(-)^B$	and $(-)^B$. But we only said how these functors worked on objects: for an arbitrary set
fong-spivak-invitation_F01307	119	■ 1.000	$X \times B$	$X \times B$	X , they return sets $X \times B$ and X^B respectively.
fong-spivak-invitation_F01308	119	■ 1.000	X^B	X^B	X , they return sets $X \times B$ and X^B respectively.
fong-spivak-invitation_F01309	119	■ 1.000	$- \times B: X \times B \rightarrow Y \times B$	$- \times B: X \times B \rightarrow Y \times B$	1. Given a morphism $f: X \rightarrow Y$, what morphism should $- \times B: X \times B \rightarrow Y \times B$
fong-spivak-invitation_F01310	119	■ 1.000	$(-)^B: X^B \rightarrow Y^B$	$(-)^B: X^B \rightarrow Y^B$	2. Given a morphism $f: X \rightarrow Y$, what morphism should $(-)^B: X^B \rightarrow Y^B$ return?
fong-spivak-invitation_F01311	119	■ 1.000	$+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$	$+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$	3. Consider the function $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, which sends $(a, b) \mapsto a + b$. Currying $+$,

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fong-spivak-invitation_F01312	119	■ 1.000	(a, b) \mapsto a+b	$(a, b) \mapsto a + b$	3. Consider the function $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, which sends $(a, b) \mapsto a + b$. Currying $+$,
fong-spivak-invitation_F01313	119	■ 1.000	p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}	$p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$	we get a certain function $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$. What is $p(3)$?
fong-spivak-invitation_F01314	119	■ 1.000	p(3)	$p(3)$	we get a certain function $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$. What is $p(3)$?
fong-spivak-invitation_F01315	119	■ 1.000	\Delta_{\mathcal{F}}	Δ_F	Given $F: \mathcal{C} \rightarrow \mathcal{D}$, the data migration functor Δ_F turns $\mathcal{D}$ -instances into $\mathcal{C}$ -instances.
fong-spivak-invitation_F01316	120	■ 1.000	\Sigma_{\mathcal{F}}	Σ_F	DDS and get a graph, the migration functors Σ_F and Π_F can be used to turn any graph
fong-spivak-invitation_F01317	120	■ 1.000	\Pi_{\mathcal{F}}	Π_F	DDS and get a graph, the migration functors Σ_F and Π_F can be used to turn any graph
fong-spivak-invitation_F01318	120	■ 1.000	J: \mathbf{Gr} \rightarrow	$J: \mathbf{Gr} \rightarrow$	into a discrete dynamical system. That is, given an instance $J: \mathbf{Gr} \rightarrow \mathbf{Set}$, we can get
fong-spivak-invitation_F01319	120	■ 1.000	\Sigma_{\mathcal{F}}(J)	$\Sigma_F(J)$	instances $\Sigma_F(J)$ and $\Pi_F(J)$ on DDS . This, however, is quite technical, and we leave
fong-spivak-invitation_F01320	120	■ 1.000	\Pi_{\mathcal{F}}(J)	$\Pi_F(J)$	instances $\Sigma_F(J)$ and $\Pi_F(J)$ on DDS . This, however, is quite technical, and we leave
fong-spivak-invitation_F01321	120	■ 0.999	(e, f)	(e, f)	The functor Π_F puts into Airline Seat the set of those pairs (e, f) where e is an Econ-
fong-spivak-invitation_F01322	121	■ 1.000	\Pi_{\mathcal{F}}(I)	$\Pi_F(I)$	I , in which case $\Pi_F(I)(\text{Airline Seat})$ would be empty. But in other uses of these same
fong-spivak-invitation_F01323	121	■ 1.000	\Delta_{\mathcal{F}}, \Sigma_{\mathcal{F}}	Δ_F, Σ_F	F between schemas and using the “induced” functors Δ_F , Σ_F , and Π_F . Indeed, in
fong-spivak-invitation_F01324	121	■ 0.989	\mathcal{C} \rightarrow \mathbf{1}	$\mathcal{C} \rightarrow \mathbf{1}$	exactly one functor $\mathcal{C} \Rightarrow \mathbf{1}$ for any $\mathcal{C}$; let’s denote it
fong-spivak-invitation_F01325	121	■ 0.856	\Sigma_{\mathcal{I}}	$\Sigma_{\mathcal{I}}$	We want to consider the data migration functors $\Sigma_{\mathcal{I}}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{1}\text{-}\mathbf{Inst}$ and
fong-spivak-invitation_F01326	121	■ 0.451	\Pi_{\mathcal{I}}: \mathcal{C}	$\Pi_{\mathcal{I}}: \mathcal{C}$	$\Pi_{\mathcal{I}}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{1}\text{-}\mathbf{Inst}$. In Example 3.44, we saw that an instance on $\mathbf{1}$ is the same thing
fong-spivak-invitation_F01327	121	■ 0.772	\Sigma_{\mathcal{I}}(I)	$\Sigma_{\mathcal{I}}(I)$	Given any schema $\mathcal{C}$ and instance $I: \mathcal{C} \rightarrow \mathbf{Set}$, we will get sets $\Sigma_{\mathcal{I}}(I)$ and $\Pi_{\mathcal{I}}(I)$. Think-
fong-spivak-invitation_F01328	121	■ 0.772	\Pi_{\mathcal{I}}(I)	$\Pi_{\mathcal{I}}(I)$	Given any schema $\mathcal{C}$ and instance $I: \mathcal{C} \rightarrow \mathbf{Set}$, we will get sets $\Sigma_{\mathcal{I}}(I)$ and $\Pi_{\mathcal{I}}(I)$. Think-
fong-spivak-invitation_F01329	121	■ 1.000	I: \mathcal{G} \rightarrow	$I: \mathcal{G} \rightarrow$	Here’s a sample instance $I: \mathcal{G} \rightarrow \mathbf{Set}$:
fong-spivak-invitation_F01330	122	■ 0.969	\mathcal{G}	$\mathcal{G}$	Exercise 3.63. Note that $\mathcal{G}$ from Eq. (3.15) is isomorphic to the schema $\mathbf{Gr}$. In Sec-
fong-spivak-invitation_F01331	122	■ 0.952	!: \mathcal{G} \rightarrow \mathbf{1}	$!: \mathcal{G} \rightarrow \mathbf{1}$	Now we have a unique functor $!: \mathcal{G} \rightarrow \mathbf{1}$, and we want to say what $\Sigma_{\mathcal{I}}(I)$ and
fong-spivak-invitation_F01332	122	■ 0.989	\Pi_{\mathcal{I}}	$\Pi_{\mathcal{I}}$	Where do these facts – that $\Pi_{\mathcal{I}}$ and $\Sigma_{\mathcal{I}}$ act the way we said – come from? Everything
fong-spivak-invitation_F01333	123	■ 0.578	!: C \rightarrow Z	$!: C \rightarrow Z$	each object C of $\mathcal{C}$, there exists a unique morphism $!: C \rightarrow Z$.
fong-spivak-invitation_F01334	123	■ 0.900	C \rightarrow \{\bullet\}	$C \rightarrow \{\bullet\}$	one function $!: C \rightarrow \{\bullet\}$. This unique function is the one that does the only thing that
fong-spivak-invitation_F01335	123	■ 1.000	c \in C	$c \in C$	can be done: it maps each element $c \in C$ to the element $\bullet \in \{\bullet\}$.
fong-spivak-invitation_F01336	123	■ 1.000	\bullet \in \{\bullet\}	$\bullet \in \{\bullet\}$	can be done: it maps each element $c \in C$ to the element $\bullet \in \{\bullet\}$.
fong-spivak-invitation_F01337	123	■ 0.825	z \in P	$z \in P$	Exercise 3.66. Let $(P, \leq)$ be a preorder, let $z \in P$ be an element, and let $\mathcal{P}$ be the
fong-spivak-invitation_F01338	123	■ 1.000	c \in P	$c \in P$	only if it is a <i>top element</i> in P : that is, if and only if for all $c \in P$ we have $c \leq z$.
fong-spivak-invitation_F01339	123	■ 1.000	c \leq z	$c \leq z$	only if it is a <i>top element</i> in P : that is, if and only if for all $c \in P$ we have $c \leq z$.

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fong-spivak-invitation_F01340	123	■ 1.000	$Z^{\{\prime\}}$	Z'	<i>Proof.</i> This is a simple, but powerful standard argument. Suppose Z and Z' are both
fong-spivak-invitation_F01341	123	■ 1.000	$a: Z \rightarrow Z^{\{\prime\}}$	$a: Z \rightarrow Z'$	terminal objects in some category $\mathcal{C}$. Then there exist (unique) maps $a: Z \rightarrow Z'$ and
fong-spivak-invitation_F01342	123	■ 1.000	$b: Z^{\{\prime\}} \rightarrow Z$	$b: Z' \rightarrow Z$	$b: Z' \rightarrow Z$. Composing these, we get a map $a \circ b: Z \rightarrow Z$. Now since Z is termi-
fong-spivak-invitation_F01343	123	■ 1.000	$a; b: Z \rightarrow Z$	$a; b: Z \rightarrow Z$	$b: Z' \rightarrow Z$. Composing these, we get a map $a \circ b: Z \rightarrow Z$. Now since Z is termi-
fong-spivak-invitation_F01344	123	■ 0.992	$Z \rightarrow Z$	$Z \rightarrow Z$	nal, this map $Z \rightarrow Z$ must be unique. But id_Z is also such a map. So we must have
fong-spivak-invitation_F01345	123	■ 0.992	id_Z	id_Z	nal, this map $Z \rightarrow Z$ must be unique. But id_Z is also such a map. So we must have
fong-spivak-invitation_F01346	123	■ 0.999	$a; b = \mathrm{id}_{\mathrm{Z}}$	$a; b = \text{id}_Z$	$a \circ b = \text{id}_Z$. Similarly, we find that $b \circ a = \text{id}_{Z'}$. Thus a is an isomorphism, with
fong-spivak-invitation_F01347	123	■ 0.999	$b; a = \mathrm{id}_{\mathrm{Z}^{\{\prime\}}}$	$b; a = \text{id}_{Z'}$	$a \circ b = \text{id}_Z$. Similarly, we find that $b \circ a = \text{id}_{Z'}$. Thus a is an isomorphism, with
fong-spivak-invitation_F01348	124	■ 1.000	X, Y	X, Y	Definition 3.71. Let $\mathcal{C}$ be a category, and let X, Y be objects in $\mathcal{C}$. A <i>product</i> of X
fong-spivak-invitation_F01349	124	■ 1.000	$p_X: X \times Y \rightarrow X$	$p_X: X \times Y \rightarrow X$	and Y is an object, denoted $X \times Y$, together with morphisms $p_X: X \times Y \rightarrow X$ and
fong-spivak-invitation_F01350	124	■ 1.000	$p_Y: X \times Y \rightarrow Y$	$p_Y: X \times Y \rightarrow Y$	$p_Y: X \times Y \rightarrow Y$ such that for all objects C together with morphisms $f: C \rightarrow X$ and
fong-spivak-invitation_F01351	124	■ 1.000	$f: C \rightarrow X$	$f: C \rightarrow X$	$p_Y: X \times Y \rightarrow Y$ such that for all objects C together with morphisms $f: C \rightarrow X$ and
fong-spivak-invitation_F01352	124	■ 1.000	$g: C \rightarrow Y$	$g: C \rightarrow Y$	$g: C \rightarrow Y$, there exists a unique morphism $C \rightarrow X \times Y$, denoted $\langle f, g \rangle$, for which the
fong-spivak-invitation_F01353	124	■ 1.000	$C \rightarrow X \times Y$	$C \rightarrow X \times Y$	$g: C \rightarrow Y$, there exists a unique morphism $C \rightarrow X \times Y$, denoted $\langle f, g \rangle$, for which the
fong-spivak-invitation_F01354	124	■ 1.000	$\langle f, g \rangle$	$\langle f, g \rangle$	$g: C \rightarrow Y$, there exists a unique morphism $C \rightarrow X \times Y$, denoted $\langle f, g \rangle$, for which the
fong-spivak-invitation_F01355	124	■ 1.000	$p_X(x, y) := x$	$p_X(x, y) := x$	$p_X(x, y) := x$ and $p_Y(x, y) := y$.
fong-spivak-invitation_F01356	124	■ 1.000	$p_Y(x, y) := y$	$p_Y(x, y) := y$	$p_X(x, y) := x$ and $p_Y(x, y) := y$.
fong-spivak-invitation_F01357	124	■ 1.000	$\langle f, g \rangle(c) := (f(c), g(c))$	$\langle f, g \rangle(c) := (f(c), g(c))$	to $X \times Y$ such that the required diagram commutes is given by $\langle f, g \rangle(c) := (f(c), g(c))$.
fong-spivak-invitation_F01358	124	■ 0.999	$\underline{6} \times \underline{4}$	$\underline{6} \times \underline{4}$	Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$.
fong-spivak-invitation_F01359	124	■ 0.999	$\underline{6}$	$\underline{6}$	Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$.
fong-spivak-invitation_F01360	124	■ 0.999	$\underline{4}$	$\underline{4}$	Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$.
fong-spivak-invitation_F01361	125	■ 0.906	$x, y \in P$	$x, y \in P$	Exercise 3.73. Let $(P, \leq)$ be a preorder, let $x, y \in P$ be elements, and let $\mathcal{P}$ be the
fong-spivak-invitation_F01362	125	■ 1.000	$x \wedge y$	$x \wedge y$	corresponding category. Show that the product $x \times y$ in $\mathcal{P}$ agrees with their meet $x \wedge y$
fong-spivak-invitation_F01363	125	■ 1.000	$\mathcal{C} \times \mathcal{D}$	$\mathcal{C} \times \mathcal{D}$	Example 3.74. Given two categories $\mathcal{C}$ and $\mathcal{D}$, their product $\mathcal{C} \times \mathcal{D}$ may be given as
fong-spivak-invitation_F01364	125	■ 0.953	(c, d)	(c, d)	follows. The objects of this category are pairs (c, d) , where c is an object of $\mathcal{C}$ and d is
fong-spivak-invitation_F01365	125	■ 0.999	$(c, d) \rightarrow (c^{\{\prime\}}, d^{\{\prime\}})$	$(c, d) \rightarrow (c', d')$	an object of $\mathcal{D}$. Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs (f, g) where $f: c \rightarrow c'$
fong-spivak-invitation_F01366	125	■ 0.999	(f, g)	(f, g)	an object of $\mathcal{D}$. Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs (f, g) where $f: c \rightarrow c'$

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fong-spivak-invitation_F01367	125	■ 0.999	$f: c \rightarrow c^{\prime}$	$f: c \rightarrow c'$	an object of $\mathcal{D}$. Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs (f, g) where $f: c \rightarrow c'$
fong-spivak-invitation_F01368	125	■ 0.487	$(f, g); \left(f^{\prime}, g^{\prime}\right)=$	$(f, g); (f', g') =$	simply given by composing each entry in the pair separately, so $(f, g) \circ (f', g') =$
fong-spivak-invitation_F01369	125	■ 0.807	$\left(f \circ g^{\prime}, g \circ g^{\prime}\right)$	$(f \circ f', g \circ g')$	$f \circ f', g \circ g'$.
fong-spivak-invitation_F01370	125	■ 1.000	$\mathbf{1} \times \mathbf{2}$	$\mathbf{1} \times \mathbf{2}$	3. What is the product category $\mathbf{1} \times \mathbf{2}$?
fong-spivak-invitation_F01371	125	■ 0.937	$\mathcal{P} \times \mathcal{Q}$	$\mathcal{P} \times \mathcal{Q}$	4. What is the product category $\mathcal{P} \times \mathcal{Q}$ when $\mathcal{P}$ and $\mathcal{Q}$ are preorders and $\mathcal{P}$ and $\mathcal{Q}$ are
fong-spivak-invitation_F01372	125	■ 1.000	$X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$	$X \xleftarrow{f} C \xrightarrow{g} Y$	pair of maps $X \xleftarrow{f} C \xrightarrow{g} Y$, there exists a unique map $C \rightarrow X \times Y$ such that certain
fong-spivak-invitation_F01373	125	■ 0.991	(X, Y)	(X, Y)	call Cone (X, Y) , the category of cones over X and Y in $\mathcal{C}$. We shall see in Exercise 3.76
fong-spivak-invitation_F01374	125	■ 0.644	$X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$	$X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$	that $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ is a terminal object in Cone (X, Y) .
fong-spivak-invitation_F01375	125	■ 0.644	$\mathbf{Cone}(X, Y)$	Cone (X, Y)	that $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ is a terminal object in Cone (X, Y) .
fong-spivak-invitation_F01376	125	■ 0.849	$X \stackrel{f'}{\leftarrow} C' \stackrel{g'}{\rightarrow} Y$	$X \xleftarrow{f'} C' \xrightarrow{g'} Y$	$X \xleftarrow{f} C \xrightarrow{g} Y$ to $X \xleftarrow{f'} C' \xrightarrow{g'} Y$ in Cone (X, Y) is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such
fong-spivak-invitation_F01377	125	■ 0.849	$a: C \rightarrow C^{\prime}$	$a: C \rightarrow C'$	$X \xleftarrow{f} C \xrightarrow{g} Y$ to $X \xleftarrow{f'} C' \xrightarrow{g'} Y$ in Cone (X, Y) is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such
fong-spivak-invitation_F01378	125	■ 0.622	$X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$	$X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$	Exercise 3.76. Check that a product $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ is exactly the same as a
fong-spivak-invitation_F01379	126	■ 0.994	(C, c_*)	(C, c_*)	Definition 3.77. Let $D: \mathcal{J} \rightarrow \mathcal{C}$ be a diagram. A cone (C, c_*) over D consists of
fong-spivak-invitation_F01380	126	■ 1.000	$C \in \mathcal{C}$	$C \in \mathcal{C}$	(i) an object $C \in \mathcal{C}$;
fong-spivak-invitation_F01381	126	■ 0.999	$j \in \mathcal{J}$	$j \in \mathcal{J}$	(ii) for each object $j \in \mathcal{J}$, a morphism $c_j: C \rightarrow D(j)$.
fong-spivak-invitation_F01382	126	■ 0.999	$c_j: C \rightarrow D(j)$	$c_j: C \rightarrow D(j)$	(ii) for each object $j \in \mathcal{J}$, a morphism $c_j: C \rightarrow D(j)$.
fong-spivak-invitation_F01383	126	■ 0.812	$f: j \rightarrow k$	$f: j \rightarrow k$	(a) for each $f: j \rightarrow k$ in $\mathcal{J}$, we have $c_k = c_j \circ D(f)$.
fong-spivak-invitation_F01384	126	■ 0.812	$c_k = c_j$	$c_k = c_j$	(a) for each $f: j \rightarrow k$ in $\mathcal{J}$, we have $c_k = c_j \circ D(f)$.
fong-spivak-invitation_F01385	126	■ 0.812	$D(f)$	$D(f)$	(a) for each $f: j \rightarrow k$ in $\mathcal{J}$, we have $c_k = c_j \circ D(f)$.
fong-spivak-invitation_F01386	126	■ 1.000	$(C, c_*) \rightarrow (C', c'_*)$	$(C, c_*) \rightarrow (C', c'_*)$	A morphism of cones $(C, c_*) \rightarrow (C', c'_*)$ is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such that
fong-spivak-invitation_F01387	126	■ 0.918	$c_j = a \circ c'_j$	$c_j = a \circ c'_j$	for all $j \in \mathcal{J}$ we have $c_j = a \circ c'_j$. Cones over D , and their morphisms, form a category
fong-spivak-invitation_F01388	126	■ 0.997	(D)	(D)	Cone (D) .
fong-spivak-invitation_F01389	126	■ 0.480	$\lim(D)$	$\lim(D)$	The limit of D , denoted $\lim(D)$, is the terminal object in the category Cone (D) . Say
fong-spivak-invitation_F01390	126	■ 0.480	Cone (D)	Cone (D)	The limit of D , denoted $\lim(D)$, is the terminal object in the category Cone (D) . Say
fong-spivak-invitation_F01391	126	■ 0.991	$\lim(D) = (C, c_*)$	$\lim(D) = (C, c_*)$	it is the cone $\lim(D) = (C, c_*)$; we refer to C as the limit object and the map c_j for any
fong-spivak-invitation_F01392	126	■ 0.991	c_j	c_j	it is the cone $\lim(D) = (C, c_*)$; we refer to C as the limit object and the map c_j for any

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fong-spivak-invitation_F01393	126	0.854	$j \in \mathcal{J}$	$j \in \mathfrak{J}$	$j \in \mathcal{J}$ as the j th projection map.
fong-spivak-invitation_F01394	126	1.000	$D: \mathcal{J} \rightarrow$	$D: \mathcal{J} \rightarrow$	with five objects and five non-identity morphisms, then we may draw a diagram $D: \mathcal{J} \rightarrow$
fong-spivak-invitation_F01395	126	0.950	$\mathcal{J}=\varnothing$	$\mathcal{J} = \varnothing$	$\mathcal{J} = \varnothing$.
fong-spivak-invitation_F01396	127	0.159	v, w	v, w	v, w and no arrows, $\mathcal{J} = \left \begin{smallmatrix} v & w \\ \bullet & \bullet \end{smallmatrix} \right $.
fong-spivak-invitation_F01397	127	0.159	$\mathfrak{d}=\stackrel{v}{\bullet} \quad \stackrel{w}{\bullet}$	$\mathfrak{d} = \begin{smallmatrix} v & w \\ \bullet & \bullet \end{smallmatrix}$	v, w and no arrows, $\mathcal{J} = \left \begin{smallmatrix} v & w \\ \bullet & \bullet \end{smallmatrix} \right $.
fong-spivak-invitation_F01398	127	0.762	V, A, s, t	V, A, s, t	Theorem 3.80. Let $\mathcal{J}$ be a category presented by the finite graph (V, A, s, t) together
fong-spivak-invitation_F01399	127	0.989	$V=$	$V =$	with some equations, and let $D: \mathcal{J} \rightarrow \mathbf{Set}$ be a set-valued functor. Write $V =$
fong-spivak-invitation_F01400	127	1.000	$\{v_1, \dots, v_n\}$	$\{v_1, \dots, v_n\}$	$\{v_1, \dots, v_n\}$. The set
fong-spivak-invitation_F01401	127	0.999	$p_i: \left(\lim_{\mathcal{J}} D \right) \rightarrow D(v_i)$	$p_i: \left(\lim_{\mathcal{J}} D \right) \rightarrow D(v_i)$	together with the projection maps $p_i: (\lim_{\mathcal{J}} D) \rightarrow D(v_i)$ given by $p_i(d_1, \dots, d_n) :=$
fong-spivak-invitation_F01402	127	0.999	$p_i(d_1, \dots, d_n) :=$	$p_i(d_1, \dots, d_n) :=$	together with the projection maps $p_i: (\lim_{\mathcal{J}} D) \rightarrow D(v_i)$ given by $p_i(d_1, \dots, d_n) :=$
fong-spivak-invitation_F01403	127	1.000	d_i	d_i	d_i is a limit of D .
fong-spivak-invitation_F01404	127	0.958	J	J	Example 3.81. If J is the empty graph $\square$, then $n = 0$: there are no vertices. There is
fong-spivak-invitation_F01405	127	0.958	$n=0$	$n = 0$	Example 3.81. If J is the empty graph $\square$, then $n = 0$: there are no vertices. There is
fong-spivak-invitation_F01406	127	1.000	$D: \mathbf{1} \rightarrow$	$D: \mathbf{1} \rightarrow$	Exercise 3.83. If $D: \mathbf{1} \rightarrow \mathbf{Set}$ is a functor, what is the limit of D ? Compute it using
fong-spivak-invitation_F01407	127	0.997	$(d_1, \dots, d_n)$	$(d_1, \dots, d_n)$	In particular, the condition that the limit of $D: \mathcal{J} \rightarrow \mathbf{Set}$ selects tuples $(d_1, \dots, d_n)$ such
fong-spivak-invitation_F01408	127	0.942	$D(a)(d_i)=d_j$	$D(a)(d_i) = d_j$	that $D(a)(d_i) = d_j$ for each morphism $a: i \rightarrow j$ in $\mathcal{J}$ allows us to use limits to select
fong-spivak-invitation_F01409	127	0.942	$a: i \rightarrow j$	$a: i \rightarrow j$	that $D(a)(d_i) = d_j$ for each morphism $a: i \rightarrow j$ in $\mathcal{J}$ allows us to use limits to select
fong-spivak-invitation_F01410	127	0.942	$\mathcal{J}$	$\mathfrak{J}$	that $D(a)(d_i) = d_j$ for each morphism $a: i \rightarrow j$ in $\mathcal{J}$ allows us to use limits to select
fong-spivak-invitation_F01411	128	0.282	$\xrightarrow{x} \xrightarrow{a} \xrightarrow{g}$	$\xrightarrow{x} \xrightarrow{a} \xrightarrow{g}$	Example 3.84. If $\mathcal{J}$ is presented by the <i>cospan</i> graph $\left \begin{smallmatrix} x & \xrightarrow{f} & a & \xleftarrow{g} & y \\ \bullet & & \bullet & & \bullet \end{smallmatrix} \right $, then its limit is
fong-spivak-invitation_F01412	128	1.000	$X \xrightarrow{f} A \xleftarrow{g} Y$	$X \xrightarrow{f} A \xleftarrow{g} Y$	known as a <i>pullback</i> . Given the diagram $X \xrightarrow{f} A \xleftarrow{g} Y$, the pullback is the cone shown
fong-spivak-invitation_F01413	128	1.000	c_a	c_a	The fact that the diagram commutes means that the diagonal arrow c_a is in some
fong-spivak-invitation_F01414	128	0.492	$X \times_A Y$	$X \times_A Y$	naming the cone point $X \times_A Y$, and adding the $\lrcorner$ symbol, as shown to the right
fong-spivak-invitation_F01415	128	1.000	$X=\underline{6}, Y=\underline{4}$	$X = \underline{6}, Y = \underline{4}$	Here is a picture to help us unpack the definition in Set . We take $X = \underline{6}, Y = \underline{4}$, and
fong-spivak-invitation_F01416	128	1.000	$f: \underline{6} \rightarrow A$	$f: \underline{6} \rightarrow A$	The functions $f: \underline{6} \rightarrow A$ and $g: \underline{4} \rightarrow A$ are expressed in the coloring of the dots:
fong-spivak-invitation_F01417	128	1.000	$g: \underline{4} \rightarrow A$	$g: \underline{4} \rightarrow A$	The functions $f: \underline{6} \rightarrow A$ and $g: \underline{4} \rightarrow A$ are expressed in the coloring of the dots:
fong-spivak-invitation_F01418	128	1.000	$g(2)=g(4)=$	$g(2) = g(4) =$	for example, $g(2) = g(4) =$ red, while $f(5) =$ black. The pullback selects pairs

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fong-spivak-invitation_F01419	128	■ 1.000	$f(5)=$	$f(5) =$	for example, $g(2) = g(4) = \text{red}$, while $f(5) = \text{black}$. The pullback selects pairs
fong-spivak-invitation_F01420	128	■ 1.000	$(i, j) \in \underline{6} \times \underline{4}$	$(i, j) \in \underline{6} \times \underline{4}$	$(i, j) \in \underline{6} \times \underline{4}$ such that $f(i)$ and $g(j)$ have the same color.
fong-spivak-invitation_F01421	128	■ 1.000	$f(i)$	$f(i)$	$(i, j) \in \underline{6} \times \underline{4}$ such that $f(i)$ and $g(j)$ have the same color.
fong-spivak-invitation_F01422	128	■ 1.000	$g(j)$	$g(j)$	$(i, j) \in \underline{6} \times \underline{4}$ such that $f(i)$ and $g(j)$ have the same color.
fong-spivak-invitation_F01423	129	■ 0.989	F^{op}	F^{op}	$F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$? That is, how should F^{op} act on objects, and how should it act on
fong-spivak-invitation_F01424	129	■ 0.989	$\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$	$\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$	$F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$? That is, how should F^{op} act on objects, and how should it act on
fong-spivak-invitation_F01425	129	■ 0.999	$\mathcal{D}^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$	$\mathcal{D}^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$	Given a diagram $D: \mathcal{J} \rightarrow \mathcal{C}$, we may take the limit of the functor $D^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$.
fong-spivak-invitation_F01426	130	■ 0.988	Δ, Σ, Π	Δ, Σ, Π	three data migration functors Δ, Σ, Π ; see [Spi12; SW15b]. The version of this story
fong-spivak-invitation_F01427	132	■ 0.699	$\leq s$	$\leq s$	the $\leq s$ on each wire; they indicate that if box A requires a resource that box B produces,
fong-spivak-invitation_F01428	132	■ 1.000	$\$ 10 \leq \$ 20, 5 \text{ W} \leq 6 \text{ W}$	$\$10 \leq \$20, 5 \text{ W} \leq 6 \text{ W}$	$\$10 < \$20, 5\text{W} < 6\text{W}$ and so on.
fong-spivak-invitation_F01429	132	■ 1.000	$(p, r) \in P \times R$	$(p, r) \in P \times R$	pair $(p, r) \in P \times R$, where P is the preorder of resources to be produced and R is
fong-spivak-invitation_F01430	132	■ 1.000	$\Phi: P \times R \rightarrow$	$\Phi: P \times R \rightarrow$	Feasibility relations hence define a function $\Phi: P \times R \rightarrow \mathbf{Bool}$. For a function
fong-spivak-invitation_F01431	132	■ 1.000	$\Phi(p, r)=$	$\Phi(p, r) =$	(a) If $\Phi(p, r) = \text{true}$ and $p' \leq p$, then $\Phi(p', r) = \text{true}$.
fong-spivak-invitation_F01432	132	■ 1.000	$p^{\text{prime}} \leq p$	$p' \leq p$	(a) If $\Phi(p, r) = \text{true}$ and $p' \leq p$, then $\Phi(p', r) = \text{true}$.
fong-spivak-invitation_F01433	132	■ 1.000	$\Phi(p^{\text{prime}}, r)=$	$\Phi(p', r) =$	(a) If $\Phi(p, r) = \text{true}$ and $p' \leq p$, then $\Phi(p', r) = \text{true}$.
fong-spivak-invitation_F01434	132	■ 0.631	$r \leq r^{\text{prime}}$	$r \leq r'$	(b) If $\Phi(p, r) = \text{true}$ and $r \leq r'$ then $\Phi(p, r') = \text{true}$.
fong-spivak-invitation_F01435	132	■ 0.631	$\Phi(p, r^{\text{prime}})=$	$\Phi(p, r') =$	(b) If $\Phi(p, r) = \text{true}$ and $r \leq r'$ then $\Phi(p, r') = \text{true}$.
fong-spivak-invitation_F01436	133	■ 1.000	$r^{\text{prime}} \geq r$	$r' \geq r$	(b) also produce p given more resources $r' \geq r$. We shall see that these two conditions
fong-spivak-invitation_F01437	133	■ 0.600	$P^{\text{op}} \times R \rightarrow$	$P^{\text{op}} \times R \rightarrow$	are formalized by requiring Φ to be a monotone map $P^{\text{op}} \times R \rightarrow \mathbf{Bool}$.
fong-spivak-invitation_F01438	133	■ 0.606	$\$$	$\$$	$\$ -$ is structured as a preorder. The order $x \leq y$ represents the <i>availability of x given y</i> ,
fong-spivak-invitation_F01439	133	■ 0.712	$5 \text{ W} \leq 10 \text{ W}$	$5 \text{ W} \leq 10 \text{ W}$	if $5\text{W} \leq 10\text{W}$, it means that whenever we are provided 10W , we implicitly also have
fong-spivak-invitation_F01440	134	■ 0.792	$\mathcal{X}^{\text{op}} = (X, \geq)$	$\mathcal{X}^{\text{op}} = (X, \geq)$	opposite $\mathcal{X}^{\text{op}} = (X, \geq)$ has $x \geq y$ iff $y \leq x$.
fong-spivak-invitation_F01441	134	■ 0.792	$x \geq y$	$x \geq y$	opposite $\mathcal{X}^{\text{op}} = (X, \geq)$ has $x \geq y$ iff $y \leq x$.
fong-spivak-invitation_F01442	134	■ 0.940	$\mathcal{X} = \left(X, \leq_X\right)$	$\mathcal{X} = (X, \leq_X)$	Definition 4.1. Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A <i>feasibility relation</i>
fong-spivak-invitation_F01443	134	■ 0.940	$\mathcal{Y} = \left(Y, \leq_Y\right)$	$\mathcal{Y} = (Y, \leq_Y)$	Definition 4.1. Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A <i>feasibility relation</i>
fong-spivak-invitation_F01444	134	■ 0.616	$\Phi: \mathcal{X} \rightarrow \mathcal{Y}$	$\Phi: \mathcal{X} \rightarrow \mathcal{Y}$	We denote this by $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$.
fong-spivak-invitation_F01445	134	■ 1.000	$\Phi(x, y)=$	$\Phi(x, y) =$	Given $x \in X$ and $y \in Y$, if $\Phi(x, y) = \text{true}$ we say x can be obtained given y .

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fong-spivak-invitation_F01446	134	0.832	$x^{\prime} \leq_X x$	$x' \leq_X x$	$x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$. In other words, if x can be
fong-spivak-invitation_F01447	134	0.832	$y \leq_Y y^{\prime}$	$y \leq_Y y'$	$x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$. In other words, if x can be
fong-spivak-invitation_F01448	134	0.832	$\Phi(x, y) \leq_{\text{Bool}} \Phi(x^{\prime}, y^{\prime})$	$\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$	$x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$. In other words, if x can be
fong-spivak-invitation_F01449	134	1.000	$x^{\prime}$	x'	obtained given y , and if x' is available given x , then x' can be obtained given y . And if
fong-spivak-invitation_F01450	134	1.000	$y^{\prime}$	y'	furthermore y is available given y' , then x' can also be obtained given y' .
fong-spivak-invitation_F01451	134	0.387	$\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$	$\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$	1. Draw the Hasse diagram for the preorder $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$.
fong-spivak-invitation_F01452	134	0.710	$\Lambda: \mathcal{X} \rightarrow \mathcal{Y}$	$\Lambda: \mathcal{X} \rightarrow \mathcal{Y}$	2. Write down a profunctor $\Lambda: \mathcal{X} \nrightarrow \mathcal{Y}$ and, reading $\Lambda(x, y) = \text{true}$ as “my aunt can
fong-spivak-invitation_F01453	134	0.710	$\Lambda(x, y) =$	$\Lambda(x, y) =$	2. Write down a profunctor $\Lambda: \mathcal{X} \nrightarrow \mathcal{Y}$ and, reading $\Lambda(x, y) = \text{true}$ as “my aunt can
fong-spivak-invitation_F01454	134	0.335	$\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$	$\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$	forms an upper set in $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$.
fong-spivak-invitation_F01455	134	1.000	$\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	and a hom-operation, which we’ll write $\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$. By definition, this operation
fong-spivak-invitation_F01456	134	1.000	$b, c, d \in \mathbb{B}$	$b, c, d \in \mathbb{B}$	satisfies the property that for all $b, c, d \in \mathbb{B}$ one has
fong-spivak-invitation_F01458	135	0.311	$\Phi: \mathcal{X} \mapsto \mathcal{Y}$	$\Phi: \mathcal{X} \mapsto \mathcal{Y}$	and $\mathcal{Y}$ be $\mathcal{V}$ -categories. A $\mathcal{V}$ -profunctor from $\mathcal{X}$ to $\mathcal{Y}$, denoted $\Phi: \mathcal{X} \nrightarrow \mathcal{Y}$, is a $\mathcal{V}$ -functor
fong-spivak-invitation_F01459	135	0.322	$\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$	$\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$	$\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following
fong-spivak-invitation_F01460	135	0.322	$x, x^{\prime} \in \mathcal{X}$	$x, x' \in \mathcal{X}$	$\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following
fong-spivak-invitation_F01461	135	0.322	$y, y^{\prime} \in \mathcal{Y}$	$y, y' \in \mathcal{Y}$	$\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following
fong-spivak-invitation_F01462	135	0.698	$\Phi: X \rightarrow Y$	$\Phi: X \rightarrow Y$	tion 4.4 in the case $\mathcal{V} = \mathbf{Bool}$. One way to imagine a Bool -profunctor $\Phi: X \nrightarrow Y$
fong-spivak-invitation_F01463	136	1.000	$W \leq N$	$W \leq N$	Both X and Y are preorders, e.g. with $W \leq N$ and $b \leq a$. With bridges coming
fong-spivak-invitation_F01464	136	0.837	N	N	a path from N to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$.
fong-spivak-invitation_F01465	136	0.837	E	E	a path from N to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$.
fong-spivak-invitation_F01466	136	0.837	$\Phi(N, e) =$	$\Phi(N, e) =$	a path from N to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$.
fong-spivak-invitation_F01467	136	0.837	$\Phi(E, a) =$	$\Phi(E, a) =$	a path from N to e and E to a , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$.
fong-spivak-invitation_F01468	136	1.000	W	W	On the other hand, since there is no path from W to d , we have $\Phi(W, d) =$
fong-spivak-invitation_F01469	136	1.000	$\Phi(W, d) =$	$\Phi(W, d) =$	On the other hand, since there is no path from W to d , we have $\Phi(W, d) =$
fong-spivak-invitation_F01470	136	1.000	$W \leq N \leq c \leq a$	$W \leq N \leq c \leq a$	$W \leq N \leq c \leq a$, etc. This is called the <i>collage</i> of Φ ; we’ll explore this in more detail
fong-spivak-invitation_F01471	136	1.000	$\Phi(m, n) \in \mathbb{B}$	$\Phi(m, n) \in \mathbb{B}$	$\Phi(m, n) \in \mathbb{B}$. Copy and complete the Bool -matrix:
fong-spivak-invitation_F01472	137	0.722	$\Phi: X \mapsto Y$	$\Phi: X \mapsto Y$	Here is a depiction of a Cost -profunctor $\Phi: X \nrightarrow Y$:
fong-spivak-invitation_F01474	137	1.000	$d_X * M_{\Phi} * d_Y$	$d_X * M_{\Phi} * d_Y$	The matrix of distances for the profunctor Φ will be equal to $d_X * M_{\Phi} * d_Y$. In fact, since

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fong-spivak-invitation_F01475	137	■ 1.000	$\Phi = M_X^3 * M_\Phi * M_Y^2$	$\Phi = M_X^3 * M_\Phi * M_Y^2$	X has four elements and Y has three, we also know that $\Phi = M_X^3 * M_\Phi * M_Y^2$.
fong-spivak-invitation_F01476	138	■ 1.000	$M_X^3 * M_\Phi * M_Y^2$	$M_X^3 * M_\Phi * M_Y^2$	Exercise 4.12. Calculate $M_X^3 * M_\Phi * M_Y^2$, remembering to do matrix multiplication
fong-spivak-invitation_F01477	138	■ 0.484	$[0, \infty], \leq$	$[0, \infty], \leq$	ple, load might be the preorder $([0, \infty], \leq)$, where an element $x \in [0, \infty]$ represents
fong-spivak-invitation_F01478	138	—	100, more_than_\	(not rendered)	—
fong-spivak-invitation_F01479	138	■ 0.941	$\Phi: (T \times E) \mapsto \$$	$\Phi: (T \times E) \mapsto \$$	obtained at each cost; as such, it is described by a feasibility relation $\Phi: (T \times E) \multimap \$$.
fong-spivak-invitation_F01480	138	■ 0.626	T, E	T, E	where T , E , and $\$$ are the preorders drawn below:
fong-spivak-invitation_F01481	139	—	500K given \	(not rendered)	—
fong-spivak-invitation_F01482	139	■ 0.997	$\$ 500 \text{ \textit{K}}$	$\$500 \text{ K}$	movie given \$500K,” and “I can provide a mean and boring movie given a good-
fong-spivak-invitation_F01483	139	■ 0.988	$\mathcal{P}, Q$	$\mathcal{P}, Q$	posing two of them in series. Imagine you have cities $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{R}$ and you have
fong-spivak-invitation_F01484	139	■ 0.988	$\mathcal{R}$	$\mathcal{R}$	posing two of them in series. Imagine you have cities $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{R}$ and you have
fong-spivak-invitation_F01485	139	■ 0.936	$\Phi: \mathcal{P} \mapsto Q$	$\Phi: \mathcal{P} \mapsto Q$	bridges – and hence feasibility matrices – connecting these cities, say $\Phi: \mathcal{P} \multimap Q$ and
fong-spivak-invitation_F01486	139	■ 0.586	$\Psi: Q \mapsto \mathcal{R}$	$\Psi: Q \mapsto \mathcal{R}$	$\Psi: Q \multimap \mathcal{R}$.
fong-spivak-invitation_F01487	140	■ 1.000	Ψ	Ψ	The feasibility matrices for Φ (in blue) and Ψ (in red) are:
fong-spivak-invitation_F01488	140	■ 1.000	$\Phi; \Psi$	$\Phi; \Psi$	trying to give a feasibility matrix $\Phi; \Psi$ for getting from $\mathcal{P}$ to $\mathcal{R}$. How should this be
fong-spivak-invitation_F01489	140	■ 0.844	$p \in \mathcal{P}$	$p \in \mathcal{P}$	done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$, the navigator searches through $\mathcal{Q}$ for a
fong-spivak-invitation_F01490	140	■ 0.844	$r \in \mathcal{R}$	$r \in \mathcal{R}$	done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$, the navigator searches through $\mathcal{Q}$ for a
fong-spivak-invitation_F01492	141	■ 0.776	$\mathcal{X}, \mathcal{Y}$	$\mathcal{X}, \mathcal{Y}$	Definition 4.14. Let $\mathcal{V}$ be a quantale, let $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \multimap \mathcal{Y}$
fong-spivak-invitation_F01493	141	■ 0.776	$\mathcal{Z}$	$\mathcal{Z}$	Definition 4.14. Let $\mathcal{V}$ be a quantale, let $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \multimap \mathcal{Y}$
fong-spivak-invitation_F01494	141	■ 0.521	$\Psi: y \rightarrow z$	$\Psi: y \rightarrow z$	and $\Psi: \mathcal{Y} \multimap \mathcal{Z}$ be $\mathcal{V}$ -profunctors. We define their <i>composite</i> , denoted $\Phi; \Psi: \mathcal{X} \multimap \mathcal{Z}$
fong-spivak-invitation_F01495	141	■ 0.521	$\Phi: \Psi: \mathcal{X} \mapsto z$	$\Phi: \Psi: \mathcal{X} \mapsto z$	and $\Psi: \mathcal{Y} \multimap \mathcal{Z}$ be $\mathcal{V}$ -profunctors. We define their <i>composite</i> , denoted $\Phi; \Psi: \mathcal{X} \multimap \mathcal{Z}$
fong-spivak-invitation_F01496	141	■ 0.496	$_{\mathcal{V}}$	$\mathcal{V}$	Theorem 4.16. For any skeletal quantale $\mathcal{V}$, there is a category Prof $_{\mathcal{V}}$ whose objects
fong-spivak-invitation_F01497	141	■ 0.411	$\mathcal{X} \mapsto 1$	$\mathcal{X} \mapsto 1$	are $\mathcal{V}$ -categories $\mathcal{X}$, whose morphisms are $\mathcal{V}$ -profunctors $\mathcal{X} \multimap \mathcal{Y}$, and with composition
fong-spivak-invitation_F01498	141	■ 0.877	$_{\text{Bool}}$	Bool	Definition 4.17. We define Feas $:= \mathbf{Prof}_{\text{Bool}}$.
fong-spivak-invitation_F01499	142	■ 0.661	$\mathrm{U}_X: \mathcal{X} \mapsto X$	$U_X: \mathcal{X} \mapsto X$	Define the <i>unit profunctor</i> $U_X: \mathcal{X} \multimap X$ on a $\mathcal{V}$ -category $\mathcal{X}$ by the formula
fong-spivak-invitation_F01500	142	■ 0.798	$\mathcal{X}(x, y) \in \mathcal{V}$	$\mathcal{X}(x, y) \in \mathcal{V}$	of elements $x, y \in \mathcal{X}$ a hom-object $\mathcal{X}(x, y) \in \mathcal{V}$. The unit profunctor U_X just assigns
fong-spivak-invitation_F01501	142	■ 0.798	U_X	U_X	of elements $x, y \in \mathcal{X}$ a hom-object $\mathcal{X}(x, y) \in \mathcal{V}$. The unit profunctor U_X just assigns
fong-spivak-invitation_F01502	142	■ 0.582	$U_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$	$U_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$	for the unit profunctor $U_X: \mathcal{X} \multimap X$.

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fong-spivak-invitation_F01503	142	0.752	$\mathrm{U}_{\mathcal{P}}$	$U_{\mathcal{P}}$	Lemma 4.19. Composing any profunctor $\Phi: \mathcal{P} \multimap \mathcal{Q}$ with either unit profunctor, $U_{\mathcal{P}}$
fong-spivak-invitation_F01504	142	0.412	$U_{\mathcal{Q}}$	$U_{\mathcal{Q}}$	or $U_{\mathcal{Q}}$, returns Φ :
fong-spivak-invitation_F01505	142	0.524	$U_{\mathcal{P}}; \Phi = \Phi$	$U_{\mathcal{P}}; \Phi = \Phi$	<i>Proof.</i> We show that $U_{\mathcal{P}} \circ \Phi = \Phi$ holds; proving $\Phi = \Phi \circ U_{\mathcal{Q}}$ is similar. Fix $p \in P$
fong-spivak-invitation_F01506	142	0.524	$\Phi = \Phi; U_{\mathcal{Q}}$	$\Phi = \Phi; U_{\mathcal{Q}}$	<i>Proof.</i> We show that $U_{\mathcal{P}} \circ \Phi = \Phi$ holds; proving $\Phi = \Phi \circ U_{\mathcal{Q}}$ is similar. Fix $p \in P$
fong-spivak-invitation_F01507	142	0.998	$\Phi \leq \mathrm{U}_{\mathcal{P}}; \Phi$	$\Phi \leq U_{\mathcal{P}}; \Phi$	and $q \in Q$. Since $\mathcal{V}$ is skeletal, to prove the equality it's enough to show $\Phi \leq U_{\mathcal{P}} \circ \Phi$
fong-spivak-invitation_F01508	142	0.519	$U_{\mathcal{P}}; \Phi \leq \Phi$	$U_{\mathcal{P}}; \Phi \leq \Phi$	and $U_{\mathcal{P}} \circ \Phi \leq \Phi$. We have one direction:
fong-spivak-invitation_F01509	142	1.000	$\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$	$\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$	For the other direction, we must show $\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$. But by
fong-spivak-invitation_F01510	142	1.000	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$	$\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$	definition of join, this holds iff $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) < \Phi(p, q)$ is true for each $p_1 \in P$.
fong-spivak-invitation_F01511	142	1.000	$p_1 \in P$	$p_1 \in P$	definition of join, this holds iff $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$ is true for each $p_1 \in P$.
fong-spivak-invitation_F01512	143	0.676	$(=, \leq, \leq, =)$	$(=, \leq, \leq, =)$	1. Justify each of the four steps $\equiv, <, <, \equiv$ in Eq. (4.9).
fong-spivak-invitation_F01513	143	0.915	$\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$	$\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$	tors $\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$, and $\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$, we have
fong-spivak-invitation_F01514	143	0.915	$\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$	$\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$	tors $\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$, and $\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$, we have
fong-spivak-invitation_F01515	144	0.744	C	C	when defining the category $\mathbf{Cospan}_{\mathcal{C}}$ in Definition 6.39. An alternative approach is to
fong-spivak-invitation_F01516	144	1.000	$F: \mathcal{P} \rightarrow \mathcal{Q}$	$F: \mathcal{P} \rightarrow \mathcal{Q}$	Definition 4.24. Let $F: \mathcal{P} \rightarrow \mathcal{Q}$ be a $\mathcal{V}$ -functor. The <i>companion of F</i> , denoted
fong-spivak-invitation_F01517	144	0.592	$\widehat{F}: \mathcal{P} \mapsto \mathcal{Q}$	$\widehat{F}: \mathcal{P} \mapsto \mathcal{Q}$	$\widehat{F}: \mathcal{P} \multimap \mathcal{Q}$, and the <i>conjoint of F</i> , denoted $\check{F}: \mathcal{Q} \multimap \mathcal{P}$, are defined to be the following
fong-spivak-invitation_F01518	144	0.592	$\check{F}: \mathcal{Q} \rightarrow \mathcal{P}$	$\check{F}: \mathcal{Q} \rightarrow \mathcal{P}$	$\widehat{F}: \mathcal{P} \multimap \mathcal{Q}$, and the <i>conjoint of F</i> , denoted $\check{F}: \mathcal{Q} \multimap \mathcal{P}$, are defined to be the following
fong-spivak-invitation_F01519	144	1.000	$F(p) \in Q$	$F(p) \in Q$	$F(p) \in Q$.
fong-spivak-invitation_F01520	144	1.000	$\widehat{F}$	$\widehat{F}$	picture in that light, we see the companion $\widehat{F}$. But now mentally reverse every dotted
fong-spivak-invitation_F01521	144	0.866	$Q \mapsto \mathcal{P}$	$Q \mapsto \mathcal{P}$	arrow, and the result would be bridges $\mathcal{Q}$ to $\mathcal{P}$. This is a profunctor $\mathcal{Q} \multimap \mathcal{P}$! We call it $\check{F}$.
fong-spivak-invitation_F01522	144	0.866	$\check{F}$	$\check{F}$	arrow, and the result would be bridges $\mathcal{Q}$ to $\mathcal{P}$. This is a profunctor $\mathcal{Q} \multimap \mathcal{P}$! We call it $\check{F}$.
fong-spivak-invitation_F01523	144	0.999	$\mathcal{P} \rightarrow \mathcal{P}$	$\mathcal{P} \rightarrow \mathcal{P}$	Example 4.25. For any preorder $\mathcal{P}$, there is an identity functor $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$. Its com-
fong-spivak-invitation_F01524	144	0.993	$\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}: \mathcal{P} \mapsto \mathcal{P}$	$\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}: \mathcal{P} \mapsto \mathcal{P}$	panion and conjoint agree $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}: \mathcal{P} \multimap \mathcal{P}$. The resulting profunctor is in fact the unit
fong-spivak-invitation_F01525	145	0.951	$\widehat{\mathrm{id}}$	$\widehat{\mathrm{id}}$	Exercise 4.26. Check that the companion $\widehat{\mathrm{id}}$ of $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$ really has the unit profunctor
fong-spivak-invitation_F01526	145	0.895	$+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	$+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	Example 4.27. Consider the function $+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, sending a triple (a, b, c)
fong-spivak-invitation_F01527	145	0.969	$a+b+c \in \mathbb{R}$	$a+b+c \in \mathbb{R}$	of real numbers to $a+b+c \in \mathbb{R}$. This function is monotonic, because if $(a, b, c) \leq$
fong-spivak-invitation_F01528	145	0.969	$(a, b, c) \leq$	$(a, b, c) \leq$	of real numbers to $a+b+c \in \mathbb{R}$. This function is monotonic, because if $(a, b, c) \leq$

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fong-spivak-invitation_F01529	145	■ 0.982	$a^{\wedge\{\prime\}}, b^{\wedge\{\prime\}}, c^{\wedge\{\prime\}}$	a', b', c'	(a', b', c') – i.e. if $a \leq a', b \leq b'$, and $c \leq c'$ – then obviously $a + b + c \leq a' + b' + c'$.
fong-spivak-invitation_F01530	145	■ 0.982	$a \leq a^{\wedge\{\prime\}}, b \leq b^{\wedge\{\prime\}}$	$a \leq a', b \leq b'$	(a', b', c') – i.e. if $\boxed{a \leq a', b \leq b'}$, and $c \leq c'$ – then obviously $a + b + c \leq a' + b' + c'$.
fong-spivak-invitation_F01531	145	■ 0.982	$c \leq c^{\wedge\{\prime\}}$	$c \leq c'$	(a', b', c') – i.e. if $a \leq a', b \leq b'$, and $c \leq c'$ – then obviously $a + b + c \leq a' + b' + c'$.
fong-spivak-invitation_F01532	145	■ 0.982	$a+b+c \leq a^{\wedge\{\prime\}}+b^{\wedge\{\prime\}}+c^{\wedge\{\prime\}}$	$a + b + c \leq a' + b' + c'$	(a', b', c') – i.e. if $a \leq a', b \leq b'$, and $c \leq c'$ – then obviously $\boxed{a + b + c \leq a' + b' + c'}$.
fong-spivak-invitation_F01533	145	■ 0.513	$\widehat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \dashv \mathbb{R}$	$\hat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \dashv \mathbb{R}$	Its companion $\boxed{\hat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \rightarrow \mathbb{R}}$ is the function that sends (a, b, c, d) to <code>true</code>
fong-spivak-invitation_F01534	145	■ 0.513	a, b, c, d	a, b, c, d	Its companion $\hat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \rightarrow \mathbb{R}$ is the function that sends (a, b, c, d) to <code>true</code>
fong-spivak-invitation_F01535	145	■ 1.000	$a+b+c \leq d$	$a + b + c \leq d$	if $a + b + c \leq d$ and to <code>false</code> otherwise.
fong-spivak-invitation_F01536	145	■ 0.472	$\check{+}$	$\check{+}$	joint $\check{+}$?
fong-spivak-invitation_F01537	145	■ 0.964	$\mathcal{Q}$	$\mathcal{Q}$	tion between preorders $\mathcal{P}$ and $\mathcal{Q}$. The definition of adjoint can be extended from the
fong-spivak-invitation_F01538	145	■ 0.706	$G: \mathcal{Q} \rightarrow \mathcal{P}$	$G : \mathcal{Q} \rightarrow \mathcal{P}$	$\mathcal{V}$ -functors $F: \mathcal{P} \rightarrow \mathcal{Q}$ and $G: \mathcal{Q} \rightarrow \mathcal{P}$ such that the following holds for all $p \in \mathcal{P}$ and
fong-spivak-invitation_F01539	145	■ 1.000	$\widehat{F} = \widetilde{G}$	$\widehat{F} = \widetilde{G}$	the former equals the conjoint of the latter: $\widehat{F} = \widetilde{G}$.
fong-spivak-invitation_F01540	145	■ 1.000	$\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$	$\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$	2. Use this to prove that $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$, as was stated in Example 4.25.
fong-spivak-invitation_F01541	145	■ 0.611	$\Phi: \mathcal{X} \rightarrow \mathcal{Y}$	$\Phi : \mathcal{X} \rightarrow \mathcal{Y}$	Definition 4.31. Let $\mathcal{V}$ be a quantale, let $\mathcal{X}$ and $\mathcal{Y}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$
fong-spivak-invitation_F01542	145	■ 0.969	$\varnothing := \bigvee \varnothing$	$\varnothing := \bigvee \varnothing$	be a $\mathcal{V}$ -profunctor. Recall the notation $\varnothing := \bigvee \varnothing$. The <i>collage of Φ</i> , denoted $\mathbf{Col}(\Phi)$ is
fong-spivak-invitation_F01543	145	■ 0.969	$\mathbf{Col}(\Phi)$	$\mathbf{Col}(\Phi)$	be a $\mathcal{V}$ -profunctor. Recall the notation $\varnothing := \bigvee \varnothing$. The <i>collage of Φ</i> , denoted $\mathbf{Col}(\Phi)$ is
fong-spivak-invitation_F01544	145	■ 0.778	$\mathrm{Ob}(\mathbf{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$	$\mathrm{Ob}(\mathbf{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$	(i) $\boxed{\mathrm{Ob}(\mathbf{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})}$.
fong-spivak-invitation_F01545	145	■ 0.964	$a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))$	$a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))$	(ii) For any $\boxed{a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))}$, define $\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}$ to be
fong-spivak-invitation_F01546	145	■ 0.964	$\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}$	$\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}$	(ii) For any $a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))$, define $\boxed{\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}}$ to be
fong-spivak-invitation_F01547	146	■ 0.660	$i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$	$i_x : \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$	There are obvious functors $i_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$ and $i_{\mathcal{Y}}: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$, sending each
fong-spivak-invitation_F01548	146	■ 0.660	$i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$	$i_y : \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$	There are obvious functors $i_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$ and $i_{\mathcal{Y}}: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$, sending each
fong-spivak-invitation_F01549	146	■ 0.311	$\Phi: \mathcal{X} \rightarrow \mathcal{Y}$	$\Phi : \mathcal{X} \rightarrow$	Example 4.32. Consider the following picture of a Cost -profunctor $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$:
fong-spivak-invitation_F01551	146	■ 0.999	$\mathbf{Col}(\Phi)$	$\mathbf{Col}(\Phi)$	profunctor Φ , we draw the Hasse diagram for $\mathbf{Col}(\Phi)$ below left, and the Cost -matrix
fong-spivak-invitation_F01553	147	■ 1.000	$5+3=8$	$5 + 3 = 8$	that $5 + 3 = 8$. One way to categorify $\mathbb{N}$ is to use the category FinSet of finite
fong-spivak-invitation_F01554	147	■ 0.840	$\overline{5} =$	$\overline{5} =$	sets of that many elements, say $\overline{5} = \{\text{apple, banana, cherry, dragonfruit, elephant}\}$,
fong-spivak-invitation_F01555	147	■ 0.539	$\overline{3} =$	$\overline{3} =$	$\overline{3} = \{\text{apple, tomato, cantaloupe}\}$, and $\overline{8} = \{\text{Ali, Bob, Carl, Deb, Eli, Fritz, Gem, Helen}\}$
fong-spivak-invitation_F01556	147	■ 0.539	$\overline{8} = \{$	$\overline{8} = \{$	$\overline{3} = \{\text{apple, tomato, cantaloupe}\}$, and $\overline{8} = \{\text{Ali, Bob, Carl, Deb, Eli, Fritz, Gem, Helen}\}$

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fong-spivak-invitation_F01557	147	■ 1.000	$\overline{5} \sqcup \overline{3} \cong \overline{8}$	$\overline{5} \sqcup \overline{3} \cong \overline{8}$	$\overline{5} \sqcup \overline{3} \cong \overline{8}$.
fong-spivak-invitation_F01558	147	■ 1.000	$\overline{5} \sqcup \overline{3}$	$\overline{5} \sqcup \overline{3}$	Thus it's not the case that $\overline{5} \sqcup \overline{3}$ is <i>equal</i> to our chosen eight-element set $\overline{8}$, but more
fong-spivak-invitation_F01559	147	■ 1.000	$\overline{8}$	$\overline{8}$	Thus it's not the case that $\overline{5} \sqcup \overline{3}$ is <i>equal</i> to our chosen eight-element set $\overline{8}$, but more
fong-spivak-invitation_F01560	148	■ 0.964	$\mathcal{P}(a, b) =$	$\mathcal{P}(a, b) =$	a morphism $a \rightarrow b$," denoted $a \leq b$ or $\mathcal{P}(a, b) = \text{true}$, we instead say "here is a set
fong-spivak-invitation_F01561	148	■ 1.000	$x_{\{0\}} \leq x_{\{1\}}$	$x_0 \leq x_1$	whenever $x_0 \leq x_1$. Chaining these together, we could prove facts in our preorder. For
fong-spivak-invitation_F01562	148	■ 1.000	$x_{\{0\}} \leq x_{\{3\}}$	$x_0 \leq x_3$	provides a proof that $x_0 \leq x_3$ (the exterior box) using three facts (the interior boxes),
fong-spivak-invitation_F01563	148	■ 1.000	$x_{\{0\}} \leq x_{\{1\}}, x_{\{1\}} \leq x_{\{2\}}$	$x_0 \leq x_1, x_1 \leq x_2$	$x_0 \leq x_1, x_1 \leq x_2$, and $x_2 \leq x_3$.
fong-spivak-invitation_F01564	148	■ 1.000	$x_{\{2\}} \leq x_{\{3\}}$	$x_2 \leq x_3$	$x_0 \leq x_1, x_1 \leq x_2$, and $x_2 \leq x_3$.
fong-spivak-invitation_F01565	148	■ 1.000	$g: B \rightarrow C$	$g: B \rightarrow C$	Suppose we are given additional morphisms $g: B \rightarrow C$, and $h: C \rightarrow D$. Representing
fong-spivak-invitation_F01566	148	■ 1.000	$h: C \rightarrow D$	$h: C \rightarrow D$	Suppose we are given additional morphisms $g: B \rightarrow C$, and $h: C \rightarrow D$. Representing
fong-spivak-invitation_F01567	148	■ 0.305	$f; (g; h)$	$f; (g; h)$	We don't know which morphism it is. Is it $f \circ (g \circ h)$? Or $(f \circ g) \circ h$? It's good that
fong-spivak-invitation_F01568	148	■ 0.305	$(f; g) \circ h$	$(f; g) \circ h$	We don't know which morphism it is. Is it $f \circ (g \circ h)$? Or $(f \circ g) \circ h$? It's good that
fong-spivak-invitation_F01569	149	■ 0.440	$f; (g; h) = (f; g); h$	$f; (g; h) = (f; g); h$	Associativity says $f \circ (g \circ h) = (f \circ g) \circ h$, so it doesn't matter which way we chose to
fong-spivak-invitation_F01570	149	■ 0.994	id_x	id_x	shall see that it is not harmful to draw id_x in any of the following three ways:
fong-spivak-invitation_F01571	149	■ 0.486	$\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$	$\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$	property and the associativity property. The unitality says that $\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$ for
fong-spivak-invitation_F01572	149	■ 1.000	$f: x \rightarrow y$	$f: x \rightarrow y$	any $f: x \rightarrow y$. In terms of diagrams this would say
fong-spivak-invitation_F01573	150	■ 1.000	$\{1\}, \times$	$\{1\}, \times$	We will soon meet the monoidal category $(\mathbf{Set}, \{1\}, \times)$. The monoidal product will
fong-spivak-invitation_F01574	150	■ 1.000	$S \times T = \{(s, t) \mid s \in S, t \in T\}$	$S \times T = \{(s, t) \mid s \in S, t \in T\}$	take two sets, S and T , and return the set $S \times T = \{(s, t) \mid s \in S, t \in T\}$. But whereas
fong-spivak-invitation_F01575	150	■ 0.997	$(p \otimes q) \otimes r = p \otimes (q \otimes r)$	$(p \otimes q) \otimes r = p \otimes (q \otimes r)$	for monoidal preorders we had the brute associative property $(p \otimes q) \otimes r = p \otimes (q \otimes r)$,
fong-spivak-invitation_F01576	150	■ 1.000	$T \times U$	$T \times U$	and an element in $T \times U$, while the second contains pairs consisting of an element in
fong-spivak-invitation_F01577	150	■ 1.000	$S \times T$	$S \times T$	$S \times T$ and an element in U . The sets are not equal, but they are clearly isomorphic, i.e.
fong-spivak-invitation_F01578	150	■ 1.000	$\alpha_{s, t, u}$	$\alpha_{s, t, u}$	example, above we replaced the associativity equation with an isomorphism $\alpha_{s, t, u}$, but
fong-spivak-invitation_F01579	150	■ 0.913	$\alpha \mathrm{mathrm{s}}$	$\alpha \mathrm{s}$	now we need a more complex property to ensure that all these α s behave reasonably!
fong-spivak-invitation_F01580	150	■ 0.907	$I \in \mathrm{Ob}(\mathcal{C})$	$I \in \mathrm{Ob}(\mathcal{C})$	(i) an object $I \in \mathrm{Ob}(\mathcal{C})$ called the <i>monoidal unit</i> ,
fong-spivak-invitation_F01581	150	■ 1.000	$\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	$\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	(ii) a functor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, called the <i>monoidal product</i>
fong-spivak-invitation_F01582	151	■ 0.863	$\lambda_c: I \otimes c \cong c$	$\lambda_c: I \otimes c \cong c$	(a) $\lambda_c: I \otimes c \cong c$ for every $c \in \mathrm{Ob}(\mathcal{C})$,
fong-spivak-invitation_F01583	151	■ 0.997	$\rho_c: c \otimes I \cong c$	$\rho_c: c \otimes I \cong c$	(b) $\rho_c: c \otimes I \cong c$ for every $c \in \mathrm{Ob}(\mathcal{C})$,

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fong-spivak-invitation_F01584	151	■ 0.984	$\alpha_{c,d,e}:(c \otimes d) \otimes e \cong c \otimes (d \otimes e)$	$\alpha_{c,d,e}:(c \otimes d) \otimes e \cong c \otimes (d \otimes e)$	(c) $\alpha_{c,d,e}:(c \otimes d) \otimes e \cong c \otimes (d \otimes e)$ for every $c, d, e \in \text{Ob}(\mathcal{C})$,
fong-spivak-invitation_F01585	151	■ 0.999	$\sigma_{c,d}: c \otimes d \cong d \otimes c$	$\sigma_{c,d}: c \otimes d \cong d \otimes c$	(d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \text{Ob}(\mathcal{C})$, called the <i>swap map</i> , such that
fong-spivak-invitation_F01586	151	■ 0.999	$c, d \in \text{Ob}(\mathcal{C})$	$c, d \in \text{Ob}(\mathcal{C})$	(d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \text{Ob}(\mathcal{C})$, called the <i>swap map</i> , such that
fong-spivak-invitation_F01587	151	■ 1.000	$\sigma \circ \sigma = \text{id}$	$\sigma \circ \sigma = \text{id}$	$\sigma \circ \sigma = \text{id}$.
fong-spivak-invitation_F01588	151	■ 0.998	$(\mathcal{P}, I, \otimes)$	$(\mathcal{P}, I, \otimes)$	a monoidal preorder is a monoidal category $(\mathcal{P}, I, \otimes)$ where, for every $p, q \in \mathcal{P}$, the set
fong-spivak-invitation_F01589	151	■ 0.998	$p, q \in \mathcal{P}$	$p, q \in \mathcal{P}$	a monoidal preorder is a monoidal category $(\mathcal{P}, I, \otimes)$ where, for every $p, q \in \mathcal{P}$, the set
fong-spivak-invitation_F01590	151	■ 1.000	$\mathcal{P}(p, q)$	$\mathcal{P}(p, q)$	$\mathcal{P}(p, q)$ has at most one element.
fong-spivak-invitation_F01591	151	■ 1.000	$I := \{1\}$	$I := \{1\}$	monoidal unit is some choice of singleton set, say $I := \{1\}$, and the monoidal product
fong-spivak-invitation_F01592	151	■ 0.990	$\otimes := \times$	$\otimes := \times$	is $\otimes := \times$. What it means that $\times$ is a functor is that:
fong-spivak-invitation_F01593	151	■ 0.600	$(S, T) \in \text{Ob}(\mathcal{C})$	$(S, T) \in \text{Ob}(\mathcal{C})$	• For any pair of objects, i.e. sets $(S, T) \in \text{Ob}(\mathbf{Set} \times \mathbf{Set})$, one obtains a set $(S \times T) \in$
fong-spivak-invitation_F01594	151	■ 0.600	$(S \times T) \in$	$(S \times T) \in$	• For any pair of objects, i.e. sets $(S, T) \in \text{Ob}(\mathbf{Set} \times \mathbf{Set})$, one obtains a set $(S \times T) \in$
fong-spivak-invitation_F01595	151	■ 0.600	$(S \times T) \in$	$(S \times T) \in$	• For any pair of objects, i.e. sets $(S, T) \in \text{Ob}(\mathbf{Set} \times \mathbf{Set})$, one obtains a set $(S \times T) \in$
fong-spivak-invitation_F01596	151	■ 0.992	$\{(s, t) \mid s \in S, t \in T\}$	$\{(s, t) \mid s \in S, t \in T\}$	$\text{Ob}(\mathbf{Set})$. We know what it is: the set of pairs $\{(s, t) \mid s \in S, t \in T\}$.
fong-spivak-invitation_F01597	151	■ 1.000	$f: S \rightarrow S'$	$f: S \rightarrow S'$	• For any pair of morphisms, i.e. functions $f: S \rightarrow S'$ and $g: T \rightarrow T'$, one obtains
fong-spivak-invitation_F01598	151	■ 1.000	$g: T \rightarrow T'$	$g: T \rightarrow T'$	• For any pair of morphisms, i.e. functions $f: S \rightarrow S'$ and $g: T \rightarrow T'$, one obtains
fong-spivak-invitation_F01599	151	■ 1.000	$(f \times g): (S \times T) \rightarrow (S' \times T')$	$(f \times g): (S \times T) \rightarrow (S' \times T')$	a function $(f \times g): (S \times T) \rightarrow (S' \times T')$. It works pointwise: $(f \times g)(s, t) :=$
fong-spivak-invitation_F01600	151	■ 1.000	$(f \times g)(s, t) :=$	$(f \times g)(s, t) :=$	a function $(f \times g): (S \times T) \rightarrow (S' \times T')$. It works pointwise: $(f \times g)(s, t) :=$
fong-spivak-invitation_F01601	151	■ 0.991	$(f(s), g(t))$	$(f(s), g(t))$	$(f(s), g(t))$.
fong-spivak-invitation_F01602	151	■ 0.975	$\text{id}_S \times \text{id}_T = \text{id}_{S \times T}$	$\text{id}_S \times \text{id}_T = \text{id}_{S \times T}$	• These should preserve identities: $\text{id}_S \times \text{id}_T = \text{id}_{S \times T}$ for any sets S, T .
fong-spivak-invitation_F01603	152	■ 1.000	$S \xrightarrow{f} S' \xrightarrow{f'} S''$	$S \xrightarrow{f} S' \xrightarrow{f'} S''$	• These should preserve composition: for any functions $S \xrightarrow{f} S' \xrightarrow{f'} S''$ and $T \xrightarrow{g}$
fong-spivak-invitation_F01604	152	■ 1.000	$T \xrightarrow{g}$	$T \xrightarrow{g}$	• These should preserve composition: for any functions $S \xrightarrow{f} S' \xrightarrow{f'} S''$ and $T \xrightarrow{g}$
fong-spivak-invitation_F01605	152	■ 1.000	$T' \xrightarrow{g'} T''$	$T' \xrightarrow{g'} T''$	$T' \xrightarrow{g'} T''$, one has
fong-spivak-invitation_F01606	152	■ 1.000	$\{1\} \times S \cong S$	$\{1\} \times S \cong S$	The four conditions, (a), (b), (c), and (d) give isomorphisms $\{1\} \times S \cong S$, etc. These
fong-spivak-invitation_F01607	152	■ 1.000	$\{(1, s) \mid s \in S\} \rightarrow S$	$\{(1, s) \mid s \in S\} \rightarrow S$	maps are obvious in the case of $\mathbf{Set}$, e.g. the function $\{(1, s) \mid s \in S\} \rightarrow S$ sending
fong-spivak-invitation_F01608	152	■ 1.000	$(1, s)$	$(1, s)$	$(1, s)$ to s . We have been calling such things bookkeeping.
fong-spivak-invitation_F01609	152	■ 1.000	$1, \times$	$1, \times$	Exercise 4.39. Consider the monoidal category $(\mathbf{Set}, 1, \times)$, together with the diagram
fong-spivak-invitation_F01610	152	■ 0.740	$A=B=C=D=F=G=\mathbb{Z}$	$A = B = C = D = F = G = \mathbb{Z}$	Suppose that $A = B = C = D = F = G = \mathbb{Z}$ and $E = \mathbb{B} = \{\text{true}, \text{false}\}$, and

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fong-spivak-invitation_F01611	152	0.740	$E=\mathbb{B}=\$	$E = \mathbb{B} =$	Suppose that $A = B = C = D = F = G = \mathbb{Z}$ and $E = \mathbb{B} = \{\text{true}, \text{false}\}$, and
fong-spivak-invitation_F01612	152	0.766	$f_{\{C\}}(a)= a $, $f_{\{D\}}(a)=a * 5$, $g_{\{E\}}(d, b)="\ d \leq b$, " $g_{\{F\}}(d, b)=d-b$	$f_C(a) = a $, $f_D(a) = a * 5$, $g_E(d, b) = "d \leq b,$ " $g_F(d, b) = d - b$	suppose that $f_C(a) = a $, $f_D(a) = a * 5$, $g_F(d, b) = "d < b,"$ $g_F(d, b) = d - b$, and
fong-spivak-invitation_F01613	152	0.999	$h(c, e)=$	$h(c, e) =$	$h(c, e) \equiv$ if e then c else $1 - c$.
fong-spivak-invitation_F01614	152	0.999	$1-c$	$1 - c$	$h(c, e) =$ if e then c else $1 - c$.
fong-spivak-invitation_F01615	152	1.000	$g_{\{E\}}(5,3)$	$g_E(5,3)$	1. What are $g_E(5, 3)$ and $g_F(5, 3)$?
fong-spivak-invitation_F01616	152	1.000	$g_{\{F\}}(5,3)$	$g_F(5,3)$	1. What are $g_E(5, 3)$ and $g_F(5, 3)$?
fong-spivak-invitation_F01617	152	1.000	$g_{\{E\}}(3,5)$	$g_E(3,5)$	2. What are $g_E(3, 5)$ and $g_F(3, 5)$?
fong-spivak-invitation_F01618	152	1.000	$g_{\{F\}}(3,5)$	$g_F(3,5)$	2. What are $g_E(3, 5)$ and $g_F(3, 5)$?
fong-spivak-invitation_F01619	152	0.997	$h(5$	$h(5$	3. What is $h(5, \text{true})$?
fong-spivak-invitation_F01620	152	0.930	$h(-5$	$h(-5$	4. What is $h(-5, \text{true})$?
fong-spivak-invitation_F01621	152	1.000	$A \times B \rightarrowtail G \times F$	$A \times B \rightarrow G \times F$	The whole diagram now defines a function $A \times B \rightarrowtail G \times F$; call it q .
fong-spivak-invitation_F01622	152	1.000	$q_{\{G\}}(-2,3)$	$q_G(-2,3)$	6. What are $q_G(-2, 3)$ and $q_F(-2, 3)$?
fong-spivak-invitation_F01623	152	1.000	$q_{\{F\}}(-2,3)$	$q_F(-2,3)$	6. What are $q_G(-2, 3)$ and $q_F(-2, 3)$?
fong-spivak-invitation_F01624	152	1.000	$q_{\{G\}}(2,3)$	$q_G(2,3)$	7. What are $q_G(2, 3)$ and $q_F(2, 3)$?
fong-spivak-invitation_F01625	152	1.000	$q_{\{F\}}(2,3)$	$q_F(2,3)$	7. What are $q_G(2, 3)$ and $q_F(2, 3)$?
fong-spivak-invitation_F01626	153	0.298	$x, y \in \mathrm{Ob}(\mathcal{X})$	$x, y \in \mathrm{Ob}(\mathcal{X})$	(ii) for every pair $x, y \in \mathrm{Ob}(\mathcal{X})$, one specifies an object $\mathcal{X}(x, y) \in \mathcal{V}$, called the <i>hom-</i>
fong-spivak-invitation_F01627	153	0.420	$\mathrm{id}_x: I \rightarrowtail \mathcal{X}(x, x)$	$\mathrm{id}_x : I \rightarrow \mathcal{X}(x, x)$	(iii) for every $x \in \mathrm{Ob}(\mathcal{X})$, one specifies a morphism $\mathrm{id}_x: I \rightarrow \mathcal{X}(x, x)$ in $\mathcal{V}$, called the
fong-spivak-invitation_F01628	153	0.596	$x, y, z \in \mathrm{operatorname{Ob}}(\mathcal{X})$	$x, y, z \in \mathrm{Ob}(\mathcal{X})$	(iv) for each $x, y, z \in \mathrm{Ob}(\mathcal{X})$, one specifies a morphism $\mathbin{\circ}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow$
fong-spivak-invitation_F01629	153	0.596	$\{ \}_9: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrowtail$	$\mathbin{\circ}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow$	(iv) for each $x, y, z \in \mathrm{Ob}(\mathcal{X})$, one specifies a morphism $\mathbin{\circ}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow$
fong-spivak-invitation_F01630	153	0.809	$\mathcal{X}(x, z)$	$\mathcal{X}(x, z)$	$\mathcal{X}(x, z)$, called the <i>composition morphism</i> .
fong-spivak-invitation_F01631	153	0.922	$\mathcal{V}=($	$\mathcal{V} = ($	Exercise 4.41. Recall from Example 4.38 that $\mathcal{V} = (\mathbf{Set}, \{1\}, \times)$ is a symmetric
fong-spivak-invitation_F01632	153	0.922	$, \{1\}, \times)$	$, \{1\}, \times)$	Exercise 4.41. Recall from Example 4.38 that $\mathcal{V} = (\mathbf{Set}, \{1\}, \times)$ is a symmetric
fong-spivak-invitation_F01633	153	0.496	$\mathrm{operatorname{id}}_x: I \rightarrowtail \mathcal{X}(x, x)$	$\mathrm{id}_x : I \rightarrow \mathcal{X}(x, x)$	$\mathrm{id}_x: I \rightarrow \mathcal{X}(x, x)$ for the object x simply becomes the requirement (a) that $I \leq \mathcal{X}(x, x)$
fong-spivak-invitation_F01634	155	0.650	$(\mathcal{C}, I, \otimes)$	$(\mathcal{C}, I, \otimes)$	Definition 4.44. Let $(\mathcal{C}, I, \otimes)$ be a symmetric monoidal category, and $c \in \mathrm{Ob}(\mathcal{C})$ an
fong-spivak-invitation_F01635	155	0.999	$c^* \in \mathrm{operatorname{Ob}}(\mathcal{C})$	$c^* \in \mathrm{Ob}(\mathcal{C})$	(i) an object $c^* \in \mathrm{Ob}(\mathcal{C})$, called the <i>dual of c</i> ,
fong-spivak-invitation_F01636	155	1.000	$\eta_c: I \rightarrowtail c^* \otimes c$	$\eta_c : I \rightarrow c^* \otimes c$	(ii) a morphism $\eta_c: I \rightarrowtail c^* \otimes c$, called the <i>unit for c</i> ,
fong-spivak-invitation_F01637	155	1.000	$\epsilon_c: c \otimes c^* \rightarrowtail I$	$\epsilon_c : c \otimes c^* \rightarrow I$	(iii) a morphism $\epsilon_c: c \otimes c^* \rightarrowtail I$, called the <i>counit for c</i> .

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fong-spivak-invitation_F01638	155	0.881	$c^{\{*\}}$	c^*	If for every object $c \in \text{Ob}(\mathcal{C})$ there exists a dual c^* for c , then we say that $(\mathcal{C}, I, \otimes)$
fong-spivak-invitation_F01639	155	1.000	$\xrightarrow{c}$	$\xrightarrow{c}$	labeled c^* , i.e. $\xrightarrow{c}$ is the same as $\xleftarrow{c^*}$. The cup and cap discussed above are in fact the
fong-spivak-invitation_F01640	155	1.000	$\xleftarrow{c^{\{*\}}}$	$\xleftarrow{c^*}$	labeled c^* , i.e. $\xrightarrow{c}$ is the same as $\xleftarrow{c^*}$. The cup and cap discussed above are in fact the
fong-spivak-invitation_F01641	156	0.844	ϵ_{sound}	ϵ_{sound}	Note that the pictures in Eq. (4.14) correspond to ϵ_{sound} and η_{sound^*} .
fong-spivak-invitation_F01642	156	0.844	η_{sound^*}	η_{sound^*}	Note that the pictures in Eq. (4.14) correspond to ϵ_{sound} and η_{sound^*} .
fong-spivak-invitation_F01643	156	0.738	$c \multimap d$	$c \multimap d$	object $c \multimap d$ and an isomorphism $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$, natural in b . While we
fong-spivak-invitation_F01644	156	0.738	$\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$	$\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$	object $c \multimap d$ and an isomorphism $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$, natural in b . While we
fong-spivak-invitation_F01645	156	1.000	$c^{\{\prime\}}$	c'	2. if c^* and c' are both duals to c then there is an isomorphism $c^* \cong c'$; and
fong-spivak-invitation_F01646	156	1.000	$c^{\{*\}} \cong c^{\{\prime\}}$	$c^* \cong c'$	2. if c^* and c' are both duals to c then there is an isomorphism $c^* \cong c'$; and
fong-spivak-invitation_F01647	156	1.000	$c \cong c^{**}$	$c \cong c^{**}$	3. there is an isomorphism between c and its double-dual, $c \cong c^{**}$.
fong-spivak-invitation_F01648	156	0.668	$c^{\{*\}} \otimes d$	$c^* \otimes d$	To prove part 1, the key idea is that, for any c and d , the object $c \multimap d$ is given by $c^* \otimes d$,
fong-spivak-invitation_F01649	156	0.959	$\text{id}_b \otimes \eta_c$	$\text{id}_b \otimes \eta_c$	$\text{id}_b \otimes \eta_c$.
fong-spivak-invitation_F01650	157	1.000	$\alpha: A \rightarrow B$	$\alpha: A \rightarrow B$	look at than to write down formally. ² If in addition to the corelation $\alpha: A \rightarrow B$ above
fong-spivak-invitation_F01651	157	1.000	$\beta: B \rightarrow C$	$\beta: B \rightarrow C$	we have another corelation $\beta: B \rightarrow C$
fong-spivak-invitation_F01652	157	0.999	$\beta \circ \alpha$	$\beta \circ \alpha$	then the composite $\beta \circ \alpha$ of our two corelations is given by
fong-spivak-invitation_F01653	157	1.000	β	β	one to the other staying within equivalence classes of either α or β .
fong-spivak-invitation_F01654	157	1.000	$A \sqcup A :=$	$A \sqcup A :=$	Indeed, note that for any finite set A there is an equivalence relation on $A \sqcup A :=$
fong-spivak-invitation_F01655	157	0.998	$\{(a, 1), (a, 2) \mid a \in A\}$	$\{(a, 1), (a, 2) \mid a \in A\}$	$\{(a, 1), (a, 2) \mid a \in A\}$ where each part simply consists of the two elements $(a, 1)$ and
fong-spivak-invitation_F01656	157	0.998	$(a, 1)$	$(a, 1)$	$\{(a, 1), (a, 2) \mid a \in A\}$ where each part simply consists of the two elements $(a, 1)$ and
fong-spivak-invitation_F01657	157	1.000	$(a, 2)$	$(a, 2)$	$(a, 2)$ for each $a \in A$. The unit on a finite set A is the corelation $\eta_A: \emptyset \rightarrow A \sqcup A$
fong-spivak-invitation_F01658	157	1.000	$\eta_A: \emptyset \rightarrow A \sqcup A$	$\eta_A: \emptyset \rightarrow A \sqcup A$	$(a, 2)$ for each $a \in A$. The unit on a finite set A is the corelation $\eta_A: \emptyset \rightarrow A \sqcup A$
fong-spivak-invitation_F01659	157	1.000	$\epsilon_A: A \sqcup A \rightarrow \emptyset$	$\epsilon_A: A \sqcup A \rightarrow \emptyset$	$\epsilon_A: A \sqcup A \rightarrow \emptyset$ specified by this same equivalence relation.
fong-spivak-invitation_F01660	157	0.992	$\emptyset \rightarrow \underline{3} \sqcup \underline{3}$	$\emptyset \rightarrow \underline{3} \sqcup \underline{3}$	1. Draw a picture of the unit corelation $\emptyset \rightarrow \underline{3} \sqcup \underline{3}$.
fong-spivak-invitation_F01661	157	0.989	$\underline{3} \sqcup \underline{3} \rightarrow \emptyset$	$\underline{3} \sqcup \underline{3} \rightarrow \emptyset$	2. Draw a picture of the counit corelation $\underline{3} \sqcup \underline{3} \rightarrow \emptyset$.
fong-spivak-invitation_F01662	157	0.907	$\alpha; \beta$	$\alpha; \beta$	² To compose corelations $\alpha: A \rightarrow B$ and $\beta: B \rightarrow C$, we need to construct an equivalence relation $\alpha \circ \beta$ on
fong-spivak-invitation_F01663	157	1.000	$A \sqcup C$	$A \sqcup C$	$A \sqcup C$. To do so requires three steps: (i) consider α and β as relations on $A \sqcup B \sqcup C$, (ii) take the transitive
fong-spivak-invitation_F01664	157	1.000	$A \sqcup B \sqcup C$	$A \sqcup B \sqcup C$	$A \sqcup C$. To do so requires three steps: (i) consider α and β as relations on $A \sqcup B \sqcup C$, (ii) take the transitive

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_F01665	157	■ 1.000	$\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$	$\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$	description. Note that, as binary relations, we have $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$, and $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$.
fong-spivak-invitation_F01666	157	■ 1.000	$\beta \subseteq (B \sqcup C) \times (B \sqcup C)$	$\beta \subseteq (B \sqcup C) \times (B \sqcup C)$	description. Note that, as binary relations, we have $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$, and $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$.
fong-spivak-invitation_F01667	157	■ 1.000	$\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$	$\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$	We also have three inclusions: $\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$, and
fong-spivak-invitation_F01668	157	■ 0.778	$\{ \}^{\iota} A \sqcup C: A \sqcup C \rightarrow A \sqcup B \sqcup C$	$\iota_{A \sqcup C}: A \sqcup C \rightarrow A \sqcup B \sqcup C$	$\iota_{A \sqcup C}: A \sqcup C \rightarrow A \sqcup B \sqcup C$. Recalling our notation from Section 1.4, we define
fong-spivak-invitation_F01669	158	■ 0.544	$\{ \}_{\text{Bool}}$	Bool	Instead of just detailing this compact closed structure for Feas = Prof _{Bool} , it's no
fong-spivak-invitation_F01670	158	■ 0.992	$(\mathcal{V}, I, \otimes)$	$(\mathcal{V}, I, \otimes)$	extra work to prove that for any skeletal (unital, commutative) quantale $(\mathcal{V}, I, \otimes)$ the
fong-spivak-invitation_F01671	158	■ 0.529	$\operatorname{Prof}_{\mathcal{V}}$	$\operatorname{Prof}_{\mathcal{V}}$	Theorem 4.48. Let $\mathcal{V}$ be a skeletal quantale. The category Prof _{$\mathcal{V}$} can be given the
fong-spivak-invitation_F01672	158	■ 0.583	$\{ \}_{\mathcal{V}}$	$\mathcal{V}$	But monoidal products need to be given on morphisms also, and the morphisms in Prof _{$\mathcal{V}$}
fong-spivak-invitation_F01673	158	■ 0.843	$\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$	$\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$	are $\mathcal{V}$ -profunctors. So, given $\mathcal{V}$ -profunctors $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$, one defines
fong-spivak-invitation_F01674	158	■ 0.843	$\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$	$\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$	are $\mathcal{V}$ -profunctors. So, given $\mathcal{V}$ -profunctors $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$, one defines
fong-spivak-invitation_F01675	158	■ 0.787	$(\Phi \times \Psi): \mathcal{X}_1 \times \mathcal{Y}_1 \mapsto \mathcal{X}_2 \times \mathcal{Y}_2$	$(\Phi \times \Psi): \mathcal{X}_1 \times \mathcal{Y}_1 \mapsto \mathcal{X}_2 \times \mathcal{Y}_2$	a $\mathcal{V}$ -profunctor $\Phi \times \Psi: \mathcal{X}_1 \times \mathcal{Y}_1 \mapsto \mathcal{X}_2 \times \mathcal{Y}_2$ by
fong-spivak-invitation_F01676	159	■ 0.977	$\Phi \times \Psi$	$\Phi \times \Psi$	the monoidal product of resource preorders $\mathcal{X}$ and $\mathcal{Y}$ and why $\Phi \times \Psi$ makes sense as the
fong-spivak-invitation_F01677	159	■ 0.665	$(1, 1) = I$	$(1, 1) = I$	1, and $(1, 1) = I$ is the monoidal unit of $\mathcal{V}$. We take 1 to be the monoidal unit of Prof _{$\mathcal{V}$} .
fong-spivak-invitation_F01678	159	■ 0.400	$X \times \mathbf{1} \mapsto X$	$X \times \mathbf{1} \mapsto X$	morphisms $X \times \mathbf{1} \mapsto X$ and $\mathbf{1} \times X \mapsto X$ in Prof _{$\mathcal{V}$} , for any $\mathcal{V}$ -category X . What are
fong-spivak-invitation_F01679	159	■ 0.400	$\mathbf{1} \times X \mapsto X$	$\mathbf{1} \times X \mapsto X$	morphisms $X \times \mathbf{1} \mapsto X$ and $\mathbf{1} \times X \mapsto X$ in Prof _{$\mathcal{V}$} , for any $\mathcal{V}$ -category X . What are
fong-spivak-invitation_F01680	159	■ 0.465	$\eta_X: \mathbf{1} \mapsto \mathcal{X}^{\text{op}} \times X$	$\eta_X: \mathbf{1} \mapsto \mathcal{X}^{\text{op}} \times X$	elaborate, the unit is a $\mathcal{V}$ -profunctor $\eta_X: \mathbf{1} \mapsto \mathcal{X}^{\text{op}} \times X$. By definition, this is a $\mathcal{V}$ -functor
fong-spivak-invitation_F01681	159	■ 0.973	$\eta_X(1, x, x') := X(x, x')$	$\eta_X(1, x, x') := X(x, x')$	we define it by $\eta_X(1, x, x') := X(x, x')$. Similarly, the counit is the profunctor $\epsilon_X: (X \times$
fong-spivak-invitation_F01682	159	■ 0.973	$\epsilon_X: (X \times$	$\epsilon_X: (X \times$	we define it by $\eta_X(1, x, x') := X(x, x')$. Similarly, the counit is the profunctor $\epsilon_X: (X \times$
fong-spivak-invitation_F01683	159	■ 0.239	$\mathcal{X}^{\text{op}}) \rightarrow \mathbf{1}$	$\mathcal{X}^{\text{op}}) \rightarrow \mathbf{1}$	$\mathcal{X}^{\text{op}}) \mapsto \mathbf{1}$, defined by $\epsilon_X(x, x', 1) := X(x, x')$.
fong-spivak-invitation_F01684	159	■ 0.239	$\epsilon_X(x, x', 1) := X(x, x')$	$\epsilon_X(x, x', 1) := X(x, x')$	$\mathcal{X}^{\text{op}}) \mapsto \mathbf{1}$, defined by $\epsilon_X(x, x', 1) := X(x, x')$.
fong-spivak-invitation_F01685	162	■ 1.000	$u = 15x$	$u = 15x$	– produces two output signals: $u = 15x$ (upper right) and $v = 3x + 21y$ (lower right).
fong-spivak-invitation_F01686	162	■ 1.000	$v = 3x + 21y$	$v = 3x + 21y$	– produces two output signals: $u = 15x$ (upper right) and $v = 3x + 21y$ (lower right).
fong-spivak-invitation_F01687	162	■ 1.000	$x + 7y$	$x + 7y$	the second copy of x to the lower signal to get $x + 7y$, and so on.
fong-spivak-invitation_F01688	163	■ 0.608	$(\mathcal{C}, 0, +)$	$(\mathcal{C}, 0, +)$	Definition 5.1. A <i>prop</i> is a symmetric strict monoidal category $(\mathcal{C}, 0, +)$ for which
fong-spivak-invitation_F01689	163	■ 0.876	$\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$	$\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$	$\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$, the monoidal unit is $0 \in \mathbb{N}$, and the monoidal product on objects is given
fong-spivak-invitation_F01690	163	■ 0.876	$0 \in \mathbb{N}$	$0 \in \mathbb{N}$	$\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$, the monoidal unit is $0 \in \mathbb{N}$, and the monoidal product on objects is given
fong-spivak-invitation_F01691	163	■ 1.000	$\mathcal{C}(m, n)$	$\mathcal{C}(m, n)$	(i) a set $\mathcal{C}(m, n)$ of morphisms $m \rightarrow n$, for $m, n \in \mathbb{N}$;

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fong-spivak-invitation_F01692	163	■ 0.341	$\operatorname{map}_{\mathrm{id}}: n \rightarrow n$	$\operatorname{map}_{\mathrm{id}}: n \rightarrow n$	(ii) for all $n \in \mathbb{N}$, an identity map $\operatorname{id}_n: n \rightarrow n$;
fong-spivak-invitation_F01693	163	■ 1.000	$\sigma_{m,n}: m+n \rightarrow n+m$	$\sigma_{m,n}: m+n \rightarrow n+m$	(iii) for all $m, n \in \mathbb{N}$, a symmetry map $\sigma_{m,n}: m+n \rightarrow n+m$;
fong-spivak-invitation_F01694	163	■ 1.000	$f: m \rightarrow n$	$f: m \rightarrow n$	(iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$; a map $(f \circ g): m \rightarrow p$;
fong-spivak-invitation_F01695	163	■ 1.000	$g: n \rightarrow p$	$g: n \rightarrow p$	(iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$; a map $(f \circ g): m \rightarrow p$;
fong-spivak-invitation_F01696	163	■ 1.000	$(f \circ g): m \rightarrow p$	$(f \circ g): m \rightarrow p$	(iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$; a map $(f \circ g): m \rightarrow p$;
fong-spivak-invitation_F01697	163	■ 1.000	$f: m \rightarrow m^{\prime}$	$f: m \rightarrow m^{\prime}$	(v) a monoidal product on morphisms: given $f: m \rightarrow m^{\prime}$ and $g: n \rightarrow n^{\prime}$, a map
fong-spivak-invitation_F01698	163	■ 1.000	$g: n \rightarrow n^{\prime}$	$g: n \rightarrow n^{\prime}$	(v) a monoidal product on morphisms: given $f: m \rightarrow m^{\prime}$ and $g: n \rightarrow n^{\prime}$, a map
fong-spivak-invitation_F01699	163	■ 0.987	$(f+g): m+n \rightarrow m^{\prime}+n^{\prime}$	$(f+g): m+n \rightarrow m^{\prime}+n^{\prime}$	$(f+g): m+n \rightarrow m^{\prime}+n^{\prime}$.
fong-spivak-invitation_F01700	163	■ 1.000	$\underline{m}=\{1, \ldots, m\}$	$\underline{m}=\{1, \ldots, m\}$	from $\underline{m}=\{1, \ldots, m\}$ to $\underline{n}=\{1, \ldots, n\}$. (The identities, symmetries, and composition
fong-spivak-invitation_F01701	163	■ 1.000	$\underline{n}=\{1, \ldots, n\}$	$\underline{n}=\{1, \ldots, n\}$	from $\underline{m}=\{1, \ldots, m\}$ to $\underline{n}=\{1, \ldots, n\}$. (The identities, symmetries, and composition
fong-spivak-invitation_F01702	163	■ 1.000	$f+g: m+n \rightarrow m^{\prime}+n^{\prime}$	$f+g: m+n \rightarrow m^{\prime}+n^{\prime}$	functions: that is, given $f: m \rightarrow m^{\prime}$ and $g: n \rightarrow n^{\prime}$, we define $f+g: m+n \rightarrow m^{\prime}+n^{\prime}$
fong-spivak-invitation_F01703	164	■ 1.000	$f: 3 \rightarrow 2$	$f: 3 \rightarrow 2$	1. Draw a morphism $f: 3 \rightarrow 2$ and a morphism $g: 2 \rightarrow 4$ in FinSet .
fong-spivak-invitation_F01704	164	■ 1.000	$g: 2 \rightarrow 4$	$g: 2 \rightarrow 4$	1. Draw a morphism $f: 3 \rightarrow 2$ and a morphism $g: 2 \rightarrow 4$ in FinSet .
fong-spivak-invitation_F01705	164	■ 1.000	$f+g$	$f+g$	2. Draw $f+g$.
fong-spivak-invitation_F01706	164	■ 0.997	$\sigma_{m,n}$	$\sigma_{m,n}$	5. Choose $m, n \in \mathbb{N}$, and draw the symmetry map $\sigma_{m,n}$ in FinSet ?
fong-spivak-invitation_F01707	164	■ 0.995	$\underline{m} \rightarrow \underline{n}$	$\underline{m} \rightarrow \underline{n}$	$\underline{m} \rightarrow \underline{n}$. Note that in this case morphisms $m \rightarrow n$ only exist when $m = n$; when $m \neq n$
fong-spivak-invitation_F01708	164	■ 0.995	$m \neq n$	$m \neq n$	$\underline{m} \rightarrow \underline{n}$. Note that in this case morphisms $m \rightarrow n$ only exist when $m = n$; when $m \neq n$
fong-spivak-invitation_F01709	164	■ 0.762	$\mathbf{Bij}(m, n)$	$\mathbf{Bij}(m, n)$	the homset $\mathbf{Bij}(m, n)$ is empty. Since Bij is a subcategory of FinSet , we can define the
fong-spivak-invitation_F01710	164	■ 0.762	$\mathbf{Bij}$	Bij	the homset $\mathbf{Bij}(m, n)$ is empty. Since Bij is a subcategory of FinSet , we can define the
fong-spivak-invitation_F01711	164	■ 1.000	$\underline{m} \sqcup \underline{n}$	$\underline{m} \sqcup \underline{n}$	are partitions on $\underline{m} \sqcup \underline{n}$ (see Example 4.46), is a prop.
fong-spivak-invitation_F01712	164	■ 0.994	$R \subseteq \underline{m} \times \underline{n}$	$R \subseteq \underline{m} \times \underline{n}$	Example 5.6. There is a prop Rel for which morphisms $m \rightarrow n$ are relations, $R \subseteq \underline{m} \times \underline{n}$.
fong-spivak-invitation_F01713	164	■ 1.000	$S \subseteq \underline{n} \times \underline{p}$	$S \subseteq \underline{n} \times \underline{p}$	The composition of R with $S \subseteq \underline{n} \times \underline{p}$ is
fong-spivak-invitation_F01714	164	■ 0.859	$(\mathbb{N}, \preceq)$	$(\mathbb{N}, \preceq)$	symmetric monoidal preorder of the form $(\mathbb{N}, \preceq)$, for some poset relation $\preceq$ on $\mathbb{N}$, where
fong-spivak-invitation_F01715	164	■ 0.859	$\preceq$	$\preceq$	symmetric monoidal preorder of the form $(\mathbb{N}, \preceq)$, for some poset relation $\preceq$ on $\mathbb{N}$, where
fong-spivak-invitation_F01716	164	■ 0.999	$R_1 + R_2$	$R_1 + R_2$	² The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ and $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ is given by
fong-spivak-invitation_F01717	164	■ 0.999	$R_1 \subseteq \underline{m}_1 \times \underline{n}_1$	$R_1 \subseteq \underline{m}_1 \times \underline{n}_1$	² The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ and $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ is given by
fong-spivak-invitation_F01718	164	■ 0.999	$R_2 \subseteq \underline{m}_2 \times \underline{n}_2$	$R_2 \subseteq \underline{m}_2 \times \underline{n}_2$	² The monoidal product $R_1 + R_2$ of relations $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ and $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ is given by

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fong-spivak-invitation_F01719	164	■ 0.226	$R_1 \sqcup R_2 \subseteq \left(\underline{m_1} \times \underline{n_1} \right) \sqcup \left(\underline{m_2} \times \underline{n_2} \right) \subseteq \left(\underline{m_1 \sqcup m_2} \right) \times \left(\underline{n_1 \sqcup n_2} \right)$	$R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1 \sqcup m_2}) \times (\underline{n_1 \sqcup n_2})$	$R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1 \sqcup m_2}) \times (\underline{n_1 \sqcup n_2})$
fong-spivak-invitation_F01720	165	■ 0.727	$F(n)=n$	$F(n) = n$	(a) F is identity-on-objects, i.e. $F(n) = n$ for all $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$, and
fong-spivak-invitation_F01721	165	■ 0.727	$n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$	$n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$	(a) F is identity-on-objects, i.e. $F(n) = n$ for all $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$, and
fong-spivak-invitation_F01722	165	■ 1.000	$f: m_1 \rightarrow m_2$	$f: m_1 \rightarrow m_2$	(b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$, we have $F(f) + F(g) = F(f + g)$
fong-spivak-invitation_F01723	165	■ 1.000	$g: n_1 \rightarrow n_2$	$g: n_1 \rightarrow n_2$	(b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$, we have $F(f) + F(g) = F(f + g)$
fong-spivak-invitation_F01724	165	■ 1.000	$F(f)+F(g)=F(f+g)$	$F(f) + F(g) = F(f + g)$	(b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$, we have $F(f) + F(g) = F(f + g)$
fong-spivak-invitation_F01725	165	■ 0.943	$i: \mathbf{Bij} \rightarrow \mathbf{FinSet}$	$i: \mathbf{Bij} \rightarrow \mathbf{FinSet}$	Example 5.10. The inclusion $i: \mathbf{Bij} \rightarrow \mathbf{FinSet}$ is a prop functor. Perhaps more interest-
fong-spivak-invitation_F01726	165	■ 0.262	$f: \underline{m} \rightarrow \underline{n}$	$f: \underline{m} \rightarrow \underline{n}$	ingly, there is a prop functor $F: \mathbf{FinSet} \rightarrow \mathbf{Rel}_{\mathbf{Fin}}$. It sends a function $f: \underline{m} \rightarrow \underline{n}$ to the
fong-spivak-invitation_F01727	165	■ 1.000	$F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$	$F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$	relation $F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$
fong-spivak-invitation_F01728	165	■ 0.991	ι	ι	Definition 5.11. For $m, n \in \mathbb{N}$, an (m, n) -port graph (V, in, out, ι) is specified by
fong-spivak-invitation_F01729	165	■ 0.896	$V \rightarrow \mathbb{N}$	$V \rightarrow \mathbb{N}$	(ii) functions $in, out: V \rightarrow \mathbb{N}$, where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>
fong-spivak-invitation_F01730	165	■ 0.896	$\text{in}(v)$	$in(v)$	(ii) functions $in, out: V \rightarrow \mathbb{N}$, where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>
fong-spivak-invitation_F01731	165	■ 0.896	$\text{out}(v)$	$out(v)$	(ii) functions $in, out: V \rightarrow \mathbb{N}$, where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>
fong-spivak-invitation_F01732	165	■ 0.959	$\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$	$\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$	(iii) a bijection $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$, where $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$ is the
fong-spivak-invitation_F01733	165	■ 0.959	$I = \{(v, i) \mid v \in V, 1 \leq i \leq \text{in}(v)\}$	$I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$	(iii) a bijection $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$, where $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$ is the
fong-spivak-invitation_F01734	165	■ 1.000	$O = \{(v, i) \mid v \in V, 1 \leq i \leq \text{out}(v)\}$	$O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$	set of <i>vertex inputs</i> , and $O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$ is the set of <i>vertex</i>
fong-spivak-invitation_F01735	165	■ 1.000	$e_{v,j}^{u,i}: u \rightarrow v$	$e_{v,j}^{u,i}: u \rightarrow v$	struct the graph with vertices V and with an arrow $e_{v,j}^{u,i}: u \rightarrow v$ for every $i, j \in \mathbb{N}$ such
fong-spivak-invitation_F01736	165	■ 1.000	$i, j \in \mathbb{N}$	$i, j \in \mathbb{N}$	struct the graph with vertices V and with an arrow $e_{v,j}^{u,i}: u \rightarrow v$ for every $i, j \in \mathbb{N}$ such
fong-spivak-invitation_F01737	165	■ 1.000	$\iota(u, i) = (v, j)$	$\iota(u, i) = (v, j)$	that $\iota(u, i) = (v, j)$; call it the <i>internal flow graph</i> . If the internal flow graph is acyclic
fong-spivak-invitation_F01738	165	■ 0.997	$m=2$	$m = 2$	Example 5.12. Here is an example of a $(2, 3)$ -port graph, i.e. with $m = 2$ and $n = 3$:
fong-spivak-invitation_F01739	165	■ 0.997	$n=3$	$n = 3$	Example 5.12. Here is an example of a $(2, 3)$ -port graph, i.e. with $m = 2$ and $n = 3$:
fong-spivak-invitation_F01740	166	■ 1.000	$V = \{a, b, c\}$	$V = \{a, b, c\}$	outer box, and three on the right. The vertex set is $V = \{a, b, c\}$ and, for example
fong-spivak-invitation_F01741	166	■ 1.000	$\text{in}(a)=1$	$in(a) = 1$	$in(a) = 1$ and $out(a) = 3$, so we draw one port on the left-hand side and three ports on
fong-spivak-invitation_F01742	166	■ 1.000	$\text{out}(a)=3$	$out(a) = 3$	$in(a) = 1$ and $out(a) = 3$, so we draw one port on the left-hand side and three ports on
fong-spivak-invitation_F01743	166	■ 0.359	(m, n)	(m, n)	Given an (m, n) -port graph (V, in, out, ι) and an (n, p) -port graph (V', in', out', ι') , we

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_F01771	168	■ 1.000	$R(x, y)$	$R(x, y)$	1. Suppose that for every $x, y \in P$, if $R(x, y)$ then $f(x) \leq f(y)$. Show that f defines
fong-spivak-invitation_F01772	168	■ 1.000	$f(x) \leq_Q f(y)$	$f(x) \leq_Q f(y)$	every $x, y \in P$, if $R(x, y)$ then $f(x) \leq_Q f(y)$.
fong-spivak-invitation_F01773	168	■ 1.000	P, Q, R	P, Q, R	Exercise 5.17. Let P, Q, R , etc. be as in Exercise 5.16. We want to see that the universal
fong-spivak-invitation_F01774	168	■ 1.000	$a, b \in Q$	$a, b \in Q$	1. Suppose that for every $a, b \in Q$, if $a \leq b$ then $(g(a), g(b)) \in R$. Is it automatically
fong-spivak-invitation_F01775	168	■ 1.000	$(g(a), g(b)) \in R$	$(g(a), g(b)) \in R$	1. Suppose that for every $a, b \in Q$, if $a \leq b$ then $(g(a), g(b)) \in R$. Is it automatically
fong-spivak-invitation_F01776	168	■ 1.000	$g: (Q, \leq_Q) \rightarrow (P, \leq_P)$	$g: (Q, \leq_Q) \rightarrow (P, \leq_P)$	true that g defines a monotone map $g: (Q, \leq_Q) \rightarrow (P, \leq_P)$?
fong-spivak-invitation_F01777	168	■ 1.000	$\operatorname{Mor}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{C})$	Let $\mathcal{C}$ be another category whose set of morphisms is denoted $\operatorname{Mor}(\mathcal{C})$.
fong-spivak-invitation_F01778	168	■ 1.000	$\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$; interpret this statement.
fong-spivak-invitation_F01779	168	■ 1.000	$\mathcal{G} \rightarrow \mathcal{C}$	$\mathcal{G} \rightarrow \mathcal{C}$	2. Show that the set of functors $\mathcal{G} \rightarrow \mathcal{C}$ are in one-to-one correspondence with the set
fong-spivak-invitation_F01780	168	■ 1.000	$f: V \rightarrow \operatorname{Ob}(\mathcal{C})$	$f: V \rightarrow \operatorname{Ob}(\mathcal{C})$	of pairs of functions (f, g) , where $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$ and $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$, for which
fong-spivak-invitation_F01781	168	■ 1.000	$g: A \rightarrow \operatorname{Mor}(\mathcal{C})$	$g: A \rightarrow \operatorname{Mor}(\mathcal{C})$	of pairs of functions (f, g) , where $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$ and $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$, for which
fong-spivak-invitation_F01782	168	■ 1.000	$\operatorname{dom}(g(a)) = f(s(a))$	$\operatorname{dom}(g(a)) = f(s(a))$	$\operatorname{dom}(g(a)) = f(s(a))$ and $\operatorname{cod}(g(a)) = f(t(a))$ for all a .
fong-spivak-invitation_F01783	168	■ 1.000	$\operatorname{cod}(g(a)) = f(t(a))$	$\operatorname{cod}(g(a)) = f(t(a))$	$\operatorname{dom}(g(a)) = f(s(a))$ and $\operatorname{cod}(g(a)) = f(t(a))$ for all a .
fong-spivak-invitation_F01784	169	■ 1.000	$\operatorname{Loop}(A)$	$\operatorname{Loop}(A)$	one-object categories. For any set A , there is a graph $\operatorname{Loop}(A)$ with one vertex and with
fong-spivak-invitation_F01785	169	■ 1.000	$A = \{a\}$	$A = \{a\}$	1. What are the elements of the free monoid on the set $A = \{a\}$?
fong-spivak-invitation_F01786	169	■ 1.000	$\{a\}$	$\{a\}$	2. Can you find a well-known monoid that is isomorphic to the free monoid on $\{a\}$?
fong-spivak-invitation_F01787	169	■ 0.993	G, s, t	G, s, t	Definition 5.21. A <i>prop signature</i> is a tuple (G, s, t) , where G is a set and $s, t: G \rightarrow \mathbb{N}$
fong-spivak-invitation_F01788	169	■ 0.993	$s, t: G \rightarrow \mathbb{N}$	$s, t: G \rightarrow \mathbb{N}$	Definition 5.21. A <i>prop signature</i> is a tuple (G, s, t) , where G is a set and $s, t: G \rightarrow \mathbb{N}$
fong-spivak-invitation_F01789	169	■ 1.000	$g \in G$	$g \in G$	are functions; each element $g \in G$ is called a <i>generator</i> and $s(g), t(g) \in \mathbb{N}$ are called
fong-spivak-invitation_F01790	169	■ 1.000	$s(g), t(g) \in \mathbb{N}$	$s(g), t(g) \in \mathbb{N}$	are functions; each element $g \in G$ is called a <i>generator</i> and $s(g), t(g) \in \mathbb{N}$ are called
fong-spivak-invitation_F01791	169	■ 0.844	s, t	s, t	its <i>in-arity and out-arity</i> . We often denote (G, s, t) simply by G , taking s, t to be
fong-spivak-invitation_F01792	169	■ 0.990	$\Gamma = (V$	$\Gamma = (V$	A <i>G-labeling</i> of a port graph $\Gamma = (V, in, out, \iota)$ is a function $\ell: V \rightarrow G$ such that
fong-spivak-invitation_F01793	169	■ 0.990	$\ell: V \rightarrow G$	$\ell: V \rightarrow G$	A <i>G-labeling</i> of a port graph $\Gamma = (V, in, out, \iota)$ is a function $\ell: V \rightarrow G$ such that
fong-spivak-invitation_F01794	169	■ 0.982	$s(\ell(v)) = \operatorname{in}(v)$	$s(\ell(v)) = \operatorname{in}(v)$	the arities agree: $s(\ell(v)) = \operatorname{in}(v)$ and $t(\ell(v)) = \operatorname{out}(v)$ for each $v \in V$.
fong-spivak-invitation_F01795	169	■ 0.982	$t(\ell(v)) = \operatorname{out}(v)$	$t(\ell(v)) = \operatorname{out}(v)$	the arities agree: $s(\ell(v)) = \operatorname{in}(v)$ and $t(\ell(v)) = \operatorname{out}(v)$ for each $v \in V$.
fong-spivak-invitation_F01796	169	■ 1.000	ℓ	ℓ	port graphs PG (see Eq. (5.4)); the labelings (the ℓ 's) are just carried along.
fong-spivak-invitation_F01797	169	■ 1.000	$(m, n) \in \mathbb{N} \times \mathbb{N}$	$(m, n) \in \mathbb{N} \times \mathbb{N}$	³ The arity of a prop morphism is a pair $(m, n) \in \mathbb{N} \times \mathbb{N}$, where m is the number of inputs and n is the

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fong-spivak-invitation_F01798	170	■ 1.000	$G=\{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$	$G = \{f : 1 \rightarrow 1, g : 2 \rightarrow 2, h : 2 \rightarrow 1\}$	with the same generator. For example, if $G = \{f : 1 \rightarrow 1, g : 2 \rightarrow 2, h : 2 \rightarrow 1\}$, then
fong-spivak-invitation_F01799	170	■ 1.000	$3 \rightarrow 2$	$3 \rightarrow 2$	the following is a morphism $3 \rightarrow 2$ in $\mathbf{Free}(G)$:
fong-spivak-invitation_F01800	170	■ 1.000	$V \rightarrow \varnothing$	$V \rightarrow \varnothing$	must have a labeling function of the form $V \rightarrow \varnothing$, and hence we must have $V =$
fong-spivak-invitation_F01801	170	■ 1.000	(n, m)	(n, m)	$\varnothing$; see Exercise 1.20. Thus the only morphisms (n, m) are those given by port graphs
fong-spivak-invitation_F01802	170	■ 0.992	$(\varnothing, !, !, \sigma)$	$(\varnothing, !, !, \sigma)$	$(\varnothing, !, !, \sigma)$, where $\sigma : n \rightarrow m$ is a bijection.
fong-spivak-invitation_F01803	170	■ 0.992	$\sigma: n \rightarrow m$	$\sigma : n \rightarrow m$	$(\varnothing, !, !, \sigma)$, where $\sigma: n \rightarrow m$ is a bijection.
fong-spivak-invitation_F01804	170	■ 0.959	$(m, n) \in \mathbb{N}^2$	$(m, n) \in \mathbb{N}^2$	i.e. having one generating morphism for each $(m, n) \in \mathbb{N}^2$. Show that $\mathbf{Free}(G)$ is the
fong-spivak-invitation_F01805	170	■ 0.860	(G, s, t)	(G, s, t)	Proposition 5.24. The free prop $\mathbf{Free}(G)$ on a signature (G, s, t) has the property that,
fong-spivak-invitation_F01806	170	■ 0.974	$\mathbf{Free}(G) \rightarrow \mathcal{C}$	$\mathbf{Free}(G) \rightarrow \mathcal{C}$	for any prop $\mathcal{C}$, the prop functors $\mathbf{Free}(G) \rightarrow \mathcal{C}$ are in one-to-one correspondence with
fong-spivak-invitation_F01807	170	■ 1.000	$G \rightarrow \mathcal{C}$	$G \rightarrow \mathcal{C}$	functions $G \rightarrow \mathcal{C}$ that send each $g \in G$ to a morphism $s(g) \rightarrow t(g)$ in $\mathcal{C}$.
fong-spivak-invitation_F01808	170	■ 1.000	$s(g) \rightarrow t(g)$	$s(g) \rightarrow t(g)$	functions $G \rightarrow \mathcal{C}$ that send each $g \in G$ to a morphism $s(g) \rightarrow t(g)$ in $\mathcal{C}$.
fong-spivak-invitation_F01810	171	■ 1.000	$e: m \rightarrow n$	$e : m \rightarrow n$	a G -generated prop expression, or simply expression $e: m \rightarrow n$, where $m, n \in \mathbb{N}$,
fong-spivak-invitation_F01811	171	■ 1.000	$\mathrm{id}_0: 0 \rightarrow 0$	$\mathrm{id}_0 : 0 \rightarrow 0$	• The empty morphism $\mathrm{id}_0: 0 \rightarrow 0$, the identity morphism $\mathrm{id}_1: 1 \rightarrow 1$, and the
fong-spivak-invitation_F01812	171	■ 1.000	$\mathrm{id}_1: 1 \rightarrow 1$	$\mathrm{id}_1 : 1 \rightarrow 1$	• The empty morphism $\mathrm{id}_0: 0 \rightarrow 0$, the identity morphism $\mathrm{id}_1: 1 \rightarrow 1$, and the
fong-spivak-invitation_F01813	171	■ 1.000	$\sigma: 2 \rightarrow 2$	$\sigma : 2 \rightarrow 2$	symmetry $\sigma : 2 \rightarrow 2$ are expressions. ⁴
fong-spivak-invitation_F01814	171	■ 1.000	$g: s(g) \rightarrow t(g)$	$g : s(g) \rightarrow t(g)$	• The generators $g \in G$ are expressions $g: s(g) \rightarrow t(g)$.
fong-spivak-invitation_F01815	171	■ 1.000	$\alpha: m \rightarrow n$	$\alpha : m \rightarrow n$	• If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an
fong-spivak-invitation_F01816	171	■ 1.000	$\beta: p \rightarrow q$	$\beta : p \rightarrow q$	• If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an
fong-spivak-invitation_F01817	171	■ 1.000	$\alpha + \beta: m + p \rightarrow n + q$	$\alpha + \beta : m + p \rightarrow n + q$	• If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an
fong-spivak-invitation_F01818	171	■ 0.988	$\beta: n \rightarrow p$	$\beta : n \rightarrow p$	• If $\alpha: m \rightarrow n$ and $\beta: n \rightarrow p$ are expressions, then $\alpha \circ \beta: m \rightarrow p$ is an expression.
fong-spivak-invitation_F01819	171	■ 0.988	$\alpha ; \beta: m \rightarrow p$	$\alpha ; \beta : m \rightarrow p$	• If $\alpha: m \rightarrow n$ and $\beta: n \rightarrow p$ are expressions, then $\alpha \circ \beta: m \rightarrow p$ is an expression.
fong-spivak-invitation_F01820	171	■ 1.000	$\mathbf{Expr}(G)$	$\mathbf{Expr}(G)$	We write $\mathbf{Expr}(G)$ for the set of expressions in G . If $e: m \rightarrow n$ is an expression, we
fong-spivak-invitation_F01821	171	■ 1.000	$f: 1 \rightarrow 1$	$f : 1 \rightarrow 1$	• $f: 1 \rightarrow 1$,
fong-spivak-invitation_F01822	171	■ 0.988	$f: \mathrm{id}_1: 1 \rightarrow 1$	$f : \mathrm{id}_1 : 1 \rightarrow 1$	• $f \circ \mathrm{id}_1: 1 \rightarrow 1$,
fong-spivak-invitation_F01823	171	■ 0.999	$h + \mathrm{id}_1: 3 \rightarrow 2$	$h + \mathrm{id}_1 : 3 \rightarrow 2$	• $h + \mathrm{id}_1: 3 \rightarrow 2$, and
fong-spivak-invitation_F01824	171	■ 0.866	$\left(h + \mathrm{id}_1\right) ; \sigma ; g ; \sigma: 3 \rightarrow 2$	$(h + \mathrm{id}_1) ; \sigma ; g ; \sigma : 3 \rightarrow 2$	• $(h + \mathrm{id}_1) \circ \sigma \circ g \circ \sigma: 3 \rightarrow 2$
fong-spivak-invitation_F01825	171	■ 0.678	$f \circ \mathrm{id}_1$	$f \circ \mathrm{id}_1$	phism. For example, we want to consider $f \circ \mathrm{id}_1$ and f to be the same morphism, since

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fong-spivak-invitation_F01826	171	■ 0.953	$f ; \mathrm{id}_1 = f$	$f; \mathrm{id}_1 = f$	the unitality axiom for categories says $f \circ \mathrm{id}_1 = f$. Nonetheless, we only consider
fong-spivak-invitation_F01827	171	■ 0.997	σ	σ	⁴ One can think of σ as the “swap” icon $\bowtie: 2 \rightarrow 2$.
fong-spivak-invitation_F01828	171	■ 0.997	$\mathrm{X}: 2 \rightarrow 2$	$X: 2 \rightarrow 2$	⁴ One can think of σ as the “swap” icon $\bowtie: 2 \rightarrow 2$.
fong-spivak-invitation_F01829	172	■ 0.616	$\left(g + \mathrm{id}_1\right) \stackrel{\circ}{\circ} \left(\mathrm{id}_1 + \sigma\right) \stackrel{\circ}{\circ} \left(\mathrm{id}_1 + g\right) \stackrel{\circ}{\circ} \left(h + \mathrm{id}_1\right)$	$(g + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + \sigma) \overset{\circ}{\circ} (\mathrm{id}_1 + g) \overset{\circ}{\circ} (h + \mathrm{id}_1)$	Here the result is $(g + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + \sigma) \overset{\circ}{\circ} (\mathrm{id}_1 + g) \overset{\circ}{\circ} (h + \mathrm{id}_1)$.
fong-spivak-invitation_F01830	172	■ 1.000	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$	Exercise 5.27. Consider again the free prop on generators $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$
fong-spivak-invitation_F01831	172	■ 0.605	$h: 2 \rightarrow 1\}$	$h: 2 \rightarrow 1\}$	$2, h: 2 \rightarrow 1\}$. Draw a picture of $(f + \mathrm{id}_1 + \mathrm{id}_1) \overset{\circ}{\circ} (\sigma + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + h) \overset{\circ}{\circ} \sigma \overset{\circ}{\circ} g$, where
fong-spivak-invitation_F01832	172	■ 0.605	$\left(f + \mathrm{id}_1 + \mathrm{id}_1\right) \stackrel{\circ}{\circ} \left(\sigma + \mathrm{id}_1\right) \stackrel{\circ}{\circ} \left(\mathrm{id}_1 + h\right) \stackrel{\circ}{\circ} \sigma \stackrel{\circ}{\circ} g$	$(f + \mathrm{id}_1 + \mathrm{id}_1) \overset{\circ}{\circ} (\sigma + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + h) \overset{\circ}{\circ} \sigma \overset{\circ}{\circ} g$	$2, h: 2 \rightarrow 1\}$. Draw a picture of $(f + \mathrm{id}_1 + \mathrm{id}_1) \overset{\circ}{\circ} (\sigma + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + h) \overset{\circ}{\circ} \sigma \overset{\circ}{\circ} g$, where
fong-spivak-invitation_F01833	172	■ 0.666	(G, s, t, E)	(G, s, t, E)	Rough Definition 5.28. A <i>presentation</i> (G, s, t, E) for a prop is a set G , functions
fong-spivak-invitation_F01834	172	■ 1.000	$E \subseteq \mathrm{Expr}(G) \times \mathrm{Expr}(G)$	$E \subseteq \mathrm{Expr}(G) \times \mathrm{Expr}(G)$	$s, t: G \rightarrow \mathbb{N}$, and a set $E \subseteq \mathrm{Expr}(G) \times \mathrm{Expr}(G)$ of pairs of G -generated prop expres-
fong-spivak-invitation_F01835	172	■ 1.000	e_1	e_1	sions, such that e_1 and e_2 have the same arity for each $(e_1, e_2) \in E$. We refer to G as
fong-spivak-invitation_F01836	172	■ 1.000	e_2	e_2	sions, such that e_1 and e_2 have the same arity for each $(e_1, e_2) \in E$. We refer to G as
fong-spivak-invitation_F01837	172	■ 1.000	$\left(e_1, e_2\right) \in E$	$(e_1, e_2) \in E$	sions, such that e_1 and e_2 have the same arity for each $(e_1, e_2) \in E$. We refer to G as
fong-spivak-invitation_F01838	172	■ 0.998	$e_1 = e_2$	$e_1 = e_2$	are elements in $\mathrm{Expr}(G)$, quotiented by both the equations $e_1 = e_2$ where $(e_1, e_2) \in E$,
fong-spivak-invitation_F01839	173	■ 1.000	$g \in G, f(g)$	$g \in G, f(g)$	• for all $g \in G$, $f(g)$ is a morphism $s(g) \rightarrow t(g)$, and
fong-spivak-invitation_F01840	173	■ 1.000	$f\left(e_1\right)=f\left(e_2\right)$	$f\left(e_1\right)=f\left(e_2\right)$	• for all $(e_1, e_2) \in E$, we have that $f\left(e_1\right)=f\left(e_2\right)$ in $\mathcal{C}$, where $f(e)$ denotes the
fong-spivak-invitation_F01841	173	■ 1.000	$f(e)$	$f(e)$	• for all $(e_1, e_2) \in E$, we have that $f\left(e_1\right)=f\left(e_2\right)$ in $\mathcal{C}$, where $f(e)$ denotes the
fong-spivak-invitation_F01842	173	■ 0.658	$(G, s, t, \varnothing)$	$(G, s, t, \emptyset)$	nition 5.21, is the same thing as the prop presented by $(G, s, t, \emptyset)$, having no relations,
fong-spivak-invitation_F01843	173	■ 0.976	$R, 0,+, 1, *$	$R, 0, +, 1, *$	Definition 5.31. A <i>rig</i> is a tuple $(R, 0, +, 1, *)$, where R is a set, $0, 1 \in R$ are elements,
fong-spivak-invitation_F01844	173	■ 0.976	$0,1 \in R$	$0, 1 \in R$	Definition 5.31. A <i>rig</i> is a tuple $(R, 0, +, 1, *)$, where R is a set, $0, 1 \in R$ are elements,
fong-spivak-invitation_F01845	173	■ 0.986	$+, *: R \times R \rightarrow R$	$+, *: R \times R \rightarrow R$	and $+, *: R \times R \rightarrow R$ are functions, such that
fong-spivak-invitation_F01846	173	■ 0.989	$(R,+, 0)$	$(R, +, 0)$	(a) $(R, +, 0)$ is a commutative monoid,
fong-spivak-invitation_F01847	173	■ 0.615	$R, *, 1$	$R, *, 1$	(b) $(R, *, 1)$ is a monoid, ⁷
fong-spivak-invitation_F01848	173	■ 1.000	$a *(b+c)=a * b+a * c$	$a * (b + c) = a * b + a * c$	(c) $a * (b + c) = a * b + a * c$ and $(a + b) * c = a * c + b * c$ for all $a, b, c \in R$,
fong-spivak-invitation_F01849	173	■ 1.000	$(a+b) * c=a * c+b * c$	$(a + b) * c = a * c + b * c$	(c) $a * (b + c) = a * b + a * c$ and $(a + b) * c = a * c + b * c$ for all $a, b, c \in R$,
fong-spivak-invitation_F01850	173	■ 1.000	$a, b, c \in R$	$a, b, c \in R$	(c) $a * (b + c) = a * b + a * c$ and $(a + b) * c = a * c + b * c$ for all $a, b, c \in R$,

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fong-spivak-invitation_F01851	173	■ 1.000	$a * 0 = 0 = 0 * a$	$a * 0 = 0 = 0 * a$	(d) $a * 0 = 0 = 0 * a$ for all $a \in R$.
fong-spivak-invitation_F01852	173	■ 1.000	$a \in R$	$a \in R$	(d) $a * 0 = 0 = 0 * a$ for all $a \in R$.
fong-spivak-invitation_F01853	173	■ 0.733	$(\mathbb{N}, 0, +, 1, *)$	$(\mathbb{N}, 0, +, 1, *)$	Example 5.32. The natural numbers form a rig $(\mathbb{N}, 0, +, 1, *)$.
fong-spivak-invitation_F01854	173	■ 0.999	$(R, *, 1)$	$(R, *, 1)$	⁷ Note that we did not demand that $(R, *, 1)$ be commutative; we shall see a naturally arising example where
fong-spivak-invitation_F01855	174	■ 0.898	$(V, 0, \vee, I, \otimes)$	$(V, 0, \vee, I, \otimes)$	Example 5.34. Any quantale $\mathcal{V} = (V, \leq, I, \otimes)$ determines a rig $(V, 0, \vee, I, \otimes)$, where
fong-spivak-invitation_F01856	174	■ 1.000	$0 = \bigvee \varnothing$	$0 = \bigvee \varnothing$	$0 = \bigvee \varnothing$ is the empty join. See Definition 2.57.
fong-spivak-invitation_F01857	174	■ 1.000	$\operatorname{Mat}_n(R)$	$\operatorname{Mat}_n(R)$	Example 5.35. If R is a rig and $n \in \mathbb{N}$ is any natural number, then the set $\operatorname{Mat}_n(R)$ of
fong-spivak-invitation_F01858	174	■ 1.000	$(n \times n)$	$(n \times n)$	$(n \times n)$ -matrices in R forms a rig. A matrix $M \in \operatorname{Mat}_n(R)$ is a function $M : \underline{n} \times \underline{n} \rightarrow R$.
fong-spivak-invitation_F01859	174	■ 1.000	$M \in \operatorname{Mat}_n(R)$	$M \in \operatorname{Mat}_n(R)$	$(n \times n)$ -matrices in R forms a rig. A matrix $M \in \operatorname{Mat}_n(R)$ is a function $M : \underline{n} \times \underline{n} \rightarrow R$.
fong-spivak-invitation_F01860	174	■ 1.000	$M : \underline{n} \times \underline{n} \rightarrow R$	$M : \underline{n} \times \underline{n} \rightarrow R$	$(n \times n)$ -matrices in R forms a rig. A matrix $M \in \operatorname{Mat}_n(R)$ is a function $M : \underline{n} \times \underline{n} \rightarrow R$.
fong-spivak-invitation_F01861	174	■ 1.000	$M + N$	$M + N$	Addition $M + N$ of matrices is given by $(M + N)(i, j) := M(i, j) + N(i, j)$ and
fong-spivak-invitation_F01862	174	■ 1.000	$(M + N)(i, j) := M(i, j) + N(i, j)$	$(M + N)(i, j) := M(i, j) + N(i, j)$	Addition $M + N$ of matrices is given by $(M + N)(i, j) := M(i, j) + N(i, j)$ and
fong-spivak-invitation_F01863	174	■ 0.999	$(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$	$(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$	multiplication $M * N$ is given by $(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$. The 0-matrix
fong-spivak-invitation_F01864	174	■ 1.000	$0(i, j) := 0$	$0(i, j) := 0$	is $0(i, j) := 0$ for all $i, j \in \underline{n}$. Note that $\operatorname{Mat}_n(R)$ is generally not commutative.
fong-spivak-invitation_F01865	174	■ 1.000	$i, j \in \underline{n}$	$i, j \in \underline{n}$	is $0(i, j) := 0$ for all $i, j \in \underline{n}$. Note that $\operatorname{Mat}_n(R)$ is generally not commutative.
fong-spivak-invitation_F01866	174	■ 1.000	$1 \in \operatorname{Mat}_n(R)$	$1 \in \operatorname{Mat}_n(R)$	multiplicative identity $1 \in \operatorname{Mat}_n(R)$?
fong-spivak-invitation_F01867	174	■ 1.000	$\operatorname{Mat}_n(\mathbb{N})$	$\operatorname{Mat}_n(\mathbb{N})$	$\operatorname{Mat}_n(\mathbb{N})$ is not commutative, where $\mathbb{N}$ is as in Example 5.32.
fong-spivak-invitation_F01868	174	■ 0.998	$(\mathbb{R}, 0, +, 1, *)$	$\mathbb{R}, 0, +, 1, *$	Example 5.37. Any ring forms a rig. In particular, the real numbers $(\mathbb{R}, 0, +, 1, *)$ are a
fong-spivak-invitation_F01869	174	■ 1.000	$M = R$	$M = R$	R -module M . We will not discuss this here, keeping to the simpler requirement that $M = R$.
fong-spivak-invitation_F01870	175	■ 1.000	$R = \mathbb{N}$	$R = \mathbb{N}$	For example, consider the rig $R = \mathbb{N}$ from Example 5.32, where the scalars are the
fong-spivak-invitation_F01871	175	■ 1.000	$a = 15x$	$a = 15x$	$a = 15x$ and $b = 3x + 21y$.
fong-spivak-invitation_F01872	175	■ 1.000	$b = 3x + 21y$	$b = 3x + 21y$	$a = 15x$ and $b = 3x + 21y$.
fong-spivak-invitation_F01873	176	■ 1.000	$15x$	$15x$	amplified by 3. Thus, if x is the first input signal, then this contributes $15x$ to the first
fong-spivak-invitation_F01874	176	■ 1.000	$3x = x + 2x$	$3x = x + 2x$	two paths to it from each input. The top input thus contributes $3x = x + 2x$, whereas
fong-spivak-invitation_F01875	176	■ 0.954	$*7$	$*7$	the bottom input, passing through an additional $*7$ amplification, contributes $21y$.
fong-spivak-invitation_F01876	176	■ 0.954	$21y$	$21y$	the bottom input, passing through an additional $*7$ amplification, contributes $21y$.
fong-spivak-invitation_F01877	176	■ 0.999	(x, y, z)	(x, y, z)	the movement of a particle. For example, consider a particle whose position (x, y, z) in

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fong-spivak-invitation_F01878	177	■ 1.000	$s, t: G_{\{R\}} \rightarrow \mathbb{N}$	$s, t: G_R \rightarrow \mathbb{N}$	and let $s, t: G_R \rightarrow \mathbb{N}$ be given by the number of dangling wires on the left and right
fong-spivak-invitation_F01879	177	■ 0.403	$\boldsymbol{\operatorname{Free}}(G_{\{R\}})$	$\mathbf{Free}(G_R)$	free prop $\mathbf{Free}(G_R)$ on this set G_R of generators. We define $\mathbf{SFG}_R := \mathbf{Free}(G_R)$.
fong-spivak-invitation_F01880	177	■ 0.403	$G_{\{R\}}$	G_R	free prop $\mathbf{Free}(G_R)$ on this set G_R of generators. We define $\mathbf{SFG}_R := \mathbf{Free}(G_R)$.
fong-spivak-invitation_F01881	177	■ 0.403	$\operatorname{SFG}_{\{R\}} := \boldsymbol{\operatorname{Free}}(G_{\{R\}})$	$\mathbf{SFG}_R := \mathbf{Free}(G_R)$	free prop $\mathbf{Free}(G_R)$ on this set G_R of generators. We define $\mathbf{SFG}_R := \mathbf{Free}(G_R)$.
fong-spivak-invitation_F01882	177	■ 0.576	$\operatorname{Free}(G_{\{R\}})$	$\mathbf{Free}(G_R)$	port graphs, morphisms in $\mathbf{Free}(G_R)$ should be drawn slightly differently. For example,
fong-spivak-invitation_F01883	178	■ 0.996	$(m \times n)$	$(m \times n)$	An $(m \times n)$ -matrix M with values in R is a function $M: (\underline{m} \times \underline{n}) \rightarrow R$. Given an
fong-spivak-invitation_F01884	178	■ 0.996	$M: (\underline{m} \times \underline{n}) \rightarrow R$	$M: (\underline{m} \times \underline{n}) \rightarrow R$	An $(m \times n)$ -matrix M with values in R is a function $M: (\underline{m} \times \underline{n}) \rightarrow R$. Given an
fong-spivak-invitation_F01885	178	■ 0.982	$(n \times p)$	$(n \times p)$	$(m \times n)$ -matrix M and an $(n \times p)$ -matrix N , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$
fong-spivak-invitation_F01886	178	■ 0.982	$(m \times p)$	$(m \times p)$	$(m \times n)$ -matrix M and an $(n \times p)$ -matrix N , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$
fong-spivak-invitation_F01887	178	■ 1.000	$a \in \underline{m}$	$a \in \underline{m}$	defined as follows for any $a \in \underline{m}$ and $c \in \underline{p}$:
fong-spivak-invitation_F01888	178	■ 1.000	$c \in \underline{p}$	$c \in \underline{p}$	defined as follows for any $a \in \underline{m}$ and $c \in \underline{p}$:
fong-spivak-invitation_F01889	178	■ 0.748	$\sum_{b \in \underline{n}}$	$\sum_{b \in \underline{n}}$	Here the $\sum_{b \in \underline{n}}$ just means repeated addition (using the rig R 's $+$ operation), as usual.
fong-spivak-invitation_F01890	178	■ 1.000	$A v$	Av	v by multiplication in the order Av , where v is a column vector. In keeping with our
fong-spivak-invitation_F01891	178	■ 0.997	$v ; A$	$v; A$	composition convention, we use the opposite order, $v \circ A$, where v is a row vector. See
fong-spivak-invitation_F01892	178	■ 0.918	$\operatorname{Mat}(R)$	$\mathbf{Mat}(R)$	Definition 5.43. Let R be a rig. We define the <i>prop of R-matrices</i> , denoted $\mathbf{Mat}(R)$,
fong-spivak-invitation_F01893	178	■ 0.999	$A: m \rightarrow n$	$A: m \rightarrow n$	monoidal product is given by the direct sum of matrices: given matrices $A: m \rightarrow n$ and
fong-spivak-invitation_F01894	178	■ 1.000	$b: p \rightarrow q$	$b: p \rightarrow q$	$b: p \rightarrow q$, we define $A + B: m + p \rightarrow n + q$ to be the block matrix
fong-spivak-invitation_F01895	178	■ 1.000	$A+B: m+p \rightarrow n+q$	$A + B: m + p \rightarrow n + q$	$b: p \rightarrow q$, we define $A + B: m + p \rightarrow n + q$ to be the block matrix
fong-spivak-invitation_F01896	178	■ 0.979	$m \times q$	$m \times q$	where each 0 represents a matrix of zeros of the appropriate dimension $(m \times q$ and $n \times p)$.
fong-spivak-invitation_F01897	178	■ 0.979	$n \times p$	$n \times p$	where each 0 represents a matrix of zeros of the appropriate dimension $(m \times q$ and $n \times p)$.
fong-spivak-invitation_F01898	178	■ 1.000	$A+B$	$A + B$	What is the direct sum matrix $A + B$?
fong-spivak-invitation_F01900	179	■ 1.000	(1×1)	(1×1)	(1×1) -matrix $(a): 1 \rightarrow 1$: it is an operation that takes an input $x \in R$ and returns $a * x$.
fong-spivak-invitation_F01901	179	■ 1.000	$(a): 1 \rightarrow 1$	$(a): 1 \rightarrow 1$	(1×1) -matrix $(a): 1 \rightarrow 1$: it is an operation that takes an input $x \in R$ and returns $a * x$.
fong-spivak-invitation_F01902	179	■ 1.000	$x \in R$	$x \in R$	(1×1) -matrix $(a): 1 \rightarrow 1$: it is an operation that takes an input $x \in R$ and returns $a * x$.
fong-spivak-invitation_F01903	179	■ 1.000	$a * x$	$a * x$	(1×1) -matrix $(a): 1 \rightarrow 1$: it is an operation that takes an input $x \in R$ and returns $a * x$.
fong-spivak-invitation_F01904	179	■ 0.588	$\supset$	$\supset$	Similarly, we can interpret $\supset$ as the (2×1) -matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: it is an operation that takes a
fong-spivak-invitation_F01905	179	■ 0.588	(2×1)	(2×1)	Similarly, we can interpret $\supset$ as the (2×1) -matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: it is an operation that takes a

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fong-spivak-invitation_F01906	179	0.588	$\binom{1}{1}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	Similarly, we can interpret $\triangleright$ as the (2×1) -matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: it is an operation that takes a
fong-spivak-invitation_F01907	179	1.000	$x+y$	$x + y$	row vector consisting of two inputs, x and y , and returns $x + y$. Here is a table showing
fong-spivak-invitation_F01908	222	0.793	$1 \rightarrow 1$	$1 \rightarrow 1$	Since this and the circuit in (6.17) are both morphisms $\underline{1} \rightarrow \underline{1}$, we may compose them
fong-spivak-invitation_F01913	179	1.000	$0 \times n$	$0 \times n$	dimensions. In linear algebra it is unusual to consider matrices of the form $0 \times n$ or $n \times 0$
fong-spivak-invitation_F01914	179	1.000	$n \times 0$	$n \times 0$	dimensions. In linear algebra it is unusual to consider matrices of the form $0 \times n$ or $n \times 0$
fong-spivak-invitation_F01915	179	0.996	(0×3)	(0×3)	can multiply a (0×3) -matrix by a $(3 \times n)$ -matrix for any n , but you can not multiply it
fong-spivak-invitation_F01916	179	0.996	$(3 \times n)$	$(3 \times n)$	can multiply a (0×3) -matrix by a $(3 \times n)$ -matrix for any n , but you can not multiply it
fong-spivak-invitation_F01917	179	1.000	$(2 \times n)$	$(2 \times n)$	by a $(2 \times n)$ -matrix.
fong-spivak-invitation_F01918	180	0.999	$S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$	$S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$	Theorem 5.45. There is a prop functor $S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$ that sends the generators
fong-spivak-invitation_F01919	180	1.000	$S(g)$	$S(g)$	We have now constructed a matrix $S(g)$ from any signal flow graph g . But how can we
fong-spivak-invitation_F01920	180	1.000	$m \times n$	$m \times n$	for the generators in Definition 5.40, the associated matrix has dimension $m \times n$, where
fong-spivak-invitation_F01921	180	0.999	(i, j)	(i, j)	m is the number of inputs and n the number of outputs, with (i, j) th entry describing the
fong-spivak-invitation_F01922	180	1.000	$\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$	$\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$	built from the morphisms $\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$, and the generators in
fong-spivak-invitation_F01923	180	1.000	$(\alpha+\beta):(m+p) \rightarrow (n+q)$	$(\alpha + \beta): (m + p) \rightarrow (n + q)$	• If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $(\alpha + \beta): (m + p) \rightarrow (n + q)$ is
fong-spivak-invitation_F01924	180	1.000	$\mathrm{id}_0, \mathrm{id}_1$	$\mathrm{id}_0, \mathrm{id}_1$	true when g is equal to $\mathrm{id}_0, \mathrm{id}_1$, and σ .
fong-spivak-invitation_F01925	180	0.853	$0 \rightarrow 0$	$0 \rightarrow 0$	$(): 0 \rightarrow 0$. The morphisms id_1, σ , and $a \in R$ map to the identity matrix, the swap
fong-spivak-invitation_F01926	180	0.853	id_1, σ	id_1, σ	$(): 0 \rightarrow 0$. The morphisms id_1, σ , and $a \in R$ map to the identity matrix, the swap
fong-spivak-invitation_F01927	180	0.607	$n=p$	$n = p$	holds for (i) $\alpha \circ \beta$ (when $n = p$) and for (ii) $\alpha + \beta$ (in general).
fong-spivak-invitation_F01928	180	0.607	$\alpha+\beta$	$\alpha + \beta$	holds for (i) $\alpha \circ \beta$ (when $n = p$) and for (ii) $\alpha + \beta$ (in general).
fong-spivak-invitation_F01929	181	1.000	$\beta: n \rightarrow q$	$\beta: n \rightarrow q$	Here $\alpha: m \rightarrow n$ and $\beta: n \rightarrow q$ are signal flow graphs, assumed to obey the proposition.
fong-spivak-invitation_F01930	181	0.798	$j \in \underline{n}$	$j \in \underline{n}$	to k – of the amplification along that path. So let’s also fix some $j \in \underline{n}$, and consider
fong-spivak-invitation_F01931	181	0.627	$\alpha /$	$\alpha /$	some j th output of α /input of β , the amplification that i contributes to k is
fong-spivak-invitation_F01932	181	0.983	$S(\alpha) ; S(\beta)$	$S(\alpha); S(\beta)$	This is exactly the formula for matrix multiplication, which is composition $S(\alpha) \circ S(\beta)$
fong-spivak-invitation_F01933	181	0.959	$\mathbf{Mat}(R)$	$\mathbf{Mat}(R)$	in the prop $\mathbf{Mat}(R)$; see Definition 5.43. So $\alpha \circ \beta$ obeys the proposition when α and
fong-spivak-invitation_F01934	181	0.666	$(m+i)$	$(m + i)$	output of $\alpha + \beta$, the i th input of β is the $(m + i)$ th output of $\alpha + \beta$, and the j th out-
fong-spivak-invitation_F01935	181	0.556	$n+j$	$n + j$	put of β is the $(n + j)$ th output of $\alpha + \beta$. This means that the matrix with (i, j) th
fong-spivak-invitation_F01936	181	1.000	$S(\alpha)+S(\beta)=S(\alpha+\beta)$	$S(\alpha) + S(\beta) = S(\alpha + \beta)$	$S(\alpha) + S(\beta) = S(\alpha + \beta)$, as in Definition 5.43. This proves the proposition.

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fong-spivak-invitation_F01937	183	■ 1.000	$M \in \operatorname{Mat}(R)$	$M \in \operatorname{Mat}(R)$	Proposition 5.48. Given any matrix $M \in \operatorname{Mat}(R)$, there exists a signal flow graph
fong-spivak-invitation_F01938	183	■ 1.000	$g \in \mathbf{SFG}_R$	$g \in \mathbf{SFG}_R$	$g \in \mathbf{SFG}_R$ such that $S(g) = M$.
fong-spivak-invitation_F01939	183	■ 1.000	$S(g)=M$	$S(g) = M$	$g \in \mathbf{SFG}_R$ such that $S(g) = M$.
fong-spivak-invitation_F01940	183	■ 1.000	$S(g)(i, j)$	$S(g)(i, j)$	such that $S(g) = M$. In particular, to compute $S(g)(i, j)$, we know that we can sim-
fong-spivak-invitation_F01941	183	■ 1.000	$M(i, j)$	$M(i, j)$	$M(i, j)$.
fong-spivak-invitation_F01942	183	■ 0.488	$\left(\begin{array}{cc} a & b \\ c & d \end{array}\right)$	$\left(\begin{array}{cc} a & b \\ c & d \end{array}\right)$	$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then we define g to be the signal flow graph
fong-spivak-invitation_F01943	183	■ 1.000	$\left(\begin{array}{c} 0 \\ 1 \\ 2 \end{array}\right)$	$\left(\begin{array}{c} 0 \\ 1 \\ 2 \end{array}\right)$	1. $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$
fong-spivak-invitation_F01944	183	■ 1.000	$\left(\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right)$	$\left(\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right)$	2. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$
fong-spivak-invitation_F01945	183	■ 1.000	$\left(\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right)$	$\left(\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right)$	3. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-invitation_F01946	184	■ 1.000	$\mathbf{SFG}_R$	$\mathbf{SFG}_R$	the same as the set of generators for $\mathbf{SFG}_R$; see Definition 5.40.
fong-spivak-invitation_F01947	184	■ 1.000	$a, b \in R$	$a, b \in R$	We have the following equations for any $a, b \in R$:
fong-spivak-invitation_F01948	185	■ 0.974	$\binom{0}{6}$	$\binom{0}{6}$	Example 5.52. Both of the signal flow graphs below represent the same matrix, $\binom{0}{6}$:
fong-spivak-invitation_F01949	185	■ 0.665	$R=(\mathbb{N}, 0, +, 1, *)$	$R = (\mathbb{N}, 0, +, 1, *)$	1. Let $R = (\mathbb{N}, 0, +, 1, *)$. By examining the presentation of $\operatorname{Mat}(R)$ in Theorem 5.51,
fong-spivak-invitation_F01950	185	■ 1.000	$R=\mathbb{N} / 3 \mathbb{N}$	$R = \mathbb{N} / 3 \mathbb{N}$	2. Now suppose the rig is $R = \mathbb{N} / 3 \mathbb{N}$; if you do not know what this means, just replace
fong-spivak-invitation_F01951	186	■ 0.780	(M, μ, η)	(M, μ, η)	Definition 5.55. A monoid object (M, μ, η) in a symmetric monoidal category
fong-spivak-invitation_F01952	186	■ 1.000	$\mu: M \otimes M \rightarrow M$	$\mu: M \otimes M \rightarrow M$	$(\mathcal{C}, I, \otimes)$ is an object M of $\mathcal{C}$ together with morphisms $\mu: M \otimes M \rightarrow M$ and $\eta: I \rightarrow M$
fong-spivak-invitation_F01953	186	■ 1.000	$\eta: I \rightarrow M$	$\eta: I \rightarrow M$	$(\mathcal{C}, I, \otimes)$ is an object M of $\mathcal{C}$ together with morphisms $\mu: M \otimes M \rightarrow M$ and $\eta: I \rightarrow M$
fong-spivak-invitation_F01954	186	■ 0.993	$(\mu \otimes \operatorname{id}) ; \mu = (\operatorname{id} \otimes \mu) ; \mu$	$(\mu \otimes \operatorname{id}); \mu = (\operatorname{id} \otimes \mu); \mu$	(a) $\mu \otimes \operatorname{id} \circ \mu = (\operatorname{id} \otimes \mu) \circ \mu$,
fong-spivak-invitation_F01955	186	■ 1.000	$(\eta \otimes \operatorname{id}) ; \mu = \operatorname{id} = (\operatorname{id} \otimes \eta) ; \mu$	$(\eta \otimes \operatorname{id}); \mu = \operatorname{id} = (\operatorname{id} \otimes \eta); \mu$	(b) $\eta \otimes \operatorname{id} \circ \mu = \operatorname{id} = (\operatorname{id} \otimes \eta) \circ \mu$.
fong-spivak-invitation_F01956	186	■ 0.528	$\sigma_{M, M}: \mu = \mu$	$\sigma_{M, M}: \mu = \mu$	(c) $\sigma_{M, M} \circ \mu = \mu$,
fong-spivak-invitation_F01957	186	■ 1.000	$\sigma_{M, M}$	$\sigma_{M, M}$	where $\sigma_{M, M}$ is the swap map on M in $\mathcal{C}$. We often denote it simply by σ .
fong-spivak-invitation_F01958	186	■ 1.000	$\mu: M \times M \rightarrow M$	$\mu: M \times M \rightarrow M$	Example 2.5; see also Example 3.7. That is, it is a set M , a function $\mu: M \times M \rightarrow M$,
fong-spivak-invitation_F01959	186	■ 1.000	$\eta(1) \in M$	$\eta(1) \in M$	which we denote by infix notation $*$, and an element $\eta(1) \in M$, which we denote by e ,
fong-spivak-invitation_F01960	186	■ 1.000	$(a * b) * c = a * (b * c)$	$(a * b) * c = a * (b * c)$	satisfying $a * b * c = a * (b * c)$ and $a * e = a = e * a$.
fong-spivak-invitation_F01961	186	■ 1.000	$a * e = a = e * a$	$a * e = a = e * a$	satisfying $(a * b) * c = a * (b * c)$ and $a * e = a = e * a$.

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fong-spivak-invitation_F01962	186	■ 1.000	$\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	$\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$	1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to
fong-spivak-invitation_F01963	186	■ 1.000	$\mu(a, b) = a * b$	$\mu(a, b) = a * b$	1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to
fong-spivak-invitation_F01964	186	■ 1.000	$\eta \in \mathbb{R}$	$\eta \in \mathbb{R}$	1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to
fong-spivak-invitation_F01965	186	■ 0.997	$\eta = 1$	$\eta = 1$	be $\eta = 1$, then $(\mathbb{R}, *, 1)$ satisfies all three conditions of Definition 5.55.
fong-spivak-invitation_F01966	186	■ 0.997	$(\mathbb{R}, *, 1)$	$(\mathbb{R}, *, 1)$	be $\eta = 1$, then $(\mathbb{R}, *, 1)$ satisfies all three conditions of Definition 5.55.
fong-spivak-invitation_F01967	186	■ 0.999	$\mu(a, b) = a + b$	$\mu(a, b) = a + b$	2. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a + b$ and if $\eta \in \mathbb{R}$ is defined
fong-spivak-invitation_F01968	186	■ 0.892	$\eta = 0$	$\eta = 0$	to be $\eta = 0$, then $(\mathbb{R}, +, 0)$ satisfies all three conditions of Definition 5.55.
fong-spivak-invitation_F01969	186	■ 0.892	$(\mathbb{R}, +, 0)$	$(\mathbb{R}, +, 0)$	to be $\eta = 0$, then $(\mathbb{R}, +, 0)$ satisfies all three conditions of Definition 5.55.
fong-spivak-invitation_F01970	186	■ 0.361	$\mu = \supset -$	$\mu = \supset -$	Example 5.58. Graphically, we can depict $\mu = \supset -$ and $\eta = \circ$. Then axioms (a), (b), and
fong-spivak-invitation_F01971	186	■ 0.361	$\eta = \circ$	$\eta = \circ$	Example 5.58. Graphically, we can depict $\mu = \supset -$ and $\eta = \circ$. Then axioms (a), (b), and
fong-spivak-invitation_F01972	187	■ 0.998	$U: \operatorname{Mat}(R) \rightarrow$	$U: \operatorname{Mat}(R) \rightarrow$	Exercise 5.59. For any rig R , there is a functor $U: \mathbf{Mat}(R) \rightarrow \mathbf{Set}$, sending the object
fong-spivak-invitation_F01973	187	■ 1.000	R^n	R^n	$n \in \mathbb{N}$ to the set R^n , and sending a morphism (matrix) $M: m \rightarrow n$ to the function
fong-spivak-invitation_F01974	187	■ 1.000	$M: m \rightarrow n$	$M: m \rightarrow n$	$n \in \mathbb{N}$ to the set R^n , and sending a morphism (matrix) $M: m \rightarrow n$ to the function
fong-spivak-invitation_F01975	187	■ 1.000	$R^m \rightarrow R^n$	$R^m \rightarrow R^n$	$R^m \rightarrow R^n$ given by vector-matrix multiplication.
fong-spivak-invitation_F01976	187	■ 0.932	$\{1\}$	$\{1\}$	is $\{1\}$, a set with one element, and the monoidal product is $\times$ (see Example 4.38).
fong-spivak-invitation_F01977	187	■ 0.773	$(U(M), U(\mu), U(\eta))$	$(U(M), U(\mu), U(\eta))$	2. Show that if (M, μ, η) is a monoid object in $\mathbf{Mat}(R)$ then $(U(M), U(\mu), U(\eta))$ is a
fong-spivak-invitation_F01978	187	■ 0.537	$\operatorname{prop} \mathbf{Mat}(R)$	$\operatorname{prop} \mathbf{Mat}(R)$	in the prop $\mathbf{Mat}(R)$. If $R = \mathbb{R}$ is the rig of real numbers, this means that we have
fong-spivak-invitation_F01979	187	■ 0.537	$R = \mathbb{R}$	$R = \mathbb{R}$	in the prop $\mathbf{Mat}(R)$. If $R = \mathbb{R}$ is the rig of real numbers, this means that we have
fong-spivak-invitation_F01980	187	■ 0.739	$(1, -\subset, \rightarrow)$	$(1, -\subset, \rightarrow)$	Example 5.60. The triple $(1, -\subset, \rightarrow)$ in $\mathbf{Mat}(R)$ forms a commutative monoid object in
fong-spivak-invitation_F01981	187	■ 0.428	$\operatorname{Mat}(R)^{\operatorname{op}}$	$\operatorname{Mat}(R)^{\operatorname{op}}$	$\mathbf{Mat}(R)^{\operatorname{op}}$. We hence also say that $(1, -\subset, \rightarrow)$ forms a <i>co-commutative comonoid object</i>
fong-spivak-invitation_F01982	187	■ 1.000	$\mu: 2 \rightarrow 1$	$\mu: 2 \rightarrow 1$	Remark 5.64. If we present a prop $\mathcal{M}$ using two generators $\mu: 2 \rightarrow 1$ and $\eta: 0 \rightarrow 1$,
fong-spivak-invitation_F01983	187	■ 1.000	$\eta: 0 \rightarrow 1$	$\eta: 0 \rightarrow 1$	Remark 5.64. If we present a prop $\mathcal{M}$ using two generators $\mu: 2 \rightarrow 1$ and $\eta: 0 \rightarrow 1$,
fong-spivak-invitation_F01984	188	■ 1.000	$\mathcal{M} \rightarrow \mathcal{C}$	$\mathcal{M} \rightarrow \mathcal{C}$	in $\mathcal{C}$ correspond to strict monoidal functors $\mathcal{M} \rightarrow \mathcal{C}$. This sort of idea leads to the
fong-spivak-invitation_F01985	188	■ 0.503	$-\subset$	$-\subset$	normally draw the copy icon like so, $-\subset$, we could just as easily reverse it and draw an
fong-spivak-invitation_F01986	188	■ 1.000	$g: m \rightarrow n$	$g: m \rightarrow n$	A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and gives an output $y \in R^n$.
fong-spivak-invitation_F01987	188	■ 1.000	$x \in R^m$	$x \in R^m$	A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and gives an output $y \in R^n$.
fong-spivak-invitation_F01988	188	■ 1.000	$y \in R^n$	$y \in R^n$	A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and gives an output $y \in R^n$.

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fong-spivak-invitation_F01989	188	■ 0.998	$\mathrm{B}(g) \subseteq R^m \times R^n$	$B(g) \subseteq R^m \times R^n$	<i>behavior</i> of g and denote it $B(g) \subseteq R^m \times R^n$. For example, the “copy” flow graph
fong-spivak-invitation_F01990	188	■ 1.000	$(x, (x, x))$	$(x, (x, x))$	behavior. Similarly, $(x, (x, x))$ is copy behavior for every $x \in R$, thus we have
fong-spivak-invitation_F01991	188	■ 0.986	g^{op}	g^{op}	the inputs with the outputs, so if we denote this reversed icon by g^{op} , we must have
fong-spivak-invitation_F01992	188	■ 0.869	$g^{\mathrm{op}}: n \rightarrow m$	$g^{\mathrm{op}}: n \rightarrow m$	$g^{\mathrm{op}}: n \rightarrow m$. Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\mathrm{op}}) \subseteq R^n \times R^m$. One simple
fong-spivak-invitation_F01993	188	■ 0.869	$\mathrm{B}(\left(g^{\mathrm{op}}\right) \subseteq R^n \times R^m \times R^m$	$B(g^{\mathrm{op}}) \subseteq R^n \times R^m$	$g^{\mathrm{op}}: n \rightarrow m$. Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\mathrm{op}}) \subseteq R^n \times R^m$. One simple
fong-spivak-invitation_F01994	188	■ 0.997	(b, a)	(b, a)	way to do this is to replace each (a, b) with (b, a) , so we would have
fong-spivak-invitation_F01995	188	■ 0.473	$\mathrm{B}(-\subseteq)$	$B(-\subseteq)$	1. What is the behavior $B(-\subseteq)$ of the reversed addition icon $\subseteq: 1 \rightarrow 2$?
fong-spivak-invitation_F01996	188	■ 0.473	$-\subseteq: 1 \rightarrow 2$	$-\subseteq: 1 \rightarrow 2$	1. What is the behavior $B(-\subseteq)$ of the reversed addition icon $\subseteq: 1 \rightarrow 2$?
fong-spivak-invitation_F01997	188	■ 0.237	$\mathrm{B}(-)$	$B(-)$	2. What is the behavior $B(\supset)$ of the reversed copy icon $\supset: 2 \rightarrow 1$?
fong-spivak-invitation_F01998	188	■ 0.237	$-: 2 \rightarrow 1$	$-: 2 \rightarrow 1$	2. What is the behavior $B(\supset)$ of the reversed copy icon $\supset: 2 \rightarrow 1$?
fong-spivak-invitation_F01999	189	■ 1.000	$B \subseteq R^m \times R^n$	$B \subseteq R^m \times R^n$	of a signal flow graph $m \rightarrow n$ is a subset $B \subseteq R^m \times R^n$, i.e. a relation. Why not try to
fong-spivak-invitation_F02000	189	■ 1.000	$N: n \rightarrow p$	$N: n \rightarrow p$	flow graphs with matrices $M: m \rightarrow n$ and $N: n \rightarrow p$, we see that their behaviors are
fong-spivak-invitation_F02001	189	■ 1.000	$B_1 := \left\{ (x, Mx) \mid x \in R^m \right\}$	$B_1 := \{(x, Mx) \mid x \in R^m\}$	the relations $B_1 := \{(x, Mx) \mid x \in R^m\}$ and $B_2 := \{(y, Ny) \mid y \in R^n\}$, while the
fong-spivak-invitation_F02002	189	■ 1.000	$B_2 := \left\{ (y, Ny) \mid y \in R^n \right\}$	$B_2 := \{(y, Ny) \mid y \in R^n\}$	the relations $B_1 := \{(x, Mx) \mid x \in R^m\}$ and $B_2 := \{(y, Ny) \mid y \in R^n\}$, while the
fong-spivak-invitation_F02003	189	■ 0.412	$M \circ_9 N$	$M \circ_9 N$	behavior of $M \circ N$ is the relation $\{(x, x \circ M \circ N) \mid x \in R^m\}$. This is a case of relation
fong-spivak-invitation_F02004	189	■ 0.412	$\left(\left(x, x \circ_9 M \circ_9 N \right) \mid x \in R^m \right)$	$\{(x, x \circ_9 M \circ_9 N) \mid x \in R^m\}$	behavior of $M \circ N$ is the relation $\{(x, x \circ M \circ N) \mid x \in R^m\}$. This is a case of relation
fong-spivak-invitation_F02005	189	■ 1.000	$B_1 \subseteq R^m \times R^n$	$B_1 \subseteq R^m \times R^n$	composition. Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$, their composite
fong-spivak-invitation_F02006	189	■ 1.000	$B_2 \subseteq R^n \times R^p$	$B_2 \subseteq R^n \times R^p$	composition. Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$, their composite
fong-spivak-invitation_F02007	189	■ 0.493	$B_1 \circ B_2 \subseteq R^m \times R^p$	$B_1 \circ B_2 \subseteq R^m \times R^p$	$B_1 \circ B_2 \subseteq R^m \times R^p$ is given by
fong-spivak-invitation_F02008	189	■ 0.999	$\mathbf{Rel}_R$	$\mathbf{Rel}_R$	Definition 5.66. Let R be a rig. We define the prop $\mathbf{Rel}_R$ of R -relations to have sub-
fong-spivak-invitation_F02009	189	■ 1.000	$C \subseteq R^p \times R^q$	$C \subseteq R^p \times R^q$	prop $\mathbf{Rel}_R$. If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$, write down
fong-spivak-invitation_F02010	189	■ 1.000	$B + C$	$B + C$	$B + C$ in set notation.
fong-spivak-invitation_F02011	189	■ 0.689	$g \in G_R$	$g \in G_R$	flow graphs. In Section 5.4.3, we wrote g^{op} for the mirror image of g , for each $g \in G_R$.
fong-spivak-invitation_F02012	189	■ 1.000	$G_R^{\mathrm{op}} := \{g^{\mathrm{op}} \mid g \in G_R\}$	$G_R^{\mathrm{op}} := \{g^{\mathrm{op}} \mid g \in G_R\}$	So let’s write $G_R^{\mathrm{op}} := \{g^{\mathrm{op}} \mid g \in G_R\}$ for the set of all the mirror images of generators.
fong-spivak-invitation_F02013	189	■ 1.000	$\mathbf{SFG}_R^+$	$\mathbf{SFG}_R^+$	We call a morphism in the prop $\mathbf{SFG}_R^+$ a (<i>non-simplified</i>) <i>signal flow graph</i> : these extend
fong-spivak-invitation_F02014	190	■ 1.000	$g: m \rightarrow n, h: \ell \rightarrow n$	$g: m \rightarrow n, h: \ell \rightarrow n$	Exercise 5.68. Let $g: m \rightarrow n, h: \ell \rightarrow n$ be signal flow graphs. Note that $h^{\mathrm{op}}: n \rightarrow \ell$
fong-spivak-invitation_F02015	190	■ 1.000	$h^{\mathrm{op}}: n \rightarrow \ell$	$h^{\mathrm{op}}: n \rightarrow \ell$	Exercise 5.68. Let $g: m \rightarrow n, h: \ell \rightarrow n$ be signal flow graphs. Note that $h^{\mathrm{op}}: n \rightarrow \ell$

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fong-spivak-invitation_F02016	190	■ 0.882	$\left(h^{\mathrm{op}}\right)$	(h^{op})	is a signal flow graph, and we can form the composite $g \circ (h^{\mathrm{op}})$:
fong-spivak-invitation_F02017	190	■ 0.615	$\left(h^{\mathrm{op}}\right) \subseteq R^m \times R^\ell$	$(h^{\mathrm{op}}) \subseteq R^m \times R^\ell$	Show that the behavior of $g \circ (h^{\mathrm{op}}) \subseteq R^m \times R^\ell$ is equal to
fong-spivak-invitation_F02018	190	■ 0.986	$g: m \rightarrow n, h: m \rightarrow p$	$g: m \rightarrow n, h: m \rightarrow p$	Exercise 5.69. Let $g: m \rightarrow n, h: m \rightarrow p$ be signal flow graphs. Note that $(g^{\mathrm{op}}): n \rightarrow$
fong-spivak-invitation_F02019	190	■ 0.986	$\left(g^{\mathrm{op}}\right): n \rightarrow$	$(g^{\mathrm{op}}): n \rightarrow$	Exercise 5.69. Let $g: m \rightarrow n, h: m \rightarrow p$ be signal flow graphs. Note that $(g^{\mathrm{op}}): n \rightarrow$
fong-spivak-invitation_F02020	190	■ 0.971	$g^{\mathrm{op}}; h$	$g^{\mathrm{op}}; h$	m is a signal flow graph, and we can form the composite $g^{\mathrm{op}} \circ h$
fong-spivak-invitation_F02021	190	■ 0.864	$S(g) \in \operatorname{Mat}(R)$	$S(g) \in \operatorname{Mat}(R)$	$S(g) \in \operatorname{Mat}(R)$ be the associated $(m \times n)$ -matrix (see Theorem 5.45).
fong-spivak-invitation_F02022	191	■ 0.981	$b_1, b_2 \in \operatorname{B}(g)$	$b_1, b_2 \in \operatorname{B}(g)$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-invitation_F02023	191	■ 0.981	$b_1 + b_2$	$b_1 + b_2$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-invitation_F02024	191	■ 0.981	$r * b_1$	$r * b_1$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-invitation_F02025	191	■ 0.981	$r \in R$	$r \in R$	subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$, for any $r \in R$. $\diamond$
fong-spivak-invitation_F02026	191	■ 0.840	$\mathbf{LinRel}_R$	$\mathbf{LinRel}_R$	Exercise 5.71. One might want to show that linear relations on R form a prop, $\mathbf{LinRel}_R$.
fong-spivak-invitation_F02027	191	■ 0.997	$(x, y) \in B$	$(x, y) \in B$	that B is linear. In other words, where for any $(x, y) \in B$ and $r \in R$ the element
fong-spivak-invitation_F02028	191	■ 0.987	$(r * x, r * y) \in R^m \times R^n$	$(r * x, r * y) \in R^m \times R^n$	$(r * x, r * y) \in R^m \times R^n$ is in B , and for any $(x', y') \in B$, the element $(x + x', y + y')$
fong-spivak-invitation_F02029	191	■ 0.987	$\left(x^{\prime}, y^{\prime}\right) \in B$	$(x', y') \in B$	$(r * x, r * y) \in R^m \times R^n$ is in B , and for any $(x', y') \in B$, the element $(x + x', y + y')$
fong-spivak-invitation_F02030	191	■ 0.987	$\left(x+x^{\prime}, y+y^{\prime}\right)$	$(x + x', y + y')$	$(r * x, r * y) \in R^m \times R^n$ is in B , and for any $(x', y') \in B$, the element $(x + x', y + y')$
fong-spivak-invitation_F02031	191	■ 0.644	$G_R \sqcup G_R^{\mathrm{op}}$	$G_R \sqcup G_R^{\mathrm{op}}$	R . Moreover, it is possible to give such a presentation whose generating set is $G_R \sqcup G_R^{\mathrm{op}}$
fong-spivak-invitation_F02032	192	■ 1.000	η_1	η_1	from Eq. (5.15) serve as the η_1 and ϵ_1 from Definition 4.44. Using monoidal products
fong-spivak-invitation_F02033	192	■ 1.000	ϵ_1	ϵ_1	from Eq. (5.15) serve as the η_1 and ϵ_1 from Definition 4.44. Using monoidal products
fong-spivak-invitation_F02034	192	■ 1.000	$n = n^*$	$n = n^*$	$n \in \mathbb{N}$ is dual to itself, $n = n^*$.
fong-spivak-invitation_F02035	192	■ 0.999	$\mathbb{R}[s, s^{-1}]$	$\mathbb{R}[s, s^{-1}]$	speed u . We'll work in signal flow diagrams over the rig $\mathbb{R}[s, s^{-1}]$ of polynomials in s
fong-spivak-invitation_F02036	192	■ 1.000	s^{-1}	s^{-1}	and s^{-1} with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$. This is standard in control
fong-spivak-invitation_F02037	192	■ 1.000	$ss^{-1} = s^{-1}s = 1$	$ss^{-1} = s^{-1}s = 1$	and s^{-1} with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$. This is standard in control
fong-spivak-invitation_F02038	192	■ 1.000	$a * u + b * v$	$a * u + b * v$	will take some linear combination $a * u + b * v$ of the desired speed and actual speed,
fong-spivak-invitation_F02039	192	■ 1.000	$\{(F, u, v) \mid u = v\}$	$\{(F, u, v) \mid u = v\}$	this signal flow graph close to the relation $\{(F, u, v) \mid u = v\}$? By phrasing problems
fong-spivak-invitation_F02040	196	■ 0.824	$\operatorname{Cospan}_{\mathcal{C}}$	$\operatorname{Cospan}_{\mathcal{C}}$	that of <i>cospan</i> s in $\mathcal{C}$, denoted $\mathbf{Cospan}_{\mathcal{C}}$ – that can conveniently package $\mathcal{C}$'s colimits in
fong-spivak-invitation_F02041	196	■ 1.000	$s_{m, n}: m \rightarrow n$	$s_{m, n}: m \rightarrow n$	an icon $s_{m, n}: m \rightarrow n$. These icons, known as <i>spiders</i> , ³ are drawn as follows:
fong-spivak-invitation_F02042	197	■ 0.988	$\varnothing \in \mathcal{C}$	$\varnothing \in \mathcal{C}$	Definition 6.1. Let $\mathcal{C}$ be a category. An <i>initial object</i> in $\mathcal{C}$ is an object $\varnothing \in \mathcal{C}$ such that

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fong-spivak-invitation_F02043	197	■ 1.000	$!_T: \varnothing \rightarrow T$	$!_T: \varnothing \rightarrow T$	for each object T in $\mathcal{C}$ there exists a unique morphism $!_T: \varnothing \rightarrow T$.
fong-spivak-invitation_F02044	198	■ 0.992	$A, \leq$	$A, \leq$	1. $(A, \leq)$ has no initial object.
fong-spivak-invitation_F02045	198	■ 0.996	$(A, \leq)$	$(A, \leq)$	3. $(A, \leq)$ has two initial objects.
fong-spivak-invitation_F02046	198	■ 1.000	$\varnothing \rightarrow T$	$\varnothing \rightarrow T$	is a unique function $\varnothing \rightarrow T$, since $\varnothing$ has no elements.
fong-spivak-invitation_F02047	198	■ 0.860	$T \in \mathrm{Ob}(\mathcal{C})$	$T \in \mathrm{Ob}(\mathcal{C})$	need to show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$)
fong-spivak-invitation_F02048	198	■ 0.860	$T=A$	$T = A$	need to show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$)
fong-spivak-invitation_F02049	198	■ 0.860	$T=B$	$T = B$	need to show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$)
fong-spivak-invitation_F02050	198	■ 0.791	$\varnothing =$	$\varnothing =$	there is a unique morphism $\varnothing \rightarrow T$. Trying the case $\varnothing =^? A$ this condition fails
fong-spivak-invitation_F02051	198	■ 1.000	$\varnothing =^? B$	$\varnothing =^? B$	$\varnothing =^? B$ this condition fails when $T = A$: there are zero morphisms $B \rightarrow A$, not
fong-spivak-invitation_F02052	198	■ 1.000	$B \rightarrow A$	$B \rightarrow A$	$\varnothing =^? B$ this condition fails when $T = A$: there are zero morphisms $B \rightarrow A$, not
fong-spivak-invitation_F02053	199	■ 0.977	$\left(R, 0_R, +_R, 1_R, *_R \right)$	$(R, 0_R, +_R, 1_R, *_R)$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$
fong-spivak-invitation_F02054	199	■ 0.977	$\left(S, 0_S, +_S, 1_S, *_S \right)$	$(S, 0_S, +_S, 1_S, *_S)$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$
fong-spivak-invitation_F02055	199	■ 0.977	$f: R \rightarrow S$	$f: R \rightarrow S$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$
fong-spivak-invitation_F02056	199	■ 0.977	$f(0_R) =$	$f(0_R) =$	$(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$
fong-spivak-invitation_F02057	199	■ 1.000	$0_S, f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$	$0_S, f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$	$0_S, f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$, etc.
fong-spivak-invitation_F02058	199	■ 1.000	c_1	c_1	Exercise 6.10. Let $\mathcal{C}$ be a category, and suppose that c_1 and c_2 are initial objects. Find
fong-spivak-invitation_F02059	199	■ 1.000	c_2	c_2	Exercise 6.10. Let $\mathcal{C}$ be a category, and suppose that c_1 and c_2 are initial objects. Find
fong-spivak-invitation_F02060	199	■ 0.998	$\iota_A: A \rightarrow$	$\iota_A: A \rightarrow$	is an object, which we denote $A + B$, together with a pair of morphisms $(\iota_A: A \rightarrow$
fong-spivak-invitation_F02061	199	■ 0.900	$A+B, \iota_B: B \rightarrow A+B$	$A + B, \iota_B: B \rightarrow A + B$	$(A + B, \iota_B: B \rightarrow A + B)$ such that, for all objects T and pairs of morphisms $(f: A \rightarrow$
fong-spivak-invitation_F02062	199	■ 0.900	$f: A \rightarrow$	$f: A \rightarrow$	$A + B, \iota_B: B \rightarrow A + B)$ such that, for all objects T and pairs of morphisms $(f: A \rightarrow$
fong-spivak-invitation_F02063	199	■ 0.999	$T, g: B \rightarrow T$	$T, g: B \rightarrow T$	$T, g: B \rightarrow T)$, there exists a unique morphism $[f, g]: A + B \rightarrow T$ such that the
fong-spivak-invitation_F02064	199	■ 0.999	$[f, g]: A+B \rightarrow T$	$[f, g]: A + B \rightarrow T$	$T, g: B \rightarrow T)$, there exists a unique morphism $[f, g]: A + B \rightarrow T$ such that the
fong-spivak-invitation_F02065	199	■ 1.000	$[f, g]$	$[f, g]$	We call $[f, g]$ the <i>copairing</i> of f and g .
fong-spivak-invitation_F02066	200	■ 0.997	$\iota_A: A \rightarrow A \sqcup B$	$\iota_A: A \rightarrow A \sqcup B$	the disjoint union $A \sqcup B$ together with the inclusion functions $\iota_A: A \rightarrow A \sqcup B$ and
fong-spivak-invitation_F02067	200	■ 1.000	$\iota_B: B \rightarrow A \sqcup B$	$\iota_B: B \rightarrow A \sqcup B$	$\iota_B: B \rightarrow A \sqcup B$.
fong-spivak-invitation_F02068	200	■ 1.000	$f: A \rightarrow T$	$f: A \rightarrow T$	Suppose we have functions $f: A \rightarrow T$ and $g: B \rightarrow T$ for some other set
fong-spivak-invitation_F02069	200	■ 1.000	$g: B \rightarrow T$	$g: B \rightarrow T$	Suppose we have functions $f: A \rightarrow T$ and $g: B \rightarrow T$ for some other set

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fong-spivak-invitation_F02876	200	■ 0.804	$[f, g]: A \sqcup B \rightarrow T$	$[f, g]: A \sqcup B \rightarrow T$	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any
fong-spivak-invitation_F02871	200	■ 0.804	$\iota_A\{A\}$	ι_A	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any
fong-spivak-invitation_F02872	200	■ 0.804	$[f, g]=f$	$[f, g] = f$	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any
fong-spivak-invitation_F02873	200	■ 0.804	$\iota_A\{B\}$	ι_B	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any
fong-spivak-invitation_F02874	200	■ 0.804	$[f, g]=g$	$[f, g] = g$	$[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$. What is it? Any
fong-spivak-invitation_F02875	200	■ 0.997	$x \in A \sqcup B$	$x \in A \sqcup B$	element $x \in A \sqcup B$ is either “from A ” or “from B ,” i.e. either there is some $a \in A$ with
fong-spivak-invitation_F02876	200	■ 0.998	$x=\iota_A\{a\}$	$x = \iota_A(a)$	$x = \iota_A(a)$ or there is some $b \in B$ with $x = \iota_B(b)$. By Eq. (6.1), we must have:
fong-spivak-invitation_F02877	200	■ 0.998	$b \in B$	$b \in B$	$x = \iota_A(a)$ or there is some $b \in B$ with $x = \iota_B(b)$. By Eq. (6.1), we must have:
fong-spivak-invitation_F02878	200	■ 0.998	$x=\iota_B\{b\}$	$x = \iota_B(b)$	$x = \iota_A(a)$ or there is some $b \in B$ with $x = \iota_B(b)$. By Eq. (6.1), we must have:
fong-spivak-invitation_F02879	200	■ 1.000	$T=\{a, b, c, \ldots, z\}$	$T = \{a, b, c, \ldots, z\}$	Exercise 6.14. Suppose $T = \{a, b, c, \ldots, z\}$ is the set of letters in the alphabet, and
fong-spivak-invitation_F02880	200	■ 0.999	$f($	$f($	element of A to the first letter of its label, e.g. $f(\text{apple}) = a$. Let $g: B \rightarrow T$ be the
fong-spivak-invitation_F02881	200	■ 0.999	$)=a$	$) = a$	element of A to the first letter of its label, e.g. $f(\text{apple}) = a$. Let $g: B \rightarrow T$ be the
fong-spivak-invitation_F02882	200	■ 0.993	$g($	$g($	function sending each element of B to the last letter of its label, e.g. $g(\text{apple}) = e$.
fong-spivak-invitation_F02883	200	■ 0.993	$)=e$	$) = e$	function sending each element of B to the last letter of its label, e.g. $g(\text{apple}) = e$.
fong-spivak-invitation_F02884	200	■ 1.000	$[f, g](x)$	$[f, g](x)$	Write down the function $[f, g](x)$ for all eight elements of $A \sqcup B$.
fong-spivak-invitation_F02885	200	■ 1.000	$f: A \rightarrow C, g: B \rightarrow C$	$f: A \rightarrow C, g: B \rightarrow C$	Exercise 6.15. Let $f: A \rightarrow C, g: B \rightarrow C$, and $h: C \rightarrow D$ be morphisms in a
fong-spivak-invitation_F02886	200	■ 0.359	$\iota_A\{A\}; [f, g]=f$	$\iota_A; [f, g] = f$	1. $\boxed{\iota_A \circ [f, g] = f}$,
fong-spivak-invitation_F02887	200	■ 0.859	$[f, g]; h=[f; h, g; h]$	$[f, g]; h = [f; h, g; h]$	3. $\boxed{[f, g] \circ h = [f \circ h, g \circ h]}$,
fong-spivak-invitation_F02888	200	■ 1.000	$\left[\iota_A\{A\}, \iota_B\{B\}\right]=\mathrm{id}_{A+B}$	$[\iota_A, \iota_B] = \mathrm{id}_{A+B}$	4. $\boxed{[\iota_A, \iota_B] = \mathrm{id}_{A+B}}$.
fong-spivak-invitation_F02889	200	■ 0.981	$(\mathcal{C}, +, \varnothing)$	$(\mathcal{C}, +, \varnothing)$	denoted $\varnothing$. Then $(\mathcal{C}, +, \varnothing)$ is a symmetric monoidal category (recall Definition 4.34).
fong-spivak-invitation_F02890	200	■ 0.807	$\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	$\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	1. Show that $+$ extends to a functor $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$. In particular, how does it act on
fong-spivak-invitation_F02891	200	■ 0.963	$\mathcal{C} \times \mathcal{C}$	$\mathcal{C} \times \mathcal{C}$	morphisms in $\mathcal{C} \times \mathcal{C}$?
fong-spivak-invitation_F02892	201	■ 1.000	$A+\varnothing \rightarrow A$	$A + \varnothing \rightarrow A$	isomorphisms $A + \varnothing \rightarrow A$ and $\varnothing + A \rightarrow A$.
fong-spivak-invitation_F02893	201	■ 1.000	$\varnothing+A \rightarrow A$	$\varnothing + A \rightarrow A$	isomorphisms $A + \varnothing \rightarrow A$ and $\varnothing + A \rightarrow A$.
fong-spivak-invitation_F02894	201	■ 1.000	$(A+B)+C \rightarrow A+(B+C)$	$(A + B) + C \rightarrow A + (B + C)$	(a) $\boxed{(A + B) + C \rightarrow A + (B + C)}$,
fong-spivak-invitation_F02895	201	■ 1.000	$A+B \rightarrow B+A$	$A + B \rightarrow B + A$	(b) $\boxed{A + B \rightarrow B + A}$.
fong-spivak-invitation_F02896	201	■ 1.000	$f: A \rightarrow X$	$f: A \rightarrow X$	Definition 6.17. Let $\mathcal{C}$ be a category and let $f: A \rightarrow X$ and $g: A \rightarrow Y$ be morphisms

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fong-spivak-invitation_F02097	201	■ 1.000	$g: A \rightarrow Y$	$g: A \rightarrow Y$	Definition 6.17. Let $\mathcal{C}$ be a category and let $f: A \rightarrow X$ and $g: A \rightarrow Y$ be morphisms
fong-spivak-invitation_F02098	201	■ 0.996	$X + \{ \}_{A} Y$	$X + {}_A Y$	in $\mathcal{C}$ that have a common domain. The <i>pushout</i> $X + {}_A Y$ is the colimit of the diagram
fong-spivak-invitation_F02099	201	■ 0.712	$X + \{A\} Y$	$X + {}_A Y$	In more detail, a pushout consists of (i) an object $X + {}_A Y$ and (ii) morphisms $\iota_X: X \rightarrow$
fong-spivak-invitation_F02100	201	■ 0.712	$\iota_X: X \rightarrow$	$\iota_X: X \rightarrow$	In more detail, a pushout consists of (i) an object $X + {}_A Y$ and (ii) morphisms $\iota_X: X \rightarrow$
fong-spivak-invitation_F02101	201	■ 0.997	$\iota_Y: Y \rightarrow X + \{ \}_{A} Y$	$\iota_Y: Y \rightarrow X + {}_A Y$	$X + {}_A Y$ and $\iota_Y: Y \rightarrow X + {}_A Y$ satisfying (a) and (b) below.
fong-spivak-invitation_F02102	201	■ 0.989	$\ulcorner$	$\ulcorner$	commutes. (We will explain the ‘ $\ulcorner$ ’ symbol below.)
fong-spivak-invitation_F02103	201	■ 1.000	$x: X \rightarrow T, y: Y \rightarrow T$	$x: X \rightarrow T, y: Y \rightarrow T$	(b) For all objects T and morphisms $x: X \rightarrow T, y: Y \rightarrow T$, if the diagram
fong-spivak-invitation_F02104	201	■ 0.955	$t: X + \{ \}_{A} Y \rightarrow T$	$t: X + {}_A Y \rightarrow T$	commutes, then there exists a unique morphism $t: X + {}_A Y \rightarrow T$ such that
fong-spivak-invitation_F02105	202	■ 1.000	ι_X	ι_X	We further call ι_X the <i>pushout of g along f</i> , and similarly ι_Y the <i>pushout of f along g</i> .
fong-spivak-invitation_F02106	202	■ 1.000	ι_Y	ι_Y	We further call ι_X the <i>pushout of g along f</i> , and similarly ι_Y the <i>pushout of f along g</i> .
fong-spivak-invitation_F02107	202	■ 1.000	$B \leftarrow A \rightarrow C$	$B \leftarrow A \rightarrow C$	pushout of a diagram $B \leftarrow A \rightarrow C$ is equal to the coproduct $B \sqcup C$: namely, both
fong-spivak-invitation_F02108	202	■ 1.000	$B \sqcup C$	$B \sqcup C$	pushout of a diagram $B \leftarrow A \rightarrow C$ is equal to the coproduct $B \sqcup C$: namely, both
fong-spivak-invitation_F02109	202	■ 1.000	$B \vee C$	$B \vee C$	are equal to the join $B \vee C$.
fong-spivak-invitation_F02110	202	■ 1.000	$i: A \rightarrow A^{\prime}$	$i: A \rightarrow A'$	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X + {}_A A'$, i.e. the following
fong-spivak-invitation_F02111	202	■ 1.000	$j: X \rightarrow X^{\prime}$	$j: X \rightarrow X'$	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X + {}_A A'$, i.e. the following
fong-spivak-invitation_F02112	202	■ 1.000	$X^{\prime}$	X'	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X + {}_A A'$, i.e. the following
fong-spivak-invitation_F02113	202	■ 1.000	$X + \{ \}_{A} A^{\prime}$	$X + {}_A A'$	$i: A \rightarrow A'$ and $j: X \rightarrow X'$, we can take X' to be the pushout $X + {}_A A'$, i.e. the following
fong-spivak-invitation_F02114	202	■ 0.659	$f^{\prime} := i^{-1} \circ f \circ j$	$f' := i^{-1} \circ f \circ j$	where $f' := i^{-1} \circ f \circ j$. To see this, observe that if there is any object T such that the
fong-spivak-invitation_F02115	202	■ 0.568	$x=i$	$x = i$	then $f \circ x = i \circ a$, and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$. This
fong-spivak-invitation_F02116	202	■ 0.568	$x^{\prime}: X \rightarrow T$	$x': X \rightarrow T$	then $f \circ x = i \circ a$, and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$. This
fong-spivak-invitation_F02117	202	■ 0.568	$x^{\prime}:=j^{-1} ; x$	$x' := j^{-1}; x$	then $f \circ x = i \circ a$, and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$. This
fong-spivak-invitation_F02118	202	■ 0.978	$f^{\prime}; x^{\prime}=i^{-1} ; f ; j ; j^{-1} ; x=i^{-1} ; i ; a=a$	$f'; x' = i^{-1}; f; j; j^{-1}; x = i^{-1}; i; a = a$	because $f' \circ x' = i^{-1} \circ f \circ j \circ j^{-1} \circ x = i^{-1} \circ i \circ a = a$.
fong-spivak-invitation_F02119	203	■ 1.000	$\{f(a) \sim g(a) \mid a \in A\}$	$\{f(a) \sim g(a) \mid a \in A\}$	relation $\{f(a) \sim g(a) \mid a \in A\}$.
fong-spivak-invitation_F02120	203	■ 1.000	$f(a)$	$f(a)$	a connection between $f(a)$ in X and $g(a)$ in Y . The pushout is the set of connected
fong-spivak-invitation_F02121	203	■ 1.000	$g(a)$	$g(a)$	a connection between $f(a)$ in X and $g(a)$ in Y . The pushout is the set of connected
fong-spivak-invitation_F02122	203	■ 1.000	$f: \underline{4} \rightarrow \underline{5}$	$f: \underline{4} \rightarrow \underline{5}$	Exercise 6.22. What is the pushout of the functions $f: \underline{4} \rightarrow \underline{5}$ and $g: \underline{4} \rightarrow \underline{3}$ pictured
fong-spivak-invitation_F02123	203	■ 1.000	$g: \underline{4} \rightarrow \underline{3}$	$g: \underline{4} \rightarrow \underline{3}$	Exercise 6.22. What is the pushout of the functions $f: \underline{4} \rightarrow \underline{5}$ and $g: \underline{4} \rightarrow \underline{3}$ pictured

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fong-spivak-invitation_F02124	203	■ 1.000	$X, Y \in$	$X, Y \in$	Example 6.23. Suppose a category $\mathcal{C}$ has an initial object $\emptyset$. For any two objects $X, Y \in$
fong-spivak-invitation_F02125	203	■ 0.884	$f: \varnothing \rightarrow X$	$f: \emptyset \rightarrow X$	$\text{Ob } \mathcal{C}$, there is a unique morphism $f: \emptyset \rightarrow X$ and a unique morphism $g: \emptyset \rightarrow Y$; this
fong-spivak-invitation_F02126	203	■ 0.884	$g: \varnothing \rightarrow Y$	$g: \emptyset \rightarrow Y$	$\text{Ob } \mathcal{C}$, there is a unique morphism $f: \emptyset \rightarrow X$ and a unique morphism $g: \emptyset \rightarrow Y$; this
fong-spivak-invitation_F02127	203	■ 0.923	$X \stackrel{f}{\leftarrow} \varnothing \stackrel{g}{\rightarrow} Y$	$X \xleftarrow{f} \emptyset \xrightarrow{g} Y$	The diagram $X \xleftarrow{f} \emptyset \xrightarrow{g} Y$ has a pushout in $\mathcal{C}$ iff X and Y have a coproduct in $\mathcal{C}$,
fong-spivak-invitation_F02128	203	■ 1.000	$X+Y$	$X + Y$	coproduct $X + Y$; then the diagram to the left below
fong-spivak-invitation_F02129	203	■ 1.000	$X+Y \rightarrow T$	$X + Y \rightarrow T$	right, there is a unique map $X + Y \rightarrow T$ making the diagram as in Eq. (6.4) commute
fong-spivak-invitation_F02130	203	■ 0.968	$\{ \}_{2}$	2	(why? ₂). This shows that $X + Y$ is a pushout, $X +_{\emptyset} Y \cong X + Y$.
fong-spivak-invitation_F02131	203	■ 0.968	$X+\varnothing Y \cong X+Y$	$X + \emptyset Y \cong X + Y$	(why? ₂). This shows that $X + Y$ is a pushout, $X +_{\emptyset} Y \cong X + Y$.
fong-spivak-invitation_F02132	203	■ 1.000	$X+\varnothing Y$	$X + \emptyset Y$	Similarly, if a pushout $X +_{\emptyset} Y$ exists, then it satisfies the universal property of the
fong-spivak-invitation_F02133	203	■ 0.997	$\{ \}_{3}$	3	coproduct (why? ₃).
fong-spivak-invitation_F02134	204	■ 1.000	$A=X=Y=\mathbb{N}$	$A = X = Y = \mathbb{N}$	Example 6.25. Let $A = X = Y = \mathbb{N}$. Consider the functions $f: A \rightarrow X$ and $g: A \rightarrow Y$
fong-spivak-invitation_F02135	204	■ 0.988	$f(a):=\lfloor a / 2 \rfloor$	$f(a) := \lfloor a/2 \rfloor$	given by the “floor” functions, $f(a) := \lfloor a/2 \rfloor$ and $g(a) := \lfloor (a+1)/2 \rfloor$.
fong-spivak-invitation_F02136	204	■ 0.988	$g(a):=\lfloor (a+1) / 2 \rfloor$	$g(a) := \lfloor (a+1)/2 \rfloor$	given by the “floor” functions, $f(a) := \lfloor a/2 \rfloor$ and $g(a) := \lfloor (a+1)/2 \rfloor$.
fong-spivak-invitation_F02137	204	■ 1.000	$x: X \rightarrow T$	$x: X \rightarrow T$	If T is any other set and we have maps $x: X \rightarrow T$ and $y: Y \rightarrow T$ that commute with
fong-spivak-invitation_F02138	204	■ 1.000	$y: Y \rightarrow T$	$y: Y \rightarrow T$	If T is any other set and we have maps $x: X \rightarrow T$ and $y: Y \rightarrow T$ that commute with
fong-spivak-invitation_F02139	204	■ 1.000	$f \cong x=g \cong y$	$f \cong x = g \cong y$	f and g , i.e. $f \cong x = g \cong y$, then this commutativity implies that
fong-spivak-invitation_F02140	204	■ 1.000	$f(1)=0$	$f(1) = 0$	In other words, Y ’s 0 and X ’s 0 go to the same place in T , say t . But since $f(1) = 0$
fong-spivak-invitation_F02141	204	■ 1.000	$g(1)=1$	$g(1) = 1$	and $g(1) = 1$, we also have that $t = x(0) = x(f(1)) = y(g(1)) = y(1)$. This means
fong-spivak-invitation_F02142	204	■ 1.000	$t=x(0)=x(f(1))=y(g(1))=y(1)$	$t = x(0) = x(f(1)) = y(g(1)) = y(1)$	and $g(1) = 1$, we also have that $t = x(0) = x(f(1)) = y(g(1)) = y(1)$. This means
fong-spivak-invitation_F02143	204	■ 1.000	$g(2)=1$	$g(2) = 1$	Y ’s 1 goes to t also. But since $g(2) = 1$ and $f(2) = 1$, we also have that $t = g(1) =$
fong-spivak-invitation_F02144	204	■ 1.000	$f(2)=1$	$f(2) = 1$	Y ’s 1 goes to t also. But since $g(2) = 1$ and $f(2) = 1$, we also have that $t = g(1) =$
fong-spivak-invitation_F02145	204	■ 1.000	$t=g(1)=$	$t = g(1) =$	Y ’s 1 goes to t also. But since $g(2) = 1$ and $f(2) = 1$, we also have that $t = g(1) =$
fong-spivak-invitation_F02146	204	■ 0.986	$y(g(2))=x(f(2))=x(1)$	$y(g(2)) = x(f(2)) = x(1)$	$y(g(2)) = x(f(2)) = x(1)$, which means that X ’s 1 also goes to t . One can keep
fong-spivak-invitation_F02147	204	■ 1.000	$X \sqcup_A Y \cong \{1\}$	$X \sqcup_A Y \cong \{1\}$	$X \sqcup_A Y \cong \{1\}$.
fong-spivak-invitation_F02148	204	■ 0.862	$j \in \operatorname{Ob} \mathcal{J}$	$j \in \text{Ob } \mathcal{J}$	have finitely many objects too, because each object $j \in \text{Ob } \mathcal{J}$ has its own identity morphism id_j .
fong-spivak-invitation_F02149	204	■ 0.862	id_j	id_j	have finitely many objects too, because each object $j \in \text{Ob } \mathcal{J}$ has its own identity morphism id_j .
fong-spivak-invitation_F02150	205	■ 0.997	$\operatorname{colim}_{\mathcal{J}} D$	$\text{colim}_{\mathcal{J}} D$	Definition 6.26. We say that a category $\mathcal{C}$ has finite colimits if a colimit, $\text{colim}_{\mathcal{J}} D$,

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fong-spivak-invitation_F02151	205	0.820	$\mathfrak{d}$	$\mathfrak{d}$	exists whenever $\mathcal{J}$ is a finite category and $D: \mathcal{J} \rightarrow \mathcal{C}$ is a diagram.
fong-spivak-invitation_F02152	205	0.963	$!: \mathbf{0} \rightarrow \mathcal{C}$	$!: \mathbf{0} \rightarrow \mathcal{C}$	$!: \mathbf{0} \rightarrow \mathcal{C}$, where $\mathbf{0}$ is the category with no objects and no morphisms, and $!$ is the unique
fong-spivak-invitation_F02153	206	0.997	$(v, d) \sim (w, e)$	$(v, d) \sim (w, e)$	generated by putting $(v, d) \sim (w, e)$ if there is an arrow $a: v \rightarrow w$ in J such that
fong-spivak-invitation_F02154	206	0.997	$a: v \rightarrow w$	$a: v \rightarrow w$	generated by putting $(v, d) \sim (w, e)$ if there is an arrow $a: v \rightarrow w$ in J such that
fong-spivak-invitation_F02155	206	0.991	$D(a)(d)=e$	$D(a)(d) = e$	$D(a)(d) = e$. Then this set, together with the functions $\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$ given
fong-spivak-invitation_F02156	206	0.991	$\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$	$\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$	$D(a)(d) = e$. Then this set, together with the functions $\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$ given
fong-spivak-invitation_F02157	206	0.996	$d \in D(v)$	$d \in D(v)$	by sending $d \in D(v)$ to its equivalence class, constitutes a colimit of D .
fong-spivak-invitation_F02158	206	0.508	$\mathfrak{J} = \begin{array}{ cc } \hline v_1 & v_2 \\ \hline \bullet & \bullet \\ \hline \end{array}$	$\mathfrak{J} = \begin{array}{ cc } \hline v_1 & v_2 \\ \hline \bullet & \bullet \\ \hline \end{array}$	Example 6.34. A coproduct is a colimit on the graph $\mathcal{J} = \begin{array}{ cc } \hline v_1 & v_2 \\ \hline \bullet & \bullet \\ \hline \end{array}$. A functor
fong-spivak-invitation_F02159	206	0.990	$X := D(\overline{v_1})$	$X := D(v_1)$	$D: \mathcal{J} \rightarrow \mathbf{Set}$ can be identified with a choice of two sets, $X := D(\overline{v_1})$ and $Y := D(v_2)$.
fong-spivak-invitation_F02160	206	0.990	$Y := D(\overline{v_2})$	$Y := D(v_2)$	$D: \mathcal{J} \rightarrow \mathbf{Set}$ can be identified with a choice of two sets, $X := D(\overline{v_1})$ and $Y := D(v_2)$.
fong-spivak-invitation_F02161	207	0.479	$\mathbf{1} = \boxed{\bullet}$	$\mathbf{1} = \boxed{\bullet}$	Example 6.35. If $\mathcal{J}$ is the category $\mathbf{1} = \boxed{\bullet}$, the formula in Theorem 6.32 yields the set
fong-spivak-invitation_F02162	207	1.000	$D(v)$	$D(v)$	This is isomorphic to the set $D(v)$. In other words, if X is a set considered as a diagram
fong-spivak-invitation_F02163	207	0.995	$X: \mathbf{1} \rightarrow$	$X: \mathbf{1} \rightarrow$	$X: \mathbf{1} \rightarrow \mathbf{Set}$, then its colimit (like its limit) is just X again.
fong-spivak-invitation_F02164	207	1.000	$X \stackrel{f}{\leftarrow} N \stackrel{g}{\rightarrow} Y$	$X \stackrel{f}{\leftarrow} N \stackrel{g}{\rightarrow} Y$	diagram $X \stackrel{f}{\leftarrow} N \stackrel{g}{\rightarrow} Y$ – agree with the description we gave in Example 6.21.
fong-spivak-invitation_F02165	207	0.985	$X \underset{g}{\stackrel{f}{\rightrightarrows}} Y$	$X \underset{g}{\stackrel{f}{\rightrightarrows}} Y$	Consider some diagram $X \underset{g}{\stackrel{f}{\rightrightarrows}} Y$ on this graph in $\mathbf{Set}$. The coequalizer of
fong-spivak-invitation_F02166	207	1.000	$y \sim y'$	$y \sim y'$	erated by declaring $y \sim y'$ whenever there exists x in X such that $f(x) = y$ and
fong-spivak-invitation_F02167	207	1.000	$f(x)=y$	$f(x) = y$	erated by declaring $y \sim y'$ whenever there exists x in X such that $f(x) = y$ and
fong-spivak-invitation_F02168	207	1.000	$g(x)=y'$	$g(x) = y'$	$g(x) = y'$.
fong-spivak-invitation_F02169	207	0.883	$f, g: X \rightarrow Y$	$f, g: X \rightarrow Y$	$\bullet$, the blue and red arrows respectively define functions $f, g: X \rightarrow Y$. Let's leave the
fong-spivak-invitation_F02170	208	1.000	$A \rightarrow N \leftarrow B$	$A \rightarrow N \leftarrow B$	common object $A \rightarrow N \leftarrow B$. The common object N is called the <i>apex</i> of the cospan
fong-spivak-invitation_F02171	208	1.000	$N \stackrel{g}{\leftarrow} B \stackrel{h}{\rightarrow} P$	$N \stackrel{g}{\leftarrow} B \stackrel{h}{\rightarrow} P$	Then, if a pushout of $N \stackrel{g}{\leftarrow} B \stackrel{h}{\rightarrow} P$ exists in $\mathcal{C}$, as shown on the left, we can extract a
fong-spivak-invitation_F02172	208	0.822	$C \in \operatorname{Ob} \mathcal{C}$	$C \in \operatorname{Ob} \mathcal{C}$	we need an identity on every object $C \in \operatorname{Ob} \mathcal{C}$; but that's not hard: use $C \rightarrow C \leftarrow C$
fong-spivak-invitation_F02173	208	0.822	$C \rightarrow C \leftarrow C$	$C \rightarrow C \leftarrow C$	we need an identity on every object $C \in \operatorname{Ob} \mathcal{C}$; but that's not hard: use $C \rightarrow C \leftarrow C$
fong-spivak-invitation_F02174	208	1.000	$A \rightarrow B \rightarrow C$	$A \rightarrow B \rightarrow C$	pushouts, so we don't know that every two sequential morphisms $A \rightarrow B \rightarrow C$ can be
fong-spivak-invitation_F02175	208	1.000	$A \rightarrow P \leftarrow B$	$A \rightarrow P \leftarrow B$	between their respective apexes. That is, the cospan $A \rightarrow P \leftarrow B$ and $A \rightarrow P' \leftarrow B$ in

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fong-spivak-invitation_F02176	208	■ 1.000	$A \rightarrowtail P^{\prime} \leftarrowtail B$	$A \rightarrow P' \leftarrow B$	between their respective apexes. That is, the cospan $A \rightarrow P \leftarrow B$ and $A \rightarrowtail P' \leftarrowtail B$ in
fong-spivak-invitation_F02177	208	■ 1.000	$P \cong P^{\prime}$	$P \cong P'$	the diagram shown left below are equivalent iff there is an isomorphism $P \cong P'$ making
fong-spivak-invitation_F02178	208	■ 0.824	$\{ \}_{{\text{\texttt{C}}}}$	$\mathcal{C}$	business: $\mathbf{Cospan}_{\mathcal{C}}$ will form a category. But in fact, we are very close to getting more.
fong-spivak-invitation_F02179	209	■ 0.500	$_{\mathcal{C}}$	$\mathcal{C}$	$\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$, i.e. $\text{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \text{Ob}(\mathcal{C})$, where the morphisms
fong-spivak-invitation_F02180	209	■ 0.500	$\operatorname{Ob}\{\left(\operatorname{\mathbf{Cospan}}_{\mathcal{C}}\right)\}_{\mathcal{C}}=\operatorname{Ob}\{\left(\operatorname{\mathcal{C}}\right)\}_{\mathcal{C}}$	$\text{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \text{Ob}(\mathcal{C})$	$\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$, i.e. $\text{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \text{Ob}(\mathcal{C})$, where the morphisms
fong-spivak-invitation_F02181	209	■ 0.907	$_{\mathcal{C}}, \varnothing, +$	$\mathcal{C}, \varnothing, +$	$(\mathbf{Cospan}_{\mathcal{C}}, \varnothing, +)$. The monoidal unit is the initial object $\varnothing \in \mathcal{C}$ and the monoidal
fong-spivak-invitation_F02182	209	■ 1.000	$A+\varnothing \cong A$	$A + \varnothing \cong A$	product is given by coproduct. The coherence isomorphisms, e.g. $A + \varnothing \cong A$, can be
fong-spivak-invitation_F02183 [conf 0.098]	209	■ 0.098	$\operatorname{\mathbf{Cospan}}_{\mathcal{C}}, \varnothing, +$	$\mathbf{Cospan}_{\mathcal{C}}, \varnothing, +$	It is a straightforward but time-consuming exercise to verify that $(\mathbf{Cospan}_{\mathcal{C}}, \varnothing, +)$
fong-spivak-invitation_F02184	209	■ 0.749	$_{\text{\texttt{FinSet}}}$	FinSet	define a symmetric monoidal category $\mathbf{Cospan}_{\text{FinSet}}$. What does it look like? It looks a
fong-spivak-invitation_F02185	209	■ 0.986	$\{ \}_{{\text{\texttt{FinSet}}}}$	FinSet	The objects of $\mathbf{Cospan}_{\text{FinSet}}$ are finite sets; here let's draw them as collections of •s.
fong-spivak-invitation_F02186	209	■ 1.000	$A \xrightarrow{f} N \stackrel{g}{\leftarrow} B$	$A \xrightarrow{f} N \xleftarrow{g} B$	six-element set. Below are two depictions of a cospan $A \xrightarrow{f} N \xleftarrow{g} B$.
fong-spivak-invitation_F02187	209	■ 1.000	$g(b)$	$g(b)$	arrows from each $a \in A$ to $f(a)$ and each $b \in B$ to $g(b)$. In the depiction on the
fong-spivak-invitation_F02188	210	■ 0.997	$A+B \rightarrowtail B+C$	$A + B \rightarrowtail B + C$	$\mathbf{Cospan}_{\text{FinSet}}$. Draw their monoidal product as a morphism $A + B \rightarrowtail B + C$ in
fong-spivak-invitation_F02189	211	■ 0.995	(X, μ, η)	(X, μ, η)	To make this precise, recall from Definition 5.55 that a commutative monoid (X, μ, η)
fong-spivak-invitation_F02190	211	■ 0.531	$\mathcal{C}, I, \otimes$	$\mathcal{C}, I, \otimes$	in symmetric monoidal category $(\mathcal{C}, I, \otimes)$ is an object X of $\mathcal{C}$ together with (noteworthy)
fong-spivak-invitation_F02191	211	—	$\mu: X \otimes X \rightarrowtail X$	$\mu: X \otimes X \rightarrow X$	
fong-spivak-invitation_F02192	211	—	$\eta: I \rightarrowtail X$	$\eta: I \rightarrow X$	—
fong-spivak-invitation_F02193	211	■ 0.786	$X \otimes X$	$X \otimes X$	where $\bowtie$ is the symmetry on $X \otimes X$. A cocommutative comonoid (X, δ, ϵ) is an object
fong-spivak-invitation_F02194	211	■ 0.786	(X, δ, ϵ)	(X, δ, ϵ)	where $\bowtie$ is the symmetry on $X \otimes X$. A cocommutative comonoid (X, δ, ϵ) is an object
fong-spivak-invitation_F02195	211	■ 1.000	$\delta: X \rightarrowtail X \otimes X, \epsilon: X \rightarrowtail I$	$\delta: X \rightarrow X \otimes X, \epsilon: X \rightarrow I$	X with maps $\delta: X \rightarrow X \otimes X, \epsilon: X \rightarrow I$, obeying the mirror images of the laws in
fong-spivak-invitation_F02196	211	■ 1.000	μ, η, δ	μ, η, δ	comonoid, and consider a wiring diagram built only from the icons μ, η, δ , and ϵ ,
fong-spivak-invitation_F02197	211	■ 1.000	ϵ	ϵ	comonoid, and consider a wiring diagram built only from the icons μ, η, δ , and ϵ ,
fong-spivak-invitation_F02198	211	■ 1.000	μ	μ	how the morphisms μ and δ , the merger and the splitter, interact.
fong-spivak-invitation_F02199	211	■ 1.000	δ	δ	how the morphisms μ and δ , the merger and the splitter, interact.
fong-spivak-invitation_F02200	211	■ 0.986	$(\mathcal{C}, \otimes, I)$	$(\mathcal{C}, \otimes, I)$	Definition 6.43. Let X be an object in a symmetric monoidal category $(\mathcal{C}, \otimes, I)$. A
fong-spivak-invitation_F02201	211	■ 0.902	$\mu, \eta, \delta, \epsilon$	$\mu, \eta, \delta, \epsilon$	<i>Frobenius structure</i> on X consists of a 4-tuple $(\mu, \eta, \delta, \epsilon)$ such that (X, μ, η) is a
fong-spivak-invitation_F02202	211	■ 0.902	X, μ, η	X, μ, η	<i>Frobenius structure</i> on X consists of a 4-tuple $(\mu, \eta, \delta, \epsilon)$ such that (X, μ, η) is a

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fong-spivak-invitation_F02283	212	■ 1.000	$X^{\otimes m} \rightarrow X^{\otimes n}$	$X^{\otimes m} \rightarrow X^{\otimes n}$	With these two equations, it turns out that two morphisms $X^{\otimes m} \rightarrow X^{\otimes n}$ – defined
fong-spivak-invitation_F02284	212	■ 1.000	μ, δ	μ, δ	by composing and tensoring identities on X and the noteworthy morphisms μ, δ , etc. –
fong-spivak-invitation_F02285	212	■ 0.889	$X, \mu, \eta, \delta, \epsilon$	$X, \mu, \eta, \delta, \epsilon$	Definition 6.44. Let $(X, \mu, \eta, \delta, \epsilon)$ be a Frobenius monoid in a monoidal category $(\mathcal{C},$
fong-spivak-invitation_F02286	212	■ 1.000	$I, \otimes)$	$I, \otimes)$	$I, \otimes)$. Let $m, n \in \mathbb{N}$. Define $s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$ to be the following morphism
fong-spivak-invitation_F02287	212	■ 1.000	$s_{\{m, n\}}: X^{\otimes m} \rightarrow X^{\otimes n}$	$s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$	$I, \otimes)$. Let $m, n \in \mathbb{N}$. Define $s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$ to be the following morphism
fong-spivak-invitation_F02288	212	■ 0.998	$(m-1) \mu \mathrm{~}\mathit{\mathrm{s}}$	$(m-1)\mu s$	It can be written formally as $(m-1) \mu s$ followed by $(n-1) \delta s$, with special cases when
fong-spivak-invitation_F02289	212	■ 0.998	$(n-1) \delta \mathrm{~}\mathit{\mathrm{s}}$	$(n-1)\delta s$	It can be written formally as $(m-1) \mu s$ followed by $(n-1) \delta s$, with special cases when
fong-spivak-invitation_F02210	212	■ 1.000	$m=0$	$m = 0$	$m = 0$ or $n = 0$.
fong-spivak-invitation_F02211	212	■ 0.993	$s_{\{m, n\}}$	$s_{m,n}$	We call $s_{m,n}$ the <i>spider of type (m, n)</i> , and can draw it more simply as the icon
fong-spivak-invitation_F02212	212	■ 1.000	$\mu, \eta, \delta, \epsilon, \sigma$	$\mu, \eta, \delta, \epsilon, \sigma$	a wire is spiderable, it means that we have a bunch of morphisms $\mu, \eta, \delta, \epsilon, \sigma$ that we
fong-spivak-invitation_F02213	212	■ 0.973	$f: X^{\otimes m} \rightarrow X^{\otimes n}$	$f: X^{\otimes m} \rightarrow X^{\otimes n}$	$I, \otimes)$. Suppose that we have a map $f: X^{\otimes m} \rightarrow X^{\otimes n}$ constructed from spiders and the
fong-spivak-invitation_F02214	212	■ 1.000	$\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$	$\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$	symmetry map $\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$ using composition and the monoidal product, and such
fong-spivak-invitation_F02215	212	■ 1.000	$f=s_{\{m, n\}}$	$f = s_{m,n}$	$f = s_{m,n}$.
fong-spivak-invitation_F02216	212	■ 1.000	$X \otimes X \rightarrow X \otimes X \otimes X$	$X \otimes X \rightarrow X \otimes X \otimes X$	morphisms $X \otimes X \rightarrow X \otimes X \otimes X$ in the following list are necessarily equal?
fong-spivak-invitation_F02217	213	■ 0.999	$G:=\{\mu, \eta, \delta, \epsilon\}$	$G := \{\mu, \eta, \delta, \epsilon\}$	Theorem 6.48. Consider the four-element set $G := \{\mu, \eta, \delta, \epsilon\}$ and define
fong-spivak-invitation_F02218	213	■ 1.000	$G \rightarrow \mathbb{N}$	$G \rightarrow \mathbb{N}$	$in, out: G \rightarrow \mathbb{N}$ as follows:
fong-spivak-invitation_F02219	213	■ 0.970	(G, E)	(G, E)	Then the free prop on (G, E) is equivalent, as a symmetric monoidal category, ⁵ to
fong-spivak-invitation_F02220	215	■ 0.970	$X, \mu_X, \delta_X, \eta_X, \epsilon_X$	$X, \mu_X, \delta_X, \eta_X, \epsilon_X$	which each object X is equipped with a Frobenius structure $(X, \mu_X, \delta_X, \eta_X, \epsilon_X)$ such
fong-spivak-invitation_F02221	215	■ 0.996	$\eta_I=\mathrm{id}_I, \epsilon_I=\epsilon_I$	$\eta_I = \mathrm{id}_I = \epsilon_I$	for all objects X, Y , and such that $\eta_I = \mathrm{id}_I = \epsilon_I$.
fong-spivak-invitation_F02222	215	■ 1.000	$\mu_X, \delta_X, \eta_X, \epsilon_X$	$\mu_X, \delta_X, \eta_X, \epsilon_X$	The Frobenius morphisms $\mu_X, \delta_X, \eta_X, \epsilon_X$ for each object X are constructed using the
fong-spivak-invitation_F02223	215	■ 0.532	$\operatorname{Cospan}_{\text{FinSet}}$	$\operatorname{Cospan}_{\text{FinSet}}$	Exercise 6.52. By Example 6.51, the category $\operatorname{Cospan}_{\text{FinSet}}$ is a hypergraph cate-
fong-spivak-invitation_F02224	215	■ 0.571	$\varnothing, \sqcup)$	$\varnothing, \sqcup)$	Example 6.54. Recall the monoidal category $(\mathbf{Corel}, \varnothing, \sqcup)$ from Example 4.46; its
fong-spivak-invitation_F02225	215	■ 1.000	$\mu_X: X \sqcup X \rightarrow X$	$\mu_X: X \sqcup X \rightarrow X$	the corelation $\mu_X: X \sqcup X \rightarrow X$ such that two elements of $X \sqcup X \sqcup X$ are equivalent if
fong-spivak-invitation_F02226	215	■ 1.000	$X \sqcup X \sqcup X$	$X \sqcup X \sqcup X$	the corelation $\mu_X: X \sqcup X \rightarrow X$ such that two elements of $X \sqcup X \sqcup X$ are equivalent if
fong-spivak-invitation_F02227	215	■ 1.000	$\delta_X: X \rightarrow X \sqcup X$	$\delta_X: X \rightarrow X \sqcup X$	and only if they come from the same underlying element of X . Define $\delta_X: X \rightarrow X \sqcup X$
fong-spivak-invitation_F02228	215	■ 1.000	$\eta_X: \varnothing \rightarrow X$	$\eta_X: \varnothing \rightarrow X$	in the same way, and define $\eta_X: \varnothing \rightarrow X$ and $\epsilon_X: X \rightarrow \varnothing$ such that no two elements
fong-spivak-invitation_F02229	215	■ 1.000	$\epsilon_X: X \rightarrow \varnothing$	$\epsilon_X: X \rightarrow \varnothing$	in the same way, and define $\eta_X: \varnothing \rightarrow X$ and $\epsilon_X: X \rightarrow \varnothing$ such that no two elements

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fong-spivak-invitation_F02230	215	■ 1.000	$X=\varnothing \sqcup X=X \sqcup \varnothing$	$X = \varnothing \sqcup X = X \sqcup \varnothing$	of $X = \varnothing \sqcup X = X \sqcup \varnothing$ are equivalent.
fong-spivak-invitation_F02231	216	■ 1.000	$\left(X, \mu_X, \eta_X, \delta_X, \epsilon_X \right)$	$(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$	These maps define a special commutative Frobenius monoid $(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$,
fong-spivak-invitation_F02232	217	■ 0.456	$\left(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{P}} \right)$	$(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{P}})$	Rough Definition 6.58. Let $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$ and $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$ be symmetric monoidal
fong-spivak-invitation_F02233	217	■ 0.456	$\left(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}} \right)$	$(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$	Rough Definition 6.58. Let $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$ and $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$ be symmetric monoidal
fong-spivak-invitation_F02234	217	■ 0.992	(F, φ)	(F, φ)	categories. To specify a <i>symmetric monoidal functor</i> (F, φ) between them,
fong-spivak-invitation_F02235	217	■ 0.986	$\varphi_I\colon I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$	$\varphi_I : I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$	(ii) one specifies a morphism $\varphi_I : I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$,
fong-spivak-invitation_F02236	217	■ 0.865	$c_1, c_2 \in \mathrm{Ob}(\mathcal{C})$	$c_1, c_2 \in \mathrm{Ob}(\mathcal{C})$	(iii) for each $c_1, c_2 \in \mathrm{Ob}(\mathcal{C})$, one specifies a morphism
fong-spivak-invitation_F02237	217	■ 1.000	φ	φ	We call the various maps φ <i>coherence maps</i> . We require the coherence maps to obey
fong-spivak-invitation_F02238	217	■ 0.990	φ_I	φ_I	monoidal structures on $\mathcal{C}$ and $\mathcal{D}$. If φ_I and φ_{c_1, c_2} are isomorphisms for all c_1, c_2 , we say
fong-spivak-invitation_F02239	217	■ 0.990	φ_{c_1, c_2}	φ_{c_1, c_2}	monoidal structures on $\mathcal{C}$ and $\mathcal{D}$. If φ_I and φ_{c_1, c_2} are isomorphisms for all c_1, c_2 , we say
fong-spivak-invitation_F02240	217	■ 0.990	c_1, c_2	c_1, c_2	monoidal structures on $\mathcal{C}$ and $\mathcal{D}$. If φ_I and φ_{c_1, c_2} are isomorphisms for all c_1, c_2 , we say
fong-spivak-invitation_F02241	217	■ 1.000	$S \in$	$S \in$	sending a set $S \in \mathbf{Set}$ to its set of subsets $P(S) := \{R \subseteq S\}$. It acts on morphisms
fong-spivak-invitation_F02242	217	■ 1.000	$\mathrm{P}(S) := \{R \subseteq S\}$	$P(S) := \{R \subseteq S\}$	sending a set $S \in \mathbf{Set}$ to its set of subsets $P(S) := \{R \subseteq S\}$. It acts on morphisms
fong-spivak-invitation_F02243	217	■ 0.954	$\mathrm{im}_f\colon \mathrm{P}(S) \rightarrow \mathrm{P}(T)$	$\mathrm{im}_f : P(S) \rightarrow P(T)$	by sending a function $f : S \rightarrow T$ to the image map $\mathrm{im}_f : P(S) \rightarrow P(T)$, which maps
fong-spivak-invitation_F02244	217	■ 1.000	$R \subseteq S$	$R \subseteq S$	$R \subseteq S$ to $\{f(r) \mid r \in R\} \subseteq T$.
fong-spivak-invitation_F02245	217	■ 1.000	$\{f(r) \mid r \in R\} \subseteq T$	$\{f(r) \mid r \in R\} \subseteq T$	$R \subseteq S$ to $\{f(r) \mid r \in R\} \subseteq T$.
fong-spivak-invitation_F02246	217	■ 0.983	$(\{1\}, \times)$	$(\{1\}, \times)$	Now consider the symmetric monoidal structure $(\{1\}, \times)$ on $\mathbf{Set}$ from Example 4.38.
fong-spivak-invitation_F02247	217	■ 1.000	$\varphi_I\colon \{1\} \rightarrow$	$\varphi_I : \{1\} \rightarrow$	To make P a symmetric monoidal functor, we need to specify a function $\varphi_I : \{1\} \rightarrow$
fong-spivak-invitation_F02248	217	■ 1.000	$\mathrm{P}(\{1\})$	$P(\{1\})$	$P(\{1\})$ and for all sets S and T , a function $\varphi_{S, T} : P(S) \times P(T) \rightarrow P(S \times T)$. One possibility
fong-spivak-invitation_F02249	217	■ 1.000	$\varphi_{S, T}\colon \mathrm{P}(S) \times \mathrm{P}(T) \rightarrow \mathrm{P}(S \times T)$	$\varphi_{S, T} : P(S) \times P(T) \rightarrow P(S \times T)$	$P(\{1\})$ and for all sets S and T , a function $\varphi_{S, T} : P(S) \times P(T) \rightarrow P(S \times T)$. One possibility
fong-spivak-invitation_F02250	217	■ 1.000	$\varphi_I(1)$	$\varphi_I(1)$	is to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and
fong-spivak-invitation_F02251	217	■ 1.000	$\{1\} \subseteq \{1\}$	$\{1\} \subseteq \{1\}$	is to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and
fong-spivak-invitation_F02252	217	■ 1.000	$A \subseteq S$	$A \subseteq S$	is to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$, and given subsets $A \subseteq S$ and
fong-spivak-invitation_F02253	217	■ 1.000	$B \subseteq T$	$B \subseteq T$	$B \subseteq T$, to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$. With these
fong-spivak-invitation_F02254	217	■ 1.000	$\varphi_{S, T}(A, B)$	$\varphi_{S, T}(A, B)$	$B \subseteq T$, to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$. With these
fong-spivak-invitation_F02255	217	■ 1.000	$A \times B \subseteq S \times T$	$A \times B \subseteq S \times T$	$B \subseteq T$, to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$. With these
fong-spivak-invitation_F02256	217	■ 0.969	P, φ	P, φ	definitions, (P, φ) is a symmetric monoidal functor.

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fong-spivak-invitation_F02257	217	■ 1.000	$\varphi_{S, T}$	$\varphi_{S, T}$	Exercise 6.60. Check that the maps $\varphi_{S, T}$ defined in Example 6.59 are natural in S and
fong-spivak-invitation_F02258	218	■ 1.000	$2 \ \Omega$	2Ω	as a resistor of resistance 2Ω). For this circuit to be a morphism in some category, i.e.
fong-spivak-invitation_F02259	218	■ 1.000	$A \rightarrowtail N$	$A \rightarrow N$	functions $A \rightarrow N$ and $B \rightarrow N$ are indicated by the gray arrows. This forms a cospan
fong-spivak-invitation_F02260	218	■ 1.000	$B \rightarrowtail N$	$B \rightarrow N$	functions $A \rightarrow N$ and $B \rightarrow N$ are indicated by the gray arrows. This forms a cospan
fong-spivak-invitation_F02261	219	■ 0.997	$(F, \varphi): (\mathcal{C}, +) \longrightarrow$	$(F, \varphi): (\mathcal{C}, +) \longrightarrow$	Definition 6.61. Let $\mathcal{C}$ be a category with finite colimits, and $(F, \varphi): (\mathcal{C}, +) \longrightarrow$
fong-spivak-invitation_F02262	219	■ 0.621	$A \rightarrowtail N \stackrel{o}{\longleftarrow} B$	$A \xrightarrow{i} N \xleftarrow{o} B$	ting of a cospan $A \xrightarrow{i} N \xleftarrow{o} B$ in $\mathcal{C}$ together with an element $s \in F(N)$. ⁶ We call
fong-spivak-invitation_F02263	219	■ 0.621	$s \in F(N)$	$s \in F(N)$. ⁶	ting of a cospan $A \xrightarrow{i} N \xleftarrow{o} B$ in $\mathcal{C}$ together with an element $s \in F(N)$. ⁶ We call
fong-spivak-invitation_F02264	219	■ 0.637	$\mathcal{C} =$	$\mathcal{C} =$	The intuition here is to use $\mathcal{C} = \mathbf{FinSet}$, and, for each object $N \in \mathbf{FinSet}$, the functor
fong-spivak-invitation_F02265	219	■ 0.637	$N \in$	$N \in$	The intuition here is to use $\mathcal{C} = \mathbf{FinSet}$, and, for each object $N \in \mathbf{FinSet}$, the functor
fong-spivak-invitation_F02266	219	■ 1.000	$F(N)$	$F(N)$	available decorations on N nodes, taken from the set $F(N)$.
fong-spivak-invitation_F02267	219	■ 1.000	$(A \rightarrowtail N \stackrel{g}{\longleftarrow} B, s)$	$(A \xrightarrow{f} N \xleftarrow{g} B, s)$	Let $(A \xrightarrow{f} N \xleftarrow{g} B, s)$ and $(B \xrightarrow{h} P \xleftarrow{k} C, t)$ represent decorated cospans. Their
fong-spivak-invitation_F02268	219	■ 1.000	$(B \rightarrowtail P \stackrel{k}{\longleftarrow} C, t)$	$(B \xrightarrow{h} P \xleftarrow{k} C, t)$	Let $(A \xrightarrow{f} N \xleftarrow{g} B, s)$ and $(B \xrightarrow{h} P \xleftarrow{k} C, t)$ represent decorated cospans. Their
fong-spivak-invitation_F02269	219	■ 1.000	$B \rightarrowtail h$	$B \xrightarrow{h}$	composite is represented by the composite of the cospans $A \xrightarrow{f} N \xleftarrow{g} B$ and $B \xrightarrow{h}$
fong-spivak-invitation_F02270	219	■ 1.000	$P \stackrel{k}{\longleftarrow} C$	$P \xleftarrow{k} C$	$P \xleftarrow{k} C$, paired with the following element of $F(N +_B P)$:
fong-spivak-invitation_F02271	219	■ 1.000	$F(N + \{ \}_B P)$	$F(N +_B P)$	$P \xleftarrow{k} C$, paired with the following element of $F(N +_B P)$:
fong-spivak-invitation_F02272	220	■ 0.559	$(F, \varphi): (\mathcal{C}, +) \longrightarrow ($	$(F, \varphi): (\mathcal{C}, +) \longrightarrow ($	functor $(F, \varphi): (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$, there is a hypergraph category $\mathbf{Cospan}_F$
fong-spivak-invitation_F02273	220	■ 0.559	$\times)$	$\times)$	functor $(F, \varphi): (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$, there is a hypergraph category $\mathbf{Cospan}_F$
fong-spivak-invitation_F02274	220	■ 0.559	$_F\}$	F	functor $(F, \varphi): (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$, there is a hypergraph category $\mathbf{Cospan}_F$
fong-spivak-invitation_F02275	220	■ 0.660	$\operatorname{Cospan}_F\}$	Cospan_F	tion $\mathbf{Cospan}_F$, even though they’re very different. An expert tells you “they’re not so
fong-spivak-invitation_F02276	220	■ 1.000	$F(c) := \{*\}$	$F(c) := \{*\}$	different; one is a special case of the other. Just use the constant functor $F(c) := \{*\}$.”
fong-spivak-invitation_F02277	220	■ 1.000	Ω	Ω	To be clear, the Ω are just labels; the above set is isomorphic to $\{\text{light}, \text{switch},$
fong-spivak-invitation_F02278	220	■ 0.926	$\sqcup \mathbb{R}^+$	$\sqcup \mathbb{R}^+$	battery $\} \sqcup \mathbb{R}^+$. But we write C this way to remind us that it consists of circuit
fong-spivak-invitation_F02279	220	■ 1.000	$\ell: A \rightarrowtail C$	$\ell: A \rightarrow C$	source and target functions, together with a function $\ell: A \rightarrow C$ labeling each edge with
fong-spivak-invitation_F02280	220	■ 1.000	$V = \{1, 2\}, A = \{e\}, s(e) = 1$	$V = \{1, 2\}, A = \{e\}, s(e) = 1$	For example, we might have the simple case of $V = \{1, 2\}, A = \{e\}, s(e) = 1$,
fong-spivak-invitation_F02281	220	■ 0.808	$t(e) = 2$	$t(e) = 2$	$t(e) = 2$ – so e is an edge from 1 to 2 – and $\ell(e) = 3\Omega$. This represents a resistor with
fong-spivak-invitation_F02282	220	■ 0.808	$\ell(e) = 3 \ \Omega$	$\ell(e) = 3\Omega$	$t(e) = 2$ – so e is an edge from 1 to 2 – and $\ell(e) = 3\Omega$. This represents a resistor with

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fong-spivak-invitation_F02283	220	■ 1.000	$3 \ \Omega$	3Ω	resistance 3Ω :
fong-spivak-invitation_F02284	220	■ 1.000	$V=\{1,2\}, \ A=\{e\}$	$V = \{1, 2\}, A = \{e\}$	could also be represented by the ‘reversed graph’, with data $V = \{1, 2\}, A = \{e\}$,
fong-spivak-invitation_F02285	220	■ 1.000	$s(e)=2, \ t(e)=1$	$s(e) = 2, t(e) = 1$	$s(e) = 2, t(e) = 1$, and $\ell(e) = 3\Omega$.
fong-spivak-invitation_F02286	221	■ 0.978	$V, \ A, \ s, \ t, \ \ell$	V, A, s, t, ℓ	Exercise 6.64. Write a tuple (V, A, s, t, ℓ) that represents the circuit in Eq. (6.11). $\diamond$
fong-spivak-invitation_F02287	221	■ 0.996	ψ	ψ	as in Definition 6.61. We’ll call the functor (Circ, ψ) . We start by defining the functor part
fong-spivak-invitation_F02288	221	■ 0.995	$f: V \rightarrow V^{\prime}$	$f : V \rightarrow V'$	On morphisms, Circ sends a function $f: V \rightarrow V'$ to the function
fong-spivak-invitation_F02289	221	■ 0.989	$c \in \text{Circ}(4)$	$c \in \text{Circ}(4)$	Exercise 6.65. To understand this functor better, let $c \in \text{Circ}(4)$ be the circuit
fong-spivak-invitation_F02290	221	■ 1.000	$f: \underline{4} \rightarrow \underline{3}$	$f : \underline{4} \rightarrow \underline{3}$	and let $f: \underline{4} \rightarrow \underline{3}$ be the function
fong-spivak-invitation_F02291	221	■ 0.998	$\text{Circ}(f)(c)$	$\text{Circ}(f)(c)$	Draw a picture of the circuit $\text{Circ}(f)(c)$.
fong-spivak-invitation_F02292	221	■ 1.000	$\psi_{V, V^{\prime}}$	$\psi_{V, V'}$	coherence maps $\psi_{V, V'}$ for finite sets V, V' , we define
fong-spivak-invitation_F02293	221	■ 1.000	$V, \ V^{\prime}$	V, V'	coherence maps $\psi_{V, V'}$ for finite sets V, V' , we define
fong-spivak-invitation_F02294	221	■ 1.000	$V+V^{\prime}$	$V + V'$	considers them together as a circuit on the disjoint union of vertices $V + V'$.
fong-spivak-invitation_F02295	221	■ 0.978	$\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$	$\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$	$\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$ should be, and draw a picture.
fong-spivak-invitation_F02296	222	■ 0.885	Circ	Circ	Our notation for this category is Cospan _{Circ} . Following Theorem 6.62, the objects of
fong-spivak-invitation_F02297	222	■ 0.668	$\underline{2}$	$\underline{2}$	tions of white circles $\circ$. For example, we’ll represent the object $\underline{2}$ of Cospan _{Circ} as two
fong-spivak-invitation_F02298	222	■ 0.668	$\mathbf{Cospan}_{\text{Circ}}$	Cospan _{Circ}	tions of white circles $\circ$. For example, we’ll represent the object $\underline{2}$ of Cospan _{Circ} as two
fong-spivak-invitation_F02299	222	■ 0.692	$\text{Cospan}_{\text{Circ}}$	$\text{Cospan}_{\text{Circ}}$	More interesting than the objects, however, are the morphisms in Cospan _{Circ} . These
fong-spivak-invitation_F02300	222	■ 1.000	$\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$	$\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$	is, cospan $\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$ together with an element c of $\text{Circ}(\underline{p})$. As an example, consider
fong-spivak-invitation_F02301	222	■ 1.000	$\text{Circ}(\underline{p})$	$\text{Circ}(\underline{p})$	is, cospan $\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$ together with an element c of $\text{Circ}(\underline{p})$. As an example, consider
fong-spivak-invitation_F02302	222	■ 0.998	$\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$	$\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$	the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$, equipped with the battery
fong-spivak-invitation_F02303	222	■ 0.998	$i_1(1)=1$	$i_1(1) = 1$	the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$, equipped with the battery
fong-spivak-invitation_F02304	222	■ 0.998	$i_2(1)=2$	$i_2(1) = 2$	the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$, equipped with the battery
fong-spivak-invitation_F02305	222	■ 0.987	$(V, \ A, \ s, \ t, \ \ell) \in \text{Circ}(2)$	$(V, A, s, t, \ell) \in \text{Circ}(2)$	some C -circuit $(V, A, s, t, \ell) \in \text{Circ}(2)$. What is it?
fong-spivak-invitation_F02307	222	■ 0.795	$\underline{1}$	$\underline{1}$	First we’ll consider composition. Consider the following decorated cospan from $\underline{1}$ to $\underline{1}$:
fong-spivak-invitation_F02308	222	■ 1.000	$\text{Circ}([\iota_N, \iota_P])(\psi_{N, P}(s, t))$	$\text{Circ}([\iota_N, \iota_P])(\psi_{N, P}(s, t))$	oration, we use the formula $\text{Circ}([\iota_N, \iota_P])(\psi_{N, P}(s, t))$ from (6.15). Again, this is rather
fong-spivak-invitation_F02309	223	■ 0.971	$\underline{1}=\{\circ\}$	$\underline{1} = \{\circ\}$	(We simply ignore the decorations for now.) If we pushout over the common set $\underline{1} = \{\circ\}$,
fong-spivak-invitation_F02310	223	■ 1.000	$\psi_{\underline{2}, \underline{2}}$	$\psi_{\underline{2}, \underline{2}}$	tion: you told us what the map $\psi_{\underline{2}, \underline{2}}$ does in Exercise 6.66. It takes the two decorations,

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fong-spivak-invitation_F02311	223	■ 0.912	$\psi_{\{N, P\}}(s, t)$	$\psi_{N,P}(s, t)$	in $\text{Circ}(\underline{4})$. So this is what the $\psi_{N,P}(s, t)$ part means. What does $[\iota_N, \iota_P]$ mean? Recall
fong-spivak-invitation_F02312	223	■ 0.912	$\text{left}[\iota_{\{N\}}, \iota_{\{P\}}\text{right}]$	$[\iota_N, \iota_P]$	in $\text{Circ}(\underline{4})$. So this is what the $\psi_{N,P}(s, t)$ part means. What does $[\iota_N, \iota_P]$ mean? Recall
fong-spivak-invitation_F02313	223	■ 1.000	$A \rightarrowtail M \leftarrowtail B$	$A \rightarrow M \leftarrow B$	given two cospans $A \rightarrow M \leftarrow B$ and $C \rightarrow N \leftarrow D$, we take their coproduct cospan
fong-spivak-invitation_F02314	223	■ 1.000	$C \rightarrowtail N \leftarrowtail D$	$C \rightarrow N \leftarrow D$	given two cospans $A \rightarrow M \leftarrow B$ and $C \rightarrow N \leftarrow D$, we take their coproduct cospan
fong-spivak-invitation_F02315	223	■ 1.000	$A+C \rightarrowtail M+N \leftarrowtail B+D$	$A + C \rightarrow M + N \leftarrow B + D$	$A + C \rightarrow M + N \leftarrow B + D$. And for decorations, we use the map $\psi_{M,N} : \text{Circ}(M) \times$
fong-spivak-invitation_F02316	223	■ 1.000	$\psi_{\{M, N\}} : \operatorname{Circ}(M) \times$	$\psi_{M,N} : \text{Circ}(M) \times$	$A + C \rightarrow M + N \leftarrow B + D$. And for decorations, we use the map $\psi_{M,N} : \text{Circ}(M) \times$
fong-spivak-invitation_F02317	224	■ 1.000	$\operatorname{Circ}(N) \rightarrowtail \operatorname{Circ}(M+N)$	$\text{Circ}(N) \rightarrow \text{Circ}(M + N)$	$\text{Circ}(N) \rightarrow \text{Circ}(M + N)$. So, for example, suppose we want to take the monoidal product
fong-spivak-invitation_F02318	224	■ 1.000	$\eta : \emptyset \rightarrowtail 2$	$\eta : 0 \rightarrow 2$	Exercise 6.69. Write x for the open circuit in (6.19). Also define cospans $\eta : 0 \rightarrow 2$
fong-spivak-invitation_F02319	224	■ 0.848	$\eta : 2 \rightarrowtail \emptyset$	$\eta : 2 \rightarrow 0$	and $\eta : 2 \rightarrow 0$ as follows:
fong-spivak-invitation_F02320	224	■ 0.878	$(\underline{1}, \emptyset, !, !, !) \in \operatorname{Circ}(\underline{1})$.	$(\underline{1}, \emptyset, !, !, !) \in \text{Circ}(\underline{1})$.	where each of these are decorated by the empty circuit $(\underline{1}, \emptyset, !, !, !) \in \text{Circ}(\underline{1})$.
fong-spivak-invitation_F02321	224	■ 0.956	$\eta ; x ; \epsilon$	$\eta; x; \epsilon$	Compute the composite $\eta \circ x \circ \epsilon \in \mathbf{Cospan}_{\text{Circ}}$. This is a morphism $\underline{0} \rightarrow \underline{0}$; we call
fong-spivak-invitation_F02322	224	■ 0.956	$\mathbf{Cospan}_{\text{Circ}}$	$\mathbf{Cospan}_{\text{Circ}}$	Compute the composite $\eta \circ x \circ \epsilon \in \mathbf{Cospan}_{\text{Circ}}$. This is a morphism $\underline{0} \rightarrow \underline{0}$; we call
fong-spivak-invitation_F02323	224	■ 0.956	$\underline{\emptyset} \rightarrowtail \underline{\emptyset}$	$\underline{0} \rightarrow \underline{0}$	Compute the composite $\eta \circ x \circ \epsilon \in \mathbf{Cospan}_{\text{Circ}}$. This is a morphism $\underline{0} \rightarrow \underline{0}$; we call
fong-spivak-invitation_F02324	225	■ 0.890	$\mathcal{O} \rightarrowtail$	$\mathcal{O} \rightarrow$	structures from operad functors $\mathcal{O} \rightarrow \mathbf{Set}$.
fong-spivak-invitation_F02325	226	■ 1.000	f, g, h, k	f, g, h, k	The cost is that the “boxes” f, g, h, k in Eq. (6.21) no longer have a left/right
fong-spivak-invitation_F02326	226	■ 0.999	$\mathcal{M} \rightarrowtail$	$\mathcal{M} \rightarrow$	theory of monoids as a prop $\mathcal{M}$, and defined monoids using functors $\mathcal{M} \rightarrow \mathbf{Set}$; see
fong-spivak-invitation_F02327	226	■ 0.578	$\mathbf{Cospan}_{\text{FinSet}}$	$\mathbf{Cospan}_{\text{FinSet}}$	Remark 5.64. In the same way, we can view the category $\mathbf{Cospan}_{\text{FinSet}}$ as some sort
fong-spivak-invitation_F02328	226	■ 0.989	$\text{FinSet} \rightarrowtail$	$\text{FinSet} \rightarrow$	$\mathbf{Cospan}_{\text{FinSet}} \rightarrow \mathbf{Set}$.
fong-spivak-invitation_F02329	226	■ 1.000	$\mathcal{O}$	$\mathcal{O}$	So that’s the idea. An operad $\mathcal{O}$ will define a theory or grammar of composition, and
fong-spivak-invitation_F02330	226	■ 0.864	$t_1, \dots, t_n, t$	$t_1, \dots, t_n, t$	(ii) for each tuple $(t_1, \dots, t_n, t)$ of types, one specifies a set $\mathcal{O}(t_1, \dots, t_n, t)$, whose
fong-spivak-invitation_F02331	226	■ 0.864	$\mathcal{O}(t_1, \dots, t_n, t)$	$\mathcal{O}(t_1, \dots, t_n, t)$	(ii) for each tuple $(t_1, \dots, t_n, t)$ of types, one specifies a set $\mathcal{O}(t_1, \dots, t_n, t)$, whose
fong-spivak-invitation_F02332	226	■ 0.990	$t_1, \dots, t_n ; t$	$t_1, \dots, t_n ; t$	elements are called <i>operations of arity</i> $(t_1, \dots, t_n ; t)$;
fong-spivak-invitation_F02333	226	■ 0.882	$s_1, \dots, s_m, t_i$	$s_1, \dots, s_m, t_i$	(iii) for each pair of tuples $(s_1, \dots, s_m, t_i)$ and $(t_1, \dots, t_n, t)$, one specifies a function
fong-spivak-invitation_F02334	227	■ 0.991	$\text{id}_t \in \mathcal{O}(t ; t)$	$\text{id}_t \in \mathcal{O}(t; t)$	(iv) for each type t , one specifies an operation $\text{id}_t \in \mathcal{O}(t; t)$ called the <i>identity</i>
fong-spivak-invitation_F02335	227	■ 1.000	$m+n-1$	$m + n - 1$	of arity $m + n - 1$: we have m arguments to fill in m , and the remaining $n - 1$ arguments
fong-spivak-invitation_F02336	227	■ 1.000	$n-1$	$n - 1$	of arity $m + n - 1$: we have m arguments to fill in m , and the remaining $n - 1$ arguments
fong-spivak-invitation_F02337	227	■ 0.648	$\circ_i$	$\circ_i$	substitution function $\circ_i$, which says we obtain the operation $f \circ_i g \in \mathcal{O}(m + n - 1)$. The
fong-spivak-invitation_F02338	227	■ 0.648	$f \circ_i g \in \mathcal{O}(m+n-1)$	$f \circ g \in \mathcal{O}(m + n - 1)$	substitution function $\circ_i$, which says we obtain the operation $f \circ_i g \in \mathcal{O}(m + n - 1)$. The

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_F02339	228	■ 0.998	$X_{\{1\}} \times \cdots \times X_{\{n\}} \rightarrow Y$	$X_1 \times \cdots \times X_n \rightarrow Y$	(ii) functions $X_1 \times \cdots \times X_n \rightarrow Y$ as operations of arity $(X_1, \dots, X_n; Y)$,
fong-spivak-invitation_F02340	228	■ 0.998	$X_{\{1\}}, \ldots, X_{\{n\}} ; Y$	$X_1, \dots, X_n; Y$	(ii) functions $X_1 \times \cdots \times X_n \rightarrow Y$ as operations of arity $(X_1, \dots, X_n; Y)$,
fong-spivak-invitation_F02341	228	■ 0.929	$f \in \operatorname{Set}(W_{\{1\}}, \ldots, W_{\{m\}} ; X_{\{i\}})$, $g \in \operatorname{Set}(X_{\{1\}}, \ldots, X_{\{n\}} ; Y)$	$f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$	where $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$, and hence $g \circ_i f$ is a
fong-spivak-invitation_F02342	228	■ 0.929	$g \circ_{\{i\}} f$	$g \circ_i f$	where $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$, and hence $g \circ_i f$ is a
fong-spivak-invitation_F02343	228	■ 0.780	$\operatorname{identities} \operatorname{id}_X \in \operatorname{Set}(X; X)$	identities $\operatorname{id}_X \in \operatorname{Set}(X; X)$	(iv) identities $\operatorname{id}_X \in \operatorname{Set}(X; X)$ given by the identity function $\operatorname{id}_X : X \rightarrow X$.
fong-spivak-invitation_F02344	228	■ 0.780	$\operatorname{id}_X : X \rightarrow X$	$\operatorname{id}_X : X \rightarrow X$	(iv) identities $\operatorname{id}_X \in \operatorname{Set}(X; X)$ given by the identity function $\operatorname{id}_X : X \rightarrow X$.
fong-spivak-invitation_F02345	228	■ 1.000	$a \in \mathbb{N}$	$a \in \mathbb{N}$	(i) natural numbers $a \in \mathbb{N}$ as types,
fong-spivak-invitation_F02346	228	■ 0.810	$\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$	$\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$	(ii) cospans $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$ of finite sets as operations of arity $(a_1, \dots, a_n; b)$,
fong-spivak-invitation_F02347	228	■ 0.810	$\left(\underline{a_1}, \ldots, \underline{a_n} ; b \right)$	$(a_1, \dots, a_n; b)$	(ii) cospans $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$ of finite sets as operations of arity $(a_1, \dots, a_n; b)$,
fong-spivak-invitation_F02348	228	■ 0.185	$\operatorname{id}_a \in \operatorname{Set}(a ; a)$	$\operatorname{id}_a \in \operatorname{Set}(a; a)$	(iv) identities $\operatorname{id}_a \in \operatorname{Set}(a; a)$ just given by the identity $\operatorname{id}_a : a \rightarrow a$.
fong-spivak-invitation_F02349	228	■ 0.185	$\operatorname{cospans} \operatorname{id}_a \in \operatorname{Set}(a; a)$	(not rendered)	(iv) identities $\operatorname{id}_a \in \operatorname{Set}(a; a)$ just given by the identity $\operatorname{id}_a : a \rightarrow a$.
fong-spivak-invitation_F02350	228	■ 0.922	$\operatorname{FinSet}, 0, +$	$\operatorname{FinSet}, 0, +$	This is the operadic analogue of the monoidal category $(\operatorname{Cospan}_{\operatorname{FinSet}}, 0, +)$.
fong-spivak-invitation_F02351	229	■ 0.536	$\underline{3}, \underline{3}, \underline{4}, \underline{2} ; \underline{3}$	$\underline{3}, \underline{3}, \underline{4}, \underline{2} ; \underline{3}$	This is an operation of arity $(\underline{3}, \underline{3}, \underline{4}, \underline{2}; \underline{3})$. Why? The circles marked f and g have 3
fong-spivak-invitation_F02352	229	■ 0.999	$\underline{3} + \underline{3} + \underline{4} + \underline{2} \rightarrow \underline{p} \leftarrow \underline{3}$	$\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$	arity is, by (ii), a cospan $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$. In the diagram above, the apex $\underline{p}$ is
fong-spivak-invitation_F02353	229	■ 0.999	$\underline{p}$	$\underline{p}$	arity is, by (ii), a cospan $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$. In the diagram above, the apex $\underline{p}$ is
fong-spivak-invitation_F02354	229	■ 0.973	$\underline{7}$	$\underline{7}$	the set $\underline{7}$, because there are 7 nodes $\bullet$ in the diagram. The function a sends each port on
fong-spivak-invitation_F02355	229	■ 0.972	$f \in \operatorname{Cospan}(2, 2 ; 2)$	$f \in \operatorname{Cospan}(2, 2; 2)$	1. Consider the following cospan $f \in \operatorname{Cospan}(2, 2; 2)$:
fong-spivak-invitation_F02356	229	■ 1.000	$g \in$	$g \in$	2. Draw the wiring diagram corresponding to the following cospan $g \in$
fong-spivak-invitation_F02357	229	■ 0.999	$\operatorname{Cospan}(2, 2, 2 ; 0)$	$\operatorname{Cospan}(2, 2, 2; 0)$	$\operatorname{Cospan}(2, 2, 2; 0)$:
fong-spivak-invitation_F02358	229	■ 0.706	$g \circ_{\{1\}} f$	$g \circ_1 f$	3. Compute the cospan $g \circ_1 f$. What is its arity?
fong-spivak-invitation_F02359	229	■ 0.589	$\mathcal{O}_{\mathcal{C}}$	$\mathcal{O}_{\mathcal{C}}$	$\mathcal{O}_{\mathcal{C}}$, called the <i>operad underlying</i> $\mathcal{C}$, defined as having:
fong-spivak-invitation_F02360	229	■ 0.957	$\operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{C})$	(i) $\operatorname{Ob}(\mathcal{C})$ as types,
fong-spivak-invitation_F02361	229	■ 0.940	$C_{\{1\}} \otimes \cdots \otimes C_{\{n\}} \rightarrow D$	$C_1 \otimes \cdots \otimes C_n \rightarrow D$	(ii) morphisms $C_1 \otimes \cdots \otimes C_n \rightarrow D$ in $\mathcal{C}$ as the operations of arity $(C_1, \dots, C_n; D)$,
fong-spivak-invitation_F02362	229	■ 0.940	$C_{\{1\}}, \ldots, C_{\{n\}} ; D$	$C_1, \dots, C_n; D$	(ii) morphisms $C_1 \otimes \cdots \otimes C_n \rightarrow D$ in $\mathcal{C}$ as the operations of arity $(C_1, \dots, C_n; D)$,
fong-spivak-invitation_F02363	230	■ 0.800	$\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a ; a)$	$\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$	(iv) identities $\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$ defined by id_a .

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fong-spivak-invitation_F02364	230	0.800	$\mathrm{id}_{\{a\}}$	id_a	(iv) identities $\mathrm{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$ defined by id_a .
fong-spivak-invitation_F02365	230	0.997	$F: \mathcal{O} \rightarrow \mathcal{P}$	$F: \mathcal{O} \rightarrow \mathcal{P}$	collections T and U respectively. To specify an operad functor $F: \mathcal{O} \rightarrow \mathcal{P}$,
fong-spivak-invitation_F02366	230	1.000	$f: T \rightarrow U$	$f: T \rightarrow U$	(i) one specifies a function $f: T \rightarrow U$,
fong-spivak-invitation_F02367	230	1.000	$F: \mathcal{O} \rightarrow$	$F: \mathcal{O} \rightarrow$	Definition 6.77. An algebra for an operad $\mathcal{O}$ is an operad functor $F: \mathcal{O} \rightarrow \mathbf{Set}$.
fong-spivak-invitation_F02368	230	1.000	$F(t)$	$F(t)$	operad $\mathcal{O}$ and an algebra $F: \mathcal{O} \rightarrow \mathbf{Set}$ will take a type t and return a set $F(t)$ of fillers
fong-spivak-invitation_F02369	230	1.000	$f \in \mathcal{O}(\mathrm{t}_1, \dots, \mathrm{t}_n; \mathrm{t})$	$f \in \mathcal{O}(t_1, \dots, t_n; t)$	for box t . Moreover, given an operation (i.e. a wiring diagram) $f \in \mathcal{O}(t_1, \dots, t_n; t)$, we
fong-spivak-invitation_F02370	230	1.000	$F(f)$	$F(f)$	get a function $F(f)$ that takes an element of each set $F(t_i)$, and returns an element of
fong-spivak-invitation_F02371	230	1.000	$F(\mathrm{t}_i)$	$F(t_i)$	get a function $F(f)$ that takes an element of each set $F(t_i)$, and returns an element of
fong-spivak-invitation_F02372	230	1.000	$\mathrm{t}_1, \dots, \mathrm{t}_n$	$t_1, \dots, t_n$	$F(t)$. For example, it takes n circuits with interface $t_1, \dots, t_n$, respectively, and returns
fong-spivak-invitation_F02373	230	0.998	$T = \mathrm{Ob}$	$T = \mathrm{Ob}$	Example 6.78. For electric circuits, the types are again finite sets, $T = \mathrm{Ob}(\mathbf{FinSet})$,
fong-spivak-invitation_F02374	230	0.691	$\mathrm{Circ}(t)$	$\mathrm{Circ}(t)$	have a set $\mathrm{Circ}(t)$ of fillers, namely the set of electric circuits with that t -marked ter-
fong-spivak-invitation_F02375	231	0.623	$\mathrm{Circ}(\varphi) \in \mathrm{Set}(\mathrm{Circ}(2))$	$\mathrm{Circ}(\varphi) \in \mathrm{Set}(\mathrm{Circ}(2))$	above-drawn case, we would get a morphism $\mathrm{Circ}(\varphi) \in \mathbf{Set}(\mathrm{Circ}(2), \mathrm{Circ}(2), \mathrm{Circ}(2))$;
fong-spivak-invitation_F02376	231	0.623	$\mathrm{Circ}(2)$	$\mathrm{Circ}(2)$	above-drawn case, we would get a morphism $\mathrm{Circ}(\varphi) \in \mathbf{Set}(\mathrm{Circ}(2), \mathrm{Circ}(2), \mathrm{Circ}(2))$;
fong-spivak-invitation_F02377	231	0.490	$\mathrm{Circ}(0)$	$\mathrm{Circ}(0)$	$\mathrm{Circ}(0)$, i.e. a function
fong-spivak-invitation_F02378	233	1.000	$\in \mathbb{R}$	$\in \mathbb{R}$	safe $\in \mathbb{R}$ from other objects. To do so, several components must interact: a sensor
fong-spivak-invitation_F02379	233	1.000	$S^{\prime}$	$S^{\prime}$	that approximates the real distance by an internal variable $S^{\prime}$, a controller that uses $S^{\prime}$ to
fong-spivak-invitation_F02380	234	0.953	$S \geq$	$S \geq$	In order for the system to ensure $S \geq \text{safe}$, we need each of the components
fong-spivak-invitation_F02381	234	0.922	$S < \text{safe}$	$S < \text{safe}$	signal A , e.g. that if at any time $S < \text{safe}$, then within one second it will emit the sig-
fong-spivak-invitation_F02382	234	0.984	$A =$	$A =$	nal $A = \text{go}$. The motor maintains a relationship between A and S , e.g. that A dictates
fong-spivak-invitation_F02383	235	0.911	$[a, b]$	$[a, b]$	• S (wire): The behavior of S over a time-interval $[a, b]$ is that of all continuous real-
fong-spivak-invitation_F02384	235	1.000	$[a, b] \rightarrow \mathbb{R}$	$[a, b] \rightarrow \mathbb{R}$	valued functions $[a, b] \rightarrow \mathbb{R}$.
fong-spivak-invitation_F02385	235	0.998	$\left\{ \left(S^{\prime}, A \right) \mid \right.$	$\{(S^{\prime}, A) \mid$	• controller (box): The relation $\{(S^{\prime}, A) \mid \text{Eq. (7.2)}\}$, i.e. all behavioral pairs $(S^{\prime}, A)$
fong-spivak-invitation_F02386	235	0.998	$\left. \left(S^{\prime}, A \right) \right\}$	$\left. (S^{\prime}, A) \right\}$	• controller (box): The relation $\{(S^{\prime}, A) \mid \text{Eq. (7.2)}\}$, i.e. all behavioral pairs $(S^{\prime}, A)$
fong-spivak-invitation_F02387	236	0.929	$\in P$	$\in P$	people to the set $\mathbb{B} = \{\text{false}, \text{true}\}$ of truth values, where Brendan $\in P$ maps to
fong-spivak-invitation_F02388	236	1.000	$\mathcal{E}$	$\mathcal{E}$	in many ways. Here are some facts that are true of any topos $\mathcal{E}$:
fong-spivak-invitation_F02389	236	1.000	$1 \rightarrow \Omega$	$1 \xrightarrow{\text{true}} \Omega$	5. $\mathcal{E}$ has a subobject classifier $1 \xrightarrow{\text{true}} \Omega$.
fong-spivak-invitation_F02390	236	0.996	$\mathcal{E} =$	$\mathcal{E} =$	In particular, since $\mathbf{Set}$ is a topos, all of the above facts are true for $\mathcal{E} = \mathbf{Set}$. Our first

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fong-spivak-invitation_F02391	237	0.833	$\lrcorner$	$\lrcorner$	In the left-hand square, the corner symbol $\lrcorner$ unambiguously means that the square
fong-spivak-invitation_F02392	237	0.999	(B, C, E, F)	(B, C, E, F)	(B, C, E, F) is a pullback. But in the right-hand square, does the corner symbol mean
fong-spivak-invitation_F02393	237	0.876	(A, B, D, E)	(A, B, D, E)	that (A, B, D, E) is a pullback or that (A, C, D, F) is a pullback? It's ambiguous but,
fong-spivak-invitation_F02394	237	0.876	(A, C, D, F)	(A, C, D, F)	that (A, B, D, E) is a pullback or that (A, C, D, F) is a pullback? It's ambiguous but,
fong-spivak-invitation_F02395	237	0.989	$\left(B, C, B^{\prime}, C^{\prime}\right)$	(B, C, B', C')	Proposition 7.1. In the commutative diagram below, suppose that the (B, C, B', C')
fong-spivak-invitation_F02396	237	0.811	$A, B, A^{\prime}, B^{\prime}$	A, B, A', B'	Then the (A, B, A', B') square is a pullback iff the (A, C, A', C') rectangle is a
fong-spivak-invitation_F02397	237	0.811	$A, C, A^{\prime}, C^{\prime}$	A, C, A', C'	Then the (A, B, A', B') square is a pullback iff the (A, C, A', C') rectangle is a
fong-spivak-invitation_F02398	238	1.000	$i: B^{\prime} \rightarrow B$	$i: B' \rightarrow B$	That is, suppose that $i: B' \rightarrow B$ is an isomorphism and $f: A \rightarrow B$ is any morphism.
fong-spivak-invitation_F02399	238	1.000	$i^{\prime}$	i'	Show that i' is an isomorphism, in the following diagram:
fong-spivak-invitation_F02400	238	1.000	$f^{\prime}$	f'	is f' .
fong-spivak-invitation_F02401	238	1.000	$f: C \rightarrow D$	$f: C \rightarrow D$	mono factorizations are. We say that a morphism $f: C \rightarrow D$ in $\mathcal{E}$ has an epi-mono
fong-spivak-invitation_F02402	238	1.000	$\operatorname{im}(f)$	$\operatorname{im}(f)$	factorization if it has an “image”; that is, there is an object $\operatorname{im}(f)$, an epimorphism
fong-spivak-invitation_F02403	238	0.811	$C \rightarrow \operatorname{im}(f)$	$C \rightarrow \operatorname{im}(f)$	$C \twoheadrightarrow \operatorname{im}(f)$, and a monomorphism $\operatorname{im}(f) \hookrightarrow D$, whose composite is f .
fong-spivak-invitation_F02404	238	0.811	$\operatorname{im}(f) \rightarrowtail D$	$\operatorname{im}(f) \hookrightarrow D$	$C \twoheadrightarrow \operatorname{im}(f)$, and a monomorphism $\operatorname{im}(f) \hookrightarrow D$, whose composite is f .
fong-spivak-invitation_F02405	238	0.938	$\operatorname{im}(f)=\{f(c) \mid$	$\operatorname{im}(f)=\{f(c) \mid$	tion $f: C \rightarrow D$ may be factored as a surjective function onto its image $\operatorname{im}(f)=\{f(c)$
fong-spivak-invitation_F02406	238	1.000	$c \in C\}$	$c \in C\}$	$c \in C\}$, followed by the inclusion of this image into the codomain D . Moreover, this
fong-spivak-invitation_F02407	239	1.000	$f: \underline{3} \rightarrow \underline{3}$	$f: \underline{3} \rightarrow \underline{3}$	Exercise 7.7. Factor the following function $f: \underline{3} \rightarrow \underline{3}$ as an epimorphism followed by
fong-spivak-invitation_F02408	239	1.000	$f=(e ; m)$	$f=(e ; m)$	epimorphism e and a monomorphism m such that $f=(e \circ m)$ is their composite.
fong-spivak-invitation_F02409	239	1.000	$C, D \in \mathcal{C}$	$C, D \in \mathcal{C}$	two objects $C, D \in \mathcal{C}$, there is a “hom-object” $D^C \in \mathcal{C}$ and a natural isomorphism for
fong-spivak-invitation_F02410	239	1.000	$D^{\mathcal{C}} \in \mathcal{C}$	$D^C \in \mathcal{C}$	two objects $C, D \in \mathcal{C}$, there is a “hom-object” $D^C \in \mathcal{C}$ and a natural isomorphism for
fong-spivak-invitation_F02411	239	1.000	$A \in \mathcal{C}$	$A \in \mathcal{C}$	any $A \in \mathcal{C}$:
fong-spivak-invitation_F02412	239	1.000	$f(-,-): A \times C \rightarrow D$	$f(-,-): A \times C \rightarrow D$	game $f(-,-): A \times C \rightarrow D$: for whatever action $a \in A$ that you take and action $c \in C$
fong-spivak-invitation_F02413	239	1.000	$f(a, c)$	$f(a, c)$	that I take, $f(a, c)$ is some value in D . Since you're self-centered but loving, you think
fong-spivak-invitation_F02414	239	1.000	$f(a,-): C \rightarrow D$	$f(a,-): C \rightarrow D$	you make a game $f(a,-): C \rightarrow D$ for me alone. In the formalism, D^C represents the
fong-spivak-invitation_F02415	239	1.000	$D^{\mathcal{C}}$	D^C	you make a game $f(a,-): C \rightarrow D$ for me alone. In the formalism, D^C represents the
fong-spivak-invitation_F02416	239	1.000	$v, w, x \in V$	$v, w, x \in V$	Definition 2.66 – and suppose it satisfies the following for all $v, w, x \in V$:
fong-spivak-invitation_F02417	239	1.000	$v \leq I$	$v \leq I$	• $v \leq I$,

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fong-spivak-invitation_F02418	239	■ 1.000	$v \otimes w \leq v$	$v \otimes w \leq v$	$v \otimes w \leq v$ and $v \otimes w \leq w$,
fong-spivak-invitation_F02419	239	■ 1.000	$v \otimes w \leq w$	$v \otimes w \leq w$	$v \otimes w \leq v$ and $v \otimes w \leq w$,
fong-spivak-invitation_F02420	239	■ 1.000	$x \leq v$	$x \leq v$	if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$.
fong-spivak-invitation_F02421	239	■ 1.000	$x \leq w$	$x \leq w$	if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$.
fong-spivak-invitation_F02422	239	■ 1.000	$x \leq v \otimes w$	$x \leq v \otimes w$	if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$.
fong-spivak-invitation_F02423	240	■ 0.781	$X \hookrightarrow Y$	$X \hookrightarrow Y$	are injections, and any injection $X \hookrightarrow Y$ is isomorphic to a subset of Y . This gives
fong-spivak-invitation_F02424	240	■ 0.822	$\Omega \in \mathcal{E}$	$\Omega \in \mathcal{E}$	nal object 1 . A <i>subobject classifier</i> in $\mathcal{E}$ consists of an object $\Omega \in \mathcal{E}$, together with a
fong-spivak-invitation_F02425	240	■ 1.000	$1 \rightarrow \Omega$	$1 \rightarrow \Omega$	monomorphism $\text{true}: 1 \rightarrow \Omega$, satisfying the following property: for any objects X
fong-spivak-invitation_F02426	240	■ 0.496	$m: X \rightsquigarrow Y$	$m: X \rightsquigarrow Y$	and Y and monomorphism $m: X \rightarrow Y$ in $\mathcal{E}$, there is a unique morphism $\ulcorner m \urcorner: Y \rightarrow \Omega$
fong-spivak-invitation_F02427	240	■ 0.496	$\ulcorner m \urcorner: Y \rightarrow \Omega$	$\ulcorner m \urcorner: Y \rightarrow \Omega$	and Y and monomorphism $m: X \rightarrow Y$ in $\mathcal{E}$, there is a unique morphism $\ulcorner m \urcorner: Y \rightarrow \Omega$
fong-spivak-invitation_F02428	240	■ 0.999	$\ulcorner m \urcorner$	$\ulcorner m \urcorner$	We refer to $\ulcorner m \urcorner$ as the <i>characteristic map</i> of m , or we say that $\ulcorner m \urcorner$ <i>classifies</i> m . Con-
fong-spivak-invitation_F02429	240	■ 0.999	$p: Y \rightarrow \Omega$	$p: Y \rightarrow \Omega$	versely, given any map $p: Y \rightarrow \Omega$, we denote the pullback of true as on the right of
fong-spivak-invitation_F02430	240	■ 1.000	$Y \rightarrow \Omega$	$Y \rightarrow \Omega$	A <i>predicate</i> on Y is a morphism $Y \rightarrow \Omega$.
fong-spivak-invitation_F02431	241	■ 1.000	$1 \rightarrow$	$1 \rightarrow$	generically called $\text{true}: 1 \rightarrow \Omega$; in the case of Set it is played by the function $1 \rightarrow$
fong-spivak-invitation_F02432	241	■ 0.976	$m: X \hookrightarrow Y$	$m: X \hookrightarrow Y$	tion 7.9 says that, for any injective function $m: X \hookrightarrow Y$ between sets, we are supposed
fong-spivak-invitation_F02433	241	■ 0.968	$\ulcorner m \urcorner: Y \rightarrow \{$	$\ulcorner m \urcorner: Y \rightarrow \{$	to be able to find a characteristic function $\ulcorner m \urcorner: Y \rightarrow \{\text{true}, \text{false}\}$ with some sort
fong-spivak-invitation_F02434	241	■ 1.000	$\ulcorner m \urcorner(y)$	$\ulcorner m \urcorner(y)$	In other words, if we think of X as a subset of Y , then we make $\ulcorner m \urcorner(y)$ equal to true
fong-spivak-invitation_F02435	241	■ 1.000	$y \in X$	$y \in X$	iff $y \in X$.
fong-spivak-invitation_F02436	241	■ 1.000	$p: Y \rightarrow \mathbb{B}$	$p: Y \rightarrow \mathbb{B}$	$p: Y \rightarrow \mathbb{B}$, and vice versa. How it works is encoded in Definition 7.9, but the basic idea
fong-spivak-invitation_F02437	241	■ 1.000	$\{Y \mid p\}$	$\{Y \mid p\}$	This might help explain our abstract notation $\{Y \mid p\}$ in Eq. (7.4).
fong-spivak-invitation_F02438	241	■ 1.000	$X = \mathbb{N} = \{0, 1, 2, \dots\}$	$X = \mathbb{N} = \{0, 1, 2, \dots\}$	Exercise 7.10. Let $X = \mathbb{N} = \{0, 1, 2, \dots\}$ and $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$; we
fong-spivak-invitation_F02439	241	■ 1.000	$Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$	$Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$	Exercise 7.10. Let $X = \mathbb{N} = \{0, 1, 2, \dots\}$ and $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$; we
fong-spivak-invitation_F02440	241	■ 0.995	$m: X \succ Y$	$m: X \succ Y$	have $X \subseteq Y$, so consider it as a monomorphism $m: X \hookrightarrow Y$. It has a characteristic
fong-spivak-invitation_F02441	241	■ 1.000	$\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$	$\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$	function $\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$, as in Definition 7.9.
fong-spivak-invitation_F02442	241	■ 1.000	$\ulcorner m \urcorner(-5) \in \mathbb{B}$	$\ulcorner m \urcorner(-5) \in \mathbb{B}$	1. What is $\ulcorner m \urcorner(-5) \in \mathbb{B}$?
fong-spivak-invitation_F02443	241	■ 1.000	$\ulcorner m \urcorner(0) \in \mathbb{B}$	$\ulcorner m \urcorner(0) \in \mathbb{B}$	2. What is $\ulcorner m \urcorner(0) \in \mathbb{B}$?
fong-spivak-invitation_F02444	241	■ 0.998	$\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$	$\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$	1. Consider the identity function $\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$. It is an injection, so it has a characteristic

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fong-spivak-invitation_F02445	241	0.808	$\left\ulcorner \mathrm{id}_{\mathbb{N}} \right\urcorner : \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$	function $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$. Give a concrete description of $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner$, i.e. its exact value for
fong-spivak-invitation_F02446	241	0.808	$\left\ulcorner \mathrm{id}_{\mathbb{N}} \right\urcorner$	$\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner$	function $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$. Give a concrete description of $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner$, i.e. its exact value for
fong-spivak-invitation_F02447	241	1.000	$!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$	$!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$	2. Consider the unique function $!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$ from the empty set. Give a concrete
fong-spivak-invitation_F02448	241	0.654	$\left\ulcorner !_{\mathbb{N}} \right\urcorner : \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$	description of $\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$.
fong-spivak-invitation_F02449	242	1.000	$1 \rightarrow \mathbb{B} \times \mathbb{B}$	$1 \rightarrow \mathbb{B} \times \mathbb{B}$	Consider the function $1 \rightarrow \mathbb{B} \times \mathbb{B}$ picking out the element $(\mathrm{true}, \mathrm{true})$. This is a
fong-spivak-invitation_F02450	242	0.994	$\urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	monomorphism, so it defines a characteristic function $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$.
fong-spivak-invitation_F02451	242	1.000	$\mathbb{B} \times \mathbb{B}$	$\mathbb{B} \times \mathbb{B}$	What function is it? By Eq. (7.6) the only element of $\mathbb{B} \times \mathbb{B}$ that can be sent to true
fong-spivak-invitation_F02452	242	0.944	$\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$	$\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$	is $(\mathrm{true}, \mathrm{true})$. Thus $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$ must be given by the following
fong-spivak-invitation_F02453	242	0.944	$\urcorner (P, Q) \in \mathbb{B}$	$\urcorner (P, Q) \in \mathbb{B}$	is $(\mathrm{true}, \mathrm{true})$. Thus $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$ must be given by the following
fong-spivak-invitation_F02455	242	1.000	$P \wedge Q$	$P \wedge Q$	This is exactly the truth table for the AND of P and Q , i.e. for $P \wedge Q$. In other words,
fong-spivak-invitation_F02456	242	0.929	$\urcorner = \wedge$	$\urcorner = \wedge$	$\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner = \wedge$. Note that this defines $\wedge$ as a function $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$, and we
fong-spivak-invitation_F02457	242	0.929	$\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner = \wedge$. Note that this defines $\wedge$ as a function $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$, and we
fong-spivak-invitation_F02458	242	1.000	$x \wedge y := \wedge(x, y)$	$x \wedge y := \wedge(x, y)$	use the usual infix notation $x \wedge y := \wedge(x, y)$.
fong-spivak-invitation_F02459	242	1.000	$\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	function $\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ defining OR, is:
fong-spivak-invitation_F02461	242	1.000	$m : X \subseteq \mathbb{B} \times \mathbb{B}$	$m : X \subseteq \mathbb{B} \times \mathbb{B}$	set $m : X \subseteq \mathbb{B} \times \mathbb{B}$, what subset would X be? By Eq. (7.6), X should be the set of
fong-spivak-invitation_F02462	242	0.912	$\mathbb{B} \times$	$\mathbb{B} \times$	tion 7.2.1. In fact, one can think of X as the union of $\{\mathrm{true}\} \times \mathbb{B}$ and $\mathbb{B} \times$
fong-spivak-invitation_F02463	242	1.000	$\Omega \times \Omega$	$\Omega \times \Omega$	gous subobject of $\Omega \times \Omega$, and hence classify a map $\Omega \times \Omega \rightarrow \Omega$, in any
fong-spivak-invitation_F02464	242	1.000	$\Omega \times \Omega \rightarrow \Omega$	$\Omega \times \Omega \rightarrow \Omega$	gous subobject of $\Omega \times \Omega$, and hence classify a map $\Omega \times \Omega \rightarrow \Omega$, in any
fong-spivak-invitation_F02465	243	1.000	$\neg : \mathbb{B} \rightarrow \mathbb{B}$	$\neg : \mathbb{B} \rightarrow \mathbb{B}$	false . The function $\neg : \mathbb{B} \rightarrow \mathbb{B}$ is the characteristic function of some thing, (*?*)
fong-spivak-invitation_F02466	243	1.000	$P \Rightarrow Q$	$P \Rightarrow Q$	Exercise 7.13. Given two booleans P, Q , define $P \Rightarrow Q$ to mean $P = (P \wedge Q)$.
fong-spivak-invitation_F02467	243	1.000	$P = (P \wedge Q)$	$P = (P \wedge Q)$	Exercise 7.13. Given two booleans P, Q , define $P \Rightarrow Q$ to mean $P = (P \wedge Q)$.
fong-spivak-invitation_F02468	243	1.000	$m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	3. What is the characteristic function $m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ for $P \Rightarrow Q$?
fong-spivak-invitation_F02469	243	0.999	$E := \{n \in \mathbb{N} \mid n \text{ is even}\}$	$E := \{n \in \mathbb{N} \mid n$	Exercise 7.14. Consider the sets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$, $P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$,
fong-spivak-invitation_F02470	243	0.999	$\}, P := \{n \in \mathbb{N} \mid n$	$\}, P := \{n \in \mathbb{N} \mid n$	Exercise 7.14. Consider the sets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$, $P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$,
fong-spivak-invitation_F02471	243	1.000	$T := \{n \in \mathbb{N} \mid n \geq 10\}$	$T := \{n \in \mathbb{N} \mid n \geq 10\}$	and $T := \{n \in \mathbb{N} \mid n > 10\}$. Each is a subset of $\mathbb{N}$, so defines a function $\mathbb{N} \rightarrow \mathbb{B}$.
fong-spivak-invitation_F02472	243	1.000	$\mathbb{N} \rightarrow \mathbb{B}$	$\mathbb{N} \rightarrow \mathbb{B}$	and $T := \{n \in \mathbb{N} \mid n \geq 10\}$. Each is a subset of $\mathbb{N}$, so defines a function $\mathbb{N} \rightarrow \mathbb{B}$.
fong-spivak-invitation_F02473	243	0.992	$\ulcorner E \urcorner(17)$	$\ulcorner E \urcorner(17)$	1. What is $\ulcorner E \urcorner(17)$?
fong-spivak-invitation_F02474	243	0.995	$\ulcorner P \urcorner(17)$	$\ulcorner P \urcorner(17)$	2. What is $\ulcorner P \urcorner(17)$?

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fong-spivak-invitation_F02475	243	■ 1.000	$\ulcorner T \urcorner(17)$	$\lceil T \rceil(17)$	3. What is $\lceil T \rceil(17)$?
fong-spivak-invitation_F02476	243	■ 1.000	$(\ulcorner E \urcorner \wedge \ulcorner P \urcorner) \vee \ulcorner T \urcorner$	$(\lceil E \rceil \wedge \lceil P \rceil) \vee \lceil T \rceil$	4. Name the smallest three elements in the set classified by $\lceil E \rceil \wedge \lceil P \rceil \vee \lceil T \rceil$.
fong-spivak-invitation_F02477	243	■ 1.000	$\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$	$\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$	how to understand logic in terms of Ω : there are various maps $\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$
fong-spivak-invitation_F02478	243	■ 1.000	$\neg: \Omega \rightarrow \Omega$	$\neg: \Omega \rightarrow \Omega$	and $\neg: \Omega \rightarrow \Omega$, etc., which serve as logical connectives. These are operations on truth
fong-spivak-invitation_F02479	244	■ 1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	of the plane $\mathbb{R}^2$ as a space, we might examine only points and regions in it. But if we
fong-spivak-invitation_F02480	244	■ 0.163	$\mathcal{C}^{\mathrm{op}} \rightarrow$	$\mathcal{C}^{\mathrm{op}} \rightarrow$	$\mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$. ⁵ Such functors are called <i>presheaves on $\mathcal{C}$</i> .
fong-spivak-invitation_F02481	244	■ 0.605	$P: \mathcal{C}^{\mathrm{op}} \rightarrow$	$P: \mathcal{C}^{\mathrm{op}} \rightarrow$	$P: \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$. To each object $c \in \mathcal{C}$, we refer to the set $P(c)$ as <i>the set of</i>
fong-spivak-invitation_F02482	244	■ 0.605	$P(c)$	$P(c)$	$P: \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$. To each object $c \in \mathcal{C}$, we refer to the set $P(c)$ as <i>the set of</i>
fong-spivak-invitation_F02483	244	■ 1.000	$P(f): P(c) \rightarrow P(c')$	$P(f): P(c) \rightarrow P(c')$	$P(f): P(c) \rightarrow P(c')$ as the <i>restriction map along f</i> . For any section $s \in P(c)$,
fong-spivak-invitation_F02484	244	■ 1.000	$s \in P(c)$	$s \in P(c)$	$P(f): P(c) \rightarrow P(c')$ as the <i>restriction map along f</i> . For any section $s \in P(c)$,
fong-spivak-invitation_F02485	244	■ 0.458	$\mathcal{C}^{\mathrm{op}}$	$\mathcal{C}^{\mathrm{op}}$	⁵ The category of functors $\mathcal{C} \rightarrow \mathbf{Set}$ is also a topos: use $\mathcal{C}^{\mathrm{op}}$ as the defining site.
fong-spivak-invitation_F02486	245	■ 1.000	$P(f)(s) \in P(c')$	$P(f)(s) \in P(c')$	we may denote $P(f)(s) \in P(c')$, i.e. its restriction along f , by $s _f$. If P and Q
fong-spivak-invitation_F02487	245	■ 1.000	$s _f$	$s _f$	we may denote $P(f)(s) \in P(c')$, i.e. its restriction along f , by $s _f$. If P and Q
fong-spivak-invitation_F02488	245	■ 1.000	$\alpha: P \rightarrow Q$	$\alpha: P \rightarrow Q$	are presheaves, a <i>morphism</i> $\alpha: P \rightarrow Q$ between them is a natural transformation of
fong-spivak-invitation_F02489	245	■ 0.402	$\mathbf{ArShp}^{\mathrm{op}} \rightarrow$	$I: \mathbf{ArShp}^{\mathrm{op}} \rightarrow$	A presheaf on $\mathbf{ArShp}$ is a functor $I: \mathbf{ArShp}^{\mathrm{op}} \rightarrow \mathbf{Set}$, which is a database instance on
fong-spivak-invitation_F02490	246	■ 1.000	$P(X) = \{U \subseteq X\}$	$P(X) = \{U \subseteq X\}$	Definition 7.17. Let X be a set, and let $P(X) = \{U \subseteq X\}$ denote its set of subsets. A
fong-spivak-invitation_F02491	246	■ 0.999	$\mathbf{Op} \subseteq P(X)$	$\mathbf{Op} \subseteq P(X)$	<i>topology</i> on X is a subset $\mathbf{Op} \subseteq P(X)$, elements of which we call <i>open sets</i> , ⁶ satisfying
fong-spivak-invitation_F02492	246	■ 1.000	$X \in \mathbf{Op}$	$X \subseteq X$	(a) Whole set: the subset $X \subseteq X$ is open, i.e. $X \in \mathbf{Op}$.
fong-spivak-invitation_F02493	246	■ 1.000	$X \in \mathbf{Op}$	$X \in \mathbf{Op}$	(a) Whole set: the subset $X \subseteq X$ is open, i.e. $X \in \mathbf{Op}$.
fong-spivak-invitation_F02494	246	■ 0.984	$U, V \in \mathbf{Op}$	$U, V \in \mathbf{Op}$	(b) Binary intersections: if $U, V \in \mathbf{Op}$ then $(U \cap V) \in \mathbf{Op}$.
fong-spivak-invitation_F02495	246	■ 0.984	$(U \cap V) \in \mathbf{Op}$	$(U \cap V) \in \mathbf{Op}$	(b) Binary intersections: if $U, V \in \mathbf{Op}$ then $(U \cap V) \in \mathbf{Op}$.
fong-spivak-invitation_F02496	246	■ 1.000	$U_i \in \mathbf{Op}$	$U_i \in \mathbf{Op}$	(c) Arbitrary unions: if I is a set and if we are given an open set $U_i \in \mathbf{Op}$ for each
fong-spivak-invitation_F02497	246	■ 1.000	$\bigcup_{i \in I} U_i \in \mathbf{Op}$	$\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$	i , then their union is also open, $\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$. We interpret the particular case
fong-spivak-invitation_F02498	246	■ 0.915	$I = \emptyset$	$I = \emptyset$	where $I = \emptyset$ to mean that the empty set is open: $\emptyset \in \mathbf{Op}$.
fong-spivak-invitation_F02499	246	■ 0.915	$\emptyset \in \mathbf{Op}$	$\emptyset \in \mathbf{Op}$	where $I = \emptyset$ to mean that the empty set is open: $\emptyset \in \mathbf{Op}$.
fong-spivak-invitation_F02500	246	■ 1.000	$U = \bigcup_{i \in I} U_i$	$U = \bigcup_{i \in I} U_i$	If $U = \bigcup_{i \in I} U_i$, we say that $(U_i)_{i \in I}$ <i>covers</i> U .

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fong-spivak-invitation_F02501	246	■ 1.000	$\left(U_{\{i\}}\right)_{\{i \in I\}}$	$(U_i)_{i \in I}$	If $U = \bigcup_{i \in I} U_i$, we say that $(U_i)_{i \in I}$ covers U .
fong-spivak-invitation_F02502	246	■ 0.932	$X, \mathbf{0\ p}$	$X, \mathbf{Op}$	A pair $(X, \mathbf{Op})$, where X is a set and $\mathbf{Op}$ is a topology on X , is called a <i>topological</i>
fong-spivak-invitation_F02503	246	■ 0.932	$\mathbf{0\ p}$	$\mathbf{Op}$	A pair $(X, \mathbf{Op})$, where X is a set and $\mathbf{Op}$ is a topology on X , is called a <i>topological</i>
fong-spivak-invitation_F02504	246	■ 1.000	$X, \mathbf{0\ p}_{\{X\}}$	$X, \mathbf{Op}_X$	A <i>continuous function</i> between topological spaces $(X, \mathbf{Op}_X)$ and $(Y, \mathbf{Op}_Y)$ is a func-
fong-spivak-invitation_F02505	246	■ 1.000	$Y, \mathbf{0\ p}_{\{Y\}}$	$Y, \mathbf{Op}_Y$	A <i>continuous function</i> between topological spaces $(X, \mathbf{Op}_X)$ and $(Y, \mathbf{Op}_Y)$ is a func-
fong-spivak-invitation_F02506	246	■ 0.999	$U \in \mathbf{0\ p}_{\{Y\}}$	$U \in \mathbf{Op}_Y$	tion $f: X \rightarrow Y$ such that, for every $U \in \mathbf{Op}_Y$, the preimage $f^{-1}(U)$ is in
fong-spivak-invitation_F02507	246	■ 0.999	$f^{-1}(U)$	$f^{-1}(U)$	tion $f: X \rightarrow Y$ such that, for every $U \in \mathbf{Op}_Y$, the preimage $f^{-1}(U)$ is in
fong-spivak-invitation_F02508	246	■ 1.000	$\mathbf{0\ p}_{\{X\}}$	$\mathbf{Op}_X$	$\mathbf{Op}_X$.
fong-spivak-invitation_F02509	246	■ 0.999	$U \in \mathbf{0\ p}$	$U \in \mathbf{Op}$	⁶ In other words, we refer to a subset $U \subseteq X$ as <i>open</i> if $U \in \mathbf{Op}$.
fong-spivak-invitation_F02510	247	■ 0.936	$\epsilon \in \mathbb{R}$	$\epsilon \in \mathbb{R}$	Example 7.18. The usual topology $\mathbf{Op}$ on $\mathbb{R}^2$ is based on “ ϵ -balls.” For any $\epsilon \in \mathbb{R}$
fong-spivak-invitation_F02511	247	■ 0.985	$\epsilon > 0$	$\epsilon > 0$	with $\epsilon > 0$, and any point $p = (x, y) \in \mathbb{R}^2$, define the ϵ -ball centered at p
fong-spivak-invitation_F02512	247	■ 0.985	$p=(x, y) \in \mathbb{R}^{\{2\}}$	$p = (x, y) \in \mathbb{R}^2$	with $\epsilon > 0$, and any point $p = (x, y) \in \mathbb{R}^2$, define the ϵ -ball centered at p
fong-spivak-invitation_F02513	247	■ 1.000	$B(x, y ; \epsilon)$	$B(x, y; \epsilon)$	In other words, $B(x, y; \epsilon)$ is the set of all points within ϵ of (x, y) .
fong-spivak-invitation_F02514	247	■ 0.796	$U \subseteq \mathbb{R}^{\{2\}}$	$U \subseteq \mathbb{R}^2$	For an arbitrary subset $U \subseteq \mathbb{R}^2$, we call it open and put it in $\mathbf{Op}$ if, for every $(x, y) \in$
fong-spivak-invitation_F02515	247	■ 0.796	$(x, y) \in$	$(x, y) \in$	For an arbitrary subset $U \subseteq \mathbb{R}^2$, we call it open and put it in $\mathbf{Op}$ if, for every $(x, y) \in$
fong-spivak-invitation_F02516	247	■ 1.000	$B(x, y ; \epsilon) \subseteq U$	$B(x, y; \epsilon) \subseteq U$	U , there exists a (small enough) $\epsilon > 0$ such that $B(x, y; \epsilon) \subset U$.
fong-spivak-invitation_F02517	247	■ 1.000	$d\left(x_{\{1\}}, x_{\{2\}}\right):=\left x_{\{1\}}-x_{\{2\}}\right $	$d\left(x_1, x_2\right):=\left x_1-x_2\right $	Exercise 7.19. Consider the set $\mathbb{R}$. It is a metric space with $d\left(x_1, x_2\right):=\left x_1-x_2\right $.
fong-spivak-invitation_F02518	247	■ 1.000	$x \in \mathbb{R}$	$x \in \mathbb{R}$	for each $x \in \mathbb{R}$, define $B(x, \epsilon)$.
fong-spivak-invitation_F02519	247	■ 1.000	$B(x, \epsilon)$	$B(x, \epsilon)$	for each $x \in \mathbb{R}$, define $B(x, \epsilon)$.
fong-spivak-invitation_F02520	247	■ 1.000	$U \subseteq \mathbb{R}$	$U \subseteq \mathbb{R}$	2. When is an arbitrary subset $U \subseteq \mathbb{R}$ called open, in analogy with Example 7.18?
fong-spivak-invitation_F02521	247	■ 1.000	$U_{\{1\}}, U_{\{2\}}$	U_1, U_2	3. Find three open sets U_1, U_2 , and U in $\mathbb{R}$, such that $(U_i)_{i \in \{1,2\}}$ covers U .
fong-spivak-invitation_F02522	247	■ 1.000	$\left(U_{\{i\}}\right)_{\{i \in\{1,2\}\}}$	$(U_i)_{i \in \{1,2\}}$	3. Find three open sets U_1, U_2 , and U in $\mathbb{R}$, such that $(U_i)_{i \in \{1,2\}}$ covers U .
fong-spivak-invitation_F02523	247	■ 1.000	$\mathbf{0\ p}_{\{\text {crse }\}}=(\varnothing, X)$	$\mathbf{Op}_{\text{crse}} = (\varnothing, X)$	possible: $\mathbf{Op}_{\text{crse}} = (\varnothing, X)$. There is also a “finest” topology, having as many open sets
fong-spivak-invitation_F02524	247	■ 0.992	$\mathbf{0\ p}_{\{\text {fine }\}}=\mathrm{P}(X)$	$\mathbf{Op}_{\text{fine}} = \mathrm{P}(X)$	as possible: $\mathbf{Op}_{\text{fine}} = \mathrm{P}(X)$. The latter, $(X, \mathrm{P}(X))$, is called the <i>discrete space on the</i>
fong-spivak-invitation_F02525	247	■ 0.992	$(X, \mathrm{P}(X))$	$(X, \mathrm{P}(X))$	as possible: $\mathbf{Op}_{\text{fine}} = \mathrm{P}(X)$. The latter, $(X, \mathrm{P}(X))$, is called the <i>discrete space on the</i>
fong-spivak-invitation_F02526	247	■ 1.000	$d\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right):=\sqrt{\left(x-x^{\prime}\right)^2+\left(y-y^{\prime}\right)^2}$	$d\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right):=\sqrt{\left(x-x^{\prime}\right)^2+\left(y-y^{\prime}\right)^2}$	⁷ Here, $d((x, y), (x', y')) := \sqrt{(x - x')^2 + (y - y')^2}$ is the usual “Euclidean distance” between two points.

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fong-spivak-invitation_F02527	248	■ 1.000	$\mathbf{0\ p}_{\{\text{crse}\}}$	$\mathbf{Op}_{\text{crse}}$	1. Verify that, for any set X , what we called $\mathbf{Op}_{\text{crse}}$ in Example 7.20 really is a topology,
fong-spivak-invitation_F02528	248	■ 1.000	$\mathbf{0\ p}_{\{\text{fine}\}}$	$\mathbf{Op}_{\text{fine}}$	2. Verify also that $\mathbf{Op}_{\text{fine}}$ really is a topology.
fong-spivak-invitation_F02529	248	■ 0.899	$\left(Y, \mathbf{0}_{\{Y\}}\right)$	$(Y, \mathbf{O}_Y)$	3. Show that if $(X, \mathbf{P}(X))$ is discrete and $(Y, \mathbf{Op}_Y)$ is any topological space, then every
fong-spivak-invitation_F02530	248	■ 1.000	$X \rightarrowtail Y$	$X \rightarrow Y$	function $X \rightarrow Y$ is continuous.
fong-spivak-invitation_F02531	248	■ 1.000	$X=\{1,2\}$	$X = \{1, 2\}$	Example 7.22. There are four topologies possible on $X = \{1, 2\}$. Two are $\mathbf{Op}_{\text{crse}}$ and
fong-spivak-invitation_F02532	248	■ 0.333	$\{1,2\}, \mathbf{0} \ \mathbf{0}_{\{p\}_1}$	$\{1, 2\}, \mathbf{Op}_1$	The two topological spaces $(\{1, 2\}, \mathbf{Op}_1)$ and $(\{1, 2\}, \mathbf{Op}_2)$ are isomorphic; either one
fong-spivak-invitation_F02533	248	■ 0.333	$\{1,2\}, \mathbf{0} \ \mathbf{0}_{\{p\}_2}$	$\{1, 2\}, \mathbf{Op}_2$	The two topological spaces $(\{1, 2\}, \mathbf{Op}_1)$ and $(\{1, 2\}, \mathbf{Op}_2)$ are isomorphic; either one
fong-spivak-invitation_F02534	248	■ 0.459	$(X, \mathbf{0\ p})$	$(X, \mathbf{Op})$	Given a topological space $(X, \mathbf{Op})$, the set $\mathbf{Op}$ has the structure of a preorder using
fong-spivak-invitation_F02535	248	■ 0.794	$\mathbf{0\ p}, \subseteq$	$\mathbf{Op}, \subseteq$	the subset relation, $(\mathbf{Op}, \subseteq)$. It is reflexive because $U \subseteq U$ for any $U \in \mathbf{Op}$, and it is
fong-spivak-invitation_F02536	248	■ 0.794	$U \subseteq U$	$U \subseteq U$	the subset relation, $(\mathbf{Op}, \subseteq)$. It is reflexive because $U \subseteq U$ for any $U \in \mathbf{Op}$, and it is
fong-spivak-invitation_F02537	248	■ 1.000	$U \subseteq V$	$U \subseteq V$	transitive because if $U \subseteq V$ and $V \subseteq W$ then $U \subseteq W$.
fong-spivak-invitation_F02538	248	■ 1.000	$V \subseteq W$	$V \subseteq W$	transitive because if $U \subseteq V$ and $V \subseteq W$ then $U \subseteq W$.
fong-spivak-invitation_F02539	248	■ 1.000	$U \subseteq W$	$U \subseteq W$	transitive because if $U \subseteq V$ and $V \subseteq W$ then $U \subseteq W$.
fong-spivak-invitation_F02540	248	■ 0.936	$U, \ V$	U, V	gory: its objects are the open sets U and for any U, V the set of morphisms $\mathbf{Op}(U, V)$
fong-spivak-invitation_F02541	248	■ 0.936	$\mathbf{0\ p}(U, \ V)$	$\mathbf{Op}(U, V)$	gory: its objects are the open sets U and for any U, V the set of morphisms $\mathbf{Op}(U, V)$
fong-spivak-invitation_F02542	248	■ 1.000	$U \nsubseteq V$	$U \not\subseteq V$	is empty if $U \not\subseteq V$ and it has one element if $U \subseteq V$.
fong-spivak-invitation_F02543	248	■ 0.623	$X, \mathbf{0\ p}_{\{1\}}$	$X, \mathbf{Op}_1$	Exercise 7.23. Recall the Sierpinski space, say $(X, \mathbf{Op}_1)$ from Example 7.22.
fong-spivak-invitation_F02544	248	■ 0.558	$Y \subseteq X$	$Y \subseteq X$	Exercise 7.24. Given any topological space $(X, \mathbf{Op})$, any subset $Y \subseteq X$ can be given
fong-spivak-invitation_F02545	248	■ 0.971	$\mathbf{0\ p}_{\{?\cap Y\}}$	$\mathbf{Op}_{? \cap Y}$	the <i>subspace topology</i> , call it $\mathbf{Op}_{? \cap Y}$. This topology defines any $A \subseteq Y$ to be open,
fong-spivak-invitation_F02546	248	■ 0.971	$A \subseteq Y$	$A \subseteq Y$	the <i>subspace topology</i> , call it $\mathbf{Op}_{? \cap Y}$. This topology defines any $A \subseteq Y$ to be open,
fong-spivak-invitation_F02547	248	■ 0.999	$A \in \mathbf{0\ p}_{\{?\cap Y\}}$	$A \in \mathbf{Op}_{? \cap Y}$	$A \in \mathbf{Op}_{? \cap Y}$, if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$.
fong-spivak-invitation_F02548	248	■ 0.999	$B \in \mathbf{0\ p}$	$B \in \mathbf{Op}$	$A \in \mathbf{Op}_{? \cap Y}$, if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$.
fong-spivak-invitation_F02549	248	■ 0.999	$A=B \cap Y$	$A = B \cap Y$	$A \in \mathbf{Op}_{? \cap Y}$, if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$.
fong-spivak-invitation_F02550	248	■ 1.000	$Y \in \mathbf{0\ p}_{\{?\cap Y\}}$	$Y \in \mathbf{Op}_{? \cap Y}$	1. Find a $B \in \mathbf{Op}$ that shows that the whole set Y is open, i.e. $Y \in \mathbf{Op}_{? \cap Y}$.
fong-spivak-invitation_F02551	248	■ 1.000	$Y \hookrightarrowtail X$	$Y \hookrightarrow X$	3. Show that the inclusion function $Y \hookrightarrow X$ is a continuous function.
fong-spivak-invitation_F02552	248	■ 0.958	$(\mathbf{0\ p}, \subseteq)$	$(\mathbf{Op}, \subseteq)$	$(\mathbf{Op}, \subseteq)$ of open sets. It turns out that $(\mathbf{Op}, \subseteq, X, \cap)$ is always a quantale in the sense
fong-spivak-invitation_F02553	248	■ 0.958	$(\mathbf{0\ p}, \subseteq, X, \cap)$	$(\mathbf{Op}, \subseteq, X, \cap)$	$(\mathbf{Op}, \subseteq)$ of open sets. It turns out that $(\mathbf{Op}, \subseteq, X, \cap)$ is always a quantale in the sense
fong-spivak-invitation_F02554	248	■ 0.585	$U_{\{i\}} \subseteq X$	$U_i \subseteq X$	⁸ Hint 1: for any set I , collection of sets $(U_i)_{i \in I}$ with $U_i \subseteq X$, and set $V \subseteq X$, one has $(\bigcup_{i \in I} U_i) \cap V =$

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fong-spivak-invitation_F02555	248	0.585	$\left(\bigcup_{i \in I} U_i\right) \cap V =$	$\left(\bigcup_{i \in I} U_i\right) \cap V =$	⁸ Hint 1: for any set I , collection of sets $(U_i)_{i \in I}$ with $U_i \subseteq X$, and set $V \subseteq X$, one has $\left(\bigcup_{i \in I} U_i\right) \cap V =$
fong-spivak-invitation_F02556	248	1.000	$\bigcup_{i \in I} (U_i \cap V)$	$\bigcup_{i \in I} (U_i \cap V)$	$\bigcup_{i \in I} (U_i \cap V)$. Hint 2: for any $U, V, W \subseteq X$, one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap \bar{W}$.
fong-spivak-invitation_F02557	248	1.000	$U, V, W \subseteq X$	$U, V, W \subseteq X$	$\bigcup_{i \in I} (U_i \cap V)$. Hint 2: for any $U, V, W \subseteq X$, one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap \bar{W}$.
fong-spivak-invitation_F02558	248	1.000	$(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$	$(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$	$\bigcup_{i \in I} (U_i \cap V)$. Hint 2: for any $U, V, W \subseteq X$, one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap \bar{W}$.
fong-spivak-invitation_F02559	249	0.945	$P: \mathbf{0}^{\mathrm{op}} \rightarrow$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$. Thus to every open set $U \in \mathbf{Op}$ we have a set $P(U)$, called the set
fong-spivak-invitation_F02560	249	0.945	$P(U)$	$P(U)$	$P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$. Thus to every open set $U \in \mathbf{Op}$ we have a set $P(U)$, called the set
fong-spivak-invitation_F02561	249	1.000	$V \subseteq U$	$V \subseteq U$	of sections over U , and to every inclusion of open sets $V \subseteq U$ we have a function
fong-spivak-invitation_F02562	249	1.000	$P(U) \rightarrow P(V)$	$P(U) \rightarrow P(V)$	$P(U) \rightarrow P(V)$ called the <i>restriction</i> . If $s \in P(U)$ is a section over U , we may denote
fong-spivak-invitation_F02563	249	1.000	$s \in P(U)$	$s \in P(U)$	$P(U) \rightarrow P(V)$ called the <i>restriction</i> . If $s \in P(U)$ is a section over U , we may denote
fong-spivak-invitation_F02564	249	1.000	$s _V$	$s _V$	its restriction to V by $s _V$. Recall that we say a collection of open sets $(U_i)_{i \in I}$ covers an
fong-spivak-invitation_F02565	249	1.000	$(s_i)_{i \in I}$	$(s_i)_{i \in I}$	Let $(U_i)_{i \in I}$ be a collection of open sets $U_i \in \mathbf{Op}$ covering U . A <i>matching family</i> $(s_i)_{i \in I}$
fong-spivak-invitation_F02566	249	1.000	$s_i \in P(U_i)$	$s_i \in P(U_i)$	of P -sections over $(U_i)_{i \in I}$ consists of a section $s_i \in P(U_i)$ for each $i \in I$, such that for
fong-spivak-invitation_F02567	249	1.000	$i \in I$	$i \in I$	of P -sections over $(U_i)_{i \in I}$ consists of a section $s_i \in P(U_i)$ for each $i \in I$, such that for
fong-spivak-invitation_F02568	249	1.000	$i, j \in I$	$i, j \in I$	every $i, j \in I$, we have
fong-spivak-invitation_F02569	249	1.000	$s _{U_i} = s_i$	$s _{U_i} = s_i$	is a <i>gluing</i> , or <i>glued section</i> , of the matching family if $s _{U_i} = s_i$ holds for all $i \in I$.
fong-spivak-invitation_F02570	249	0.997	$\mathbf{Shv}(X, \mathbf{0}^{\mathrm{op}})$	$\mathbf{Shv}(X, \mathbf{Op})$	We denote by $\mathbf{Shv}(X, \mathbf{Op})$, or more briefly $\mathbf{Shv}(X)$, the category of sheaves on
fong-spivak-invitation_F02571	249	0.997	$\mathbf{Shv}(X)$	$\mathbf{Shv}(X)$	We denote by $\mathbf{Shv}(X, \mathbf{Op})$, or more briefly $\mathbf{Shv}(X)$, the category of sheaves on
fong-spivak-invitation_F02572	250	1.000	$U = \emptyset \subseteq X$	$U = \emptyset \subseteq X$	take $U = \emptyset \subseteq X$ and $I = \emptyset$, then the empty collection of subsets (one for each $i \in I$,
fong-spivak-invitation_F02573	250	0.948	$P(\emptyset) \cong \{()\}$	$P(\emptyset) \cong \{()\}$	(but rarely sufficient) condition is that $P(\emptyset) \cong \{()\}$, i.e. $P(\emptyset)$ must be a set with one
fong-spivak-invitation_F02574	250	0.948	$P(\emptyset)$	$P(\emptyset)$	(but rarely sufficient) condition is that $P(\emptyset) \cong \{()\}$, i.e. $P(\emptyset)$ must be a set with one
fong-spivak-invitation_F02575	250	0.984	$f(a_1) = f(a_2) = a, f(b_1) = b$	$f(a_1) = f(a_2) = a, f(b_1) = b$	the element of Y immediately below it. For example, $f(a_1) = f(a_2) = a, f(b_1) = b$,
fong-spivak-invitation_F02576	250	1.000	$f^{-1}(y) \subseteq X$	$f^{-1}(y) \subseteq X$	For each point $y \in Y$, the preimage set $f^{-1}(y) \subseteq X$ above it is often called the <i>fiber</i>
fong-spivak-invitation_F02577	250	1.000	$f': X \rightarrow Y$	$f': X \rightarrow Y$	4. Give an example of a function $f': X \rightarrow Y$ for which every fiber has either one or
fong-spivak-invitation_F02578	251	0.952	$\mathrm{Sec}_f: \mathbf{0}^{\mathrm{op}}(Y)^{\mathrm{op}} \rightarrow$	$\mathrm{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow$	we begin by building a presheaf – i.e. a functor $\mathrm{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow$ $\mathbf{Set}$ – and then we'll
fong-spivak-invitation_F02579	251	1.000	Sec_f	Sec_f	Define the presheaf Sec_f on an arbitrary subset $U \subseteq Y$ by:
fong-spivak-invitation_F02580	251	1.000	$U \subseteq Y$	$U \subseteq Y$	Define the presheaf Sec_f on an arbitrary subset $U \subseteq Y$ by:
fong-spivak-invitation_F02581	251	1.000	$\mathrm{Sec}_f(U)$	$\mathrm{Sec}_f(U)$	One might describe $\mathrm{Sec}_f(U)$ as the set of all ways to pick a “cross-section” of the f

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fong-spivak-invitation_F02582	251	■ 1.000	$s \in \operatorname{Sec}_f(U)$	$s \in \operatorname{Sec}_f(U)$	arrangement over U . That is, an element $s \in \operatorname{Sec}_f(U)$ is a choice of one element per
fong-spivak-invitation_F02583	251	■ 1.000	$U=\{a, b\}$	$U = \{a, b\}$	As an example, let's say $U = \{a, b\}$. How many such s 's are there in $\operatorname{Sec}_f(U)$? To
fong-spivak-invitation_F02584	251	■ 0.570	$V_1=\{a, b, c\}$	$V_1 = \{a, b, c\}$	1. Let $V_1 = \{a, b, c\}$. Draw all the sections over it, i.e. all elements of $\operatorname{Sec}_f(V_1)$, as we
fong-spivak-invitation_F02585	251	■ 0.570	$\operatorname{Sec}_f(V_1)$	$\operatorname{Sec}_f(V_1)$	1. Let $V_1 = \{a, b, c\}$. Draw all the sections over it, i.e. all elements of $\operatorname{Sec}_f(V_1)$, as we
fong-spivak-invitation_F02586	251	■ 0.993	$V_2=\{a, b, c, d\}$	$V_2 = \{a, b, c, d\}$	2. Let $V_2 = \{a, b, c, d\}$. Again draw all the sections, $\operatorname{Sec}_f(V_2)$.
fong-spivak-invitation_F02587	251	■ 0.993	$\operatorname{Sec}_f(V_2)$	$\operatorname{Sec}_f(V_2)$	2. Let $V_2 = \{a, b, c, d\}$. Again draw all the sections, $\operatorname{Sec}_f(V_2)$.
fong-spivak-invitation_F02588	251	■ 0.990	$V_3=\{a, b, d, e\}$	$V_3 = \{a, b, d, e\}$	3. Let $V_3 = \{a, b, d, e\}$. How many sections (elements of $\operatorname{Sec}_f(V_3)$) are there?
fong-spivak-invitation_F02589	251	■ 0.990	$\operatorname{Sec}_f(V_3)$	$\operatorname{Sec}_f(V_3)$	3. Let $V_3 = \{a, b, d, e\}$. How many sections (elements of $\operatorname{Sec}_f(V_3)$) are there?
fong-spivak-invitation_F02590	251	■ 1.000	$V=\{a\}$	$V = \{a\}$	restriction maps are easy: if $V \subseteq U$, say $V = \{a\}$ and $U = \{a, b\}$, then given a section
fong-spivak-invitation_F02591	251	■ 1.000	$\operatorname{Sec}_f(\{a, b, c\})$	$\operatorname{Sec}_f(\{a, b, c\})$	1. Write out the sets of sections $\operatorname{Sec}_f(\{a, b, c\})$ and $\operatorname{Sec}_f(\{a, c\})$.
fong-spivak-invitation_F02592	251	■ 1.000	$\operatorname{Sec}_f(\{a, c\})$	$\operatorname{Sec}_f(\{a, c\})$	1. Write out the sets of sections $\operatorname{Sec}_f(\{a, b, c\})$ and $\operatorname{Sec}_f(\{a, c\})$.
fong-spivak-invitation_F02593	252	■ 1.000	$U_1=\{a, b\}$	$U_1 = \{a, b\}$	condition, consider the set $U_1 = \{a, b\}$ and $U_2 = \{b, e\}$. These cover the set $U =$
fong-spivak-invitation_F02594	252	■ 1.000	$U_2=\{b, e\}$	$U_2 = \{b, e\}$	condition, consider the set $U_1 = \{a, b\}$ and $U_2 = \{b, e\}$. These cover the set $U =$
fong-spivak-invitation_F02595	252	■ 1.000	$U=$	$U =$	condition, consider the set $U_1 = \{a, b\}$ and $U_2 = \{b, e\}$. These cover the set $U =$
fong-spivak-invitation_F02596	252	■ 1.000	$\{a, b, e\}=U_1 \cup U_2$	$\{a, b, e\} = U_1 \cup U_2$	$\{a, b, e\} = U_1 \cup U_2$. By Definition 7.27, a matching family for this cover consists of a
fong-spivak-invitation_F02597	252	■ 1.000	U_1	U_1	section over U_1 and a section over U_2 that agree on the overlap set, $U_1 \cap U_2 = \{b\}$.
fong-spivak-invitation_F02598	252	■ 1.000	U_2	U_2	section over U_1 and a section over U_2 that agree on the overlap set, $U_1 \cap U_2 = \{b\}$.
fong-spivak-invitation_F02599	252	■ 1.000	$U_1 \cap U_2 = \{b\}$	$U_1 \cap U_2 = \{b\}$	section over U_1 and a section over U_2 that agree on the overlap set, $U_1 \cap U_2 = \{b\}$.
fong-spivak-invitation_F02600	252	■ 1.000	$s_1 \in \operatorname{Sec}_f(U_1)$	$s_1 \in \operatorname{Sec}_f(U_1)$	So consider $s_1 \in \operatorname{Sec}_f(U_1)$ and $s_2 \in \operatorname{Sec}_f(U_2)$ shown below.
fong-spivak-invitation_F02601	252	■ 1.000	$s_2 \in \operatorname{Sec}_f(U_2)$	$s_2 \in \operatorname{Sec}_f(U_2)$	So consider $s_1 \in \operatorname{Sec}_f(U_1)$ and $s_2 \in \operatorname{Sec}_f(U_2)$ shown below.
fong-spivak-invitation_F02602	252	■ 1.000	g_1	g_1	Since sections g_1 and g_2 agree on the overlap – they both send b to b_2 – the two sections
fong-spivak-invitation_F02603	252	■ 1.000	b_2	b_2	Since sections g_1 and g_2 agree on the overlap – they both send b to b_2 – the two sections
fong-spivak-invitation_F02604	252	■ 1.000	$U=\{a, b, e\}$	$U = \{a, b, e\}$	shown in Eq. (7.12) can be glued to form a single section over $U = \{a, b, e\}$:
fong-spivak-invitation_F02605	252	■ 0.524	$\left(s_1, s_2\right)$	(s_1, s_2)	2. For your answer (s_1, s_2) in part 1, can you find a section $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that
fong-spivak-invitation_F02606	252	■ 0.524	$s \in \operatorname{Sec}_f(U_1 \cup U_2)$	$s \in \operatorname{Sec}_f(U_1 \cup U_2)$	2. For your answer (s_1, s_2) in part 1, can you find a section $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that
fong-spivak-invitation_F02607	252	■ 1.000	$s _{U_1}=s_1$	$s _{U_1} = s_1$	$s _{U_1} = s_1$ and $s _{U_2} = s_2$?
fong-spivak-invitation_F02608	252	■ 1.000	$s _{U_2}=s_2$	$s _{U_2} = s_2$	$s _{U_1} = s_1$ and $s _{U_2} = s_2$?

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fong-spivak-invitation_F02609	252	■ 0.982	$h_{\{1\}} \in \operatorname{Sec}_{\{f\}}(\left(U_{\{1\}}\right))$	$h_1 \in \operatorname{Sec}_f(U_1)$	3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the
fong-spivak-invitation_F02610	252	■ 0.982	$h_{\{2\}} \in \operatorname{Sec}_{\{f\}}(\left(U_{\{2\}}\right))$	$h_2 \in \operatorname{Sec}_f(U_2)$	3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the
fong-spivak-invitation_F02611	252	■ 0.982	d o	<i>do</i>	3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the
fong-spivak-invitation_F02612	252	■ 1.000	$h \in \operatorname{Sec}_{\{f\}}(\left(U_{\{1\}} \cup U_{\{2\}}\right))$	$h \in \operatorname{Sec}_f(U_1 \cup U_2)$	4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$? $\diamond$
fong-spivak-invitation_F02613	252	■ 1.000	$\left.h\right _{U_{\{1\}}}=h_{\{1\}}$	$h _{U_1} = h_1$	4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$? $\diamond$
fong-spivak-invitation_F02614	252	■ 1.000	$\left.h\right _{U_{\{2\}}}=h_{\{2\}}$	$h _{U_2} = h_2$	4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$? $\diamond$
fong-spivak-invitation_F02615	253	■ 1.000	$f:\left(X, \mathbf{p}_{\{X\}}\right) \rightarrow\left(Y, \mathbf{p}_{\{Y\}}\right)$	$f:(X, \mathbf{Op}_X) \rightarrow (Y, \mathbf{Op}_Y)$	Example 7.33. Let $f:(X, \mathbf{Op}_X) \rightarrow (Y, \mathbf{Op}_Y)$ be a continuous function. Consider the
fong-spivak-invitation_F02616	253	■ 0.965	$\operatorname{Sec}_{\{f\}}: \mathbf{p}_{\{Y\}}^{\mathrm{op}} \rightarrow$	$\operatorname{Sec}_f: \mathbf{Op}_Y^{\mathrm{op}} \rightarrow$	functor $\operatorname{Sec}_f: \mathbf{Op}_Y^{\mathrm{op}} \rightarrow \mathbf{Set}$ given by
fong-spivak-invitation_F02617	253	■ 1.000	$\mathbf{p}_{\{Y\}}$	$\mathbf{Op}_Y$	The morphisms of $\mathbf{Op}_Y$ are inclusions $V \subseteq U$. Given $g: U \rightarrow X$ and $V \subseteq U$, what
fong-spivak-invitation_F02618	253	■ 1.000	$g: U \rightarrow X$	$g: U \rightarrow X$	The morphisms of $\mathbf{Op}_Y$ are inclusions $V \subseteq U$. Given $g: U \rightarrow X$ and $V \subseteq U$, what
fong-spivak-invitation_F02619	253	■ 1.000	$T M$	TM	TM whose points are pairs (m, v) , where $m \in M$ is a point in the manifold and v is a
fong-spivak-invitation_F02620	253	■ 1.000	(m, v)	(m, v)	TM whose points are pairs (m, v) , where $m \in M$ is a point in the manifold and v is a
fong-spivak-invitation_F02621	253	■ 1.000	$m \in M$	$m \in M$	TM whose points are pairs (m, v) , where $m \in M$ is a point in the manifold and v is a
fong-spivak-invitation_F02622	253	■ 1.000	$\pi: T M \rightarrow$	$\pi: TM \rightarrow$	The tangent bundle TM on a manifold M comes with a continuous map $\pi: TM \rightarrow$
fong-spivak-invitation_F02623	253	■ 1.000	$(m, v) \mapsto m$	$(m, v) \mapsto m$	M back down to the manifold, sending $(m, v) \mapsto m$. One might say that π
fong-spivak-invitation_F02624	253	■ 1.000	π	π	M back down to the manifold, sending $(m, v) \mapsto m$. One might say that π
fong-spivak-invitation_F02625	253	■ 0.846	$\operatorname{Sec}_{\{\pi\}}$	Sec_π	Example 7.33, π defines a sheaf Sec_π . It could be called the sheaf of “tangent
fong-spivak-invitation_F02626	254	■ 1.000	$U \subseteq M$	$U \subseteq M$	Earth, above. Given an open subset $U \subseteq M$, an element $v \in \operatorname{Sec}_\pi(U)$ is called
fong-spivak-invitation_F02627	254	■ 1.000	$v \in \operatorname{Sec}_{\{\pi\}}(U)$	$v \in \operatorname{Sec}_\pi(U)$	Earth, above. Given an open subset $U \subseteq M$, an element $v \in \operatorname{Sec}_\pi(U)$ is called
fong-spivak-invitation_F02628	254	■ 1.000	$v(u)$	$v(u)$	a vector field over U because it continuously assigns a tangent vector $v(u)$ to each
fong-spivak-invitation_F02629	254	■ 1.000	$U \neq M$	$U \neq M$	possible to prove that for every open set U , as long as $U \neq M$, there is a vector field
fong-spivak-invitation_F02630	254	■ 1.000	$U \neq X$	$U \neq X$	that hair can be well combed whenever $U \neq X$ but cannot be well combed when $U = X$
fong-spivak-invitation_F02631	254	■ 1.000	$U=X$	$U = X$	that hair can be well combed whenever $U \neq X$ but cannot be well combed when $U = X$
fong-spivak-invitation_F02632	254	■ 0.997	$\mathbf{Shv}\{\mathbf{S h v}\}(M)$	$\mathbf{Shv}(M)$	that we can consider the category $\mathbf{Shv}(M)$ of sheaves on M ; in fact, such categories are
fong-spivak-invitation_F02633	254	■ 0.884	$\{1,2\}, \mathbf{p}_{\{1\}}$	$\{1,2\}, \mathbf{Op}_1$	Exercise 7.37. Consider the Sierpinski space $(\{1,2\}, \mathbf{Op}_1)$ from Example 7.22.
fong-spivak-invitation_F02634	255	■ 1.000	$X=\mathbf{1}$	$X = \mathbf{1}$	one-point space $X = \mathbf{1}$ with its unique topology, we find that the category $\mathbf{Set}$ is a topos,
fong-spivak-invitation_F02635	255	■ 1.000	$\wedge, \vee, \neg, \rightarrow, \exists, \forall$	$\wedge, \vee, \neg, \Rightarrow, \exists, \forall$	a topological space. There, the same logical symbols $\wedge, \vee, \neg, \Rightarrow, \exists, \forall$ become opera-

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fong-spivak-invitation_F02636	255	■ 0.934	$\Omega=\mathbb{B}$	$\Omega = \mathbb{B}$	classifier is the set of booleans $\Omega = \mathbb{B}$. In a sheaf topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, the object
fong-spivak-invitation_F02637	255	■ 0.934	$\mathcal{E}=\operatorname{Shv}(X, \mathbf{Op})$	$\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$	classifier is the set of booleans $\Omega = \mathbb{B}$. In a sheaf topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, the object
fong-spivak-invitation_F02638	255	■ 0.809	$\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow$	$\Omega : \mathbf{Op}^{\mathrm{op}} \rightarrow$	presheaf that satisfies the sheaf condition. As a presheaf it is just a functor $\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow$
fong-spivak-invitation_F02639	255	■ 1.000	$\Omega(U)$	$\Omega(U)$	Set ; it assigns a set $\Omega(U)$ to each open $U \subseteq X$ and comes with a restriction map
fong-spivak-invitation_F02640	255	■ 1.000	$\Omega(U) \rightarrow \Omega(V)$	$\Omega(U) \rightarrow \Omega(V)$	$\Omega(U) \rightarrow \Omega(V)$ whenever $V \subseteq U$. So in our quest to understand Ω , we first ask the
fong-spivak-invitation_F02641	256	■ 1.000	$X=\{1\}$	$X = \{1\}$	Exercise 7.38. Let $X = \{1\}$ be the one-point space. We said above that the subobject
fong-spivak-invitation_F02642	256	■ 0.966	$\mathbf{Shv}(1)$	$\mathbf{Shv}(1)$	classifier of $\mathbf{Shv}(1)$ is the set $\mathbb{B}$ of booleans, but how does that align with the definition
fong-spivak-invitation_F02643	256	■ 1.000	$W \subseteq V \subseteq U$	$W \subseteq V \subseteq U$	i.e. that whenever $W \subseteq V \subseteq U$, the restriction map $\Omega(U) \rightarrow \Omega(V)$ fol-
fong-spivak-invitation_F02644	256	■ 1.000	$\Omega(V) \rightarrow \Omega(W)$	$\Omega(V) \rightarrow \Omega(W)$	lowed by the restriction map $\Omega(V) \rightarrow \Omega(W)$ is the same as the restriction map
fong-spivak-invitation_F02645	256	■ 1.000	$\Omega(U) \rightarrow \Omega(W)$	$\Omega(U) \rightarrow \Omega(W)$	$\Omega(U) \rightarrow \Omega(W)$.
fong-spivak-invitation_F02646	256	■ 1.000	$V_i \in \Omega(U_i)$	$V_i \in \Omega(U_i)$	an element $V_i \in \Omega(U_i)$, i.e. an open set $V_i \subseteq U_i$, for each $i \in I$. Suppose further
fong-spivak-invitation_F02647	256	■ 1.000	$V_i \subseteq U_i$	$V_i \subseteq U_i$	an element $V_i \in \Omega(U_i)$, i.e. an open set $V_i \subseteq U_i$, for each $i \in I$. Suppose further
fong-spivak-invitation_F02648	256	■ 1.000	$V_i \cap U_j = V_j \cap U_i$	$V_i \cap U_j = V_j \cap U_i$	that, for all $i, j \in I$, it is the case that $V_i \cap U_j = V_j \cap U_i$, i.e. that the elements form
fong-spivak-invitation_F02649	256	■ 1.000	$V := \bigcup_{i \in I} V_i$	$V := \bigcup_{i \in I} V_i$	a matching family. Define $V := \bigcup_{i \in I} V_i$; it is an open subset of U , so we can con-
fong-spivak-invitation_F02650	256	■ 1.000	$(V_i)_{i \in I}$	$(V_i)_{i \in I}$	the $(V_i)_{i \in I}$:
fong-spivak-invitation_F02651	256	■ 1.000	$V \cap U_j = V_j$	$V \cap U_j = V_j$	In other words $V \cap U_j = V_j$ for any $j \in I$. So our Ω has been upgraded from presheaf
fong-spivak-invitation_F02652	256	■ 1.000	$j \in I$	$j \in I$	In other words $V \cap U_j = V_j$ for any $j \in I$. So our Ω has been upgraded from presheaf
fong-spivak-invitation_F02653	256	■ 0.908	$\{1\} \rightarrow \Omega$	$\{1\} \rightarrow \Omega$	standing, we also need to give a sheaf morphism $\mathrm{true}: \{1\} \rightarrow \Omega$. Here $\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$
fong-spivak-invitation_F02654	256	■ 0.908	$\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$	$\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$	standing, we also need to give a sheaf morphism $\mathrm{true}: \{1\} \rightarrow \Omega$. Here $\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$
fong-spivak-invitation_F02655	256	■ 1.000	$\{1\} = \{1\}(U) \rightarrow \Omega(U)$	$\{1\} = \{1\}(U) \rightarrow \Omega(U)$	that assigns, for every $U \in \mathbf{Op}$, the function $\{1\} = \{1\}(U) \rightarrow \Omega(U)$ sending $1 \mapsto U$,
fong-spivak-invitation_F02656	256	■ 1.000	$1 \mapsto U$	$1 \mapsto U$	that assigns, for every $U \in \mathbf{Op}$, the function $\{1\} = \{1\}(U) \rightarrow \Omega(U)$ sending $1 \mapsto U$,
fong-spivak-invitation_F02657	256	■ 1.000	$U=\varnothing$	$U = \varnothing$	true ; if U is nothing, $U = \varnothing$, then P is really false. But, in general, it's just true in some
fong-spivak-invitation_F02658	257	■ 0.944	$(V, V; A)$	$(V, V; A)$	$\mathrm{true}: 1 \rightarrow \Omega$ sends that loop to $(V, V; A)$.
fong-spivak-invitation_F02659	257	■ 1.000	$i: H \subseteq G$	$i: H \subseteq G$	Given any graph G and subgraph $i: H \subseteq G$, we need to construct a graph homo-
fong-spivak-invitation_F02660	257	■ 0.546	$\lceil H \rceil: G \rightarrow \Omega$	$\lceil H \rceil: G \rightarrow \Omega$	morphism $\lceil H \rceil: G \rightarrow \Omega$ classifying H . The idea is that, for each part of G , we decide
fong-spivak-invitation_F02661	257	■ 0.577	$(V, 0; 0), (0, V; 0), (V, V; 0)$	$(V, 0; 0), (0, V; 0), (V, V; 0)$	we send a to $(V, 0; 0), (0, V; 0), (V, V; 0)$, or $(0, 0; 0)$, respectively.
fong-spivak-invitation_F02662	257	■ 0.577	$0; 0$	$0; 0$	we send a to $(V, 0; 0), (0, V; 0), (V, V; 0)$, or $(0, 0; 0)$, respectively.

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fong-spivak-invitation_F02663	257	■ 1.000	$H \subseteq G$	$H \subseteq G$	Exercise 7.41. Consider the subgraph $H \subset G$ shown here:
fong-spivak-invitation_F02664	257	■ 1.000	$U, V \in \Omega(X)$	$U, V \in \Omega(X)$	ues $U, V \in \Omega(X)$, then their conjunction “ U AND V ” is their intersection, and their
fong-spivak-invitation_F02665	257	■ 1.000	$\cap$	$\cap$	These formulas are easy to remember, because $\wedge$ looks like $\cap$ and $\vee$ looks like $\cup$. The
fong-spivak-invitation_F02666	257	■ 1.000	$\cup$	$\cup$	These formulas are easy to remember, because $\wedge$ looks like $\cap$ and $\vee$ looks like $\cup$. The
fong-spivak-invitation_F02667	257	■ 1.000	$U \Rightarrow V$	$U \Rightarrow V$	implication $U \Rightarrow V$ is the largest open set R such that $R \cap U \subseteq V$, i.e.
fong-spivak-invitation_F02668	257	■ 1.000	$R \cap U \subseteq V$	$R \cap U \subseteq V$	implication $U \Rightarrow V$ is the largest open set R such that $R \cap U \subseteq V$, i.e.
fong-spivak-invitation_F02669	258	■ 0.933	$\neg U :=$	$\neg U :=$	Finally, the negation of U is given by $\neg U := (U \Rightarrow \text{false})$, and this turns out to
fong-spivak-invitation_F02670	258	■ 0.933	$U \Rightarrow$	$U \Rightarrow$	Finally, the negation of U is given by $\neg U := (U \Rightarrow \text{false})$, and this turns out to
fong-spivak-invitation_F02671	258	■ 1.000	$R \cap U = \varnothing$	$R \cap U = \varnothing$	be relatively simple. By Eq. (7.16), it is the union of all R such that $R \cap U = \varnothing$, i.e.
fong-spivak-invitation_F02672	258	■ 1.000	$\neg U$	$\neg U$	recognize that $\neg U$ is the “interior of the complement of U .”
fong-spivak-invitation_F02673	258	■ 1.000	$X = \mathbb{R}$	$X = \mathbb{R}$	Example 7.42. Consider the real line $X = \mathbb{R}$ as a topological space (see Exercise 7.19).
fong-spivak-invitation_F02674	258	■ 1.000	$U = \{x \in \mathbb{R} \mid x < 3\}$	$U = \{x \in \mathbb{R} \mid x < 3\}$	Let $U, V \in \Omega(X)$ be the open sets $U = \{x \in \mathbb{R} \mid x < 3\}$ and $V = \{x \in \mathbb{R} \mid -4 < x <$
fong-spivak-invitation_F02675	258	■ 1.000	$V = \{x \in \mathbb{R} \mid -4 < x <$	$V = \{x \in \mathbb{R} \mid -4 < x <$	Let $U, V \in \Omega(X)$ be the open sets $U = \{x \in \mathbb{R} \mid x < 3\}$ and $V = \{x \in \mathbb{R} \mid -4 < x <$
fong-spivak-invitation_F02676	258	■ 1.000	$U = (-\infty, 3)$	$U = (-\infty, 3)$	4). Using interval notation, $U = (-\infty, 3)$ and $V = (-4, 4)$. Then
fong-spivak-invitation_F02677	258	■ 1.000	$V = (-4, 4)$	$V = (-4, 4)$	4). Using interval notation, $U = (-\infty, 3)$ and $V = (-4, 4)$. Then
fong-spivak-invitation_F02678	258	■ 1.000	$U \wedge V = (-4, 3)$	$U \wedge V = (-4, 3)$	• $U \wedge V = (-4, 3)$,
fong-spivak-invitation_F02679	258	■ 1.000	$U \vee V = (-\infty, 4)$	$U \vee V = (-\infty, 4)$	• $U \vee V = (-\infty, 4)$,
fong-spivak-invitation_F02680	258	■ 1.000	$\neg U = (3, \infty)$	$\neg U = (3, \infty)$	• $\neg U = (3, \infty)$,
fong-spivak-invitation_F02681	258	■ 1.000	$\neg V = (-\infty, -4) \cup (4, \infty)$	$\neg V = (-\infty, -4) \cup (4, \infty)$	• $\neg V = (-\infty, -4) \cup (4, \infty)$,
fong-spivak-invitation_F02682	258	■ 1.000	$(U \Rightarrow V) = (-4, \infty)$	$(U \Rightarrow V) = (-4, \infty)$	• $U \Rightarrow V = (-4, \infty)$,
fong-spivak-invitation_F02683	258	■ 1.000	$(V \Rightarrow U) = U$	$(V \Rightarrow U) = U$	• $V \Rightarrow U = U$.
fong-spivak-invitation_F02684	258	■ 1.000	$U = \mathbb{R} - \{0\}$	$U = \mathbb{R} - \{0\}$	subset $U = \mathbb{R} - \{0\}$.
fong-spivak-invitation_F02685	258	■ 1.000	$\neg \neg U$	$\neg \neg U$	2. What open subset is $\neg \neg U$?
fong-spivak-invitation_F02686	258	■ 1.000	$U \subseteq \neg \neg U$	$U \subseteq \neg \neg U$	3. Is it true that $U \subset \neg \neg U$?
fong-spivak-invitation_F02687	258	■ 1.000	$\neg \neg U \subseteq U$	$\neg \neg U \subseteq U$	4. Is it true that $\neg \neg U \subseteq U$?
fong-spivak-invitation_F02688	258	■ 0.983	$\wedge U = U$	$\wedge U = U$	$U \in \mathbf{Op}$, we have $(\text{true} \wedge U) = U$. Which open set is it?
fong-spivak-invitation_F02689	258	■ 0.748	$\vee U = \text{true}$	$\vee U = \text{true}$	2. Other things we should expect from true include $(\text{true} \vee U) = \text{true}$ and $(U \Rightarrow$
fong-spivak-invitation_F02690	258	■ 0.748	$(U \Rightarrow$	$(U \Rightarrow$	2. Other things we should expect from true include $(\text{true} \vee U) = \text{true}$ and $(U \Rightarrow$

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fong-spivak-invitation_F02691	258	■ 0.840	)=	) =	$\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$. Do these hold for your answer to 1?
fong-spivak-invitation_F02692	258	■ 0.840	(	(	$\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$. Do these hold for your answer to 1?
fong-spivak-invitation_F02693	258	■ 0.840	$\rightarrow U)=U$	$\Rightarrow U) = U$	$\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$. Do these hold for your answer to 1?
fong-spivak-invitation_F02694	258	■ 0.999	$\vee U)=U$	$\vee U) = U$	$U \in \mathbf{Op}$, we have $(\text{false} \vee U) = U$. Which open set is it?
fong-spivak-invitation_F02695	258	■ 0.900	$\wedge U$	$\wedge U$	4. Other things we should expect from false include $(\text{false} \wedge U) =$
fong-spivak-invitation_F02696	259	■ 1.000	$\pi: E \rightarrow X$	$\pi: E \rightarrow X$	Example 7.45. For a vector bundle $\pi: E \rightarrow X$ over a space X , the corresponding sheaf
fong-spivak-invitation_F02697	259	■ 0.988	$i_U: U \subseteq X$	$i_U: U \subseteq X$	is Sec_π corresponding to its sections: to each open set $i_U: U \subseteq X$, we associate the set
fong-spivak-invitation_F02698	259	■ 0.691	$s: U \rightarrow E$	$s: U \rightarrow E$	of functions $s: U \rightarrow E$ for which $s \circ \pi = i_U$. For example, in the case of the tangent
fong-spivak-invitation_F02699	259	■ 0.691	$s; \pi = i_U$	$s; \pi = i_U$	of functions $s: U \rightarrow E$ for which $s \circ \pi = i_U$. For example, in the case of the tangent
fong-spivak-invitation_F02700	259	■ 1.000	$\pi: TM \rightarrow M$	$\pi: TM \rightarrow M$	bundle $\pi: TM \rightarrow M$ (see Example 7.34), the corresponding sheaf, call it $\mathbf{VF}$, associates
fong-spivak-invitation_F02701	259	■ 0.998	$\mathbf{VF}(U)$	$\mathbf{VF}(U)$	to each U the set $\mathbf{VF}(U)$ of vector fields on U .
fong-spivak-invitation_F02702	259	■ 0.813	$\rightarrow \Omega$	$\rightarrow \Omega$	For example, one could have a predicate $\text{Grad}: \mathbf{VF} \rightarrow \Omega$ that asks for the largest sub-
fong-spivak-invitation_F02703	259	■ 0.998	$\text{Grad}(v)$	$\text{Grad}(v)$	space $\text{Grad}(v)$ on which a given vector field v comes from the gradient of some scalar
fong-spivak-invitation_F02704	259	■ 0.887	$\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$	$\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$	In a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, a predicate is a sheaf morphism $p: S \rightarrow \Omega$ where
fong-spivak-invitation_F02705	259	■ 0.887	$p: S \rightarrow \Omega$	$p: S \rightarrow \Omega$	In a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$, a predicate is a sheaf morphism $p: S \rightarrow \Omega$ where
fong-spivak-invitation_F02706	259	■ 1.000	$S \in \mathcal{E}$	$S \in \mathcal{E}$	$S \in \mathcal{E}$ is a sheaf and $\Omega \in \mathcal{E}$ is the subobject classifier, the sheaf of truth values. By
fong-spivak-invitation_F02707	259	■ 1.000	$p(U): S(U) \rightarrow \Omega(U)$	$p(U): S(U) \rightarrow \Omega(U)$	Definition 7.27 we get a function $p(U): S(U) \rightarrow \Omega(U)$ for any open set $U \subseteq X$. In
fong-spivak-invitation_F02708	259	■ 0.972	$\text{Bob} \in S(U)$	$\text{Bob} \in S(U)$	sheaf of people (people come and go over time), $\text{Bob} \in S(U)$ is a person existing over a
fong-spivak-invitation_F02709	259	■ 0.998	$s \in S(U)$	$s \in S(U)$	the weather,” acts on sections $s \in S(U)$, say $s = \text{Bob}$. But, by Definition 7.9, any
fong-spivak-invitation_F02710	259	■ 0.998	$s =$	$s =$	the weather,” acts on sections $s \in S(U)$, say $s = \text{Bob}$. But, by Definition 7.9, any
fong-spivak-invitation_F02711	260	■ 1.000	$\{S \mid p\} \subseteq S$	$\{S \mid p\} \subseteq S$	predicate $p: S \rightarrow \Omega$ also defines a subobject of $\{S \mid p\} \subseteq S$. Describe the sections of
fong-spivak-invitation_F02712	260	■ 0.763	$ \Omega^E =$	$ \Omega^E =$	For a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ and object (sheaf) $S \in \mathcal{E}$, the set of S -predicates $ \Omega^E =$
fong-spivak-invitation_F02713	260	■ 1.000	$\mathcal{E}(S, \Omega)$	$\mathcal{E}(S, \Omega)$	$\mathcal{E}(S, \Omega)$ is naturally given the structure of a poset, which we denote
fong-spivak-invitation_F02714	260	■ 1.000	$p, q: S \rightarrow \Omega$	$p, q: S \rightarrow \Omega$	Given two predicates $p, q: S \rightarrow \Omega$, we say that $p \leq^S q$ if the first implies the second.
fong-spivak-invitation_F02715	260	■ 1.000	$p \leq^S q$	$p \leq^S q$	Given two predicates $p, q: S \rightarrow \Omega$, we say that $p \leq^S q$ if the first implies the second.
fong-spivak-invitation_F02716	260	■ 1.000	$p(s) \subseteq U$	$p(s) \subseteq U$	$p(s) \subseteq U$ and $q(s) \subseteq U$. We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and
fong-spivak-invitation_F02717	260	■ 1.000	$q(s) \subseteq U$	$q(s) \subseteq U$	$p(s) \subseteq U$ and $q(s) \subseteq U$. We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and

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fong-spivak-invitation_F02718	260	■ 1.000	$p(s) \subseteq q(s)$	$p(s) \subseteq q(s)$	$p(s) \subseteq U$ and $q(s) \subseteq U$. We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and
fong-spivak-invitation_F02719	260	■ 1.000	$\leq^S$	$\leq^S$	$s \in S(U)$. We often drop the superscript from $\leq^S$ and simply write $\leq$. In formal logic
fong-spivak-invitation_F02720	260	■ 1.000	$S=1$	$S = 1$	In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $, and refer to elements
fong-spivak-invitation_F02721	260	■ 1.000	$ \Omega^S $	$ \Omega^S $	In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $, and refer to elements
fong-spivak-invitation_F02722	260	■ 1.000	$ \Omega $	$ \Omega $	In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $, and refer to elements
fong-spivak-invitation_F02723	260	■ 1.000	$p \in \Omega $	$p \in \Omega $	$p \in \Omega $ as <i>propositions</i> . They are just morphisms $p: 1 \rightarrow \Omega$.
fong-spivak-invitation_F02724	260	■ 1.000	$p: 1 \rightarrow \Omega$	$p: 1 \rightarrow \Omega$	$p \in \Omega $ as <i>propositions</i> . They are just morphisms $p: 1 \rightarrow \Omega$.
fong-spivak-invitation_F02725	260	■ 1.000	$p=q$	$p = q$	$p = q$. The reason is that for any subsets $U, V \subseteq X$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$.
fong-spivak-invitation_F02726	260	■ 1.000	$U, V \subseteq X$	$U, V \subseteq X$	$p = q$. The reason is that for any subsets $U, V \subseteq X$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$.
fong-spivak-invitation_F02727	260	■ 1.000	$U=V$	$U = V$	$p = q$. The reason is that for any subsets $U, V \subseteq X$, if $U \subseteq V$ and $V \subseteq U$ then $U = V$.
fong-spivak-invitation_F02728	260	■ 0.813	$S \in \mathbf{Shv}(X)$	$S \in \mathbf{Shv}(X)$	Exercise 7.47. Give an example of a space X , a sheaf $S \in \mathbf{Shv}(X)$, and two predicates
fong-spivak-invitation_F02729	260	■ 0.977	$p(s) \vdash_{s:S} q(s)$	$p(s) \vdash_{s:S} q(s)$	$p, q: S \rightarrow \Omega$ for which $p(s) \vdash_{s:S} q(s)$ holds. You do not have to be formal.
fong-spivak-invitation_F02730	260	■ 0.999	$\wedge, \vee, \rightarrow, \neg$	$\wedge, \vee, \Rightarrow, \neg$	All of the logical symbols (<code>true</code> , <code>false</code> , $\wedge, \vee, \Rightarrow, \neg$) from Section 7.4.2 make
fong-spivak-invitation_F02731	260	■ 1.000	$(p \wedge q): S \rightarrow \Omega$	$(p \wedge q): S \rightarrow \Omega$	$(p \wedge q): S \rightarrow \Omega$ by $(p \wedge q)(s) := p(s) \wedge q(s)$, and similarly for $\vee$. Thus one says
fong-spivak-invitation_F02732	260	■ 1.000	$(p \wedge q)(s) := p(s) \wedge q(s)$	$(p \wedge q)(s) := p(s) \wedge q(s)$	$(p \wedge q): S \rightarrow \Omega$ by $(p \wedge q)(s) := p(s) \wedge q(s)$, and similarly for $\vee$. Thus one says
fong-spivak-invitation_F02733	260	■ 1.000	$n+1$	$n + 1$	exists.” Each takes in a predicate of $n + 1$ variables and returns a predicate of n variables.
fong-spivak-invitation_F02734	260	■ 0.979	$S, T \in \mathbf{Shv}(X, \mathbf{Op})$	$S, T \in \mathbf{Shv}(X, \mathbf{Op})$	Example 7.48. Suppose we have two sheaves $S, T \in \mathbf{Shv}(X, \mathbf{Op})$ and a predicate
fong-spivak-invitation_F02735	260	■ 1.000	$p: S \times T \rightarrow \Omega$	$p: S \times T \rightarrow \Omega$	$p: S \times T \rightarrow \Omega$. Let’s say T represents what’s considered newsworthy and S is again
fong-spivak-invitation_F02736	260	■ 1.000	$t \in T(U)$	$t \in T(U)$	the set of people. So for a subset of time U , a section $t \in T(U)$ is something that’s
fong-spivak-invitation_F02737	261	■ 1.000	$p(s, t)$	$p(s, t)$	subset of times on which $p(s, t)$ is true.
fong-spivak-invitation_F02738	261	■ 0.992	$T, \dots$	$T, \dots$	“For all t in T , ... is worried about t ” is itself a predicate on just one variable, S ,
fong-spivak-invitation_F02739	261	■ 0.998	$\exists(t: T). p(s, t)$	$\exists(t: T). p(s, t)$	also a predicate on S , which we denote $\exists(t: T). p(s, t)$. If we apply this predicate to
fong-spivak-invitation_F02740	261	■ 0.999	$p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$	$p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$	Exercise 7.49. In the topos Set , where $\Omega = \mathbb{B}$, consider the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$
fong-spivak-invitation_F02741	261	■ 1.000	$\forall(z: \mathbb{Z}). p(n, z)$	$\forall(z: \mathbb{Z}). p(n, z)$	1. What is the set of $n \in \mathbb{N}$ for which the predicate $\forall(z: \mathbb{Z}). p(n, z)$ holds?
fong-spivak-invitation_F02742	261	■ 1.000	$\exists(z: \mathbb{Z}). p(n, z)$	$\exists(z: \mathbb{Z}). p(n, z)$	2. What is the set of $n \in \mathbb{N}$ for which the predicate $\exists(z: \mathbb{Z}). p(n, z)$ holds?
fong-spivak-invitation_F02743	261	■ 0.999	$z \in \mathbb{Z}$	$z \in \mathbb{Z}$	3. What is the set of $z \in \mathbb{Z}$ for which the predicate $\forall(n: \mathbb{N}). p(n, z)$ holds?
fong-spivak-invitation_F02744	261	■ 0.999	$\forall(n: \mathbb{N}). p(n, z)$	$\forall(n: \mathbb{N}). p(n, z)$	3. What is the set of $z \in \mathbb{Z}$ for which the predicate $\forall(n: \mathbb{N}). p(n, z)$ holds?

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fong-spivak-invitation_F02745	261	■ 0.979	$\exists(n : \mathbb{N}) . p(n, z)$	$\exists(n : \mathbb{N}).p(n, z)$	4. What is the set of $z \in \mathbb{Z}$ for which the predicate $\exists(n : \mathbb{N}). p(n, z)$ holds?
fong-spivak-invitation_F02746	261	■ 0.817	$\forall(t :$	$\forall(t :$	So, given p , we have a universally and an existentially quantified predicate $\forall(t :$
fong-spivak-invitation_F02747	261	■ 1.000	$S \rightarrow \Omega$	$S \rightarrow \Omega$	morphisms $S \rightarrow \Omega$ or, equivalently, as subsheaves of S ?
fong-spivak-invitation_F02748	261	■ 1.000	$\forall(t : T) . p(s, t)$	$\forall(t : T).p(s, t)$	Given a predicate $p : S \times T \rightarrow \Omega$, the universally quantified predicate $\forall(t : T). p(s, t)$
fong-spivak-invitation_F02749	261	■ 1.000	$V \in \Omega(U)$	$V \in \Omega(U)$	takes a section $s \in S(U)$, for any open set U , and returns a certain open set $V \in \Omega(U)$.
fong-spivak-invitation_F02750	261	■ 1.000	$p(\text{left}(\text{left.s} \mid_{\{V\}}, t \mid_{\text{right}}))=V$	$p(s _V, t) = V$	Namely, it returns the largest open set $V \subseteq U$ for which $p(s _V, t) = V$ holds for all
fong-spivak-invitation_F02751	261	■ 1.000	$t \in T(V)$	$t \in T(V)$	$t \in T(V)$.
fong-spivak-invitation_F02752	261	■ 0.998	$p(s, t)=$	$p(s, t) =$	above definition to the example $p(s, t) =$ “person s is worried about news t ” from
fong-spivak-invitation_F02753	261	■ 1.000	$T(V)$	$T(V)$	Example 7.48. Here, $T(V)$ is the set of items that are in the news throughout the
fong-spivak-invitation_F02754	262	■ 0.516	$p^{\{\prime\}}: S \rightarrow \Omega^T$	$p' : S \rightarrow \Omega^T$	where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and true^T is the currying of the
fong-spivak-invitation_F02755	262	■ 0.516	$S \times T \rightarrow \Omega$	$S \times T \rightarrow \Omega$	where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and true^T is the currying of the
fong-spivak-invitation_F02756	262	■ 0.516	$\{ \}^T$	T	where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and true^T is the currying of the
fong-spivak-invitation_F02757	262	■ 0.817	$1 \times T \stackrel{!}{\rightarrow} 1 \stackrel{\text{true}}{\rightarrow} \Omega$	$1 \times T \xrightarrow{!} 1 \xrightarrow{\text{true}} \Omega$	composite $1 \times T \xrightarrow{!} 1 \xrightarrow{\text{true}} \Omega$. See Eq. (7.3).
fong-spivak-invitation_F02758	262	■ 1.000	$V = \bigcup_i V_i$	$V = \bigcup_i V_i$	namely the union $V = \bigcup_i V_i$ of all the open sets V_i for which there exists some $t_i \in$
fong-spivak-invitation_F02759	262	■ 1.000	V_i	V_i	namely the union $V = \bigcup_i V_i$ of all the open sets V_i for which there exists some $t_i \in$
fong-spivak-invitation_F02760	262	■ 1.000	$t_i \in$	$t_i \in$	namely the union $V = \bigcup_i V_i$ of all the open sets V_i for which there exists some $t_i \in$
fong-spivak-invitation_F02761	262	■ 0.999	$T(\text{left}(V_i \mid_{\text{right}}))$	$T(V_i)$	$T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$. If the result is U itself, you might be tempted to
fong-spivak-invitation_F02762	262	■ 0.999	$p(\text{left}(\text{left.s} \mid_{\{V_i\}}, t_i \mid_{\text{right}}))=V_i$	$p(s _{V_i}, t_i) = V_i$	$T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$. If the result is U itself, you might be tempted to
fong-spivak-invitation_F02763	262	■ 1.000	$p(t)$	$p(t)$	think “ah, so there exists some $t \in T(U)$ satisfying $p(t)$,” but that is not necessarily so.
fong-spivak-invitation_F02764	262	■ 1.000	$U = \bigcup U_i$	$U = \bigcup U_i$	There is just a cover of $U = \bigcup U_i$ and local sections $t_i \in T(U_i)$, each satisfying p , as
fong-spivak-invitation_F02765	262	■ 1.000	$t_i \in T(\text{left}(U_i \mid_{\text{right}}))$	$t_i \in T(U_i)$	There is just a cover of $U = \bigcup U_i$ and local sections $t_i \in T(U_i)$, each satisfying p , as
fong-spivak-invitation_F02766	262	■ 1.000	$\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$	$\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$	with the subobject classified by p , namely $\boxed{\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T}$ compose
fong-spivak-invitation_F02767	262	■ 1.000	$\pi_S : S \times T \rightarrow S$	$\pi_S : S \times T \rightarrow S$	with the projection $\pi_S : S \times T \rightarrow S$ as on the upper right; then take the epi-mono
fong-spivak-invitation_F02768	262	■ 0.983	$(\Omega^S , \leq^S)$	$(\Omega^S , \leq^S)$	poset $(\Omega^S , \leq^S)$ of predicates on S , where $ \Omega^S = \mathcal{E}(S, \Omega)$ is just the set of morphisms
fong-spivak-invitation_F02769	262	■ 0.983	$ \Omega^S = \mathcal{E}(S, \Omega)$	$ \Omega^S = \mathcal{E}(S, \Omega)$	poset $(\Omega^S , \leq^S)$ of predicates on S , where $ \Omega^S = \mathcal{E}(S, \Omega)$ is just the set of morphisms
fong-spivak-invitation_F02770	263	■ 0.996	$j : \Omega \rightarrow \Omega$	$j : \Omega \rightarrow \Omega$	Definition 7.52. A modality in $\mathbf{Shv}(X)$ is a sheaf morphism $j : \Omega \rightarrow \Omega$ satisfying three
fong-spivak-invitation_F02771	263	■ 1.000	$p, q \in \Omega(U)$	$p, q \in \Omega(U)$	properties for all $U \subseteq X$ and $p, q \in \Omega(U)$:

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fong-spivak-invitation_F02772	263	■ 0.965	$(j ; j)(p) \leq j(p)$	$(j;j)(p) \leq j(p)$	(b) $(\overline{j} \circ j)(p) < j(p)$,
fong-spivak-invitation_F02773	263	■ 1.000	$j(p \wedge q)=j(p) \wedge j(q)$	$j(p \wedge q) = j(p) \wedge j(q)$	(c) $\overline{j}(p \wedge q) = j(p) \wedge j(q)$.
fong-spivak-invitation_F02774	263	■ 1.000	$p \in \Omega(U)$	$p \in \Omega(U)$	holds for all $U \subseteq X$ and $p \in \Omega(U)$. Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq$
fong-spivak-invitation_F02775	263	■ 1.000	$q \in \Omega(U)$	$q \in \Omega(U)$	holds for all $U \subseteq X$ and $p \in \Omega(U)$. Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq$
fong-spivak-invitation_F02776	263	■ 1.000	$j(j(q)) \leq$	$j(j(q)) \leq$	holds for all $U \subseteq X$ and $p \in \Omega(U)$. Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq$
fong-spivak-invitation_F02777	263	■ 1.000	$j(q)$	$j(q)$	$j(q)$ iff $j(j(q)) = j(q)$.
fong-spivak-invitation_F02778	263	■ 1.000	$j(j(q))=j(q)$	$j(j(q)) = j(q)$	$j(q)$ iff $\overline{j}(j(q)) = j(q)$.
fong-spivak-invitation_F02779	263	■ 0.999	$p, \ldots$	$p, \dots$	Diego,” there is a modal operator “assuming $p, \dots$ ” Now we are in a position to make
fong-spivak-invitation_F02780	263	■ 1.000	$\Omega \rightarrow \Omega$	$\Omega \rightarrow \Omega$	(a) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending q to $p \Rightarrow q$ is a modality,
fong-spivak-invitation_F02781	263	■ 1.000	$p \vee q$	$p \vee q$	(b) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending q to $p \vee q$ is a modality,
fong-spivak-invitation_F02782	263	■ 0.431	$q \Rightarrow p$	$q \Rightarrow p$	(c) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending q to $(q \Rightarrow p) \Rightarrow p$ is a modality.
fong-spivak-invitation_F02783	263	■ 1.000	$p(s)$	$p(s)$	$p(s)$, and what does it mean?
fong-spivak-invitation_F02784	263	■ 0.999	$j(p(s))$	$j(p(s))$	3. What sort of thing is $j(p(s))$ and what does it mean?
fong-spivak-invitation_F02785	263	■ 1.000	$p(s) \leq j(p(s))$	$p(s) \leq j(p(s))$	4. Is it true that $\overline{p}(s) < j(p(s))$? Explain briefly.
fong-spivak-invitation_F02786	263	■ 1.000	$j(j(p(s)))=j(p(s))$	$j(j(p(s))) = j(p(s))$	5. Is it true that $\overline{j}(j(p(s))) = j(p(s))$? Explain briefly.
fong-spivak-invitation_F02787	263	■ 1.000	$q: S \rightarrow \Omega$	$q: S \rightarrow \Omega$	6. Choose another predicate $q: S \rightarrow \Omega$. Is it true that $j(p \wedge q) = j(p) \wedge j(q)$? Explain
fong-spivak-invitation_F02788	264	■ 0.515	$t \in T(c)$	$t \in T(c)$	Is it true that for every table $c \in \text{Ob}(\mathcal{C})$ and every row $t \in T(c)$ there exists a row $s \in S(c)$ such
fong-spivak-invitation_F02789	264	■ 0.515	$s \in S(c)$	$s \in S(c)$	Is it true that for every table $c \in \text{Ob}(\mathcal{C})$ and every row $t \in T(c)$ there exists a row $s \in S(c)$ such
fong-spivak-invitation_F02790	264	■ 1.000	$f(s)=t$	$f(s) = t$	that $f(s) = t$?
fong-spivak-invitation_F02791	264	■ 1.000	$t \in T(u)$	$t \in T(u)$	Is it true that for every open set $U \subseteq X$ and every section $t \in T(u)$ of the bundle T , there exists
fong-spivak-invitation_F02792	264	■ 0.999	$\left(U_{\{i\}} \subseteq U\right)_{i \in I}$	$(U_i \subseteq U)_{i \in I}$	an open covering of $\overline{U_i \subseteq U}_{i \in I}$ of U and a section $s_i \in S(U_i)$ of the bundle S for each $i \in I$,
fong-spivak-invitation_F02793	264	■ 0.999	$s_{\{i\}} \in S\left(U_{\{i\}}\right)$	$s_i \in S\left(U_i\right)$	an open covering of $(U_i \subseteq U)_{i \in I}$ of U and a section $s_i \in S(U_i)$ of the bundle S for each $i \in I$,
fong-spivak-invitation_F02794	264	■ 0.999	$f\left(s_{\{i\}}\right)=\left.t\right _{u_{\{i\}}}$	$f\left(s_i\right)=t _{u_i}$	such that $f(s_i) = t _{u_i}$ is the restriction of s to U_i ?
fong-spivak-invitation_F02795	264	■ 0.999	$U_{\{i\}}$	U_i	such that $f(s_i) = t _{u_i}$ is the restriction of s to U_i ?
fong-spivak-invitation_F02796	265	■ 1.000	$\mathbb{I} \mathbb{R}$	$\mathbb{I}\mathbb{R}$	The interval domain $\mathbb{I}\mathbb{R}$ is a specific topological space, which we will use to model inter-
fong-spivak-invitation_F02797	265	■ 0.680	$\mathbf{Shv}(\mathbb{I} \mathbb{R})$	$\mathbf{Shv}(\mathbb{I}\mathbb{R})$	vals of time. In other words, we will be interested in the category $\mathbf{Shv}(\mathbb{I}\mathbb{R})$ of sheaves
fong-spivak-invitation_F02798	265	■ 0.898	$a < b$	$a < b$	For $a < b$ in $\mathbb{R}$, let $o_{[a,b]}$ denote the set $o_{[a,b]} := \{[d,u] \in \mathbb{I}\mathbb{R} \mid a < d \leq u < b\}$; these

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fong-spivak-invitation_F02799	265	0.898	$\mathsf{o}_{\{[a, b]\}}$	$\mathcal{O}_{[a,b]}$	For $a < b$ in $\mathbb{R}$, let $\mathcal{O}_{[a,b]}$ denote the set $\mathcal{O}_{[a,b]} := \{[d, u] \in \mathbb{IR} \mid a < d \leq u < b\}$; these
fong-spivak-invitation_F02800	265	0.898	$\mathsf{o}_{\{[a, b]\}} := \{[d, u] \in \mathbb{IR} \mid a < d \leq u < b\}$	$\mathcal{O}_{[a,b]} := \{[d, u] \in \mathbb{R} \mid a < d \leq u < b\}$	For $a < b$ in $\mathbb{R}$, let $\mathcal{O}_{[a,b]}$ denote the set $\mathcal{O}_{[a,b]} := \{[d, u] \in \mathbb{IR} \mid a < d < u < b\}$; these
fong-spivak-invitation_F02801	265	1.000	$\mathsf{o}_{\{[0,5]\}}$	$\mathcal{O}_{[0,5]}$	Thus for example, $\mathcal{O}_{[0,5]}$ is an open set: it contains every $[d, u]$ contained in the open
fong-spivak-invitation_F02802	265	1.000	$[d, u]$	$[d, u]$	Thus for example, $\mathcal{O}_{[0,5]}$ is an open set: it contains every $[d, u]$ contained in the open
fong-spivak-invitation_F02803	265	0.979	$\{x \in \mathbb{IR} \mid 0 < x < 5\}$	$\{x \in \mathbb{R} \mid 0 < x < 5\}$	interval $\{x \in \mathbb{R} \mid 0 < x < 5\}$. Similarly $\mathcal{O}_{[4,8]}$ is an open set, but note that $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$
fong-spivak-invitation_F02804	265	0.979	$\mathsf{o}_{\{[4,8]\}}$	$\mathcal{O}_{[4,8]}$	interval $\{x \in \mathbb{R} \mid 0 < x < 5\}$. Similarly $\mathcal{O}_{[4,8]}$ is an open set, but note that $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$
fong-spivak-invitation_F02805	265	0.979	$\mathsf{o}_{\{[0,5]\}} \cup \mathsf{o}_{\{[4,8]\}} \neq$	$\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$	interval $\{x \in \mathbb{R} \mid 0 < x < 5\}$. Similarly $\mathcal{O}_{[4,8]}$ is an open set, but note that $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$
fong-spivak-invitation_F02806	265	0.993	$\mathsf{o}_{\{[0,8]\}}$	$\mathcal{O}_{[0,8]}$	$\mathcal{O}_{[0,8]}$. Indeed, the interval $[2, 6]$ is in the right-hand side but not the left.
fong-spivak-invitation_F02807	265	1.000	$[2,6] \in \mathsf{o}_{\{[0,8]\}}$	$[2, 6] \in \mathcal{O}_{[0,8]}$	1. Explain why $[2, 6] \in \mathcal{O}_{[0,8]}$.
fong-spivak-invitation_F02808	265	0.977	$\mathsf{notin} \mathsf{o}_{\{[0,5]\}} \cup \mathsf{o}_{\{[4,8]\}}$	$\notin \mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]}$	2. Explain why $[2, 6] \notin \mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]}$.
fong-spivak-invitation_F02809	265	0.505	$\mathbf{B\,T} := \operatorname{Shv}(\mathbb{IR} \mid \mathbb{R}, \mathbf{Op})$	$\mathbf{BT} := \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$	Let $\mathbf{Op}$ denote the open sets of $\mathbb{IR}$, as described above, and let $\mathbf{BT} := \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$
fong-spivak-invitation_F02810	265	0.963	$\mathbb{R} \subseteq \mathbb{R}$	$\mathbb{R} \subseteq \mathbb{R}$	Exercise 7.59. Show that a subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{IR}$
fong-spivak-invitation_F02811	265	1.000	$U \cap \mathbb{R}$	$U \cap \mathbb{R}$	iff $U \cap \mathbb{R}$ is open in the usual topology on $\mathbb{R}$ defined in Example 7.18.
fong-spivak-invitation_F02812	265	0.912	$\mathbf{B\,T} = \operatorname{Shv}(\mathbb{IR} \mid \mathbb{R}, \mathbf{Op})$	$\mathbf{BT} = \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$	We cannot go into much depth about the sheaf topos $\mathbf{BT} = \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$, for reasons
fong-spivak-invitation_F02813	266	0.569	$\mathbb{IR}$	$\mathbb{IR}$	discuss what it means to be a sheaf on $\mathbb{IR}$, giving a few examples including that of the
fong-spivak-invitation_F02814	266	0.576	$\mathbb{IR}, \mathbf{Op}$	$\mathbb{IR}, \mathbf{Op}$	A sheaf S on the interval domain $(\mathbb{IR}, \mathbf{Op})$ is a functor $S: \mathbf{Op}^{\operatorname{op}} \rightarrow \mathbf{Set}$: it assigns to
fong-spivak-invitation_F02815	266	0.576	$S: \mathbf{Op}^{\operatorname{op}} \rightarrow$	$S: \mathbf{Op}^{\operatorname{op}} \rightarrow$	A sheaf S on the interval domain $(\mathbb{IR}, \mathbf{Op})$ is a functor $S: \mathbf{Op}^{\operatorname{op}} \rightarrow \mathbf{Set}$: it assigns to
fong-spivak-invitation_F02816	266	1.000	$S(U)$	$S(U)$	each open set U a set $S(U)$; how should we interpret this? An element $s \in S(U)$ is
fong-spivak-invitation_F02817	266	1.000	$\left(s_i\right)$	(s_i)	we can find matching U_i -events (s_i) for each $i \in I$, then the sheaf condition (Defini-
fong-spivak-invitation_F02818	266	0.871	$U \subseteq \mathbb{R}$	$U \subseteq \mathbb{R}$	We said in Section 7.5.1 that every open set $U \subseteq \mathbb{IR}$ can be written as the union
fong-spivak-invitation_F02819	266	1.000	$S(\mathsf{o}_{\{[a, b]\}})$	$S(\mathcal{O}_{[a,b]})$	$S(\mathcal{O}_{[a,b]})$ on these basic open sets. The sheaf condition furthermore implies that these
fong-spivak-invitation_F02820	266	0.991	$\operatorname{Shv}(\mathbb{IR} \mid \mathbb{R})$	$A \in \operatorname{Shv}(\mathbb{IR})$	Example 7.60. For any set A there is a sheaf $A \in \operatorname{Shv}(\mathbb{IR})$ that assigns to each open set
fong-spivak-invitation_F02821	266	1.000	$\operatorname{Shv}(\mathbb{IR} \mid \mathbb{R})(U) := A$	$A(U) := A$	U the set $A(U) := A$. This allows us to refer to integers, or real numbers, or letters of an
fong-spivak-invitation_F02822	266	1.000	$7 \in \mathbb{N}$	$7 \in \mathbb{N}$	alphabet, as though they were behaviors. What sort of behavior is $7 \in \mathbb{N}$? It is the sort of
fong-spivak-invitation_F02823	266	0.758	F_X	F_X	Example 7.61. Fix any topological space $(X, \mathbf{Op}_X)$. Then there is a sheaf F_X of local
fong-spivak-invitation_F02824	266	0.787	$\mathbb{R}$	$\mathbb{R}$	functions from $\mathbb{IR}$ to X . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$, we assign the set $F_X(U) :=$
fong-spivak-invitation_F02825	266	0.787	$U \in \mathbf{Op}_{\mathbb{R}}$	$U \in \mathbf{Op}_{\mathbb{R}}$	functions from $\mathbb{IR}$ to X . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$, we assign the set $F_X(U) :=$

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fong-spivak-invitation_F02826	266	0.787	$F_{\{X\}}(U) :=$	$F_X(U) :=$	<i>functions from $\mathbb{IR}$ to X. That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$, we assign the set $F_X(U) :=$</i>
fong-spivak-invitation_F02827	266	0.759	$\{f: U \rightarrowtail X \mid f$	$\{f: U \rightarrow X \mid f$	$\{f: U \rightarrow X \mid f$ is continuous}. There is also the sheaf G_X of local functions on the
fong-spivak-invitation_F02828	266	0.759	$G_{\{X\}}$	G_X	$\{f: U \rightarrow X \mid f$ is continuous}. There is also the sheaf G_X of local functions on the
fong-spivak-invitation_F02829	266	0.519	$\mathbb{R} \subseteq \mathbb{R}$	$\mathbb{R} \subseteq \mathbb{R}$	subspace $\mathbb{R} \subseteq \mathbb{IR}$. That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$, we assign the set $G_X(U) :=$
fong-spivak-invitation_F02830	266	0.519	$U \in \mathbf{Op}_{\mathbb{R}}$	$U \in \mathbf{Op}_{\mathbb{R}}$	subspace $\mathbb{R} \subseteq \mathbb{IR}$. That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$, we assign the set $G_X(U) :=$
fong-spivak-invitation_F02831	266	0.519	$G_{\{X\}}(U) :=$	$G_X(U) :=$	subspace $\mathbb{R} \subseteq \mathbb{IR}$. That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$, we assign the set $G_X(U) :=$
fong-spivak-invitation_F02832	266	0.985	$\{f: U \cap \mathbb{R} \rightarrowtail X \mid f$	$\{f: U \cap \mathbb{R} \rightarrow X \mid f$	$\{f: U \cap \mathbb{R} \rightarrow X \mid f$ is continuous}.
fong-spivak-invitation_F02833	266	0.846	$(X, \mathbf{Op}_X)$	$(X, \mathbf{Op}_X)$	$(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) := \{f: U \cap$
fong-spivak-invitation_F02834	266	0.846	$R \subseteq \mathbb{R}$	$R \subseteq \mathbb{R}$	$(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) := \{f: U \cap$
fong-spivak-invitation_F02835	266	0.846	$H_{\{X\}}(U) := \{f: U \cap$	$H_X(U) := \{f: U \cap$	$(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) := \{f: U \cap$
fong-spivak-invitation_F02836	266	1.000	$R \rightarrowtail X \mid f$	$R \rightarrow X \mid f$	$R \rightarrow X \mid f$ is continuous}.
fong-spivak-invitation_F02837	266	1.000	$H_{\{X\}}$	H_X	1. Is H_X a presheaf? If not, why not; if so, what are the restriction maps?
fong-spivak-invitation_F02838	267	0.740	$(d, d+3)$	$(d, d+3)$	idea is straightforward: for any interval of time, say a three-day interval $[d, d+3]$, let
fong-spivak-invitation_F02839	267	1.000	$L(d, d+3)$	$L(d, d+3)$	$L(d, d+3)$ denote the set of all possible behaviors a person who is in love could possibly
fong-spivak-invitation_F02840	267	0.458	$\mathbb{I}\mathbb{R}, \mathbf{Op}$	$\mathbb{I}\mathbb{R}, \mathbf{Op}$	BT is the topos of sheaves on the space $(\mathbb{IR}, \mathbf{Op})$, as defined in Section 7.5.1. As
fong-spivak-invitation_F02841	267	1.000	$(0, 5)$	$(0, 5)$	Suppose Bob likes the weather throughout the interval $(0, 5)$ and throughout the
fong-spivak-invitation_F02842	267	0.806	$(4, 8)$	$(4, 8)$	remained watching position p .” Then just because it’s true on $(0, 5)$ and on $(4, 8)$, does
fong-spivak-invitation_F02843	267	1.000	$(0, 8)$	$(0, 8)$	be true on $(0, 5)$ and true on $(4, 8)$ but not on $(0, 8)$. In order to capture the semantics of
fong-spivak-invitation_F02844	268	1.000	$\theta \in \mathbb{R}_{\geq 0}$	$\theta \in \mathbb{R}_{\geq 0}$	behavior types: the altimeter is a variable whose value $\theta \in \mathbb{R}_{\geq 0}$ is changing continuously
fong-spivak-invitation_F02845	268	1.000	$p(D, T)$	$p(D, T)$	p , one could imagine the predicate $p(D, T)$ for $D: \text{dials}$ and $T: \text{thrusters}$ as
fong-spivak-invitation_F02846	268	0.968	$@_t$	$@_t$	Here $@_t$ is a modality, as we discussed in Definition 7.52; in fact it turns out to be one
fong-spivak-invitation_F02847	268	1.000	$@_t(q)$	$@_t(q)$	statement $@_t(q)$ says that q is true in some small enough neighborhood around t . So
fong-spivak-invitation_F02848	268	0.999	$D \in$	$D \in$	Given an actual playing-out-of-events over a time period U , i.e. actual section $D \in$
fong-spivak-invitation_F02849	268	1.000	$\text{dials}(U)$	$\text{dials}(U)$	$\text{dials}(U)$ and section $T \in \text{thrusters}(U)$, the predicate Eq. (7.19) will hold on
fong-spivak-invitation_F02850	268	1.000	$T \in \text{thrusters}(U)$	$T \in \text{thrusters}(U)$	$\text{dials}(U)$ and section $T \in \text{thrusters}(U)$, the predicate Eq. (7.19) will hold on
fong-spivak-invitation_F02851	270	1.000	$f(x)=x+5$	$f(x) = x + 5$	Order-preserving: Take $f(x) = x + 5$; if $x \leq y$ then $x + 5 \leq y + 5$, so f is order-preserving.
fong-spivak-invitation_F02852	270	1.000	$x+5 \leq y+5$	$x + 5 \leq y + 5$	Order-preserving: Take $f(x) = x + 5$; if $x \leq y$ then $x + 5 \leq y + 5$, so f is order-preserving.

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fong-spivak-invitation_F02853	270	■ 0.498	$g(x) := -x$	$g(x) := -x$	Take $g(x) := -x$; even though $1 \leq 2$, the required inequality $-1 \leq^? -2$ does not hold, so g
fong-spivak-invitation_F02854	270	■ 0.498	$1 \leq 2$	$1 \leq 2$	Take $g(x) := -x$; even though $1 \leq 2$, the required inequality $-1 \leq^? -2$ does not hold, so g
fong-spivak-invitation_F02855	270	■ 0.498	$-1 \leq ?$	$-1 \leq ?$	Take $g(x) := -x$; even though $1 \leq 2$, the required inequality $-1 \leq^? -2$ does not hold, so g
fong-spivak-invitation_F02856	270	■ 0.992	$f(x) := x+5$	$f(x) := x+5$	Metric-preserving: Take $f(x) := x+5$; for any x, y we have $ x-y = (x+5)-(y+5) $
fong-spivak-invitation_F02857	270	■ 0.992	$ x-y = (x+5)-(y+5) $	$ x-y = (x+5)-(y+5) $	Metric-preserving: Take $f(x) := x+5$; for any x, y we have $ x-y = (x+5)-(y+5) $
fong-spivak-invitation_F02858	270	■ 1.000	$g(x) := 2 * x$	$g(x) := 2 * x$	$g(x) := 2 * x$; then with $x = 1$ and $y = 2$ we have $ x-y = 1$ but $ 2x-2y = 2$, so g does
fong-spivak-invitation_F02859	270	■ 1.000	$x=1$	$x = 1$	$g(x) := 2 * x$; then with $x = 1$ and $y = 2$ we have $ x-y = 1$ but $ 2x-2y = 2$, so g does
fong-spivak-invitation_F02860	270	■ 1.000	$y=2$	$y = 2$	$g(x) := 2 * x$; then with $x = 1$ and $y = 2$ we have $ x-y = 1$ but $ 2x-2y = 2$, so g does
fong-spivak-invitation_F02861	270	■ 1.000	$ x-y =1$	$ x-y = 1$	$g(x) := 2 * x$; then with $x = 1$ and $y = 2$ we have $ x-y = 1$ but $ 2x-2y = 2$, so g does
fong-spivak-invitation_F02862	270	■ 1.000	$ 2x-2y =2$	$ 2x-2y = 2$	$g(x) := 2 * x$; then with $x = 1$ and $y = 2$ we have $ x-y = 1$ but $ 2x-2y = 2$, so g does
fong-spivak-invitation_F02863	270	■ 1.000	$f(x) := 3 * x$	$f(x) := 3 * x$	Addition-preserving: Take $f(x) := 3 * x$; for any x, y we have $3 * (x+y) = (3 * x) + (3 * y)$, so
fong-spivak-invitation_F02864	270	■ 1.000	$3 * (x+y) = (3 * x) + (3 * y)$	$3 * (x+y) = (3 * x) + (3 * y)$	Addition-preserving: Take $f(x) := 3 * x$; for any x, y we have $3 * (x+y) = (3 * x) + (3 * y)$, so
fong-spivak-invitation_F02865	270	■ 1.000	$g(x) := x+1$	$g(x) := x+1$	f preserves addition. Take $g(x) := x+1$; then with $x = 0$ and $y = 0$, we have $g(x+y) = 1$,
fong-spivak-invitation_F02866	270	■ 1.000	$x=0$	$x = 0$	f preserves addition. Take $g(x) := x+1$; then with $x = 0$ and $y = 0$, we have $g(x+y) = 1$,
fong-spivak-invitation_F02867	270	■ 1.000	$y=0$	$y = 0$	f preserves addition. Take $g(x) := x+1$; then with $x = 0$ and $y = 0$, we have $g(x+y) = 1$,
fong-spivak-invitation_F02868	270	■ 1.000	$g(x+y)=1$	$g(x+y) = 1$	f preserves addition. Take $g(x) := x+1$; then with $x = 0$ and $y = 0$, we have $g(x+y) = 1$,
fong-spivak-invitation_F02869	270	■ 1.000	$g(x)+g(y)=2$	$g(x) + g(y) = 2$	but $g(x) + g(y) = 2$, so g does not preserve addition.
fong-spivak-invitation_F02870	271	■ 0.995	$\{1,2,3,4\}$	$\{1,2,3,4\}$	2. Here is a picture (using text, rather than circles) for partitions of the set $\{1, 2, 3, 4\}$:
fong-spivak-invitation_F02871	271	■ 1.000	$A=(12)(3)(4)$	$A = (12)(3)(4)$	For the remaining parts, we choose $A = (12)(3)(4)$ and $B = (13)(2)(4)$.
fong-spivak-invitation_F02872	271	■ 1.000	$B=(13)(2)(4)$	$B = (13)(2)(4)$	For the remaining parts, we choose $A = (12)(3)(4)$ and $B = (13)(2)(4)$.
fong-spivak-invitation_F02873	271	■ 1.000	$A \vee B = (123)(4)$	$A \vee B = (123)(4)$	3. $A \vee B = (123)(4)$.
fong-spivak-invitation_F02874	271	■ 1.000	$0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$	$0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$	2. This is false: $0 \in \mathbb{N}$ but $0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$.
fong-spivak-invitation_F02875	271	■ 0.936	$\{1,2,3\}$	$\{1,2,3\}$	2. The union of $\{1, 2, 3\}$ and $\{1\}$ is $\{1, 2, 3\} \cup \{1\} = \{1, 2, 3\}$.
fong-spivak-invitation_F02876	271	■ 0.936	$\{1,2,3\} \cup \{1\} = \{1,2,3\}$	$\{1,2,3\} \cup \{1\} = \{1,2,3\}$	2. The union of $\{1, 2, 3\}$ and $\{1\}$ is $\{1, 2, 3\} \cup \{1\} = \{1, 2, 3\}$.
fong-spivak-invitation_F02877	271	■ 0.994	$\{h, 1\} \times \{1,2,3\}$	$\{h, 1\} \times \{1,2,3\}$	3. The six elements of $\{h, 1\} \times \{1, 2, 3\}$ are
fong-spivak-invitation_F02878	271	■ 0.996	$\{h, 1\} \sqcup \{1,2,3\}$	$\{h, 1\} \sqcup \{1,2,3\}$	4. The five elements of $\{h, 1\} \sqcup \{1, 2, 3\}$ are
fong-spivak-invitation_F02879	271	■ 1.000	$\{h, 1\} \cup \{1,2,3\}$	$\{h, 1\} \cup \{1,2,3\}$	5. The four elements of $\{h, 1\} \cup \{1, 2, 3\}$ are
fong-spivak-invitation_F02880	272	■ 1.000	$p_1^{\prime}, p_2^{\prime} \in P^{\prime}$	$p_1^{\prime}, p_2^{\prime} \in P^{\prime}$	1. Given $p \in P$, suppose we had $p_1^{\prime}, p_2^{\prime} \in P^{\prime}$ such that $A_p = A_{p_1^{\prime}}^{\prime}$ and $A_p = A_{p_2^{\prime}}^{\prime}$. Well then

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fong-spivak-invitation_F02881	272	■ 1.000	$A_{\{p\}}=A_{\{p_1\}^{\wedge\{\prime\prime}\}}^{\wedge\{\prime\prime\}}$	$A_p = A'_{p'_1}$	1. Given $p \in P$, suppose we had $p'_1, p'_2 \in P'$ such that $A_p = A'_{p'_1}$ and $A_p = A'_{p'_2}$. Well then
fong-spivak-invitation_F02882	272	■ 1.000	$A_{\{p\}}=A_{\{p_2\}^{\wedge\{\prime\prime}\}}^{\wedge\{\prime\prime\}}$	$A_p = A'_{p'_2}$	1. Given $p \in P$, suppose we had $p'_1, p'_2 \in P'$ such that $A_p = A'_{p'_1}$ and $A_p = A'_{p'_2}$. Well then
fong-spivak-invitation_F02883	272	■ 1.000	$A_{\{p_1\}^{\wedge\{\prime\prime}\}}=A_{\{p_2\}^{\wedge\{\prime\prime}\}}$	$A_{p'_1} = A_{p'_2}$	$A_{p'_1} = A_{p'_2}$, so in particular $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$. By Definition 1.10 of partition, $A_{p'_1} \neq \emptyset$,
fong-spivak-invitation_F02884	272	■ 1.000	$A_{\{p_1\}^{\wedge\{\prime\prime}\}} \cap A_{\{p_2\}^{\wedge\{\prime\prime}\}}=A_{\{p_1\}^{\wedge\{\prime\prime\}}}$	$A_{p'_1} \cap A_{p'_2} = A_{p'_1}$	$A_{p'_1} = A_{p'_2}$, so in particular $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$. By Definition 1.10 of partition, $A_{p'_1} \neq \emptyset$,
fong-spivak-invitation_F02885	272	■ 1.000	$A_{\{p_1\}^{\wedge\{\prime\prime}\}} \neq \varnothing$	$A_{p'_1} \neq \emptyset$	$A_{p'_1} = A_{p'_2}$, so in particular $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$. By Definition 1.10 of partition, $A_{p'_1} \neq \emptyset$,
fong-spivak-invitation_F02886	272	■ 1.000	$p_1 \neq p_2$	$p_1 \neq p_2$	and yet if $p_1 \neq p_2$ then $A_{p'_1} \cap A_{p'_2} = \emptyset$. This can't be, so we must have $p'_1 = p'_2$, as
fong-spivak-invitation_F02887	272	■ 1.000	$A_{\{p_1\}^{\wedge\{\prime\prime}\}} \cap A_{\{p_2\}^{\wedge\{\prime\prime}\}}=\varnothing$	$A_{p'_1} \cap A_{p'_2} = \emptyset$	and yet if $p_1 \neq p_2$ then $A_{p'_1} \cap A_{p'_2} = \emptyset$. This can't be, so we must have $p'_1 = p'_2$, as
fong-spivak-invitation_F02888	272	■ 1.000	$p_1^{\wedge\{\prime\prime\}}=p_2^{\wedge\{\prime\prime\}}$	$p'_1 = p'_2$	and yet if $p_1 \neq p_2$ then $A_{p'_1} \cap A_{p'_2} = \emptyset$. This can't be, so we must have $p'_1 = p'_2$, as
fong-spivak-invitation_F02889	272	■ 1.000	$A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}} \neq \varnothing$	$A'_{p'} \neq \emptyset$	Since $A'_{p'} \neq \emptyset$ is nonempty by definition, we can pick some $a \in A'_{p'}$; since $A'_{p'} \subseteq A$, we
fong-spivak-invitation_F02890	272	■ 1.000	$a \in A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}}$	$a \in A'_{p'}$	Since $A'_{p'} \neq \emptyset$ is nonempty by definition, we can pick some $a \in A'_{p'}$; since $A'_{p'} \subseteq A$, we
fong-spivak-invitation_F02891	272	■ 1.000	$A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}} \subseteq A$	$A'_{p'} \subseteq A$	Since $A'_{p'} \neq \emptyset$ is nonempty by definition, we can pick some $a \in A'_{p'}$; since $A'_{p'} \subseteq A$, we
fong-spivak-invitation_F02892	272	■ 1.000	$p^{\wedge\{\prime\prime\}} \in P^{\wedge\{\prime\prime\}}$	$p'' \in P'$	p ; now we show that $A_p = A'_{p'}$. By assumption there is some $p'' \in P'$ with $A_p = A'_{p''}$,
fong-spivak-invitation_F02893	272	■ 1.000	$A_{\{p\}}=A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}}$	$A_p = A'_{p''}$	p ; now we show that $A_p = A'_{p'}$. By assumption there is some $p'' \in P'$ with $A_p = A'_{p''}$,
fong-spivak-invitation_F02894	272	■ 1.000	$a \in A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}}$	$a \in A'_{p''}$	so now $a \in A'_{p''}$ and $a \in A'_{p'}$, so $a \in A'_{p'} \cap A'_{p''}$. Again by definition, having a nonempty
fong-spivak-invitation_F02895	272	■ 1.000	$a \in A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}} \cap A_{\{p^{\wedge\{\prime\prime\}}\}^{\wedge\{\prime\prime\}}}$	$a \in A'_{p'} \cap A'_{p''}$	so now $a \in A'_{p''}$ and $a \in A'_{p'}$, so $a \in A'_{p'} \cap A'_{p''}$. Again by definition, having a nonempty
fong-spivak-invitation_F02896	272	■ 1.000	$p^{\wedge\{\prime\prime\}}=p^{\wedge\{\prime\prime\}}^{\wedge\{\prime\prime\}}$	$p' = p''$	intersection means $p' = p''$. So we conclude that $A_p = A_{p'}$.
fong-spivak-invitation_F02897	272	■ 1.000	$A_{\{p\}}=A_{\{p^{\wedge\{\prime\prime\}}\}}$	$A_p = A_{p'}$	intersection means $p' = p''$. So we conclude that $A_p = A_{p'}$.
fong-spivak-invitation_F02898	272	■ 1.000	$a \in A_{\{p\}} \cap A_{\{q\}}$	$a \in A_p \cap A_q$	suppose otherwise and derive a contradiction. So suppose there exists $a \in A_p \cap A_q$; we will
fong-spivak-invitation_F02899	272	■ 1.000	$A_{\{p\}}=A_{\{q\}}$	$A_p = A_q$	show that $A_p = A_q$, which contradicts an earlier hypothesis. To show that these two subsets
fong-spivak-invitation_F02900	272	■ 1.000	$a^{\wedge\{\prime\prime\}} \in A_{\{p\}}$	$a' \in A_p$	are equal, it suffices to show that $a' \in A_p$ iff $a' \in A_q$ for all $a' \in A$. Suppose $a' \in A_p$;
fong-spivak-invitation_F02901	272	■ 1.000	$a^{\wedge\{\prime\prime\}} \in A_{\{q\}}$	$a' \in A_q$	are equal, it suffices to show that $a' \in A_p$ iff $a' \in A_q$ for all $a' \in A$. Suppose $a' \in A_p$;
fong-spivak-invitation_F02902	272	■ 1.000	$a^{\wedge\{\prime\prime\}} \in A$	$a' \in A$	are equal, it suffices to show that $a' \in A_p$ iff $a' \in A_q$ for all $a' \in A$. Suppose $a' \in A_p$;
fong-spivak-invitation_F02903	272	■ 1.000	$a \sim a^{\wedge\{\prime\prime\}}$	$a \sim a'$	then because A_p is connected, we have $a \sim a'$. And because A_q is closed, $a' \in A_q$. In a
fong-spivak-invitation_F02904	272	■ 1.000	$X:=\left\{a^{\wedge\{\prime\prime\}} \in A \mid a^{\wedge\{\prime\prime\}} \sim a\right\}$	$X := \{a' \in A \mid a' \sim a\}$	show that there is some closed and connected subset containing a . Let $X := \{a' \in A \mid a' \sim a\}$;
fong-spivak-invitation_F02905	272	■ 0.962	$a^{\wedge\{\prime\prime\}} \in X$	$a' \in X$	we claim it is closed and connected and contains a . To see X is closed, suppose $a' \in X$ and

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fong-spivak-invitation_F02906	272	■ 0.999	$b \sim a^{\prime}$	$b \sim a'$	$b \sim a'$; then $b \sim a$ by transitivity and symmetry of $\sim$, so $b \in X$. To see that X is connected,
fong-spivak-invitation_F02907	272	■ 0.999	$b \sim a$	$b \sim a$	$b \sim a'$; then $b \sim a$ by transitivity and symmetry of $\sim$, so $b \in X$. To see that X is connected,
fong-spivak-invitation_F02908	272	■ 0.999	$b \in X$	$b \in X$	$b \sim a'$; then $b \sim a$ by transitivity and symmetry of $\sim$, so $b \in X$. To see that X is connected,
fong-spivak-invitation_F02909	272	■ 0.999	$b, c \in X$	$b, c \in X$	suppose $b, c \in X$; then $b \sim a$ and $c \sim a$ so $b \sim c$ by the transitivity and symmetry of $\sim$.
fong-spivak-invitation_F02910	272	■ 0.999	$c \sim a$	$c \sim a$	suppose $b, c \in X$; then $b \sim a$ and $c \sim a$ so $b \sim c$ by the transitivity and symmetry of $\sim$.
fong-spivak-invitation_F02911	272	■ 0.896	$a \in X$	$a \in X$	Finally, $a \in X$ by the reflexivity of $\sim$.
fong-spivak-invitation_F02912	272	■ 1.000	$\varnothing \rightarrow \{1\}$	$\emptyset \rightarrow \{1\}$	1. The unique function $\emptyset \rightarrow \{1\}$ is injective but not surjective.
fong-spivak-invitation_F02913	272	■ 1.000	$\{a, b\} \rightarrow \{1\}$	$\{a, b\} \rightarrow \{1\}$	2. The unique function $\{a, b\} \rightarrow \{1\}$ is surjective but not injective.
fong-spivak-invitation_F02914	273	■ 1.000	$F \subseteq A \times \varnothing$	$F \subseteq A \times \emptyset$	By Definition 1.17, a function $f: A \rightarrow \emptyset$ is a subset $F \subseteq A \times \emptyset$ such that, for all $a \in A$, there
fong-spivak-invitation_F02915	273	■ 0.862	$b \in \varnothing$	$b \in \emptyset$	exists a unique $b \in \emptyset$ with $(a, b) \in F$. But there are no elements $b \in \emptyset$, so if F is to have the
fong-spivak-invitation_F02916	273	■ 0.862	$(a, b) \in F$	$(a, b) \in F$	exists a unique $b \in \emptyset$ with $(a, b) \in F$. But there are no elements $b \in \emptyset$, so if F is to have the
fong-spivak-invitation_F02917	273	■ 0.862	$b \in^? \varnothing$	$b \in^? \emptyset$	exists a unique $b \in \emptyset$ with $(a, b) \in F$. But there are no elements $b \in \emptyset$, so if F is to have the
fong-spivak-invitation_F02918	273	■ 0.999	$\{\bullet, *, \circ\}$	$\{\bullet, *, \circ\}$	Below each partition, we draw a corresponding surjection out of $\{\bullet, *, \circ\}$:
fong-spivak-invitation_F02919	273	■ 0.962	$1 \rightarrow 3$	$1 \rightarrow 3$	The graph G from Exercise 1.33 is a strange Hasse diagram because it has two arrows $1 \rightarrow 3$ and
fong-spivak-invitation_F02920	273	■ 0.582	$P := V = \{1, 2, 3, 4\}$	$P := V = \{1, 2, 3, 4\}$	formula from working. The preorder $(P, \leq)$ is given by taking $P := V = \{1, 2, 3, 4\}$ and writing
fong-spivak-invitation_F02921	274	■ 1.000	$(\bullet)(\circ)(*) \leq (\bullet)(\circ)(*)$	$(\bullet)(\circ)(*) \leq (\bullet)(\circ)(*)$	$(\bullet)(\circ)(*) < (\bullet)(\circ)(*)$,
fong-spivak-invitation_F02922	274	■ 1.000	$(\bullet)(\circ)(*) \leq (\bullet\circ)(*)$	$(\bullet)(\circ)(*) \leq (\bullet\circ)(*)$	$(\bullet)(\circ)(*) < (\bullet\circ)(*)$,
fong-spivak-invitation_F02923	274	■ 0.982	$(\bullet)(\circ)(*) \leq (\bullet*)(\circ)$	$(\bullet)(\circ)(*) \leq (\bullet*)(\circ)$	$(\bullet)(\circ)(*) < (\bullet*)(\circ)$,
fong-spivak-invitation_F02924	274	■ 0.950	$(\bullet)(\circ)(*) \leq (\bullet)(\circ*)$	$(\bullet)(\circ)(*) \leq (\bullet)(\circ*)$	$(\bullet)(\circ)(*) < (\bullet)(\circ*)$,
fong-spivak-invitation_F02925	274	■ 0.651	$(\bullet)(\circ)(*) \leq (\bullet\circ*)$	$(\bullet)(\circ)(*) \leq (\bullet\circ*)$	$(\bullet)(\circ)(*) < (\bullet\circ*)$,
fong-spivak-invitation_F02926	274	■ 1.000	$(\bullet\circ)(*) \leq (\bullet\circ)(*)$	$(\bullet\circ)(*) \leq (\bullet\circ)(*)$	$(\bullet\circ)(*) < (\bullet\circ)(*)$,
fong-spivak-invitation_F02927	274	■ 1.000	$(\bullet\circ)(*) \leq (\bullet\circ*)$	$(\bullet\circ)(*) \leq (\bullet\circ*)$	$(\bullet\circ)(*) < (\bullet\circ*)$,
fong-spivak-invitation_F02928	274	■ 1.000	$(\bullet*)(\circ) \leq (\bullet*)(\circ)$	$(\bullet*)(\circ) \leq (\bullet*)(\circ)$	$(\bullet*)(\circ) < (\bullet*)(\circ)$,
fong-spivak-invitation_F02929	274	■ 1.000	$(\bullet*)(\circ) \leq (\bullet\circ*)$	$(\bullet*)(\circ) \leq (\bullet\circ*)$	$(\bullet*)(\circ) < (\bullet\circ*)$,
fong-spivak-invitation_F02930	274	■ 0.999	$(\bullet)(\circ*) \leq (\bullet)(\circ*)$	$(\bullet)(\circ*) \leq (\bullet)(\circ*)$	$(\bullet)(\circ*) < (\bullet)(\circ*)$,
fong-spivak-invitation_F02931	274	■ 0.896	$(\bullet)(\circ*) \leq (\bullet\circ*)$	$(\bullet)(\circ*) \leq (\bullet\circ*)$	$(\bullet)(\circ*) < (\bullet\circ*)$,
fong-spivak-invitation_F02932	274	■ 0.860	$(\bullet\circ*) \leq (\bullet\circ*)$	$(\bullet\circ*) \leq (\bullet\circ*)$	$(\bullet\circ*) < (\bullet\circ*)$.
fong-spivak-invitation_F02933	274	■ 1.000	$4 \not\leq 6$	$4 \not\leq 6$	No, it is not a total order; for example $4 \not\leq 6$ and $6 \not\leq 4$.
fong-spivak-invitation_F02934	274	■ 1.000	$6 \not\leq 4$	$6 \not\leq 4$	No, it is not a total order; for example $4 \not\leq 6$ and $6 \not\leq 4$.

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fong-spivak-invitation_F02935	274	■ 1.000	$a, b \in \mathbb{R}$	$a, b \in \mathbb{R}$	Yes, the usual $\leq$ ordering is a total order on $\mathbb{R}$: for every $a, b \in \mathbb{R}$ either $a \leq b$ or $b \leq a$.
fong-spivak-invitation_F02936	274	■ 1.000	$!: S \rightarrow \{1\}$	$!: S \rightarrow \{1\}$	The coarsest partition on S corresponds to the unique function $!: S \rightarrow \{1\}$. The finest partition on
fong-spivak-invitation_F02937	274	■ 0.953	$\mathrm{U}(X)$	$\mathrm{U}(X)$	q such that $p \leq q$ is p itself, so q is definitely in U ! This means that $\mathrm{U}(X)$ contains all subsets of
fong-spivak-invitation_F02938	274	■ 1.000	$\mathrm{U}(X) = \mathrm{P}(X)$	$\mathrm{U}(X) = \mathrm{P}(X)$	X , so it's exactly the power set, $\mathrm{U}(X) = \mathrm{P}(X)$.
fong-spivak-invitation_F02939	275	■ 1.000	$0 \leq \cdots \leq 3$	$0 \leq \cdots \leq 3$	With $X = \{0, 1, 2\}$, the Hasse diagram for $\mathrm{P}(X)$, the preorder $0 \leq \cdots \leq 3$, and the cardinality
fong-spivak-invitation_F02940	275	■ 1.000	$q \in \uparrow p$	$q \in \uparrow p$	1. Let $q \in \uparrow p$, and suppose $q \leq q'$. Since $q \in \uparrow p$, we have $p \leq q$. Thus by transitivity $p \leq q'$,
fong-spivak-invitation_F02941	275	■ 1.000	$p \leq q^{\prime}$	$p \leq q'$	1. Let $q \in \uparrow p$, and suppose $q \leq q'$. Since $q \in \uparrow p$, we have $p \leq q$. Thus by transitivity $p \leq q'$,
fong-spivak-invitation_F02942	275	■ 0.835	$q^{\prime} \in \uparrow p$	$q' \in \uparrow p$	so $q' \in \uparrow p$. Thus $\uparrow p$ is an upper set.
fong-spivak-invitation_F02943	275	■ 0.835	$\uparrow p$	$\uparrow p$	so $q' \in \uparrow p$. Thus $\uparrow p$ is an upper set.
fong-spivak-invitation_F02944	275	■ 0.966	$q \leq^{\mathrm{op}} p$	$q \leq^{\mathrm{op}} p$	2. Suppose $p \leq q$ in P ; this means that $q \leq^{\mathrm{op}} p$ in P^{op} . We must show that $\uparrow q \subseteq \uparrow p$. Take
fong-spivak-invitation_F02945	275	■ 0.966	P^{op}	P^{op}	2. Suppose $p \leq q$ in P ; this means that $q \leq^{\mathrm{op}} p$ in P^{op} . We must show that $\uparrow q \subseteq \uparrow p$. Take
fong-spivak-invitation_F02946	275	■ 0.966	$\uparrow q \subseteq \uparrow p$	$\uparrow q \subseteq \uparrow p$	2. Suppose $p \leq q$ in P ; this means that $q \leq^{\mathrm{op}} p$ in P^{op} . We must show that $\uparrow q \subseteq \uparrow p$. Take
fong-spivak-invitation_F02947	275	■ 1.000	$q^{\prime} \in \uparrow q$	$q' \in \uparrow q$	any $q' \in \uparrow q$. Then $q \leq q'$, so by transitivity $p \leq q'$, and hence $q' \in \uparrow p$. Thus $\uparrow q \subseteq \uparrow p$.
fong-spivak-invitation_F02948	275	■ 1.000	$p \not\leq p^{\prime}$	$p \not\leq p'$	that if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$. This is straightforward, since by reflexivity we always have
fong-spivak-invitation_F02949	275	■ 1.000	$\uparrow(p') \not\subseteq \uparrow(p)$	$\uparrow(p') \not\subseteq \uparrow(p)$	that if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$. This is straightforward, since by reflexivity we always have
fong-spivak-invitation_F02950	275	■ 1.000	$p' \in \uparrow(p')$	$p' \in \uparrow(p')$	$p' \in \uparrow(p')$, but if $p \not\leq p'$, then $p' \notin \uparrow(p)$, so $\uparrow(p') \not\subseteq \uparrow(p)$.
fong-spivak-invitation_F02951	275	■ 1.000	$p' \notin \uparrow(p)$	$p' \notin \uparrow(p)$	$p' \in \uparrow(p')$, but if $p \not\leq p'$, then $p' \notin \uparrow(p)$, so $\uparrow(p') \not\subseteq \uparrow(p)$.
fong-spivak-invitation_F02952	276	■ 0.506	$P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$	$P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$	4. The map $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$ can be depicted:
fong-spivak-invitation_F02953	276	■ 1.000	$p_1 \leq_P p_2$	$p_1 \leq_P p_2$	every function $f: P \rightarrow Q$ is monotone, i.e. that if $p_1 \leq_P p_2$ then $f(p_1) \leq_Q f(p_2)$. But in
fong-spivak-invitation_F02954	276	■ 1.000	$f(p_1) \leq_Q f(p_2)$	$f(p_1) \leq_Q f(p_2)$	every function $f: P \rightarrow Q$ is monotone, i.e. that if $p_1 \leq_P p_2$ then $f(p_1) \leq_Q f(p_2)$. But in
fong-spivak-invitation_F02955	276	■ 1.000	$f(p_1) = f(p_2)$	$f(p_1) = f(p_2)$	$f(p_1) = f(p_2)$, so $f(p_1) \leq f(p_2)$.
fong-spivak-invitation_F02956	276	■ 1.000	$f(p_1) \leq f(p_2)$	$f(p_1) \leq f(p_2)$	$f(p_1) = f(p_2)$, so $f(p_1) \leq f(p_2)$.
fong-spivak-invitation_F02957	276	■ 0.999	$X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$	$X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$	Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$; let $f: X \rightarrow$
fong-spivak-invitation_F02958	276	■ 0.999	$Y = \{n, z, p\}$	$Y = \{n, z, p\}$	Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$; let $f: X \rightarrow$
fong-spivak-invitation_F02959	276	■ 0.999	$f: X \rightarrow$	$f: X \rightarrow$	Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$; let $f: X \rightarrow$
fong-spivak-invitation_F02960	276	■ 1.000	$P := (nz)(p)$	$P := (nz)(p)$	We consider two partitions of Y , namely $P := (nz)(p)$ and $Q := (np)(z)$. Technically, these are
fong-spivak-invitation_F02961	276	■ 1.000	$Q := (np)(z)$	$Q := (np)(z)$	We consider two partitions of Y , namely $P := (nz)(p)$ and $Q := (np)(z)$. Technically, these are
fong-spivak-invitation_F02962	276	■ 1.000	$\{\{n, z\}, \{p\}\}$	$\{\{n, z\}, \{p\}\}$	notation for $\{\{n, z\}, \{p\}\}$ and $\{\{n, p\}, \{z\}\}$ as sets of disjoint subsets whose union is Y . Their pulled

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fong-spivak-invitation_F02963	276	■ 1.000	$\{\{n, p\}, \{z\}\}$	$\{\{n, p\}, \{z\}\}$	notation for $\{\{n, z\}, \{p\}\}$ and $\{\{n, p\}, \{z\}\}$ as sets of disjoint subsets whose union is Y . Their pulled
fong-spivak-invitation_F02964	276	■ 1.000	$f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$	$f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$	back partitions are $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$ and $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$, or
fong-spivak-invitation_F02965	276	■ 1.000	$f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$	$f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$	back partitions are $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$ and $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$, or
fong-spivak-invitation_F02966	276	■ 1.000	$\left(P, \leq_P\right), \left(Q, \leq_Q\right)$	$(P, \leq_P), (Q, \leq_Q)$	We have preorders $(P, \leq_P)$, $(Q, \leq_Q)$, and $(R, \leq_R)$, and we have monotone maps $f: P \rightarrow Q$
fong-spivak-invitation_F02967	276	■ 0.812	id_P	id_P	1. To see that id_P is monotone, we need to show that if $p_1 \leq_P p_2$ then $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$. But
fong-spivak-invitation_F02968	276	■ 0.812	$\operatorname{id}_P(p_1) \leq \operatorname{id}_P(p_2)$	$\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$	1. To see that id_P is monotone, we need to show that if $p_1 \leq_P p_2$ then $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$. But
fong-spivak-invitation_F02969	276	■ 1.000	$\operatorname{id}_P(p) = p$	$\mathrm{id}_P(p) = p$	$\mathrm{id}_P(p) = p$ for all $p \in P$, so this is clear.
fong-spivak-invitation_F02970	276	■ 0.988	$q_1 \leq_Q q_2$	$q_1 \leq_Q q_2$	2. We have that $p_1 \leq_P p_2$ implies $f(p_1) \leq_Q f(p_2)$ and that $q_1 \leq_Q q_2$ implies $g(q_1) \leq_R$
fong-spivak-invitation_F02971	276	■ 0.988	$g(q_1) \leq_R$	$g(q_1) \leq_R$	2. We have that $p_1 \leq_P p_2$ implies $f(p_1) \leq_Q f(p_2)$ and that $q_1 \leq_Q q_2$ implies $g(q_1) \leq_R$
fong-spivak-invitation_F02972	276	■ 1.000	$g(q_2)$	$g(q_2)$	$g(q_2)$. By substitution, $p_1 \leq_P p_2$ implies $g(f(p_1)) \leq_R g(f(p_2))$ which is exactly what is
fong-spivak-invitation_F02973	276	■ 1.000	$g(f(p_1)) \leq_R g(f(p_2))$	$g(f(p_1)) \leq_R g(f(p_2))$	$g(q_2)$. By substitution, $p_1 \leq_P p_2$ implies $g(f(p_1)) \leq_R g(f(p_2))$ which is exactly what is
fong-spivak-invitation_F02974	276	■ 0.679	$(f; g)$	$(f; g)$	required for $(f \circ g)$ to be monotone.
fong-spivak-invitation_F02975	276	■ 0.998	$p_2 \leq p_1$	$p_2 \leq p_1$	skeletal, i.e. $p_1 \leq p_2$ and $p_2 \leq p_1$ implies $p_1 = p_2$. And suppose it is dagger, i.e. $p_1 \leq p_2$
fong-spivak-invitation_F02976	277	■ 1.000	$\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow \mathbb{B}$	$\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow \mathbb{B}$	$\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow \mathbb{B}$. So suppose P, Q are partitions with $P \leq Q$; we need to show that if
fong-spivak-invitation_F02977	277	■ 0.994	$\Phi(P) =$	$\Phi(P) =$	$\Phi(P) = \text{true}$ then $\Phi(Q) = \text{true}$.
fong-spivak-invitation_F02978	277	■ 0.994	$\Phi(Q) =$	$\Phi(Q) =$	$\Phi(P) = \text{true}$ then $\Phi(Q) = \text{true}$.
fong-spivak-invitation_F02979	277	■ 1.000	$x \sim_P y$	$x \sim_P y$	lumps more stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying
fong-spivak-invitation_F02980	277	■ 1.000	$x \sim_Q y$	$x \sim_Q y$	lumps more stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying
fong-spivak-invitation_F02981	277	■ 1.000	$x, y \in \{\bullet, *, \circ\}$	$x, y \in \{\bullet, *, \circ\}$	lumps more stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$. Applying
fong-spivak-invitation_F02982	277	■ 0.980	$U \mapsto f^{-1}(U)$	$U \mapsto f^{-1}(U)$	Given a function $f: P \rightarrow Q$, we have $f^*: \mathcal{U}(Q) \rightarrow \mathcal{U}(P)$ given by $U \mapsto f^{-1}(U)$. But upper
fong-spivak-invitation_F02983	277	■ 1.000	$u: Q \rightarrow \mathbb{B}$	$u: Q \rightarrow \mathbb{B}$	sets in Q are classified by monotone maps $u: Q \rightarrow \mathbb{B}$, and similarly for P ; our job is to show that
fong-spivak-invitation_F02984	277	■ 1.000	$f^*(U)$	$f^*(U)$	$f^*(U)$ is given by composing the classifier u with f .
fong-spivak-invitation_F02985	277	■ 0.904	$q \mapsto$	$q \mapsto$	$q \mapsto \text{true}$ iff $q \in U$. Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$; it corresponds to
fong-spivak-invitation_F02986	277	■ 0.904	$p \mapsto$	$p \mapsto$	$q \mapsto \text{true}$ iff $q \in U$. Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$; it corresponds to
fong-spivak-invitation_F02987	277	■ 0.904	$f(p) \in U$	$f(p) \in U$	$q \mapsto \text{true}$ iff $q \in U$. Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$; it corresponds to
fong-spivak-invitation_F02988	277	■ 1.000	$\{p \in P \mid f(p) \in U\}$	$\{p \in P \mid f(p) \in U\}$	the upper set $\{p \in P \mid f(p) \in U\}$ which is exactly $f^{-1}(U)$.
fong-spivak-invitation_F02989	277	■ 1.000	$S = \left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\}$	$S = \left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\}$	1. 0 is a lower bound for $S = \{\frac{1}{n+1} \mid n \in \mathbb{N}\}$ because $0 \leq \frac{1}{n+1}$ for any $n \in \mathbb{N}$.
fong-spivak-invitation_F02990	277	■ 1.000	$0 \leq \frac{1}{n+1}$	$0 \leq \frac{1}{n+1}$	1. 0 is a lower bound for $S = \{\frac{1}{n+1} \mid n \in \mathbb{N}\}$ because $0 \leq \frac{1}{n+1}$ for any $n \in \mathbb{N}$.

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fong-spivak-invitation_F03019	278	■ 1.000	$f(p)=1$	$f(p) = 1$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and
fong-spivak-invitation_F03020	278	■ 1.000	$g(q)=2$	$g(q) = 2$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and
fong-spivak-invitation_F03021	278	■ 1.000	$f(p)=1 \leq 1=q$	$f(p) = 1 \leq 1 = q$	When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$, so both $f(p) = 1 \leq 1 = q$ and
fong-spivak-invitation_F03022	278	■ 1.000	$p=1 \leq g(q)$	$p = 1 \leq g(q)$	$p = 1 \leq g(q)$; it works! The same sort of story happens when (p, q) is (1, 2), (1, 3), (2, 1),
fong-spivak-invitation_F03023	278	■ 1.000	(p, q)	(p, q)	$p = 1 \leq g(q)$; it works! The same sort of story happens when (p, q) is (1, 2), (1, 3), (2, 1),
fong-spivak-invitation_F03024	278	■ 1.000	$(1,2),(1,3),(2,1)$	$(1,2),(1,3),(2,1)$	$p = 1 \leq g(q)$; it works! The same sort of story happens when (p, q) is (1,2),(1,3),(2,1) ,
fong-spivak-invitation_F03025	278	■ 0.991	$p=3, q=1$	$p = 3, q = 1$	(2, 2), (2, 3), and (3, 3). A different story happens for $p = 3, q = 1$ and $p = 3, q = 2$. In
fong-spivak-invitation_F03026	278	■ 0.991	$p=3, q=2$	$p = 3, q = 2$	(2, 2), (2, 3), and (3, 3). A different story happens for $p = 3, q = 1$ and $p = 3, q = 2$. In
fong-spivak-invitation_F03027	278	■ 1.000	$f(p)=3$	$f(p) = 3$	those cases $f(p) = 3$ and $g(q) = 2$, and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$.
fong-spivak-invitation_F03028	278	■ 1.000	$f(p) \not\leq q$	$f(p) \not\leq q$	those cases $f(p) = 3$ and $g(q) = 2$, and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$.
fong-spivak-invitation_F03029	278	■ 1.000	$p \not\leq g(q)$	$p \not\leq g(q)$	those cases $f(p) = 3$ and $g(q) = 2$, and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$.
fong-spivak-invitation_F03030	278	■ 1.000	$f(2) \not\leq 1$	$f(2) \not\leq 1$	Here f is <i>not</i> left adjoint to g because $f(2) \not\leq 1$ but $2 \leq g(1)$.
fong-spivak-invitation_F03031	278	■ 1.000	$2 \leq g(1)$	$2 \leq g(1)$	Here f is <i>not</i> left adjoint to g because $f(2) \not\leq 1$ but $2 \leq g(1)$.
fong-spivak-invitation_F03032	278	■ 1.000	$C(r):=\lceil r / 3\rceil$	$C(r) := \lceil r/3 \rceil$	happens. Writing $C(r) := \lceil r/3 \rceil$, then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$
fong-spivak-invitation_F03033	278	■ 1.000	$L(z) \leq r$	$L(z) \leq r$	happens. Writing $C(r) := \lceil r/3 \rceil$, then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$
fong-spivak-invitation_F03034	278	■ 1.000	$z \leq C(r)$	$z \leq C(r)$	happens. Writing $C(r) := \lceil r/3 \rceil$, then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$
fong-spivak-invitation_F03035	278	■ 1.000	$z=1$	$z = 1$	by definition of adjunction. So take $z = 1$ and $r = .01$; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$, and
fong-spivak-invitation_F03036	278	■ 1.000	$r=.01$	$r = .01$	by definition of adjunction. So take $z = 1$ and $r = .01$; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$, and
fong-spivak-invitation_F03037	278	■ 1.000	$\lceil r / 3\rceil=1$	$\lceil r/3 \rceil = 1$	by definition of adjunction. So take $z = 1$ and $r = .01$; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$, and
fong-spivak-invitation_F03038	278	■ 1.000	$L(1) \leq 0.01$	$L(1) \leq 0.01$	hence $L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By
fong-spivak-invitation_F03039	278	■ 1.000	$L(1) \leq r$	$L(1) \leq r$	hence $L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By
fong-spivak-invitation_F03040	278	■ 1.000	$r>0$	$r > 0$	hence $L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By
fong-spivak-invitation_F03041	278	■ 1.000	$L(1) \leq 0$	$L(1) \leq 0$	hence $L(z) \leq r$, i.e. $L(1) \leq 0.01$. In the same way $L(1) \leq r$ for all $r > 0$, so $L(1) \leq 0$. By
fong-spivak-invitation_F03042	278	■ 1.000	$1 \leq C(0)=\lceil 0 / 3\rceil=0$	$1 \leq C(0) = \lceil 0/3 \rceil = 0$	definition of adjunction $1 \leq C(0) = \lceil 0/3 \rceil = 0$, a contradiction.
fong-spivak-invitation_F03043	278	■ 1.000	$c_{\{1\}}, c_{\{2\}}, c_{\{3\}}, c_{\{4\}}$	c_1, c_2, c_3, c_4	see Example 1.96. Take c_1, c_2, c_3, c_4 to be the following partitions:
fong-spivak-invitation_F03044	278	■ 0.997	$g_{\{!\}\left(c_{\{1\}}\right)}, g_{\{!\}\left(c_{\{2\}}\right)}, g_{\{!\}\left(c_{\{3\}}\right)}$	$g!(c_1), g!(c_2), g!(c_3)$	Then the induced partitions $g!(c_1), g!(c_2), g!(c_3)$, and $g!(c_4)$ on T are:
fong-spivak-invitation_F03045	278	■ 0.997	$g_{\{!\}\left(c_{\{4\}}\right)}$	$g!(c_4)$	Then the induced partitions $g!(c_1), g!(c_2), g!(c_3)$, and $g!(c_4)$ on T are:
fong-spivak-invitation_F03046	279	■ 1.000	$S=\{1,2,3,4\}$	$S = \{1, 2, 3, 4\}$	Here are the partitions on $S = \{1, 2, 3, 4\}$ induced via g^* by the five partitions on $T =$

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fong-spivak-invitation_F03047	279	■ 0.980	$c \leq$	$c \leq$	5. Comparing c , the left-hand partition in part 1., with $g^*(d)$ and $g^*(e)$, we indeed have $c \leq$
fong-spivak-invitation_F03048	279	■ 1.000	$f: P \rightarrowtail Q: g$	$f: P \rightarrowtail Q: g$	Suppose P and Q are preorders, and that $f: P \rightarrowtail Q: g$ are monotone maps.
fong-spivak-invitation_F03049	279	■ 1.000	$g(q) \leq g(q)$	$g(q) \leq g(q)$	and $q \in Q$. Then starting with the reflexivity fact $g(q) \leq g(q)$, the definition with $p := g(q)$
fong-spivak-invitation_F03050	279	■ 1.000	$p := g(q)$	$p := g(q)$	and $q \in Q$. Then starting with the reflexivity fact $g(q) \leq g(q)$, the definition with $p := g(q)$
fong-spivak-invitation_F03051	279	■ 1.000	$f(p) \leq f(g(q))$	$f(p) \leq f(g(q))$	both sides, we have $f(p) \leq f(g(q))$, and then by transitivity $f(g(q)) \leq q$ implies $f(p) \leq q$,
fong-spivak-invitation_F03052	280	■ 1.000	$g, g^{\prime}: Q \rightarrowtail P$	$g, g^{\prime}: Q \rightarrowtail P$	1. Suppose that $f: P \rightarrowtail Q$ has two right adjoints, $g, g^{\prime}: Q \rightarrowtail P$. We want to show that
fong-spivak-invitation_F03053	280	■ 1.000	$g(q) \leq g^{\prime}(q)$	$g(q) \leq g^{\prime}(q)$	$g(q) \cong g^{\prime}(q)$ for all $q \in Q$. We will prove $g(q) \leq g^{\prime}(q)$; the inequality $g^{\prime}(q) \leq g(q)$
fong-spivak-invitation_F03054	280	■ 1.000	$g^{\prime}(q) \leq g(q)$	$g^{\prime}(q) \leq g(q)$	$g(q) \cong g^{\prime}(q)$ for all $q \in Q$. We will prove $g(q) \leq g^{\prime}(q)$; the inequality $g^{\prime}(q) \leq g(q)$
fong-spivak-invitation_F03055	280	■ 1.000	$p \leq g^{\prime}(f(p))$	$p \leq g^{\prime}(f(p))$	is similar. To do this, we use the fact that $p \leq g^{\prime}(f(p))$ and $f(g(q)) \leq q$ for all p, q by
fong-spivak-invitation_F03056	280	■ 1.000	$j := \bigvee A$	$j := \bigvee A$	Suppose $f: P \rightarrowtail Q$ is left adjoint to $g: Q \rightarrowtail P$. Let $A \subseteq P$ be any subset and let $j := \bigvee A$ be
fong-spivak-invitation_F03057	280	■ 1.000	$f(a) \leq f(j)$	$f(a) \leq f(j)$	its join. Then since f is monotone $f(a) \leq f(j)$ for all $a \in A$, so $f(j)$ is an upper bound for the
fong-spivak-invitation_F03058	280	■ 1.000	$f(j)$	$f(j)$	its join. Then since f is monotone $f(a) \leq f(j)$ for all $a \in A$, so $f(j)$ is an upper bound for the
fong-spivak-invitation_F03059	280	■ 1.000	$f(a) \leq b$	$f(a) \leq b$	$f(A)$, meaning we have $f(a) \leq b$ for all $a \in A$. Then by definition of adjunction, we also have
fong-spivak-invitation_F03060	280	■ 1.000	$a \leq g(b)$	$a \leq g(b)$	$a \leq g(b)$ for all $a \in A$. By definition of join, we have $j \leq g(b)$. Again by definition of adjunction
fong-spivak-invitation_F03061	280	■ 1.000	$j \leq g(b)$	$j \leq g(b)$	$a \leq g(b)$ for all $a \in A$. By definition of join, we have $j \leq g(b)$. Again by definition of adjunction
fong-spivak-invitation_F03062	280	■ 1.000	$f(j) \leq b$	$f(j) \leq b$	$f(j) \leq b$, as desired.
fong-spivak-invitation_F03063	281	■ 1.000	$f(p)$	$f(p)$	The former uses that $f(g(q)) \leq q$, with $f(p)$ substituted for q : this gives $f(g(f(p))) \leq f(p)$,
fong-spivak-invitation_F03066	281	■ 0.977	$B_{\{1\}} := \{a, b\}$	$B_1 := \{a, b\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-invitation_F03067	281	■ 0.977	$B_{\{2\}} := \{c\}$	$B_2 := \{c\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-invitation_F03068	281	■ 0.977	$f^*(B_{\{1\}}) = \{a_1\}$	$f^*(B_1) = \{a_1\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-invitation_F03069	281	■ 0.977	$f^*(B_{\{2\}}) = \{c_1, c_2\}$	$f^*(B_2) = \{c_1, c_2\}$	1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$. Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$.
fong-spivak-invitation_F03070	281	■ 1.000	$A_{\{1\}} := \varnothing$	$A_1 := \varnothing$	2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-invitation_F03071	281	■ 1.000	$A_{\{2\}} := \{a_1, c_1\}$	$A_2 := \{a_1, c_1\}$	2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-invitation_F03072	281	■ 1.000	$f_!(A_{\{1\}}) = \varnothing$	$f_!(A_1) = \varnothing$	2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-invitation_F03073	281	■ 1.000	$f_!(A_{\{2\}}) = \{a, c\}$	$f_!(A_2) = \{a, c\}$	2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$. Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$.
fong-spivak-invitation_F03074	281	■ 0.999	$A_{\{1\}}$	A_1	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-invitation_F03075	281	■ 0.999	$A_{\{2\}}$	A_2	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-invitation_F03076	281	■ 0.999	$f_*(A_{\{1\}}) = \{b\}$	$f_*(A_1) = \{b\}$	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.

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fong-spivak-invitation_F03077	281	■ 0.999	$f_{\{*\}}\left(A_{\{2\}}\right)=\{a, b\}$	$f_*(A_2) = \{a, b\}$	3. With the same A_1 and A_2 , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$.
fong-spivak-invitation_F03078	281	■ 1.000	$g(f(p))$	$g(f(p))$	1. By Proposition 1.101 we have that $p \leq g(f(p))$, and of course $g(f(p))$ and $(f \circ g)(p)$ mean
fong-spivak-invitation_F03079	281	■ 1.000	$(f \ ; \ g)(p)$	$(f;g)(p)$	1. By Proposition 1.101 we have that $p \leq g(f(p))$, and of course $g(f(p))$ and $(f \circ g)(p)$ mean
fong-spivak-invitation_F03080	281	■ 1.000	$g(f(g(f(p)))) \ \leq g(f(p))$	$g(f(g(f(p)))) \leq g(f(p))$	2. We want to show that $g(f(g(f(p)))) < g(f(p))$ and $g(f(p)) \leq g(f(g(f(p))))$ for all p .
fong-spivak-invitation_F03081	281	■ 1.000	$g(f(p)) \ \leq g(f(g(f(p))))$	$g(f(p)) \leq g(f(g(f(p))))$	2. We want to show that $g(f(g(f(p)))) \leq g(f(p))$ and $g(f(p)) \leq g(f(g(f(p))))$ for all p .
fong-spivak-invitation_F03082	281	■ 1.000	$p^{\{\prime\}} \ \leq g\left(f\left(p^{\{\prime\}}\right)\right)$	$p' \leq g(f(p'))$	The latter is just the fact that $p' \leq g(f(p'))$ for any p' , applied with $g(f(p))$ in place of p' .
fong-spivak-invitation_F03083	281	■ 1.000	$p^{\{\prime\}}$	p'	The latter is just the fact that $p' \leq g(f(p'))$ for any p' , applied with $g(f(p))$ in place of p' .
fong-spivak-invitation_F03084	281	■ 1.000	$f(g(f(p))) \ \leq f(p)$	$f(g(f(p))) \leq f(p)$	The former uses that $f(g(q)) \leq q$, with $f(p)$ substituted for q : this gives $f(g(f(p))) < f(p)$,
fong-spivak-invitation_F03085	281	■ 0.657	$a \ b$	ab	We denote tuples (a, b) by ab for space reasons. So the relation $\{(1, 1), (1, 2), (2, 1)\}$ will be
fong-spivak-invitation_F03086	281	■ 0.657	$\{(1,1),(1,2),(2,1)\}$	$\{(1,1), (1,2), (2,1)\}$	We denote tuples (a, b) by ab for space reasons. So the relation $\{(1, 1), (1, 2), (2, 1)\}$ will be
fong-spivak-invitation_F03087	281	■ 1.000	$S:=\{1,2,3\}$	$S := \{1, 2, 3\}$	Let $S := \{1, 2, 3\}$.
fong-spivak-invitation_F03088	281	■ 0.997	$1 \ \leq 1, 2 \ \leq 2$	$1 \leq 1, 2 \leq 2$	1. Let $\leq$ be the preorder with $1 \leq 2$, and of course $1 \leq 1, 2 \leq 2$, and $3 \leq 3$. Then $U(\leq) =$
fong-spivak-invitation_F03089	281	■ 0.997	$3 \ \leq 3$	$3 \leq 3$	1. Let $\leq$ be the preorder with $1 \leq 2$, and of course $1 \leq 1, 2 \leq 2$, and $3 \leq 3$. Then $U(\leq) =$
fong-spivak-invitation_F03090	281	■ 0.997	$U(\leq)=$	$U(\leq) =$	1. Let $\leq$ be the preorder with $1 \leq 2$, and of course $1 \leq 1, 2 \leq 2$, and $3 \leq 3$. Then $U(\leq) =$
fong-spivak-invitation_F03091	281	■ 0.876	$\{(1,1),(1,2),(2,2),(3,3)\}$	$\{(1,1), (1,2), (2,2), (3,3)\}$	$\{(1, 1), (1, 2), (2, 2), (3, 3)\}$
fong-spivak-invitation_F03092	281	■ 1.000	$Q:=\{(1,1)\}$	$Q := \{(1, 1)\}$	2. Let $Q := \{(1, 1)\}$ and $Q' := \{(2, 1)\}$.
fong-spivak-invitation_F03093	281	■ 1.000	$Q^{\{\prime\}}:=\{(2,1)\}$	$Q' := \{(2, 1)\}$	2. Let $Q := \{(1, 1)\}$ and $Q' := \{(2, 1)\}$.
fong-spivak-invitation_F03094	282	■ 0.812	$\mathrm{Cl}(Q)$	$\mathrm{Cl}(Q)$	3. The closure $\mathrm{Cl}(Q)$ of Q is the smallest preorder containing $(1, 1)$, which is $\mathrm{Cl}(Q) =$
fong-spivak-invitation_F03095	282	■ 0.812	$\mathrm{Cl}(Q)=$	$\mathrm{Cl}(Q) =$	3. The closure $\mathrm{Cl}(Q)$ of Q is the smallest preorder containing $(1, 1)$, which is $\mathrm{Cl}(Q) =$
fong-spivak-invitation_F03096	282	■ 1.000	$\{(1,1),(2,2),(3,3)\}$	$\{(1,1), (2,2), (3,3)\}$	$\{(1, 1), (2, 2), (3, 3)\}$. Similarly, $\mathrm{Cl}(Q') = \{(1, 1), (2, 1), (2, 2), (3, 3)\}$. It is easy to see that
fong-spivak-invitation_F03097	282	■ 1.000	$\mathrm{Cl}(Q^{\{\prime\}})=\{(1,1),(2,1),(2,2),(3,3)\}$	$\mathrm{Cl}(Q') = \{(1, 1), (2, 1), (2, 2), (3, 3)\}$	$\{(1, 1), (2, 2), (3, 3)\}$. Similarly, $\mathrm{Cl}(Q') = \{(1, 1), (2, 1), (2, 2), (3, 3)\}$. It is easy to see that
fong-spivak-invitation_F03098	282	■ 0.995	$(2,1)$	$(2, 1)$	4. It is easy to see that $\mathrm{Cl}(Q') \not\leq$ because the ordered pair $(2, 1)$ is in $\mathrm{Cl}(Q')$ but is not in $\leq$.
fong-spivak-invitation_F03099	282	■ 0.995	$\mathrm{Cl}(Q^{\{\prime\}})$	$\mathrm{Cl}(Q')$	4. It is easy to see that $\mathrm{Cl}(Q') \not\leq$ because the ordered pair $(2, 1)$ is in $\mathrm{Cl}(Q')$ but is not in $\leq$.
fong-spivak-invitation_F03100	282	■ 1.000	$x_{\{1\}}=-1, \ x_{\{2\}}=0, \ y_{\{1\}}=-1$	$x_1 = -1, x_2 = 0, y_1 = -1$	The expert is right! The proposal violates property (a) when $x_1 = -1, x_2 = 0, y_1 = -1$, and
fong-spivak-invitation_F03101	282	■ 1.000	$y_{\{2\}}=1$	$y_2 = 1$	$y_2 = 1$. Indeed $-1 \leq -1$ and $0 \leq 1$, but $-1 * 0 = 0 \not\leq -1 = -1 * 1$.
fong-spivak-invitation_F03102	282	■ 1.000	$-1 \ \leq -1$	$-1 \leq -1$	$y_2 = 1$. Indeed $-1 \leq -1$ and $0 \leq 1$, but $-1 * 0 = 0 \not\leq -1 = -1 * 1$.
fong-spivak-invitation_F03103	282	■ 1.000	$-1 * 0=0 \ \not \leq -1=-1 * 1$	$-1 * 0 = 0 \not\leq -1 = -1 * 1$	$y_2 = 1$. Indeed $-1 \leq -1$ and $0 \leq 1$, but $-1 * 0 = 0 \not\leq -1 = -1 * 1$.
fong-spivak-invitation_F03104	282	■ 0.988	$(\operatorname{Disc}(M), =, *, e)$	$(\mathrm{Disc}(M), =, *, e)$	To check that $(\mathbf{Disc}(M), =, *, e)$ is a symmetric monoidal preorder, we need to check our pro-
fong-spivak-invitation_F03105	282	■ 1.000	$x_{\{1\}} \ \otimes x_{\{2\}}=x_{\{1\}} \ \otimes x_{\{2\}}$	$x_1 \otimes x_2 = x_1 \otimes x_2$	$x_1 \otimes x_2 = x_1 \otimes x_2$, conditions (b) and (c) are precisely the equations Eq. (2.2), and (d) is the

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fong-spivak-invitation_F03106	282	■ 1.000	$u \leq u, v \leq v$	$u \leq u, v \leq v$	2. We use reflexivity when we assert that $u \leq u, v \leq v$ and $z \leq z$, and use transitivity to assert
fong-spivak-invitation_F03107	282	■ 1.000	$z \leq z$	$z \leq z$	2. We use reflexivity when we assert that $u \leq u, v \leq v$ and $z \leq z$, and use transitivity to assert
fong-spivak-invitation_F03108	282	■ 1.000	$t+u \leq y+z$	$t + u \leq y + z$	that the above sequence of inequalities implies the single inequality $t+u \leq y+z$.
fong-spivak-invitation_F03109	282	■ 1.000	$z \rightarrow w$	$z \rightarrow w$	Condition (a), monotonicity, says that if $x \rightarrow y$ and $z \rightarrow w$ are reactions, then $x + z \rightarrow y + w$ is a
fong-spivak-invitation_F03110	282	■ 1.000	$x+z \rightarrow y+w$	$x + z \rightarrow y + w$	Condition (a), monotonicity, says that if $x \rightarrow y$ and $z \rightarrow w$ are reactions, then $x + z \rightarrow y + w$ is a
fong-spivak-invitation_F03111	283	■ 0.954	$1 \mid 1$	$1 \mid 1$	This proposed monoidal product is not monotonic: we have $1 1$ and $1 2$, but $(1 + 1) \nmid (1 + 2)$.
fong-spivak-invitation_F03112	283	■ 0.954	$1 \mid 2$	$1 \mid 2$	This proposed monoidal product is not monotonic: we have $1 1$ and $1 2$, but $(1 + 1) \nmid (1 + 2)$.
fong-spivak-invitation_F03113	283	■ 0.954	$(1+1) \nmid (1+2)$	$(1 + 1) \nmid (1 + 2)$	This proposed monoidal product is not monotonic: we have $1 1$ and $1 2$, but $(1 + 1) \nmid (1 + 2)$.
fong-spivak-invitation_F03114	283	■ 0.964	$z \leq w$	$z \leq w$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \text{yes}) =$
fong-spivak-invitation_F03115	283	■ 0.964	$\min(x, z) \leq \min(y, w)$	$\min(x, z) \leq \min(y, w)$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \text{yes}) =$
fong-spivak-invitation_F03116	283	■ 0.964	$\min(x$	$\min(x$	2. We need to show that (a) if $x \leq y$ and $z \leq w$, then $\min(x, z) \leq \min(y, w)$, (b) $\min(x, \text{yes}) =$
fong-spivak-invitation_F03117	283	■ 0.766	$x=\min(y \text{ e } s, x)$	$x = \min(\text{yes}, x)$	$x = \min(\text{yes}, x)$, (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, x)$.
fong-spivak-invitation_F03118	283	■ 0.766	$\min(\min(x, y), z)=\min(x, \min(y, z))$	$\min(\min(x, y), z) = \min(x, \min(y, z))$	$x = \min(\text{yes}, x)$, (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, x)$.
fong-spivak-invitation_F03119	283	■ 0.766	$\min(x, y)=\min(y, x)$	$\min(x, y) = \min(y, x)$	$x = \min(\text{yes}, x)$, (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$, and (d) $\min(x, y) = \min(y, x)$.
fong-spivak-invitation_F03120	283	■ 0.984	$(\mathrm{P}(S), \leq, S, \cap)$	$(P(S), \leq, S, \cap)$	Yes, $(P(S), \leq, S, \cap)$ is a symmetric monoidal preorder.
fong-spivak-invitation_F03121	283	■ 1.000	$(P \wedge Q)(n)$	$(P \wedge Q)(n)$	statements P and Q and makes the statement $P \wedge Q$, where $(P \wedge Q)(n)$ is true if $P(n)$ and
fong-spivak-invitation_F03122	283	■ 0.601	$\left(\mathrm{Prop}^{\mathbb{N}}, \leq, \right.$	$(\mathrm{Prop}^{\mathbb{N}}, \leq$	$Q(n)$ are true, and false otherwise. Then $(\mathrm{Prop}^{\mathbb{N}}, \leq, \text{true}, \wedge)$ forms a symmetric monoidal
fong-spivak-invitation_F03123	283	■ 0.601	$\left.\wedge\right)$	$\wedge)$	$Q(n)$ are true, and false otherwise. Then $(\mathrm{Prop}^{\mathbb{N}}, \leq, \text{true}, \wedge)$ forms a symmetric monoidal
fong-spivak-invitation_F03124	283	■ 0.997	$n+10 \leq 1$	$n + 10 \leq 1$	bers, such as “ $n + 10 \leq 1$ ” or “ n is made of cheese.” We can also define $\vee$ such that $(P \vee Q)(n)$
fong-spivak-invitation_F03125	283	■ 0.997	$(P \vee Q)(n)$	$(P \vee Q)(n)$	bers, such as “ $n + 10 \leq 1$ ” or “ n is made of cheese.” We can also define $\vee$ such that $(P \vee Q)(n)$
fong-spivak-invitation_F03126	283	■ 0.769	$\left.\vee\right)$	$\vee)$	is true if and only if at least one of $P(n)$ and $Q(n)$ is true. Then $(\mathrm{Prop}^{\mathbb{N}}, \leq, \text{false}, \vee)$ forms a
fong-spivak-invitation_F03127	283	■ 0.827	$I \otimes x = x =$	$I \otimes x = x =$	These conditions have nothing to do with the order in $\mathcal{X}^{\mathrm{op}}$: they simply state that $I \otimes x = x =$
fong-spivak-invitation_F03128	283	■ 1.000	$x \otimes I$	$x \otimes I$	$x \otimes I$, and $(x \otimes y) \otimes z = x \otimes (y \otimes z)$, and $x \otimes y = y \otimes x$. Since these are true in $\mathcal{X}$, they are true
fong-spivak-invitation_F03129	283	■ 0.727	X^{op}	X^{op}	in $\mathcal{X}^{\mathrm{op}}$.
fong-spivak-invitation_F03130	283	■ 0.675	$\mathrm{Cost}^{\mathrm{op}}$	$\mathrm{Cost}^{\mathrm{op}}$	1. The preorder $\mathbf{Cost}^{\mathrm{op}}$ has underlying set $[0, \infty]$, and the usual increasing order on real numbers
fong-spivak-invitation_F03131	284	■ 1.000	$=\infty \geq 0=g$	$= \infty \geq 0 = g$	1. The map g is monotonic as $g(\text{false}) = \infty \geq 0 = g(\text{true})$, satisfies condition (a) since
fong-spivak-invitation_F03132	284	■ 1.000	$0 \geq 0=g$	$0 \geq 0 = g$	$0 \geq 0 = g(\text{true})$, and satisfies condition (b) since
fong-spivak-invitation_F03133	284	■ 0.990	$f(n)=1$	$f(n) = 1$	2. The map $f(n) = 1$ for all $n \in \mathbb{N}$ defines a monoidal monotone $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$.

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fong-spivak-invitation_F03134	284	■ 0.990	$f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$	$f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$	2. The map $f(n) = 1$ for all $n \in \mathbb{N}$ defines a monoidal monotone $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$.
fong-spivak-invitation_F03135	284	■ 0.276	$\mathbb{Z}, \leq, *, 1$	$\mathbb{Z}, \leq, *, 1$	3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * 1) \not\leq 1$.
fong-spivak-invitation_F03136	284	■ 0.276	≤ 0	≤ 0	3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * 1) \not\leq 1$.
fong-spivak-invitation_F03137	284	■ 1.000	$-1 \not\leq (0 * 0)$	$-1 \not\leq (0 * 0)$	$-1 \not\leq (0 * 0)$.
fong-spivak-invitation_F03138	284	■ 0.382	$\mathcal{X}_P$	$\mathcal{X}_P$	struct a Bool -category $\mathcal{X}_P$ with P as its set of objects, and with $\mathcal{X}_P(p, q) := \text{true}$ if $p \leq q$,
fong-spivak-invitation_F03139	284	■ 0.382	$\mathcal{X}_P(p, q) := \text{true}$	$\mathcal{X}_P(p, q) := \text{true}$	struct a Bool -category $\mathcal{X}_P$ with P as its set of objects, and with $\mathcal{X}_P(p, q) := \text{true}$ if $p \leq q$,
fong-spivak-invitation_F03140	284	■ 0.579	$\mathcal{X}_P(p, q) :=$	$\mathcal{X}_P(p, q) :=$	and $\mathcal{X}_P(p, q) := \text{false}$ otherwise. How do we turn this back into a preorder? Following the
fong-spivak-invitation_F03141	284	■ 0.626	$\text{Ob}(\mathcal{X}_P) = P$	$\text{Ob}(\mathcal{X}_P) = P$	proof of Theorem 2.32, we construct a preorder with underlying set $\text{Ob}(\mathcal{X}_P) = P$, and with
fong-spivak-invitation_F03142	284	■ 0.219	$\mathcal{X}_P(p, q) = \text{true}$	$\mathcal{X}_P(p, q) = \text{true}$	$p \leq q$ if and only if $\mathcal{X}_P(p, q) = \text{true}$. This is precisely the preorder $(P, \leq)$!
fong-spivak-invitation_F03143	284	■ 0.694	$\text{Ob}(\mathcal{X})$	$(\text{Ob}(\mathcal{X}), \leq)$	2. Let $\mathcal{X}$ be a Bool -category. By the proof of Theorem 2.32, we construct a preorder $(\text{Ob}(\mathcal{X}), \leq)$,
fong-spivak-invitation_F03144	284	■ 0.897	$\text{Ob}(X)$	$\text{Ob}(X)$	ple 2.31 in part 1, we construct a Bool -category $\mathcal{X}'$ whose set of objects is $\text{Ob}(\mathcal{X})$, and such
fong-spivak-invitation_F03145	284	■ 0.727	$\mathcal{X}'(x, y) =$	$\mathcal{X}'(x, y) =$	that $\mathcal{X}'(x, y) = \text{true}$ if and only if $x \leq y$ in $(\text{Ob}(\mathcal{X}), \leq)$. But, by construction, this means
fong-spivak-invitation_F03146	284	■ 0.642	$\mathcal{X}'(x, y) = \mathcal{X}(x, y)$	$\mathcal{X}'(x, y) = \mathcal{X}(x, y)$	$\mathcal{X}'(x, y) = \mathcal{X}(x, y)$. So we get back the Bool -category we started with.
fong-spivak-invitation_F03147	284	■ 0.492	$d(\text{US})$	$d(\text{US})$	The distance $d(\text{US}, \text{Spain})$ is bigger: the distance from, for example, San Diego to anywhere in
fong-spivak-invitation_F03148	284	■ 0.971	$([0, \infty], \geq)$	$([0, \infty], \geq)$	The difference between a Lawvere metric space – that is, a category enriched over $[0, \infty]$, $\geq$,
fong-spivak-invitation_F03149	284	■ 0.558	$0, +)$	$0, +)$	$0, +)$ – and a category enriched over $(\mathbb{R}_{\geq 0}, \geq, 0, +)$ is that, in the latter, infinite distances
fong-spivak-invitation_F03150	285	■ 0.575	$\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$	$\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$	$\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$ for all x, y, z . So a NMY -category can be thought of as set
fong-spivak-invitation_F03151	285	■ 0.575	x, y, z	x, y, z	$\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$ for all x, y, z . So a NMY -category can be thought of as set
fong-spivak-invitation_F03152	286	■ 0.684	$\mathcal{X}(C, D)$	$\mathcal{X}(C, D)$	For example, to compute the hom-object $\mathcal{X}(C, D)$, we notice that there are two paths: $C \rightarrow$
fong-spivak-invitation_F03153	286	■ 0.684	$C \rightarrow$	$C \rightarrow$	For example, to compute the hom-object $\mathcal{X}(C, D)$, we notice that there are two paths: $C \rightarrow$
fong-spivak-invitation_F03154	286	■ 1.000	$A \rightarrow B \rightarrow D$	$A \rightarrow B \rightarrow D$	$A \rightarrow B \rightarrow D$ and $C \rightarrow D$. For the first path, the intersection is the set $\{\text{boat}\}$. For the second
fong-spivak-invitation_F03155	286	■ 1.000	$C \rightarrow D$	$C \rightarrow D$	$A \rightarrow B \rightarrow D$ and $C \rightarrow D$. For the first path, the intersection is the set $\{\text{boat}\}$. For the second
fong-spivak-invitation_F03156	286	■ 0.515	$\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$	$\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$	paths, we ensure that $\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$ for all x, y, z .
fong-spivak-invitation_F03158	286	■ 1.000	$\min(M(x, y), M(y, z)) \leq M(x, y)$	$\min(M(x, y), M(y, z)) \leq M(x, y)$	monoidal unit ∞ , and because $\min(M(x, y), M(y, z)) \leq M(x, y)$ for all $x, y, z \in \{A, B, C\}$.
fong-spivak-invitation_F03159	286	■ 1.000	$x, y, z \in \{A, B, C\}$	$x, y, z \in \{A, B, C\}$	monoidal unit ∞ , and because $\min(M(x, y), M(y, z)) \leq M(x, y)$ for all $x, y, z \in \{A, B, C\}$.
fong-spivak-invitation_F03160	286	■ 0.507	$d(A, B) = d(B, A)$	$d(A, B) = d(B, A)$	2. Let $\mathcal{X}$ be the Lawvere metric space with two objects A and B , such that $d(A, B) = d(B, A) =$
fong-spivak-invitation_F03161	286	■ 0.501	$X_f = \bullet \bullet$	$X_f = \overset{A}{\bullet} \overset{B}{\bullet}$	5. Then we have $X_f = \left \overset{A}{\bullet} \quad \overset{B}{\bullet} \right $ while $X_g = \left \overset{A}{\bullet} \longrightarrow \overset{B}{\bullet} \right $.
fong-spivak-invitation_F03162	286	■ 0.501	$X_g = \bullet \longrightarrow$	$X_g = \overset{A}{\bullet} \longrightarrow$	5. Then we have $X_f = \left \overset{A}{\bullet} \quad \overset{B}{\bullet} \right $ while $X_g = \left \overset{A}{\bullet} \longrightarrow \overset{B}{\bullet} \right $.

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fong-spivak-invitation_F03163	287	■ 1.000	$d(x, y) \leq d(y, x)$	$d(x, y) \leq d(y, x)$	so for all x, y we must have $d(x, y) \leq d(y, x)$. But this means also that $d(y, x) \leq d(x, y)$,
fong-spivak-invitation_F03164	287	■ 1.000	$d(y, x) \leq d(x, y)$	$d(y, x) \leq d(x, y)$	so for all x, y we must have $d(x, y) \leq d(y, x)$. But this means also that $d(y, x) \leq d(x, y)$,
fong-spivak-invitation_F03165	287	■ 1.000	$d(y, x) =$	$d(y, x) =$	Now let's consider the skeletality condition. This says that if $d(x, y) = 0$ and $d(y, x) =$
fong-spivak-invitation_F03166	287	■ 0.791	$(x, y) \in \mathcal{X} \times \mathcal{Y}$	$(x, y) \in \mathcal{X} \times \mathcal{Y}$	1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$.
fong-spivak-invitation_F03167	287	■ 0.791	$I \leq y(y, y)$	$I \leq y(y, y)$	1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$.
fong-spivak-invitation_F03168	287	■ 0.850	$I = I \otimes I \leq X(x, x) \otimes y(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$	$I = I \otimes I \leq X(x, x) \otimes y(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$	Thus $I = I \otimes I \leq \mathcal{X}(x, x) \otimes \mathcal{Y}(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$.
fong-spivak-invitation_F03169	287	■ 0.665	$y(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes$	$y(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes$	3. In particular, we use the symmetry to conclude that $\mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes$
fong-spivak-invitation_F03170	287	■ 0.998	$y(y_1, y_2)$	$y(y_1, y_2)$	$\mathcal{Y}(y_1, y_2)$.
fong-spivak-invitation_F03171	287	■ 0.993	$(\mathbb{R} \times \mathbb{R})(5, 6), (-1, 4) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$	$(\mathbb{R} \times \mathbb{R})(5, 6), (-1, 4) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$	We just apply Definition 2.53(ii): $(\mathbb{R} \times \mathbb{R})(5, 6), (-1, 4) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$.
fong-spivak-invitation_F03172	287	■ 0.999	$u \leq u'$	$u \leq u'$	1. The function $- \otimes v: V \rightarrow V$ is monotone, because if $u \leq u'$ then $u \otimes v \leq u' \otimes v$ by the
fong-spivak-invitation_F03173	287	■ 0.999	$u \otimes v \leq u' \otimes v$	$u \otimes v \leq u' \otimes v$	1. The function $- \otimes v: V \rightarrow V$ is monotone, because if $u \leq u'$ then $u \otimes v \leq u' \otimes v$ by the
fong-spivak-invitation_F03174	287	■ 0.997	$a := (v \multimap w)$	$a := (v \multimap w)$	2. Let $a := (v \multimap w)$ in Eq. (2.23). The right-hand side is thus $(v \multimap w) \leq (v \multimap w)$, which is
fong-spivak-invitation_F03175	287	■ 0.975	$(v \multimap u) \otimes v \leq u \leq u'$	$(v \multimap u) \otimes v \leq u \leq u'$	3. Let $u \leq u'$. Then, using part 2, $(v \multimap u) \otimes v \leq u \leq u'$. Applying Eq. (2.23), we thus have
fong-spivak-invitation_F03176	287	■ 1.000	$(v \multimap u) \leq (v \multimap u')$	$(v \multimap u) \leq (v \multimap u')$	$(v \multimap u) \leq (v \multimap u')$. This shows that the map $(v \multimap -): V \rightarrow V$ is monotone.
fong-spivak-invitation_F03177	287	■ 1.000	$(v \multimap -): V \rightarrow V$	$(v \multimap -): V \rightarrow V$	$(v \multimap u) \leq (v \multimap u')$. This shows that the map $(v \multimap -): V \rightarrow V$ is monotone.
fong-spivak-invitation_F03178	287	■ 0.998	$(v \multimap -)$	$(v \multimap -)$	do not know $(- \otimes v)$ and $(v \multimap -)$ are monotone maps. We proved this, however, in parts 1
fong-spivak-invitation_F03179	288	■ 1.000	$x \Rightarrow y$	$x \Rightarrow y$	We need to find the hom-element. This is given by implication. That is, define the function $x \Rightarrow y$
fong-spivak-invitation_F03180	288	■ 0.987	$(a \wedge v) \leq w$	$(a \wedge v) \leq w$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$,
fong-spivak-invitation_F03181	288	■ 0.987	$a \leq (v \Rightarrow w)$	$a \leq (v \Rightarrow w)$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$,
fong-spivak-invitation_F03182	288	■ 0.987	$v =$	$v =$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$,
fong-spivak-invitation_F03183	288	■ 0.987	$a \wedge$	$a \wedge$	Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$. Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$,
fong-spivak-invitation_F03184	288	■ 0.570	$\Rightarrow w$	$\Rightarrow w$	and so the left-hand side is always true. But $(\text{false} \Rightarrow w) = \text{true}$, so the right-hand side
fong-spivak-invitation_F03185	288	■ 0.672	$v = \text{true}$	$v = \text{true}$	is always true too. If $v = \text{true}$, then $a \wedge \text{true} = a$ so the left-hand side says $a \leq w$. But
fong-spivak-invitation_F03186	288	■ 0.672	$= a$	$= a$	is always true too. If $v = \text{true}$, then $a \wedge \text{true} = a$ so the left-hand side says $a \leq w$. But
fong-spivak-invitation_F03187	288	■ 0.672	$a \leq w$	$a \leq w$	is always true too. If $v = \text{true}$, then $a \wedge \text{true} = a$ so the left-hand side says $a \leq w$. But
fong-spivak-invitation_F03188	288	■ 0.969	$\Rightarrow w) = w$	$\Rightarrow w) = w$	$(\text{true} \Rightarrow w) = w$, so the right-hand side is the same. Thus $\Rightarrow$ defines a hom-element as per
fong-spivak-invitation_F03189	288	■ 0.672	$(\bigvee \varnothing) = \text{false}$	$(\bigvee \varnothing) = \text{false}$	1a. In Bool , $(\bigvee \varnothing) = \text{false}$, the least element.
fong-spivak-invitation_F03190	288	■ 0.999	$(\bigvee \varnothing) = \infty$	$(\bigvee \varnothing) = \infty$	1b. In Cost , $(\bigvee \varnothing) = \infty$. This is because we use the opposite order $\geq$ on $[0, \infty]$, so $\bigvee \varnothing$ is the

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fong-spivak-invitation_F03191	288	■ 1.000	$(\bigvee \varnothing)=\emptyset$	$(\bigvee \emptyset) = 0$	we denote $(\bigvee \emptyset) = 0$, is a bit confusing! Beware!
fong-spivak-invitation_F03192	288	■ 1.000	$\min (x, y)$	$\min(x, y)$	2b. In Cost , $x \vee y$ is the minimum $\min(x, y)$. Again because we use the opposite order on $[0, \infty]$,
fong-spivak-invitation_F03193	288	■ 0.764	$(\mathrm{P}(S), \subseteq, S, \cap)$	$(P(S), \subseteq, S, \cap)$	Yes, the power-set monoidal preorder $(P(S), \subseteq, S, \cap)$ is a quantale. The hom-object $B \multimap C$ is
fong-spivak-invitation_F03194	288	■ 0.764	$B \multimap C$	$B \multimap C$	Yes, the power-set monoidal preorder $(P(S), \subseteq, S, \cap)$ is a quantale. The hom-object $B \multimap C$ is
fong-spivak-invitation_F03195	288	■ 1.000	$\bar{B} \cup C$	$\bar{B} \cup C$	given by $\bar{B} \cup C$, where $\bar{B}$ is the <i>complement</i> of B : it contains all elements of S not contained in
fong-spivak-invitation_F03196	288	■ 1.000	$\bar{B}$	$\bar{B}$	given by $\bar{B} \cup C$, where $\bar{B}$ is the <i>complement</i> of B : it contains all elements of S not contained in
fong-spivak-invitation_F03197	288	■ 1.000	$(A \cap B) \subseteq C$	$(A \cap B) \subseteq C$	B . To see that this satisfies Eq. (2.23), note that if $(A \cap B) \subseteq C$, then
fong-spivak-invitation_F03198	288	■ 1.000	$A \subseteq (\bar{B} \cup C)$	$A \subseteq (\bar{B} \cup C)$	On the other hand, if $A \subseteq (\bar{B} \cup C)$, then
fong-spivak-invitation_F03199	288	■ 1.000	$v \in V, 0 \otimes v = 0$	$v \in V, 0 \otimes v = 0$	1. We first use Proposition 2.64 (2) and symmetry to show that, for all $v \in V, 0 \otimes v = 0$.
fong-spivak-invitation_F03200	289	■ 0.998	$\left(v, a_1, \ldots, a_n\right)$	$(v, a_1, \ldots, a_n)$	define a path in G to be a tuple of the form $(v, a_1, \ldots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$,
fong-spivak-invitation_F03201	289	■ 0.998	$s\left(a_1\right)=v$	$s\left(a_1\right)=v$	define a path in G to be a tuple of the form $(v, a_1, \ldots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$,
fong-spivak-invitation_F03202	290	■ 1.000	$t\left(a_i\right)=s\left(a_{i+1}\right)$	$t\left(a_i\right)=s\left(a_{i+1}\right)$	and $t(a_i) = s(a_{i+1})$ for all $i \in \{1, \ldots, n-1\}$; the length of this path is n , and this definition
fong-spivak-invitation_F03203	290	■ 1.000	$i \in\{1, \ldots, n-1\}$	$i \in \{1, \ldots, n-1\}$	and $t(a_i) = s(a_{i+1})$ for all $i \in \{1, \ldots, n-1\}$; the length of this path is n , and this definition
fong-spivak-invitation_F03204	290	■ 1.000	$s(p):=v$	$s(p) := v$	makes sense for any $n \in \mathbb{N}$. We say that the source of p is $s(p) := v$ and the target of p is defined
fong-spivak-invitation_F03205	290	■ 0.998	$p=\left(v, a_1, \ldots, a_m\right)$	$p = (v, a_1, \ldots, a_m)$	Two paths $p = (v, a_1, \ldots, a_m)$ and $q = (w, b_1, \ldots, b_n)$ can be concatenated if $t(p) = s(q)$, in
fong-spivak-invitation_F03206	290	■ 0.998	$q=\left(w, b_1, \ldots, b_n\right)$	$q = (w, b_1, \ldots, b_n)$	Two paths $p = (v, a_1, \ldots, a_m)$ and $q = (w, b_1, \ldots, b_n)$ can be concatenated if $t(p) = s(q)$, in
fong-spivak-invitation_F03207	290	■ 0.998	$t(p)=s(q)$	$t(p) = s(q)$	Two paths $p = (v, a_1, \ldots, a_m)$ and $q = (w, b_1, \ldots, b_n)$ can be concatenated if $t(p) = s(q)$, in
fong-spivak-invitation_F03208	290	■ 0.991	$p=(v)$	$p = (v)$	$p = (v)$ for some $v \in V$. It is easy to see from the above that (v) and $(w, b_1, \ldots, b_n)$ can be
fong-spivak-invitation_F03209	290	■ 0.991	(v)	(v)	$p = (v)$ for some $v \in V$. It is easy to see from the above that (v) and $(w, b_1, \ldots, b_n)$ can be
fong-spivak-invitation_F03210	290	■ 0.991	$\left(w, b_1, \ldots, b_n\right)$	$(w, b_1, \ldots, b_n)$	$p = (v)$ for some $v \in V$. It is easy to see from the above that (v) and $(w, b_1, \ldots, b_n)$ can be
fong-spivak-invitation_F03211	290	■ 1.000	$v=w$	$v = w$	concatenated iff $v = w$, in which case the result is $(w, b_1, \ldots, b_n)$. Similarly $(v, a_1, \ldots, a_m)$
fong-spivak-invitation_F03212	290	■ 1.000	$\left(v, a_1, \ldots, a_m\right)$	$(v, a_1, \ldots, a_m)$	concatenated iff $v = w$, in which case the result is $(w, b_1, \ldots, b_n)$. Similarly $(v, a_1, \ldots, a_m)$
fong-spivak-invitation_F03213	290	■ 0.896	(w)	(w)	and (w) can be concatenated iff $w = t(a_m)$, in which case the result is $(v, a_1, \ldots, a_m)$. Finally,
fong-spivak-invitation_F03214	290	■ 0.896	$w=t\left(a_m\right)$	$w = t\left(a_m\right)$	and (w) can be concatenated iff $w = t(a_m)$, in which case the result is $(v, a_1, \ldots, a_m)$. Finally,
fong-spivak-invitation_F03215	290	■ 0.940	$r=\left(x, c_1, \ldots, c_o\right)$	$r = (x, c_1, \ldots, c_o)$	for associativity with p and q as above and $r = (x, c_1, \ldots, c_o)$, the formula readily reads that
fong-spivak-invitation_F03216	290	■ 0.379	$(p ; q)\{\}_9^{\circ} r$	$(p; q)_9^{\circ} r$	whichever way they are concatenated, $(p \circ q) \circ r$ or $p \circ (q \circ r)$, the result is
fong-spivak-invitation_F03217	290	■ 0.379	$p \circ(q ; r)$	$p \circ (q; r)$	whichever way they are concatenated, $(p \circ q) \circ r$ or $p \circ (q \circ r)$, the result is
fong-spivak-invitation_F03219	293	■ 1.000	$f_{\{2\}}$	f_2	The other way reverses f_1 and f_2 .

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fong-spivak-invitation_F03250	293	0.421	$\stackrel{a'}{\text{f}_1} \xrightarrow{\text{f}_2} \text{b}'$ $\stackrel{a'}{\text{f}_1} \xrightarrow{\text{f}_2} \text{b}'$	$\stackrel{a'}{\text{f}_1} \xrightarrow{\text{f}_2} \text{b}'$	We need to give two functors F, G from $\bullet \xrightarrow{f} \bullet$ to $\bullet \xrightarrow{\frac{f_1}{f_2}} \bullet$ whose on-objects parts are the
fong-spivak-invitation_F03251	293	0.929	$\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$	$\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$	1. Let $\mathcal{C}$ be a category. Then defining $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ by $\operatorname{id}_{\mathcal{C}}(x) = x$ for every object and morphism
fong-spivak-invitation_F03252	293	0.929	$\operatorname{id}_{\mathcal{C}}(x) = x$	$\operatorname{id}_{\mathcal{C}}(x) = x$	1. Let $\mathcal{C}$ be a category. Then defining $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ by $\operatorname{id}_{\mathcal{C}}(x) = x$ for every object and morphism
fong-spivak-invitation_F03253	293	0.108	$\operatorname{id}_{\mathcal{C}}(\operatorname{id}_{\mathcal{C}}) = \operatorname{id}_{\mathcal{C}} = \operatorname{id}_{\operatorname{id}_{\mathcal{C}}(\mathcal{C})}$	$\operatorname{id}_{\mathcal{C}}(\operatorname{id}_{\mathcal{C}}) = \operatorname{id}_{\mathcal{C}} = \operatorname{id}_{\operatorname{id}_{\mathcal{C}}(\mathcal{C})}$	in $\mathcal{C}$ is a functor because it preserves identities $\operatorname{id}_{\mathcal{C}}(\operatorname{id}_{\mathcal{C}}) = \operatorname{id}_{\mathcal{C}} = \operatorname{id}_{\operatorname{id}_{\mathcal{C}}(\mathcal{C})}$ for each object
fong-spivak-invitation_F03254	293	0.166	$\operatorname{id}_{\mathcal{C}}(f; g) = f; g = \operatorname{id}_{\mathcal{C}}(f)g \operatorname{id}_{\mathcal{U}(g)}$	$\operatorname{id}_{\mathcal{C}}(f; g) = f; g = \operatorname{id}_{\mathcal{C}}(f)g \operatorname{id}_{\mathcal{U}(g)}$	$c \in \operatorname{Ob}(\mathcal{C})$, and it preserves composition $\operatorname{id}_{\mathcal{C}}(f \circ g) = \operatorname{id}_{\mathcal{C}}(f) \circ \operatorname{id}_{\mathcal{C}}(g) = f \circ g = \operatorname{id}_{\mathcal{C}}(f) \circ \operatorname{id}_{\mathcal{C}}(g)$ for each pair
fong-spivak-invitation_F03255	293	0.468	$F; G$	$F; G$	2. Given functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$, we need to show that $F \circ G$ is a functor, i.e.
fong-spivak-invitation_F03256	293	0.970	$(F; G)(\operatorname{id}_{\mathcal{C}}) =$	$(F; G)(\operatorname{id}_{\mathcal{C}}) =$	that it preserves identities and compositions. If $c \in \mathcal{C}$ is an object then $(F \circ G)(\operatorname{id}_c) =$
fong-spivak-invitation_F03257	293	0.452	$G(F(\operatorname{id}_{\mathcal{C}})) = G(\operatorname{id}_{F(\mathcal{C})}) = \operatorname{id}_{G(F(\mathcal{C}))}$	$G(F(\operatorname{id}_{\mathcal{C}})) = G(\operatorname{id}_{F(\mathcal{C})}) = \operatorname{id}_{G(F(\mathcal{C}))}$	$G(F(\operatorname{id}_c)) = G(\operatorname{id}_{F(c)}) = \operatorname{id}_{G(F(c))}$ because F and G preserve identities. If f, g are
fong-spivak-invitation_F03258	293	0.904	$\operatorname{id}_{\mathcal{C}}$	$\operatorname{id}_{\mathcal{C}}$	functor and we pre-compose it as above with $\operatorname{id}_{\mathcal{C}}$; it is easy to see that the result will again be
fong-spivak-invitation_F03259	293	0.987	$\operatorname{id}_{\mathcal{D}}$	$\operatorname{id}_{\mathcal{D}}$	F , and similarly if we post-compose F with $\operatorname{id}_{\mathcal{D}}$. This gives unitality, and associativity is just
fong-spivak-invitation_F03260	293	1.000	$H: \mathcal{E} \rightarrow \mathcal{F}$	$H: \mathcal{E} \rightarrow \mathcal{F}$	as easy, though more wordy. Given F as above and $G: \mathcal{D} \rightarrow \mathcal{E}$ and $H: \mathcal{E} \rightarrow \mathcal{F}$, we need to
fong-spivak-invitation_F03261	293	0.638	$(F; G) \circ H = F; (G; H)$	$(F; G) \circ H = F; (G; H)$	show that $(F \circ G) \circ H = F \circ (G \circ H)$. It's a simple application of the definition: for any $x \in \mathcal{C}$,
fong-spivak-invitation_F03262	293	0.638	$x \in \mathcal{C}$	$x \in \mathcal{C}$	show that $(F \circ G) \circ H = F \circ (G \circ H)$. It's a simple application of the definition: for any $x \in \mathcal{C}$,
fong-spivak-invitation_F03263	293	0.999	$F_S(\operatorname{id}_1) = \operatorname{id}_S$	$F_S(\operatorname{id}_1) = \operatorname{id}_S$	Let $S \in \mathbf{Set}$ be a set. Define $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ by $F_S(1) = S$ and $F_S(\operatorname{id}_1) = \operatorname{id}_S$. With this definition,
fong-spivak-invitation_F03264	293	0.883	F_S	F_S	F_S preserves identities and compositions (the only compositions in $\mathbf{1}$ is the composite of the
fong-spivak-invitation_F03265	293	0.883	$\mathbf{1}$	$\mathbf{1}$	F_S preserves identities and compositions (the only compositions in $\mathbf{1}$ is the composite of the
fong-spivak-invitation_F03266	294	1.000	$D(z)$	$D(z)$	1. Data on this schema, i.e. a set-valued functor, assigns a set $D(z)$ and a function $D(s): D(z) \rightarrow$
fong-spivak-invitation_F03267	294	1.000	$D(s): D(z) \rightarrow$	$D(s): D(z) \rightarrow$	1. Data on this schema, i.e. a set-valued functor, assigns a set $D(z)$ and a function $D(s): D(z) \rightarrow$
fong-spivak-invitation_F03268	294	0.877	D_Z	D_Z	<i>involution</i> of the set D_z :
fong-spivak-invitation_F03269	294	1.000	$D(c)$	$D(c)$	2. We could make $D(c)$ the set of people at a “secret Santa” Christmas party, where everyone
fong-spivak-invitation_F03270	294	0.884	$D(b)$	$D(b)$	gives a gift to someone, possibly themselves. Take $D(b)$ to be the set of gifts, g the giver
fong-spivak-invitation_F03271	294	1.000	$D(a)$	$D(a)$	person), $D(a)$ is the set of people who give a gift to themselves, and $d(f): D(a) \rightarrow D(b)$ is
fong-spivak-invitation_F03272	294	1.000	$d(f): D(a) \rightarrow D(b)$	$d(f): D(a) \rightarrow D(b)$	person), $D(a)$ is the set of people who give a gift to themselves, and $d(f): D(a) \rightarrow D(b)$ is
fong-spivak-invitation_F03273	294	1.000	$F, G, H \in \mathcal{D}^{\mathcal{C}}$	$F, G, H \in \mathcal{D}^{\mathcal{C}}$	$F, G, H \in \mathcal{D}^{\mathcal{C}}$; these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$. Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$
fong-spivak-invitation_F03274	294	1.000	$F, G, H: \mathcal{C} \rightarrow \mathcal{D}$	$F, G, H: \mathcal{C} \rightarrow \mathcal{D}$	$F, G, H \in \mathcal{D}^{\mathcal{C}}$; these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$. Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$
fong-spivak-invitation_F03275	294	1.000	$\beta: G \rightarrow H$	$\beta: G \rightarrow H$	$F, G, H \in \mathcal{D}^{\mathcal{C}}$; these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$. Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$
fong-spivak-invitation_F03276	294	0.983	$c \in \mathcal{C}, \alpha$	$c \in \mathcal{C}, \alpha$	So for each $c \in \mathcal{C}$, $\alpha_c: F(c) \rightarrow G(c)$ and β has a component $\beta_c: G(c) \rightarrow$

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fong-spivak-invitation_F03277	294	■ 0.983	$\backslash\mathrm{beta}_{\{c\}}: G(c) \rightarrowtail$	$\beta_c : G(c) \rightarrow$	So for each $c \in \mathcal{C}$, α has a component $\alpha_c: F(c) \rightarrow G(c)$ and β has a component $\beta_c: G(c) \rightarrow$
fong-spivak-invitation_F03278	294	■ 0.606	$H(c)$	$H(c)$	$H(c)$ in $\mathcal{D}$. The expert has told us to define $(\alpha \circ \beta)_c := (\alpha_c \circ \beta_c)$, and indeed that is a morphism
fong-spivak-invitation_F03279	294	■ 0.606	$(\alpha ; \beta)_{\{C\}}:=\left(\alpha_{\{C\}} ; \beta_{\{C\}}\right)$	$(\alpha; \beta)_C := (\alpha_C; \beta_C)$	$H(c)$ in $\mathcal{D}$. The expert has told us to define $(\alpha \circ \beta)_c := (\alpha_c \circ \beta_c)$, and indeed that is a morphism
fong-spivak-invitation_F03280	294	■ 1.000	$F(c) \rightarrowtail H(c)$	$F(c) \rightarrow H(c)$	$F(c) \rightarrow H(c)$.
fong-spivak-invitation_F03281	295	■ 0.997	$\backslash\mathrm{operatorname{id}}_{\{F\}}$	id_F	2. We propose that the identity natural transformation id_F on a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ has as its
fong-spivak-invitation_F03282	295	■ 0.599	$\left(\backslash\mathrm{operatorname{id}}_{\{F\}}\right)_{\{C\}}:=\backslash\mathrm{operatorname{id}}_{\{F(c)\}}$	$(\mathrm{id}_F)_C := \mathrm{id}_{F(c)}$	c -component the morphism $(\mathrm{id}_F)_c := \mathrm{id}_{F(c)}$ in $\mathcal{D}$, for any c . The naturality square
fong-spivak-invitation_F03283	295	■ 0.992	$\mathrm{Id}_{\{F\}}$	id_F	obviously commutes for any $f: c \rightarrow c'$. And it is unital: post-composing id_F with any
fong-spivak-invitation_F03284	295	■ 1.000	$\backslash\mathrm{beta}: F \rightarrowtail G$	$\beta: F \rightarrow G$	$\beta: F \rightarrow G$ (and similarly for precomposing with any $\alpha: E \rightarrow F$) results in a natural trans-
fong-spivak-invitation_F03285	295	■ 1.000	$\backslash\mathrm{alpha}: E \rightarrowtail F$	$\alpha: E \rightarrow F$	$\beta: F \rightarrow G$ (and similarly for precomposing with any $\alpha: E \rightarrow F$) results in a natural trans-
fong-spivak-invitation_F03286 [conf 0.093]	295	■ 0.093	$\left(\backslash\mathrm{operatorname{id}}_{\{F\}}\right)_{\{C\}}$	$(\mathrm{id}_F)_C$	formation $\mathrm{id}_F \circ \beta$ with components $(\mathrm{id}_F)_c \circ \beta_c = (\mathrm{id}_{F(c)} \circ \beta_c) = \beta_c$, and this is just β as
fong-spivak-invitation_F03287 [conf 0.093]	295	■ 0.093	$\backslash\mathrm{beta}_{\{C\}}:=\left(\backslash\mathrm{operatorname{id}}_{\{F(c)\}}\right)^{\circ}\backslash\mathrm{beta}_{\{C\}}\rightarrowtail\backslash\mathrm{beta}_{\{C\}}$	$\beta_C = (\mathrm{id}_{F(c)} \circ \beta_C) = \beta_C$	formation $\mathrm{id}_F \circ \beta$ with components $(\mathrm{id}_F)_c \circ \beta_c = (\mathrm{id}_{F(c)} \circ \beta_c) = \beta_c$, and this is just β as
fong-spivak-invitation_F03288	295	■ 1.000	$\backslash\mathrm{alpha}, \backslash\mathrm{beta}: F \rightarrowtail G$	$\alpha, \beta: F \rightarrow G$	1. Suppose that $F, G: \mathcal{C} \rightarrow \mathcal{P}$ are functors and $\alpha, \beta: F \rightarrow G$ are natural transformations; we
fong-spivak-invitation_F03289	295	■ 0.981	$\backslash\mathrm{alpha}=\backslash\mathrm{beta}$	$\alpha = \beta$	need to show that $\alpha = \beta$. It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \mathrm{Ob}(\mathcal{C})$. But α_c
fong-spivak-invitation_F03290	295	■ 0.981	$\backslash\mathrm{alpha}_{\{c\}}=\backslash\mathrm{beta}_{\{c\}}$	$\alpha_c = \beta_c$	need to show that $\alpha = \beta$. It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \mathrm{Ob}(\mathcal{C})$. But α_c
fong-spivak-invitation_F03291	295	■ 0.728	$\backslash\mathrm{beta}_{\{C\}}$	β_C	and β_c are morphisms $F(c) \rightarrow G(c)$ in $\mathcal{P}$, which is a preorder, and the definition of a preorder
fong-spivak-invitation_F03292	295	■ 0.728	$F(c) \rightarrowtail G(c)$	$F(c) \rightarrow G(c)$	and β_c are morphisms $F(c) \rightarrow G(c)$ in $\mathcal{P}$, which is a preorder, and the definition of a preorder
fong-spivak-invitation_F03293	295	■ 0.999	$\backslash\mathrm{alpha}_{\{C\}}=\backslash\mathrm{beta}_{\{C\}}$	$\alpha_C = \beta_C$	$\alpha_c = \beta_c$, as desired.
fong-spivak-invitation_F03294	295	■ 0.717	$\backslash\mathrm{mathcal{P}}:=\backslash\mathrm{mathbf{1}}$	$\mathcal{P} := \mathbf{1}$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \begin{array}{c} a \\ \bullet \xrightarrow{f_1} \bullet \\ f_2 \end{array} \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and
fong-spivak-invitation_F03295	295	■ 0.717	$\backslash\mathrm{mathcal{C}}:=\backslash\mathrm{stackrel{a}{\bullet}}\backslash\mathrm{bullet}\rightarrowtail\backslash\mathrm{f}_{\{2\}}\{\backslash\mathrm{f}_{\{1\}}\}\backslash\mathrm{stackrel{b}{\bullet}}\backslash\mathrm{bullet}$	$\mathcal{C} := \begin{array}{c} a \\ \bullet \xrightarrow{f_1} \bullet \\ f_2 \end{array}$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \begin{array}{c} a \\ \bullet \xrightarrow{f_1} \bullet \\ f_2 \end{array} \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and
fong-spivak-invitation_F03296	295	■ 0.717	$F(1):=a$	$F(1) := a$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \begin{array}{c} a \\ \bullet \xrightarrow{f_1} \bullet \\ f_2 \end{array} \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and
fong-spivak-invitation_F03297	295	■ 0.717	$G(1):=b$	$G(1) := b$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \begin{array}{c} a \\ \bullet \xrightarrow{f_1} \bullet \\ f_2 \end{array} \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and
fong-spivak-invitation_F03298	295	■ 0.717	$\backslash\mathrm{alpha}_{\{1\}}:=\backslash\mathrm{f}_{\{1\}}$	$\alpha_1 := f_1$	2. This is false. Let $\mathcal{P} := \mathbf{1}$, let $\mathcal{C} := \left \begin{array}{c} a \\ \bullet \xrightarrow{f_1} \bullet \\ f_2 \end{array} \right $, let $F(1) := a$, let $G(1) := b$, let $\alpha_1 := f_1$, and
fong-spivak-invitation_F03299	295	■ 1.000	$\backslash\mathrm{beta}_{\{1\}}:=\backslash\mathrm{g}_{\{2\}}$	$\beta_1 := g_2$	let $\beta_1 := g_2$.
fong-spivak-invitation_F03300	296	■ 1.000	$\backslash\mathrm{alpha}_{\{\backslash\mathrm{text}{Arrow}\}}(b)=e$	$\alpha_{\mathrm{Arrow}}(b) = e$	1. We have $\alpha_{\mathrm{Arrow}}(b) = e$ and $\alpha_{\mathrm{Vertex}}(1) = 4$, $\alpha_{\mathrm{Vertex}}(2) = 5$, and $\alpha_{\mathrm{Vertex}}(3) = 5$.
fong-spivak-invitation_F03301	296	■ 1.000	$\backslash\mathrm{alpha}_{\{\backslash\mathrm{text}{Vertex}\}}(1)=4, \backslash\mathrm{alpha}_{\{\backslash\mathrm{text}{Vertex}\}}(2)=5$	$\alpha_{\mathrm{Vertex}}(1) = 4, \alpha_{\mathrm{Vertex}}(2) = 5$	1. We have $\alpha_{\mathrm{Arrow}}(b) = e$ and $\alpha_{\mathrm{Vertex}}(1) = 4, \alpha_{\mathrm{Vertex}}(2) = 5$, and $\alpha_{\mathrm{Vertex}}(3) = 5$.

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fong-spivak-invitation_F03302	296	■ 1.000	$\alpha_{\text{Vertex}}(3)=5$	$\alpha_{\text{Vertex}}(3) = 5$	1. We have $\alpha_{\text{Arrow}}(b) = e$ and $\alpha_{\text{Vertex}}(1) = 4$, $\alpha_{\text{Vertex}}(2) = 5$, and $\alpha_{\text{Vertex}}(3) = 5$.
fong-spivak-invitation_F03303	297	■ 1.000	$X \times B \rightarrowtail Y \times B$	$X \times B \rightarrow Y \times B$	1. Given an arbitrary function $f: X \rightarrow Y$, we need a function $X \times B \rightarrowtail Y \times B$. We suggest
fong-spivak-invitation_F03304	297	■ 0.965	$f \times B$	$f \times B$	the function which might be denoted $f \times B$; it sends (x, b) to $(f(x), b)$. This assignment is
fong-spivak-invitation_F03305	297	■ 0.965	(x, b)	(x, b)	the function which might be denoted $f \times B$; it sends (x, b) to $(f(x), b)$. This assignment is
fong-spivak-invitation_F03306	297	■ 0.965	$(f(x), b)$	$(f(x), b)$	the function which might be denoted $f \times B$; it sends (x, b) to $(f(x), b)$. This assignment is
fong-spivak-invitation_F03307	297	■ 0.860	$\mathrm{id}_{X \times B}$	$\mathrm{id}_{X \times B}$	functorial: applied to id_X it returns $\mathrm{id}_{X \times B}$ and it preserves composition.
fong-spivak-invitation_F03308	297	■ 1.000	$X^B \rightarrowtail Y^B$	$X^B \rightarrow Y^B$	2. Given a function $f: X \rightarrow Y$, we need a function $X^B \rightarrow Y^B$. The canonical function would
fong-spivak-invitation_F03309	297	■ 0.885	f^B	f^B	be denoted f^B ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$. This is
fong-spivak-invitation_F03310	297	■ 0.885	$g: B \rightarrowtail X$	$g: B \rightarrow X$	be denoted f^B ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$. This is
fong-spivak-invitation_F03311	297	■ 0.885	$(g \circ f): B \rightarrowtail X \rightarrowtail Y$	$(g \circ f): B \rightarrow X \rightarrow Y$	be denoted f^B ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$. This is
fong-spivak-invitation_F03312	297	■ 0.984	$f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$	$f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$	functorial: applied to id_X it sends g to g , i.e. $f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$, and applied to the composite
fong-spivak-invitation_F03313	297	■ 0.773	$(f_1 \circ f_2): X \rightarrowtail Y \rightarrowtail Z$	$(f_1 \circ f_2): X \rightarrow Y \rightarrow Z$	$(f_1 \circ f_2): X \rightarrow Y \rightarrow Z$, we have
fong-spivak-invitation_F03314	297	■ 1.000	$g \in X^B$	$g \in X^B$	for any $g \in X^B$.
fong-spivak-invitation_F03315	297	■ 0.993	$\mathbb{N}^{\mathbb{N}}$	$\mathbb{N}^{\mathbb{N}}$	3. If $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$ is the result of currying $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, then $p(3)$ is an element of $\mathbb{N}^{\mathbb{N}}$, i.e. we
fong-spivak-invitation_F03316	297	■ 0.974	$p(3): \mathbb{N} \rightarrowtail \mathbb{N}$	$p(3): \mathbb{N} \rightarrow \mathbb{N}$	have $p(3): \mathbb{N} \rightarrow \mathbb{N}$; what function is it? It is the function that adds 3. That is $p(3)(n) := n + 3$.
fong-spivak-invitation_F03317	297	■ 0.974	$p(3)(n) := n + 3$	$p(3)(n) := n + 3$	have $p(3): \mathbb{N} \rightarrow \mathbb{N}$; what function is it? It is the function that adds 3. That is $p(3)(n) := n + 3$.
fong-spivak-invitation_F03318	297	■ 0.989	$!: \mathcal{C} \rightarrowtail \mathbf{1}$	$!: \mathcal{C} \rightarrow \mathbf{1}$	The functor $!: \mathcal{C} \rightarrowtail \mathbf{1}$ from Eq. (3.14) sends each object $c \in \mathcal{C}$ to the unique object $1 \in \mathbf{1}$ and
fong-spivak-invitation_F03319	297	■ 0.989	$1 \in \mathbf{1}$	$1 \in \mathbf{1}$	The functor $!: \mathcal{C} \rightarrow \mathbf{1}$ from Eq. (3.14) sends each object $c \in \mathcal{C}$ to the unique object $1 \in \mathbf{1}$ and
fong-spivak-invitation_F03320	297	■ 0.987	$\mathrm{B} = \mathrm{Bob}$, $3 = \mathrm{Em}_3$	$B = \text{Bob}, 3 = \text{Em}_3$	Here it is, with names and emails shortened (e.g. B=Bob, 3=Em_3):
fong-spivak-invitation_F03321	297	■ 1.000	$c \rightarrowtail z$	$c \rightarrow z$	$\Rightarrow$. When $\mathcal{C}$ is the category underlying a preorder, there is at most one morphism between any
fong-spivak-invitation_F03322	298	■ 0.969	$\mathcal{C} \in \mathbf{Cat}$	$\mathcal{C} \in \mathbf{Cat}$	$\mathcal{C} \rightarrow \mathbf{1}$ for any object (category) $\mathcal{C} \in \mathbf{Cat}$.
fong-spivak-invitation_F03323	298	■ 0.944	$2V := \bullet \bullet$	$2V := \bullet \bullet$	Consider the graph $2V := \boxed{\bullet \bullet}$ with two vertices and no arrows, and let $\mathcal{C} = \mathbf{Free}(2V)$; it has
fong-spivak-invitation_F03324	298	■ 0.944	$\mathcal{C} = \mathbf{Free}(2V)$	$\mathcal{C} = \mathbf{Free}(2V)$	Consider the graph $2V := \boxed{\bullet \bullet}$ with two vertices and no arrows, and let $\mathcal{C} = \mathbf{Free}(2V)$; it has
fong-spivak-invitation_F03325	298	■ 1.000	$z \in \mathcal{P}$	$z \in \mathcal{P}$	A product of x and y in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that
fong-spivak-invitation_F03326	298	■ 1.000	$z \rightarrowtail x$	$z \rightarrow x$	A product of x and y in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that
fong-spivak-invitation_F03327	298	■ 1.000	$z \rightarrowtail y$	$z \rightarrow y$	A product of x and y in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that
fong-spivak-invitation_F03328	298	■ 1.000	$z^{\prime}$	z'	for any other object z' and maps $z' \rightarrow x$ and $z' \rightarrow y$, there is a unique morphism $z' \rightarrow z$ making
fong-spivak-invitation_F03329	298	■ 1.000	$z^{\prime} \rightarrowtail x$	$z' \rightarrow x$	for any other object z' and maps $z' \rightarrow x$ and $z' \rightarrow y$, there is a unique morphism $z' \rightarrow z$ making

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fong-spivak-invitation_F03382	301	■ 0.995	$\Phi:(T \times E) \longrightarrow \$$	$\Phi:(T \times E) \longrightarrow \$$	Yes, this is valid: it just means that the profunctor $\Phi:(T \times E) \multimap \$$ does not relate (good-natured,
fong-spivak-invitation_F03383	301	■ 0.996	$\Phi(($	$\Phi(($	funny) to any element of \$. More formally, it means that $\Phi((\text{good-natured, funny}), p) = \text{false}$
fong-spivak-invitation_F03384	301	■ 0.996	$), p) =$	$), p) =$	funny) to any element of \$. More formally, it means that $\Phi((\text{good-natured, funny}), p) = \text{false}$
fong-spivak-invitation_F03385	301	■ 1.000	$p \in \$ = \{\$100 \mathrm{~K}, \$500 \mathrm{~K}, \$1 \mathrm{~M}\}$	$p \in \$ = \{\$100 \mathrm{~K}, \$500 \mathrm{~K}, \$1 \mathrm{~M}\}$	for all $p \in \$ = \{\$100\text{K}, \$500\text{K}, \$1\text{M}\}$. We can interpret this to mean that it is not feasible to
fong-spivak-invitation_F03386	301	■ 0.932	$(\Phi: \Psi)(-, -) =$	$(\Phi: \Psi)(-, -) =$	shortest paths go through the bridges from D to y and y to r , so in this case $\Phi(\Psi)(-, -) \equiv$
fong-spivak-invitation_F03387	301	■ 0.753	$X(-, D) + 9 + Z(r, -)$	$X(-, D) + 9 + Z(r, -)$	$X(-, D) + 9 + Z(r, -)$. This gives:
fong-spivak-invitation_F03389	302	■ 0.951	$U_{\mathcal{X}}$	$U_{\mathcal{X}}$	We choose the Cost -category $\mathcal{X}$ from Eq. (2.18). The unit profunctor $U_{\mathcal{X}}$ on $\mathcal{X}$ is described by the
fong-spivak-invitation_F03390	302	■ 0.971	$I \leq \mathcal{P}(p, p)$	$I \leq \mathcal{P}(p, p)$	city of $\otimes$ (Definition 2.1(a)) applied to the inequalities $I \leq \mathcal{P}(p, p)$ (the identity law for $\mathcal{P}$ at
fong-spivak-invitation_F03391	302	■ 1.000	$\Phi(p, q) \leq \Phi(p, q)$	$\Phi(p, q) \leq \Phi(p, q)$	p , Definition 2.30(a)) and $\Phi(p, q) \leq \Phi(p, q)$ (reflexivity of preorder $\mathcal{V}$, Definition 1.25(a)).
fong-spivak-invitation_F03392	302	■ 1.000	$i_{\{0\}} \in I$	$i_0 \in I$	The third step uses the definition of join: for any set I element $i_0 \in I$, a subset $s_{i \in I}$ at a poset,
fong-spivak-invitation_F03393	302	■ 1.000	$s_{\{i \in I\}}$	$s_{i \in I}$	The third step uses the definition of join: for any set I element $i_0 \in I$, a subset $s_{i \in I}$ at a poset,
fong-spivak-invitation_F03394	302	■ 0.998	$s_{\{i_{\{0\}}\}} \leq \bigvee_{i \in I} s$	$s_{i_0} \leq \bigvee_{i \in I} s$	we have $s_{i_0} \leq \bigvee_{i \in I} s$. In our case, $Ii_0 = p$, and $s_i = \mathcal{P}(p, i) \otimes \Phi$. The final equality is just
fong-spivak-invitation_F03395	302	■ 0.998	$I i_{\{0\}} = p$	$Ii_0 = p$	we have $s_{i_0} \leq \bigvee_{i \in I} s$. In our case, $Ii_0 = p$, and $s_i = \mathcal{P}(p, i) \otimes \Phi$. The final equality is just
fong-spivak-invitation_F03396	302	■ 0.998	$s_{\{i\}} = \mathcal{P}(p, i) \otimes \Phi$	$s_i = \mathcal{P}(p, i) \otimes \Phi$	we have $s_{i_0} \leq \bigvee_{i \in I} s$. In our case, $Ii_0 = p$, and $s_i = \mathcal{P}(p, i) \otimes \Phi$. The final equality is just
fong-spivak-invitation_F03397	302	■ 0.558	$I = \text{true}$	$I = \text{true}$	2. Note that, in Bool , $I = \text{true}$. Since the identity law at p says $\text{true} \leq \mathcal{P}(p, p)$, and true is
fong-spivak-invitation_F03398	302	■ 0.558	$\text{true} \leq \mathcal{P}(p, p)$	$\text{true} \leq \mathcal{P}(p, p)$	2. Note that, in Bool , $I = \text{true}$. Since the identity law at p says $\text{true} \leq \mathcal{P}(p, p)$, and true is
fong-spivak-invitation_F03399	302	■ 0.912	$\mathcal{P}(p, p) = \text{tr}$	$\mathcal{P}(p, p) = \text{tr}$	the largest element of the preorder Bool , we thus have $\mathcal{P}(p, p) = \text{true}$ for all p . This shows
fong-spivak-invitation_F03400	302	■ 0.888	$\Phi(p, q)$	$\Phi(p, q)$	side of the inequality is equal to $\Phi(p, q)$. We split into two cases. Suppose $\Phi(p, q) = \text{true}$.
fong-spivak-invitation_F03401	302	■ 0.888	$\Phi(p, q) = \text{true}$	$\Phi(p, q) = \text{true}$	side of the inequality is equal to $\Phi(p, q)$. We split into two cases. Suppose $\Phi(p, q) = \text{true}$.
fong-spivak-invitation_F03402	302	■ 0.449	$\Phi(p, q) =$	$\Phi(p, q) =$	Next, suppose $\Phi(p, q) \equiv \text{false}$. Note that since Φ is a monotone map $\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow \mathbf{Bool}$, if
fong-spivak-invitation_F03403	302	■ 0.449	$\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow$	$\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow$	Next, suppose $\Phi(p, q) = \text{false}$. Note that since Φ is a monotone map $\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow \mathbf{Bool}$, if
fong-spivak-invitation_F03404	302	■ 0.987	$p \leq p_1$	$p \leq p_1$	$p \leq p_1$ in $\mathcal{P}$, then $\Phi(p_1, q) \leq \Phi(p, q)$ in Bool . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$
fong-spivak-invitation_F03405	302	■ 0.987	$\Phi(p_1, q) \leq \Phi(p, q)$	$\Phi(p_1, q) \leq \Phi(p, q)$	$p \leq p_1$ in $\mathcal{P}$, then $\Phi(p_1, q) \leq \Phi(p, q)$ in Bool . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$
fong-spivak-invitation_F03406	302	■ 0.987	$\mathcal{P}(p, p_1) = \text{true}$	$\mathcal{P}(p, p_1) = \text{true}$	$p \leq p_1$ in $\mathcal{P}$, then $\Phi(p_1, q) \leq \Phi(p, q)$ in Bool . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$
fong-spivak-invitation_F03407	302	■ 0.987	$\Phi(p_1, q) =$	$\Phi(p_1, q) =$	$p \leq p_1$ in $\mathcal{P}$, then $\Phi(p_1, q) \leq \Phi(p, q)$ in Bool . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$
fong-spivak-invitation_F03408	302	■ 0.672	$\mathcal{P}(p, p_1) =$	$\mathcal{P}(p, p_1) =$	$\Phi(p, q) = \text{false}$. This implies that for all $p_1 \in \mathcal{P}$, we have either $\mathcal{P}(p, p_1) = \text{false}$ or
fong-spivak-invitation_F03409	302	■ 0.775	$\bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q) = \bigvee_{p_1 \in \mathcal{P}} \text{false} = \text{false}$	$\bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q) = \bigvee_{p_1 \in \mathcal{P}} \text{false} = \text{false}$	$\Phi(p_1, q) = \text{false}$, and hence $\bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q) = \bigvee_{p_1 \in \mathcal{P}} \text{false} = \text{false}$.
fong-spivak-invitation_F03410	302	■ 1.000	$\Phi(p, q) = \bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q)$	$\Phi(p, q) = \bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q)$	Thus, in either case, we see that $\Phi(p, q) = \bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q)$, as required.

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fong-spivak-invitation_F03411	302	■ 1.000	$v \otimes I = v$	$v \otimes I = v$	3. The first equation is unitality in monoidal categories, $v \otimes I = v$ for any $v \in V$. The second
fong-spivak-invitation_F03412	302	■ 1.000	$I \leq Q(q, q)$	$I \leq Q(q, q)$	is that $I \leq Q(q, q)$ by unitality of enriched categories, see Definition 2.30, together with
fong-spivak-invitation_F03413	302	■ 1.000	$v_{\{1\}} \leq v_{\{2\}}$	$v_1 \leq v_2$	monotonicity of monoidal product: $v_1 \leq v_2$ implies $v \otimes v_1 \leq v \otimes v_2$. The third was shown in
fong-spivak-invitation_F03414	302	■ 1.000	$v \otimes v_{\{1\}} \leq v \otimes v_{\{2\}}$	$v \otimes v_1 \leq v \otimes v_2$	monotonicity of monoidal product: $v_1 \leq v_2$ implies $v \otimes v_1 \leq v \otimes v_2$. The third was shown in
fong-spivak-invitation_F03415	303	■ 0.888	$\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$	$\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$	This is very straightforward. We wish to check $\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$ has the formula $\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$.
fong-spivak-invitation_F03416	303	■ 0.888	$\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$	$\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$	This is very straightforward. We wish to check $\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$ has the formula $\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$.
fong-spivak-invitation_F03417	303	■ 0.928	$\widehat{\mathrm{id}}(p, q) := \mathcal{P}(\mathrm{id}(p), q) = \mathcal{P}(p, q)$	$\widehat{\mathrm{id}}(p, q) := \mathcal{P}(\mathrm{id}(p), q) = \mathcal{P}(p, q)$	By Definition 4.24, $\widehat{\mathrm{id}}(p, q) := \mathcal{P}(\mathrm{id}(p), q) = \mathcal{P}(p, q)$. So they're the same.
fong-spivak-invitation_F03418	303	■ 0.743	$\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$	$\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$	The conjoint $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ sends (a, b, c, d) to $\mathbb{R}(a, b + c + d)$, which is true if $a \leq$
fong-spivak-invitation_F03419	303	■ 0.743	$\mathbb{R}(a, b + c + d)$	$\mathbb{R}(a, b + c + d)$	The conjoint $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ sends (a, b, c, d) to $\mathbb{R}(a, b + c + d)$, which is true if $a \leq$
fong-spivak-invitation_F03420	303	■ 0.743	$a \leq$	$a \leq$	The conjoint $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ sends (a, b, c, d) to $\mathbb{R}(a, b + c + d)$, which is true if $a \leq$
fong-spivak-invitation_F03421	303	■ 1.000	$b + c + d$	$b + c + d$	$b + c + d$, and false otherwise.
fong-spivak-invitation_F03422	303	■ 0.863	$\widehat{F}(p, q) = Q(F(p), q)$	$\widehat{F}(p, q) = Q(F(p), q)$	1. By Definition 4.24, $\widehat{F}(p, q) = Q(F(p), q)$ and $\check{G}(p, q) = Q(p, G(q))$. Since $\mathcal{V}$ is skeletal, F
fong-spivak-invitation_F03423	303	■ 0.863	$\check{G}(p, q) = Q(p, G(q))$	$\check{G}(p, q) = Q(p, G(q))$	1. By Definition 4.24, $\widehat{F}(p, q) = Q(F(p), q)$ and $\check{G}(p, q) = Q(p, G(q))$. Since $\mathcal{V}$ is skeletal, F
fong-spivak-invitation_F03424	303	■ 0.766	$Q(F(p), q) = Q(p, G(q))$	$Q(F(p), q) = Q(p, G(q))$	and G are $\mathcal{V}$ adjoints if and only if $Q(F(p), q) = Q(p, G(q))$. Thus F and G are adjoints if
fong-spivak-invitation_F03425	303	■ 0.948	$\widehat{F} = \check{G}$	$\widehat{F} = \check{G}$	and only if $\widehat{F} = \check{G}$.
fong-spivak-invitation_F03426	303	■ 0.999	$\otimes: \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$	$\otimes: \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$	Definition 2.1 calls for merely a function $\otimes: \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$, while Definition 4.34 calls for a <i>functor</i> .
fong-spivak-invitation_F03427	303	■ 0.999	$(x_1, x_2) \leq (y_1, y_2)$	$(x_1, x_2) \leq (y_1, y_2)$	Definition 2.1: it says that if $(x_1, x_2) \leq (y_1, y_2)$ in $\mathcal{P} \otimes \mathcal{P}$, then we must have $x_1 \otimes x_2 \leq y_1 \otimes y_2$
fong-spivak-invitation_F03428	303	■ 0.999	$\mathcal{P} \otimes \mathcal{P}$	$\mathcal{P} \otimes \mathcal{P}$	Definition 2.1: it says that if $(x_1, x_2) \leq (y_1, y_2)$ in $\mathcal{P} \otimes \mathcal{P}$, then we must have $x_1 \otimes x_2 \leq y_1 \otimes y_2$
fong-spivak-invitation_F03429	303	■ 1.000	$2.1 \otimes$	$2.1 \otimes$	in $\mathcal{P}$. So it is also the case that in Definition 2.1 $\otimes$ is a functor.
fong-spivak-invitation_F03430	303	■ 0.998	$g_E(5, 3) =$	$g_E(5, 3) =$	1. $g_E(5, 3) = \text{false}$, $g_F(5, 3) = 2$.
fong-spivak-invitation_F03431	303	■ 0.998	$g_F(5, 3) = 2$	$g_F(5, 3) = 2$	1. $g_E(5, 3) = \text{false}$, $g_F(5, 3) = 2$.
fong-spivak-invitation_F03432	303	■ 0.976	$g_E(3, 5) =$	$g_E(3, 5) =$	2. $g_E(3, 5) = \text{true}$, $g_F(3, 5) = -2$.
fong-spivak-invitation_F03433	303	■ 0.976	$g_F(3, 5) = -2$	$g_F(3, 5) = -2$	2. $g_E(3, 5) = \text{true}$, $g_F(3, 5) = -2$.
fong-spivak-invitation_F03434	304	■ 0.733	$) = 5$	$) = 5$	3. $h(5, \text{true}) = 5$.
fong-spivak-invitation_F03435	304	■ 0.885	$) = -5$	$) = -5$	4. $h(-5, \text{true}) = -5$.
fong-spivak-invitation_F03436	304	■ 0.693	$) = 6$	$) = 6$	5. $h(-5, \text{false}) = 6$.
fong-spivak-invitation_F03437	304	■ 1.000	$q_G(-2, 3) = 2, q_F(-2, 3) = -13$	$q_G(-2, 3) = 2, q_F(-2, 3) = -13$	6. $q_G(-2, 3) = 2, q_F(-2, 3) = -13$.
fong-spivak-invitation_F03438	304	■ 1.000	$q_G(2, 3) = -1, q_F(2, 3) = 7$	$q_G(2, 3) = -1, q_F(2, 3) = 7$	7. $q_G(2, 3) = -1, q_F(2, 3) = 7$.

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fong-spivak-invitation_F03439	304	0.970	$\mathcal{C}(x, y)$	$\mathcal{C}(x, y)$	$\mathcal{C}(x, y)$ of morphisms. Moreover, recall that the monoidal unit I is the one-element set $\{1\}$. This
fong-spivak-invitation_F03440	304	0.483	$\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$	$\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$	means a morphism $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$ is a function $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$. This is the same data as
fong-spivak-invitation_F03441	304	0.483	$\mathrm{id}_x:\{1\} \rightarrow \mathcal{C}(x, x)$	$\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$	means a morphism $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$ is a function $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$. This is the same data as
fong-spivak-invitation_F03442	304	0.919	$\mathrm{id}_x=\mathrm{id}_x(1) \in \mathcal{C}(x, x)$	$\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$	simply an element $\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$; we call this data the identity morphism on x . Finally,
fong-spivak-invitation_F03443	304	0.960	$\{ \}_9: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	$g: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	a morphism $\S: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ is a function $\S: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$; this is
fong-spivak-invitation_F03444	304	0.960	$\{ \}_9: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	$g: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$	a morphism $\S: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ is a function $\S: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$; this is
fong-spivak-invitation_F03445	304	0.662	$I \rightarrow \mathcal{X}(x, x)$	$I \rightarrow \mathcal{X}(x, x)$	An identity element in a Cost -category $\mathcal{X}$ is a morphism $I \rightarrow \mathcal{X}(x, x)$ in Cost = $([0, \infty], \geq,$
fong-spivak-invitation_F03446	304	0.662	$=([0, \infty], \geq$	$=([0, \infty], \geq$	An identity element in a Cost -category $\mathcal{X}$ is a morphism $I \rightarrow \mathcal{X}(x, x)$ in Cost $\equiv ([0, \infty], \geq,$
fong-spivak-invitation_F03447	304	0.802	$0 \geq \mathfrak{X}(x, x)$	$0 \geq \mathfrak{X}(x, x)$	0, +), and hence the condition that $0 \geq \mathcal{X}(x, x)$. This implies that $\mathcal{X}(x, x) = 0$. In terms of
fong-spivak-invitation_F03448	304	0.802	$\mathcal{X}(x, x)=0$	$\mathcal{X}(x, x) = 0$	0, +), and hence the condition that $0 \geq \mathcal{X}(x, x)$. This implies that $\mathcal{X}(x, x) = 0$. In terms of
fong-spivak-invitation_F03449	305	0.801	$x \in \mathcal{X}$	$x \in \mathcal{X}$	resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available
fong-spivak-invitation_F03450	305	0.801	$y \in \mathcal{Y}$	$y \in \mathcal{Y}$	resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available
fong-spivak-invitation_F03451	305	0.801	$(x, y) \leq (x', y')$	$(x, y) \leq (x', y')$	resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available
fong-spivak-invitation_F03452	305	0.801	$y \leq y'$	$y \leq y'$	resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$. That is, if x is available
fong-spivak-invitation_F03453	305	0.920	$\Phi: X_1 \mapsto X_2$	$\Phi: X_1 \mapsto X_2$	Given two profunctors $\Phi: \mathcal{X}_1 \multimap \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \multimap \mathcal{Y}_2$, the profunctor $\Phi \times \Psi$ represents their
fong-spivak-invitation_F03454	305	1.000	y_1	y_1	conjunction, i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained
fong-spivak-invitation_F03455	305	1.000	x_1	x_1	conjunction, i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained
fong-spivak-invitation_F03456	305	1.000	y_2	y_2	conjunction, i.e. AND. In other words, if y_1 can be obtained given x_1 AND y_2 can be obtained
fong-spivak-invitation_F03457	305	0.627	x_2	x_2	given x_2 , then (y_1, y_2) can be obtained given (x_1, x_2) .
fong-spivak-invitation_F03458	305	0.627	(y_1, y_2)	(y_1, y_2)	given x_2 , then (y_1, y_2) can be obtained given (x_1, x_2) .
fong-spivak-invitation_F03459	305	0.627	(x_1, x_2)	(x_1, x_2)	given x_2 , then (y_1, y_2) can be obtained given (x_1, x_2) .
fong-spivak-invitation_F03460	305	0.398	$\mathcal{X} \times \mathbf{1} - \mathfrak{X}$	$\mathcal{X} \times \mathbf{1} - \mathfrak{X}$	The profunctor $\mathcal{X} \times \mathbf{1} \multimap \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) :=$
fong-spivak-invitation_F03461	305	0.398	$\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$	$\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$	The profunctor $\mathcal{X} \times \mathbf{1} \multimap \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) :=$
fong-spivak-invitation_F03462	305	0.398	$\alpha((x, 1), y) :=$	$\alpha((x, 1), y) :=$	The profunctor $\mathcal{X} \times \mathbf{1} \multimap \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) :=$
fong-spivak-invitation_F03463	305	0.732	$X(x, y)$	$X(x, y)$	$\mathcal{X}(x, y)$ is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \multimap \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$.
fong-spivak-invitation_F03464	305	0.732	$\alpha^{-1}: X \mapsto X \times \mathbf{1}$	$\alpha^{-1}: X \mapsto X \times \mathbf{1}$	$\mathcal{X}(x, y)$ is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \multimap \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$.
fong-spivak-invitation_F03465	305	0.732	$\alpha^{-1}(x, (y, 1)) := X(x, y)$	$\alpha^{-1}(x, (y, 1)) := X(x, y)$	$\mathcal{X}(x, y)$ is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \multimap \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$.
fong-spivak-invitation_F03466	305	0.844	$\alpha^{-1}; \alpha = \mathrm{U}_{\mathcal{X}}$	$\alpha^{-1}; \alpha = \mathrm{U}_{\mathcal{X}}$	To see that $\alpha^{-1} \circ \alpha = \mathrm{U}_{\mathcal{X}}$, note first that the unit law for $\mathcal{X}$ at z and the definition of join imply

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fong-spivak-invitation_F03467	305	0.753	$\alpha ; \alpha^{-1} = \mathrm{U}_{X \times \mathbf{1}}$	$\alpha; \alpha^{-1} = U_{X \times \mathbf{1}}$	Similarly we can show $\alpha \circ \alpha^{-1} = U_{X \times \mathbf{1}}$, and hence that α is an isomorphism $X \times \mathbf{1} \rightarrowtail X$.
fong-spivak-invitation_F03468	305	0.984	$\beta((1, x), y) := \mathrm{mathcal{X}}(x, y)$	$\beta((1, x), y) := \mathcal{X}(x, y)$	Moreover, we can similarly show that $\beta((1, x), y) := X(x, y)$ defines an isomorphism $\beta: \mathbf{1} \times$
fong-spivak-invitation_F03469	305	0.984	$\beta: \mathbf{1} \times$	$\beta: \mathbf{1} \times$	Moreover, we can similarly show that $\beta((1, x), y) := X(x, y)$ defines an isomorphism $\beta: \mathbf{1} \times$
fong-spivak-invitation_F03470	305	0.641	$x \rightarrowtail x$	$x \rightarrow x$	$X \rightarrowtail X$.
fong-spivak-invitation_F03471	306	0.823	α^{-1}	α^{-1}	is itself the identity (i.e. the unit profunctor on X), where α and α^{-1} are the isomorphisms defined
fong-spivak-invitation_F03472	306	0.997	$\Phi(x, y)$	$\Phi(x, y)$	Freely using the distributivity of $\otimes$ over $\vee$, the value $\Phi(x, y)$ of this composite at $(x, y) \in$
fong-spivak-invitation_F03473	306	0.443	$\mathrm{mathscr{X}}^{\mathrm{OP}} \times \mathrm{mathcal{X}}$	$\mathcal{X}^{\mathrm{OP}} \times \mathcal{X}$	$X^{\mathrm{op}} \times X$ is given by
fong-spivak-invitation_F03474	306	0.996	$\mathrm{U}_{\mathrm{mathcal{X}}}(a, b) = \mathrm{mathcal{X}}(a, b)$	$U_{\mathcal{X}}(a, b) = \mathcal{X}(a, b)$	with the unit profunctor $U_{\mathcal{X}}(a, b) = X(a, b)$ is the identity.
fong-spivak-invitation_F03475	306	1.000	$m \rightarrowtail p$	$m \rightarrow p$	$m \rightarrow p$ given by $(f \circ g)(i) = g(f(i))$ for all $1 \leq i \leq m$.
fong-spivak-invitation_F03476	306	1.000	$(f \circ g)(i) = g(f(i))$	$(f; g)(i) = g(f(i))$	$m \rightarrow p$ given by $(f \circ g)(i) = g(f(i))$ for all $1 \leq i \leq m$.
fong-spivak-invitation_F03477	306	1.000	$1 \leq i \leq m$	$1 \leq i \leq m$	$m \rightarrow p$ given by $(f \circ g)(i) = g(f(i))$ for all $1 \leq i \leq m$.
fong-spivak-invitation_F03478	306	0.426	$\mathrm{id}_m: m \rightarrowtail m$	$\mathrm{id}_m: m \rightarrow m$	4. The identity $\mathrm{id}_m: m \rightarrow m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2
fong-spivak-invitation_F03479	306	0.426	$\mathrm{id}_m(i) = i$	$\mathrm{id}_m(i) = i$	4. The identity $\mathrm{id}_m: m \rightarrow m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2
fong-spivak-invitation_F03480	306	0.426	id_2	id_2	4. The identity $\mathrm{id}_m: m \rightarrow m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$. Here is a picture of id_2
fong-spivak-invitation_F03481	306	1.000	id_8	id_8	and id_8 :
fong-spivak-invitation_F03482	307	1.000	$\sigma_{3,5}: 8 \rightarrowtail 8$	$\sigma_{3,5}: 8 \rightarrow 8$	5. Here is a picture of the symmetry $\sigma_{3,5}: 8 \rightarrow 8$:
fong-spivak-invitation_F03483	307	0.996	$m \preceq n$	$m \preceq n$	whose order is denoted $m \preceq n$, and with the property that whenever $m_1 \preceq n_1$ and $m_2 \preceq n_2$ hold
fong-spivak-invitation_F03484	307	0.996	$m_1 \preceq n_1$	$m_1 \preceq n_1$	whose order is denoted $m \preceq n$, and with the property that whenever $m_1 \preceq n_1$ and $m_2 \preceq n_2$ hold
fong-spivak-invitation_F03485	307	0.996	$m_2 \preceq n_2$	$m_2 \preceq n_2$	whose order is denoted $m \preceq n$, and with the property that whenever $m_1 \preceq n_1$ and $m_2 \preceq n_2$ hold
fong-spivak-invitation_F03486	307	1.000	$m_1 + m_2 \preceq n_1 + n_2$	$m_1 + m_2 \preceq n_1 + n_2$	then $m_1 + m_2 \preceq n_1 + n_2$ does too.
fong-spivak-invitation_F03487	307	1.000	$d \in \mathbb{N}$	$d \in \mathbb{N}$	2. Take $\preceq$ to be the usual order, $m \preceq n$ iff there exists $d \in \mathbb{N}$ with $d + m = n$.
fong-spivak-invitation_F03488	307	1.000	$d + m = n$	$d + m = n$	2. Take $\preceq$ to be the usual order, $m \preceq n$ iff there exists $d \in \mathbb{N}$ with $d + m = n$.
fong-spivak-invitation_F03489	307	0.999	$m = n + d$	$m = n + d$	3. Take $\preceq$ to be the reverse of the usual order, $m \preceq n$ iff there exists $d \in \mathbb{N}$ with $m = n + d$.
fong-spivak-invitation_F03490	307	1.000	$q \in \mathbb{N}$	$q \in \mathbb{N}$	5. (Non-example.) Take $\preceq$ to be the division order, $m \preceq n$ iff there exists $q \in \mathbb{N}$ with $m * q = d$.
fong-spivak-invitation_F03491	307	1.000	$m * q = d$	$m * q = d$	5. (Non-example.) Take $\preceq$ to be the division order, $m \preceq n$ iff there exists $q \in \mathbb{N}$ with $m * q = d$.
fong-spivak-invitation_F03492	307	1.000	$2 \preceq 4$	$2 \preceq 4$	This is a perfectly good poset, but it does not satisfy the monotonicity property: we have $2 \preceq 4$
fong-spivak-invitation_F03493	307	0.958	$3 \preceq 3$	$3 \preceq 3$	and $5 \preceq 7$ but not $5 \preceq 7$.
fong-spivak-invitation_F03494	307	0.958	$5 \preceq 7$	$5 \preceq 7$	and $3 \preceq 3$ but not $5 \preceq 7$.

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fong-spivak-invitation_F03495	307	■ 0.492	$\mathbf{B i j}(m, n):=\{f: \underline{m} \rightarrow \underline{n} \mid f$	$\mathbf{Bij}(m, n) := \{f : \underline{m} \rightarrow \underline{n} \mid f$	1. $\mathbf{Bij}(m, n) := \{f : \underline{m} \rightarrow \underline{n} \mid f$ is a bijection}. Note that $\mathbf{Bij}(m, n) = \emptyset$ if $m \neq n$ and it
fong-spivak-invitation_F03496	307	■ 0.492	$\mathbf{B i j}(m, n)=\varnothing$	$\mathbf{Bij}(m, n) = \emptyset$	1. $\mathbf{Bij}(m, n) := \{f : \underline{m} \rightarrow \underline{n} \mid f$ is a bijection}. Note that $\mathbf{Bij}(m, n) = \emptyset$ if $m \neq n$ and it
fong-spivak-invitation_F03497	307	■ 1.000	$n \rightarrow n$	$n \rightarrow n$	2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$.
fong-spivak-invitation_F03498	307	■ 1.000	$\underline{n} \rightarrow \underline{n}$	$\underline{n} \rightarrow \underline{n}$	2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$.
fong-spivak-invitation_F03499	307	■ 1.000	$i \mapsto i$	$i \mapsto i$	2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$.
fong-spivak-invitation_F03500	307	■ 1.000	$m+n \rightarrow n+m$	$m+n \rightarrow n+m$	3. The symmetry map $m+n \rightarrow n+m$ is the bijection $\sigma_{m,n} : \underline{m+n} \rightarrow \underline{n+m}$ given by
fong-spivak-invitation_F03501	307	■ 1.000	$\sigma_{m, n}: \underline{m+n} \rightarrow \underline{n+m}$	$\sigma_{m,n} : \underline{m+n} \rightarrow \underline{n+m}$	3. The symmetry map $m+n \rightarrow n+m$ is the bijection $\sigma_{m,n} : \underline{m+n} \rightarrow \underline{n+m}$ given by
fong-spivak-invitation_F03502	307	■ 1.000	$n \rightarrow p$	$n \rightarrow p$	4. Composition of bijections $m \rightarrow n$ and $n \rightarrow p$ is just their composition as functions,
fong-spivak-invitation_F03503	307	■ 1.000	$(f+g):(m+$	$(f+g):(m+$	5. Given bijections $f: m \rightarrow m'$ and $g: n \rightarrow n'$, their monoidal product $(f+g):(m+$
fong-spivak-invitation_F03504	307	■ 1.000	$n) \rightarrow (m^{\prime}+n^{\prime})$	$n) \rightarrow (m' + n')$	$\underline{n}) \rightarrow (m' + n')$ is given by
fong-spivak-invitation_F03505	307	■ 1.000	$i \sim j$	$i \sim j$	reflexive relation, i.e. where $i \sim j$ iff $i = j$.
fong-spivak-invitation_F03506	307	■ 1.000	$i=j$	$i = j$	reflexive relation, i.e. where $i \sim j$ iff $i = j$.
fong-spivak-invitation_F03507	308	■ 0.868	$\underline{m+n+n+m}$	$\underline{m+n+n+m}$	3. The symmetry map $\sigma_{m,n}$, as an equivalence relation on $\underline{m+n+n+m}$ is “the obvious
fong-spivak-invitation_F03508	308	■ 1.000	$ i-j =m+n+n$	$ i-j = m+n+n$	• $\underline{i-j} \equiv m+n+n$, or
fong-spivak-invitation_F03509	308	■ 0.996	$m+1 \leq i \leq m+n+n$	$m+1 \leq i \leq m+n+n$	• $\underline{m+1 \leq i \leq m+n+n}$ and $m+1 \leq j \leq m+n+n$ and $ i-j = n$.
fong-spivak-invitation_F03510	308	■ 0.996	$m+1 \leq j \leq m+n+n$	$m+1 \leq j \leq m+n+n$	• $m+1 \leq i \leq m+n+n$ and $\underline{m+1 \leq j \leq m+n+n}$ and $ i-j = n$.
fong-spivak-invitation_F03511	308	■ 0.996	$ i-j =n$	$ i-j = n$	• $m+1 \leq i \leq m+n+n$ and $m+1 \leq j \leq m+n+n$ and $\underline{i-j} \equiv n$.
fong-spivak-invitation_F03512	308	■ 0.808	$\underline{m+n}$	$\underline{m+n}$	5. Given equivalence relations $\sim$ on $\underline{m+n}$ and $\sim'$ on $\underline{m'+n'}$, we need an equivalence rela-
fong-spivak-invitation_F03513	308	■ 0.808	$\sim^{\prime}$	$\sim'$	5. Given equivalence relations $\sim$ on $\underline{m+n}$ and $\sim'$ on $\underline{m'+n'}$, we need an equivalence rela-
fong-spivak-invitation_F03514	308	■ 0.808	$\underline{m^{\prime}+n^{\prime}}$	$\underline{m'+n'}$	5. Given equivalence relations $\sim$ on $\underline{m+n}$ and $\sim'$ on $\underline{m'+n'}$, we need an equivalence rela-
fong-spivak-invitation_F03515	308	■ 0.777	$\left(\sim+\sim^{\prime}\right)$	$(\sim + \sim')$	tion $(\sim + \sim')$ on $\underline{m+n+m'+n'}$. We take it to be “the obvious thing,” namely “using
fong-spivak-invitation_F03516	308	■ 0.777	$m+n+m^{\prime}+n^{\prime}$	$m+n+m'+n'$	tion $(\sim + \sim')$ on $\underline{m+n+m'+n'}$. We take it to be “the obvious thing,” namely “using
fong-spivak-invitation_F03517	308	■ 0.990	$\quad i \leq m+n$	$i \leq m+n$	• $i \leq m+n$ and $j \leq m+n$ and $i \sim j$, or
fong-spivak-invitation_F03518	308	■ 0.990	$j \leq m+n$	$j \leq m+n$	• $i \leq m+n$ and $j \leq m+n$ and $i \sim j$, or
fong-spivak-invitation_F03519	308	■ 0.793	$m+n+1 \leq i$	$m+n+1 \leq i$	• $m+n+1 \leq i$ and $m+n+1 \leq j$ and $i \sim' j$.
fong-spivak-invitation_F03520	308	■ 0.793	$m+n+1 \leq j$	$m+n+1 \leq j$	• $m+n+1 \leq i$ and $m+n+1 \leq j$ and $i \sim' j$.
fong-spivak-invitation_F03521	308	■ 0.793	$i \sim^{\prime} j$	$i \sim' j$	• $m+n+1 \leq i$ and $m+n+1 \leq j$ and $i \sim' j$.
fong-spivak-invitation_F03522	308	■ 0.998	$\boldsymbol{\operatorname{Rel}}(m, n)$	$\mathbf{Rel}(m, n)$	1. $\mathbf{Rel}(m, n)$ is the set of relations on the set $\underline{m} \times \underline{n}$, i.e. the set of subsets of $\underline{m} \times \underline{n}$, i.e. its

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fong-spivak-invitation_F03523	308	■ 0.998	$\underline{m} \times \underline{n}$	$\underline{m} \times \underline{n}$	1. $\mathbf{Rel}(m, n)$ is the set of relations on the set $\underline{m} \times \underline{n}$, i.e. the set of subsets of $\underline{m} \times \underline{n}$, i.e. its
fong-spivak-invitation_F03524	308	■ 1.000	$\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$	$\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$	2. The identity map $n \rightarrow n$ is the subset $\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$.
fong-spivak-invitation_F03525	308	■ 1.000	$(i, j) \in (\underline{m+n}) \times (\underline{n+m})$	$(i, j) \in (\underline{m+n}) \times (\underline{n+m})$	3. The symmetry map $m+n \rightarrow n+m$ is the subset of pairs $(i, j) \in (\underline{m+n}) \times (\underline{n+m})$ such
fong-spivak-invitation_F03526	308	■ 0.998	$i \leq m$	$i \leq m$	• $i \leq m$ and $m+1 \leq j$ and $i+m = j$, or
fong-spivak-invitation_F03527	308	■ 0.998	$m+1 \leq j$	$m+1 \leq j$	• $i \leq m$ and $m+1 \leq j$ and $i+m = j$, or
fong-spivak-invitation_F03528	308	■ 0.998	$i+m=j$	$i+m=j$	• $i \leq m$ and $m+1 \leq j$ and $i+m = j$, or
fong-spivak-invitation_F03529	308	■ 0.974	$m+1 \leq i$	$m+1 \leq i$	• $\underline{m+1} \leq$ and $j \leq m$ and $j+m = i$.
fong-spivak-invitation_F03530	308	■ 0.974	$j \leq m$	$j \leq m$	• $m+1 \leq i$ and $j \leq m$ and $j+m = i$.
fong-spivak-invitation_F03531	308	■ 0.974	$j+m=i$	$j+m=i$	• $m+1 \leq i$ and $j \leq m$ and $j+m = i$.
fong-spivak-invitation_F03532	308	■ 0.667	$R^{\prime} \subseteq \underline{m}^{\prime} \times \underline{n}^{\prime}$	$R' \subseteq \underline{m}' \times \underline{n}'$	5. Given a relation $R \subseteq \underline{m} \times \underline{n}$ and a relation $R' \subseteq \underline{m}' \times \underline{n}'$, we need a relation $(R + R') \subseteq$
fong-spivak-invitation_F03533	308	■ 0.667	$(R + R^{\prime}) \subseteq$	$(R + R') \subseteq$	5. Given a relation $R \subseteq \underline{m} \times \underline{n}$ and a relation $R' \subseteq \underline{m}' \times \underline{n}'$, we need a relation $(R + R') \subseteq$
fong-spivak-invitation_F03534	308	■ 1.000	$\underline{m+m^{\prime}} \times \underline{n+n^{\prime}}$	$\underline{m+m'} \times \underline{n+n'}$	$\underline{m+m'} \times \underline{n+n'}$. As stated in the example (footnote), this can be given by a universal
fong-spivak-invitation_F03535	308	■ 0.580	$R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$	$R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$	given by $\underline{R_1} \sqcup \underline{R_2} \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$.
fong-spivak-invitation_F03536	308	■ 0.818	(n, p)	(n, p)	Composition of an (m, n) -port graph G and an (n, p) -port graph H looks visually like sticking
fong-spivak-invitation_F03537	309	■ 1.000	$\leq_P$	$\leq_P$	We have a relation $R \subseteq P \times P$ which generates a preorder $\leq_P$ on P , we have an arbitrary preorder
fong-spivak-invitation_F03538	309	■ 0.999	$x \leq_P y$	$x \leq_P y$	monotone, i.e. that for every $x \leq_P y$ we have $f(x) \leq_Q f(y)$. By definition of P being the
fong-spivak-invitation_F03539	309	■ 0.993	$x_{\{0\}}, \ldots, x_{\{n\}}$	$x_0, \dots, x_n$	reflexive, transitive closure of R , we have $x \leq_P y$ iff there exists $n \in \mathbb{N}$ and $\underline{x_0, \dots, x_n}$ in P
fong-spivak-invitation_F03540	309	■ 1.000	$x_{\{0\}}=x$	$x_0 = x$	with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$. (The case $n = 0$ handles
fong-spivak-invitation_F03541	309	■ 1.000	$x_{\{n\}}=y$	$x_n = y$	with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$. (The case $n = 0$ handles
fong-spivak-invitation_F03542	309	■ 1.000	$R(x_{\{i\}}, x_{\{i+1\}})$	$R(x_i, x_{i+1})$	with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$. (The case $n = 0$ handles
fong-spivak-invitation_F03543	309	■ 1.000	$0 \leq i \leq n-1$	$0 \leq i \leq n-1$	with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $\underline{0 \leq i \leq n-1}$. (The case $n = 0$ handles
fong-spivak-invitation_F03544	309	■ 1.000	$f(x_{\{i\}}) \leq_Q f(x_{\{i+1\}})$	$f(x_i) \leq_Q f(x_{i+1})$	reflexivity.) But then, by assumption, $R(x_i, x_{i+1})$ implies $f(x_i) \leq_Q f(x_{i+1})$ for each i . By
fong-spivak-invitation_F03545	309	■ 1.000	$f(x_{\{0\}}) \leq_Q f(x_{\{i\}})$	$f(x_0) \leq_Q f(x_i)$	induction on i we show that $f(x_0) \leq_Q f(x_i)$ for all $0 \leq i \leq n$, at which point we are done.
fong-spivak-invitation_F03546	309	■ 0.725	$x \leq \leq_P y$	$x \leq \leq_P y$	2. Suppose now that f is monotone, and take $x, y \in P$ for which $R(x, y)$ holds. Then $x \leq_P y$
fong-spivak-invitation_F03547	309	■ 0.945	$n=1, x_{\{0\}}=x$	$n = 1, x_0 = x$	the above description is with $\underline{n=1, x_0=x}$, and $x_n = y$, which we said implies $x \leq_P y$.)
fong-spivak-invitation_F03548	309	■ 0.983	$R(g(a), g(b))$	$R(g(a), g(b))$	1. If $R(g(a), g(b))$ holds for all $a \leq_Q b$ then g is monotone, because $R(x, y)$ implies $x \leq_P y$.
fong-spivak-invitation_F03549	309	■ 0.983	$a \leq_Q b$	$a \leq_Q b$	1. If $R(g(a), g(b))$ holds for all $a \leq_Q b$ then g is monotone, because $R(x, y)$ implies $x \leq_P y$.

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fong-spivak-invitation_F03550	309	■ 1.000	$(g(a), g(b)) \notin R$	$(g(a), g(b)) \notin R$	$a \leq_Q b$ and $\boxed{g(a), g(b)} \notin R$. Indeed, take $Q := \{1\}$ to be the free preorder on one element,
fong-spivak-invitation_F03551	309	■ 1.000	$Q := \{1\}$	$Q := \{1\}$	$a \leq_Q b$ and $(g(a), g(b)) \notin R$. Indeed, take $Q := \{1\}$ to be the free preorder on one element,
fong-spivak-invitation_F03552	309	■ 1.000	$P := \{1\}$	$P := \{1\}$	and take $P := \{1\}$ with $R = \emptyset$. Then the unique function $g: Q \rightarrow P$ is monotone (because
fong-spivak-invitation_F03553	309	■ 1.000	$R = \text{\varnothing}$	$R = \emptyset$	and take $P := \{1\}$ with $R = \emptyset$. Then the unique function $g: Q \rightarrow P$ is monotone (because
fong-spivak-invitation_F03554	309	■ 1.000	$(g(1), g(1)) \notin R$	$(g(1), g(1)) \notin R$	$\leq_P$ is reflexive even though R is empty), and yet $\boxed{g(1), g(1)} \notin R$.
fong-spivak-invitation_F03555	309	■ 0.991	$q \in \operatorname{Mor}(\mathcal{C})$	$q \in \operatorname{Mor}(\mathcal{C})$	exactly one element of $\operatorname{Ob}(\mathcal{C})$. So for dom we take any $q \in \operatorname{Mor}(\mathcal{C})$, view it as a morphism
fong-spivak-invitation_F03556	309	■ 0.997	$q: y \rightarrow z$	$q: y \rightarrow z$	$\boxed{q: y \Rightarrow z}$, and send it to its domain y . Similarly for cod : we put $\operatorname{cod}(q) := z$.
fong-spivak-invitation_F03557	309	■ 0.997	$\operatorname{cod}(q) := z$	$\operatorname{cod}(q) := z$	$q: y \rightarrow z$, and send it to its domain y . Similarly for cod : we put $\operatorname{cod}(q) := z$.
fong-spivak-invitation_F03558	309	■ 1.000	$F: \mathcal{G} \rightarrow \mathcal{C}$	$F: \mathcal{G} \rightarrow \mathcal{C}$	2. Suppose first that we are given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$. On objects we have a function
fong-spivak-invitation_F03559	309	■ 0.990	$\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$	$\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$, and this defines f since $\operatorname{Ob}(\mathcal{G}) = V$. On morphisms, first note that
fong-spivak-invitation_F03560	309	■ 0.990	$\operatorname{Ob}(\mathcal{G}) = V$	$\operatorname{Ob}(\mathcal{G}) = V$	$\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$, and this defines f since $\operatorname{Ob}(\mathcal{G}) = V$. On morphisms, first note that
fong-spivak-invitation_F03561	309	■ 1.000	$\operatorname{Mor}(\mathcal{G})$	$\operatorname{Mor}(\mathcal{G})$	the arrows of graph G are exactly the length=1 paths in G , whereas $\operatorname{Mor}(\mathcal{G})$ is the set of
fong-spivak-invitation_F03562	309	■ 1.000	$A \subseteq \operatorname{Mor}(\mathcal{G})$	$A \subseteq \operatorname{Mor}(\mathcal{G})$	all paths in G , so we have an inclusion $A \subseteq \operatorname{Mor}(\mathcal{G})$. The functor F provides a function
fong-spivak-invitation_F03563	310	■ 1.000	$\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$, which we can restrict to A to obtain $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$. All functors satisfy
fong-spivak-invitation_F03564	310	■ 1.000	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$ for any $r: w \rightarrow x$. In particular when
fong-spivak-invitation_F03565	310	■ 1.000	$\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$	$\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$ for any $r: w \rightarrow x$. In particular when
fong-spivak-invitation_F03566	310	■ 1.000	$r: w \rightarrow x$	$r: w \rightarrow x$	$\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$ for any $r: w \rightarrow x$. In particular when
fong-spivak-invitation_F03567	310	■ 0.976	$r \in A$	$r \in A$	$r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$. Thus we have found (f, g) with
fong-spivak-invitation_F03568	310	■ 0.976	$\operatorname{dom}(r) = s(r)$	$\operatorname{dom}(r) = s(r)$	$r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$. Thus we have found (f, g) with
fong-spivak-invitation_F03569	310	■ 0.976	$\operatorname{cod}(r) = t(r)$	$\operatorname{cod}(r) = t(r)$	$r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$. Thus we have found (f, g) with
fong-spivak-invitation_F03570	310	■ 1.000	$p := (v_0, a_1, a_2, \dots, a_n)$	$p := (v_0, a_1, a_2, \dots, a_n)$	$\boxed{p := (v_0, a_1, a_2, \dots, a_n)}$ in G , where $v_0 \in V$, $a_i \in A$, $v_0 = s(a_1)$, and $t(a_i) = s(a_{i+1})$
fong-spivak-invitation_F03571	310	■ 1.000	$v_0 \in V, a_i \in A, v_0 = s(a_1)$	$v_0 \in V, a_i \in A, v_0 = s(a_1)$	$p := (v_0, a_1, a_2, \dots, a_n)$ in G , where $\boxed{v_0 \in V, a_i \in A, v_0 = s(a_1)}$ and $t(a_i) = s(a_{i+1})$
fong-spivak-invitation_F03572	310	■ 1.000	$1 \leq i \leq n-1$	$1 \leq i \leq n-1$	for all $1 \leq i \leq n-1$. Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose domain is $f(v_0)$ and the
fong-spivak-invitation_F03573	310	■ 1.000	$g(a_i)$	$g(a_i)$	for all $1 \leq i \leq n-1$. Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose domain is $f(v_0)$ and the
fong-spivak-invitation_F03574	310	■ 1.000	$f(v_0)$	$f(v_0)$	for all $1 \leq i \leq n-1$. Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose domain is $f(v_0)$ and the
fong-spivak-invitation_F03575	310	■ 1.000	$g(a_{i+1})$	$g(a_{i+1})$	morphisms $g(a_i)$ and $g(a_{i+1})$ are composable for every $1 \leq i \leq n-1$. We then take
fong-spivak-invitation_F03576	310	■ 0.442	$F(p) := \operatorname{id}_{f(v_0)} ; g(a_1) ; \dots ; g(a_n)$	$F(p) := \operatorname{id}_{f(v_0)} ; g(a_1) ; \dots ; g(a_n)$	$F(p) := \operatorname{id}_{f(v_0)} \circ g(a_1) \circ \dots \circ g(a_n)$ to be the composite. It is easy to check that this is
fong-spivak-invitation_F03577	310	■ 1.000	(f', g')	(f', g')	turn them into a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ and then extract the new pair of functions $\boxed{f', g'}$, then

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fong-spivak-invitation_F03578	310	■ 0.985	$f=f^{\prime}$	$f = f'$	$f = f'$ and $g = g'$. Finally, given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$, we extract the pair of functions (f, g)
fong-spivak-invitation_F03579	310	■ 0.985	$g=g^{\prime}$	$g = g'$	$f = f'$ and $g = g'$. Finally, given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$, we extract the pair of functions (f, g)
fong-spivak-invitation_F03580	310	■ 1.000	$F^{\prime}: \mathcal{G} \rightarrow \mathcal{C}$	$F' : \mathcal{G} \rightarrow \mathcal{C}$	as above and then turn them into a new functor $F': \mathcal{G} \rightarrow \mathcal{C}$. It is clear that F and F' act the
fong-spivak-invitation_F03581	310	■ 1.000	$F^{\prime}$	F'	as above and then turn them into a new functor $F': \mathcal{G} \rightarrow \mathcal{C}$. It is clear that F and F' act the
fong-spivak-invitation_F03582	310	■ 0.366	$\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C})$	$\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C})$	3. $(\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C}), \operatorname{dom}, \operatorname{cod})$ is a graph; let's denote it $U(\mathcal{C}) \in \mathbf{Grph}$. We have functors
fong-spivak-invitation_F03583	310	■ 0.366	$U(\mathcal{C}) \in$	$U(\mathcal{C}) \in$	3. $(\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C}), \operatorname{dom}, \operatorname{cod})$ is a graph; let's denote it $U(\mathcal{C}) \in \mathbf{Grph}$. We have functors
fong-spivak-invitation_F03584	310	■ 1.000	$a^i * a^j = a^{i+j}$	$a^i * a^j = a^{i+j}$	$a^i * a^j = a^{i+j}$.
fong-spivak-invitation_F03585	310	■ 1.000	$a^i \mapsto i$	$a^i \mapsto i$	2. This is isomorphic to $\mathbb{N}$, by sending $a^i \mapsto i$.
fong-spivak-invitation_F03586	310	■ 0.600	$\{a, b\}$	$\{a, b\}$	3. The elements of the free monoid on the set $\{a, b\}$ are 'words in a and b ,' each of which we will
fong-spivak-invitation_F03587	310	■ 0.714	$\operatorname{Free}(G, s, t)$	$\mathbf{Free}(G, s, t)$	We have two props: the prop of port graphs and the free prop $\mathbf{Free}(G, s, t)$ where
fong-spivak-invitation_F03588	310	■ 0.698	(Γ, ℓ)	(Γ, ℓ)	(Γ, ℓ) , where $\Gamma = (V, in, out, \iota)$ is an (m, n) -port graph and $\ell: V \rightarrow G$ is a function, such
fong-spivak-invitation_F03589	311	■ 1.000	$\ell(v) \in G$	$\ell(v) \in G$	box with some left-hand ports and some right-hand ports – an arity – and $\ell(v) \in G$ is sup-
fong-spivak-invitation_F03590	311	■ 0.512	Γ	Γ	words, a morphism in our particular $\mathbf{Free}(G)$ can be identified with an (m, n) -port graph Γ ,
fong-spivak-invitation_F03591	311	■ 0.186	$(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{9} (\sigma + \operatorname{id}_1) \stackrel{\circ}{9} (\operatorname{id}_1 + h) \stackrel{\circ}{\sigma} \circ g$	$(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{9} (\sigma + \operatorname{id}_1) \stackrel{\circ}{9} (\operatorname{id}_1 + h) \stackrel{\circ}{\sigma} \circ g$	Here is a picture of $(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{9} (\sigma + \operatorname{id}_1) \stackrel{\circ}{9} (\operatorname{id}_1 + h) \stackrel{\circ}{\sigma} \circ g$, in the free prop on generators
fong-spivak-invitation_F03592	311	■ 0.990	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$	$G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$
fong-spivak-invitation_F03593	311	■ 0.999	$\{1\}, \{2\}, \{3\}$	$\{1\}, \{2\}, \{3\}$	is the set $\{1\}, \{2\}, \{3\}$. It is subtly different from S , but the two are naturally isomorphic, and
fong-spivak-invitation_F03594	311	■ 0.554	$(R, 0, +, 1, *)$	$(R, 0, +, 1, *)$	1. If $(R, 0, +, 1, *)$ is a rig, then the multiplicative identity $1 \in \operatorname{Mat}_n(R)$ is the usual n -by- n
fong-spivak-invitation_F03595	311	■ 1.000	$n=4$	$n = 4$	those elements of R). So for $n = 4$ it is:
fong-spivak-invitation_F03596	311	■ 1.000	$A, B \in \operatorname{Mat}_2(\mathbb{N})$	$A, B \in \operatorname{Mat}_2(\mathbb{N})$	2. We choose $n = 2$ and hence need to find two elements $A, B \in \operatorname{Mat}_2(\mathbb{N})$ such that $A * B \neq$
fong-spivak-invitation_F03597	311	■ 1.000	$A * B \neq$	$A * B \neq$	2. We choose $n = 2$ and hence need to find two elements $A, B \in \operatorname{Mat}_2(\mathbb{N})$ such that $A * B \neq$
fong-spivak-invitation_F03598	311	■ 1.000	$B * A$	$B * A$	$B * A$.
fong-spivak-invitation_F03599	311	■ 0.937	$(A * B)(1, 1) = 0 * 0 + 1 * 1 = 1$	$(A * B)(1, 1) = 0 * 0 + 1 * 1 = 1$	One can calculate from the multiplication formula (recalled in Example 5.35) that $(A * B)(1, 1) = 0 * 0 + 1 * 1 = 1$ and $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$, which are not
fong-spivak-invitation_F03600	311	■ 1.000	$(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$	$(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$	$(B * A)(1, 1) = 0 * 0 + 1 * 1 = 1$ and $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$, which are not
fong-spivak-invitation_F03601	311	■ 1.000	$(16x + 4y, x + 4y)$	$(16x + 4y, x + 4y)$	the resulting output signal is $(16x + 4y, x + 4y)$.
fong-spivak-invitation_F03602	312	■ 0.997	$(16x + 4y, x + 4y)$	$(16x + 4y, x + 4y)$	
fong-spivak-invitation_F03603	312	■ 0.673	$A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$	$A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$	The monoidal product of $A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 5 & 6 & 1 \end{pmatrix}$ is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_F03604	312	0.673	$B=\left(\begin{array}{llll}2 & 5 & 6 & 1\end{array}\right)$	$B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right)$	The monoidal product of $A = \left(\begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right)$ and $B = \left(\begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right)$ is
fong-spivak-invitation_F03605	313	1.000	$c_{\{n\}}$	c_n	• For the first layer g_1 , take the monoidal product of m copies of c_n ,
fong-spivak-invitation_F03606	313	1.000	$s_{\{a\}}: 1 \rightarrow 1$	$s_a: 1 \rightarrow 1$	where $s_a: 1 \rightarrow 1$ is the signal flow graph generator “scalar multiplication by a .” This layer
fong-spivak-invitation_F03607	313	1.000	$g_{\{3\}}: m \times n \rightarrow m \times n$	$g_3: m \times n \rightarrow m \times n$	signal flow graph $g_3: m \times n \rightarrow m \times n$, i.e. the composite and monoidal product of swap and
fong-spivak-invitation_F03608	313	1.000	$(i-1) m+j$	$(i-1)m+j$	identity maps, such that the $(i-1)m+j$ th input is sent to the $(j-1)n+i$ th output, for all
fong-spivak-invitation_F03609	313	1.000	$(j-1) n+i$	$(j-1)n+i$	identity maps, such that the $(i-1)m+j$ th input is sent to the $(j-1)n+i$ th output, for all
fong-spivak-invitation_F03610	313	1.000	$1 \leq j \leq m$	$1 \leq j \leq m$	$1 \leq i \leq n$ and $1 \leq j \leq m$.
fong-spivak-invitation_F03611	313	0.999	$a_{\{m\}}$	a_m	where a_m is the signal flow graph that adds m inputs to produce a single output:
fong-spivak-invitation_F03612	313	0.528	$g=g_{\{1\}} \, g_{\{2\}}$	$g = g_1 g_2$	Using Proposition 5.46, it is a straightforward but tedious calculation to show that $g = g_1 \circ g_2 \circ$
fong-spivak-invitation_F03613	313	0.735	$g_{\{3\}}$	g_3	$g_3 \circ g_4: m \rightarrow n$ has the property that $S(g) = M$.
fong-spivak-invitation_F03614	313	0.735	$g_{\{4\}}: m \rightarrow n$	$g_4: m \rightarrow n$	$g_3 \circ g_4: m \rightarrow n$ has the property that $S(g) = M$.
fong-spivak-invitation_F03615	315	0.723	$(\mu \otimes \mathrm{id}) ; \mu = (\mathrm{id} \otimes \mu) \circ \mu$ $\stackrel{\circ}{\circ} \mu$	$(\mu \otimes \mathrm{id}); \mu = (\mathrm{id} \otimes \mu) \circ \mu$	(a) $(\mu \otimes \mathrm{id}) \circ \mu = (\mathrm{id} \otimes \mu) \circ \mu$
fong-spivak-invitation_F03616	315	1.000	$R^{\{0\}}=\{()\}$	$R^0 = \{()\}$	R . In particular, $R^0 = \{()\}$ consists of a single vector of dimension 0.
fong-spivak-invitation_F03617	315	0.549	$\operatorname{prop}(\operatorname{Mat}(R))$	$\operatorname{prop}(\operatorname{Mat}(R))$	1. U preserves the monoidal unit because 0 is the monoidal unit of any prop ($\mathbf{Mat}(R)$ is a prop),
fong-spivak-invitation_F03618	315	0.927	$R^{\{0\}}$	R^0	$\{1\}$ is the monoidal unit of $\mathbf{Set}$, and R^0 is canonically isomorphic to $\{1\}$. U also preserves the
fong-spivak-invitation_F03619	315	0.927	$\{1\} \cdot U$	$\{1\}.U$	$\{1\}$ is the monoidal unit of $\mathbf{Set}$, and R^0 is canonically isomorphic to $\{1\}$. U also preserves the
fong-spivak-invitation_F03620	315	1.000	$R^{\{m\}} \times R^{\{n\}} \cong R^{\{m+n\}}$	$R^m \times R^n \cong R^{m+n}$	monoidal product because there is a canonical isomorphism $R^m \times R^n \cong R^{m+n}$.
fong-spivak-invitation_F03621	315	0.876	(m, μ, η)	(m, μ, η)	2. A monoid object in $\mathbf{Mat}(R)$ is a tuple (m, μ, η) where $m \in \mathbb{N}$, $\mu: m+m \rightarrow m$, and $\eta: 0 \rightarrow m$
fong-spivak-invitation_F03622	315	0.876	$m \in \mathbb{N}, \mu: m+m \rightarrow m$	$m \in \mathbb{N}, \mu: m+m \rightarrow m$	2. A monoid object in $\mathbf{Mat}(R)$ is a tuple (m, μ, η) where $m \in \mathbb{N}$, $\mu: m+m \rightarrow m$, and $\eta: 0 \rightarrow m$
fong-spivak-invitation_F03623	315	0.876	$\eta: 0 \rightarrow m$	$\eta: 0 \rightarrow m$	2. A monoid object in $\mathbf{Mat}(R)$ is a tuple (m, μ, η) where $m \in \mathbb{N}$, $\mu: m+m \rightarrow m$, and $\eta: 0 \rightarrow m$
fong-spivak-invitation_F03624	315	0.896	$\mu(\eta, x)=x=\mu(x, \eta)$	$\mu(\eta, x) = x = \mu(x, \eta)$	satisfy the properties $\mu(\eta, x) = x = \mu(x, \eta)$ and $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$. Note that
fong-spivak-invitation_F03625	315	0.896	$\mu(x, \mu(y, z))=\mu(\mu(x, y), z)$	$\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$	satisfy the properties $\mu(\eta, x) = x = \mu(x, \eta)$ and $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$. Note that
fong-spivak-invitation_F03626	315	1.000	$0 \rightarrow m$	$0 \rightarrow m$	there is only one morphism $0 \rightarrow m$ in $\mathbf{Mat}(R)$ for any m . It is not hard to show that for any
fong-spivak-invitation_F03627	315	1.000	$m \in \mathbb{N}$	$m \in \mathbb{N}$	$m \in \mathbb{N}$ there is only one monoid structure. For example, when $m = 2$, μ must be the following
fong-spivak-invitation_F03628	315	1.000	$m=2, \mu$	$m = 2, \mu$	$m \in \mathbb{N}$ there is only one monoid structure. For example, when $m = 2$, μ must be the following
fong-spivak-invitation_F03629	315	1.000	$U(\eta): R^{\{0\}} \rightarrow R^{\{m\}}$	$U(\eta): R^0 \rightarrow R^m$	Anyway, for any monoid (m, μ, η) , the morphism $U(\eta): R^0 \rightarrow R^m$ is given by $U(\eta)(1) :=$
fong-spivak-invitation_F03630	315	1.000	$U(\eta)(1) :=$	$U(\eta)(1) :=$	Anyway, for any monoid (m, μ, η) , the morphism $U(\eta): R^0 \rightarrow R^m$ is given by $U(\eta)(1) :=$
fong-spivak-invitation_F03631	315	0.508	$(0, \ldots, 0)$	$(0, \dots, 0)$	$(0, \dots, 0)$, and the morphism $U(\mu): R^m \times R^m \rightarrow R^m$ is given by

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fong-spivak-invitation-F03632	315	0.508	$U(\mu): R^{\{m\}} \times R^{\{m\}} \rightarrow R^{\{m\}}$	$U(\mu): R^m \times R^m \rightarrow R^m$	$(0, \dots, 0)$, and the morphism $U(\mu): R^m \times R^m \rightarrow R^m$ is given by
fong-spivak-invitation-F03633	315	1.000	$R^{\{m\}}$	R^m	These give R^m the structure of a monoid.
fong-spivak-invitation-F03634	315	0.510	$(1, \circ, -c)$	$(1, \circ, -c)$	3. The triple $(1, \circ, -c)$ corresponds to the additive monoid structure on $\mathbb{R}$, e.g. with $(5, 3) \mapsto 8$.
fong-spivak-invitation-F03635	315	0.510	$(5, 3) \mapsto 8$	$(5, 3) \mapsto 8$	3. The triple $(1, \circ, -c)$ corresponds to the additive monoid structure on $\mathbb{R}$, e.g. with $(5, 3) \mapsto 8$.
fong-spivak-invitation-F03636	316	0.732	$\mathrm{B}(-<)$	$B(-<)$	1. The behavior $B(-<)$ of the reversed addition icon $-<: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid$
fong-spivak-invitation-F03637	316	0.732	$-<: 1 \rightarrow 2$	$-<: 1 \rightarrow 2$	1. The behavior $B(-<)$ of the reversed addition icon $-<: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid$
fong-spivak-invitation-F03638	316	0.732	$\left\{(x, y, z) \in R^{\{3\}} \mid \text{mid} \right\}$	$\{(x, y, z) \in R^3 \mid$	1. The behavior $B(-<)$ of the reversed addition icon $-<: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid$
fong-spivak-invitation-F03639	316	0.982	$x=y+z\}$	$x = y + z\}$	$x = y + z\}$.
fong-spivak-invitation-F03640	316	0.141	$\mathrm{B}(\supset^{\{-\}})$	$B(\supset^-)$	2. The behavior $B(\supset^-)$ of the reversed copy icon, $\supset^-: 2 \rightarrow 1$ is the relation $\{(x, y, z) \in R^3 \mid$
fong-spivak-invitation-F03641	316	0.141	$\exists^{\{-\}}: 2 \rightarrow 1$	$\exists^-: 2 \rightarrow 1$	2. The behavior $B(\supset^-)$ of the reversed copy icon, $\supset^-: 2 \rightarrow 1$ is the relation $\{(x, y, z) \in R^3 \mid$
fong-spivak-invitation-F03642	316	0.968	$x=y=z\}$	$x = y = z\}$	$x = y = z\}$.
fong-spivak-invitation-F03643	316	1.000	$B+C \subseteq R^{\{m+p\}} \times R^{\{n+q\}}$	$B + C \subseteq R^{m+p} \times R^{n+q}$	If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$, then take $B + C \subseteq R^{m+p} \times R^{n+q}$
fong-spivak-invitation-F03644	316	0.410	$\mathrm{B}(\leftarrow g \rightarrow)$	$B(g$	respectively, and, by Eq. (5.13), the composite $B(g \circ (h^{\mathrm{op}})) = B(g) \circ B(h^{\mathrm{op}})$ is:
fong-spivak-invitation-F03645	316	0.410	$\leftarrow \leftarrow (h^{\mathrm{op}} \rightarrow) \rightarrow = \mathrm{B}(g) ; \rightarrow ; \leftarrow \leftarrow (h^{\mathrm{op}} \rightarrow) \rightarrow$	$(h^{\mathrm{op}})) = B(g); B(h^{\mathrm{op}})$	respectively, and, by Eq. (5.13), the composite $B(g \circ (h^{\mathrm{op}})) = B(g) \circ B(h^{\mathrm{op}})$ is:
fong-spivak-invitation-F03646	316	1.000	$S(h)$	$S(h)$	Since $S(g)$ and $S(h)$ are functions, the above immediately reduces to the desired formula:
fong-spivak-invitation-F03647	316	0.981	$h: m \rightarrow p$	$h: m \rightarrow p$	The behavior of $g^{\mathrm{op}}: n \rightarrow m$ and $h: m \rightarrow p$ are
fong-spivak-invitation-F03648	316	0.998	$\mathrm{B}(\leftarrow (g^{\mathrm{op}} \rightarrow) ; h \rightarrow) = \mathrm{B}(\leftarrow (g^{\mathrm{op}} \rightarrow) ; \rightarrow) ; \mathrm{B}(h)$	$B((g^{\mathrm{op}}); h) = B(g^{\mathrm{op}}); B(h)$	respectively, and, by Eq. (5.13), the composite $B((g^{\mathrm{op}}) \circ h) = B(g^{\mathrm{op}}) \circ B(h)$ is:
fong-spivak-invitation-F03649	316	0.555	$\{y \in R \mid y=0\}$	$\{y \in R \mid y = 0\}$	1. The behavior of the 0-reverse $\dashv$ is the subset $\{y \in R \mid y = 0\}$, and its n -fold tensor is similarly
fong-spivak-invitation-F03650	316	1.000	$\left\{y \in R^{\{n\}} \mid y=0 \right\}$	$\{y \in R^n \mid y = 0\}$	$\{y \in R^n \mid y = 0\}$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$,
fong-spivak-invitation-F03651	316	1.000	$S(g) \subseteq R^{\{m\}} \times R^{\{n\}}$	$S(g) \subseteq R^m \times R^n$	$\{y \in R^n \mid y = 0\}$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$,
fong-spivak-invitation-F03652	316	1.000	$\left\{x \in R^{\{m\}} \mid S(g)=0 \right\}$	$\{x \in R^m \mid S(g) = 0\}$	$\{y \in R^n \mid y = 0\}$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$,
fong-spivak-invitation-F03653	316	0.868	$\{x \in R\}$	$\{x \in R\}$	2. The behavior of the discard-inverse $\leftarrow$ is the subset $\{x \in R\}$, i.e. the largest subset of R , and
fong-spivak-invitation-F03654	316	1.000	$R^{\{n\}} \subseteq R^{\{n\}}$	$R^n \subseteq R^n$	similarly its m -fold tensor is $R^n \subseteq R^n$. Composing this relation with $S(g) \subseteq R^m \times R^n$ gives
fong-spivak-invitation-F03655	316	0.999	$\left\{y \in R^{\{n\}} \mid \text{mid} \right\}$	$\{y \in R^n \mid$	$\{y \in R^n \mid$ there exists $x \in R^m$ such that $S(g)(x) = y\}$, which is exactly the image of $S(g)$.
fong-spivak-invitation-F03656	316	0.999	$\leftarrow S(g)(x)=y \rightarrow$	$S(g)(x) = y\}$	$\{y \in R^n \mid$ there exists $x \in R^m$ such that $S(g)(x) = y\}$, which is exactly the image of $S(g)$.
fong-spivak-invitation-F03657	317	0.552	$\mathrm{B}(g)=\{(x, y) \mid S(g)(x)=y\}$	$B(g) = \{(x, y) \mid S(g)(x) = y\}$	3. For any $g: m \rightarrow n$, we first claim that the behavior $B(g) = \{(x, y) \mid S(g)(x) = y\}$ is linear, i.e.
fong-spivak-invitation-F03658	317	1.000	$S(g)(x)=y$	$S(g)(x) = y$	so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$. Thus we
fong-spivak-invitation-F03659	317	1.000	$S(g)(r\ x)=r\ y$	$S(g)(rx) = ry$	so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$. Thus we

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fong-spivak-invitation_F03660	317	■ 1.000	$S(g)\left(x_{\{1\}}+x_{\{2\}}\right)=S(g)\left(x_{\{1\}}\right)+S(g)\left(x_{\{2\}}\right)$	$S(g)\left(x_1+x_2\right)=S(g)\left(x_1\right)+S(g)\left(x_2\right)$	so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1+x_2) = S(g)(x_1) + S(g)(x_2)$. Thus we
fong-spivak-invitation_F03661	317	■ 1.000	$(x, y) \in \mathrm{B}(g)$	$(x, y) \in \mathrm{B}(g)$	conclude that $(x, y) \in \mathrm{B}(g)$ implies $(rx, ry) \in \mathrm{B}(g)$, so it's closed under scalar multiplication,
fong-spivak-invitation_F03662	317	■ 1.000	$(r x, r y) \in \mathrm{B}(g)$	$(rx, ry) \in \mathrm{B}(g)$	conclude that $(x, y) \in \mathrm{B}(g)$ implies $(rx, ry) \in \mathrm{B}(g)$, so it's closed under scalar multiplication,
fong-spivak-invitation_F03663	317	■ 1.000	$\left(x_{\{1\}}, y_{\{1\}}\right), \left(x_{\{2\}}, y_{\{2\}}\right) \in \mathrm{B}(g)$	$\left(x_1, y_1\right), \left(x_2, y_2\right) \in \mathrm{B}(g)$	and $(x_1, y_1), (x_2, y_2) \in \mathrm{B}(g)$ implies $(x_1+x_2, y_1+y_2) \in \mathrm{B}(g)$, so it's closed under addition.
fong-spivak-invitation_F03664	317	■ 1.000	$\left(x_{\{1\}}+x_{\{2\}}, y_{\{1\}}+y_{\{2\}}\right) \in \mathrm{B}(g)$	$\left(x_1+x_2, y_1+y_2\right) \in \mathrm{B}(g)$	and $(x_1, y_1), (x_2, y_2) \in \mathrm{B}(g)$ implies $(x_1+x_2, y_1+y_2) \in \mathrm{B}(g)$, so it's closed under addition.
fong-spivak-invitation_F03665	317	■ 0.999	$\mathrm{B}\left(g^{\mathrm{op}}\right)$	$\mathrm{B}\left(g^{\mathrm{op}}\right)$	Similarly, the behavior $\mathrm{B}(g^{\mathrm{op}})$ is also linear; the proof is similar.
fong-spivak-invitation_F03666	317	■ 0.386	$C \subseteq R^{\{n\}} \times R^{\{p\}}$	$C \subseteq R^m \times R^p$	$B \subseteq R^m \times R^n$ and $C \subseteq R^n \times R^p$ are linear. Take $(x_1, z_1), (x_2, z_2) \in B \circ C$ and take $r \in R$. By
fong-spivak-invitation_F03667	317	■ 0.386	$\left(x_{\{1\}}, z_{\{1\}}\right), \left(x_{\{2\}}, z_{\{2\}}\right) \in B \circ C$	$\left(x_1, z_1\right), \left(x_2, z_2\right) \in B \circ C$	$B \subseteq R^m \times R^n$ and $C \subseteq R^n \times R^p$ are linear. Take $(x_1, z_1), (x_2, z_2) \in B \circ C$ and take $r \in R$. By
fong-spivak-invitation_F03668	317	■ 0.999	$y_{\{1\}}, y_{\{2\}} \in R^{\{n\}}$	$y_1, y_2 \in R^n$	definition, there exist $y_1, y_2 \in R^n$ such that $(x_1, y_1), (x_2, y_2) \in B$ and $(y_1, z_1), (y_2, z_2) \in C$.
fong-spivak-invitation_F03669	317	■ 0.999	$\left(x_{\{1\}}, y_{\{1\}}\right), \left(x_{\{2\}}, y_{\{2\}}\right) \in B$	$\left(x_1, y_1\right), \left(x_2, y_2\right) \in B$	definition, there exist $y_1, y_2 \in R^n$ such that $(x_1, y_1), (x_2, y_2) \in B$ and $(y_1, z_1), (y_2, z_2) \in C$.
fong-spivak-invitation_F03670	317	■ 0.999	$\left(y_{\{1\}}, z_{\{1\}}\right), \left(y_{\{2\}}, z_{\{2\}}\right) \in C$	$\left(y_1, z_1\right), \left(y_2, z_2\right) \in C$	definition, there exist $y_1, y_2 \in R^n$ such that $(x_1, y_1), (x_2, y_2) \in B$ and $(y_1, z_1), (y_2, z_2) \in C$.
fong-spivak-invitation_F03671	317	■ 1.000	$\left(r x_{\{1\}}, r y_{\{1\}}\right) \in B$	$\left(r x_1, r y_1\right) \in B$	Since B and C are linear, $(rx_1, ry_1) \in B$ and $(ry_1, rz_1) \in C$, and also $(x_1+x_2, y_1+y_2) \in B$
fong-spivak-invitation_F03672	317	■ 1.000	$\left(r y_{\{1\}}, r z_{\{1\}}\right) \in C$	$\left(r y_1, r z_1\right) \in C$	Since B and C are linear, $(rx_1, ry_1) \in B$ and $(ry_1, rz_1) \in C$, and also $(x_1+x_2, y_1+y_2) \in B$
fong-spivak-invitation_F03673	317	■ 1.000	$\left(x_{\{1\}}+x_{\{2\}}, y_{\{1\}}+y_{\{2\}}\right) \in B$	$\left(x_1+x_2, y_1+y_2\right) \in B$	Since B and C are linear, $(rx_1, ry_1) \in B$ and $(ry_1, rz_1) \in C$, and also $\boxed{x_1+x_2, y_1+y_2} \in B$
fong-spivak-invitation_F03674	317	■ 0.328	$\left(y_{\{1\}}+y_{\{2\}}, z_{\{1\}}+z_{\{2\}}\right) \in C$	$\left(y_1+y_2, z_1+z_2\right) \in C$	and $(y_1+y_2, z_1+z_2) \in C$. Hence $(rx_1, rz_1) \in (B \circ C)$ and $(x_1+x_2, z_1+z_2) \in (B \circ C)$, as
fong-spivak-invitation_F03675	317	■ 0.328	$\left(r x_{\{1\}}, r z_{\{1\}}\right) \in (B ; C)$	$\left(r x_1, r z_1\right) \in (B ; C)$	and $(y_1+y_2, z_1+z_2) \in C$. Hence $\boxed{rx_1, rz_1} \in (B \circ C)$ and $(x_1+x_2, z_1+z_2) \in (B \circ C)$, as
fong-spivak-invitation_F03676	317	■ 0.328	$\left(x_{\{1\}}+x_{\{2\}}, z_{\{1\}}+z_{\{2\}}\right) \in (B \circ C)$	$\left(x_1+x_2, z_1+z_2\right) \in (B \circ C)$	and $(y_1+y_2, z_1+z_2) \in C$. Hence $(rx_1, rz_1) \in (B \circ C)$ and $(x_1+x_2, z_1+z_2) \in (B \circ C)$, as
fong-spivak-invitation_F03677	317	■ 0.902	$(B ; C) \subseteq R^{\{m\}} \times R^{\{p\}}$	$(B ; C) \subseteq R^m \times R^p$	$(B \circ C) \subseteq R^m \times R^p$ consisting of all (x, z) for which there exists $y \in R^n$ with $(x, y) \in B$ and
fong-spivak-invitation_F03678	317	■ 0.902	(x, z)	(x, z)	$(B \circ C) \subseteq R^m \times R^p$ consisting of all (x, z) for which there exists $y \in R^n$ with $(x, y) \in B$ and
fong-spivak-invitation_F03679	317	■ 0.929	$(y, z) \in C$	$(y, z) \in C$	$(y, z) \in C$. We want to show that the set $(B \circ C)$ is linear, i.e. closed under scalar multiplication and
fong-spivak-invitation_F03680	317	■ 0.929	$(B \circ C)$	$(B \circ C)$	$(y, z) \in C$. We want to show that the set $(B \circ C)$ is linear, i.e. closed under scalar multiplication and
fong-spivak-invitation_F03681	317	■ 0.956	$(x, z) \in (B ; C)$	$(x, z) \in (B ; C)$	For scalar multiplication, take an $(x, z) \in (B \circ C)$ and any $r \in R$. Since B is linear, we have
fong-spivak-invitation_F03682	317	■ 0.905	$(r * x, r * y) \in B$	$(r * x, r * y) \in B$	$(r * x, r * y) \in B$ and since C is linear we have $(r * y, r * z) \in C$, so then $(r * x, r * z) \in (B \circ C)$.
fong-spivak-invitation_F03683	317	■ 0.905	$(r * y, r * z) \in C$	$(r * y, r * z) \in C$	$(r * x, r * y) \in B$ and since C is linear we have $(r * y, r * z) \in C$, so then $(r * x, r * z) \in (B \circ C)$.
fong-spivak-invitation_F03684	317	■ 0.905	$(r * x, r * z) \in (B \circ C)$	$(r * x, r * z) \in (B \circ C)$	$(r * x, r * y) \in B$ and since C is linear we have $(r * y, r * z) \in C$, so then $(r * x, r * z) \in (B \circ C)$.
fong-spivak-invitation_F03685	317	■ 1.000	$\left(x^{\prime}, z^{\prime}\right) \in (B ; C)$	$\left(x', z'\right) \in (B ; C)$	For addition, if we also have $(x', z') \in (B \circ C)$ then there is some $y' \in R^n$ with $(x', y') \in B$ and
fong-spivak-invitation_F03686	317	■ 1.000	$y^{\prime} \in R^{\{n\}}$	$y' \in R^n$	For addition, if we also have $(x', z') \in (B \circ C)$ then there is some $y' \in R^n$ with $(x', y') \in B$ and
fong-spivak-invitation_F03687	317	■ 0.999	$\left(y^{\prime}, z^{\prime}\right) \in C$	$\left(y', z'\right) \in C$	$\boxed{y', z'} \in C$, so since B and C are linear we have $(x+x', y+y') \in B$ and $(y+y', z+z') \in C$,

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fong-spivak-invitation_F03688	317	■ 0.999	$\text{\textbackslash left}(x+x^{\text{\textbackslash prime}}, y+y^{\text{\textbackslash prime}}\text{\textbackslash right}) \text{\textbackslash in } B$	$(x + x', y + y') \in B$	$(y', z') \in C$, so since B and C are linear we have $(x + x', y + y') \in B$ and $\boxed{y + y', z + z'} \in C$.
fong-spivak-invitation_F03689	317	■ 0.999	$\text{\textbackslash left}(y+y^{\text{\textbackslash prime}}, z+z^{\text{\textbackslash prime}}\text{\textbackslash right}) \text{\textbackslash in } C$	$(y + y', z + z') \in C$	$(y', z') \in C$, so since B and C are linear we have $(x + x', y + y') \in B$ and $\boxed{y + y', z + z'} \in C$.
fong-spivak-invitation_F03690	317	■ 0.918	$\text{\textbackslash left}(x+x^{\text{\textbackslash prime}}, z+z^{\text{\textbackslash prime}}\text{\textbackslash right}) \text{\textbackslash in}(B ; C)$	$(x + x', z + z') \in (B; C)$	hence $\boxed{x + x', z + z'} \in (B \circ C)$.
fong-spivak-invitation_F03691	317	■ 1.000	$a \text{\textbackslash not } \text{\textbackslash leq } b$	$a \not\leq b$	1. The left-most (the discrete preorder on A) has no initial object, because $a \not\leq b$ and $b \not\leq a$.
fong-spivak-invitation_F03692	317	■ 1.000	$b \text{\textbackslash not } \text{\textbackslash leq } a$	$b \not\leq a$	1. The left-most (the discrete preorder on A) has no initial object, because $a \not\leq b$ and $b \not\leq a$.
fong-spivak-invitation_F03693	317	■ 0.999	$f\text{\textbackslash left}(1_{\text{\textbackslash R}}\text{\textbackslash right})=1_{\text{\textbackslash S}}$	$f(1_R) = 1_S$	1. The remaining conditions are that $\boxed{f(1_R) = 1_S}$, and that $f(r_1 *_R r_2) = f(r_1) *_S f(r_2)$.
fong-spivak-invitation_F03694	317	■ 0.999	$f\text{\textbackslash left}(r_{\text{\textbackslash 1}} *_\text{\textbackslash R} r_{\text{\textbackslash 2}}\text{\textbackslash right})=f\text{\textbackslash left}(r_{\text{\textbackslash 1}}\text{\textbackslash right}) *_\text{\textbackslash S} f\text{\textbackslash left}(r_{\text{\textbackslash 2}}\text{\textbackslash right})$	$f(r_1 *_R r_2) = f(r_1) *_S f(r_2)$	1. The remaining conditions are that $f(1_R) = 1_S$, and that $\boxed{f(r_1 *_R r_2) = f(r_1) *_S f(r_2)}$.
fong-spivak-invitation_F03695	317	■ 0.998	$R=\text{\textbackslash left}(R, \text{\textbackslash 0}_{\text{\textbackslash R}},+_{\text{\textbackslash R}}, \text{\textbackslash 1}_{\text{\textbackslash R}}, *_\text{\textbackslash R}\text{\textbackslash right})$	$R = (R, 0_R, +_R, 1_R, *_R)$	it is initial means that, for any other rig $R = (R, 0_R, +_R, 1_R, *_R)$, there is a unique rig
fong-spivak-invitation_F03696	317	■ 1.000	$f\text{: } \text{\textbackslash mathbb{N}} \text{\textbackslash rrightarrow } R$	$f : \mathbb{N} \rightarrow R$	homomorphism $\boxed{f : \mathbb{N} \rightarrow R}$.
fong-spivak-invitation_F03697	318	■ 0.773	$\text{\textbackslash 0}_{\text{\textbackslash R}}, \text{\textbackslash 1}$	$0_R, 1$	What is this homomorphism? Well, to be a rig homomorphism, f must send 0 to 0_R , 1 to 1_R .
fong-spivak-invitation_F03698	318	■ 0.773	$\text{\textbackslash 1}_{\text{\textbackslash R}}$	1_R	What is this homomorphism? Well, to be a rig homomorphism, f must send 0 to 0_R , 1 to 1_R .
fong-spivak-invitation_F03699	318	■ 0.731	$f(m+n)=f(m)+_{\text{\textbackslash R}} f(n)$	$f(m + n) = f(m) +_R f(n)$	Furthermore, we must also have $f(m + n) = f(m) +_R f(n)$, and hence
fong-spivak-invitation_F03700	318	■ 1.000	$f(m *_\text{\textbackslash n})=f(m) *_\text{\textbackslash R} f(n)$	$f(m *_n) = f(m) *_R f(n)$	It remains to check $f(m *_n) = f(m) *_R f(n)$, and this will follow from distributivity. Noting
fong-spivak-invitation_F03701	318	■ 1.000	$f(m *_\text{\textbackslash n})$	$f(m *_n)$	that $f(m *_n)$ is equal to the sum of mn copies of 1_R , we have
fong-spivak-invitation_F03702	318	■ 1.000	$m \text{\textbackslash n}$	mn	that $f(m *_n)$ is equal to the sum of mn copies of 1_R , we have
fong-spivak-invitation_F03703	318	■ 1.000	$\text{\textbackslash varnothing } \text{\textbackslash rrightarrow } c$	$\emptyset \rightarrow c$	there is a unique map $\emptyset \rightarrow c$ coming from the initial object.
fong-spivak-invitation_F03704	318	■ 1.000	$c_{\text{\textbackslash 1}} \text{\textbackslash rrightarrow } c$	$c_1 \rightarrow c$	If c_1 is initial then by the universal property, for any c there is a unique morphism $c_1 \rightarrow c$; in
fong-spivak-invitation_F03705	318	■ 1.000	$c_{\text{\textbackslash 2}} \text{\textbackslash rrightarrow } c_{\text{\textbackslash 1}}$	$c_2 \rightarrow c_1$	a unique morphism $c_2 \rightarrow c_1$, call it g . But how do we know that f and g are mutually inverse?
fong-spivak-invitation_F03706	318	■ 0.999	$c_{\text{\textbackslash 1}} \text{\textbackslash rrightarrow } c_{\text{\textbackslash 1}}$	$c_1 \rightarrow c_1$	Well since c_1 is initial there is a unique morphism $c_1 \rightarrow c_1$. But we can think of two: id_{c_1} and
fong-spivak-invitation_F03707	318	■ 0.999	$\text{\textbackslash mathrm{id}}_{\text{\textbackslash c}_{\text{\textbackslash 1}}}$	id_{c_1}	Well since c_1 is initial there is a unique morphism $c_1 \rightarrow c_1$. But we can think of two: id_{c_1} and
fong-spivak-invitation_F03708 [conf 0.078]	318	■ 0.078	$=\text{\textbackslash mathrm{id}}_{\text{\textbackslash c}_{\text{\textbackslash 1}}}$	$= \text{id}_{c_1}$	$f \circ g$. Thus they must be equal. Similarly for c_2 , so we have $f \circ g = \text{id}_{c_1}$ and $g \circ f = \text{id}_{c_2}$, which
fong-spivak-invitation_F03709 [conf 0.078]	318	■ 0.078	$f=\text{\textbackslash mathrm{id}}_{\text{\textbackslash c}_{\text{\textbackslash 2}}}$	$f = \text{id}_{c_2}$	$f \circ g$. Thus they must be equal. Similarly for c_2 , so we have $f \circ g = \text{id}_{c_1}$ and $g \circ f = \text{id}_{c_2}$, which
fong-spivak-invitation_F03710	318	■ 1.000	$p+q$	$p + q$	Translating Definition 6.11 to this case, a coproduct $p + q$ in P is an element of P such that
fong-spivak-invitation_F03711	318	■ 1.000	$p \text{\textbackslash leq } p+q$	$p \leq p + q$	$p \leq p + q$ and $q \leq p + q$, and such that for all elements $x \in P$ with maps $p \leq x$ and $q \leq x$, we
fong-spivak-invitation_F03712	318	■ 1.000	$q \text{\textbackslash leq } p+q$	$q \leq p + q$	$p \leq p + q$ and $q \leq p + q$, and such that for all elements $x \in P$ with maps $p \leq x$ and $q \leq x$, we
fong-spivak-invitation_F03713	318	■ 1.000	$p+q \text{\textbackslash leq } x$	$p + q \leq x$	have $p + q \leq x$. But this says exactly that $p + q$ is a join: it is a least element above both p and
fong-spivak-invitation_F03714	319	■ 0.996	o	o	$\text{orange1} \mapsto o$
fong-spivak-invitation_F03715	319	■ 0.366	$\text{\textbackslash iota}_{\text{\textbackslash A}}\{ \text{\textbackslash }_{\text{\textbackslash 9}}^{\text{\textbackslash circ}}[f, g]=f$	$\iota_{A9}^\circ[f, g] = f$	1. The equation $\iota_A \circ [f, g] = f$ is the commutativity of the left-hand triangle in the commutative

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fong-spivak-invitation_F03716	319	■ 0.451	$\iota_{B\{\circ\}}[f, g] = g$	$\iota_B^\circ[f, g] = g$	2. The equation $\iota_B \circ [f, g] = g$ is the commutativity of the right-hand triangle in the commutative
fong-spivak-invitation_F03717	319	■ 0.428	$[f, g]: h = [f; h, g \circ h]$	$[f, g]: h = [f; h, g \circ h]$	3. The equation $[f, g] \circ h = [f \circ h, g \circ h]$ follows from the universal property of the coproduct.
fong-spivak-invitation_F03718	319	■ 0.998	$[f; h, g; h]: A+B \rightarrow D$	$[f; h, g; h]: A + B \rightarrow D$	commutes, and the universal property says there is a unique map $[f \circ h, g \circ h]: A + B \rightarrow D$
fong-spivak-invitation_F03719	319	■ 0.966	$[f, g]; h = [f; h, g \circ h]$	$[f, g]; h = [f; h, g \circ h]$	for which this occurs. Hence we must have $[f, g] \circ h = [f \circ h, g \circ h]$.
fong-spivak-invitation_F03720	319	■ 0.997	$+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	$+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$	1. To define a functor $+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ we must define its action on objects and morphisms. In
fong-spivak-invitation_F03721	319	■ 0.943	(A, B)	(A, B)	both cases, we just take the coproduct. If (A, B) is an object of $\mathcal{C} \times \mathcal{C}$, its image $A + B$ is, as
fong-spivak-invitation_F03722	319	■ 1.000	$(f, g): (A, B) \rightarrow (C, D)$	$(f, g): (A, B) \rightarrow (C, D)$	usual, the coproduct of the two objects of $\mathcal{C}$. If $(f, g): (A, B) \rightarrow (C, D)$ is a morphism, then
fong-spivak-invitation_F03723	319	■ 0.969	$f+g=\left[f\{\circ\} \iota_{\{C\}}, g\{\circ\} \iota_{\{D\}}\right]: A+B \rightarrow C+D$	$f + g = [f^\circ \iota_C, g^\circ \iota_D]: A + B \rightarrow C + D$	we can form a morphism $f + g = [f \circ \iota_C, g \circ \iota_D]: A + B \rightarrow C + D$, where $\iota_C: C \rightarrow C + D$
fong-spivak-invitation_F03724	319	■ 0.969	$\iota_{\{C\}}: C \rightarrow C+D$	$\iota_C: C \rightarrow C + D$	we can form a morphism $f + g = [f \circ \iota_C, g \circ \iota_D]: A + B \rightarrow C + D$, where $\iota_C: C \rightarrow C + D$
fong-spivak-invitation_F03725	319	■ 0.997	$\iota_{\{D\}}: D \rightarrow C+D$	$\iota_D: D \rightarrow C + D$	and $\iota_D: D \rightarrow C + D$ are the canonical morphisms given by the definition of the coproduct
fong-spivak-invitation_F03726	320	■ 0.491	$(h, k): (C, D) \rightarrow (E, F)$	$(h, k): (C, D) \rightarrow (E, F)$	have a morphism $(h, k): (C, D) \rightarrow (E, F)$ in $\mathcal{C} \times \mathcal{C}$. We need to show that $(f + g) \circ (h + k) =$
fong-spivak-invitation_F03727	320	■ 0.491	$(f+g)\{\circ\} (h+k)=$	$(f + g) \circ (h + k) =$	have a morphism $(h, k): (C, D) \rightarrow (E, F)$ in $\mathcal{C} \times \mathcal{C}$. We need to show that $(f + g) \circ (h + k) =$
fong-spivak-invitation_F03728	320	■ 0.755	$(f; h)+(g; k)$	$(f; h) + (g; k)$	$(f \circ h) + (g \circ k)$. This is a slightly more complicated version of the argument in Exercise 6.15
fong-spivak-invitation_F03729	320	■ 1.000	$(f; h)+$	$(f; h) +$	Indeed, we again use the uniqueness of the copairing in (6.1), this time to show that $(f \circ h) +$
fong-spivak-invitation_F03730	320	■ 0.158	$(g \circ k)=\left[f\{\circ\} h\{\circ\} \iota_{\{E\}}, g\{\circ\} k\{\circ\} \iota_{\{F\}}\right]=(f+g)\{\circ\} (h+k)$	$(g \circ k) = [f \circ h \circ \iota_E, g \circ k \circ \iota_F] = (f + g) \circ (h + k)$	$(f \circ h) + (g \circ k) = [f \circ h \circ \iota_E, g \circ k \circ \iota_F] = (f + g) \circ (h + k)$, as required.
fong-spivak-invitation_F03731	320	■ 0.998	$!_A: \varnothing \rightarrow A$	$!_A: \emptyset \rightarrow A$	2. Recall that the universal property of the initial object gives a unique map $!_A: \emptyset \rightarrow A$. Then
fong-spivak-invitation_F03732	320	■ 0.738	$\left[\mathrm{id}_A, !_A\right]$	$[\mathrm{id}_A, !_A]$	the copairing $[\mathrm{id}_A, !_A]$ is a map $A + \emptyset \rightarrow A$. Moreover, it is an isomorphism with inverse
fong-spivak-invitation_F03733	320	■ 0.592	$\{\circ\} A: A \rightarrow A+\varnothing$	$\circ A: A \rightarrow A + \emptyset$	$\iota_A: A \rightarrow A + \emptyset$. Indeed, using the properties in Exercise 6.15 and the universal property of
fong-spivak-invitation_F03734	320	■ 0.679	$\left[\mathrm{id}_A, !_A\right]=\mathrm{id}_A$	$[\mathrm{id}_A, !_A] = \mathrm{id}_A$	the initial object, we have $\iota_A \circ [\mathrm{id}_A, !_A] = \mathrm{id}_A$, and
fong-spivak-invitation_F03735	320	■ 0.986	$\left[!_A, \mathrm{id}_A\right]: \varnothing+A \rightarrow A$	$[_A, \mathrm{id}_A]: \emptyset + A \rightarrow A$	An analogous argument shows $[_A, \mathrm{id}_A]: \emptyset + A \rightarrow A$ is an isomorphism.
fong-spivak-invitation_F03736	320	■ 0.644	$\left[\mathrm{id}_A+\iota_B, \iota_C\right]=\left[\left[\iota_A, \iota_B\{\circ\} \iota_{\{B+C\}}\right], \iota_C\{\circ\} \iota_{\{B+C\}}\right]: (A+B)+C \rightarrow A+(B+C)$	$[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A + B) + C \rightarrow A + (B + C)$	(a) The map $[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A + B) + C \rightarrow A + (B + C)$
fong-spivak-invitation_F03737	320	■ 0.999	$\left[\iota_A, \iota_B+\mathrm{id}_C\right]: A+(B+C) \rightarrow (A+B)+C$	$[\iota_A, \iota_B + \mathrm{id}_C]: A + (B + C) \rightarrow (A + B) + C$	is an isomorphism, with inverse $[_A, \iota_B + \mathrm{id}_C]: A + (B + C) \rightarrow (A + B) + C$.
fong-spivak-invitation_F03738	320	■ 1.000	$\left[\iota_A, \iota_B\right]: A+B \rightarrow B+A$	$[\iota_A, \iota_B]: A + B \rightarrow B + A$	(b) The map $[_A, \iota_B]: A + B \rightarrow B + A$ is an isomorphism. Note our notation here is slightly
fong-spivak-invitation_F03739	320	■ 0.991	$\iota_A: A \rightarrow A+B$	$\iota_A: A \rightarrow A + B$	confusing: there are two maps named ι_A , (i) $\iota_A: A \rightarrow A + B$, and (ii) $\iota_A: A \rightarrow B + A$,
fong-spivak-invitation_F03740	320	■ 0.991	$\iota_A: A \rightarrow B+A$	$\iota_A: A \rightarrow B + A$	confusing: there are two maps named ι_A , (i) $\iota_A: A \rightarrow A + B$, and (ii) $\iota_A: A \rightarrow B + A$,
fong-spivak-invitation_F03741	320	■ 0.980	$\left[\iota_B, \iota_A\right]: B+A \rightarrow$	$[\iota_B, \iota_A]: B + A \rightarrow$	and similarly for ι_B . In the above we mean the map (ii). It has inverse $[_B, \iota_A]: B + A \rightarrow$
fong-spivak-invitation_F03742	320	■ 1.000	$A=B=C$	$A = B = C$	$A = B = C$, and the square consisting of all identities is its pushout.
fong-spivak-invitation_F03743	320	■ 1.000	$s^{\prime}$	s'	than one object, because if s' is another object then there is a morphism $s \rightarrow s'$, but the

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fong-spivak-invitation_F03744	320	■ 1.000	$s \rightarrow s^{\prime}$	$s \rightarrow s'$	than one object, because if s' is another object then there is a morphism $s \rightarrow s'$, but the
fong-spivak-invitation_F03745	320	■ 1.000	$s=s^{\prime}$	$s = s'$	only morphisms in S are identities so $s = s'$. Hence the set S must consist of exactly one
fong-spivak-invitation_F03746	321	■ 0.990	$\underline{5} \sqcup \underline{3}$	$\underline{5} \sqcup \underline{3}$	set of equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$. If we denote
fong-spivak-invitation_F03747	321	■ 0.990	$\{f(a) \sim g(a) \mid a \in \underline{4}\}$	$\{f(a) \sim g(a) \mid a \in \underline{4}\}$	set of equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$. If we denote
fong-spivak-invitation_F03748	321	■ 1.000	$\underline{5}$	$\underline{5}$	elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$, we can redraw the functions f, g :
fong-spivak-invitation_F03749	321	■ 1.000	$\{1, \ldots, 5\}$	$\{1, \dots, 5\}$	elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$, we can redraw the functions f, g :
fong-spivak-invitation_F03750	321	■ 1.000	$\underline{3}$	$\underline{3}$	elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$, we can redraw the functions f, g :
fong-spivak-invitation_F03751	321	■ 1.000	$\left\{1^{\prime}, 2^{\prime}, 3^{\prime}\right\}$	$\{1', 2', 3'\}$	elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$, we can redraw the functions f, g :
fong-spivak-invitation_F03752	321	■ 1.000	$5 \sim 3^{\prime}$	$5 \sim 3'$	$5 \sim 3'$. The equivalence classes are $\{1, 1', 3\}$, $\{2\}$, $\{4\}$, and $\{5, 2', 3'\}$. These four are exactly the
fong-spivak-invitation_F03753	321	■ 1.000	$\left\{1, 1^{\prime}, 3\right\},\{2\},\{4\}$	$\{1, 1', 3\}, \{2\}, \{4\}$	$5 \sim 3'$. The equivalence classes are $\{1, 1', 3\}$, $\{2\}$, $\{4\}$, and $\{5, 2', 3'\}$. These four are exactly the
fong-spivak-invitation_F03754	321	■ 1.000	$\left\{5, 2^{\prime}, 3^{\prime}\right\}$	$\{5, 2', 3'\}$	$5 \sim 3'$. The equivalence classes are $\{1, 1', 3\}$, $\{2\}$, $\{4\}$, and $\{5, 2', 3'\}$. These four are exactly the
fong-spivak-invitation_F03755	321	■ 1.000	$\varnothing \rightarrow X+Y$	$\emptyset \rightarrow X + Y$	1. The diagram to the left commutes because $\emptyset$ is initial, and so has a unique map $\emptyset \rightarrow X + Y$.
fong-spivak-invitation_F03756	321	■ 0.635	$f\{ \}_{9} \iota_X=g\{ \}_{9} \iota_Y$	$f_{9\iota_X} = g_{9\iota_Y}$	This implies we must have $f \circ \iota_X = g \circ \iota_Y$.
fong-spivak-invitation_F03757	321	■ 0.479	$t: X+\varnothing Y \rightarrow T$	$t: X + \emptyset Y \rightarrow T$	map $t: X + \emptyset Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$. Thus $X + \emptyset Y$ is the coproduct
fong-spivak-invitation_F03758	321	■ 0.479	$t=x$	$t = x$	map $t: X + \emptyset Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$. Thus $X + \emptyset Y$ is the coproduct
fong-spivak-invitation_F03759	321	■ 0.479	$t=y$	$t = y$	map $t: X + \emptyset Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$. Thus $X + \emptyset Y$ is the coproduct
fong-spivak-invitation_F03760	322	■ 1.000	A, B, X, Y	A, B, X, Y	That is, we need to show that S , together with the maps from A, B, X, Y , and Z , has the
fong-spivak-invitation_F03761	322	■ 1.000	$S \rightarrow T$	$S \rightarrow T$	We need to show there is a unique map $S \rightarrow T$ making everything commute. Since Q is a
fong-spivak-invitation_F03762	322	■ 1.000	$X \leftarrow A \rightarrow Y$	$X \leftarrow A \rightarrow Y$	pushout of $X \leftarrow A \rightarrow Y$, there is a unique map $Q \rightarrow T$ making a commutative triangle with
fong-spivak-invitation_F03763	322	■ 1.000	$Q \rightarrow T$	$Q \rightarrow T$	pushout of $X \leftarrow A \rightarrow Y$, there is a unique map $Q \rightarrow T$ making a commutative triangle with
fong-spivak-invitation_F03764	322	■ 1.000	$Y \leftarrow B \rightarrow Z$	$Y \leftarrow B \rightarrow Z$	Y , and since R is the pushout of $Y \leftarrow B \rightarrow Z$, there is a unique map $R \rightarrow T$ making a com-
fong-spivak-invitation_F03765	322	■ 1.000	$R \rightarrow T$	$R \rightarrow T$	Y , and since R is the pushout of $Y \leftarrow B \rightarrow Z$, there is a unique map $R \rightarrow T$ making a com-
fong-spivak-invitation_F03766	322	■ 1.000	(Y, Q, R, T)	(Y, Q, R, T)	mutative triangle with Y . This implies that there is a commuting (Y, Q, R, T) square, and hence
fong-spivak-invitation_F03767	322	■ 0.996	$X+\{ \}_{N} Y$	$X +_N Y$	The formula in Theorem 6.32 says that the pushout $X +_N Y$ is given by the set of equivalence
fong-spivak-invitation_F03768	322	■ 1.000	$X \sqcup N \sqcup Y$	$X \sqcup N \sqcup Y$	classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$
fong-spivak-invitation_F03769	322	■ 1.000	$x \sim n$	$x \sim n$	classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$
fong-spivak-invitation_F03770	322	■ 1.000	$x=f(n)$	$x = f(n)$	classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$
fong-spivak-invitation_F03771	322	■ 1.000	$y \sim n$	$y \sim n$	classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$, and $y \sim n$

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fong-spivak-invitation_F03772	322	■ 1.000	y=g(n)	$y = g(n)$	if $y = g(n)$, where $x \in X, y \in Y, n \in N$. Since for every $n \in N$ there exists an $x \in X$ such that
fong-spivak-invitation_F03773	322	■ 1.000	$x \in X, y \in Y, n \in N$	$x \in X, y \in Y, n \in N$	if $y = g(n)$, where $x \in X, y \in Y, n \in N$. Since for every $n \in N$ there exists an $x \in X$ such that
fong-spivak-invitation_F03774	322	■ 1.000	$n \in N$	$n \in N$	if $y = g(n)$, where $x \in X, y \in Y, n \in N$. Since for every $n \in N$ there exists an $x \in X$ such that
fong-spivak-invitation_F03775	323	■ 1.000	$B \rightarrowtail D \otimes D$	$B \rightarrow D \otimes D$	morphism $B \rightarrowtail D \otimes D$.
fong-spivak-invitation_F03776	323	■ 1.000	$g: X \rightarrowtail Y$	$g: X \rightarrow Y$	$f: X \rightarrow Y$ and $g: X \rightarrow Y$ such that
fong-spivak-invitation_F03777	324	■ 0.998	$\iota_1: X \rightarrowtail X+X$	$\iota_1: X \rightarrow X + X$	Write $\iota_1: X \rightarrow X + X$ for the map into the first copy of X in $X + X$, given by the definition of
fong-spivak-invitation_F03778	324	■ 0.998	$X+X$	$X + X$	Write $\iota_1: X \rightarrow X + X$ for the map into the first copy of X in $X + X$, given by the definition of
fong-spivak-invitation_F03779	324	■ 0.446	$\iota_1 \circ \{\mathrm{id}, \mathrm{id}\} = \mathrm{id}$	$\iota_1 \circ [\mathrm{id}, \mathrm{id}] = \mathrm{id}$	coproduct. Then, using the fact that $\iota_1 \circ [\mathrm{id}, \mathrm{id}] = \mathrm{id}$ from Exercise 6.15 1, and the commutativity
fong-spivak-invitation_F03780 [conf 0.080]	324	■ 0.080	$f = \iota_1$	$f = \iota_1$	of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] g = g$. This means that $f: X \rightarrow T$
fong-spivak-invitation_F03781 [conf 0.080]	324	■ 0.080	$\{\circlearrowleft\} f = \iota_1$	$\circlearrowleft f = \iota_1$	of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] g = g$. This means that $f: X \rightarrow T$
fong-spivak-invitation_F03782 [conf 0.080]	324	■ 0.080	$\left[\right]$	$[$	of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] g = g$. This means that $f: X \rightarrow T$
fong-spivak-invitation_F03783 [conf 0.080]	324	■ 0.080	$\{\circlearrowleft\} g = g$	$\circlearrowleft g = g$	of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] g = g$. This means that $f: X \rightarrow T$
fong-spivak-invitation_F03784 [conf 0.080]	324	■ 0.080	$f: X \rightarrowtail T$	$f: X \rightarrow T$	of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] g = g$. This means that $f: X \rightarrow T$
fong-spivak-invitation_F03785	324	■ 0.971	$X \rightarrowtail N \rightarrowtail Y$	$X \leftarrow N \rightarrow Y$	the objects of $\mathcal{C}$. Next, observe that a morphism in $\mathbf{Cospan}_F$ is a cospan $X \leftarrow N \rightarrow Y$ in $\mathcal{C}$
fong-spivak-invitation_F03786	324	■ 0.980	$FN = \{*\}$	$FN = \{*\}$	together with an element of $FN = \{*\}$. But FN also has a unique element, $*$! So there's no
fong-spivak-invitation_F03787	324	■ 0.980	FN	FN	together with an element of $FN = \{*\}$. But FN also has a unique element, $*$! So there's no
fong-spivak-invitation_F03788	324	■ 0.966	$\operatorname{Cospan}_F = \operatorname{Cospan}_{\mathcal{C}}$	$\operatorname{Cospan}_F = \operatorname{Cospan}_{\mathcal{C}}$	so $\operatorname{Cospan}_F = \operatorname{Cospan}_{\mathcal{C}}$.
fong-spivak-invitation_F03789	324	■ 0.974	$\operatorname{Cospan}_{\mathcal{C}} \rightarrowtail \operatorname{Cospan}_F$	$\operatorname{Cospan}_{\mathcal{C}} \rightarrow \operatorname{Cospan}_F$	morphism is the identity-on-objects functor $\mathbf{Cospan}_{\mathcal{C}} \rightarrow \mathbf{Cospan}_F$ that simply decorates each
fong-spivak-invitation_F03790	325	■ 0.848	$V = \{\mathrm{ul}, \mathrm{ur}, \mathrm{dl}, \mathrm{dr}\}$	$V = \{\mathrm{ul}, \mathrm{ur}, \mathrm{dl}, \mathrm{dr}\}$	We can represent the circuit in Eq. (6.11) by the tuple (V, A, s, t, ℓ) where $V = \{\mathrm{ul}, \mathrm{ur}, \mathrm{dl}, \mathrm{dr}\}$.
fong-spivak-invitation_F03791	325	■ 0.928	$A = \{\mathrm{r}1, \mathrm{r}2, \mathrm{r}3, \mathrm{c}1, \mathrm{i}1\}$	$A = \{\mathrm{r}1, \mathrm{r}2, \mathrm{r}3, \mathrm{c}1, \mathrm{i}1\}$	$A = \{\mathrm{r}1, \mathrm{r}2, \mathrm{r}3, \mathrm{c}1, \mathrm{i}1\}$, and s, t , and ℓ are defined by the table
fong-spivak-invitation_F03798	325	■ 1.000	$\psi_{\underline{2}, \underline{2}}(b, s)$	$\psi_{\underline{2}, \underline{2}}(b, s)$	The circuit $\psi_{\underline{2}, \underline{2}}(b, s)$ is the disjoint union of the two labeled graphs b and s :
fong-spivak-invitation_F03799	325	■ 0.555	$1 \rightarrowtail \underline{2} \stackrel{g}{\leftarrow} \underline{1}$	$1 \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$	The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$, where $f(1) = 1$ and $g(1) = 2$. The decoration is the
fong-spivak-invitation_F03800	325	■ 0.555	$f(1)=1$	$f(1) = 1$	The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$, where $f(1) = 1$ and $g(1) = 2$. The decoration is the
fong-spivak-invitation_F03801	325	■ 0.555	$g(1)=2$	$g(1) = 2$	The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$, where $f(1) = 1$ and $g(1) = 2$. The decoration is the
fong-spivak-invitation_F03802	325	■ 0.923	$\underline{2}, \{a\}, s, t, \ell$	$\underline{2}, \{a\}, s, t, \ell$	C -circuit $(\underline{2}, \{a\}, s, t, \ell)$, where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$.
fong-spivak-invitation_F03803	325	■ 0.923	$s(a)=1, t(a)=2$	$s(a) = 1, t(a) = 2$	C -circuit $(\underline{2}, \{a\}, s, t, \ell)$, where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$.
fong-spivak-invitation_F03804	325	■ 0.923	$\ell(a)=$	$\ell(a) =$	C -circuit $(\underline{2}, \{a\}, s, t, \ell)$, where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$.
fong-spivak-invitation_F03805	325	■ 1.000	$C := (V, A, s, t, \ell)$	$C := (V, A, s, t, \ell)$	Recall the circuit $C := (V, A, s, t, \ell)$ from the solution to Exercise 6.64. Then the first decorated

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fong-spivak-invitation_F03806	325	■ 0.940	$\underline{1} \xrightarrow{\quad} V \stackrel{\text{g}}{\leftarrow} \underline{2}, f(1) = \text{ul}, g(1) =$ $\underline{2}, f(1)=\mathrm{ul}, g(1)=$	$\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) =$	cospan is given by the cospan $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) =$ ur, and $g(2) = \text{ur}$, decorated
fong-spivak-invitation_F03807	325	■ 0.940	$g(2)=$	$g(2) =$	cospan is given by the cospan $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) = \text{ur}$, and $g(2) = \text{ur}$, decorated
fong-spivak-invitation_F03808	325	■ 0.975	$\underline{2} \xrightarrow{\quad} V^{\prime} \stackrel{\text{g}^{\prime}}{\leftarrow} \underline{2}$ $\stackrel{\text{g}^{\prime}}{\leftarrow} \underline{2}$	$\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$	by circuit C. The second decorated cospan is given by the cospan $\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$ and the circuit
fong-spivak-invitation_F03809	325	■ 1.000	$C^{\prime}:=\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)$ $C^{\prime}:=\left(V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime}\right)$	$C' := (V', A', s', t', \ell')$	$C' := (V', A', s', t', \ell')$, where $V' = \{l, r, d\}$, $A' = \{r1', r2'\}$, and the functions are given by the
fong-spivak-invitation_F03810	325	■ 1.000	$V^{\prime}=\left\{l, r, d\right\}, A^{\prime}=\left\{\mathrm{r} 1^{\prime}, \mathrm{r} 2^{\prime}\right\}$ $V^{\prime}=\{l, r, d\}, A^{\prime}=\{\mathrm{r} 1', \mathrm{r} 2'\}$	$V' = \{l, r, d\}, A' = \{r1', r2'\}$	$C' := (V', A', s', t', \ell')$, where $V' = \{l, r, d\}, A' = \{r1', r2'\}$, and the functions are given by the
fong-spivak-invitation_F03815	325	■ 0.989	$V \stackrel{\text{g}}{\leftarrow} \underline{2} \xrightarrow{\quad} V^{\prime}$ $V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$	$V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$	To compose these, we first take the pushout of $V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$. This gives the a new apex
fong-spivak-invitation_F03816	325	■ 0.998	$V^{\prime} \backslash \text{prime} \text{prime}=\left\{\mathrm{ul}, \mathrm{dl}, \mathrm{dr}, \mathrm{m}, \mathrm{r}\right\}$ $V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$	$V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$	$V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) =$
fong-spivak-invitation_F03817	325	■ 0.998	$\underline{1} \xrightarrow{\quad} V^{\prime} \stackrel{\text{k}}{\leftarrow} \underline{2}$ $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$	$\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$	$V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) =$
fong-spivak-invitation_F03818	325	■ 0.998	$h(1)=$	$h(1) =$	$V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) =$
fong-spivak-invitation_F03819	325	■ 0.388	$k(1)=\mathrm{r}$	$k(1) = \text{r}$	ul, $k(1) = \text{r}$ and $k(2) = \text{m}$. The new circuit is given by $(V, '' A + A', s, '' t, '' \ell'')$ where the
fong-spivak-invitation_F03820	325	■ 0.388	$k(2)=\mathrm{m}$	$k(2) = \text{m}$	ul, $k(1) = \text{r}$ and $k(2) = \text{m}$. The new circuit is given by $(V, '' A + A', s, '' t, '' \ell'')$ where the
fong-spivak-invitation_F03821	325	■ 0.388	$\left(V, '' A + A^{\prime}, s, '' t, '' \ell^{\prime \prime}\right)$ $(V, '' A + A', s, '' t, '' \ell'')$	$(V, '' A + A', s, '' t, '' \ell'')$	ul, $k(1) = \text{r}$ and $k(2) = \text{m}$. The new circuit is given by $(V, '' A + A', s, '' t, '' \ell'')$ where the
fong-spivak-invitation_F03825	326	■ 1.000	η	η	Composing η and x we have
fong-spivak-invitation_F03826	327	■ 0.637	$2, 2, 2, 2 ; \emptyset$	$2, 2, 2, 2; 0$	3. The composite $g \circ_1 f$ has arity $(2, 2, 2, 2; 0)$; there is a depiction on the left:
fong-spivak-invitation_F03827	327	■ 0.898	$\left(A, B, A^{\prime}, B^{\prime}\right)$	(A, B, A', B')	We need to show that the (A, B, A', B') square is a pullback iff the (A, C, A', C') rectangle is a
fong-spivak-invitation_F03828	327	■ 0.898	$\left(A, C, A^{\prime}, C^{\prime}\right)$	(A, C, A', C')	We need to show that the (A, B, A', B') square is a pullback iff the (A, C, A', C') rectangle is a
fong-spivak-invitation_F03829	327	■ 0.608	X, p, q	X, p, q	Suppose first that (A, B, A', B') is a pullback, and take any (X, p, q) as in the following
fong-spivak-invitation_F03830	327	■ 0.274	$g^{\prime}=p$	$g' = p$	where $q \circ f' \circ g' = p \circ h_3$. Then by the universal property of the (B, C, B', C') pullback, we get
fong-spivak-invitation_F03831	327	■ 0.274	$h_{\{3\}}$	h_3	where $q \circ f' \circ g' = p \circ h_3$. Then by the universal property of the (B, C, B', C') pullback, we get
fong-spivak-invitation_F03832	327	■ 0.229	$r ; h_{\{2\}}=g$	$r; h_2 = g$	In other words $r \circ h_2 = g \circ f'$ and $r \circ g = p$. Then by the universal property of the (A, B, A', B')
fong-spivak-invitation_F03833	327	■ 0.229	$r ; g=p$	$r; g = p$	In other words $r \circ h_2 = g \circ f'$ and $r \circ g = p$. Then by the universal property of the (A, B, A', B')
fong-spivak-invitation_F03834	327	■ 1.000	$r^{\prime}: X \rightarrow A$	$r' : X \rightarrow A$	pullback, we get a unique dotted arrow $r' : X \rightarrow A$ making the right-hand diagram commute, i.e.
fong-spivak-invitation_F03835	327	■ 0.999	$r^{\prime} ; f=r$	$r'; f = r$	$r' \circ f = r$ and $r' \circ h_1 = q$. This gives the existence of an r with the required property, $r' \circ f = r$
fong-spivak-invitation_F03836	327	■ 0.999	$r^{\prime} ; h_{\{1\}}=q$	$r'; h_1 = q$	$r' \circ f = r$ and $r' \circ h_1 = q$. This gives the existence of an r with the required property, $r' \circ f = r$
fong-spivak-invitation_F03837	328	■ 1.000	$r^{\prime} ; f ; g=r ; g=p$	$r'; f; g = r; g = p$	and $r' \circ f \circ g = r \circ g = p$. To see uniqueness, suppose given another morphisms r_0 such that
fong-spivak-invitation_F03838	328	■ 1.000	$r_{\{0\}}$	r_0	and $r' \circ f \circ g = r \circ g = p$. To see uniqueness, suppose given another morphisms r_0 such that

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fong-spivak-invitation_F03839	328	■ 0.279	$r_{\{0\}} \stackrel{\circ}{\circ} \text{\circ} f \text{\circ} g = p$	$r_0 \stackrel{\circ}{\circ} f \text{\circ} g = p$	$r_0 \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$:
fong-spivak-invitation_F03840	328	■ 0.279	$r_{\{0\}} \stackrel{\circ}{\circ} \text{\circ} h_1 = q$	$r_0 \stackrel{\circ}{\circ} h_1 = q$	$r_0 \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$:
fong-spivak-invitation_F03841	328	■ 0.413	$r_{\{0\}} \stackrel{\circ}{\circ} \text{\circ} f = r$	$r_0 \stackrel{\circ}{\circ} f = r$	Then by the uniqueness of r , we must have $r_0 \circledcirc f = r$, and then by the uniqueness of r' , we must
fong-spivak-invitation_F03842	328	■ 0.413	$r^{\wedge\{\prime\}}$	r'	Then by the uniqueness of r , we must have $r_0 \circledcirc f = r$, and then by the uniqueness of r' , we must
fong-spivak-invitation_F03843	328	■ 1.000	$r_{\{0\}} = r^{\wedge\{\prime\}}$	$r_0 = r'$	have $r_0 = r'$. This proves the first result.
fong-spivak-invitation_F03844	328	■ 0.823	$B, C, B^{\wedge\{\prime\}}, C^{\wedge\{\prime\}}$	B, C, B', C'	The second is similar. Suppose that (A, C, A', C') and (B, C, B', C') are pullbacks and
fong-spivak-invitation_F03845	328	■ 0.824	$r ; h_{\{2\}} = q ; f^{\wedge\{\prime\}}$	$r; h_2 = q; f'$	i.e. where $r \circledcirc h_2 = q \circledcirc f'$. Then letting $p := r \circledcirc g$, we have
fong-spivak-invitation_F03846	328	■ 0.824	$p := r ; g$	$p := r; g$	i.e. where $r \circledcirc h_2 = q \circledcirc f'$. Then letting $p := r \circledcirc g$, we have
fong-spivak-invitation_F03847	328	■ 0.666	$r^{\wedge\{\prime\}}: X \rightarrow$	$r': X \rightarrow$	so by the universal property of the (A, C, A', C') pullback, there is a unique morphism $r': X \rightarrow$
fong-spivak-invitation_F03848	328	■ 0.427	$r^{\wedge\{\prime\}} ; f ; g = p$	$r'; f; g = p$	A such that $r' \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$, as shown:
fong-spivak-invitation_F03849	328	■ 0.427	$r_{\{0\}}: h_{\{1\}} = q$	$r_0: h_1 = q$	A such that $r' \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$, as shown:
fong-spivak-invitation_F03850	328	■ 0.104	$r_{\{0\}} := r^{\wedge\{\prime\}} \{ \}_{9}^{\wedge\{\circ\}} f$	$r_0 := r'^{\circ}_9 f$	But now let $r_0 := r' \circledcirc f$. It satisfies $r_0 \circledcirc g = p$ and $r_0 \circledcirc h_2 = q \circledcirc f'$, and r satisfies the same
fong-spivak-invitation_F03851	328	■ 0.104	$r_{\{0\}} \{ \}_{9}^{\wedge\{\circ\}} g = p$	$r_0 \stackrel{\circ}{\circ} g = p$	But now let $r_0 := r' \circledcirc f$. It satisfies $r_0 \circledcirc g = p$ and $r_0 \circledcirc h_2 = q \circledcirc f'$, and r satisfies the same
fong-spivak-invitation_F03852	328	■ 0.104	$r_{\{0\}} \{ \}_{9}^{\wedge\{\circ\}} h_{\{2\}} = q$	$r_0 \stackrel{\circ}{\circ} h_2 = q$	But now let $r_0 := r' \circledcirc f$. It satisfies $r_0 \circledcirc g = p$ and $r_0 \circledcirc h_2 = q \circledcirc f'$, and r satisfies the same
fong-spivak-invitation_F03853	328	■ 1.000	$a_{\{1\}}, a_{\{2\}} \in A$	$a_1, a_2 \in A$	A function $f: A \rightarrow B$ is injective iff for all $\boxed{a_1, a_2 \in A}$, if $f(a_1) = f(a_2)$ then $a_1 = a_2$. It is a
fong-spivak-invitation_F03854	328	■ 1.000	$f \text{\left(} a_{\{1\}} \text{\right)} = f \text{\left(} a_{\{2\}} \text{\right)}$	$f(a_1) = f(a_2)$	A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$, if $\boxed{f(a_1) = f(a_2)}$ then $a_1 = a_2$. It is a
fong-spivak-invitation_F03855	328	■ 1.000	$a_{\{1\}} = a_{\{2\}}$	$a_1 = a_2$	A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$ then $a_1 = a_2$. It is a
fong-spivak-invitation_F03856	328	■ 0.908	$g_{\{1\}}, g_{\{2\}}: X \rightarrow A$	$g_1, g_2: X \rightarrow A$	monomorphism iff for all sets X and functions $g_1, g_2: X \rightarrow A$, if $g_1 \circledcirc f = g_2 \circledcirc f$ then $g_1 = g_2$.
fong-spivak-invitation_F03857	328	■ 0.908	$f = g_{\{2\}}$	$f = g_2$	monomorphism iff for all sets X and functions $g_1, g_2: X \rightarrow A$, if $g_1 \circledcirc f = g_2 \circledcirc f$ then $g_1 = g_2$.
fong-spivak-invitation_F03858	328	■ 0.908	$g_{\{1\}} = g_{\{2\}}$	$g_1 = g_2$	monomorphism iff for all sets X and functions $g_1, g_2: X \rightarrow A$, if $g_1 \circledcirc f = g_2 \circledcirc f$ then $g_1 = g_2$.
fong-spivak-invitation_F03859	329	■ 1.000	$X = \{*\}$	$X = \{*\}$	Let $X = \{*\}$ be a one-element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and
fong-spivak-invitation_F03860	329	■ 1.000	$g_{\{1\}}(*) := a_{\{1\}}$	$g_1(*) := a_1$	Let $X = \{*\}$ be a one-element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and
fong-spivak-invitation_F03861	329	■ 0.509	$g_{\{2\}}(*) := a_{\{2\}}$	$g_2(*) := a_2$	$\boxed{g_2(*) := a_2}$. Then $g_1 \circledcirc f = g_2 \circledcirc f$, so $g_1 = g_2$, so $a_1 = a_2$.
fong-spivak-invitation_F03862	329	■ 0.378	$f =$	$f =$	2. Suppose that f is an injection, let X be any set, and let $g_1, g_2: X \rightarrow A$ be such that $g_1 \circledcirc f =$
fong-spivak-invitation_F03863	329	■ 0.653	$g_{\{1\}}(x) = g_{\{2\}}(x)$	$g_1(x) = g_2(x)$	$g_2 \circledcirc f$. We will have $g_1 = g_2$ if we can show that $\boxed{g_1(x) = g_2(x)}$ for every $x \in X$. So take any
fong-spivak-invitation_F03864	329	■ 1.000	$f \text{\left(} g_{\{1\}}(x) \text{\right)} = f \text{\left(} g_{\{2\}}(x) \text{\right)}$	$f(g_1(x)) = f(g_2(x))$	$x \in X$; since $\boxed{f(g_1(x)) = f(g_2(x))}$ and f is injective, we have $g_1(x) = g_2(x)$ as desired.
fong-spivak-invitation_F03865	329	■ 0.605	$j := i^{\wedge\{-1\}}$	$j := i^{-1}$	Let $j := i^{-1}$ be the inverse of i , and consider $g := (f \circledcirc j): A \rightarrow B'$. Then $g \circledcirc i = f$, so by the
fong-spivak-invitation_F03866	329	■ 0.605	$g := (f: j): A \rightarrow B^{\wedge\{\prime\}}$	$g := (f: j): A \rightarrow B'$	Let $j := i^{-1}$ be the inverse of i , and consider $g := (f \circledcirc j): A \rightarrow B'$. Then $g \circledcirc i = f$, so by the

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fong-spivak-invitation_F03867	329	0.605	$g: i=f$	$g : i = f$	Let $j := i^{-1}$ be the inverse of i , and consider $g := (f \circ j) : A \rightarrow B'$. Then $g \circ i = f$, so by the
fong-spivak-invitation_F03868	329	0.990	$j^{\wedge\prime} : A \rightarrow A^{\prime}$	$j' : A \rightarrow A'$	existence part of the universal property, there is a map $j' : A \rightarrow A'$ such that $j' \circ i' = \text{id}_A$ and
fong-spivak-invitation_F03869	329	0.990	$j^{\wedge\prime} ; i^{\wedge\prime} = \text{id}_{A^{\prime}}$	$j'; i' = \text{id}_A$	existence part of the universal property, there is a map $j' : A \rightarrow A'$ such that $j' \circ i' = \text{id}_A$ and
fong-spivak-invitation_F03870	329	0.357	$j^{\wedge\prime} \{ \}_9 f^{\wedge\prime} = f \{ \}_9^{\circ} j$	$j'_9 f' = f'_9 j$	$j' \circ f' = f \circ j$. We will be done if we can show $i' \circ j' = \text{id}_{A'}$. One checks that $(i' \circ j') \circ i' = i'$ and
fong-spivak-invitation_F03871	329	0.357	$i^{\wedge\prime} \{ \}_9 j^{\wedge\prime} = \text{id}_{A^{\prime}}$	$i'_9 j' = \text{id}_{A'}$	$j' \circ f' = f \circ j$. We will be done if we can show $i' \circ j' = \text{id}_{A'}$. One checks that $(i' \circ j') \circ i' = i'$ and
fong-spivak-invitation_F03872	329	0.357	$\left(i^{\wedge\prime} \{ \}_9 j^{\wedge\prime} \right) \{ \}_9 i^{\wedge\prime} = i^{\wedge\prime}$	$(i'_9 j')_9 i' = i'$	$j' \circ f' = f \circ j$. We will be done if we can show $i' \circ j' = \text{id}_{A'}$. One checks that $(i' \circ j') \circ i' = i'$ and
fong-spivak-invitation_F03873	329	0.184	$\left(i^{\wedge\prime} \{ \}_9 j^{\wedge\prime} \right) \{ \}_9^{\circ} f^{\wedge\prime} = i^{\wedge\prime} f^{\wedge\prime} \{ \}_9^{\circ} j = f^{\wedge\prime} \{ \}_9^{\circ} i^{\wedge\prime} = f^{\wedge\prime}$	$(i'_9 j')_9 f' = i'_9 f'_9 j = f'_9 i'_9 j = f'$	that $(i' \circ j') \circ f' = i' \circ f \circ j = f \circ i \circ j = f'$. But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$
fong-spivak-invitation_F03874	329	0.184	$\text{id}_{A^{\prime}}$	$\text{id}_{A'}$	that $(i' \circ j') \circ f' = i' \circ f \circ j = f \circ i \circ j = f'$. But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$
fong-spivak-invitation_F03875	329	0.184	$\text{id}_{A^{\prime}} \{ \}_9 i^{\wedge\prime} = i^{\wedge\prime}$	$\text{id}_{A'} \circ i' = i'$	that $(i' \circ j') \circ f' = i' \circ f \circ j = f \circ i \circ j = f'$. But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$
fong-spivak-invitation_F03876	329	0.774	$\text{id}_{A^{\prime}} \{ \}_9 f^{\wedge\prime} = f^{\wedge\prime}$	$\text{id}_{A'} \circ f' = f'$	and $\text{id}_{A'} \circ f' = f'$, so by the uniqueness part of the universal property, $(i' \circ j') = \text{id}_{A'}$.
fong-spivak-invitation_F03877	329	0.774	$\left(i^{\wedge\prime} ; j^{\wedge\prime} \right) = \text{id}_{A^{\prime}}$	$(i'; j') = \text{id}_{A'}$	and $\text{id}_{A'} \circ f' = f'$, so by the uniqueness part of the universal property, $(i' \circ j') = \text{id}_{A'}$.
fong-spivak-invitation_F03878	329	0.669	$g: X \rightarrow A$	$g : X \rightarrow A$	So take any object X and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$.
fong-spivak-invitation_F03879	329	0.669	$h: X \rightarrow B$	$h : X \rightarrow B$	So take any object X and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$.
fong-spivak-invitation_F03880	329	0.669	$f=h$	$f = h$	So take any object X and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$.
fong-spivak-invitation_F03881	329	0.669	id_B	id_B	So take any object X and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$.
fong-spivak-invitation_F03882	329	0.650	$r: X \rightarrow A$	$r : X \rightarrow A$	We need to show there is a unique morphism $r : X \rightarrow A$ such that $r \circ \text{id}_A = g$ and $r \circ f = h$.
fong-spivak-invitation_F03883	329	0.650	$r \{ \}_9^{\circ} \text{id}_A = g$	$r'_9 \text{id}_A = g$	We need to show there is a unique morphism $r : X \rightarrow A$ such that $r \circ \text{id}_A = g$ and $r \circ f = h$.
fong-spivak-invitation_F03884	329	0.650	$r: f=h$	$r : f = h$	We need to show there is a unique morphism $r : X \rightarrow A$ such that $r \circ \text{id}_A = g$ and $r \circ f = h$.
fong-spivak-invitation_F03885	329	1.000	$r=g$	$r = g$	That's easy: the first requirement forces $r = g$ and the second requirement is then fulfilled.
fong-spivak-invitation_F03886	330	1.000	$v \otimes w$	$v \otimes w$	• $v \otimes w$ serves as a meet operation, i.e. it satisfies the same universal property as $\wedge$, namely
fong-spivak-invitation_F03887	330	0.958	$v \leq (w \multimap x)$	$v \leq (w \multimap x)$	does, namely $v \leq (w \multimap x)$ iff $v \otimes w \leq x$. So $\mathcal{V}$ is a cartesian closed category, and of course
fong-spivak-invitation_F03888	330	0.958	$v \otimes W \leq x$	$v \otimes W \leq x$	does, namely $v \leq (w \multimap x)$ iff $v \otimes W \leq x$. So $\mathcal{V}$ is a cartesian closed category, and of course
fong-spivak-invitation_F03889	330	0.991	$\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$	$\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$	This is the product preorder $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$: its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq$
fong-spivak-invitation_F03890	330	0.991	$(a, b) \in \mathbb{N} \times \mathbb{N}$	$(a, b) \in \mathbb{N} \times \mathbb{N}$	This is the product preorder $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$: its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq$
fong-spivak-invitation_F03891	330	0.991	$(a, b) \leq$	$(a, b) \leq$	This is the product preorder $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$: its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq$
fong-spivak-invitation_F03892	330	0.600	$\left(a^{\wedge\prime}, b^{\wedge\prime} \right)$	(a', b')	(a', b') iff, in the usual ordering on $\mathbb{N}$, we have $a' \leq a$ and $b' \leq b$. But you can just look at the
fong-spivak-invitation_F03893	330	0.600	$a^{\wedge\prime} \leq a$	$a' \leq a$	(a', b') iff, in the usual ordering on $\mathbb{N}$, we have $a' \leq a$ and $b' \leq b$. But you can just look at the

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fong-spivak-invitation_F03894	330	0.600	$b^{\{\prime\}}$ $\leq b$	$b' \leq b$	(a', b') iff, in the usual ordering on $\mathbb{N}$, we have $a' \leq a$ and $b' \leq b$. But you can just look at the
fong-spivak-invitation_F03895	330	0.859	$(a, b) \wedge (a^{\{\prime\}}, b^{\{\prime\}}) = (\max(a, a^{\{\prime\}}), \max(b, b^{\{\prime\}}))$	$(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$	It has a top element, $(0, 0)$, and it has binary meets, $(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$.
fong-spivak-invitation_F03896	330	0.998	$x \multimap y$	$x \multimap y$	for each x, y , the hom-object $x \multimap y$ exists. The formula for it is $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$,
fong-spivak-invitation_F03897	330	0.998	$x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$	$x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$	for each x, y , the hom-object $x \multimap y$ exists. The formula for it is $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$.
fong-spivak-invitation_F03898	330	1.000	$y \wedge x \leq y$	$y \wedge x \leq y$	i.e. we need these particular joins to exist. Since $y \wedge x \leq y$, we have $y \leq x \multimap y$. So we can
fong-spivak-invitation_F03899	330	1.000	$y \leq x \multimap y$	$y \leq x \multimap y$	i.e. we need these particular joins to exist. Since $y \wedge x \leq y$, we have $y \leq x \multimap y$. So we can
fong-spivak-invitation_F03900	330	1.000	$x \multimap y = \bigvee \{w \mid y \leq w \text{ and } w \wedge x \leq y\}$	$x \multimap y = \bigvee \{w \mid y \leq w$	replace the formula with $x \multimap y = \bigvee \{w \mid y \leq w \text{ and } w \wedge x \leq y\}$. But the set of elements
fong-spivak-invitation_F03901	330	1.000	$w \wedge x \leq y\}$	$w \wedge x \leq y\}$	replace the formula with $x \multimap y = \bigvee \{w \mid y \leq w \text{ and } w \wedge x \leq y\}$. But the set of elements
fong-spivak-invitation_F03902	330	0.543	$\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$	$\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$	in $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$ that are bigger than y is finite and nonempty. ² So this is a finite nonempty join,
fong-spivak-invitation_F03903	330	0.651	$\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$	$\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$	and $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$ has all finite nonempty joins: they are given by inf.
fong-spivak-invitation_F03904	330	1.000	$y = (a, b)$	$y = (a, b)$	² If $y = (a, b)$ then there are exactly $(a + 1) * (b + 1)$ elements y' for which $y \leq y'$.
fong-spivak-invitation_F03905	330	1.000	$(a+1) * (b+1)$	$(a + 1) * (b + 1)$	² If $y = (a, b)$ then there are exactly $(a + 1) * (b + 1)$ elements y' for which $y \leq y'$.
fong-spivak-invitation_F03906	331	0.997	$m: \mathbb{Z} \rightarrow \mathbb{B}$	$m: \mathbb{Z} \rightarrow \mathbb{B}$	Let $m: \mathbb{Z} \rightarrow \mathbb{B}$ be the characteristic function of the inclusion $\mathbb{N} \subseteq \mathbb{Z}$.
fong-spivak-invitation_F03907	331	0.997	$\mathbb{N} \subseteq \mathbb{Z}$	$\mathbb{N} \subseteq \mathbb{Z}$	Let $m: \mathbb{Z} \rightarrow \mathbb{B}$ be the characteristic function of the inclusion $\mathbb{N} \subseteq \mathbb{Z}$.
fong-spivak-invitation_F03908	331	0.992	$\ulcorner m \urcorner(-5) =$	$\ulcorner m \urcorner(-5) =$	1. $\ulcorner m \urcorner(-5) = \text{false}$.
fong-spivak-invitation_F03909	331	1.000	$\ulcorner m \urcorner(0) =$	$\ulcorner m \urcorner(0) =$	2. $\ulcorner m \urcorner(0) = \text{true}$.
fong-spivak-invitation_F03910	331	0.615	$\ulcorner !\mathbb{N} \urcorner: \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner !\mathbb{N} \urcorner: \mathbb{N} \rightarrow \mathbb{B}$	2. Let $!\mathbb{N}: \emptyset \rightarrow \mathbb{N}$ be the inclusion of the empty set. The characteristic function $\ulcorner !\mathbb{N} \urcorner: \mathbb{N} \rightarrow \mathbb{B}$
fong-spivak-invitation_F03911	331	1.000	$A \subseteq \mathbb{B}$	$A \subseteq \mathbb{B}$	1. The sort of thing $(*)$ we're looking for is a subobject of $\mathbb{B}$, say $A \subseteq \mathbb{B}$. This would have a
fong-spivak-invitation_F03912	331	0.999	$\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	$\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$	3. The characteristic function for $P \Rightarrow Q$ is the function $\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ given by the first,
fong-spivak-invitation_F03913	331	0.859	$\subseteq \mathbb{B} \times \mathbb{B}$	$\subseteq \mathbb{B} \times \mathbb{B}$	4. It classifies the subset $\{(true, true), (false, true), (false, false)\}$ $\subseteq \mathbb{B} \times \mathbb{B}$.
fong-spivak-invitation_F03914	331	0.980	$\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$	$\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$	Say that $\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$ classify respectively the subsets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$,
fong-spivak-invitation_F03915	331	1.000	$P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$	$P := \{n \in \mathbb{N} \mid n$	$P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$, and $T := \{n \in \mathbb{N} \mid n \geq 10\}$ of $\mathbb{N}$.
fong-spivak-invitation_F03916	331	0.999	$\ulcorner E \urcorner(17) =$	$\ulcorner E \urcorner(17) =$	1. $\ulcorner E \urcorner(17) = \text{false}$ because 17 is not even.
fong-spivak-invitation_F03917	331	0.865	$\ulcorner P \urcorner(17) =$	$\ulcorner P \urcorner(17) =$	2. $\ulcorner P \urcorner(17) = \text{true}$ because 17 is prime.
fong-spivak-invitation_F03918	331	0.690	$\ulcorner T \urcorner(17) = \text{true}$	$\ulcorner T \urcorner(17) = \text{true}$	3. $\ulcorner T \urcorner(17) = \text{true}$ because $17 \geq 10$.
fong-spivak-invitation_F03919	331	0.690	$17 \geq 10$	$17 \geq 10$	3. $\ulcorner T \urcorner(17) = \text{true}$ because $17 \geq 10$.
fong-spivak-invitation_F03920	331	0.995	$B(x, \epsilon) := \{x' \in \mathbb{R} \mid$	$B(x, \epsilon) := \{x' \in \mathbb{R} \mid$	1. The one-dimensional analogue of an ϵ -ball around a point $x \in \mathbb{R}$ is $B(x, \epsilon) := \{x' \in \mathbb{R} \mid$
fong-spivak-invitation_F03921	331	1.000	$ x - x' < \epsilon\}$	$ x - x' < \epsilon\}$	$ x - x' < \epsilon$, i.e. the set of all points within ϵ of x .

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fong-spivak-invitation_F03922	331	■ 1.000	$x \in U$	$x \in U$	2. A subset $U \subseteq \mathbb{R}$ is open if, for every $x \in U$, there is some $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$.
fong-spivak-invitation_F03923	331	■ 1.000	$B(x, \epsilon) \subseteq U$	$B(x, \epsilon) \subseteq U$	2. A subset $U \subseteq \mathbb{R}$ is open if, for every $x \in U$, there is some $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$.
fong-spivak-invitation_F03924	331	■ 1.000	$U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$	$U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$	3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$. Then $U := U_1 \cup U_2 = \{x \in$
fong-spivak-invitation_F03925	331	■ 1.000	$U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$	$U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$	3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$. Then $U := U_1 \cup U_2 = \{x \in$
fong-spivak-invitation_F03926	331	■ 1.000	$U := U_1 \cup U_2 = \{x \in$	$U := U_1 \cup U_2 = \{x \in$	3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$. Then $U := U_1 \cup U_2 = \{x \in$
fong-spivak-invitation_F03927	331	■ 1.000	$\mathbb{R} \mid 0 < x < 3\}$	$\mathbb{R} \mid 0 < x < 3\}$	$\mathbb{R} \mid 0 < x < 3\}$.
fong-spivak-invitation_F03928	332	■ 1.000	$I = \{1, 2, 3, 4, \dots\}$	$I = \{1, 2, 3, 4, \dots\}$	4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$, so we have
fong-spivak-invitation_F03929	332	■ 1.000	$U_i := \left\{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\right\}$	$U_i := \left\{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\right\}$	4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$, so we have
fong-spivak-invitation_F03930	332	■ 1.000	$U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$	$U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$	$U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$. Their union is $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$.
fong-spivak-invitation_F03931	332	■ 1.000	$U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$	$U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$	$U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$. Their union is $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$.
fong-spivak-invitation_F03932	332	■ 1.000	$\varnothing \subseteq X$	$\varnothing \subseteq X$	1. The coarse topology on X is the one whose only open sets are $X \subseteq X$ and $\varnothing \subseteq X$. This is a
fong-spivak-invitation_F03933	332	■ 1.000	$\bigcup_{i \in I} A_i$	$\bigcup_{i \in I} A_i$	(the union $\bigcup_{i \in I} A_i$ is X iff $A_i = X$ for some $i \in I$).
fong-spivak-invitation_F03934	332	■ 1.000	$A_i = X$	$A_i = X$	(the union $\bigcup_{i \in I} A_i$ is X iff $A_i = X$ for some $i \in I$).
fong-spivak-invitation_F03935	332	■ 1.000	$A \subseteq X$	$A \subseteq X$	2. The fine topology on X is the one where every subset $A \subseteq X$ is considered open. All the
fong-spivak-invitation_F03936	332	■ 1.000	$f^{-1}(U) \subseteq X$	$f^{-1}(U) \subseteq X$	$f^{-1}(U) \subseteq X$ is open, and everything in X is open.
fong-spivak-invitation_F03937	332	■ 1.000	$\varnothing \rightarrow \{1\} \rightarrow \{1, 2\}$	$\varnothing \rightarrow \{1\} \rightarrow \{1, 2\}$	1. The Hasse diagram for the Sierpinski topology is $\varnothing \rightarrow \{1\} \rightarrow \{1, 2\}$.
fong-spivak-invitation_F03938	332	■ 1.000	$U_i = U$	$U_i = U$	• or $U_i = U$ for some $i \in I$.
fong-spivak-invitation_F03939	332	■ 0.996	$Y = B \cap Y$	$Y = B \cap Y$	1. We want to show that $Y \in \mathbf{Op}_{\mathcal{T} \cap Y}$. We need to find $B \in \mathbf{Op}$ such that $Y = B \cap Y$; this is easy,
fong-spivak-invitation_F03940	332	■ 1.000	$B = X$	$B = X$	take $B = X$.
fong-spivak-invitation_F03941	332	■ 1.000	$\varnothing = \varnothing \cap Y$	$\varnothing = \varnothing \cap Y$	trary union. Note $\varnothing = \varnothing \cap Y$, so according to the formula, $\varnothing \in \mathbf{Op}_{\mathcal{T} \cap Y}$. Suppose that
fong-spivak-invitation_F03942	332	■ 1.000	$\varnothing \in \mathbf{Op}_{\mathcal{T} \cap Y}$	$\varnothing \in \mathbf{Op}_{\mathcal{T} \cap Y}$	trary union. Note $\varnothing = \varnothing \cap Y$, so according to the formula, $\varnothing \in \mathbf{Op}_{\mathcal{T} \cap Y}$. Suppose that
fong-spivak-invitation_F03943	332	■ 0.929	$A_1, A_2 \in \mathbf{Op}_{\mathcal{T} \cap Y}$	$A_1, A_2 \in \mathbf{Op}_{\mathcal{T} \cap Y}$	$A_1, A_2 \in \mathbf{Op}_{\mathcal{T} \cap Y}$. Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$.
fong-spivak-invitation_F03944	332	■ 0.929	$B_1, B_2 \in \mathbf{Op}$	$B_1, B_2 \in \mathbf{Op}$	$A_1, A_2 \in \mathbf{Op}_{\mathcal{T} \cap Y}$. Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$.
fong-spivak-invitation_F03945	332	■ 0.929	$A_1 = B_1 \cap Y$	$A_1 = B_1 \cap Y$	$A_1, A_2 \in \mathbf{Op}_{\mathcal{T} \cap Y}$. Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$.
fong-spivak-invitation_F03946	332	■ 0.929	$A_2 = B_2 \cap Y$	$A_2 = B_2 \cap Y$	$A_1, A_2 \in \mathbf{Op}_{\mathcal{T} \cap Y}$. Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$.
fong-spivak-invitation_F03947	332	■ 0.892	$A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) = (B_1 \cap B_2) \cap Y$	$A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) = (B_1 \cap B_2) \cap Y$	But then $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) = (B_1 \cap B_2) \cap Y$, so it is in $\mathbf{Op}_{\mathcal{T} \cap Y}$ since
fong-spivak-invitation_F03948	332	■ 0.999	$B_1 \cap B_2 \in \mathbf{Op}$	$B_1 \cap B_2 \in \mathbf{Op}$	$B_1 \cap B_2 \in \mathbf{Op}$. The same idea works for arbitrary unions: given a set I and A_i for each $i \in I$,

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fong-spivak-invitation_F03949	332	■ 0.999	$A_{\{i\}}$	A_i	$B_1 \cap B_2 \in \mathbf{Op}$. The same idea works for arbitrary unions: given a set I and A_i for each $i \in I$,
fong-spivak-invitation_F03950	332	■ 1.000	$A_{\{i\}} = B_{\{i\}} \cap Y$	$A_i = B_i \cap Y$	we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$, and
fong-spivak-invitation_F03951	332	■ 1.000	$B_{\{i\}} \in \mathbf{Op}$	$B_i \in \mathbf{Op}$	we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$, and
fong-spivak-invitation_F03952	332	■ 1.000	$U \subseteq \mathbf{Op}$	$U \subseteq \mathbf{Op}$	each labeled by an open set $U \subseteq \mathbf{Op}$. It might look something like this:
fong-spivak-invitation_F03953	333	■ 1.000	$\mathcal{C}(B, C) = \left((U_3 \wedge U_1) \vee (U_4 \wedge U_2) \right)$	$\mathcal{C}(B, C) = (U_3 \wedge U_1) \vee (U_4 \wedge U_2)$	$\mathcal{C}(B, C) = (U_3 \wedge U_1) \vee (U_4 \wedge U_2)$, because $\wedge$ is the monoidal product in $\mathcal{V}$.
fong-spivak-invitation_F03954	333	■ 1.000	$\mathcal{C}(a, b)$	$\mathcal{C}(a, b)$	In general, we can thus imagine the open set $\mathcal{C}(a, b)$ as a kind of “size restriction” for getting
fong-spivak-invitation_F03955	333	■ 1.000	$\left\{ a_{\{1\}}, a_{\{2\}} \right\}$	$\{a_1, a_2\}$	1. The fiber of f over a is $\{a_1, a_2\}$.
fong-spivak-invitation_F03956	333	■ 1.000	$\left\{ c_{\{1\}} \right\}$	$\{c_1\}$	2. The fiber of f over c is $\{c_1\}$.
fong-spivak-invitation_F03957	334	■ 0.999	$\operatorname{Sec}_f(V_2) = \varnothing$	$\operatorname{Sec}_f(V_2) = \varnothing$	2. When $V_2 = \{a, b, c, d\}$, there are no sections: $\operatorname{Sec}_f(V_2) = \varnothing$.
fong-spivak-invitation_F03958	334	■ 0.827	$2 * 3 * 1 * 2 = 12$	$2 * 3 * 1 * 2 = 12$	3. When $V_3 = \{a, b, d, e\}$, the set $\operatorname{Sec}_f(V_3)$ has $2 * 3 * 1 * 2 = 12$ elements.
fong-spivak-invitation_F03959	334	■ 0.998	$g_{\{1\}} := \left(a_{\{1\}}, b_{\{1\}} \right)$	$g_1 := (a_1, b_1)$	1. Let $g_1 := (a_1, b_1)$ and $g_2 := (b_2, e_1)$; these do not agree on the overlap.
fong-spivak-invitation_F03960	334	■ 0.998	$g_{\{2\}} := \left(b_{\{2\}}, e_{\{1\}} \right)$	$g_2 := (b_2, e_1)$	1. Let $g_1 := (a_1, b_1)$ and $g_2 := (b_2, e_1)$; these do not agree on the overlap.
fong-spivak-invitation_F03961	334	■ 1.000	$g \in \operatorname{Sec}_f(U_1 \cup U_2)$	$g \in \operatorname{Sec}_f(U_1 \cup U_2)$	2. No, there’s no section $g \in \operatorname{Sec}_f(U_1 \cup U_2)$ for which $g _{U_1} = g_1$ and $g _{U_2} = g_2$.
fong-spivak-invitation_F03962	334	■ 1.000	$g _{U_1} = g_1$	$g _{U_1} = g_1$	2. No, there’s no section $g \in \operatorname{Sec}_f(U_1 \cup U_2)$ for which $g _{U_1} = g_1$ and $g _{U_2} = g_2$.
fong-spivak-invitation_F03963	334	■ 1.000	$g _{U_2} = g_2$	$g _{U_2} = g_2$	2. No, there’s no section $g \in \operatorname{Sec}_f(U_1 \cup U_2)$ for which $g _{U_1} = g_1$ and $g _{U_2} = g_2$.
fong-spivak-invitation_F03964	335	■ 0.997	$F(\{1, 2\}) \rightarrow F(\{1\}) \rightarrow$	$F(\{1, 2\}) \rightarrow F(\{1\}) \rightarrow$	2. A presheaf F on $\mathbf{Op}$ consists of any three sets and any two functions $F(\{1, 2\}) \rightarrow F(\{1\}) \rightarrow$
fong-spivak-invitation_F03965	335	■ 1.000	$F(\varnothing)$	$F(\varnothing)$	$F(\varnothing)$ between them.
fong-spivak-invitation_F03966	335	■ 1.000	$F(\varnothing) \cong \{1\}$	$F(\varnothing) \cong \{1\}$	4. As explained in Example 7.28, F will be a sheaf iff $F(\varnothing) \cong \{1\}$. Thus the category of sheaves
fong-spivak-invitation_F03967	335	■ 1.000	$F(\{1, 2\}) \rightarrow F(\{1\})$	$F(\{1, 2\}) \rightarrow F(\{1\})$	is equivalent to that of just two sets and one function $F(\{1, 2\}) \rightarrow F(\{1\})$.
fong-spivak-invitation_F03968	335	■ 0.748	$S \in \mathbf{Shv}(X)$	$S \in \mathbf{Shv}(X)$	The one-point space $X = \{1\}$ has two open sets, $\varnothing$ and $\{1\}$, and every sheaf $S \in \mathbf{Shv}(X)$ assigns
fong-spivak-invitation_F03969	335	■ 0.999	$S(\varnothing) = \{()\}$	$S(\varnothing) = \{()\}$	$S(\varnothing) = \{()\}$ by the sheaf condition (see Example 7.28). So the only data in a sheaf $S \in \mathbf{Shv}(X)$ is
fong-spivak-invitation_F03970	335	■ 1.000	$S(\{1\})$	$S(\{1\})$	the set $S(\{1\})$. This is how we get the correspondence between sets and sheaves on the one-point
fong-spivak-invitation_F03971	335	■ 0.862	$\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow$	$\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow$	According to Eq. (7.13), the subobject classifier $\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$ in $\mathbf{Shv}(X)$ should be the
fong-spivak-invitation_F03972	335	■ 0.862	$\mathbf{Shv}(X)$	$\mathbf{Shv}(X)$	According to Eq. (7.13), the subobject classifier $\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$ in $\mathbf{Shv}(X)$ should be the
fong-spivak-invitation_F03973	335	■ 0.945	$\Omega(\{1\})$	$\Omega(\{1\})$	functor where $\Omega(\{1\})$ is the set of open sets of $\{1\}$. So we’re hoping to see that there is a one-
fong-spivak-invitation_F03974	335	■ 0.445	$\mathbf{Op}(\{1\})$	$\mathbf{Op}(\{1\})$	to-one correspondence between the set $\mathbf{Op}(\{1\})$ and the set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans.
fong-spivak-invitation_F03975	335	■ 1.000	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$	By Eqs. (7.13) and (7.14) the definition of $\Omega(U)$ is $\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$, and the
fong-spivak-invitation_F03976	335	■ 1.000	$U' \mapsto U' \cap V$	$U' \mapsto U' \cap V$	definition of the restriction map for $V \subseteq U$ is $U' \mapsto U' \cap V$.

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fong-spivak-invitation_F03977	335	■ 0.996	$U^{\{\prime\}}$ $\subseteq U$	$U' \subseteq U$	1. It is functorial: given $W \subseteq V \subseteq U$ and $U' \subseteq U$, we indeed have $(U' \cap V) \cap W = U' \cap W$,
fong-spivak-invitation_F03978	335	■ 0.996	$\left(U^{\{\prime\}} \cap V\right) \cap W = U^{\{\prime\}} \cap W$	$(U' \cap V) \cap W = U' \cap W$	1. It is functorial: given $W \subseteq V \subseteq U$ and $U' \subseteq U$, we indeed have $\boxed{U' \cap V \cap W = U' \cap W}$,
fong-spivak-invitation_F03979	335	■ 1.000	$W \subseteq V$	$W \subseteq V$	since $W \subseteq V$. For functoriality, we also need preservation of identities, and this amounts to
fong-spivak-invitation_F03980	335	■ 1.000	$U^{\{\prime\}} \cap U = U^{\{\prime\}}$	$U' \cap U = U'$	$U' \cap U = U'$ for all $U' \subseteq U$.
fong-spivak-invitation_F03981	335	■ 1.000	$G^{\{\prime\}}$	G'	There is only one that classifies G' , and here it is. Let's write $\gamma := \ulcorner G' \urcorner$.
fong-spivak-invitation_F03982	335	■ 1.000	$\gamma := \left\langle G^{\{\prime\}} \right\rangle$	$\gamma := \ulcorner G' \urcorner$	There is only one that classifies G' , and here it is. Let's write $\gamma := \ulcorner G' \urcorner$.
fong-spivak-invitation_F03983	335	■ 1.000	$\gamma(D) = 0$	$\gamma(D) = 0$	• Since D is missing from G' , we have $\gamma(D) = 0$ (vertex: missing).
fong-spivak-invitation_F03984	335	■ 1.000	A, B, C	A, B, C	• Since vertices A, B, C are present in G' we have $\gamma(A) = \gamma(B) = \gamma(C) = V$ (vertex: present).
fong-spivak-invitation_F03985	335	■ 1.000	$\gamma(A) = \gamma(B) = \gamma(C) = V$	$\gamma(A) = \gamma(B) = \gamma(C) = V$	• Since vertices A, B, C are present in G' we have $\boxed{\gamma(A) = \gamma(B) = \gamma(C) = V}$ (vertex: present).
fong-spivak-invitation_F03986	335	■ 1.000	$\gamma(i) = (V, 0; 0)$	$\gamma(i) = (V, 0; 0)$	• The above forces $\boxed{\gamma(i) = (V, 0; 0)}$ (arrow from present vertex to missing vertex: missing).
fong-spivak-invitation_F03987	335	■ 1.000	$\gamma(f) = (V, V; A)$	$\gamma(f) = (V, V; A)$	• Since the arrow f is in G' , we have $\boxed{\gamma(f) = (V, V; A)}$ (arrow from present vertex to present
fong-spivak-invitation_F03988	336	■ 1.000	$\gamma(g) = \gamma(h) = (V, V; 0)$	$\gamma(g) = \gamma(h) = (V, V; 0)$	• Since the arrows g and h are missing in G' , we have $\gamma(g) = \gamma(h) = (V, V; 0)$ (arrow from
fong-spivak-invitation_F03989	336	■ 1.000	$U = \mathbb{R} - \{0\} \subseteq \mathbb{R}$	$U = \mathbb{R} - \{0\} \subseteq \mathbb{R}$	With $\boxed{U = \mathbb{R} - \{0\} \subseteq \mathbb{R}}$, we have the following.
fong-spivak-invitation_F03990	336	■ 1.000	$\mathbb{R} - U = \{0\}$	$\mathbb{R} - U = \{0\}$	1. The complement of U is $\mathbb{R} - U = \{0\}$ and $\neg U$ is its interior, which is $\neg U = \emptyset$.
fong-spivak-invitation_F03991	336	■ 1.000	$\neg U = \varnothing$	$\neg U = \emptyset$	1. The complement of U is $\mathbb{R} - U = \{0\}$ and $\neg U$ is its interior, which is $\neg U = \emptyset$.
fong-spivak-invitation_F03992	336	■ 1.000	$\mathbb{R} - \neg U = \mathbb{R}$	$\mathbb{R} - \emptyset = \mathbb{R}$	2. The complement of $\neg U$ is $\mathbb{R} - \emptyset = \mathbb{R}$, and this is open, so $\neg \neg U = \mathbb{R}$.
fong-spivak-invitation_F03993	336	■ 1.000	$\neg \neg U = \mathbb{R}$	$\neg \neg U = \mathbb{R}$	2. The complement of $\neg U$ is $\mathbb{R} - \emptyset = \mathbb{R}$, and this is open, so $\neg \neg U = \mathbb{R}$.
fong-spivak-invitation_F03994	336	■ 0.869	$\neg \neg U \subseteq U$	$\neg \neg U \subseteq U$	4. It is false that $\neg \neg U \subseteq U$.
fong-spivak-invitation_F03995	336	■ 0.976	$V \in \mathbf{Op}$	$V \in \mathbf{Op}$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$,
fong-spivak-invitation_F03996	336	■ 0.976	$\top \wedge V = V$	$\top \wedge V = V$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$,
fong-spivak-invitation_F03997	336	■ 0.976	$V = X$	$V = X$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$,
fong-spivak-invitation_F03998	336	■ 0.976	$\top \wedge X := \top \cap X = X$	$\top \wedge X := \top \cap X = X$	1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$,
fong-spivak-invitation_F03999	336	■ 1.000	$\top = \top \cap X = X$	$\top = \top \cap X = X$	but anything intersected with X is itself, so $\top = \top \cap X = X$.
fong-spivak-invitation_F04000	336	■ 0.897	$(\top \vee V) := (X \cup V) = X$	$(\top \vee V) := (X \cup V) = X$	2. $(\top \vee V) := (X \cup V) = X$ holds, $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because
fong-spivak-invitation_F04001	336	■ 0.897	$(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$	$(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$	2. $(\top \vee V) := (X \cup V) = X$ holds, $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because
fong-spivak-invitation_F04002	336	■ 1.000	$(X \cap V) \subseteq X$	$(X \cap V) \subseteq X$	$(X \cap V) \subseteq X$, and $(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$ holds because $R \cap X = R$.
fong-spivak-invitation_F04003	336	■ 1.000	$(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$	$(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$	$(X \cap V) \subseteq X$, and $\boxed{X \Rightarrow V = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y}$ holds because $R \cap X = R$.
fong-spivak-invitation_F04004	336	■ 1.000	$R \cap X = R$	$R \cap X = R$	$(X \cap V) \subseteq X$, and $(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$ holds because $R \cap X = R$.

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fong-spivak-invitation-F04005	336	■ 0.992	$(\perp \vee V) = V$	$(\perp \vee V) = V$	3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$, then when $V = \emptyset$ we have $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$
fong-spivak-invitation-F04006	336	■ 0.992	$V = \emptyset$	$V = \emptyset$	3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$, then when $V = \emptyset$ we have $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$
fong-spivak-invitation-F04007	336	■ 0.992	$(\perp \vee \varnothing) := (\perp \cup \varnothing) =$	$(\perp \vee \emptyset) := (\perp \cup \emptyset) =$	3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$, then when $V = \emptyset$ we have $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$
fong-spivak-invitation-F04008	336	■ 0.679	$\perp = \perp \cup \varnothing = \varnothing$	$\perp = \perp \cup \emptyset = \emptyset$	$\emptyset$, but anything in a union with $\emptyset$ is itself, so $\perp = \perp \cup \emptyset = \emptyset$.
fong-spivak-invitation-F04009	336	■ 0.995	$(\perp \wedge V) = (\varnothing \cap V) = \varnothing$	$(\perp \wedge V) = (\emptyset \cap V) = \emptyset$	4. $\perp \wedge V = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op} R \cap \emptyset \subseteq V} R = X$ holds because
fong-spivak-invitation-F04010	336	■ 0.995	$(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op} R \cap \emptyset \subseteq V} R = X$	$(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op} R \cap \emptyset \subseteq V} R = X$	4. $(\perp \wedge V) = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op} R \cap \emptyset \subseteq V} R = X$ holds because
fong-spivak-invitation-F04011	336	■ 0.835	$(X \cap \varnothing) \subseteq V$	$(X \cap \emptyset) \subseteq V$	$(X \cap \emptyset) \subseteq V$.
fong-spivak-invitation-F04012	336	■ 1.000	$\{S \mid p\}$	$\{S \mid p\}$	time is a person who is alive throughout that interval. A section in the subobject $\{S \mid p\}$ over any
fong-spivak-invitation-F04013	336	■ 0.508	$S = \mathbb{N}$	$S = \mathbb{N}$	ing to the set $S = \mathbb{N}$, let $p(s)$ be the predicate “ $24 \leq s \leq 28$,” and let $q(s)$ be the predicate “ s is
fong-spivak-invitation-F04014	336	■ 0.508	$\leq s \leq 28$	$\leq s \leq 28$	ing to the set $S = \mathbb{N}$, let $p(s)$ be the predicate “ $24 \leq s \leq 28$,” and let $q(s)$ be the predicate “ s is
fong-spivak-invitation-F04015	336	■ 0.508	$q(s)$	$q(s)$	ing to the set $S = \mathbb{N}$, let $p(s)$ be the predicate “ $24 \leq s \leq 28$,” and let $q(s)$ be the predicate “ s is
fong-spivak-invitation-F04016	336	■ 1.000	$p(n, z)$	$p(n, z)$	We have the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ given by $p(n, z)$ iff $n \leq z $.
fong-spivak-invitation-F04017	336	■ 1.000	$n \leq z $	$n \leq z $	We have the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ given by $p(n, z)$ iff $n \leq z $.
fong-spivak-invitation-F04018	336	■ 1.000	$\{0\} \subseteq \mathbb{N}$	$\{0\} \subseteq \mathbb{N}$	1. The predicate $\forall (z: \mathbb{Z}). p(n, z)$ holds for $\{0\} \subseteq \mathbb{N}$.
fong-spivak-invitation-F04019	336	■ 1.000	$\mathbb{N} \subseteq \mathbb{N}$	$\mathbb{N} \subseteq \mathbb{N}$	2. The predicate $\exists (z: \mathbb{Z}). p(n, z)$ holds for $\mathbb{N} \subseteq \mathbb{N}$.
fong-spivak-invitation-F04020	336	■ 1.000	$\varnothing \subseteq \mathbb{Z}$	$\emptyset \subseteq \mathbb{Z}$	3. The predicate $\forall (n: \mathbb{N}). p(n, z)$ holds for $\emptyset \subseteq \mathbb{Z}$.
fong-spivak-invitation-F04021	336	■ 1.000	$\mathbb{Z} \subseteq \mathbb{Z}$	$\mathbb{Z} \subseteq \mathbb{Z}$	4. The predicate $\exists (n: \mathbb{N}). p(n, z)$ holds for $\mathbb{Z} \subseteq \mathbb{Z}$.
fong-spivak-invitation-F04022	337	■ 0.532	$\forall (t: T)$	$\forall (t: T)$	1. The formula says that $\forall (t: T). p(s, t)$ “returns the largest open set $V \subseteq U$ for which
fong-spivak-invitation-F04023	337	■ 1.000	$t_i \in T(V_i)$	$t_i \in T(V_i)$	for which there exists some $t_i \in T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$.” Substituting, this becomes
fong-spivak-invitation-F04024	337	■ 1.000	t_i	t_i	“the union of all time intervals V_i for which there is some item t_i in the news about which s
fong-spivak-invitation-F04025	337	■ 1.000	$j(j(q)) \leq j(q)$	$j(j(q)) \leq j(q)$	It is clear that if $j(j(q)) = j(q)$ then $j(j(q)) \leq j(q)$ by reflexivity. On the other hand, assume
fong-spivak-invitation-F04026	337	■ 0.999	$p := j(q)$	$p := j(q)$	$p := j(q)$ we have both $j(p) \leq p$ and $p \leq j(p)$. This means $p \cong j(p)$, but Ω is a poset (not
fong-spivak-invitation-F04027	337	■ 0.999	$j(p) \leq p$	$j(p) \leq p$	$p := j(q)$ we have both $j(p) \leq p$ and $p \leq j(p)$. This means $p \cong j(p)$, but Ω is a poset (not
fong-spivak-invitation-F04028	337	■ 0.999	$p \cong j(p)$	$p \cong j(p)$	$p := j(q)$ we have both $j(p) \leq p$ and $p \leq j(p)$. This means $p \cong j(p)$, but Ω is a poset (not
fong-spivak-invitation-F04029	337	■ 1.000	$p = j(p)$	$p = j(p)$	just a preorder) so $p = j(p)$, i.e. $j(j(q)) = j(q)$ as desired.
fong-spivak-invitation-F04030	337	■ 0.912	$s \in S(U), p(s)$	$s \in S(U), p(s)$	2. Let U be the interval 2019/01/01 – 2019/02/01. For an arbitrary person $s \in S(U)$, $p(s)$ is a
fong-spivak-invitation-F04031	338	■ 0.916	$j(j(p(s))) = j(p(s))$	$j(j(p(s))) = j(p(s))$	5. It is true that $j(j(p(s))) = j(p(s))$, because suppose we are given a time during which “if Bob
fong-spivak-invitation-F04032	338	■ 0.890	$0 \leq 2 \leq 6 \leq 8$	$0 \leq 2 \leq 6 \leq 8$	1. Since $0 \leq 2 \leq 6 \leq 8$, we have $[2, 6] \in o_{[0,8]}$ by the above formula.

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fong-spivak-invitation_F04033	338	■ 0.890	$[2,6] \in 0_{[0,8]}$	$[2,6] \in O_{[0,8]}$	1. Since $0 \leq 2 \leq 6 \leq 8$, we have $[2,6] \in o_{[0,8]}$ by the above formula.
fong-spivak-invitation_F04034	338	■ 0.603	$\text{\texttt{\textbackslash in^{\text{? } } o_{\{[0,5]\} \cup o_{[4,8]}}}$	$\in^? o_{[0,5]} \cup o_{[4,8]}$	2. In order to have $[2,6] \in^? o_{[0,5]} \cup o_{[4,8]}$, we would need to have either $[2,6] \in^? o_{[0,5]}$ or
fong-spivak-invitation_F04035	338	■ 0.603	$\text{\texttt{\textbackslash in^{\text{? } } o_{\{[0,5]\}}}$	$\in^? o_{[0,5]}$	2. In order to have $[2,6] \in^? o_{[0,5]} \cup o_{[4,8]}$, we would need to have either $[2,6] \in^? o_{[0,5]}$ or
fong-spivak-invitation_F04036	338	■ 1.000	$[2,6] \in^{\text{?}} o_{[4,8]}$	$[2,6] \in^? o_{[4,8]}$	$[2,6] \in^? o_{[4,8]}$. But the formula does not hold in either case.
fong-spivak-invitation_F04037	338	■ 0.984	$\mathbb{R} \subseteq \mathbb{I}$	$\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$	A subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$ iff there is an open set $U' \subseteq \mathbb{I}\mathbb{R}$
fong-spivak-invitation_F04038	338	■ 0.984	$U^{\prime} \subseteq \mathbb{I}$	$U' \subseteq \mathbb{I}\mathbb{R}$	A subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{I}\mathbb{R}$ iff there is an open set $U' \subseteq \mathbb{I}\mathbb{R}$
fong-spivak-invitation_F04039	338	■ 1.000	$U=U^{\prime} \cap \mathbb{R}$	$U = U' \cap \mathbb{R}$	with $U = U' \cap \mathbb{R}$. We want to show that this is the case iff U is open in the usual topology.
fong-spivak-invitation_F04040	338	■ 1.000	$U^{\prime}=\bigcup_{i \in I} o_{\left[a_i, b_i\right]}$	$U' = \bigcup_{i \in I} o_{[a_i, b_i]}$	union of some basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$, where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{I}\mathbb{R} \mid a_i < d < u <$
fong-spivak-invitation_F04041	338	■ 1.000	$o_{\left[a_i, b_i\right]}=\left\{[d, u] \in \mathbb{R} \mid a_i<d<u<\right.$	$o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u <$	union of some basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$, where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{I}\mathbb{R} \mid a_i < d < u <$
fong-spivak-invitation_F04042	338	■ 0.956	$\left. b_i\right\}$	$b_i\}$	$b_i\}$. Since $\mathbb{R} = \{[x, x] \in \mathbb{I}\mathbb{R}\}$, the intersection $U' \cap \mathbb{R}$ will then be
fong-spivak-invitation_F04043	338	■ 0.956	$\mathbb{R}=\{[x, x] \in \mathbb{I} \mathbb{R}\}$	$\mathbb{R} = \{[x, x] \in \mathbb{I}\mathbb{R}\}$	$b_i\}$. Since $\mathbb{R} = \{[x, x] \in \mathbb{I}\mathbb{R}\}$, the intersection $U' \cap \mathbb{R}$ will then be
fong-spivak-invitation_F04044	338	■ 0.956	$U^{\prime} \cap \mathbb{R}$	$U' \cap \mathbb{R}$	$b_i\}$. Since $\mathbb{R} = \{[x, x] \in \mathbb{I}\mathbb{R}\}$, the intersection $U' \cap \mathbb{R}$ will then be
fong-spivak-invitation_F04045	338	■ 1.000	$B\left(m_i, r_i\right)$	$B\left(m_i, r_i\right)$	and this is just the union of open balls $B\left(m_i, r_i\right)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i :=$
fong-spivak-invitation_F04046	338	■ 1.000	$m_i:=\frac{a_i+b_i}{2}$	$m_i := \frac{a_i + b_i}{2}$	and this is just the union of open balls $B\left(m_i, r_i\right)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i :=$
fong-spivak-invitation_F04047	338	■ 1.000	$r_i:=$	$r_i :=$	and this is just the union of open balls $B\left(m_i, r_i\right)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i :=$
fong-spivak-invitation_F04048	338	■ 0.500	$\frac{b_i-a_i}{2}$	$\frac{b_i - a_i}{2}$	$\frac{b_i-a_i}{2}$ is the radius of the interval (a_i, b_i) . The open balls $B\left(m_i, r_i\right)$ are open in the usual topology
fong-spivak-invitation_F04049	338	■ 0.500	$\left(a_i, b_i\right)$	$\left(a_i, b_i\right)$	$\frac{b_i-a_i}{2}$ is the radius of the interval (a_i, b_i) . The open balls $B\left(m_i, r_i\right)$ are open in the usual topology
fong-spivak-invitation_F04050	338	■ 1.000	$U=\bigcup_{j \in J} B\left(m_j, \epsilon_j\right)$	$U = \bigcup_{j \in J} B\left(m_j, \epsilon_j\right)$	Suppose that U is open in the usual topology. Then $U = \bigcup_{j \in J} B\left(m_j, \epsilon_j\right)$ for some set J . Let
fong-spivak-invitation_F04051	338	■ 1.000	$a_j:=m_j-\epsilon_j$	$a_j := m_j - \epsilon_j$	$a_j := m_j - \epsilon_j$ and $b_j := m_j + \epsilon_j$. Then
fong-spivak-invitation_F04052	338	■ 1.000	$b_j:=m_j+\epsilon_j$	$b_j := m_j + \epsilon_j$	$a_j := m_j - \epsilon_j$ and $b_j := m_j + \epsilon_j$. Then
fong-spivak-invitation_F04053	338	■ 0.991	$H_X(U):=\{f: U \cap R \rightarrow X \mid f$	$H_X(U) := \{f : U \cap R \rightarrow X \mid f$	$H_X(U) := \{f : U \cap R \rightarrow X \mid f$ is continuous}.
fong-spivak-invitation_F04054	338	■ 1.000	$f: U \cap R \rightarrow$	$f : U \cap R \rightarrow$	1. H_X is a presheaf: given $V \subseteq U$ the restriction map sends the continuous function $f : U \cap R \rightarrow$
fong-spivak-invitation_F04055	338	■ 1.000	$V \cap R \subseteq U \cap R$	$V \cap R \subseteq U \cap R$	X to its restriction along the subset $V \cap R \subseteq U \cap R$.
fong-spivak-invitation_F04056	338	■ 1.000	$U=\bigcup_i U_i$	$U = \bigcup_i U_i$	2. It is a sheaf: given any family U_i of open sets with $U = \bigcup_i U_i$ and a continuous function
fong-spivak-invitation_F04057	338	■ 1.000	$f_i: U_i \cap R \rightarrow X$	$f_i : U_i \cap R \rightarrow X$	$f_i : U_i \cap R \rightarrow X$ for each i , agreeing on overlaps, they can be glued together to give a
fong-spivak-invitation_F04058	338	■ 1.000	$U \cap R$	$U \cap R$	continuous function on all of $U \cap R$, since $U \cap R = (\bigcup_i U_i) \cap R = \bigcup_i (U_i \cap R)$.

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fong-spivak-invitation_F04659	338	■ 1.000	$U \cap R = \left(\bigcup_i U_i \right) \cap R = \bigcup_i (U_i \cap R)$	$U \cap R = \left(\bigcup_i U_i \right) \cap R = \bigcup_i (U_i \cap R)$	continuous function on all of $U \cap R$, since $U \cap R = (\bigcup_i U_i) \cap R = \bigcup_i (U_i \cap R)$.